MAgent Theory 2.0

Gauge-Covariant Agents
and Inherited Organization

Robert C. Dennis

Independent Researcher

October 2026

Abstract

An agent holds beliefs and generative models, receives records, makes predictions, and changes its laws. We formulate these objects on associated statistical bundles, with independent directed interactions that may depend on the current belief and model laws. On regular statistical families, the joint-law Fisher metric and a connection induce a covariant pullback tensor on the contextual base. A collective least-squares construction selects a connection class from declared comparison data. Exact identities distinguish information lost under a channel from changes caused by incompatible connections, and characterize when a parent retention preserves the collective connection; when both levels use collective minimizers, the two retention identities reduce to one.

Interacting agents admit three objectives, which we keep separate. In a global free energy all agents are parts of one normalized joint, and local updates are coordinate descent of one functional. In per-agent free energies each agent's generative model treats its peers' communicated laws as competing sources of prior information. With Gaussian references, affine evaluators, linear-Gaussian channels, and log-quadratic records, both free-energy flows are tangent to the Gaussian product family. A reciprocal law potential generates coupled Fisher dynamics in which sources also move toward their receivers; with identity channels its flow leaves the Gaussian family as soon as a source is attended by a receiver with a different law. With fixed positive source rows on finite alphabets, we prove that this potential is not an affine function of the free energy of any normalized model over finitely many finite-valued variables, under any recognition rule, when some attended edge is an identity channel, and that with fixed channels that are not constant in their input it is not, on any open set, an affine function of the free energy of a normalized model whose recognition is the product of the agent laws. Constant-output channels, countable auxiliary latent variables, and coupled recognition on atomless spaces give exact identifications, so the separation rests on these hypotheses. Conditional-information identities describe communication, common recovery, and the cost of discarding record history. Parent dynamics and boundary responses must descend through the retained configuration.

A two-agent conditional-prediction experiment gives an exact interacting record model before posterior projection. Each speaker samples its result prediction from private evidence and a shared transcript; the receiver conditions on that actual mechanism. The joint private-record posterior is a product tilt of the fixed preparation coupling, so its cross-agent dependence never changes and, with the preparation fixed, its two marginals determine it. With a fixed reconstruction of represented states it preserves correlated model/content beliefs and the declared future communication laws. An affine Gaussian specialization with one presentation closes exactly, while finite Gaussian-mixture closure with several presentations and retention of local beliefs alone can fail. The common preparation and protocol are supplied, and the resulting fixed-history inference is distinct from the population flows.

A formation standard requires an open admitted class, selection by a selector family fixed before the flows are analyzed, exact inheritance, compression of a distinction that a declared child-level probe resolves, interaction necessity tested against two null flows, one keeping the deleted peer

labels and one renormalizing the source rows, and reported beside a frozen-environment control, and a second stage that does not form from every handoff in a declared counterfactual class. An anchored linear consensus calibrates it: it passes interaction necessity and fails the counterfactual clause. Under the reciprocal potential, a profile construction preserves coupled identity-channel dynamics and a declared response interface through arbitrary finite binary depth on a supplied family; it is an equivariant symmetry reduction and makes no selection claim. A four-agent Wishart construction selects two parents and then their union with exact original-time inheritance, but its membership schedule is the centroid dendrogram of the prepared initial shapes at every coupling, the renormalized null reproduces it, integrations indicate that the label-preserving null, whose clocks are agent dependent already at zero coupling, changes it on an open set, and the union forms from every counterfactual handoff in the domain of the selector; it is a two-stage selection with exact inheritance. A six-agent construction admits a full open set of positive bit-circle laws, with preferential pairs, a later parent join, inherited moving-representative dynamics, adaptive response histories, and, on a declared contextual family with tied belief and model rotations, nonflat rank-two contextual geometry on a finite window. Its orientation loss is relative to the supplied interface, its block relations are transient, and interaction necessity and counterfactual dependence are not established; integrations of its limit field indicate that the null flow forms the same blocks. It is not shown to be a recursive formation. A nonsmooth extension gives finite-time fusion with exact inheritance; its explicit coalescence order is a coupling-independent linkage of the initial data, it meets interaction necessity only through its exact-equality selector, and fused parents compress nothing, so it is a two-stage selection with exact inheritance. A gated erasure construction is a single-stage selection whose interaction necessity is not determined.

For the normalized per-agent free energies, a separate construction gives persistent grouping against conflicting records; its small-signal limit is the anchored opinion pooling of Friedkin and Johnsen. Its interaction-necessity verdict depends on the declared selector constant: it holds against both nulls for ratio thresholds in $(1/8, 1/3)$ and fails for thresholds in $(1/3, 1)$, and the frozen-environment control reproduces the selected sequence. An angular extension admits a parent that quotients a hidden phase and preserves the original-time port dynamics, every admitted finite adaptive response-record history, and the actual collective connection and contextual tensor on a nonflat finite-window family. The quotiented phase is gauge relative to every declared observable algebra, so the retention fails compression; the construction is a single-stage selection with exact inheritance, and no second compressive handoff is established. No construction in this manuscript is proved to satisfy every clause of the formation standard.

A collective phase quotient preserves fixed relative probes and admits exact local two-jet evolution with finite reference memory. The common orientation it erases is gauge relative to every declared algebra, so this whole-block retention also fails compression and gives a single-stage selection with exact inheritance. A finite preparation generator gives another conditional realization. A causal local phase accumulator obeys the original-time native dynamics and preserves every admitted history; actual geometric inheritance is proved under initial collective-phase coherence across context, a sufficient condition that is not shown to be necessary. Relative to the declared marked structure group, scoped obstructions exclude order-zero rotation-invariant value-path retentions and smooth bounded-jet causal memories on a declared rich preparation class. The unrestricted

causal-local-parent question remains inconclusive; neither interaction-created invisibility nor a second compressive handoff is proved.

A separate contextual extension adds to the reciprocal potential an auxiliary alignment energy that couples neighboring laws in declared finite-dimensional regular families. On finite graphs, a regular minimizing branch gives an explicit Fisher force and local uniqueness; the reduced cost equals the minimized cost only while the branch stays globally minimizing, and for general linear frames of dimension at least three the branch hypothesis excludes consensus contexts. On a supplied flat torus, a free proper isometric action reduces the contextual cost to a quotient Dirichlet energy. The resulting shape PDE and frame and source-row ODEs have a local Sobolev solution with exact total-energy dissipation, and no locally gauge-invariant eliminated energy diffuses frames. These results retain fixed-reference forces that need not descend to the quotient. They establish no global regularity, singular-orbit passage, independent row or link diffusion, or graph-refinement limit.

A separate geometric construction builds base transport from regular contextual law, interface, and directed-link data once comparison labels, columns, weights, and a metric are declared. A spanning message matrix gives displacement identity, affine connection covariance, an explicit curvature formula, and holonomy orthogonal for the induced metric. Local alignment under a declared bi-invariant metric gives a curved example from reciprocal rotation links, with Abelian holonomy. A single marked square frame instead gives flat local transport. Link-only alignment requires an invariant target metric, which conjugation targets of the general linear group do not admit, continuous invisible gauge sectors obstruct a link-only selector, and no new evolution or inheritance result is asserted.

Selective communication introduces a further distinction between the transmitted code and the invertible dictionary used to interpret it. Conditional-information identities quantify reconstruction loss for general laws. With a known code, grounded messages identify dictionaries modulo code stabilizers, while an anchored loop prior couples their unread directions. A Gaussian tangent model gives exact recognition formulas and separates global, receiver-owned, and reciprocal link dynamics; for its three-agent reference data the coupled reciprocal flow has a local subcritical pitchfork. Integrating out Gaussian contents produces a quadratic cost of an affine coded residual on reinterpretations, with a record-dependent linear term. Its relation to directed holonomy requires explicit cycle hypotheses, and its cohomological form holds in the small-noise limit at fixed anchors and, within the tangent model, in the weak-anchor limit. Supplied rigid clusters admit a restricted parent description under normalization and connectivity conditions; this reduction supplies no new formation theorem.

Reading the construction

The mathematical objects are introduced in the order in which they are needed: agent laws, generative models and records, interactions, statistical and contextual geometry, free energies and dynamics, predictive compression, inherited organization, a contextual-coupling extension, and base transport derived from contextual agent data. The final three technical chapters develop selective communication, latent dictionary learning, and curvature costs in supplied hierarchies. The main results include their decisive proofs. The appendices give the internal history construction, conditional prediction and generative memory, exact interaction between conditional predictors,

directional links and base connections, an optional determinant-metric and circle example, and notation. Secondary calculations are collected in a separate supplement.

Definitions specify the model's objects; constitutive choices specify how those objects interact. A theorem states a consequence of explicit hypotheses. An open problem identifies a further construction that is not supplied by those hypotheses. General probability measures are used whenever measurability suffices. Finite alphabets, smooth statistical families, and vector fibers enter only where an example, a differential calculation, or a stated hypothesis requires them; the obstruction to a global free-energy identification is one such hypothesis. The general definitions assume no Gaussian closure; the fusion example states its lognormal restriction explicitly.

Contents

Abstract i

1 Agents, Possibilities, and Records 1
1.1 Statistical sections . . . 1
1.2 Presentation and probability . . . 4
1.3 Internal records . . . 6
1.4 Beliefs and generative models . . . 8
1.5 Content laws on fields . . . 11
1.6 What correlations describe . . . 12

2 Law-Dependent Interactions 14
2.1 Two sectors and their comparison spaces . . . 14
2.2 Reference consistency . . . 15
2.3 Exponential links: one constitutive example . . . 16
2.4 An explicit constitutive family for general laws . . . 19
2.5 Loops, reciprocity, and what a law-level cost can see . . . 20
2.6 Dependence, direction, and inherited responses . . . 22

3 Geometry of Statistical Sections 23
3.1 Regular law families . . . 23
3.2 Covariant statistical change . . . 24
3.3 Null directions and quotients . . . 26
3.4 A collective choice of connection . . . 28
3.5 Information retained by a channel . . . 32
3.6 Geometry across a retention . . . 33

4 Free Energies and Coupled Dynamics of Agent Laws 36
4.1 A potential for the two interacting sectors . . . 36
4.2 Source selection and reduction . . . 38
4.3 The derivative includes every interaction role . . . 39
4.4 Fisher dynamics with equal mobilities . . . 40
4.5 A global free energy and its local free energies . . . 47
4.6 Per-agent free energies . . . 49
4.7 The reciprocal potential is not a global free energy . . . 53

5 Prediction and Selective Communication 59
5.1 Model presentations and their evaluators . . . 59
5.2 Conditional prediction within an agent . . . 60
5.3 Communication between conditional predictors . . . 61
5.4 What the receiver reconstructs . . . 62
5.5 A predictive coding criterion . . . 65

6 Inherited Parents and Recursive Formation **66**
6.1 Two objects that a parent must retain . . . 66
6.2 Exact deterministic inheritance . . . 67
6.3 Exact stochastic inheritance . . . 72
6.4 Boundary responses and composition . . . 74
6.5 A finite two-level construction . . . 79
6.6 Averaged laws under smooth peer dynamics . . . 80
6.7 Memory and formation . . . 83
6.8 Statistical partitions and their limits . . . 85
6.9 Architectures whose selection cannot see the interaction . . . 87
6.10 The formation standard . . . 89

7 Finite Fusion with Exact Statistical Inheritance **97**
7.1 A law family closed under geometric averaging . . . 97
7.2 Messages and state-dependent scalar links . . . 99
7.3 The continuous subgradient dynamics . . . 100
7.4 An invariant class with explicit global solutions . . . 103
7.5 Two fusion stages without a supplied partition . . . 104
7.6 The inherited handoff and its scope . . . 106
7.7 Status against the formation standard . . . 109

8 Sufficient States for Smooth KL Evolution **111**
8.1 The force that survives marginalization . . . 111
8.2 Conditional reconstruction within the original potential . . . 113
8.3 An exact law-valued completion . . . 115
8.4 Boundary records and feedback . . . 120
8.5 What is still required for formation . . . 122

9 State-Controlled Erasure and Persistent Source Preferences **124**
9.1 A homogeneous population with internal gate coordinates . . . 124
9.2 Finite entry into an exactly erasing region . . . 126
9.3 A change of preferred sources without synchronization . . . 127
9.4 The exact inherited state . . . 130
9.5 Status against the formation standard . . . 132

10 Public Compression and Inherited Responses **134**
10.1 An asymptotic collective phase near a stable reference . . . 134
10.2 An ordinary response that the phase does not preserve . . . 138
10.3 What unrestricted Gibbs probes retain . . . 139
10.4 The remaining recursive requirement . . . 141
10.5 Status against the formation standard . . . 142

11 Boundary Closure and Recursive Composition **143**
11.1 A general-law family on a supplied tree . . . 143
11.2 The exact inherited profile equations . . . 144
11.3 The hierarchy as an equivariant symmetry reduction . . . 148
11.4 Recursive retention and the response process . . . 152
11.5 Public distinctions discarded at every depth . . . 155
11.6 Why ordinary response completion can remove all compression . . . 157

12 Two-Stage Selection with Exact Inheritance on a Wishart Family **159**
12.1 Matrix laws and the declared architecture . . . 160
12.2 The complete smooth Fisher flow . . . 161
12.3 A common relaxation clock . . . 163
12.4 Parent laws and exact joining . . . 165
12.5 Selected membership . . . 166
12.6 Exact joint response histories . . . 172
12.7 Inherited collective geometry . . . 173

13 Preferential Selection in Two Stages from Open Law Classes **177**
13.1 A homogeneous selective-channel model . . . 177
13.2 Two stages selected by native routing . . . 179
13.3 A full open domain of correlated laws . . . 185
13.4 Orientation compression and inherited dynamics . . . 187
13.5 Responses and retained histories . . . 190
13.6 Collective geometry on actual parent laws . . . 191
13.7 The combined existence statement . . . 193
13.8 Two tests of stronger inheritance claims . . . 195

14 Per-Agent Grouping with Persistent Records **197**
14.1 What changes under an own-objective derivative . . . 197
14.2 A persistent group against conflicting evidence . . . 200
14.3 A fixed-model angular extension . . . 203
14.4 A hidden-phase quotient with inherited records . . . 207
14.5 Collective geometry of the same parent . . . 208
14.6 Report against the formation standard . . . 211

15 Collective Compression and Conditional Local Parents **212**
15.1 A transmitted angle . . . 212
15.2 A collective rather than primitive quotient . . . 214
15.3 Finite reference memory and closed local jets . . . 215
15.4 Handoff frames and finite preparation generators . . . 218
15.5 A causal local phase memory . . . 219
15.6 Scoped obstructions to stronger locality . . . 223
15.7 Report against the formation standard . . . 227

16 Contextual Coupling of Statistical Laws **228**
16.1 Finite contextual graphs . . . 228
16.2 A periodic continuum base . . . 242

17 Base Transport from Contextual Agent Data **257**
17.1 Contextual data and their transformation laws . . . 257
17.2 What an agent frame does determine . . . 258
17.3 Retaining reciprocal and backtrack information . . . 259
17.4 A full-rank contextual message construction . . . 259
17.5 Reversal, finite paths, and zero displacement . . . 261
17.6 Curvature and a curved law-dependent example . . . 263
17.7 A declared structure group and joint sector data . . . 265
17.8 Direct alignment of contextual link data . . . 266
17.9 What pointwise links do not determine . . . 268
17.10 Message coordinates, dynamics, and remaining scope . . . 269

18 Communication Codes and Learned Alignment **270**
18.1 Codes and dictionaries . . . 270
18.2 Reception through a code . . . 272
18.3 What messages reveal about the dictionary . . . 274
18.4 Several modalities . . . 276
18.5 Curvature as dissonance . . . 278
18.6 Bandwidth and source rows . . . 280
18.7 Learning the dictionaries . . . 281
18.8 Extensions beyond the declared experiments . . . 290

19 Learning Directed Links under Three Objectives **292**
19.1 The tangent communication model . . . 292
19.2 What conversation determines . . . 294
19.3 The two flows . . . 296
19.4 The grounded limit . . . 299
19.5 Confidence under ownership . . . 300
19.6 The reciprocal potential . . . 301
19.7 Coupled means, covariances, and rows . . . 305
19.8 An exact stability threshold with changing uncertainty . . . 309
19.9 The local subcritical pitchfork . . . 311
19.10 Scope and open problems . . . 314

20 Curvature Costs and Supplied Hierarchies **316**
20.1 Contents, records, and supplied parents . . . 316
20.2 The action induced by contents . . . 317
20.3 Stationary reinterpretation . . . 321
20.4 Rigid clusters and their boundary dictionaries . . . 322
20.5 Remaining extensions . . . 325

21 Beyond the Existence Constructions **327**

A A Normalized Internal Record Process **332**
A.1 Prediction followed by assimilation . . . 333

B Conditional Prediction and Generative Memory **335**
B.1 A normalized cue and result model . . . 335
B.2 A Gaussian realization and its response . . . 337
B.3 Retained moments and statistical geometry . . . 339
B.4 Frames, sharing, and sequential records . . . 340

C Interacting Conditional-Prediction Agents **343**
C.1 Preparation, information, and transmission . . . 343
C.2 The history law and exact assimilation . . . 345
C.3 A fixed reconstruction and the retained state . . . 347
C.4 Free energies for the declared information . . . 350
C.5 An exact two-way binary exchange . . . 352
C.6 Gaussian communication and an exact affine closure . . . 354
C.7 Failure of finite-mixture closure . . . 357
C.8 Local beliefs do not generally determine the next update . . . 359

D Connections in a Moving Presentation **361**

E A Conditional Determinant Geometry **364**

F Notation **368**

Chapter 1

Agents, Possibilities, and Records

The program behind this theory asks whether interacting agents can produce further agents. This text supplies the objects needed to pose that question exactly, the conditions under which a parent description inherits the dynamics of its children, and constructions measured against a formation standard (definition 6.36). No construction in this manuscript is proved to satisfy every clause of that standard; each construction states which clauses it meets, which it fails, and which remain undetermined, and the construction of a recursive formation is posed as open problem 6.38. The first distinction is between an agent's probability law and the possibilities represented by that law. The second is between comparing two descriptions and specifying an interaction that changes them. A parent construction must preserve enough of both to continue the dynamics.

1.1 Statistical sections

Let $\mathcal{C}$ be a second-countable smooth base of contexts and let $P \to \mathcal{C}$ be a principal G-bundle, with G a second-countable Lie group. A context labels where a description applies; it is not identified with physical position or time. For each agent i let K_i and M_i be standard Borel spaces with jointly Borel actions $G \times \mathsf{K}_i \to \mathsf{K}_i$ and $G \times \mathsf{M}_i \to \mathsf{M}_i$. Their elements k and m represent state contents and model presentations. The actions may be different representations of the same group. They act on probability measures by pushforward. When no agent index is needed we write K, M.

Throughout, one structure group acts on both sectors. A gauge transformation of an agent is a single element of G, acting on state contents and on model presentations through the two actions; there is no separate belief group and model group. The represented contents form two associated bundles over the same principal bundle,

$$\mathcal{E}_q = P \times_G \mathsf{K}, \qquad \mathcal{E}_s = P \times_G \mathsf{M}, \tag{1.1}$$

which in a linear realization are vector bundles $P \times_{\varrho_q} V_q$ and $P \times_{\varrho_s} V_s$ for representations $\varrho_q : G \to \mathrm{GL}(V_q)$ and $\varrho_s : G \to \mathrm{GL}(V_s)$, of ranks $K = \dim V_q$ and $\tilde{K} = \dim V_s$. Equation (1.1) is written for one agent; agent i has the content bundles $P \times_G \mathsf{K}_i$ and $P \times_G \mathsf{M}_i$, all associated to the one principal bundle P. The statistical bundle defined next carries laws over the fiber product of $\mathcal{E}_q$ and $\mathcal{E}_s$, on which G acts diagonally.

Definition 1.1 (Statistical agent section). Set $\mathsf{Y}_i = \mathsf{K}_i \times \mathsf{M}_i$, with the product action, and give $\mathcal{P}(\mathsf{Y}_i)$ the sigma-algebra generated by evaluation on measurable sets. The associated statistical bundle of agent i is

$$\mathcal{B}_i = P \times_G \mathcal{P}(\mathsf{Y}_i), \qquad (ug, \zeta) \sim (u, g_{\#}\zeta). \tag{1.2}$$

The statistical component of agent i is a Borel section ζ_i of $\mathcal{B}_i$ over an open domain $\mathcal{U}_i \subseteq \mathcal{C}$. Smooth sections are defined only into admitted smooth subbundles. The full state of the agent also includes the internal variables and access mechanisms read by its update rule.

The associated-bundle construction is that of Kobayashi and Nomizu [122, Section I.5]. Its measurable form needs the joint measurability of the action. If each g acts bimeasurably but $(g, y) \mapsto gy$ is not jointly Borel, for example a translation of $\mathbb{R}$ by a discontinuous additive function of $g \in \mathbb{R}$, the Borel class of a section can depend on the trivialization in which it is written.

Lemma 1.2 (Measurable structure of the statistical bundle). Under the stated hypotheses, $(g, \zeta) \mapsto g_{\#}\zeta$ is Borel on $G \times \mathcal{P}(\mathsf{Y}_i)$, and $\mathcal{B}_i$ carries a standard Borel structure, independent of the atlas, for which every local trivialization is a Borel isomorphism onto its image. Transition maps preserve fiberwise mixtures, null sets, mutual absolute continuity, and relative entropy.

Proof. For a Borel set A, $(g_{\#}\zeta)(A) = \int \mathbf{1}_A(gy)\zeta(dy)$. The integrand is Borel in (g, y) by joint measurability, and the integral of a bounded jointly Borel function against ζ is Borel in (g, ζ): this holds for products $f(g)h(y)$ and extends by the monotone class theorem. Since $\mathcal{C}$ is second countable, countably many trivializing patches cover it, and their transition functions are Borel. The gluing maps $(c, \zeta) \mapsto (c, g_{\alpha\beta}(c)_{\#}\zeta)$ are therefore Borel bijections with Borel inverses. The space $\mathcal{P}(\mathsf{Y}_i)$ with the sigma-algebra generated by the evaluations is standard Borel [117, Section 17.E], so $\mathcal{B}_i$ is a countable Borel union of standard Borel pieces glued by Borel isomorphisms; a second atlas yields the same structure because its transitions to the first are Borel by the same argument. Pushforward is affine and preserves null sets and absolute continuity, and proposition 1.6 gives invariance of relative entropy. □

At a fixed context, disintegration gives

$$\zeta_i(dk, dm) = q_i(dk \mid m)s_i(dm), \qquad q_i(dk) = \int q_i(dk \mid m)s_i(dm). \tag{1.3}$$

Here q_i is the state marginal and s_i is the law over model presentations. Regular conditional laws exist on the stated spaces, with versions defined up to the appropriate marginal null sets [114]. Disintegration commutes with the product action: the conditional law of $g_{\#}\zeta_i$ given m' is $g_{\#}q_i(\cdot \mid g^{-1}m')$ for $g_{\#}s_i$-almost every m', so eq. (1.3) descends to bundle points. A model presentation acquires predictive content through a normalized kernel; its name or coordinate value alone supplies no prediction (section 1.4).

The product restriction $\zeta_i = q_i \otimes s_i$ removes dependence between the two sectors within an agent. A product over agents makes a further restriction on represented contents. Neither restriction prevents the update of one law from depending on other laws. Coupling is a property of the dynamics; factorization is a property of a chosen probability description. The coupled population flows developed below impose the product restriction. The general joint law enters through its Fisher metric in chapter 3 and through the local conditional-prediction construction of section 5.2 and appendix B. The restriction is not neutral, as chapter 4 shows: it changes the local optimum of a belief, and the joint-law flow does not preserve it.

Smoothness is an additional hypothesis. When derivatives of laws are required, use a declared regular statistical family stable under the relevant coordinate changes. The whole space $\mathcal{P}(\mathsf{Y}_i)$ is

not assigned a smooth manifold structure by notation. Chapter 3 equips an admitted family with the Fisher metric of the full joint law and pulls that metric back to contextual directions through a covariant derivative. The product restriction has a simpler metric, but marginal metrics alone do not measure changes of dependence in a general joint family. A collection of pointwise marginals also does not determine a probability law on fields over $\mathcal{C}$; section 1.5 supplies such a law as a separate object.

The population at a context consists of the agents whose domains contain that context. Most of the dynamics below is formulated at a common c_*. This retains the ambient bundle and local sections without requiring parallel transport between distinct base points. Independent interactions between agents at c_* are defined in chapter 2. Contextual statistical geometry is developed in chapter 3; appendix D records the transformation law of connection data in a moving presentation. Chapter 17 constructs such data from regular contextual laws, interfaces, and links with declared comparison rules; selection of those rules and continuation through singular regions remain open. The potentials of chapter 4 contain no coupling between contexts. The formation constructions retain these context-local flows and compute connections as diagnostics. The separate extension in chapter 16 adds an aligned contextual energy and derives its neighboring-context Fisher force on declared regular families.

Parents in later chapters are maps between statistical descriptions of different agents. The following definition supplies the corresponding morphism of bundles.

Definition 1.3 (Morphism of statistical bundles). Let $\tilde{f} : P \to \bar{P}$ be a principal bundle map covering $f : \mathcal{C} \to \bar{\mathcal{C}}$ with homomorphism $\phi : G \to \bar{G}$, so that $\tilde{f}(ug) = \tilde{f}(u)\phi(g)$. Let $\kappa : \mathsf{Y} \rightsquigarrow \bar{\mathsf{Y}}$ be a Markov kernel with $\kappa(gy, \cdot) = \phi(g)_{\#}\kappa(y, \cdot)$. The induced morphism is

$$\Psi_{\tilde{f},\kappa}[u, \zeta] = [\tilde{f}(u), \zeta\kappa]. \tag{1.4}$$

Proposition 1.4 (Statistical bundle morphisms). The map eq. (1.4) is well defined, fiberwise affine, and contracts relative entropy within each fiber. In a linear realization, let $A : V_q \to \bar{V}_q$ and $\tilde{A} : V_s \to \bar{V}_s$ be linear maps with $A\varrho_q(g) = \bar{\varrho}_q(\phi(g))A$ and $\tilde{A}\varrho_s(g) = \bar{\varrho}_s(\phi(g))\tilde{A}$ for all $g \in G$. Then the same principal map induces well-defined morphisms of the content bundles,

$$\Phi[u, k] = [\tilde{f}(u), Ak], \qquad \tilde{\Phi}[u, m] = [\tilde{f}(u), \tilde{A}m], \tag{1.5}$$

and the deterministic kernel $\kappa((k, m), \cdot) = \delta_{(Ak,\tilde{A}m)}$ induces the corresponding statistical morphism.

Proof. Equivariance of κ gives $(g_{\#}\zeta)\kappa = \phi(g)_{\#}(\zeta\kappa)$. Hence $[u, g_{\#}\zeta] \mapsto [\tilde{f}(u), \phi(g)_{\#}(\zeta\kappa)] = [\tilde{f}(u)\phi(g), \zeta\kappa]$, which is the image of the equivalent representative $[ug, \zeta]$. The map $\zeta \mapsto \zeta\kappa$ is affine, and $D_{\mathrm{KL}}(\zeta\kappa\|\zeta'\kappa) \le D_{\mathrm{KL}}(\zeta\|\zeta')$ is the data-processing inequality for relative entropy under a Markov kernel on standard Borel spaces [95, Chapter 5]. For the content morphisms, the representative $[ug, \varrho_q(g)^{-1}k]$ is sent to $[\tilde{f}(u)\phi(g), A\varrho_q(g)^{-1}k] = [\tilde{f}(u)\phi(g), \bar{\varrho}_q(\phi(g))^{-1}Ak] = [\tilde{f}(u), Ak]$, and likewise for $\tilde{\Phi}$. The deterministic kernel is equivariant by the two intertwining relations. □

The same argument applies verbatim to a Borel family of kernels κ_c indexed by the base point, equivariant at each c, that is, to an equivariant kernel from the sample bundle of the fine description to the pullback by f of the coarse sample bundle; the constant kernel of definition 1.3 is its simplest

case. A retention whose kernel varies over the base or depends on the configuration is a morphism only in this fibered sense, and a retention that reads several fibers at once is not a fiberwise morphism at all.

The belief and model morphisms share the principal map $\tilde{f}$. Two principal maps covering the same f with the same homomorphism ϕ differ by a gauge transformation h of $\bar{P}$, and the pair of sector maps built from them corresponds to the base-point-dependent kernel obtained by composing with $h(u)$ in one sector; it is covered by the fibered form above. Only sector maps with different homomorphisms $\phi_1 \neq \phi_2$ would make independent groups act in the two sectors, and these are not admitted.

A parent is an agent of a declared type when its retained state is again a section of such a bundle, carries an admitted family, a register with a covariant readout, a generative model in the sense of definition 1.11 (presentation prior, evaluator, and record kernel), and interacts with exterior agents through channels of definition 2.1. Proving that a retention is closed in this sense is part of the formation problem of chapter 6, not a consequence of the definition.

1.2 Presentation and probability

Fix c_* and a reference point $\sigma \in P_{c_*}$. A frame of agent i is a point $u_i \in P_{c_*}$, and $u_i = \sigma U_i^{-1}$ defines $U_i \in G$. The point u_i identifies the fiber of $\mathcal{B}_i$ at c_* with $\mathcal{P}(\mathsf{Y}_i)$, and the law displayed by agent i is the coordinate expression of its bundle point in that frame. In a linear realization $\mathsf{K}_i = V_q$ and $\mathsf{M}_i = V_s$ carry linear representations ϱ_q, ϱ_s of G, the sector frames are $U_i^q = \varrho_q(U_i)$ and $U_i^s = \varrho_s(U_i)$, and E_i^q, E_i^s denote agent i's coordinate copies of V_q, V_s. Laws written on E_i^a, as in chapter 2, are the displayed laws. The reference laws are

$$\bar{q}_i = ((U_i^q)^{-1})_{\#} q_i, \qquad \bar{s}_i = ((U_i^s)^{-1})_{\#} s_i. \tag{1.6}$$

They are the coordinate expressions in the reference frame σ, since $[\sigma U_i^{-1}, \zeta] = [\sigma, (U_i^{-1})_{\#}\zeta]$. A passive change sends u_i to $u_i g_i^{-1}$, hence q_i to $(g_i^q)_{\#} q_i$ and U_i^q to $g_i^q U_i^q$, with the corresponding model-sector rule. The reference laws are unchanged. This is a change of description, not an inference step. One frame induces both sector frames, and a passive change is one element g_i with $g_i^q = \varrho_q(g_i)$ and $g_i^s = \varrho_s(g_i)$. Independent sector changes would require the product group $G_q \times G_s$; the theory does not use them.

Because every frame-dependent object can be written through its reference law, covariance under independent passive changes at each agent holds for any rule expressed in reference laws and constrains nothing by itself. The frames act here as a dressing field in the sense of Fournel et al. [77]. A substantive symmetry requirement is a statement about the reference rules themselves, such as equivariance of an interaction under a common change of reference at one context. Chapter 2 states it as eq. (2.15), and a result uses it only where it says so. Two forms must be kept apart. If every ingredient of a rule that is not invariant, such as a metric, a code, or a fixed direction, is declared as supplied data and re-expressed together with the reference laws, the rule is unchanged by the common reference change automatically; this naturality is the dressing argument applied to the reference point σ and again constrains nothing. Active equivariance holds the rule, the references, the evaluators, and every supplied datum fixed and requires the dynamics to commute

with the common action of G. A supplied datum carried with the reference reduces the group under which the model is actively symmetric to the stabilizer of that datum, for example to an orthogonal group for a fixed inner product. A symmetry that holds only because such data are re-expressed is artificial in the sense of François [78]; a result that needs active equivariance says so.

An active transformation changes the reference laws at fixed frames, for example $\bar{q}_i \mapsto a_{\#}\bar{q}_i$ for a bimeasurable bijection a of K_i. Whether two configurations related in this way are distinguishable depends on a declared observable algebra.

Definition 1.5 (Observable algebras and gauge distinctions). The population algebra $\mathfrak{A}_{\mathrm{pop}}$ of a construction consists of the measurable functions of the configuration that the agents themselves read: their source rows, the outputs of their channels and the costs built from them, their evaluators and record kernels, and the laws of their record readouts. A response protocol is a declared family of external queries together with their response kernels; the protocol algebra $\mathfrak{A}_{\mathrm{prot}}$ adds to $\mathfrak{A}_{\mathrm{pop}}$ the response laws of the admitted queries of a declared protocol. An element of an algebra is evaluated at the configuration reached at any time along the declared dynamics from the configuration in question. Two configurations, or two configurations related by an active transformation, differ by a distinction that is gauge relative to an algebra $\mathfrak{A}$ when no element of $\mathfrak{A}$ separates them at any time. A comparison made at a single instant is called instantaneous and is not used to declare a distinction gauge.

A law functional that no agent, channel, record, or admitted query reads, for example a child-level diagnostic of a law used only to assess a retention, belongs to no observable algebra. A distinction that only such a functional resolves is gauge relative to every declared algebra; the text then names the functional rather than calling the distinction physical. When only an external protocol resolves a distinction, the text names that protocol rather than calling the distinction physical without qualification, and the words physical and public are used only relative to a named algebra. The time-evolved form matters: a law-level comparison can fail to separate two configurations at one instant while the dynamics separates them later (proposition 1.17 and the remark after it).

For probability measures μ, ν on one measurable space, relative entropy is

$$D_{\mathrm{KL}}(\mu\|\nu) = \int \log \frac{d\mu}{d\nu} d\mu, \tag{1.7}$$

when $\mu \ll \nu$, and is $+\infty$ otherwise. In this formula and below, a displayed integral is understood with its declared measure; no common Lebesgue density is required. Relative entropy compares distinguishability of laws [124].

Proposition 1.6 (Coordinate invariance). For an injective Borel map f between standard Borel spaces, in particular a bimeasurable bijection, including divergences with value $+\infty$,

$$D_{\mathrm{KL}}(f_{\#}\mu\|f_{\#}\nu) = D_{\mathrm{KL}}(\mu\|\nu). \tag{1.8}$$

Proof. By the Lusin–Souslin theorem [117, Theorem 15.1] the image of f is Borel and f is a Borel isomorphism onto it, so it suffices to treat a bimeasurable bijection onto a Borel set carrying both pushforwards. Absolute continuity is preserved in both directions. If $h = d\mu/d\nu$, the transformed density is $h \circ f^{-1}$. Integrating its logarithm against $f_{\#}\mu$ gives the original integral. If absolute

continuity fails, a set of zero ν-measure and positive μ-measure is carried to a set with the same property. □

Definition 1.7 (Normalized channel). A Markov kernel $K : X \rightsquigarrow Y$ assigns a probability measure $K(x, \cdot)$ to each x, measurably in x. Its action on a law is

$$(\mu K)(A) = \int K(x, A)\mu(dx). \tag{1.9}$$

A deterministic measurable map $f : X \to Y$ is represented by $K_f(x, \cdot) = \delta_{f(x)}$. Then $\mu K_f = f_{\#}\mu$.

An invertible frame change preserves all distinctions. A channel that discards information or adds noise need not. Both are well-defined operations on general probability laws, but their roles in an agent interaction differ.

1.3 Internal records

At a fixed context, let S_i be the admitted internal-state space of agent i, and let S be the complete configuration space. It contains the statistical laws, the source rows defined in chapter 4, interfaces, and memory variables that the dynamics reads. Write $\mathrm{pr}_i : \mathsf{S} \to \mathsf{S}_i$ for the local state projection. For the measurable constructions, take these spaces to be standard Borel. A finite collection of probability-law spaces and finite or Euclidean memory spaces supplies such a realization. A flow, retention, or response used in a measurable construction must be Borel for these structures. Smoothness in a Banach chart of log densities does not by itself ensure this, because the norm-Borel sigma-algebra of $L^\infty(\lambda)$ for nonatomic λ is strictly finer than the trace of the evaluation sigma-algebra; where such flows are used inside history laws, joint Borel measurability of $(t, x) \mapsto \Phi_t(x)$ for the evaluation structure is a stated hypothesis. It holds without further argument for continuous flows on finite-dimensional families and on separable charts whose chart map into the law space is continuous and injective, by the Lusin–Souslin theorem [117, Theorem 15.1]. Policies and record densities entering such history laws are subject to the measurability conditions of appendix A.

Definition 1.8 (Record and readout). A record readout is a measurable map

$$\rho_i : \mathsf{S}_i \longrightarrow \mathsf{O}_i, \tag{1.10}$$

where O_i is a standard Borel alphabet of internally distinguishable outcomes. If an interaction with source j produces configuration Z_{t+1}, its record variable is

$$O_{ij,t} = \rho_i(\mathrm{pr}_i\, Z_{t+1}). \tag{1.11}$$

For stochastic dynamics this is a measurable function from the probability space of realizations to O_i. A realized value is denoted $o_{ij,t}$. The source label is part of the record only when the internal state retains it.

A record can be transient. Persistence means that the distinction remains accessible through a specified later readout. A list kept outside the agent does not supply this accessibility. A memory

coordinate is therefore an explicit part of the model unless the required distinction is already represented in its state.

Proposition 1.9 (Induced record law). Let $\Gamma_{ij} : \mathsf{S} \rightsquigarrow \mathsf{S}$ be a normalized interaction transition. Then

$$\mathcal{R}_{ij}(x, A) = \Gamma_{ij}\left(x, \{x' : \rho_i(\mathrm{pr}_i\, x') \in A\}\right), \tag{1.12}$$

is a normalized record kernel. If $\Gamma_{ij}(x, \cdot) = \delta_{F_{ij}(x)}$, the record is the deterministic function $\rho_i \circ \mathrm{pr}_i \circ F_{ij}$.

Proof. The right side is the pushforward of a probability measure through the measurable map $\rho_i \circ \mathrm{pr}_i$. Its total mass is one, and measurability in x follows from the defining property of a kernel. The deterministic case follows by evaluating a Dirac measure. □

A readout that reads frame-dependent coordinates can change under a mere change of description. For example, a readout of the first coordinate of the displayed mean of q_i returns 1 for the law $\mathcal{N}(e_1, I)$ in one frame and 0 after a passive rotation by $\pi/2$, although the configuration is the same. Records therefore carry a covariance requirement. A passive change is a tuple $\mathbf{g} = (g_i)_i$ of elements of G. It acts on S by a Borel automorphism $x \mapsto \mathbf{g} \cdot x$, on laws by pushforward, on frames by left multiplication, and on interface data by their declared rule, with $\mathrm{pr}_i(\mathbf{g} \cdot x) = g_i \cdot \mathrm{pr}_i\, x$. A readout is invariant if $\rho_i(g_i \cdot x_i) = \rho_i(x_i)$, and equivariant if $\rho_i(g_i \cdot x_i) = \tau_i(g_i)\rho_i(x_i)$ for a declared jointly Borel action τ_i of G on O_i. A transition is covariant if $\Gamma_{ij}(\mathbf{g} \cdot x, \mathbf{g} \cdot B) = \Gamma_{ij}(x, B)$ for all measurable B. Every readout, record kernel, and response kernel of this text is required to be invariant or declared equivariant, and every interaction transition covariant.

By the dressing argument after eq. (1.6), a readout written through reference laws satisfies this requirement automatically: the first coordinate of the reference mean $U_i^{-1}\mu_i$ is unchanged by every passive change, yet it changes under a common change of the reference point σ at the same bundle point. A readout, record kernel, or response kernel is reference consistent if re-expressing all reference laws and supplied data under a common change of reference point leaves it unchanged, or changes its outcome by the declared action. This is the condition that hypothesis 2.3 imposes on channels; without it a record law is not a function of the bundle data. Results whose conclusions concern bundle data assume it for the records they use.

Proposition 1.10 (Covariant record laws). If Γ_{ij} is covariant and ρ_i is equivariant with action τ_i, then $\mathcal{R}_{ij}(\mathbf{g} \cdot x, A) = \mathcal{R}_{ij}(x, \tau_i(g_i)^{-1}A)$. For an invariant readout, $\mathcal{R}_{ij}(\mathbf{g} \cdot x, A) = \mathcal{R}_{ij}(x, A)$.

Proof. Write $B_A = \{x' : \rho_i(\mathrm{pr}_i\, x') \in A\}$. Covariance gives $\mathcal{R}_{ij}(\mathbf{g} \cdot x, A) = \Gamma_{ij}(x, \mathbf{g}^{-1} \cdot B_A)$, and $\mathbf{g}^{-1} \cdot B_A = \{x' : \tau_i(g_i)\rho_i(\mathrm{pr}_i\, x') \in A\} = B_{\tau_i(g_i)^{-1}A}$. □

This definition introduces no observing agent outside the population. It specifies where an outcome is available and how it can enter later dynamics. It does not derive the transition Γ_{ij} or decide whether its stochasticity represents objective chance. Those are further modeling choices. A response protocol queried from outside the population, as in chapter 13, is a separate declared device and is named wherever it is the only element of the observable algebra that resolves a distinction.

For example, a vector comparison can be represented by

$$r_{ij} : E_i^q \times E_j^q \times \mathbb{R}^r \to \mathbb{R}^r, \qquad r_{ij}(k_i, k_j, \epsilon) = A_i k_i - B_{ij} k_j + \epsilon. \tag{1.13}$$

Under passive changes $k_i \mapsto g_i k_i$ and $k_j \mapsto g_j k_j$ the comparison matrices transform as $A_i \mapsto A_i g_i^{-1}$ and $B_{ij} \mapsto B_{ij} g_j^{-1}$, so the comparison value is invariant. A noise law turns this map into a record kernel on represented contents. A joint content law determines its predicted outcomes. A transition on agent states must still specify how an outcome is registered and how the agent changes. A predictive likelihood and an internal update thus have different domains.

1.4 Beliefs and generative models

An agent holds beliefs and generative models. The joint law ζ_i describes the first; the second is the mechanism that gives a model presentation predictive content.

Definition 1.11 (Generative model of an agent). At a fixed context, a generative model of agent i consists of a proper presentation prior r_i on M_i, a normalized evaluator $G_i : \mathsf{M}_i \rightsquigarrow \mathsf{K}_i$, and a normalized record kernel $\ell_i : \mathsf{K}_i \times \mathsf{M}_i \rightsquigarrow \mathsf{O}_i$. They define the normalized joint

$$P_i(dm, dk, do) = r_i(dm) G_i(m, dk) \ell_i(k, m; do). \tag{1.14}$$

Each presentation m indexes the generative model $G_i(m, dk)\ell_i(k, m; do)$: a prior over represented contents together with a likelihood for records. The presentation prior r_i is the prior over presentations; the name reference law is reserved for the reference-frame laws eq. (1.6). When ℓ_i does not depend on m, it is the record kernel H_i of chapter 5. The model is the self-only case of the per-agent model eq. (4.26) of chapter 4, in which peers' communicated laws are added as further sources.

The model is specified in reference coordinates by $(\bar{r}_i, \bar{G}_i, \bar{\ell}_i)$ and displayed through the frame,

$$r_i = (U_i)_{\#}\bar{r}_i, \qquad G_i(m, \cdot) = (U_i)_{\#}\bar{G}_i(U_i^{-1}m, \cdot), \qquad \ell_i(k, m; \cdot) = \tau_i (U_i)_{\#}\bar{\ell}_i(U_i^{-1}k, U_i^{-1}m; \cdot), \tag{1.15}$$

with τ_i trivial for invariant readouts. The record kernel has a declared density, $\ell_i(k, m; do) = l_i(o \mid k, m)\lambda_{\mathsf{O}_i}(do)$, with l_i jointly measurable in (o, k, m) and λ_{O_i} a declared sigma-finite reference measure invariant under τ_i. Changing λ_{O_i} to $w\lambda_{\mathsf{O}_i}$ shifts the free energy below by $\log w(o)$, so the reference measure is part of the model.

Given a record value o with $0 < P_i(o) < \infty$, where $P_i(o) = \int l_i(o \mid k, m) G_i(m, dk) r_i(dm)$ is the evidence density, the recognition law ζ_i is a candidate for the posterior $P_i(dm, dk \mid o)$, and

$$\mathcal{F}_i(\zeta_i; o) = D_{\mathrm{KL}}\left(\zeta_i \| P_i(\cdot \mid o)\right) - \log P_i(o) \tag{1.16}$$

is its variational free energy [28, 80]. In this reading q_i is the agent's belief about represented contents, its perception, and s_i is its belief over generative models. The model law alone predicts nothing: by eq. (1.14), prediction requires the evaluator and the record kernel. Write

$$K_i^{\mathrm{rec}}(m, do) = \int G_i(m, dk) \ell_i(k, m; do) \tag{1.17}$$

for the composite record kernel of a presentation. An evaluator and record kernel whose composite does not depend on m make the model sector predictively inert; a constant evaluator suffices when ℓ_i does not read m, and does not suffice otherwise. Without records, the unique minimizer of $D_{\mathrm{KL}}(\zeta_i \| r_i G_i)$ is the prior joint itself, whose belief conditional $q_i(dk \mid m)$ equals the evaluator $G_i(m, dk)$; the conditional in eq. (1.3) is the agent's current departure from that prior. With records, the posterior form eq. (1.16) equals $D_{\mathrm{KL}}(\zeta_i \| r_i G_i) - \mathbb{E}_{\zeta_i} \log l_i(o \mid k, m)$ whenever the right side is not of the form $\infty - \infty$.

These generative models contain no action, policy, or expected free energy, and their presentation priors and evaluators are held fixed in every result of the text. The free energies studied below are therefore perceptual: records are supplied or registered by declared mechanisms, and neither the choice of the next record nor the learning of r_i and G_i is modeled.

Proposition 1.12 (Frame invariance of the agent free energy). Let the model be displayed by eq. (1.15). Under a passive change $U_i \mapsto g_i U_i$, with $\zeta_i \mapsto (g_i)_{\#}\zeta_i$ and $o \mapsto \tau_i(g_i)o$, the value of eq. (1.16) is unchanged. If the reference data are re-expressed under a common change of reference point, the displayed model, and hence the value, is unchanged as well.

Proof. The displayed joint transforms by the pushforward under $(m, k, o) \mapsto (g_i m, g_i k, \tau_i(g_i)o)$. Invariance of λ_{O_i} gives the transformed density $l_i(\tau_i(g_i)^{-1}o \mid g_i^{-1}k, g_i^{-1}m)$, so the evidence density at $\tau_i(g_i)o$ equals $P_i(o)$, and the posterior given the transformed record is the pushforward of $P_i(\cdot \mid o)$ under $(m, k) \mapsto (g_i m, g_i k)$. Proposition 1.6 gives invariance of the divergence. Under a common change of reference point the frames and the reference data change by inverse elements, which cancel in eq. (1.15). □

The second statement is naturality in the sense described after eq. (1.6). Holding $(\bar{r}_i, \bar{G}_i, \bar{\ell}_i)$ fixed while the reference laws change is an active transformation, and it changes $\mathcal{F}_i$ in general: with $\mathsf{K}_i = \mathbb{R}$, a standard normal evaluator, no record, and ζ_i standard normal, a translation of the reference law by a moves $\mathcal{F}_i$ from 0 to $a^2/2$.

Definition 1.13 (Shared generative model). Let agents i and j have $\mathsf{M}_i = \mathsf{M}_j$, $\mathsf{K}_i = \mathsf{K}_j$, $\mathsf{O}_i = \mathsf{O}_j$, with the same actions of G. Their generative models are isomorphic if there are bimeasurable bijections ϕ of M_i and ψ of K_i with

$$r_j = \phi_{\#} r_i, \qquad G_j(\phi(m), \cdot) = \psi_{\#} G_i(m, \cdot), \qquad \ell_j(\psi(k), \phi(m); \cdot) = \ell_i(k, m; \cdot). \tag{1.18}$$

They share a generative model up to frame if the reference models of eq. (1.15) are related by one element $g \in G$ acting on both sectors,

$$\bar{r}_j = g_{\#} \bar{r}_i, \qquad \bar{G}_j(gm, \cdot) = g_{\#} \bar{G}_i(m, \cdot), \qquad \bar{\ell}_j(gk, gm; \cdot) = \tau(g)_{\#} \bar{\ell}_i(k, m; \cdot), \tag{1.19}$$

with τ the common declared action on records. Sharing up to frame implies isomorphism, with ϕ and ψ the actions of $U_j g U_i^{-1}$ in the two sectors when records are invariant; the converse fails, since eq. (1.18) admits independent bijections of the two sectors that are not actions of a group element, and for nonatomic presentation priors the condition $r_j = \phi_{\#} r_i$ alone constrains nothing. A population whose members pairwise share a generative model up to frame is called a species. Statements that use channels, codes, or dictionaries within a species require the frame version,

because channels are covariant only under the group action. Codes and dictionaries are not components of eq. (1.14); a shared language is a separately declared datum.

Sharing a generative model is a statement about mechanisms. Agreement of beliefs over models, $D_{\mathrm{KL}}(s_j \| \phi_{\#} s_i) = 0$, is a statement about states. The two are independent, and agents with a shared model can hold different beliefs. What records can test is weaker than either form of sharing. For records that are conditionally independent given (k, m), write $K_i^{(n)}(m, do_{1:n}) = \int G_i(m, dk) \prod_{t=1}^{n} \ell_i(k, m; do_t)$ for the composite kernel of n records, so that $K_i^{(1)} = K_i^{\mathrm{rec}}$.

Proposition 1.14 (Predictive equivalence). Fix $n \geq 1$, a bimeasurable bijection ϕ of M_i, and the preparation and accessible record history.

(i) If the models are isomorphic through (ϕ, ψ), then $\int K_j^{(n)}(m', \cdot)(\phi_{\#} s)(dm') = \int K_i^{(n)}(m, \cdot) s(dm)$ for every model belief s, and after a common history of n records the posteriors correspond under $\psi \times \phi$.

(ii) Equality of the predicted laws of n records for every model belief s and its image $\phi_{\#} s$ holds exactly when $K_j^{(n)}(\phi(m), \cdot) = K_i^{(n)}(m, \cdot)$ for every m; this condition also makes the model posteriors correspond under ϕ after every history of n records.

(iii) Isomorphism is strictly stronger than the condition of (ii) for every n.

Proof. For (i), substituting the three relations of eq. (1.18) gives $P_j = (\phi \times \psi \times \mathrm{id})_{\#} P_i$ record by record, which yields both statements. For (ii), sufficiency is integration against s and Bayes' rule with the same kernel; necessity follows by taking $s = \delta_m$. For (iii), let $\mathsf{K} = \mathsf{M} = \mathsf{O} = \{0, 1\}$ with equal presentation priors, $\ell_i(k, m; \cdot) = \ell_j(k, m; \cdot) = \mathrm{Ber}(p_m)$ independent of k with $p_0 \neq p_1$, $G_i(m, \cdot)$ uniform and $G_j(m, \cdot) = \delta_0$. Then $K_i^{(n)} = K_j^{(n)}$ is the law of n independent $\mathrm{Ber}(p_m)$ records for every n, with $\phi = \mathrm{id}$, while no bijection ψ carries the uniform law to δ_0; the content posteriors stay uniform for i and equal δ_0 for j. □

Under the recognition reading, in which agent i predicts a record by $\int \ell_i(k, m; \cdot) \zeta_i(dk, dm)$, equal model beliefs do not suffice even for one shared model, since the belief conditionals also enter; predictions agree when the models are isomorphic through (ϕ, ψ) and $\zeta_j = (\psi \times \phi)_{\#} \zeta_i$. A record registered by one agent can change its posterior, and therefore its predictions, without changing those of an agent that did not register it; a record whose likelihood is constant in (k, m) changes nothing.

The free energy eq. (1.16) is a single-agent object. Interactions add a further choice. In one choice all agents are parts of one normalized joint over their contents and records. In another each agent's generative model treats the communicated laws of its peers as data. Chapter 4 develops both choices and a third, a direct interaction potential between laws. With fixed positive rows and an edge compared through an identity channel, proposition 4.24 proves two restricted statements about the third: on finite alphabets no normalized model over finitely many finite-valued variables reproduces it for any rule assigning recognition laws, and with fixed channels no normalized model reproduces it on an open set when recognition is the product of the agent laws themselves. Neither statement covers coupled recognition laws on infinite spaces or law-dependent channels; the limits of the obstruction are described after proposition 1.16.

1.5 Content laws on fields

The global choice needs a joint law over the contents of all agents and, where contexts are compared, over contents at different contexts. Write $E_i = P \times_G \mathsf{Y}_i$ for the associated sample bundle. A point $[u, \zeta]$ of $\mathcal{B}_i$ at c is identified with the law $(y \mapsto [u, y])_{\#}\zeta$ on the fiber of E_i at c; this identification does not depend on the representative.

Definition 1.15 (Content law on fields). A content model consists of a standard Borel coefficient space Z, a proper prior P_0 on Z, and jointly measurable sample-section maps $f_i : \mathsf{Z} \times \mathcal{U}_i \to E_i$ with $f_i(z, c)$ in the fiber over c. For a recognition law Q on Z, the agent sections and finite-design laws are

$$\zeta_i^Q(c) = (f_{i,c})_{\#}Q, \qquad Q_D = (F_D)_{\#}Q, \qquad F_D(z) = \big(f_i(z, c)\big)_{(i,c)\in D}, \tag{1.20}$$

where $f_{i,c} = f_i(\cdot, c)$ and D is a finite set of pairs with $c \in \mathcal{U}_i$. A record mechanism is a normalized kernel $K_a(do_a \mid z) = k_a(o_a \mid z)\lambda_a(do_a)$ with a declared sigma-finite reference measure λ_a and a density k_a jointly measurable in (o_a, z); it may read field evaluations, model presentations, communication maps, and parallel transports along declared paths. A record kernel must be specified through bundle points, equivalently it must transform with every fiber datum it reads. With conditionally independent records, $\mathbb{P}(dz, do) = P_0(dz)\prod_a K_a(do_a \mid z)$. At a record value o write $L_o(z) = \prod_a k_a(o_a \mid z)$ for the likelihood on Z; when $0 < Z_o = \int L_o dP_0 < \infty$, the posterior is $\Pi_o = L_o P_0 / Z_o$ and the free energy of a recognition law is $\mathcal{F}(Q; o) = D_{\mathrm{KL}}(Q \| \Pi_o) - \log Z_o$, as in eq. (4.19).

The sample bundle E_i carries a standard Borel structure by the argument of lemma 1.2, since its transition maps $(c, y) \mapsto (c, g_{\alpha\beta}(c)y)$ are Borel by joint measurability of the action. In a trivialization, $\zeta_i^Q(c)(B) = \int \mathbf{1}_B(f_i(z, c))Q(dz)$ is Borel in c by Tonelli's theorem, so $c \mapsto \zeta_i^Q(c)$ is a Borel section of $\mathcal{B}_i$.

Proposition 1.16 (Consistency and projected free energy). For $D \subseteq D'$ let $r_{D'D}$ delete the added coordinates. Then $(r_{D'D})_{\#}Q_{D'} = Q_D$, and the marginal of Q_D at (i, c) is $\zeta_i^Q(c)$. For a record value o with $0 < Z_o < \infty$, and whenever the free energy is finite,

$$\mathcal{F}(Q; o) = D_{\mathrm{KL}}(Q_D \| (F_D)_{\#}\Pi_o) - \log Z_o + \int D_{\mathrm{KL}}\big(Q(dz \mid F_D = w) \| \Pi_o(dz \mid F_D = w)\big) Q_D(dw). \tag{1.21}$$

Proof. Since $F_D = r_{D'D} \circ F_{D'}$, pushforwards compose, which gives consistency, and the marginal statement is the case of one pair. The second identity is the chain rule for relative entropy applied to Q and Π_o disintegrated along the measurable map F_D, which is the deterministic-kernel case of theorem 5.1, combined with $\mathcal{F}(Q; o) = D_{\mathrm{KL}}(Q \| \Pi_o) - \log Z_o$. □

The last term of eq. (1.21) is the exact cost of the information that a finite description omits. It need not vanish even when D contains every evaluation the records read, because recognition can have the wrong conditional law in unobserved directions. The law-valued configuration of the agents at the contexts in D is the list of marginals of Q_D; it does not determine Q_D. The reciprocal potential of chapter 4 is a separate constitutive object defined on such lists. Under the hypotheses of proposition 4.24, it is not $\mathcal{F}(Q; o)$ when Q is the product of these marginals and the channels are fixed, nor, on finite alphabets, for any rule assigning recognition laws. On infinite content spaces

no such statement holds for coupled recognition laws without a restriction on how Q depends on the configuration. Take Z the product of the agents' spaces, f_i the coordinate maps, P_0 the product of the reference joints of chapter 4, and no record. Then $D_{\mathrm{KL}}(Q\|P_0)$ is the sum of the marginal divergences and the total correlation of Q, and a coupling of atomless marginals can carry any prescribed nonnegative total correlation; a Gaussian copula with correlation ρ carries $-\frac{1}{2}\log(1-\rho^2)$ for two agents. Choosing the coupling of the agent laws at each configuration so that its total correlation equals the excess of the potential over its single-agent terms reproduces the potential with fixed rows, up to a constant, as $\mathcal{F}(Q(x);o)$ with the agent laws as marginals, at every configuration with atomless laws and finite potential. This construction is degenerate, since the coupling only stores the cost, but it shows that the separation of the reciprocal objective from the global one rests on the product recognition family and on the kinetic metric, not on normalization alone. On a finite alphabet the total correlation of a coupling is bounded by the logarithm of the alphabet sizes, which is the boundedness used in part (a) of the proposition.

1.6 What correlations describe

Suppose two represented bits have the independent uniform law. Register the event that they agree. Their conditional joint law assigns probability $1/2$ to each of $(0,0)$ and $(1,1)$. Each marginal remains uniform, while the conditional expectation of their product changes from $1/4$ to $1/2$. An exact joint posterior has acquired dependence that its two marginals cannot report.

This example identifies a concrete need for a joint law: predicting an observable that reads both bits after the shared record. It does not imply that every admissible agent dynamics must propagate that posterior. A coupled product-family dynamics is another precise model, provided it is declared, but it does not preserve the marginal symmetry. Against the hard agreement record, a product recognition law has finite free energy only when it is a point mass on $(0,0)$ or on $(1,1)$, at relative entropy $\log 2$ from the posterior. For a soft record $\Pi \propto e^{J\sigma_1\sigma_2}$ in spin variables, the symmetric product optimum has magnetization $\bar{\sigma}$ solving $\bar{\sigma} = \tanh(J\bar{\sigma})$, which is zero for $J \le 1$ and nonzero for $J > 1$. Similarly, two routes that inspect the same source must share its identity; replacing it by independent redraws specifies a different experiment.

The same distinction governs what a comparison of laws can detect along a loop of routes.

Proposition 1.17 (Law comparisons and shared samples). Let H, H' be bimeasurable bijections of a standard Borel space E, for example two loop transports, and let $\nu \in \mathcal{P}(E)$.

(i) If $H_\#\nu = H'_\#\nu$, every functional of the pair of laws $(\nu, H_\#\nu)$ takes the same value for H and H'. In particular $D_{\mathrm{KL}}(\nu\|H_\#\nu) = 0$ exactly when H preserves ν.

(ii) The law $(\mathrm{id}, H)_\#\nu$ of the shared-sample pair (Y, HY), $Y \sim \nu$, determines H up to ν-null sets.

Proof. Part (i) is immediate because the pair of laws is the same. For (ii), suppose $(\mathrm{id}, H)_\#\nu = (\mathrm{id}, H')_\#\nu$. The complement of the graph of H' is a Borel set B. Its measure under the first law is $\nu\{H \neq H'\}$ and under the second law is zero. □

Each interaction cost of chapters 2 and 4, and each receiver force, depends on a transport only through the transported law, so at one configuration no law-level comparison separates two transports with the same image law. A rotation R leaves a centered Gaussian $\mathcal{N}(0,\Sigma)$ unchanged

exactly when $R\Sigma R^T = \Sigma$; in dimension at least two this stabilizer always contains nonidentity rotations, for example the half-turn in the plane of two eigenvectors, and for isotropic Σ it is the whole rotation group. The dynamics is not blind in the same way. The source-role force of a cost $D_{\mathrm{KL}}(\nu_i \| H_{\#}\nu_j)$ is the receiver likelihood ratio pulled back through H itself, $y \mapsto -(d\nu_i/dH_{\#}\nu_j)(Hy)$, which depends on H pointwise. For a self channel H on a finite alphabet and an initial law ν_0 with $H_{\#}\nu_0 = \nu_0$, the first variation of $D_{\mathrm{KL}}(\nu \| H_{\#}\nu)$ vanishes at ν_0, so the initial velocity of the unit Fisher flow of $D_{\mathrm{KL}}(\nu\|r) + D_{\mathrm{KL}}(\nu\|H_{\#}\nu)$ does not depend on H within the stabilizer; with v that common initial log-velocity, $\nu(t) = \nu_0(1 + tv) + O(t^2)$ and the quadratic expansion of relative entropy give

$$D_{\mathrm{KL}}(\nu(t)\|H_{\#}\nu(t)) = \frac{t^2}{2}\sum_y \nu_0(y)\big(v(y) - v(H^{-1}y)\big)^2 + O(t^3). \tag{1.22}$$

The loop classes of H and of the identity are therefore separated along the trajectory at second order whenever the record-free drift breaks the stabilizer, and on a two-cycle through two agents it can appear at first order, because the source forces at the initial configuration already differ. A record that is sensitive to such a transport at the instant of comparison must read a coupling of the same sample along the routes, as the route mark of the supplement does; under a frame change g at the base point the pair transforms diagonally, $(Y, HY) \mapsto (gY, gHg^{-1}gY)$, so invariant records are diagonal-invariant functions of the pair. Such records require a content law in the sense of definition 1.15.

The configuration x of law-valued agents, a sample tuple $(k_i, m_i)_i$ used in a probabilistic model, and an ensemble of possible configurations are three different objects. A deterministic dynamics on x needs no probability law over configurations. A preparation ensemble can be introduced to assess predictions or compression, but it is then stated explicitly. These distinctions will matter when a parent retains a law over contents while discarding information needed to evolve that law.

Chapter 2

Law-Dependent Interactions

An interaction compares or transmits information between agents. Its law may depend on the agents' current statistical states. The general object is a normalized Markov channel between represented spaces; exponential links of matrix generators are one constitutive example of such a channel, not the definition of an interaction. Gauge covariance specifies how this dependence is represented when coordinates change; it does not require the interaction to be a frame identification or select a unique mechanism. We first construct these interactions at one context. Transport between different contexts is a separate datum, whose transformation law is recorded in appendix D.

2.1 Two sectors and their comparison spaces

Fix a context c and a finite set of agents defined there. Write

$$\nu_i^q = q_i, \qquad \nu_i^s = s_i, \tag{2.1}$$

for the belief and model marginals of agent i. The sector index a takes values in $\{q, s\}$. These laws live on standard Borel spaces E_i^a. No product assumption between the two sectors, or between different agents, is required to define an interaction between their marginals. Such a product is a separate choice when a joint recognition law is needed.

In a vector realization choose reference spaces V_q, V_s and invertible presentation maps

$$U_i^a : V_a \longrightarrow E_i^a, \qquad \bar{q}_i = (U_i^q)_\#^{-1} q_i, \qquad \bar{s}_i = (U_i^s)_\#^{-1} s_i. \tag{2.2}$$

We also write $\bar{\nu}_i^a = (U_i^a)_\#^{-1} \nu_i^a$ and $\bar{\boldsymbol{\nu}}^a = (\bar{\nu}_i^a)_i$ for the tuple of reference laws in sector a. Here E_i^a is agent i's displayed coordinate copy of V_a. At a common context the agents' statistical sections take values in the fibers over one point of bundles eq. (1.2) associated to one principal bundle; in the linear realization with common V_q, V_s these fibers coincide. The per-agent presentations are points of the frame torsor at that context, used to display each agent's law, and are not separate fibers. One structure group G acts through the representations ϱ_q, ϱ_s of eq. (1.1), so a passive change at agent i is a single element g_i, with $g_i^a = \varrho_a(g_i)$. Independent sector changes would require the product group $G_q \times G_s$, which the theory does not use; the collective connection of proposition 3.9 depends on this choice, because the group enters its normal equation. The two presentations do not identify the belief and model interaction maps. Their domains, parameters, and dependence on the laws remain separately specified.

A passive re-expression $y_i' = g_i^a y_i$ sends ν_i^a to $(g_i^a)_\# \nu_i^a$ and U_i^a to $g_i^a U_i^a$. Therefore the reference laws in eq. (2.2) are unchanged. A common reference change is one element $h \in G$; it acts on both sectors at once, $U_i^a \mapsto U_i^a h_a$ with $h_a = \varrho_a(h)$, and sends every reference law to $(h_a^{-1})_\# \bar{\nu}_i^a$. The first

operation changes an agent's displayed coordinates; the second changes the common coordinates in which interaction parameters are written.

Definition 2.1 (Normalized interaction channel). Let S be the standard Borel configuration space of chapter 1. A sector channel from source j to receiver i is a Markov kernel

$$\mathcal{K}^a_{ij} : \mathsf{S} \times E^a_j \rightsquigarrow E^a_i, \tag{2.3}$$

that is, $(x, y) \mapsto \mathcal{K}^a_{ij}[x](y, A)$ is jointly measurable for every measurable A and is a probability measure in A. At configuration x it sends a law to $(\nu^a_j \mathcal{K}^a_{ij}[x])(A) = \int \mathcal{K}^a_{ij}[x](y, A)\nu^a_j(dy)$. When the channel does not depend on the configuration we omit x. Covariance requires

$$\mathcal{K}^{a\prime}_{ij}(g^a_j y, A) = \mathcal{K}^a_{ij}(y, (g^a_i)^{-1}A). \tag{2.4}$$

Composition of sending, transmission, and receiving kernels produces another normalized channel by iterated integration. Invertibility is not required. Alternatively, two channels can send the receiver and source laws into a common message space, where their communicated predictions are compared. The primitive self channel $\mathcal{K}^a_{ii}$ is constitutive data. The identity channel is one choice; later constructions may declare a nonidentity self channel, and they state which.

Joint measurability makes $x \mapsto \nu^a_j \mathcal{K}^a_{ij}[x]$ a measurable map into $\mathcal{P}(E^a_i)$, with the sigma-algebra generated by the evaluations, by a monotone-class argument, so the divergences, rows, and potentials built from these channels in chapter 4 are measurable functions of the configuration.

Proposition 2.2 (Covariant law comparison). Under eq. (2.4), the divergence

$$D^a_{ij} = D_{\mathrm{KL}}(\nu^a_i \| \nu^a_j \mathcal{K}^a_{ij}), \tag{2.5}$$

is unchanged by passive bimeasurable coordinate changes, including when its value is infinite.

Proof. Changing the integration variable in the kernel gives $\nu^{a\prime}_j \mathcal{K}^{a\prime}_{ij} = (g^a_i)_{\#}(\nu^a_j \mathcal{K}^a_{ij})$. For any bimeasurable bijection g, absolute continuity is preserved and the transformed Radon–Nikodym derivative is $(d\mu/d\eta) \circ g^{-1}$. Its logarithmic integral is therefore unchanged. Failure of absolute continuity is also preserved, giving infinity on both sides. □

This is the coordinate invariance of relative entropy [124]. It makes the comparison intrinsic without asserting that either law is a posterior under a fixed population model.

2.2 Reference consistency

Per-agent covariance is automatic for any rule written in reference coordinates. If a channel is specified by a reference kernel $\bar{\mathcal{K}}^a_{ij} : V_a \rightsquigarrow V_a$ and displayed as $\mathcal{K}^a_{ij}(y, A) = \bar{\mathcal{K}}^a_{ij}((U^a_j)^{-1}y, (U^a_i)^{-1}A)$, then eq. (2.4) holds identically, and proposition 1.6 with $f = (U^a_i)^{-1}$ gives

$$D_{\mathrm{KL}}(\nu^a_i \| \nu^a_j \mathcal{K}^a_{ij}) = D_{\mathrm{KL}}(\bar{\nu}^a_i \| \bar{\nu}^a_j \bar{\mathcal{K}}^a_{ij}). \tag{2.6}$$

The frames therefore act as a dressing field, as stated after eq. (1.6): per-agent covariance constrains no reference rule. What remains concerns the common reference change, in the two forms distinguished there.

Hypothesis 2.3 (Reference consistency). A reference rule is reference consistent if, for every common reference change $h \in G$, acting through $h_a = \varrho_a(h)$ in both sectors, re-expressing all reference laws and supplied interface data leaves the displayed channel unchanged. A rule whose sector-a kernel reads laws of both sectors is re-expressed in both. For a reference kernel this means

$$\bar{K}^{a\prime}_{ij}\big[(h_a^{-1})_{\#}\bar{\boldsymbol{\nu}}^a\big](h_a^{-1}v, h_a^{-1}B) = \bar{K}^a_{ij}[\bar{\boldsymbol{\nu}}^a](v, B), \tag{2.7}$$

and for the exponential generators below the theory imposes the generator-level strengthening eq. (2.15). For exponential links the hypothesis itself is $e^{L'} = h_a^{-1}e^{L}h_a$, which agrees with eq. (2.15) when the generators have spectra in the strip $|\operatorname{Im} z| < \pi$, by uniqueness of the principal logarithm, and is weaker otherwise; the generator form is the one that differentiation of generators in eq. (2.23) and priors on generators require. Without reference consistency, a displayed interaction depends on the choice of reference point in the frame torsor and is not a function of the bundle data; on a nontrivial principal bundle no global choice of that point exists.

Reference consistency in this sense is naturality: any rule built from its inputs passes once every non-invariant ingredient, such as a fixed metric, code, or direction, is declared as supplied data and re-expressed with the reference laws. It excludes hidden structure and nothing more. Active equivariance holds the rule and every supplied datum fixed and requires the displayed channels to commute with the common action of G. The moment family below satisfies it with no supplied data, since the inverse bilinear form is taken from the law itself, as do invariant references and Haar channels under their declared groups; fixed codes, fixed lab directions, and fixed anchor metrics do not, and reduce the actively symmetric group to their stabilizers. A result that needs either property names it.

2.3 Exponential links: one constitutive example

Choose a message space H_a and reference interfaces

$$T^a_j : V_a \longrightarrow H_a, \qquad O^a_i : H_a \longrightarrow V_a. \tag{2.8}$$

Their direct content map is $M^a_{ij} = U^a_i O^a_i T^a_j (U^a_j)^{-1}$. A deterministic channel can push the entire law through this map even when the image is lower dimensional. A density formula for a nonsingular law cannot then be used on the original ambient space without accounting for that change of support.

Definition 2.4 (Law-dependent reference generator). Let $\mathcal{L}^a_{ij}$ be a measurable function of the tuple of reference laws $\bar{\boldsymbol{\nu}}^a$, and of any explicitly declared interface data, with values in $\operatorname{End}(V_a)$. Define

$$L^a_{ij} = \mathcal{L}^a_{ij}(\bar{\boldsymbol{\nu}}^a), \qquad \Theta^a_{ij} = U^a_i \exp(L^a_{ij})(U^a_j)^{-1}. \tag{2.9}$$

Dependence on the two endpoint laws alone is the special case $\mathcal{L}^a_{ij}(\bar{\nu}^a_i, \bar{\nu}^a_j)$. For a receiving–sending factorization, $L^a_{ij} = O^a_i(\bar{\boldsymbol{\nu}}^a)T^a_j(\bar{\boldsymbol{\nu}}^a)$, provided both maps refer to the same H_a and V_a. The

associated deterministic channel is $\mathcal{K}^a_{ij}(y,\cdot) = \delta_{\Theta^a_{ij}y}$, which is an instance of definition 2.1. For exponential links the primitive self link is the identity, $L^a_{ii} = 0$.

The exponential is taken on the reference endomorphism, not on a map with different domain and codomain. Its values lie in $\exp(\mathfrak{gl}(V_a))$, which is contained in the identity component $\mathrm{GL}^+(V_a)$ and, for $\dim V_a \geq 2$, is a proper subset of it: the matrix $\begin{pmatrix} -1 & 1 \\ 0 & -1 \end{pmatrix}$ has determinant one and no real logarithm. By Culver [50], a real invertible matrix has a real logarithm exactly when, for every negative eigenvalue λ and every size k, the Jordan blocks $J_k(\lambda)$ occur an even number of times; here $J_2(-1)$ occurs once. Orientation-reversing and discrete interactions are therefore not exponential links, although they are channels in the sense of definition 2.1. A smaller group acting on the frames can still re-express exponential links covariantly. Interpreting them as parallel transports of a connection for that smaller group would additionally require generators in its represented Lie algebra.

Definition 2.5 (Group-valued links). Let $\mathfrak{g}$ be the Lie algebra of G, and identify frames with elements $U_i \in G$ as in eq. (1.6). Group-valued links assign to each directed edge $j \to i$ a belief reference link $\bar{\theta}_{ij} \in G$ and a model reference link $\tilde{\bar{\theta}}_{ij} \in G$, either of which may depend on the laws as in definition 2.4, and set

$$\theta_{ij} = U_i \bar{\theta}_{ij} U_j^{-1}, \qquad \tilde{\theta}_{ij} = U_i \tilde{\bar{\theta}}_{ij} U_j^{-1}, \qquad \Theta^q_{ij} = \varrho_q(\theta_{ij}), \qquad \Theta^s_{ij} = \varrho_s(\tilde{\theta}_{ij}). \tag{2.10}$$

Where the reference links lie in the image of the exponential, the exponential chart writes them as $\bar{\theta}_{ij} = \exp(\omega_{ij})$ and $\tilde{\bar{\theta}}_{ij} = \exp(\tilde{\omega}_{ij})$ with generators $\omega_{ij}, \tilde{\omega}_{ij} \in \mathfrak{g}$, and the reference generators of eq. (2.9) are $L^q_{ij} = d\varrho_q(\omega_{ij})$ and $L^s_{ij} = d\varrho_s(\tilde{\omega}_{ij})$. The links are tied when $\theta_{ij} = \tilde{\theta}_{ij}$, that is, when one G-valued link acts on both sectors through the two representations. A law-dependent tied link is then a function of the reference laws of both sectors, and reference consistency is required under the joint action of one $h \in G$.

Opposite edges remain independent: θ_{ji} is not required to equal θ_{ij}^{-1}, in either sector.

Proposition 2.6 (The sector mismatch is covariant). Under a passive change $U_i \mapsto g_i U_i$ at every agent, $\theta_{ij} \mapsto g_i \theta_{ij} g_j^{-1}$ and $\tilde{\theta}_{ij} \mapsto g_i \tilde{\theta}_{ij} g_j^{-1}$. The mismatch

$$\Xi_{ij} = \tilde{\theta}_{ij} \theta_{ij}^{-1} \in G \tag{2.11}$$

transforms by conjugation at the receiver, $\Xi_{ij} \mapsto g_i \Xi_{ij} g_i^{-1}$. Its conjugacy class is therefore independent of the frames, and the tied case $\Xi_{ij} = e$ is a frame-independent condition.

Proof. The reference links are unchanged by a passive change, so

$$\theta'_{ij} = g_i U_i \bar{\theta}_{ij} U_j^{-1} g_j^{-1} = g_i \theta_{ij} g_j^{-1}, \tag{2.12}$$

and likewise for $\tilde{\theta}_{ij}$. Then

$$\Xi'_{ij} = g_i \tilde{\theta}_{ij} g_j^{-1} g_j \theta_{ij}^{-1} g_i^{-1} = g_i \Xi_{ij} g_i^{-1}. \tag{2.13}$$

□

The general theory keeps both link fields. Results that use one G-valued link for both sectors state the tied hypothesis. The channels see only $\varrho_q(\theta_{ij})$ and $\varrho_s(\tilde{\theta}_{ij})$, and these are realized by one G-valued link θ^* exactly when Ξ_{ij} lies in the normal subgroup $\ker\varrho_s \ker\varrho_q$: if $\theta^* = k_q\theta_{ij} = k_s\tilde{\theta}_{ij}$ with $k_a \in \ker\varrho_a$, then $\Xi_{ij} = k_s^{-1}k_q$, and conversely. The frame-independent invariant of the displayed interaction is therefore the conjugacy class of the image of Ξ_{ij} in $G/(\ker\varrho_s \ker\varrho_q)$, and the tied hypothesis that later results need is $\Xi_{ij} \in \ker\varrho_s \ker\varrho_q$, for which $\Xi_{ij} = e$ suffices. Only when the representations are faithful does a nonidentity mismatch mean that no single link realizes both displayed transports; for example, the congruence action $Y \mapsto gYg^T$ of $SO(2)$ has kernel $\{\pm I\}$, and $\tilde{\theta}_{ij} = -\theta_{ij}$ gives the nonidentity mismatch $-I$ with the same displayed model transport.

Proposition 2.7 (Endpoint covariance and reference consistency). The links in eq. (2.9) obey

$$\Theta^{a\prime}_{ij} = g^a_i \Theta^a_{ij} (g^a_j)^{-1}, \tag{2.14}$$

and their associated channels satisfy eq. (2.4). The links are also independent of a common reference rechoice if

$$\mathcal{L}^{a\prime}_{ij}\big((h_a^{-1})_{\#}\bar{\boldsymbol{\nu}}^a\big) = h_a^{-1} L^a_{ij} h_a, \tag{2.15}$$

where the prime includes the re-expression of any supplied interface data.

Proof. Passive site changes leave the reference laws and generator unchanged, so multiplication by the transformed endpoint frames proves eq. (2.14). For the channel, $\delta_{g_i\Theta g_j^{-1} g_j y}(A) = \delta_{\Theta y}(g_i^{-1}A)$. Under a reference rechoice, the exponential series gives $e^{h_a^{-1}Lh_a} = h_a^{-1}e^L h_a$. Substitution cancels both occurrences of h_a in eq. (2.9). □

For factored generators, $O^a_i \mapsto h_a^{-1}O^a_i$ and $T^a_j \mapsto T^a_j h_a$ suffice. Message-coordinate changes $T^a_j \mapsto r_a T^a_j$, $O^a_i \mapsto O^a_i r_a^{-1}$ cancel within the product. By eq. (2.6), eq. (2.14) holds for every generator rule and constrains nothing; eq. (2.15) is the form in which hypothesis 2.3 is imposed on exponential links. It constrains a rule only once the supplied interface data are fixed; with those data re-expressed it is the naturality condition described after hypothesis 2.3.

Proposition 2.8 (Low-rank generator and invertible transport). For finite matrices $O : H \to V$ and $T : V \to H$, the map OT has rank at most $\dim H$, whereas e^{OT} is invertible. Moreover,

$$e^{OT}v = v \quad (v \in \ker T), \qquad \operatorname{im}(e^{OT} - I) \subseteq \operatorname{im} O. \tag{2.16}$$

Proof. The rank statement follows from factorization. The exponential series and $e^{OT}e^{-OT} = I$ prove invertibility. Every positive power of OT vanishes on $\ker T$, and every such power has image in $\operatorname{im} O$. Summing the convergent series proves the remaining assertions. □

Thus the exponential link retains an identity action in unread directions. It is a different operation from sending only the selective content OTv. Because the link is invertible, proposition 1.6 gives $D_{\mathrm{KL}}(\nu_i \| (\Theta_{ij})_{\#}\nu_j) = D_{\mathrm{KL}}((\Theta_{ij}^{-1})_{\#}\nu_i \| \nu_j)$: an exponential link of a low-rank generator compares complete laws and discards nothing. Information-discarding, or selective, channels are noninvertible kernels of definition 2.1. The identity self-link also differs from a coarse internal loop: a parent may retain nonidentity loop information inherited from several child interactions.

2.4 An explicit constitutive family for general laws

The preceding definition does not select the functional $\mathcal{L}$. The following choice shows that nonconstant, reference-consistent generators can depend directly on laws without restricting them to a Gaussian family.

Definition 2.9 (Moment-dependent receiving and sending). For each sector let $V_a = \mathbb{R}^{d_a}$ and admit laws with finite second moments and positive-definite covariance. In reference coordinates write their means and covariances as μ_i^a and Σ_i^a. For a supplied scalar λ_{ij}^a on each nonself directed edge, put

$$L_{ij}^a = \lambda_{ij}^a O_i^a T_j^a, \qquad O_i^a(t) = \mu_i^a t, \qquad T_j^a(v) = (\mu_j^a)^T (\Sigma_j^a)^{-1} v. \tag{2.17}$$

Here $H_a = \mathbb{R}$, and $L_{ij}^a = \lambda_{ij}^a \mu_i^a (\mu_j^a)^T (\Sigma_j^a)^{-1}$ has rank at most one. Set $L_{ii}^a = 0$ separately. The edge coefficient is constitutive data, not a consequence of covariance.

Proposition 2.10 (Equivariance of the moment family). The family eq. (2.17) satisfies eq. (2.15) under every invertible linear reference change.

Proof. Under $U_i^a \mapsto U_i^a h_a$, the moments become $\mu_i^{a\prime} = h_a^{-1}\mu_i^a$ and $\Sigma_j^{a\prime} = h_a^{-1}\Sigma_j^a h_a^{-T}$. Consequently

$$(\mu_j^{a\prime})^T (\Sigma_j^{a\prime})^{-1} = (\mu_j^a)^T (\Sigma_j^a)^{-1} h_a. \tag{2.18}$$

Multiplication with the transformed receiving column gives $L_{ij}^{a\prime} = h_a^{-1} L_{ij}^a h_a$. □

No fixed Euclidean metric was inserted into this comparison: the source covariance supplies the inverse bilinear form and transforms with the law. Finite moments and invertible covariance are assumptions of this example alone. Centered laws give zero generators in this family, so it does not encode all distinctions between laws. The family is never reciprocal on independent means: if μ_i^a and μ_j^a are linearly independent and $\lambda_{ij}^a, \lambda_{ji}^a \neq 0$, then $\Theta_{ij}^a \Theta_{ji}^a \neq I$. Indeed, in reference coordinates $e^{L_{ij}} = I + \varphi_1(t_{ij}) L_{ij}$ with $t_{ij} = \operatorname{tr} L_{ij}$ and $\varphi_1(t) = (e^t - 1)/t > 0$, extended by $\varphi_1(0) = 1$, because $L_{ij}^2 = t_{ij} L_{ij}$ for a rank-one matrix; hence the inverse of $e^{L_{ji}}$ is $I - \varphi_1(-t_{ji}) L_{ji}$ and $e^{L_{ij}} e^{L_{ji}} = I$ would force $\varphi_1(t_{ij}) L_{ij} = -\varphi_1(-t_{ji}) L_{ji}$, an equality of two nonzero rank-one matrices with images $\operatorname{span} \mu_i$ and $\operatorname{span} \mu_j$.

For a non-Gaussian example in two dimensions, take independent centered Laplace coordinates of variance one and translate their joint law to means e_1 and e_2 for two agents. Both covariances equal I. With identity frames and unit edge coefficients,

$$L_{12} = E_{12}, \qquad L_{21} = E_{21}, \qquad \Theta_{12}\Theta_{21} = (I + E_{12})(I + E_{21}) = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}. \tag{2.19}$$

The loop is nontrivial even though both interfaces have rank one. This is a communication loop at one context, not the curvature of a base connection. Chapter 17 constructs base transport from smooth contextual fields of law and link data with a declared comparison rule.

2.5 Loops, reciprocity, and what a law-level cost can see

Opposite attending edges are independent constitutive choices. In particular, Θ_{ji} need not equal Θ_{ij}^{-1}, as eq. (2.19) demonstrates. Geometric reversal of the *same* transport is instead its inverse. Endpoint covariance and conjugation of loop products are the familiar transformation laws of lattice link variables [199], here holding in display coordinates only; neither imposes an inverse relation between two independently modeled interactions. Directed non-inverse links at one context form a representation of the interaction quiver. Under independent changes in $\mathrm{GL}(V_a)$ at each vertex, its polynomial invariants are generated by traces of products along oriented cycles [130]. The passive changes of this theory act through $\varrho_a(G)$, and for a proper subgroup the invariant ring is larger: for orthogonal G, $\mathrm{tr}(\Theta_{ij}^T\Theta_{ij})$ and the other singular-value functions of a single open edge are already invariant. The common reference change is one element of G acting by simultaneous conjugation on all reference links through ϱ_q and ϱ_s at once. When $\varrho_a(G) = \mathrm{GL}(V_a)$, the invariants of that action include traces of arbitrary words in the links [166], for example $\mathrm{tr}(e^{L_{12}}e^{L_{34}})$, which is not a path product; for every subgroup such traces remain invariant.

Proposition 2.11 (Reciprocal and backtrack parts of a directed pair). Let $i \neq j$, let Θ_{ij} and Θ_{ji} be invertible links of one sector with values in $\mathrm{GL}(V_a)$, and suppose the backtrack loop $N_{ij,j} = \Theta_{ji}\Theta_{ij}$ at j has no eigenvalue in $(-\infty, 0]$. Write N_j for $N_{ij,j}$ when the pair is fixed. With the principal square root, put

$$\Omega_{ij} = \Theta_{ij}N_j^{-1/2}, \qquad \Theta_{ij} = \Omega_{ij}N_j^{1/2}. \tag{2.20}$$

Then $\Omega_{ji} = \Omega_{ij}^{-1}$, and under passive endpoint changes $\Omega'_{ij} = g_i\Omega_{ij}g_j^{-1}$ and $N'_j = g_jN_jg_j^{-1}$. This is the unique factorization $\Theta_{ij} = \Omega B_j$, $\Theta_{ji} = \Omega^{-1}B_i$ with the balance condition $B_i = \Omega B_j\Omega^{-1}$ and with the spectrum of B_j in the open right half-plane.

Proof. The loop $N_i = \Theta_{ij}\Theta_{ji} = \Theta_{ij}N_j\Theta_{ij}^{-1}$ is similar to N_j, so it also has no eigenvalue in $(-\infty, 0]$. For a primary matrix function f defined on the common spectrum of AB and BA, $Bf(AB) = f(BA)B$ [107, Chapter 1]. With $A = \Theta_{ij}$, $B = \Theta_{ji}$ and $f(z) = z^{-1/2}$,

$$\Omega_{ji}\Omega_{ij} = \Theta_{ji}N_i^{-1/2}\Theta_{ij}N_j^{-1/2} = N_j^{-1/2}\Theta_{ji}\Theta_{ij}N_j^{-1/2} = I. \tag{2.21}$$

For the transformation law, $N'_j = g_jN_jg_j^{-1}$ by eq. (2.14), and the principal square root commutes with similarity, so $(N'_j)^{-1/2} = g_jN_j^{-1/2}g_j^{-1}$. For uniqueness, balance gives $N_j = \Omega^{-1}B_i\Omega B_j = B_j^2$, and the square root with spectrum in the open right half-plane is the principal one, so $B_j = N_j^{1/2}$ and $\Omega = \Omega_{ij}$. □

The principal square root and the identity $Bf(AB) = f(BA)B$ for primary matrix functions are in Higham [107, Sections 1.7 and 1.8]. The reciprocal parts Ω form a representation in which reversal is inversion, as for the transport of the extended general linear frame bundle in appendix D, while N_j records nonreciprocity. A real square root, and hence a balanced real split, can fail to exist: on the moment family below with identity frames, unit covariances, $\mu_1 = e_1$, $\mu_2 = e_2$, and coefficients with $\lambda_{12}\lambda_{21} \leq -4$, the backtrack has determinant one and trace $2 + \lambda_{12}\lambda_{21} \leq -2$, so it is similar to $J_2(-1)$ or has two distinct negative eigenvalues, and it has no real square root. The case $i = j$ is excluded; for a nonidentity self channel the formula would return the matrix sign of Θ_{ii}.

The split is computed in each sector separately and does not respect group-valued or tied links: the principal root does not commute with a representation when the representation moves an eigenvalue across the branch cut. For $G = SO(2)$ and a tied pair with backtrack rotation angle 0.8π, the fundamental sector has reciprocal part ϱ_q of a rotation, while a charge-two sector has backtrack angle $1.6\pi \equiv -0.4\pi$ and principal root angle -0.2π, not the image 0.8π of the fundamental root. A group-level split avoids this. If $\theta_{ji}\theta_{ij} = \exp(X)$ with $X \in \mathfrak{g}$ and every eigenvalue of $d\varrho_a(X)$ has imaginary part in $(-\pi, \pi)$ in each sector used, put $\Omega^G_{ij} = \theta_{ij}\exp(-X/2)$. Then $\varrho_a(\theta_{ji}\theta_{ij})^{1/2} = \varrho_a(\exp(X/2))$ by uniqueness of the square root with spectrum in the open right half-plane, so $\varrho_a(\Omega^G_{ij})$ is the sector split eq. (2.20) in every such sector, it lies in $\varrho_a(G)$, tied links have tied reciprocal parts, and $\Omega^G_{ji} = (\Omega^G_{ij})^{-1}$ because $\theta_{ij}\theta_{ji} = \exp(\mathrm{Ad}_{\theta_{ij}} X)$.

When $\Theta_{ji} = \Theta_{ij}^T$, eq. (2.20) is the polar decomposition. That condition depends on the frame, since the transpose is not covariant under non-orthogonal changes. A covariant form uses the law metrics $M_i = \Sigma_i^{-1}$ of laws with positive-definite covariance: call the pair law-adjoint when $\Theta_{ji} = M_j^{-1}\Theta_{ij}^T M_i$, a condition preserved by passive changes because $M_i \mapsto g_i^{-T} M_i g_i^{-1}$. For a law-adjoint pair, $N_j = M_j^{-1}(\Theta_{ij}^T M_i \Theta_{ij})$ is a product of two symmetric positive-definite matrices, so its spectrum is positive and the hypothesis of proposition 2.11 holds. Every real power of N_j is self-adjoint for M_j, and therefore $\Omega_{ij}^T M_i \Omega_{ij} = M_j$, that is, $\Omega_{ij}\Sigma_j\Omega_{ij}^T = \Sigma_i$: the reciprocal part carries the sender covariance onto the receiver covariance. In eq. (2.19), $N_2 = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$ is positive definite and $\Omega_{12} = 5^{-1/2}\begin{pmatrix} 2 & 1 \\ -1 & 2 \end{pmatrix}$ is a rotation. The nontrivial two-cycle of that example is therefore pure nonreciprocity, $\Omega_{12}\Omega_{21} = I$; cycle products of the reciprocal parts on cycles of length at least three can carry reciprocal holonomy.

The peer-law comparison terms introduced above and in chapter 4 have the form $D_{\mathrm{KL}}(\nu_i \| \nu_j \mathcal{K}_{ij})$. By proposition 1.17, a loop map H entering a comparison only through $H_{\#}\nu$ cannot be distinguished, at one configuration, from another map with the same transported law, while the shared-sample law $(\mathrm{id}, H)_{\#}\nu$ determines H up to a ν-null set. A rotation R is invisible in this sense against a centered Gaussian $\mathcal{N}(0, \Sigma)$ exactly when $R\Sigma R^T = \Sigma$; for every Σ in dimension at least two this stabilizer contains nonidentity rotations, the half-turns in planes of eigenvectors, and for isotropic Σ it is the whole rotation group. The reciprocal dynamics can still separate such loop maps along trajectories through its source forces, as described after proposition 1.17. Under a frame change g at the base vertex the shared-sample pair transforms diagonally, $(Y, HY) \mapsto (gY, (gHg^{-1})gY)$. A direct shared-sample loop experiment therefore requires a joint law coupling the routes. This is one way to identify the loop map. Grounded edge messages can also identify dictionaries, and hence their products, to the extent allowed by their codes and referents; chapter 18 states that experiment and its identifiability conditions. These observations use more information than the two marginal laws in the original comparison.

Proposition 2.12 (Covariance-dressed loop defect). Let W be the product of the links of a closed walk based at i in a linear sector, for example the two-cycle $W = \Theta_{ij}\Theta_{ji}$, with links produced by a reference-consistent rule, and let Σ_i be the covariance of ν_i in the same displayed coordinates, assumed positive definite. Then

$$Q_{ij} = \tfrac{1}{2}\operatorname{tr}\big(\Sigma_i (I - W)^T \Sigma_i^{-1} (I - W)\big) = \tfrac{1}{2}\big\|\Sigma_i^{-1/2}(I - W)\Sigma_i^{1/2}\big\|_F^2, \tag{2.22}$$

is unchanged by passive changes and by every common invertible reference change, is nonnegative, and vanishes exactly when $W = I$.

Proof. Under $g \in \mathrm{GL}(E_i)$, $W \mapsto gWg^{-1}$ and $\Sigma_i \mapsto g\Sigma_i g^T$. Substitution gives $\frac{1}{2}\operatorname{tr}(g\Sigma_i(I - W)^T\Sigma_i^{-1}(I - W)g^{-1})$, which equals the original trace by cyclicity. Because the rule is reference consistent, a common reference change acts on the reference loop and reference covariance in the same way; for a rule that is not, the displayed links themselves change and Q_{ij} changes with them. The second expression is a squared Frobenius norm of a similarity transform of $I - W$, which proves the last two assertions. □

Continuous conjugation-invariant functions of W alone cannot detect a unipotent loop such as $I + E_{12}$, whose conjugacy class contains I in its closure, whereas $Q = \frac{1}{2}$ for that loop with $\Sigma_i = I$ and $Q = 2$ with $\Sigma_i = \operatorname{diag}(1, 4)$. The defect is invariant under joint re-expression of the loop and the base covariance; it is not a function of the conjugacy class of W alone. For eq. (2.19), $Q_{12} = \frac{3}{2}$. This link defect remains positive for a nonidentity map that fixes the law, so it differs from a transported-law divergence. It is not a term of the original potential of chapter 4. Chapters 18 and 19 separately introduce latent generators with an anchored loop prior in a declared tangent model. Their recognition dynamics does not alter the fixed-link formation constructions.

2.6 Dependence, direction, and inherited responses

Covariance constrains which law-only generators are possible. If a reference transformation fixes all input laws and all supplied data, eq. (2.15) requires its matrix to commute with the resulting generator. Identical centered laws invariant under every orthogonal transformation therefore cannot select a distinguished off-diagonal direction through a fully orthogonally equivariant law-only rule. Directional interfaces can carry additional information, but that information must be declared.

When links depend on the current laws, differentiation must include that dependence. At fixed frames a variation of an objective has the form

$$d\mathcal{E} = d_\nu\mathcal{E} + \sum_{a,i,j}\langle d_{L^a_{ij}}\mathcal{E}, D\mathcal{L}^a_{ij}[d\bar{\boldsymbol{\nu}}^a]\rangle. \tag{2.23}$$

This statement uses a differentiable statistical family and differentiable constitutive maps. It includes both the sender and receiver occurrences, and every other law on which the generator depends. Freezing the link during an update defines another declared dynamics; it is not generally the derivative of the substituted objective. Nor does substituting live recognition laws into a likelihood make that likelihood part of a fixed generative model during recognition variation.

An inherited parent's state must determine its future boundary response, including the law dependence of its generators. If two child configurations with identical retained data give different future responses, those data do not support autonomous closure.

Chapter 3

Geometry of Statistical Sections

A section's statistical change induces contextual geometry relative to a connection. We construct that tensor, a collective rule for choosing the connection, and conditions for their inheritance.

The role of these objects in the rest of the manuscript is limited and should be stated at the outset. No vector field, source row, channel, or record kernel in chapter 4 through chapter 13 depends on a connection, on the collective connection selected below, or on their curvature; each context evolves by a context-local flow. The formation chapters compute connections, pullback tensors, and curvature as diagnostics of those evolved sections. The separate construction in chapter 16 adds the minimized contextual energy to the reciprocal potential and proves local well-posedness for finite graphs and a regular periodic continuum chart, while the minimizing transports stay on their regular branch. The remaining extensions are recorded in open problem 3.12. Within the formation constructions of those chapters the gauge structure therefore enters only through the group orbits that exact descent may discard, and no selection or inheritance result there depends on a connection or a holonomy.

3.1 Regular law families

Definition 3.1 (Fisher geometry on an admitted family). Let $\mathcal{N}$ be a finite-dimensional smooth manifold of probability laws on a standard Borel space Y. In each parameter chart $\theta \mapsto P_\theta$, assume differentiability in quadratic mean with centered square-integrable score $\ell_\theta : \mathsf{Y} \to \mathbb{R}^d$. At a fixed θ, this means that the Lebesgue decomposition $P_{\theta+\delta} = p_\delta P_\theta + P_\delta^\perp$ satisfies

$$P_\delta^\perp(\mathsf{Y}) = o(|\delta|^2), \qquad \left\| \sqrt{p_\delta} - 1 - \tfrac{1}{2}\delta^\top \ell_\theta \right\|_{L^2(P_\theta)} = o(|\delta|). \tag{3.1}$$

For a tangent vector v, its scalar score ℓ_v is the corresponding directional derivative in this expansion. Require the Fisher tensor

$$g_P^F(v, w) = \mathbb{E}_P[\ell_v \ell_w] \tag{3.2}$$

to be smooth and positive definite on the admitted tangent space. Redundant parameter directions must be removed or treated as null directions before imposing this last requirement.

Condition eq. (3.1) is Le Cam's differentiability in quadratic mean [131]. These family-specific hypotheses allow parameter-dependent supports: the laws of the family need not share null sets, and no common positive density is required [10]. A finite-dimensional family satisfying eq. (3.1) is nevertheless dominated. Differentiability in quadratic mean makes $\theta \mapsto P_\theta$ continuous in total variation, and for a countable dense set $\{\theta_n\}$ the mixture $\sum_n 2^{-n} P_{\theta_n}$ dominates every P_θ. The choice of the Fisher tensor is not arbitrary: up to a constant factor it is the only Riemannian metric on finite statistical simplices invariant under Markov congruent embeddings [39], with extensions

to general sample spaces under the hypotheses of Ay et al. [9]. That characterization concerns the metric on one law space; it does not fix the relative weights of separate factors, which remain constitutive below.

Suppose the finite-dimensional Lie group G acts smoothly on $\mathcal{N}$ by the sample-space pushforwards in definition 1.1. A fixed bimeasurable bijection carries a score ℓ_v to $\ell_v \circ g^{-1}$. Change of variables in eq. (3.2) therefore makes the action a Fisher isometry. Its infinitesimal action is

$$J_P\xi = \left.\frac{d}{dt}\right|_{t=0} (\exp(t\xi))_{\#}P, \qquad \xi \in \mathfrak{g}. \tag{3.3}$$

The regular family must be stable under every group element used in these comparisons.

Proposition 3.2 (Fisher metric of the joint agent law). Let $\zeta_\theta(dk, dm) = q_\theta(dk \mid m)s_\theta(dm)$ be a regular family. Assume the marginal and conditional derivatives obey the product rule, that the joint score is $\ell^\zeta_v(k, m) = \ell^s_v(m) + \ell^{q|m}_v(k)$, and that these scores are square-integrable. Assume differentiation of conditional normalization is valid, so $\int \ell^{q|m}_v(k)q_\theta(dk \mid m) = 0$ for s_θ-almost every m. Then

$$g^F_\zeta(v, w) = g^F_s(v, w) + \int g^F_{q(\cdot|m)}(v, w)s(dm). \tag{3.4}$$

Proof. Expand $\mathbb{E}_\zeta[(\ell^s_v + \ell^{q|m}_v)(\ell^s_w + \ell^{q|m}_w)]$. Each mixed term vanishes after conditioning on m. The two remaining terms are exactly those displayed. □

This is the chain rule for Fisher information [201], stated for a joint law with differentiable conditional normalization.

For the actual product family $q_\theta \otimes s_\theta$, eq. (3.4) reduces to $g^F_q + g^F_s$. The two marginal metrics do not in general recover the joint metric. For example, the law

$$\zeta_\rho(k, m) = \frac{1 + \rho km}{4}, \qquad k, m \in \{-1, 1\}, \qquad |\rho| < 1, \tag{3.5}$$

has two fixed uniform marginals, whereas its joint score is $km/(1 + \rho km)$. Its expected squared score is

$$g^F_{\zeta_\rho}(\partial_\rho, \partial_\rho) = \frac{1}{1 - \rho^2}. \tag{3.6}$$

A declared marginal metric omits this dependence direction. We use the joint law unless a restriction is stated.

3.2 Covariant statistical change

Restrict the statistical bundle of eq. (1.2) to the admitted smooth fiber $\mathcal{N}$, writing $E = P \times_G \mathcal{N}$ and $\pi : E \to \mathcal{C}$. A principal connection ω induces a horizontal lift $H^\omega_e : T_c\mathcal{C} \to T_eE$ and vertical projection $\mathrm{ver}^\omega : T_eE \to V_eE$, where $V_eE = \ker T_e\pi$ and $c = \pi(e)$. The vertical Fisher metric is well defined because the transition action is isometric.

Definition 3.3 (Covariant pullback). For a smooth statistical section $\zeta : \mathcal{U} \to E$, define

$$D^{\omega}\zeta = \mathrm{ver}^{\omega} \circ T\zeta, \qquad h^{\omega}_{\zeta}(X, Y) = g^{F}_{\zeta}(D^{\omega}\zeta X, D^{\omega}\zeta Y). \tag{3.7}$$

The vertical projection removes the base component of $T\zeta$ forced by $\pi \circ \zeta = \mathrm{id}$. The resulting base tensor is positive semidefinite.

In a local frame write A for the connection form and $p(c)$ for the law coordinates. With the convention that the horizontal lift is $(X, -J_p A(X))$,

$$D^{A}p(X) = dp(X) + J_p A(X). \tag{3.8}$$

Here $J_p A(X)$ is a statistical tangent vector.

Proposition 3.4 (Gauge covariance and connection dependence). Under a passive change $p' = g_{\#}p$ and $A' = \mathrm{Ad}_g A - (dg)g^{-1}$,

$$D^{A'}p' = T(g_{\#})D^{A}p, \qquad h^{A'}_{p'} = h^{A}_{p}. \tag{3.9}$$

For two connections on the same bundle, let $\alpha = \omega' - \omega$ denote their adjoint-bundle-valued difference. Along one fixed section put $b_X = J_{\zeta}\alpha(X)$ and $u_X = D^{\omega}\zeta X$. Then

$$D^{\omega'}\zeta X = u_X + b_X, \tag{3.10}$$

$$h^{\omega'}_{\zeta}(X, Y) = h^{\omega}_{\zeta}(X, Y) + g^{F}(u_X, b_Y) + g^{F}(b_X, u_Y) + g^{F}(b_X, b_Y). \tag{3.11}$$

Proof. Differentiation of $p' = g_{\#}p$ gives $dp' = T(g_{\#})dp + J_{p'}((dg)g^{-1})$. Equivariance of the fundamental action gives $J_{p'}\,\mathrm{Ad}_g = T(g_{\#})J_p$. Substitution in eq. (3.8) cancels the inhomogeneous terms. Fisher invariance proves eq. (3.9). Replacing A by $A + \alpha$ in that same derivative adds b; expanding the bilinear pairing gives eq. (3.11). □

Changing the connection changes the comparison. Appendix D records its transformation law in a moving presentation. Chapter 17 derives connections from regular contextual law, interface, and link data with declared comparison rules. Those selectors need not equal the Fisher selector of proposition 3.9, and passive covariance does not remove their constitutive dependence.

Passive invariance does not make the tensor a property of the section alone. Suppose G acts transitively on $\mathcal{N}$. The orbit map $g \mapsto g_{\#}p_0$ is then a submersion and has local sections, so a local gauge makes the section constant, $p(c) = p_0$. In that gauge eq. (3.8) gives $D^{A}p = J_{p_0}A$, and

$$h^{A}_{p}(X, Y) = \mathfrak{k}_{p_0}(A(X), A(Y)), \qquad \mathfrak{k}_{p_0}(\xi, \eta) = g^{F}(J_{p_0}\xi, J_{p_0}\eta). \tag{3.12}$$

As A ranges over local connection forms, this ranges over every positive semidefinite form of rank at most $\dim\mathcal{N}$. For a homogeneous fiber the per-agent contextual tensor therefore records the supplied connection and no gauge-invariant property of the section; this is the situation of a single Gaussian family under the affine group, which acts transitively.

Proposition 3.5 (Transported relative entropy). Fix a smooth curve γ through c with $\dot\gamma(0) = X$, and transport its laws back to the fiber at c:

$$\mu_t = (T^{\omega}_{\gamma(t)\to c})_{\#}\zeta(\gamma(t)). \tag{3.13}$$

Assume, in addition to definition 3.1, that μ_t has positive densities f_t with respect to one measure near $t = 0$, that $\log f_t$ is twice differentiable, and that both derivatives can be interchanged with the integrals of f_t and $f_0 \log f_t$. Assume their second derivatives are continuous at zero. Then

$$D_{\mathrm{KL}}(\zeta(c)\|\mu_t) = \frac{t^2}{2} h^{\omega}_{\zeta}(X, X) + o(t^2). \tag{3.14}$$

Proof. The derivative of the transported law is $D^{\omega}\zeta X$ by the horizontal-lift convention. Write $\ell = \partial_t \log f_t|_0$ for its score. Differentiating normalization once gives $\mathbb{E}_{\mu_0}\ell = 0$. Differentiating twice gives $\mathbb{E}_{\mu_0}[\partial_t^2 \log f_t|_0 + \ell^2] = 0$. The first derivative of the displayed relative entropy is therefore zero and its second derivative is $\mathbb{E}_{\mu_0}\ell^2 = h^{\omega}_{\zeta}(X, X)$. Taylor's theorem proves the claim. □

This is the classical second-order relation between relative entropy and Fisher information [123], applied to the transported law. Quadratic-mean differentiability alone does not justify this second-order KL expansion.

3.3 Null directions and quotients

Proposition 3.6 (Rank and quotient geometry). For the positive fiber Fisher metric in definition 3.1,

$$\operatorname{rad} h^{\omega}_{\zeta} = \ker D^{\omega}\zeta, \qquad \operatorname{rank} h^{\omega}_{\zeta} = \operatorname{rank} D^{\omega}\zeta. \tag{3.15}$$

If the latter rank is constant, $K = \ker D^{\omega}\zeta$ is a smooth subbundle and $T\mathcal{U}/K$ carries the positive-definite metric $\bar h([X], [Y]) = h^{\omega}_{\zeta}(X, Y)$.

Proof. The diagonal value of h^{ω}_{ζ} is the squared Fisher norm of $D^{\omega}\zeta X$, so it vanishes exactly on the derivative's kernel. For a positive semidefinite form, its diagonal null set is its radical. Rank-nullity gives the second identity, and the constant-rank theorem for bundle maps makes K a smooth subbundle. The pairing is independent of representatives because it annihilates K, and it is positive on every nonzero quotient class. □

A quotient vector bundle is not automatically the tangent bundle of a quotient manifold. Suppose a smooth Hausdorff manifold $\bar{\mathcal{U}}$ and a surjective submersion $q : \mathcal{U} \to \bar{\mathcal{U}}$ with connected fibers satisfy $\ker Tq = K$. The tensor descends to $\bar{\mathcal{U}}$ exactly when it is basic: besides annihilating K, it obeys $\mathcal{L}_Z h^{\omega}_{\zeta} = 0$ for every vertical field $Z \in \Gamma(K)$. Necessity follows by differentiating a pulled-back tensor along a fiber. For sufficiency, use submersion coordinates (y, z): the annihilation condition leaves only $dy^a dy^b$ components, and the Lie derivatives in the z directions make their coefficients independent of z. Connected fibers make these local definitions agree across each fiber. The resulting tensor is positive definite by eq. (3.15). A basic semidefinite tensor of this kind is a transverse metric of the foliation by the fibers; extended by any metric along the fibers, it gives a bundle-like metric [147, 169].

Integrability of the radical and basicness are both measured by the curvature. In a local frame write $V = D^A p$, $j(p, \xi) = J_p \xi$, $d_p j$ for its derivative in p in a chart of $\mathcal{N}$, and $F = dA + [A \wedge A]$ with $[A \wedge A](X, Y) = [A(X), A(Y)]$.

Proposition 3.7 (Curvature criteria for the radical). For base vector fields X, Y,

$$V([X, Y]) = \big(X(VY) + d_p j(VY, A(X))\big) - \big(Y(VX) + d_p j(VX, A(Y))\big) - J_p F(X, Y). \tag{3.16}$$

Consequently, where the rank of V is constant, the radical $K = \ker V$ is involutive exactly when $J_p F(X, Y) = 0$ for all $X, Y \in K$. For $Z \in \Gamma(K)$ and arbitrary X, Y,

$$(\mathcal{L}_Z h^A_p)(X, Y) = g^F\big(J_p F(Z, X), VY\big) + g^F\big(VX, J_p F(Z, Y)\big), \tag{3.17}$$

so an involutive radical carries a basic tensor exactly when the right side vanishes. Both conditions hold for every flat connection.

Proof. In chart coordinates, $dp([X, Y]) = X(dp(Y)) - Y(dp(X))$ and $A([X, Y]) = X(A(Y)) - Y(A(X)) - dA(X, Y)$. Differentiating $VY = dp(Y) + J_p A(Y)$ along X gives $X(VY) = X(dp(Y)) + d_p j(dp(X), A(Y)) + J_p X(A(Y))$. Combining these identities and substituting $dp = V - J_p A$ leaves the terms of eq. (3.16) together with $d_p j(J_p A(X), A(Y)) - d_p j(J_p A(Y), A(X))$. That expression is the bracket of the fundamental fields of $A(X)$ and $A(Y)$, which for a left action equals $-J_p[A(X), A(Y)]$; with the $-J_p dA(X, Y)$ term it gives $-J_p F(X, Y)$. If $X, Y \in \Gamma(K)$, then $VX = VY = 0$ identically, and eq. (3.16) reduces to $V([X, Y]) = -J_p F(X, Y)$, which proves the involutivity criterion.

For eq. (3.17), write $(\mathcal{L}_Z h)(X, Y) = Z(g_p(VX, VY)) - h([Z, X], Y) - h(X, [Z, Y])$, expand the first term with $dp(Z) = -J_p A(Z)$, and insert eq. (3.16) with $VZ = 0$. The terms containing $\xi = A(Z)$ combine into $-(\mathcal{L}_{\xi^*} g^F)(VX, VY)$, the Lie derivative of the Fisher metric along the fundamental field of ξ, which vanishes because the action is isometric. The curvature terms remain. If $F = 0$, both right sides vanish. □

The criteria are sharp. On the trivial translation bundle over $\mathbb{R}^3$, use the unit-variance normal location family, the constant section of mean zero, and $A = dz - x\,dy$. Its location Fisher metric is one, so

$$D^A p = dz - x\,dy, \qquad h^A_p = (dz - x\,dy)^2. \tag{3.18}$$

This has constant rank one, but its null fields ∂_x and $\partial_y + x\partial_z$ have bracket ∂_z, which is not null. Here $F = -dx \wedge dy$ and $V([\partial_x, \partial_y + x\partial_z]) = 1 = -F(\partial_x, \partial_y + x\partial_z)$, as eq. (3.16) requires. The connection in this example is supplied, not selected: for a single location family under translations the minimizing connection of proposition 3.9 makes the tensor vanish, and proposition 3.10 shows that the minimal tensor has an involutive radical and is basic wherever J and the minimal tensor have constant rank. Gaussian laws are not required for the preceding results.

A contextual tangent quotient differs from a group-orbit quotient or measurable parent retention, which may discard information having no Fisher tangent.

3.4 A collective choice of connection

Several law sections can determine a common comparison rule by minimizing their shared frame motion. Choose smooth comparisons S_i placing the participating laws in compatible associated fibers, and write $p_i = (\mathsf{S}_i)_{\#}\zeta_i$. For bimeasurable comparisons, independent source changes g_i and a common target change g act by

$$\zeta_i' = (g_i)_{\#}\zeta_i, \qquad \mathsf{S}_i' = g\mathsf{S}_i g_i^{-1}, \qquad p_i' = g_{\#}p_i. \tag{3.19}$$

The compositions in this formula mean composition of represented sample maps. More general comparisons may be declared, but their target laws must still form an admitted smooth equivariant family.

Definition 3.8 (Collective Fisher data). Let $\boldsymbol{p} = (p_i)_i$ lie in a finite product of regular statistical manifolds with diagonal G-action. Choose positive constants w_i and define

$$\mathsf{G}_{\boldsymbol{p}}(v, w) = \sum_i w_i g_{p_i}^F(v_i, w_i), \tag{3.20}$$

$$J_{\boldsymbol{p}}\xi = (J_{p_i}\xi)_i, \qquad \mathfrak{h}_{\boldsymbol{p}} = \ker J_{\boldsymbol{p}}, \qquad H_{\boldsymbol{p}} = (\operatorname{im} J_{\boldsymbol{p}})^{\perp_{\mathsf{G}}}, \tag{3.21}$$

$$\mathfrak{k}_{\boldsymbol{p}}(\xi, \eta) = \mathsf{G}_{\boldsymbol{p}}(J_{\boldsymbol{p}}\xi, J_{\boldsymbol{p}}\eta). \tag{3.22}$$

The weights and comparisons are constitutive data. A different declared joint statistical family may replace the product if its metric and group action satisfy the same hypotheses.

Proposition 3.9 (Collective minimizing connection class). Suppose $J_{\boldsymbol{p}(c)}$ has constant rank on $\mathcal{U}$. For each contextual direction X, the minimizers of

$$a \longmapsto \|d\boldsymbol{p}(X) + J_{\boldsymbol{p}}a\|_{\mathsf{G}}^2 \tag{3.23}$$

form an affine class $A^F(X) + \mathfrak{h}_{\boldsymbol{p}}$, characterized by

$$\mathfrak{k}(A^F(X), \xi) = -\mathsf{G}(d\boldsymbol{p}(X), J\xi), \tag{3.24}$$

$$r_X := d\boldsymbol{p}(X) + JA^F(X) = P_H d\boldsymbol{p}(X). \tag{3.25}$$

Smooth local representatives can be chosen, linear in X. With $h^{\min}(X, Y) = \mathsf{G}(r_X, r_Y)$, any other local connection form A satisfies

$$h^A(X, Y) = h^{\min}(X, Y) + \mathfrak{k}((A - A^F)(X), (A - A^F)(Y)). \tag{3.26}$$

The minimizing class transforms by the affine connection law, and r transforms by the tangent action.

Proof. Project $-d\boldsymbol{p}(X)$ orthogonally onto $\operatorname{im} J$. Its preimages under J give all minimizing a, and two preimages differ exactly by $\ker J$. Orthogonality to every $J\xi$ gives the normal equation. At constant rank, choose a smooth local complement to $\ker J$; the positive form $\mathfrak{k}$ restricted to it has a smooth inverse, providing the stated representative. Writing $d\boldsymbol{p} + JA = r + J(A - A^F)$ and using orthogonality gives eq. (3.26). Under a passive target change, the derivative and connection terms

cancel exactly as in eq. (3.9). Isometry of the action carries minimizers to minimizers and residuals to residuals. □

If infinitesimal isotropy is nonzero, the construction leaves stabilizer-valued components undetermined. An arbitrarily chosen pseudoinverse representative need not obey the affine law. Global minimizing representatives nevertheless exist on any paracompact domain of constant rank. Being minimizing at c is a frame-independent property of a connection, and at each c the minimizing values form an affine subspace. Given local minimizing connections ω_α and a subordinate partition of unity ϕ_α, the connection $\sum_\alpha \phi_\alpha \omega_\alpha$ is an affine combination of minimizers at every point and is therefore minimizing. What requires $\ker J = 0$ is uniqueness: then the local forms are unique, obey the affine law, and agree on overlaps; discrete stabilizers do not obstruct this. Curvature statements about a class should be made for $J_{\boldsymbol{p}}F$, which by the next proposition is the same for every smooth minimizing representative, or after a representative has been fixed, since the stabilizer-valued part of the full curvature depends on that choice.

Proposition 3.10 (Class curvature of the minimizing connection). Assume the hypotheses of proposition 3.9, and let A and $A+\beta$ be smooth minimizing representatives, so that β takes values in $\ker J_{\boldsymbol{p}}$ and $D^A\boldsymbol{p} = r$ takes values in H. Then $J_{\boldsymbol{p}}F^{A+\beta} = J_{\boldsymbol{p}}F^A$ on every pair of directions. Where $h^{\min}$ has constant rank, $J_{\boldsymbol{p}}F^A(Z,X) = 0$ for every Z in its radical and every X; consequently the radical is involutive and $h^{\min}$ is basic. No freeness or properness of the action is used.

Proof. Write G for the collective metric in a chart. The fundamental fields $\xi^*(p) = J_p\xi$ are Killing fields, $(D_{\xi^*}\mathsf{G})(u,w) + \mathsf{G}(d_pj(u,\xi),w) + \mathsf{G}(u,d_pj(w,\xi)) = 0$, and for a left action $[\xi^*,\eta^*] = -[\xi,\eta]^*$. For $\beta \in \ker J_p$ the field β^* vanishes at p, so $u \mapsto d_pj(u,\beta)$ is skew for G; it maps $J_p\xi$ to $d_pj(J_p\xi,\beta) - d_pj(J_p\beta,\xi) = [\xi^*,\beta^*](p) \in \operatorname{im} J_p$, so it preserves $\operatorname{im} J_p$ and hence H_p. Both representatives give the same $V = r$, so eq. (3.16) for A and $A+\beta$ gives $J(F^{A+\beta} - F^A)(X,Y) = d_pj(r_Y,\beta(X)) - d_pj(r_X,\beta(Y))$. The left side lies in $\operatorname{im} J$ and the right side in H, so both vanish.

Now let $Z \in \Gamma(K)$, so $r_Z = 0$ and $d\boldsymbol{p}(Z) = -J_p\eta$ with $\eta = A(Z)$. Differentiate $\mathsf{G}(r_X, J_p\xi) = 0$ along Z for fixed ξ. The derivative of the metric along $-\eta^*$ is given by the Killing equation, and the derivative of $J_p\xi$ is $-d_pj(J_p\eta,\xi)$, so

$$\mathsf{G}\big(Z(r_X) + d_pj(r_X,\eta), J_p\xi\big) + \mathsf{G}\big(r_X, d_pj(J_p\xi,\eta) - d_pj(J_p\eta,\xi)\big) = 0. \tag{3.27}$$

The second argument of the last pairing is $[\xi^*,\eta^*](p) \in \operatorname{im} J_p$, which is orthogonal to r_X. Hence $Z(r_X) + d_pj(r_X,\eta)$ is orthogonal to $\operatorname{im} J_p$. With $VZ = 0$, eq. (3.16) reads $r_{[Z,X]} = Z(r_X) + d_pj(r_X,\eta) - J_pF(Z,X)$; projecting onto $\operatorname{im} J_p$ gives $J_pF(Z,X) = 0$. Then proposition 3.7 gives involutivity, and the right side of eq. (3.17) vanishes, so $h^{\min}$ is basic. □

The statement restricted to the radical therefore carries no information about the class; the class curvature is $J_{\boldsymbol{p}}F$ on pairs of directions transverse to the radical.

Proposition 3.11 (Shape-space form of the minimal tensor). Assume the hypotheses of proposition 3.9 with $\ker J_{\boldsymbol{p}(c)} = 0$ on $\mathcal{U}$. Let $\theta^{\mathrm{mech}}(v) = \mathfrak{k}^{-1}\mathsf{G}(v, J\cdot)$ denote the mechanical connection of the diagonal action, defined where the action is locally free. Then $A^F = -\boldsymbol{p}^*\theta^{\mathrm{mech}}$ and $F^{A^F} = -\boldsymbol{p}^*\Omega^{\mathrm{mech}}$, where $\Omega^{\mathrm{mech}} = d\theta^{\mathrm{mech}} - [\theta^{\mathrm{mech}} \wedge \theta^{\mathrm{mech}}]$ is horizontal. Hence $F^{A^F}(Z,\cdot) = 0$ for Z in the radical of $h^{\min}$, which in this locally free case recovers the conclusions of proposition 3.10. If

the action is free and proper, so that $\pi : \mathcal{N}^n \to \mathcal{N}^n/G$ is a Riemannian submersion for the quotient metric g_{sh}, then

$$h^{\min} = (\pi \circ \boldsymbol{p})^* g_{\mathrm{sh}}, \qquad \operatorname{rad} h^{\min} = \ker d(\pi \circ \boldsymbol{p}). \tag{3.28}$$

Proof. Write $\theta = \theta^{\mathrm{mech}}$ and $\Omega = \Omega^{\mathrm{mech}}$ in this proof. The normal equation eq. (3.25) reads $A^F(X) = -\theta(d\boldsymbol{p}(X))$. Pullback commutes with d and with the bracket of Lie-algebra-valued forms, so $F^{A^F} = -\boldsymbol{p}^*(d\theta - [\theta \wedge \theta])$. For a left action $[\xi^*, \eta^*] = -[\xi, \eta]^*$, and $\theta(\xi^*) = \xi$ is constant, so $\Omega(\xi^*, \eta^*) = -\theta([\xi^*, \eta^*]) - [\xi, \eta] = 0$. For a horizontal field v, $\Omega(\xi^*, v) = -\theta([\xi^*, v]) = (\mathcal{L}_{\xi^*}\theta)(v)$, and equivariance of θ makes this Lie derivative a commutator of ξ with $\theta(v) = 0$, so it vanishes. Thus Ω is horizontal. If Z lies in the radical, then $r_Z = 0$ and $d\boldsymbol{p}(Z) = -JA^F(Z)$ is an orbit direction, so $F^{A^F}(Z, \cdot) = 0$. Under a free proper action, $d\pi$ restricted to H is an isometry onto the tangent space of the orbit space, and $d\pi$ annihilates $\operatorname{im} J$. Since $r_X = P_H d\boldsymbol{p}(X)$, $h^{\min}(X, Y) = g_{\mathrm{sh}}(d\pi(d\boldsymbol{p}(X)), d\pi(d\boldsymbol{p}(Y)))$. □

The collective construction is therefore classical. The minimizing connection is the mechanical connection of Smale [182], with the connection form and locked inertia of Kummer [125] and Marsden et al. [143], the orthogonal separation of rotations from shape change studied for molecules by Guichardet [98] and as a gauge field of the n-body problem by Littlejohn and Reinsch [139]; eq. (3.26) is the split of kinetic energy into shape and locked-rotation parts, and $h^{\min}$ is the pullback of the quotient metric of the Riemannian submersion onto shape space [154]. The construction is applied here with the Fisher metric of the declared collective family as the kinetic metric; its Hermitian analogue, in which the common phase of a state vector is removed by the same orthogonal splitting, gives the Riemannian metric of Provost and Vallee [167] on manifolds of quantum states. Rank is bounded by the shape dimension, $\operatorname{rank} h^{\min} \le \dim H = \dim \mathcal{N}^n - \dim(\text{orbit})$. A single transitive family has $H = 0$ and zero minimizing residual; a collective family need not.

The constant-rank hypothesis fails where infinitesimal isotropy increases. Agreement need not cause such an increase: any number of unit-covariance planar normal laws with common nonzero mean e_1 has a free diagonal $SO(2)$ action and positive locked inertia. If the infinitesimal stabilizer does increase, the rank of J drops; a smooth extension of the minimizing Lie-algebra-valued representative is then not guaranteed [136]. For example, let $SO(2)$ act on the unit-covariance normal family on $\mathbb{R}^2$ and take the section $\mu(c) = (\epsilon, c)$ for one agent. Then $J\xi = \xi(-c, \epsilon)$, $\mathfrak{k} = c^2 + \epsilon^2$, and

$$A^F(\partial_c) = -\frac{\epsilon}{c^2 + \epsilon^2}, \qquad \int_{\mathbb{R}} A^F(\partial_c) dc = -\pi, \tag{3.29}$$

so as $\epsilon \to 0$ the connection concentrates to $-\pi\delta(c)$ at the isotropic point. The minimal tensor stays bounded, $h^{\min}(X, X) \le \|d\boldsymbol{p}(X)\|_{\mathsf{G}}^2$ by the choice $a = 0$. This one-dimensional example exhibits connection concentration, not curvature blowup. A smooth collective connection and its curvature require the relevant regularity hypotheses; the displayed bound for $h^{\min}$ alone does not control derivatives of a connection near a change of orbit type.

The constant weights w_i in definition 3.8 have a probabilistic reading. With $\sum_i w_i = 1$, the weighted metric is the Fisher metric, in the $\boldsymbol{p}$ directions, of the labeled family $\sum_i w_i \delta_i \otimes p_i$. If the weights depend on the context, its Fisher metric acquires the term $\sum_i dw_i \otimes dw_i / w_i$. The group acts on samples and not on labels, so the weight scores have zero Fisher pairing with every orbit direction; the normal equation is unchanged and $h^{\min}$ gains that connection-independent term.

A correlated population law, whose Fisher metric sees dependence directions invisible to every marginal (proposition 3.2), is not of this weighted product form and is not treated here.

The collective connection is also the auxiliary field of a gauge-invariant contextual energy. For a Riemannian metric γ on a domain of $\mathcal{C}$ with volume form dvol_γ, put

$$\mathcal{E}_{\mathrm{ctx}}[\boldsymbol{p}, A] = \tfrac{1}{2}\int \gamma^{\mu\nu}\mathsf{G}\big(D^A_\mu \boldsymbol{p}, D^A_\nu \boldsymbol{p}\big)\mathrm{dvol}_\gamma. \tag{3.30}$$

By eq. (3.9) it is gauge invariant. Its integrand is a sum over directions of the costs minimized in proposition 3.9, so at fixed $\boldsymbol{p}$ its minimizers over A are the minimizing connections, and the minimum is $\frac{1}{2}\int \mathrm{tr}_\gamma\, h^{\min}\mathrm{dvol}_\gamma$. This is the structure of the $\mathbb{CP}^{N-1}$ sigma model, whose gauge field has no kinetic term and is eliminated algebraically [51]; the analogy is structural and does not import that model's quantum dynamics.

Open problem 3.12 (Extensions of the contextual Fisher flow). Chapter 16 supplies a finite-graph construction and a continuum local theorem on a flat torus. Auxiliary elimination diffuses the law shape and leaves frame and source-row equations without independent diffusion. Global regularity, changes of orbit type, physical boundaries, independently evolving rows and communication links, and a graph-refinement theorem remain open. The quotient target can be open and positively curved, so the local theorem alone gives no global continuation. Deriving a kinetic term for the connection from a declared ensemble of contextual sections is a separate problem; that ensemble is not supplied. A first step toward graph refinement is energy-level consistency: for the edge cost of chapter 16 between $\boldsymbol{p}(c)$ and $\boldsymbol{p}(c+\varepsilon X)$, minimized over transports near the identity, ε^{-2} times the minimum should converge to $\frac{1}{2}h^{\min}(X,X)$ on regions of constant rank and uniformly positive locked inertia, with minimizer $\exp(-\varepsilon A^F(X)+O(\varepsilon^2))$; convergence of the gradient flows is the harder part. Because an eliminated energy of the form eq. (3.30) is invariant under every local frame change, it exerts no force along frame directions, so independent frame diffusion requires an energy that is not locally gauge invariant; source rows, which are frame invariant, could be diffused by adding a row Dirichlet term, a constitutive choice that chapter 16 does not make.

Proposition 3.13 (Aggregate covariant derivative). Let $\mathcal{M}$ be a smooth equivariant map from the collective statistical family to a finite-dimensional representation W of G. For $M = \mathcal{M}(\boldsymbol{p})$ define

$$e(X) = d\mathcal{M}_{\boldsymbol{p}} r_X = dM(X) + \rho_*(A^F(X))M. \tag{3.31}$$

This W-valued one-form is independent of the representative of the minimizing class. If $\dim\mathcal{C} = \dim W$, it is an isomorphism exactly when the actual map $d\mathcal{M}_{\boldsymbol{p}} \circ r$ has full rank.

Proof. Differentiating equivariance gives $d\mathcal{M}_{\boldsymbol{p}} J\xi = \rho_*(\xi)M$. Applying this identity to eq. (3.25) proves eq. (3.31). A change in A^F belonging to $\ker J$ has zero image under $d\mathcal{M}$, proving representative independence. The rank statement is the criterion for a linear map between equal-dimensional spaces to be invertible. □

Surjectivity of $d\mathcal{M}|_H$ establishes that a full-rank first jet can be chosen at the law tuple: choose horizontal preimages of a basis of W and realize those as derivatives in a local statistical chart. It does not establish full rank for the supplied section or its preservation under inference. The

determinant specialization in appendix E uses an additional form on W; the Fisher metric above remains positive. Surjectivity itself is a restriction on the number and configuration of the agents; proposition E.2 computes it for proper complex Gaussian targets, for which an invertible aggregate jet on an open four-dimensional base needs at least four agents.

3.5 Information retained by a channel

Theorem 3.14 (Score projection and exact Fisher loss). Let $\{P_\theta\}$ satisfy eq. (3.1) at θ_0, and let $K : \mathsf{Y} \rightsquigarrow \bar{\mathsf{Y}}$ be one normalized Markov kernel independent of θ. Under the joint law $P_{\theta_0}(dy)K(y, d\bar{y})$, the output family is differentiable in quadratic mean with score and Fisher matrix

$$\bar{\ell}(\bar{Y}) = \mathbb{E}[\ell(Y) \mid \bar{Y}], \tag{3.32}$$

$$I(P_{\theta_0}) - I(P_{\theta_0}K) = \mathbb{E}\,\mathrm{Cov}(\ell(Y) \mid \bar{Y}) \succeq 0. \tag{3.33}$$

Equality in direction v holds exactly when $v^\top \ell(Y)$ is measurable from $\bar{Y}$ up to a joint null set.

Proof. Lift the decomposition in eq. (3.1) to the joint space. Its absolutely continuous square root is $V_\delta = \sqrt{p_\delta(Y)}$ with the same L^2 remainder, and its singular mass is still $o(|\delta|^2)$. For its output, put $m_\delta = \mathbb{E}[V_\delta \mid \bar{Y}]$ and $v_\delta = \mathbb{E}[(V_\delta - m_\delta)^2 \mid \bar{Y}]$. The projected density of this absolutely continuous part is $q_\delta = m_\delta^2 + v_\delta$. Conditional expectation is an L^2 contraction, so $m_\delta = 1 + \frac{1}{2}\delta^\top \mathbb{E}[\ell \mid \bar{Y}] + o_{L^2}(|\delta|)$.

To replace m_δ by $\sqrt{q_\delta}$, observe that $m_\delta \geq 0$ and

$$(\sqrt{q_\delta} - m_\delta)^2 = v_\delta \frac{v_\delta}{(\sqrt{q_\delta} + m_\delta)^2} \leq v_\delta. \tag{3.34}$$

Set the fraction to zero where its numerator and denominator both vanish. It tends to zero in probability. Moreover $v_\delta/|\delta|^2$ is uniformly integrable as $\delta \to 0$: the square-root expansion expresses $(V_\delta - 1)/|\delta|$ as a bounded finite-dimensional combination of the fixed L^2 scores plus an L^2 remainder tending to zero, and conditional variance preserves the resulting uniform integrability. Thus the last display has expectation $o(|\delta|^2)$. The projected singular part may acquire an absolutely continuous density a_δ, but its total mass is $o(|\delta|^2)$. The inequality $(\sqrt{q_\delta + a_\delta} - \sqrt{q_\delta})^2 \leq a_\delta$ shows that this change does not affect the score; any remaining singular output mass has the same small bound. This proves output differentiability and its score formula. Total covariance gives the matrix identity. Its quadratic form in v is the expected conditional variance of $v^\top \ell$, which vanishes exactly under the stated measurability condition. □

This score-projection mechanism is the one in Pollard [164, Theorem 7]; the Markov formulation is also treated in Ay et al. [10]. It identifies local score sufficiency. It does not identify a kernel recovering every law in the experiment; that an insufficient statistic can preserve Fisher information is the phenomenon of Kagan and Shepp [113]. For example, take independent Bernoulli variables with probabilities $1/2 + \theta/4$ and $1/2 + \theta^2/4$, and discard the second. At $\theta = 0$ its score vanishes, so no Fisher information is lost. A fixed recovery kernel applied to the first Bernoulli law makes every reconstructed event probability affine in θ, whereas the required second-coordinate probability is quadratic. Such a kernel cannot recover the family on a neighborhood.

Proposition 3.15 (Moving-channel correction). Let the channel instead have density $k_\theta(\bar{y} \mid y)$ with respect to a fixed output measure. Assume the joint family is differentiable in quadratic mean with square-integrable score $\ell_\theta(Y) + b_\theta(Y, \bar{Y})$, where $b_\theta = \partial_\theta \log k_\theta$, over an input family P_θ that is differentiable in quadratic mean with score ℓ_θ. Then $\mathbb{E}[b_\theta \mid Y] = 0$ almost surely, and

$$\bar{\ell}_\theta(\bar{Y}) = \mathbb{E}_\theta[\ell_\theta(Y) + b_\theta(Y, \bar{Y}) \mid \bar{Y}], \tag{3.35}$$

$$I(\bar{Y}) = I(Y) + \mathbb{E}_\theta I(K_\theta(Y, \cdot)) - \mathbb{E}_\theta \operatorname{Cov}(\ell_\theta + b_\theta \mid \bar{Y}). \tag{3.36}$$

Proof. The projection $(y, \bar{y}) \mapsto y$ is a fixed kernel whose output family is P_θ, because every channel is normalized; by theorem 3.14 its output score is $\ell_\theta + \mathbb{E}[b_\theta \mid Y]$, and uniqueness of the L^2 derivative in eq. (3.1) gives $\mathbb{E}[b_\theta \mid Y] = 0$. Apply theorem 3.14 to the fixed output projection $(y, \bar{y}) \mapsto \bar{y}$ of the joint experiment; its score is the stated conditional expectation. The chain rule proposition 3.2, with the input as conditioning variable, gives the joint Fisher matrix as the input matrix plus the average conditional channel matrix. Subtracting the conditional covariance proves the formula. □

Consequently input-to-output Fisher change has no fixed sign for a moving channel, even along one smooth family: for $Y \sim \mathcal{N}(\theta, 1)$ and $\bar{Y} \mid Y \sim \mathcal{N}(\theta Y, 1)$ the output is $\mathcal{N}(\theta^2, \theta^2 + 1)$, with $I(\bar{Y}) = \theta^2(4\theta^2 + 6)/(\theta^2 + 1)^2$, so $I(\bar{Y}) - I(Y)$ equals -1, $\frac{3}{2}$, and $\frac{139}{50}$ at $\theta = 0, 1, 3$. A constant input followed by a Bernoulli output with success probability $(1 + e^{-\theta})^{-1}$ has zero input Fisher information and output information $e^{-\theta}/(1 + e^{-\theta})^2 > 0$. A singular moving deterministic statistic need not have a channel log density at all. Its actual weak derivative, schematically $d(PK) = (dP)K + P(dK)$ when these derivatives exist, must replace that expression. These distinctions apply to the law-dependent channels used in the interaction potential.

3.6 Geometry across a retention

Let $\Psi : E \to \bar{E}$ be a smooth bundle map covering $f : \mathcal{C} \to \bar{\mathcal{C}}$, where both statistical bundles carry positive vertical Fisher metrics and connections $\omega, \bar{\omega}$. Assume the sections are related,

$$\Psi \circ \zeta = \bar{\zeta} \circ f. \tag{3.37}$$

This is a hypothesis: a map over f does not automatically make every fine section descend through f.

Such maps arise from equivariant kernels. Let $\tilde{f} : P \to \bar{P}$ be a principal bundle map covering f with homomorphism $\phi : G \to \bar{G}$, and let $\kappa : \mathrm{Y} \rightsquigarrow \bar{\mathrm{Y}}$ be a Markov kernel with $\kappa(gy, \cdot) = \phi(g)_\# \kappa(y, \cdot)$. Then $\Psi[u, \zeta] = [\tilde{f}(u), \zeta\kappa]$ is the statistical bundle morphism of proposition 1.4: it is well defined, fiberwise affine, and contracts relative entropy, and for a fixed kernel it gives $\Delta_F \succeq 0$ below by theorem 3.14.

Definition 3.16 (Horizontal transport defect). For $X \in T_c\mathcal{C}$, define

$$a_\Psi(X) = T\Psi(H^\omega_{\zeta(c)} X) - H^{\bar{\omega}}_{\Psi(\zeta(c))}(TfX). \tag{3.38}$$

Both terms project to TfX, so their difference lies in the same coarse vertical tangent space. Set $u_X = D^\omega \zeta X$, $\Psi_* = T^V \Psi$, and $\Delta_F = g^F - \Psi_*^* \bar{g}^F$ along the section.

Proposition 3.17 (Signed comparison of base tensors). Under eq. (3.37),

$$D^{\bar{\omega}}\bar{\zeta}(TfX) = \Psi_* u_X + a_\Psi(X), \tag{3.39}$$

$$h^{\omega}_{\zeta} - f^* h^{\bar{\omega}}_{\bar{\zeta}} = u^* \Delta_F - \mathcal{X}_\Psi - \mathcal{Q}_\Psi, \tag{3.40}$$

where

$$\mathcal{X}_\Psi(X,Y) = \bar{g}^F(\Psi_* u_X, a_\Psi(Y)) + \bar{g}^F(a_\Psi(X), \Psi_* u_Y), \tag{3.41}$$

$$\mathcal{Q}_\Psi(X,Y) = \bar{g}^F(a_\Psi(X), a_\Psi(Y)). \tag{3.42}$$

If each fiber map is realized by a fixed parameter-independent Markov kernel, then $\Delta_F \succeq 0$. In that case base contraction holds exactly when, for every X,

$$2\bar{g}^F(\Psi_* u_X, a_\Psi(X)) + \|a_\Psi(X)\|^2_{\bar{g}^F} \le \Delta_F(u_X, u_X). \tag{3.43}$$

Proof. Differentiate eq. (3.37), decompose $T\zeta X = H^\omega X + u_X$, and take coarse vertical parts. This proves the first identity. Expand the coarse Fisher pairing of $\Psi_* u + a_\Psi$ and subtract it from the fine pairing to obtain the second. For a fixed Markov fiber map, theorem 3.14 gives $\Delta_F \succeq 0$. Evaluating the signed identity on the diagonal proves the exact contraction criterion. □

Zero horizontal defect is sufficient, but is not necessary, for eq. (3.43). For a witness, take the unit-variance normal location family under translations on both levels, a fiber map adding independent unit-variance normal noise, so that the coarse location Fisher metric is $\frac{1}{2}$ and $\Delta_F = \frac{1}{2}$, and connections differing by a constant c, so that $a_\Psi = c$ in the location coordinate. For $u_X = 1$, the base tensors are 1 and $\frac{1}{2}(1+c)^2$. The difference $1 - \frac{1}{2}(1+c)^2$ equals $\frac{79}{200}$ at $c = \frac{1}{10}$, which is contraction with a nonzero defect, and $-\frac{1}{8}$ at $c = \frac{1}{2}$, where contraction fails. Information discarded in the fiber and mismatch of horizontal transports enter different terms. For two successive smooth statistical maps R_{01}, R_{12}, the vertical defects also compose exactly:

$$\Delta_{02} = \Delta_{01} + R^*_{01}\Delta_{12}, \qquad \Delta_{ab} = g_a - R^*_{ab} g_b. \tag{3.44}$$

This follows by inserting and subtracting $R^*_{01} g_1$ and using contravariance of pullback. Both terms are nonnegative when the two maps are fixed Markov morphisms.

Proposition 3.18 (Retention of the collective connection). Let $R : \mathcal{N} \to \bar{\mathcal{N}}$ be a smooth G-equivariant map between positive-metric collective statistical families, independent of the contextual coordinate. Denote their metrics by $g, \bar{g}$, their infinitesimal orbit maps by $J, \bar{J}$, and put $\Delta = g - R^* \bar{g}$. Let $\boldsymbol{p}(c)$ be a smooth fine section, and let A^F and $r_X = d\boldsymbol{p}(X) + JA^F(X)$ be a fine minimizing representative and residual. The same representative minimizes the cost at $\bar{\boldsymbol{p}} = R(\boldsymbol{p})$ exactly when

$$\Delta(r_X, J\xi) = 0 \quad \text{for all } X, \xi. \tag{3.45}$$

Under this condition, equality of the entire fine and coarse affine minimizing classes holds exactly when $\ker J = \ker \bar{J}$. For every possible fine first jet, the condition is $\Delta(H, \operatorname{im} J) = 0$. If $\Delta \succeq 0$, vanishing loss on every orbit direction is sufficient. This includes a fixed Markov morphism equipped

with the Fisher metrics of its input and output experiments in the same normalization; arbitrary independently reweighted metrics need not satisfy that premise.

Proof. Equivariance gives $dRJ = \bar{J}$. Thus the coarse residual using A^F is dRr_X. Its normal equation is $\bar{g}(dRr_X, \bar{J}\xi) = 0$. Fine orthogonality gives $g(r_X, J\xi) = 0$, so the coarse equation is equivalent to eq. (3.45). Once the representative minimizes both costs, their classes are $A^F + \ker J$ and $A^F + \ker \bar{J}$, proving the equality criterion. Every horizontal vector is realizable as a residual by choosing the first jet equal to that vector, which gives the all-jets statement. When $\Delta \succeq 0$, Cauchy–Schwarz for this semidefinite form makes $\Delta(v, J\xi) = 0$ whenever $\Delta(J\xi, J\xi) = 0$. □

Contraction alone is insufficient. Two independent unit-variance normal location laws with $J\xi = (\xi, \xi)$ and means $(x, -x)$ have Fisher matrix I and minimizing coefficient zero. Adding independent unit normal noise only to the first sample is a fixed equivariant channel. Its coarse Fisher matrix is $\mathrm{diag}(1/2, 1)$, whose cost $\frac{1}{2}(1+a)^2 + (-1+a)^2$ is minimized at $a = 1/3$. The minimizing connection has changed despite Fisher contraction. Its preservation is an additional requirement beyond the dynamical and response conditions in chapter 6. The minimal tensor itself nevertheless contracts, and the connection shift is measured exactly.

Corollary 3.19 (Exact contraction of the minimal tensor). In the setting of proposition 3.18, assume that the coarse orbit map $\bar{J}$ also has constant rank along $\bar{\boldsymbol{p}} = R(\boldsymbol{p})$, and let $\bar{A}^F$ be a coarse minimizing representative with minimal tensor $\bar{h}^{\min}$ and locked inertia $\bar{\mathfrak{k}}$. Then

$$h^{\min}(X, X) - \bar{h}^{\min}(X, X) = \Delta(r_X, r_X) + \bar{\mathfrak{k}}\big((A^F - \bar{A}^F)(X), (A^F - \bar{A}^F)(X)\big). \tag{3.46}$$

The right side does not depend on the coarse representative. If $\Delta \succeq 0$, in particular for a fixed Markov morphism with Fisher metrics in the same normalization, both terms are nonnegative, so $\bar{h}^{\min} \leq h^{\min}$; the second term vanishes exactly when eq. (3.45) holds.

Proof. By equivariance, the coarse residual of the fine representative is $dR(r_X)$. Applying eq. (3.26) at the coarse level to the connection A^F gives $\bar{g}(dR(r_X), dR(r_X)) = \bar{h}^{\min}(X, X) + \bar{\mathfrak{k}}((A^F - \bar{A}^F)(X), (A^F - \bar{A}^F)(X))$. The left side equals $g(r_X, r_X) - \Delta(r_X, r_X) = h^{\min}(X, X) - \Delta(r_X, r_X)$. The form $\bar{\mathfrak{k}}$ annihilates $\ker \bar{J}$, which gives representative independence. The last statement is the criterion of proposition 3.18. □

In the two-location example above, $h^{\min} = 2$, $\bar{h}^{\min} = \frac{4}{3}$, $\Delta(r, r) = \frac{1}{2}$, and the inertia term is $\frac{3}{2}(\frac{1}{3})^2 = \frac{1}{6}$. The signed identity eq. (3.40) reduces to this one when both levels use collective minimizers over one base and Ψ is induced by R. The horizontal defect is then $a_\Psi = \bar{J}(\bar{A}^F - A^F)$, the coarse derivative is the coarse residual $dR(r) + a_\Psi$, and coarse orthogonality gives $\bar{g}(dR(r), a_\Psi) = -\bar{\mathfrak{k}}(\bar{A}^F - A^F, \bar{A}^F - A^F)$, so $\mathcal{X}_\Psi + \mathcal{Q}_\Psi = -\bar{\mathfrak{k}}(\bar{A}^F - A^F, \bar{A}^F - A^F)$ on the diagonal. The information term is then evaluated on the connection-independent residual, both contributions are nonnegative when $\Delta \succeq 0$, and eq. (3.43) holds automatically. For successive equivariant retentions the identity telescopes, each step contributing its own pair of nonnegative terms. Combined with proposition 3.11, when $\Delta \succeq 0$ the map induced by R between shape spaces does not increase the quotient metric along the section.

Chapter 4

Free Energies and Coupled Dynamics of Agent Laws

Interacting agents can be given three different objectives, and this chapter keeps them apart. In the global picture all agents are parts of one normalized joint over their contents and records, and each agent's local free energy is that global free energy viewed as a function of its own recognition factor. In the per-agent picture each agent has its own normalized generative model in the sense of definition 1.11, in which its own evaluator and the communicated laws of its peers are competing sources of prior information. The third object is a direct interaction potential between laws, specified without an observation likelihood. Its gradient flow moves each source toward the receivers that attend to it. We first develop this reciprocal potential, its source rows, and its Fisher dynamics, because the constructions of chapter 7 to chapter 13 use it. We then give the two free energies and prove two restricted obstructions to identifying the reciprocal potential with a free energy: with the source rows fixed and an identity channel on some attended edge, no normalized model over finitely many finite-valued variables reproduces it on the interior of finite alphabets under any recognition rule, and, for fixed channels on finite alphabets, no normalized model whose recognition is exactly the product of the agent laws reproduces it on any open set unless every attended peer channel is constant in its input. Outside these hypotheses exact identifications exist, and remark 4.25 records them.

4.1 A potential for the two interacting sectors

Fix a context and a finite set of agents. Agent i has state and model spaces $\mathsf{K}_i, \mathsf{M}_i$, both standard Borel. In this chapter its joint law is restricted explicitly to $\zeta_i = q_i \otimes s_i$. Choose a proper model reference r_i and a normalized evaluator $G_i : \mathsf{M}_i \rightsquigarrow \mathsf{K}_i$. The reference joint is

$$R_i(dk, dm) = r_i(dm)G_i(m, dk), \qquad \mathcal{E}_0 = \sum_i D_{\mathrm{KL}}(q_i \otimes s_i \| R_i). \tag{4.1}$$

The evaluator couples the sectors even though their recognition law is a product. Where the terms are finite, disintegration gives

$$D_{\mathrm{KL}}(q_i \otimes s_i \| R_i) = D_{\mathrm{KL}}(s_i \| r_i) + \int D_{\mathrm{KL}}(q_i \| G_i(m, \cdot))s_i(dm). \tag{4.2}$$

This is a conditional relative-entropy identity: the model marginal contributes the first term and the state conditional contributes the second. In particular, the expectation of a log evaluator in this objective is different from the log of the predictive mixture $s_i G_i$.

Remark 4.1 (What the product restriction removes). Over unrestricted joint laws the unique minimizer of $D_{\mathrm{KL}}(\zeta_i\|R_i)$ is $\zeta_i = R_i$, whose belief conditional is the evaluator. Under the product restriction and an evaluator density g_i, the belief minimizer at fixed s_i is instead $q_i \propto \exp\int \log g_i(m,\cdot)s_i(dm)$, which differs from the model prediction s_iG_i. The restriction therefore carries part of the distinction between belief and model. It is also not invariant under the joint-law dynamics. For binary $k,m\in\{-1,1\}$, let the joint Fisher flow be $\partial_t \log\zeta_i = -(h-\mathbb{E}_{\zeta_i}h)$ with $h(k,m)=\log(\zeta_i/R_i)(k,m)+h^q(k)+h^s(m)$, where the peer terms h^q,h^s depend only on the marginals. The additive terms cancel in the log odds ratio $\mathrm{LOR}(\zeta)=\sum_{k,m}km\log\zeta(k,m)$, so $\partial_t\,\mathrm{LOR}(\zeta_i)=-(\mathrm{LOR}(\zeta_i)-\mathrm{LOR}(R_i))$. Dependence between the sectors is generated at unit rate whenever R_i is not a product. The product flow used below is the restriction of the dynamics to the product family, not a consequence of it.

Let $\mathbf{q}=(q_i)_i$ and $\mathbf{s}=(s_i)_i$. For each allowed directed edge $j\to i$, choose normalized channels $\mathcal{K}^q_{ij}[\mathbf{q}]:\mathsf{K}_j\rightsquigarrow\mathsf{K}_i$ and $\mathcal{K}^s_{ij}[\mathbf{s}]:\mathsf{M}_j\rightsquigarrow\mathsf{M}_i$. Their dependence on the laws is part of the specified interaction. Throughout this chapter a belief channel depends only on the belief laws and a model channel only on the model laws, not on the rows, interfaces, or memory variables that definition 2.1 also admits; a dependence on the other sector or on the rows adds the corresponding terms to eq. (4.11) and changes the row statements of sections 4.2 and 4.4. A law-dependent channel is required to be a kernel $\mathcal{K}^q_{ij}:\mathsf{S}\times\mathsf{K}_j\rightsquigarrow\mathsf{K}_i$, jointly in the configuration and the source value, and likewise in the model sector. Then $x\mapsto\nu^q_{ij}(x)$ is Borel by a monotone-class argument, relative entropy is Borel on pairs of laws through its supremum over finite partitions from a countable generating algebra [95, Lemma 5.2.2], a result Gray attributes to Dobrushin, the Borel structure on laws being the one generated by the evaluations, and the costs, Gibbs rows, and reduced potential below are Borel functions of the configuration, as proposition 1.9 and the record appendix require. Define

$$\begin{aligned}\nu^q_{ij}&=q_j\mathcal{K}^q_{ij}[\mathbf{q}],\quad D^q_{ij}=D_{\mathrm{KL}}(q_i\|\nu^q_{ij}),\\ \nu^s_{ij}&=s_j\mathcal{K}^s_{ij}[\mathbf{s}],\quad D^s_{ij}=D_{\mathrm{KL}}(s_i\|\nu^s_{ij}).\end{aligned}\tag{4.3}$$

The source sets $\mathcal{N}^q_i,\mathcal{N}^s_i$ are finite and nonempty, with positive normalized reference rows π^q_i,π^s_i. Belief routing uses β_i and model routing uses γ_i on their respective sets.

Definition 4.2 (Reciprocal law potential). For fixed $T_q,T_s>0$, set

$$\begin{aligned}\mathcal{E}=\mathcal{E}_0&+\sum_i\left[\sum_{j\in\mathcal{N}^q_i}\beta_{ij}D^q_{ij}+T_qD_{\mathrm{KL}}(\beta_i\|\pi^q_i)\right]\\&+\sum_i\left[\sum_{j\in\mathcal{N}^s_i}\gamma_{ij}D^s_{ij}+T_sD_{\mathrm{KL}}(\gamma_i\|\pi^s_i)\right].\end{aligned}\tag{4.4}$$

The reference laws, evaluators, source priors, and temperatures are held fixed in the variations below. The admissible domain consists of laws for which all displayed divergences are finite.

This potential is a constitutive choice at the level of agent laws. No distribution over these law configurations is required to define it. It contains no record; records enter its dynamics only through the assimilation events of definition 4.28. Its gradient is reciprocal: a law that serves as a source appears in the cost of every receiver that attends to it, so the flow moves sources as well as

receivers (section 4.3). Section 4.6 gives the per-agent alternative in which peers' laws are data, and proposition 4.24 shows, under its stated hypotheses, that this potential is not a global variational free energy. General probability laws enter eq. (4.4); differentiability will require stronger hypotheses. An identity self-channel has $D^q_{ii} = 0$ throughout its domain, with the same convention in the model sector. Independent opposite attending edges remain independent data.

The receiver side of eq. (4.4) is a pooling rule. With the other agents, rows, and channels frozen, and with the appearances of q_i as a source ignored, stationarity of q_i gives $\log q_i = (1+\sum_j \beta_{ij})^{-1}\big(\int \log g_i(m,\cdot)s_i(dm)+\sum_j \beta_{ij}\log \nu^q_{ij}\big)+\text{const}$, a logarithmic opinion pool of the evaluator and the compared laws [1, 88]. Iterated relative-entropy averaging over a network is the consensus scheme of Battistelli and Chisci [19], with DeGroot [58] as its linear antecedent. The source side, which proposition 4.5 identifies, is not part of these pooling rules.

The correlated local inference of section 5.2 and proposition B.2 makes a different use of the same generative-model definition: it conditions a declared joint exactly before any output projection. Its component evidence weights and finite conditioning step do not identify it with the product-family population flow considered here.

4.2 Source selection and reduction

Proposition 4.3 (Gibbs rows and the exact reduced potential). Hold the agent laws fixed and suppose every source cost is finite. Define

$$\begin{aligned} Z^q_i &= \sum_j \pi^q_{ij} e^{-D^q_{ij}/T_q}, \quad \beta^*_{ij} = \frac{\pi^q_{ij} e^{-D^q_{ij}/T_q}}{Z^q_i}, \\ Z^s_i &= \sum_j \pi^s_{ij} e^{-D^s_{ij}/T_s}, \quad \gamma^*_{ij} = \frac{\pi^s_{ij} e^{-D^s_{ij}/T_s}}{Z^s_i}. \end{aligned} \tag{4.5}$$

These are the unique minimizing rows. With

$$\mathcal{E}_{\text{red}} = \mathcal{E}_0 - T_q \sum_i \log Z^q_i - T_s \sum_i \log Z^s_i, \tag{4.6}$$

the exact identity is

$$\mathcal{E} = \mathcal{E}_{\text{red}} + T_q \sum_i D_{\text{KL}}(\beta_i \| \beta^*_i) + T_s \sum_i D_{\text{KL}}(\gamma_i \| \gamma^*_i). \tag{4.7}$$

Proof. For one row, $\log \beta^*_j = \log \pi_j - D_j/T - \log Z$. Substitute this expression into $T D_{\text{KL}}(\beta\|\beta^*)$ to obtain the row objective plus $T \log Z$. Summing proves eq. (4.7). Relative entropy is nonnegative and vanishes exactly when its two probability laws agree, which proves uniqueness, including competitors on the simplex boundary. □

Proposition 4.3 is the finite case of the Gibbs variational principle, the convex duality between relative entropy and the log moment generating function [66, 67]. What is specific here is that the costs D_{ij} are themselves divergences between agent laws, so the rows move with the laws they select.

On a differentiable family, direct differentiation of the finite sums in eq. (4.5) gives

$$d\mathcal{E}_{\mathrm{red}} = d\mathcal{E}_0 + \sum_{ij} \beta^*_{ij} dD^q_{ij} + \sum_{ij} \gamma^*_{ij} dD^s_{ij}. \tag{4.8}$$

Indeed, $-T_q d\log Z^q_i = \sum_j \beta^*_{ij} dD^q_{ij}$, and the other sector is identical. There is no derivative of an optimized row to add. Every dependence of its cost on the laws still belongs in dD. If a source prior were varied, this formula would also contain $-T_q \sum_{ij} \beta^*_{ij} d\log \pi^q_{ij}$ and its model counterpart.

Remark 4.4 (Identity self sources and learned source priors). Suppose the source set of i contains i with an identity self-channel, so $D^q_{ii} = 0$. Since every cost is nonnegative, $\pi^q_{ii} \le Z^q_i \le 1$, hence $\beta^*_{ii} \ge \pi^q_{ii}$ and $\beta^*_{ij} \le (\pi^q_{ij}/\pi^q_{ii}) e^{-D^q_{ij}/T_q}$ for $j \neq i$. A peer whose divergence exceeds $T_q \log(\pi^q_{ij}/(\pi^q_{ii}\varepsilon))$ receives row weight below ε, so at fixed priors the reduced interaction is a smooth form of bounded-confidence opinion dynamics [105]. If instead the prior row π^q_i is made a dynamical table of eq. (4.4) with unit Fisher mobility, then $\partial\mathcal{E}/\partial\pi^q_{ij} = -T_q\beta_{ij}/\pi^q_{ij}$ and its Fisher flow is exactly

$$\dot\pi^q_i = T_q(\beta_i - \pi^q_i). \tag{4.9}$$

With Gibbs rows this is $\dot\pi^q_{ij} = (T_q/Z^q_i)\pi^q_{ij}(e^{-D^q_{ij}/T_q} - Z^q_i)$, a replicator equation with fitness e^{-D/T_q} [109], slow at small T_q without an inserted multiplier. An identity self source then has $\dot\pi^q_{ii} = (T_q/Z^q_i)\pi^q_{ii}(1 - Z^q_i) \ge 0$, and the infimum of the row terms over priors and rows is approached in the limit of isolation, where each agent attends only to itself. A learned-prior mechanism therefore needs self excluded from the learnable rows or its self prior held fixed. Whether learned rows without self concentrate on a pairing is open.

4.3 The derivative includes every interaction role

To display the variation, choose densities against fixed references on a regular dominated family. Write a for the density of a receiver law and b for that of its compared law. For mass-preserving variations and justified differentiation under the integrals,

$$dD_{\mathrm{KL}}(a\|b) = \int \log(a/b)\,da - \int (a/b)\,db. \tag{4.10}$$

Here the notation $\int f\,da$ denotes integration against the signed variation of the law, rather than differentiation of the sample coordinate. The formula follows by differentiating $a\log(a/b)$; the additional integral of da is zero.

For a belief interaction the compared-law variation is

$$d\nu^q_{ij} = (dq_j)\mathcal{K}^q_{ij}[\mathbf{q}] + q_j D\mathcal{K}^q_{ij}[\mathbf{q}](d\mathbf{q}). \tag{4.11}$$

The second term differentiates the prescribed channel itself. Thus an agent contributes when it is a receiver, when it is a source, and whenever its law changes a channel elsewhere. For fixed discrete kernels, the incoming contribution from $j \to i$ to a variation at source value v is

$$-\beta_{ij} \sum_u \frac{q_i(u)}{\nu^q_{ij}(u)} \mathcal{K}^q_{ij}(v,u). \tag{4.12}$$

The model sector has the same expression with q, β replaced by s, γ. If the channels are constructed from law-dependent moments or generators, eq. (4.11) also differentiates those moments and generators. Keeping their numerical values fixed computes a different vector field.

An identity self-channel illustrates why the variable identification matters. Here $a = b$ and $da = db$, so eq. (4.10) is zero throughout the state space. Freezing the source as an old snapshot while moving the receiver replaces this zero functional by a divergence from that snapshot. Its first derivative still vanishes at equality, but its curvature and its force after displacement differ. More generally, directed source selection does not make the gradient of eq. (4.4) one-way: a source appears in the energy of every row that attends to it. Each directed edge $j \to i$ therefore carries influence in both directions. The force on the source is the receiver likelihood ratio q_i/ν_{ij}^q pulled back through the adjoint of the channel, so it is supported on what the channel reads; for a deterministic readout T it is a function of $T(v)$, and only the message marginal of the source moves. The communication model of chapter 5 sends messages from source to receiver only. A population of communicating agents realizes the reciprocal flow only if this reverse message is also modeled; otherwise each source must be able to read the laws of every receiver that attends to it. Every agent descends the same total $\mathcal{E}$, so the population plays an identical-interest game in the sense of Monderer and Shapley [148]. Section 4.6 shows what is lost when each agent descends only its own objective.

Proposition 4.5 (Receiver and source pools). Let the receivers $i \in I_j$ of a source j use a common fixed channel K_j, with weights $\beta_{ij} > 0$, $W_j = \sum_{i \in I_j} \beta_{ij}$, and pooled receiver law $\bar{q}_{\to j} = W_j^{-1} \sum_{i \in I_j} \beta_{ij} q_i$. If all displayed divergences are finite,

$$\sum_{i \in I_j} \beta_{ij} D_{\mathrm{KL}}(q_i \| q_j K_j) = \sum_{i \in I_j} \beta_{ij} D_{\mathrm{KL}}(q_i \| \bar{q}_{\to j}) + W_j D_{\mathrm{KL}}(\bar{q}_{\to j} \| q_j K_j). \tag{4.13}$$

Proof. Expand both sides with $\int \log$ of densities against a common reference. The receiver entropies agree. On the right, $-\sum_i \beta_{ij} \int \log \bar{q}_{\to j} dq_i + W_j \int \log \bar{q}_{\to j} d\bar{q}_{\to j} = 0$ because $\sum_i \beta_{ij} q_i = W_j \bar{q}_{\to j}$, and the remaining cross term equals the left cross term for the same reason. □

Equation (4.13) is the compensation identity for the relative entropy of a mixture [188], applied with the common channel. The first term on the right does not contain q_j. At fixed rows the complete source-role force on q_j is therefore the force of $W_j D_{\mathrm{KL}}(\bar{q}_{\to j} \| q_j K_j)$, which fits one source law to the arithmetic pool of its receivers in the mass-covering direction, while the receiver-role force is the logarithmic pool above. In a Gaussian covariance sector with identity channel, the source force on the variance v_j from one receiver $\mathcal{N}(m_i, v_i)$ is $(v_j - v_i - (m_i - m_j)^2)/(2v_j^2)$, which vanishes at $v_j = v_i + (m_i - m_j)^2$: a source more confident than its receivers' spread is widened.

4.4 Fisher dynamics with equal mobilities

The kinetic metric in this chapter uses its declared product recognition family. It is the product specialization of proposition 3.2, with separate Fisher blocks for the source rows. The contextual tensor in definition 3.3 instead evaluates changes along the base through a connection; it does not replace the vertical metric used for inference at a fixed context.

Theorem 4.6 (Local flow and exact dissipation on positive tables). Suppose the state and model alphabets are finite, all r_i and evaluator entries $G_i(m,k)$ are strictly positive, and the channel entries are twice continuously differentiable functions of the interior probability tables. Assume every compared-law entry is positive there. Let u^a range over all tables $q_i, s_i, \beta_i, \gamma_i$. Equip their product with the Fisher metric

$$g_u(v,w) = \sum_a \sum_z \frac{v_z^a w_z^a}{u_z^a}, \qquad \sum_z v_z^a = \sum_z w_z^a = 0. \tag{4.14}$$

For every interior initial condition there is a unique maximal interior solution of

$$\dot{u}_z^a = -u_z^a \left(h_z^a - \sum_w u_w^a h_w^a \right), \qquad h_z^a = \frac{\partial \mathcal{E}}{\partial u_z^a}. \tag{4.15}$$

Along this solution,

$$\frac{d\mathcal{E}}{dt} = -\sum_a \mathrm{Var}_{u^a}(h^a) \le 0. \tag{4.16}$$

Proof. The stated positivity and regularity make $\mathcal{E}$ twice continuously differentiable in interior simplex coordinates. The right side of eq. (4.15) is therefore locally Lipschitz, giving a unique local solution and its maximal continuation in the interior. Its entries sum to zero within each table. Centering h^a removes any ambiguity from extending $\mathcal{E}$ off the normalized tables. For every tangent vector v, the metric pairing of $u^a(h^a - \mathbb{E}_{u^a} h^a)$ with v^a equals $\sum_z h_z^a v_z^a$. This identifies the Fisher gradient. Finally, substitute eq. (4.15) into $\dot{\mathcal{E}} = \sum_{a,z} h_z^a \dot{u}_z^a$ to obtain the sum of negative variances. Interior continuation does not assert that the trajectory stays away from the boundary for all time. □

On a single simplex eq. (4.15) is the replicator equation with fitness $-h$, the metric eq. (4.14) is the Shahshahani metric, and its identification as a gradient flow is Shahshahani's theorem [109, 176]; eq. (4.16) is the corresponding form of Fisher's fundamental theorem. Theorem 4.6 is that statement on a product of simplices with law-dependent channels. For fixed channels the local statement extends to all time.

Proposition 4.7 (Global flow for fixed channels). In addition to the hypotheses of theorem 4.6, suppose every channel is independent of the laws. Positivity of every compared law on the interior then means that no channel has a null column, so $c_{\min} = \min_{i,j,z} \sum_v \mathcal{K}_{ij}(v,z) > 0$ in each sector. Then every interior solution of eq. (4.15) exists for all $t \ge 0$, its entries are bounded below by a positive constant uniformly in time, $\int_0^\infty \sum_a \mathrm{Var}_{u^a}(h^a) dt \le \mathcal{E}(u(0))$, and it converges to a single critical point of $\mathcal{E}$.

Proof. Consider the belief tables. Constants in h cancel on centering, so replace the belief force of agent i by $\tilde{h}_z = \log q_i(z) - \int \log G_i(m,z) s_i(dm) + \sum_j \beta_{ij} \log(q_i(z)/\nu_{ij}(z)) + \sigma_z$, where $\sigma_z \le 0$ collects the incoming terms eq. (4.12). Let $m(t)$ be the minimum of $\log q_i(z)$ over agents and states, attained at (i,z). Every entry of every belief table is at least e^m, so $\nu_{ij}(z) \ge c_{\min} e^m$ and $\tilde{h}_z \le m + C_G + \log^+(1/c_{\min})$, with $C_G = \max(-\log G)$. On the other hand $\mathbb{E}_{q_i} \tilde{h} \ge -\log|\mathsf{K}_i| - N$, where N is the number of agents, which bounds the total row weight $\sum_k \beta_{ki}$ of the receivers attending to i: the entropy term is at least $-\log|\mathsf{K}_i|$, the evaluator and receiver terms are nonnegative, and each

incoming term averages to $-\beta_{ki}$ under q_i. Hence the lower Dini derivative of m is at least $-m - C$ with $C = C_G + \log^+(1/c_{\min}) + \log\max_i |\mathsf{K}_i| + N$, and $m(t) \geq \min(m(0), -C)$. The model tables are treated in the same way; their force also contains the evaluator term $-\sum_k q_i(k) \log G_i(m,k) \leq C_G$, so the model constant is $C + \max_{i,m}(-\log r_i(m))$, with $\log\max_i |\mathsf{M}_i|$ in place of $\log\max_i |\mathsf{K}_i|$. With these floors every D_{ij} is bounded by a constant $\bar{D}$, and the same argument applied to $h_j = D_{ij} + T(\log(\beta_{ij}/\pi_{ij}) + 1)$ bounds the rows below by $\min(\log\beta_{ij}(0), -\bar{D}/T - \log(1/\min\pi))$ in logarithm. The trajectory therefore stays in a compact subset of the interior, so it is global. Since $\mathcal{E} \geq 0$, integrating eq. (4.16) bounds the dissipated variance. With fixed channels $\mathcal{E}$ is real analytic on the interior of the product of simplices, and on the compact set the Fisher metric is uniformly equivalent to the Euclidean one, so the Łojasiewicz gradient inequality [140] gives finite length, and hence convergence to one critical point, of the bounded trajectory; this is Łojasiewicz's convergence theorem for gradient trajectories of real-analytic functions, of which Absil et al. [2] give the discrete-iterate counterpart. □

For law-dependent channels the channel-derivative term of eq. (4.11) can be unbounded near the boundary, and no statement of this kind follows without further hypotheses.

Proposition 4.8 (Row relaxation). Hold all costs of one row fixed and write $l_j = \log(\beta_j/\beta_j^*)$ with β^* from eq. (4.5). Under eq. (4.15), $\frac{d}{dt}(l_j - l_k) = -T(l_j - l_k)$ for every pair of sources.

Proof. The row force is $h_j = D_j + T(\log(\beta_j/\pi_j) + 1) = Tl_j + T(1 - \log Z)$, and eq. (4.15) gives $\frac{d}{dt}\log\beta_j = -(h_j - \mathbb{E}_\beta h)$. □

All blocks have unit mobility, and proposition 4.8 shows its price: under unit mobility, log-odds contrasts within a row relax at rate exactly T, so rows that select sharply are slow. Exact Gibbs rows are the formal limit of infinite row mobility, in which the strictly convex row objective is slaved to its unique interior minimizer; this is a Tikhonov-type singular limit, which is not proved here, and it is not a property of the unit-mobility flow. One may optimize the source rows at each state and apply the same construction to $\mathcal{E}_{\mathrm{red}}$ and the matter tables alone; eq. (4.8) gives its complete derivative. The two flows are different vector fields with different transients, but they share their critical points, Morse indices, and nullities (proposition 4.9), hence their linear stability at nondegenerate critical points. Their local minima also correspond: $\mathcal{E} = \mathcal{E}_{\mathrm{red}} + T_q\sum_i D_{\mathrm{KL}}(\beta_i\|\beta_i^*) + T_s\sum_i D_{\mathrm{KL}}(\gamma_i\|\gamma_i^*) \geq \mathcal{E}_{\mathrm{red}}$ with equality on the graph of the Gibbs rows, which are continuous in the laws. At degenerate critical points the Hessians do not decide Lyapunov stability, and no equivalence of the two flows' stability is claimed there. Equal mobility has a justification beyond convention. Unit mobilities are the Fisher metric of the single product law of all tables, regarded as one law of all represented variables and source labels, because the Fisher metric of a product law restricted to the product family is the sum of the factor metrics. If the kinetic metric is required to be the restriction of a metric on the law space of all these variables jointly that is invariant under Markov congruent embeddings, Čencov's theorem [39] forces it, on finite sample spaces, to be a constant multiple of the Fisher metric, so equal mobility is fixed up to one global time scale. The product step holds on general spaces, since cross terms of centered scores of independent factors vanish; the uniqueness step on general spaces needs, in addition to invariance under sufficient statistics, that the metric be a local and weakly continuous tensor field on parametrized measure models, which is the setting of Ay et al. [9]. In the registered-agreement model of theorem 4.26 the configuration is a lattice

variable with a point recognition, and that model does not itself supply a kinetic metric. Imposing invariance on each table separately allows an independent constant for each table, equivalent to replicating that table. A persistent slow model sector would have to follow from the interactions and their state, rather than from an inserted small multiplier in its equation.

Proposition 4.9 (Full and reduced flows have the same critical structure). Under the hypotheses of theorem 4.6, $(u, \beta^*(u), \gamma^*(u))$ is a critical point of $\mathcal{E}$ exactly when u is a critical point of $\mathcal{E}_{\mathrm{red}}$, and at such a point the Hessians of $\mathcal{E}$ and $\mathcal{E}_{\mathrm{red}}$ on the tangent spaces have the same number of negative and zero eigenvalues.

Proof. At fixed laws the row terms are $\sum_j \beta_j D_j + T D_{\mathrm{KL}}(\beta\|\pi)$, whose Hessian on the tangent space of the simplex is T times the Fisher metric, hence positive definite; the row optimum is β^*, and eq. (4.8) identifies the critical points. The Hessian of $\mathcal{E}_{\mathrm{red}}$ is the Schur complement of the positive-definite row block in the Hessian of $\mathcal{E}$, and Haynsworth's inertia additivity [104] gives the claim. □

For a regular finite-dimensional family with positive-definite Fisher tensor $g_{ab}(\theta)$, the corresponding equation is $\dot\theta^a = -(g^{-1})^{ab}\partial_b\mathcal{E}$ [3]. Whenever the derivatives exist, contraction gives $\dot{\mathcal{E}} = -(\partial\mathcal{E})^T g^{-1}(\partial\mathcal{E})$. Bimeasurable re-expression of samples, carrying all kernels and reference laws with it, preserves relative entropy and Fisher score inner products. On a family preserved by that re-expression, the gradient therefore transforms as a tangent vector; this is passive covariance. Active equivariance would further need equivariant constitutive maps for the law-dependent channels, as in eq. (2.15), and transported references and evaluators. These statements concern fixed coordinate changes; coordinates that themselves evolve require their coordinate-velocity term.

Definition 4.10 (Restricted-family flow). Let $\theta \mapsto p_\theta$ be an admitted regular finite-dimensional family for some of the laws, with Fisher tensor $g(\theta)$. The restricted-family flow is $\dot\theta = -g(\theta)^{-1}\partial_\theta\mathcal{E}(p_\theta)$. For a Gaussian law $\mathcal{N}(\mu, \Sigma)$ on $\mathbb{R}^d$, with $\nabla_\mu\mathcal{E}$ and the symmetric gradient $\nabla_\Sigma\mathcal{E}$, it reads

$$\dot\mu = -\Sigma\nabla_\mu\mathcal{E}, \qquad \dot\Sigma = -2\Sigma(\nabla_\Sigma\mathcal{E})\Sigma. \tag{4.17}$$

The restricted velocity is $-g(\theta)^{-1}$ applied to the score pairing of the first variation, $\partial_{\theta^a}\mathcal{E} = \mathbb{E}_{p_\theta}[\varphi\,\partial_a \log p_\theta]$. When the full velocity $-(\varphi - \mathbb{E}\varphi)$ has finite Fisher norm, this is its Fisher-orthogonal projection onto the tangent space of the family. That norm can be infinite: with an identity channel the source term of a Gaussian source q_i contains q_k/q_i for each attending receiver q_k, and $\mathbb{E}_{q_i}[(q_k/q_i)^2]$ is finite exactly when $2\Sigma_k^{-1} - \Sigma_i^{-1}$ is positive definite. The score pairing remains finite, since $\mathbb{E}_{q_i}[\psi q_k/q_i] = \mathbb{E}_{q_k}\psi$ for polynomial ψ, so eq. (4.17) is defined in that case as well, as a weak projection. The restricted flow is a solution of the full equation only on families to which the full velocity is tangent, and the source role prevents this for Gaussian agents under the reciprocal potential. No result of this chapter establishes well-posedness of the full density-level flow from Gaussian initial data, since Gaussian laws with different parameters do not share a bounded log-density chart.

Proposition 4.11 (Gaussian agents are not an invariant family). Let all belief laws be Gaussian on $\mathbb{R}^d$ with identity channels and fixed rows, let the evaluators be Gaussian with means affine in m and fixed covariance, and let the model laws have finite second moments. Suppose at least one

receiver $k \neq i$ attends to i with $\beta_{ki} > 0$ and $q_k \neq q_i$. Then the Fisher velocity $-(\varphi - \mathbb{E}_{q_i}\varphi)$ of q_i, with φ the first variation of $\mathcal{E}$ in q_i, is not the logarithmic derivative of any differentiable curve of Gaussian laws.

Proof. For a curve of Gaussian laws, $\partial_t \log q$ is a polynomial of degree at most two. The local term contributes $\log q_i - \int \log g_i(m,\cdot) s_i(dm)$, and each receiver term $\beta_{ij}\log(q_i/q_j)$; both are polynomials of degree at most two. By eq. (4.12) with identity channels, the source terms sum to $-\sum_k \beta_{ki} q_k/q_i = -\sum_k \beta_{ki} e^{Q_k}$ over the attending receivers, with each $Q_k = \log(q_k/q_i)$ a polynomial of degree at most two. Choose a line on which Q_k is nonconstant for the given receiver; it exists because $q_k \neq q_i$. Restricted to it, every term with nonconstant Q_k either grows faster than every polynomial at one end of the line, when the leading coefficient of Q_k there is positive, or tends to zero at both ends, when it is a negative quadratic coefficient. If some term grows, the positive sum grows faster than every polynomial at that end. Otherwise the sum of the nonconstant terms is positive and tends to zero at both ends, which no polynomial does. In either case the source sum is not a polynomial on the line, while all other terms of φ are. Hence φ is not a polynomial of degree at most two. □

The obstruction belongs to the source role of the reciprocal potential, and the two free-energy pictures do not have it.

Proposition 4.12 (Gaussian closure of the per-agent and global flows). Let $\mathsf{K}_i, \mathsf{M}_i$ be Euclidean spaces. In the per-agent picture of definition 4.19, let the model references r_i be Gaussian, the evaluators $G_i(m,\cdot)$ Gaussian with mean affine in m and fixed covariance, every channel linear Gaussian, so that each compared law ν^q_{ij}, ν^s_{il} of a Gaussian source is Gaussian, and every record log likelihood $\log \ell_i(o \mid k, m)$ a polynomial of degree at most two in (k, m). At every configuration of Gaussian laws q_i, s_i the first variations of $\mathcal{F}_i$ in q_i and in s_i are polynomials of degree at most two, for all row values, so the per-agent Fisher velocity is tangent to the Gaussian product family. In the global picture of proposition 4.16, let the prior be Gaussian and every likelihood factor log quadratic in the contents it reads; then at every Gaussian product recognition the mean field $\bar{e}_i$ is a polynomial of degree at most two in y_i, the block minimizer Q_i^* is Gaussian, and the Fisher velocity of $\mathcal{F}$ in Q_i is tangent to the Gaussian family.

Proof. In the per-agent picture the peers' laws are data, so no source term appears. The first variation of eq. (4.27) in q_i is $\log q_i - \sum_j \beta_{ij} \int \log \kappa_{ij}(m,\cdot) s_i(dm) - \int \log \ell_i(o \mid \cdot, m) s_i(dm)$ up to a constant, using $\sum_j \beta_{ij} = 1$. The logarithm of a Gaussian evaluator with affine mean is a quadratic polynomial in (k, m), so its integral against s_i is a quadratic polynomial in k whose coefficients involve the first two moments of s_i; the compared laws are Gaussian in k; the record term is quadratic by hypothesis. The first variation in s_i is $\log s_i - \sum_l \gamma_{il} \log \rho_{il} + \beta_{ii} D_{\mathrm{KL}}(q_i \| G_i(m,\cdot)) + \mathbb{E}_{q_i}[-\log \ell_i(o \mid k, m)]$ up to constants and terms independent of m; the relative entropy between Gaussians of equal covariance with mean affine in m is quadratic in m, and the other terms are quadratic by hypothesis. In the global picture $\log p$ is a quadratic polynomial of y, so $\bar{e}_i = \int \log p \, dQ_{-i}$ is a quadratic polynomial of y_i whenever the factors Q_j have finite second moments, and the first variation $\log a_i - \bar{e}_i$ of eq. (4.21) is quadratic. A centered quadratic polynomial is the logarithmic velocity of a curve of Gaussian laws, which proves tangency. □

Under these hypotheses eq. (4.17), applied to $\mathcal{F}_i$ or to $\mathcal{F}$, gives a solution of the full density-level equation, and it is the full flow wherever that equation has unique solutions, for as long as the

covariances of the restricted solution stay positive definite. Tangency is a feature of the Gaussian family, not a property of general laws: for non-Gaussian evaluators, channels, or records, neither free-energy picture preserves a finite-dimensional family in general. Noninvariance under the reciprocal potential and its evidence-augmented form is a property of their source role.

The inheritance and formation results of chapter 8 to chapter 13 are proved for full density charts or for families declared invariant under the full flow in their own chapters. They do not transfer to the restricted Gaussian flow without a separate proof. Channels that are constant in their input, as in chapter 12, remove the source term and with it this obstruction.

The density-level flow used in those chapters needs its own hypotheses. For a law q on a standard Borel space with probability reference λ, write $q_l = e^l\lambda/\mathbb{E}_\lambda e^l$ for l in the chart $L^\infty_0(\lambda) = \{l \in L^\infty(\lambda) : \mathbb{E}_\lambda l = 0\}$, and use log-odds coordinates for rows. Suppose $\mathcal{E}$ has a first variation $\varphi_q \in L^\infty(\lambda)$ in each law, in the sense that $\frac{d}{dt}\mathcal{E}(q_{l(t)}) = \sum \int \varphi_q d\dot{q}$ along every continuously differentiable chart curve.

Definition 4.13 (Fisher flow in bounded log-density charts). The chart Fisher flow is $\dot{l} = -(\varphi_{q_l} - \mathbb{E}_\lambda \varphi_{q_l})$ for every law, together with eq. (4.15) for the rows.

Proposition 4.14 (Local well-posedness in charts). Suppose the map from chart coordinates to the first variations $\varphi_q \in L^\infty(\lambda)$ is Lipschitz on bounded sets. Then for every initial point there is a unique maximal continuously differentiable solution on $[0, t^*)$, and if $t^* < \infty$ the chart coordinates are unbounded as $t \to t^*$. Along it, $\partial_t \log q = -(\varphi_q - \mathbb{E}_q\varphi_q)$ and $d\mathcal{E}/dt = -\sum \mathrm{Var}_q(\varphi_q) \le 0$, the sum running over all laws and rows. The hypothesis holds for relative-entropy terms when the reference laws and evaluators have bounded log densities against λ and every channel is either an identity between agents sharing λ or a fixed kernel with densities bounded above and below by positive constants; for law-dependent channels it is an additional hypothesis on the first variation of the channel.

Proof. The maps $l \mapsto \mathbb{E}_{q_l} f = \mathbb{E}_\lambda[fe^l]/\mathbb{E}_\lambda e^l$ are locally Lipschitz in $(l, f) \in L^\infty \times L^\infty$, so the vector field is Lipschitz on bounded sets, and the Picard–Lindelöf theorem in a Banach space gives local existence, uniqueness, and the continuation alternative [64]. Since $\log q_l = l - \log \mathbb{E}_\lambda e^l$ and $\frac{d}{dt}\log \mathbb{E}_\lambda e^l = \mathbb{E}_q \dot{l}$, the logarithmic velocity is $-(\varphi_q - \mathbb{E}_q\varphi_q)$; integrating it against $\varphi_q dq$ gives the variance. Under the stated bounds, the receiver terms $\log(q_i/\nu_{ij})$ and evaluator terms are bounded, the source terms $-\int (q_i/\nu_{ij})(w)\mathcal{K}_{ij}(v, dw)$ are bounded, and each is a composition of exponentials, logarithms on bounded ranges, and integrals against fixed bounded kernels, hence Lipschitz on bounded sets. □

For fixed channels the maximum-principle argument of proposition 4.7 transfers to the chart, with essential infima and suprema in place of minima and maxima. Because the densities are taken against the probability reference λ, the constants do not depend on the size of the sample space.

Proposition 4.15 (Global chart flow for fixed channels). Assume the sufficient conditions of proposition 4.14 with every channel fixed: $|\log g_i| \le C_g$ and $|\log r_i| \le C_r$ for the evaluator and reference densities, and every channel either an identity between agents sharing a reference or a fixed kernel whose density lies in $[c, C_K]$ with $0 < c \le C_K < \infty$. Let the rows be fixed or follow

eq. (4.15). Put $C = C_g + C_r$, let N be the number of agents, and set

$$A = 2C + N + \max(0, -\log c), \qquad B = 2\mathcal{E}(0) + 3C + \max(0, \log C_K) + N(1 + C_K/c). \tag{4.18}$$

Then every chart solution exists for all $t \geq 0$, and the essential infimum $m(t)$ and essential supremum $M(t)$, over all agents and both sectors, of the log densities against their references satisfy $m(t) \geq \min(m(0), -A)$ and $M(t) \leq \max(M(0), B)$. In particular $\int_0^\infty \sum \mathrm{Var}_q(\varphi_q)dt \leq \mathcal{E}(0)$.

Proof. Write $u = \log(dq_i/d\lambda)$ for a belief density; $t \mapsto u$ is continuously differentiable into L^∞, and $\dot{u} = \mathbb{E}_{q_i}\varphi_i - \varphi_i$. Since λ is a probability law, $m \leq 0 \leq M$. By eq. (4.10) the first variation is, up to constants, $\varphi_i = u - \int \log g_i(m, \cdot)s_i(dm) + \sum_j \beta_{ij} \log(q_i/\nu_{ij}) - \sum_k \beta_{ki} \int (q_k/\nu_{ki})(w)\mathcal{K}_{ki}(\cdot, dw)$. For the lower bound, the evaluator term is at most C_g, a kernel receiver term is at most $u - \log c = (u - m) + (m - \log c) \leq (u - m) + \max(0, -\log c)$ because $\nu \geq c$ and $m \leq 0$, an identity receiver term $u - \log q_j$ is at most $u - m$, and the source terms are nonpositive. Since $\sum_j \beta_{ij} \leq 1$, $\varphi_i \leq 2u - m + C_g + \max(0, -\log c)$. On the other side $\mathbb{E}_{q_i}\varphi_i \geq -C_g - N$, because $D_{\mathrm{KL}}(q_i\|\lambda)$ and the receiver divergences are nonnegative and each source term integrates to $-\beta_{ki}$ under q_i. Hence $\dot{u} \geq m - A - 2u$ almost everywhere. For $0 < h < 1/2$, the expansion $u(t + h) = u + h\dot{u} + o(h)$ in L^∞ and $u \geq m$ almost everywhere give $u(t + h) \geq m - h(m + A) - o(h)$ almost everywhere, so the lower right Dini derivative of m is at least $-(m + A)$. The model sector is identical: its evaluator term $-\int \log g_i(m, k)q_i(dk)$ varies in m by at most C_g and the reference adds C_r. Comparison gives $m(t) \geq \min(m(0), -A)$.

For the upper bound, fix $\varepsilon > 0$ and consider points where $u \geq M - \varepsilon$. There a kernel receiver term is at least $u - \log C_K \geq -\varepsilon - \max(0, \log C_K)$, because $M \geq 0$, an identity receiver term is at least $u - M \geq -\varepsilon$, a kernel source term is at least $-\beta_{ki}C_K/c$ because $\int (q_k/\nu_{ki})(w)k(z, w)\lambda(dw) \leq C_K/c$, and an identity source term $-\beta_{ki}q_k/q_i$ is at least $-\beta_{ki}e^{\varepsilon}$. Since $\mathcal{E}$ is nonincreasing by proposition 4.14, $\sum_j \beta_{ij}D_{ij} \leq \mathcal{E}(0)$ and $D_{\mathrm{KL}}(q_i\|\lambda) \leq \int D_{\mathrm{KL}}(q_i\|G_i(m, \cdot))s_i(dm) + C_g \leq \mathcal{E}(0) + C_g$, so $\mathbb{E}_{q_i}\varphi_i \leq 2\mathcal{E}(0) + 2C$. Hence at those points $\dot{u} \leq B - u + \delta(\varepsilon)$ with $\delta(\varepsilon) \to 0$, while at the remaining points $u + h\dot{u} \leq M - \varepsilon + h\|\dot{u}\|_\infty \leq M$ for small h. The upper Dini derivative of M is therefore at most $B - M$, and $M(t) \leq \max(M(0), B)$; the model sector is the same with C_r. With these bounds every divergence D_{ij} is bounded, and the row argument of proposition 4.7 bounds the rows below. Since the chart coordinate is $l = \log q - \mathbb{E}_\lambda \log q$, it stays bounded, and the continuation alternative of proposition 4.14 gives $t^* = \infty$. Integrating the dissipation identity and using $\mathcal{E} \geq 0$ gives the last claim. □

The fixed-channel hypothesis enters the floor estimate. For law-dependent channels the channel-derivative term of eq. (4.11) can grow as a density tends to zero, and no global statement is made. Convergence on general spaces would need a Łojasiewicz–Simon inequality and is open. Other global statements are stated where they are used.

The Fisher–Rao metric used here is, up to scale, the metric singled out by invariance under sufficient statistics: on finite sample spaces by Čencov [39], and on general sample spaces among local and weakly continuous tensor fields on parametrized measure models by Ay et al. [9].

4.5 A global free energy and its local free energies

Choose a proper prior $P_0(dy, dJ)$ on represented contents and source labels, and a normalized record kernel $\mathscr{R}(do \mid y, J)$. Then $\mathbb{P} = P_0\mathscr{R}$ is normalized by integrating the record first. At a declared record density $\ell_o(y, J)$ with $0 < Z_o = \int \ell_o dP_0 < \infty$, its posterior is $\Pi_o = \ell_o P_0/Z_o$. Substitution proves

$$\mathcal{F}(Q; o) = D_{\mathrm{KL}}(Q\|P_0) - \mathbb{E}_Q \log \ell_o = D_{\mathrm{KL}}(Q\|\Pi_o) - \log Z_o, \tag{4.19}$$

when the split terms are defined. The right side supplies the extended definition otherwise. This is the variational free energy of a fixed model [80, 112, 153]. A product family for Q is permitted and its block updates remain coupled through the likelihood factors. Each block derivative contains every factor incident on it.

In the global picture the represented contents of all agents are variables of one such model. Write $y = (y_1, \dots, y_N)$ with $y_i \in \mathsf{Y}_i$, where Y_i collects the state and model coordinates of agent i, and let λ_i be a sigma-finite reference on Y_i. Source labels, when present, are further coordinates: they may be included in an agent's block, treated as additional blocks, or integrated out, and the proposition below holds for every finite block partition. Peer interactions are likelihood factors: for example, a comparison record eq. (1.13) between agents i and j contributes a factor that depends on (y_i, y_j).

Proposition 4.16 (Local free energies of a global model). Let Π_o have a density p with respect to $\bigotimes_j \lambda_j$, and restrict recognition to products $Q = \bigotimes_j Q_j$ with $Q_j = a_j\lambda_j$. Assume $\int |a_j \log a_j| d\lambda_j < \infty$ for every j, $\int |\log p| dQ < \infty$, and, for the fixed i below, $\int |\log p(y_i, \cdot)| dQ_{-i} < \infty$ for λ_i-almost every y_i. Fix i and the factors $Q_{-i} = \bigotimes_{j\neq i} Q_j$, and put

$$\bar{e}_i(y_i) = \int \log p(y_i, y_{-i}) Q_{-i}(dy_{-i}), \qquad Z_i = \int e^{\bar{e}_i} d\lambda_i. \tag{4.20}$$

If $0 < Z_i < \infty$, then with $Q_i^* = e^{\bar{e}_i}\lambda_i/Z_i$,

$$\mathcal{F}(Q; o) = D_{\mathrm{KL}}(Q_i\|Q_i^*) - \log Z_i + \sum_{j\neq i} \int a_j \log a_j d\lambda_j - \log Z_o. \tag{4.21}$$

Consequently the local free energy of agent i, defined as $\mathcal{F}$ viewed as a function of Q_i alone, has the unique minimizer Q_i^* among competitors $Q_i' = a_i'\lambda_i$ satisfying the same integrability hypotheses, and every update of one factor to its minimizer lowers $\mathcal{F}$ or leaves it unchanged. Along any schedule of such block updates at whose visited configurations the hypotheses hold, $\mathcal{F}$ is nonincreasing; on finite alphabets with $p > 0$ they hold at every configuration, while on general spaces their propagation must be checked for the model at hand. The peers enter $\bar{e}_i$ only through expectations, so $\bar{e}_i$ is affine in each Q_j with $j \neq i$.

Proof. By the assumed integrability, $\mathcal{F}(Q; o) = D_{\mathrm{KL}}(Q\|\Pi_o) - \log Z_o = \sum_j \int a_j \log a_j d\lambda_j - \int \log p dQ - \log Z_o$. Fubini's theorem gives $\int \log p dQ = \int \bar{e}_i dQ_i$. Since $\log(dQ_i/dQ_i^*) = \log a_i - \bar{e}_i + \log Z_i$, the terms involving Q_i equal $D_{\mathrm{KL}}(Q_i\|Q_i^*) - \log Z_i$, which proves eq. (4.21). The remaining statements follow because relative entropy is nonnegative and vanishes only at equality, and because $\bar{e}_i$ is an integral of a fixed function against the product Q_{-i}. □

This is the coordinate-ascent identity of mean-field variational inference [23, 28], here read as a statement about agents: each agent minimizing its own local free energy, with its peers' current factors given, performs coordinate descent of one collective functional. The identification requires that every agent's model be a conditional of the same joint and that peers act through expectations of their contents. The latter property is what the two other objectives of this chapter lack. In this chapter "local free energy" means the global free energy as a function of one mean-field block. The single-agent correlated objective of proposition B.2 and the marginal objectives of proposition C.7, whose minimizers are exact marginal posteriors, are different objects.

With independent source priors and one likelihood factor reading each label, a factorized source row has cost $C_{ij} = \mathbb{E}_{Q_Y}[-\log \ell_i(o_i \mid Y, j)]$ and optimizer $\beta_{ij} \propto \pi_{ij} e^{-C_{ij}}$. This expected log likelihood differs from both the peer divergence and the likelihood integrated over a prior. For an isolated source record, exact source inference gives $\mathbb{P}(J = j \mid o) \propto \pi_j \mathbb{E}_{P_Y} \ell_j(o \mid Y)$. Equality of these different rows needs additional hypotheses; proposition 4.18 and theorem 4.26 give two exact settings.

Proposition 4.17 (Entropy and the likelihood Jensen gap). On standard Borel spaces, let $k(w \mid v)$ be a density of a normalized kernel into a receiver space with sigma-finite reference λ. Let the receiver law be $q = a\lambda$ and the source law be p. Put $b(w) = \int k(w \mid v)p(dv)$. Assume $0 < b(w) < \infty$ for q-almost every w, $k > 0$ almost everywhere under $q \otimes p$, and finiteness of $\mathbb{E}_q|\log a|$, $\mathbb{E}_q|\log b|$, and $\mathbb{E}_{q\otimes p}|\log k|$. Define $p^w(dv) = k(w \mid v)p(dv)/b(w)$ on this full-measure set. Then

$$\mathbb{E}_{q\otimes p}[-\log k(W \mid V)] = H_\lambda(q) + D_{\mathrm{KL}}(q\|pK) + \int D_{\mathrm{KL}}(p\|p^w)q(dw). \tag{4.22}$$

Proof. The density ratio gives $D_{\mathrm{KL}}(p\|p^w) = \log b(w) - \int \log k(w \mid v)p(dv)$. Integrating against q and using $-\int \log b dq = H_\lambda(q) + D_{\mathrm{KL}}(q\|pK)$ proves the identity. The last term vanishes exactly when $p^w = p$ for q-almost every w. □

The entropy here is relative to the declared reference. A record likelihood normalized in o need not be normalized in the receiver coordinate w, so proposition 4.17 requires that additional kernel interpretation. Even when it applies, the entropy and conditional divergence can affect the dynamics.

The identity gives the exact relation between the Gibbs rows and a normalized source-label model in one sector.

Proposition 4.18 (Gibbs rows and a normalized label model). Let all alphabets be finite with counting reference, let the represented contents $Y_i \sim r_i$ be independent, let labels $J_i \sim \pi_i$ range over nonself sources, and let each receiver register one binary record whose success probability given $J_i = j$ is $k_{ij}(Y_i \mid Y_j)^{1/T}$, where $k_{ij}(w \mid v) = \mathcal{K}_{ij}(v, \{w\}) > 0$. With all records registered and product recognition $\bigotimes_i q_i \otimes \bigotimes_i \beta_i$, the free energy $\mathcal{F}$ of this model satisfies

$$T\mathcal{F} = \mathcal{E} + \sum_i \big[H(q_i) + (T-1)D_{\mathrm{KL}}(q_i\|r_i)\big] + \sum_{ij} \beta_{ij} G_{ij}, \qquad G_{ij} = \int D_{\mathrm{KL}}(q_j\|q_j^w)q_i(dw), \tag{4.23}$$

where $\mathcal{E}$ is eq. (4.4) in this sector with local term $\sum_i D_{\mathrm{KL}}(q_i\|r_i)$, and q_j^w is the source conditional of proposition 4.17. The recognition-optimal rows are $\beta_{ij} \propto \pi_{ij} e^{-(D_{ij}+G_{ij})/T}$, and they equal the Gibbs rows eq. (4.5) exactly when G_{ij} does not depend on j.

Proof. Since $k_{ij} \le 1$ on a finite alphabet, the record probabilities are normalized. Substitution gives $T\mathcal{F} = T\sum_i D_{\mathrm{KL}}(q_i\|r_i) + T\sum_i D_{\mathrm{KL}}(\beta_i\|\pi_i) + \sum_{ij}\beta_{ij}\mathbb{E}_{q_i\otimes q_j}[-\log k_{ij}]$. Apply eq. (4.22) to each expectation and use $\sum_j \beta_{ij} = 1$. The row statement is proposition 4.3 with costs $D_{ij} + G_{ij}$. □

This generalizes the Gaussian calculation of the supplement. A channel whose kernel vanishes on a set charged by $q_i \otimes q_j$, such as a deterministic bit-transmitting channel, makes the expected log likelihood infinite and has no reading as a label likelihood of this kind.

There is also an exact obstruction to universal equality. Freeze a channel K and receiver q. Suppose p_1K, p_2K have positive densities that differ on a set of positive q-measure, with both divergences finite. Strict convexity of $-\log$ gives

$$D_{\mathrm{KL}}(q\|(tp_1 + (1-t)p_2)K) < tD_{\mathrm{KL}}(q\|p_1K) + (1-t)D_{\mathrm{KL}}(q\|p_2K), \qquad 0 < t < 1. \tag{4.24}$$

This follows pointwise on the differing-density set and then by integration. By contrast, the expectation $\mathbb{E}_{q\otimes p}e(W,V)$ of any fixed integrable sample energy is affine in p. It cannot equal the divergence at both endpoints and their strict mixture. The obstruction concerns a fixed sample energy; it does not exclude a direct law potential, a frozen-template episode, or an explicitly different probability model whose variables are law points.

4.6 Per-agent free energies

In the per-agent picture each agent has its own normalized generative model, and the communicated laws of its peers enter that model as data. Formally the per-agent picture is a class: each agent descends the free energy of a normalized model in which the current laws of its peers are parameters and its own laws are recognition laws. This chapter develops one instance in detail, the mixture-source model below, in which the agent's own evaluator and reference become one source among several and a latent label records which source supplies the prior. Two other instances occur. A sequence of frozen-source episodes in the sense of corollary 4.27 has each agent descend its own terms of eq. (4.4) at frozen sources, with the additive local term $\mathcal{E}_0$ rather than a competing self source. The tangent models of chapter 19 enter peers through a comparison-record likelihood rather than through mixture sources. Results proved for one instance do not transfer to another without a separate argument.

Definition 4.19 (Per-agent mixture-source model). Fix agent i and the current laws of its peers. Let the finite source sets $\mathcal{N}_i^q, \mathcal{N}_i^s$ contain the self label i, with positive normalized priors π_i^q, π_i^s. Define content sources and presentation sources by

$$\kappa_{ii}(m,\cdot) = G_i(m,\cdot), \quad \kappa_{ij}(m,\cdot) = \nu_{ij}^q \ (j \ne i), \qquad \rho_{ii} = r_i, \quad \rho_{il} = \nu_{il}^s \ (l \ne i), \tag{4.25}$$

with ν_{ij}^q, ν_{il}^s as in eq. (4.3). The per-agent model on (m,k,J,L,o) is

$$P_i(dm,dk,J=j,L=l,do) = \pi_{il}^s\rho_{il}(dm)\pi_{ij}^q\kappa_{ij}(m,dk)\ell_i(k,m;do). \tag{4.26}$$

It is normalized for every value of the peer laws. Every incoming channel $\mathcal{K}_{ij}^q, \mathcal{K}_{il}^s$ of agent i is required to be independent of agent i's own laws (q_i, s_i). Those peer laws are then parameters of agent i's model, not variables over which it infers. If an incoming channel read (q_i, s_i), the model

would depend on the law it is used to score, $\mathcal{F}_i$ would not be the free energy of a fixed model in agent i's recognition, and definition 4.21 would differentiate the model itself.

Proposition 4.20 (Per-agent free energy). In eq. (4.26), fix a record value o with $0 < P_i(o) < \infty$, where the record density $l_i(o \mid k, m)$ is jointly measurable as in definition 1.11, and define $\mathcal{F}_i$ by eq. (1.16) for this model. Write P_i^0 for the record-free part of eq. (4.26), and take recognition $Q = s_i \otimes q_i \otimes \beta_i \otimes \gamma_i$ on (m, k, J, L) with $D_{\mathrm{KL}}(Q\|P_i^0) < \infty$, equivalently with every source cost of positive row weight finite. Then $\mathbb{E}_Q[(\log l_i)^+] \le D_{\mathrm{KL}}(Q\|P_i^0) + \log(1 + P_i(o))$, and in $(-\infty, \infty]$

$$\begin{aligned}\mathcal{F}_i = {} & \sum_l \gamma_{il} D_{\mathrm{KL}}(s_i\|\rho_{il}) + D_{\mathrm{KL}}(\gamma_i\|\pi_i^s) + \sum_j \beta_{ij} \int D_{\mathrm{KL}}(q_i\|\kappa_{ij}(m,\cdot)) s_i(dm) \\ & + D_{\mathrm{KL}}(\beta_i\|\pi_i^q) + \iint -\log l_i(o \mid k, m) q_i(dk) s_i(dm).\end{aligned} \tag{4.27}$$

For other recognition laws eq. (1.16) supplies the extended definition. For fixed laws with $\mathbb{E}_{s_i \otimes q_i}(\log l_i)^+ < \infty$ and at least one finite cost in each sector, the minimizing rows are the Gibbs rows of proposition 4.3 at unit temperature, with $e^{-\infty} = 0$ for infinite costs, self costs $D_{\mathrm{KL}}(s_i\|r_i)$ and $\int D_{\mathrm{KL}}(q_i\|G_i(m,\cdot)) s_i(dm)$, and peer costs D_{il}^s, D_{ij}^q.

Proof. No density of the sources is needed. Disintegrate P_i^0 successively along L, m, and J; the chain rule for relative entropy in $[0, \infty]$ [95], with $\sum_l \gamma_{il} = \sum_j \beta_{ij} = 1$, gives $D_{\mathrm{KL}}(Q\|P_i^0)$ as the sum of the four nonnegative terms of eq. (4.27) other than the record term. For the record term, apply the variational characterization of relative entropy [66] to the truncations of $g = (\log l_i)^+$ and pass to the limit by monotone convergence: $\mathbb{E}_Q g \le D_{\mathrm{KL}}(Q\|P_i^0) + \log \mathbb{E}_{P_i^0} e^g$, and $e^g \le l_i + 1$ with $\mathbb{E}_{P_i^0} l_i(o \mid K, M) = P_i(o)$. Thus the record term is defined in $(-\infty, \infty]$. Since $dP_i(\cdot \mid o)/dP_i^0 = l_i/P_i(o)$, eq. (1.16) equals $D_{\mathrm{KL}}(Q\|P_i^0) - \mathbb{E}_Q \log l_i$, which proves eq. (4.27). At fixed laws the row terms are those of proposition 4.3 with $T_q = T_s = 1$; a source of infinite cost receives weight zero at every finite value, and the proof of proposition 4.3 applies on the face of finite-cost sources. □

The extended definition matters. If $D_{\mathrm{KL}}(\Pi\|P_i^0) = \infty$ at the exact posterior Π, which can happen when the record likelihood is unbounded, the separated expression eq. (4.27) reads $\infty - \infty$ at the optimum, while eq. (1.16) gives the finite value $-\log P_i(o)$. In the per-agent model the self costs are these evaluator and reference divergences; they are not the identity-channel divergences $D_{ii}^q = 0$ of the reciprocal potential.

At unit temperatures the peer terms of eq. (4.27) are exactly the peer terms of the reciprocal potential, and the self terms reproduce the split eq. (4.2) of $\mathcal{E}_0$, now weighted by the self probabilities β_{ii}, γ_{ii}. The agent's own model competes with its peers for the role of prior, and the rows infer which source to trust. Records add a likelihood term that persists in the objective. A temperature $T \ne 1$ multiplies the label divergences. Dividing the objective by T also tempers the record factor and the entropies of q_i and s_i, so the result is not the free energy of a normalized model. Only the row block at fixed laws is the loss-based update of Bissiri et al. [24] with weight $1/T$; the whole objective is a block-weighted generalized variational objective.

Definition 4.21 (Per-agent dynamics). On finite positive tables satisfying the hypotheses of theorem 4.6, with incoming channels as in definition 4.19 and positive record likelihoods $l_i(o \mid$

$k, m) > 0$, let each table u^a belong to one agent $i(a)$. The per-agent dynamics is eq. (4.15) with $h^a_z = \partial \mathcal{F}_{i(a)}/\partial u^a_z$, where the peer laws inside $\mathcal{F}_{i(a)}$ take their current values. The Gibbs-reduced per-agent dynamics replaces each agent's rows at every state by the unit-temperature Gibbs rows of proposition 4.20 and applies eq. (4.15) to the belief and model tables alone, with the own-block first variations of the reduced objective $\mathcal{F}^{\mathrm{red}}_i = \min_{\beta_i,\gamma_i} \mathcal{F}_i$. On general standard Borel spaces, the chart per-agent dynamics is the field of definition 4.13 with $\varphi_{q_i}, \varphi_{s_i}$ the own-law first variations of $\mathcal{F}_i$ or of $\mathcal{F}^{\mathrm{red}}_i$ at the current peer laws.

The proof of theorem 4.6 applies without change to both finite-table versions and gives a unique maximal interior solution. On charts, the proof of existence, uniqueness, and the continuation alternative in proposition 4.14 uses no gradient structure, so it applies whenever the map from the chart coordinates of all laws to the own-law first variations is Lipschitz on bounded sets of L^∞. This holds when references, evaluators, and record likelihoods have bounded log densities, each incoming channel is an identity, a fixed kernel with two-sided density bounds, or a channel whose output has a logarithm that is a Lipschitz function of the source chart coordinate on bounded sets, and the Gibbs rows are then smooth functions of bounded costs. A finite-dimensional family to which this field is tangent is then invariant, by uniqueness; tangency alone gives only a solution that stays in the family. The dissipation identity does not carry over: $d\mathcal{F}_i/dt$ is minus the sum of agent i's own variances plus the pairing of the derivative of $\mathcal{F}_i$ in the peer laws with the peers' velocities, and agent i controls only the first part. Let $\mathcal{F}_{\mathrm{tot}} = \sum_i \mathcal{F}_i$, regarded as a function of all tables. The per-agent force on u^a omits from $\partial \mathcal{F}_{\mathrm{tot}}/\partial u^a$ exactly the derivatives of the other agents' free energies, namely the source-role terms eq. (4.12) and the channel derivatives eq. (4.11) of edges that $i(a)$ does not receive. The Fisher gradient flow of $\mathcal{F}_{\mathrm{tot}}$ keeps those terms. It is the Fisher gradient flow of a reciprocal potential of the evidence-augmented form eq. (4.32) at unit temperatures, with a cross-sector self cost $\int D_{\mathrm{KL}}(q_i \| G_i(m, \cdot)) s_i(dm)$ competing as a source in place of the additive term $\mathcal{E}_0$. Results proved for the reciprocal flow therefore do not transfer to the per-agent dynamics without a separate argument. For the reciprocal potential, proposition 4.9 transfers critical points and linear stability between the evolving-row and Gibbs-reduced conventions; its Schur-complement argument uses a symmetric Hessian, and no corresponding transfer is proved for the non-gradient per-agent field.

Proposition 4.22 (Per-agent dynamics need not be a gradient flow). Take three agents with Bernoulli belief laws of log odds l_1, l_2, l_3, a trivial model sector, evaluators equal to Bernoulli references of log odds ϱ_i, identity channels, and rows held fixed at $\beta_{ii} = 1/(1 + w)$ and $\beta_{i,i+1} = w/(1 + w)$ with $w > 0$ and indices modulo three. The per-agent belief dynamics at these frozen rows is

$$(1 + w)\dot{l}_i = -(l_i - \varrho_i) - w(l_i - l_{i+1}), \tag{4.28}$$

whose linear part has eigenvalues $-1/(1 + w)$ and $(-1 - 3w/2 \pm i\sqrt{3}w/2)/(1 + w)$. No Riemannian metric makes eq. (4.28) the gradient flow of a function. By contrast, the linearization of the Fisher gradient flow of eq. (4.4) at any interior critical point is self-adjoint for the Fisher metric and has real spectrum.

Proof. For a Bernoulli law the Fisher flow in log odds is $\dot{l} = -\partial F/\partial p$, and $\partial_p D_{\mathrm{KL}}(\mathrm{Ber}(p) \| \mathrm{Ber}(p')) = l - l'$, which gives eq. (4.28). Its linear part is circulant, with eigenvalues $(-1 - w + w\omega)/(1 + w)$ for the cube roots of unity ω. At an equilibrium of a gradient flow $\dot{x} = -g^{-1}\nabla F$ the linearization is

$-g^{-1}\operatorname{Hess} F$, which is self-adjoint for g and hence has real eigenvalues; the same argument applies to eq. (4.4). □

The frozen rows are not an instance of definition 4.21, in which rows evolve, but the conclusion survives there when the evaluators are equal, $\varrho_i = \varrho$, and the source priors are $(1/(1+w), w/(1+w))$. Then $l_i = \varrho$ with rows at the prior is an equilibrium, every cost vanishes there with zero gradient, the row block and the belief block decouple in the linearization, and the belief block is the linear part of eq. (4.28). The ring is not a gradient flow, but it still has a strict global Lyapunov function: its matrix is normal with negative-definite symmetric part, so $|l - l^*|^2$ decreases. The absence of a gradient structure does not imply the absence of a Lyapunov function, and for fixed rows and channels the per-agent belief dynamics is a global contraction.

Proposition 4.23 (Contraction of per-agent beliefs at fixed rows and channels). On finite alphabets, hold the model laws, the rows, and the record values fixed, with $\beta_{ii} \geq b > 0$ for every agent, every incoming belief channel fixed with no null column, and $l_i(o \mid k, m) > 0$. In coordinates $\theta_i = \log q_i$ modulo additive constants, with the seminorm $\|\theta\| = \max_i \operatorname{osc}(\theta_i)$, where $\operatorname{osc}(f) = \max f - \min f$, the per-agent belief dynamics has a unique equilibrium θ^*, every solution exists for all $t \geq 0$, and $\|\theta(t) - \theta^*\| \leq e^{-bt}\|\theta(0) - \theta^*\|$. In particular $\|\theta - \theta^*\|$ is a strict global Lyapunov function.

Proof. Up to additive constants in each agent, the per-agent belief force is $\varphi_i = \theta_i - \Phi_i(\theta)$ with $\Phi_i(\theta) = \beta_{ii} \int \log G_i(m, \cdot) s_i(dm) + \sum_{j \neq i} \beta_{ij} \log(q_j \mathcal{K}_{ij}) + \int \log l_i(o \mid \cdot, m) s_i(dm)$, so the flow is $\dot\theta = -\theta + \Phi(\theta)$ modulo constants. If $e^a q' \leq q \leq e^{a'} q'$ entrywise, the same bounds hold for $q\mathcal{K}$ and $q'\mathcal{K}$, so $\operatorname{osc}(\log q\mathcal{K} - \log q'\mathcal{K}) \leq \operatorname{osc}(\log q - \log q')$; the absence of null columns makes every $q_j \mathcal{K}_{ij}$ positive. Hence $\operatorname{osc}(\Phi_i(\theta) - \Phi_i(\theta')) \leq (1 - \beta_{ii})\|\theta - \theta'\|$, and Φ is a contraction with constant $1 - b$ on the finite-dimensional quotient by constants. It has a unique fixed point θ^* by Banach's theorem, and the field is globally Lipschitz, so solutions are global. For two solutions, $\delta = \theta - \theta'$ satisfies $\delta_i(t + h) = (1 - h)\delta_i + h[\Phi_i(\theta) - \Phi_i(\theta')] + o(h)$ modulo constants, so the upper Dini derivative of $\|\delta\|$ is at most $-\|\delta\| + (1 - b)\|\delta\|$. Taking $\theta' = \theta^*$ proves the estimate. □

The nonexpansiveness of positive kernels in the oscillation of log ratios is the elementary half of Birkhoff's contraction for the Hilbert projective metric. Evolving rows, law-dependent channels, and a coupled model sector are outside this proposition, and whether some instance of the per-agent class has no Lyapunov function at all is not decided here.

The cross-derivative criterion of Monderer and Shapley [148] says the same as proposition 4.22 in game-theoretic terms. On alphabets with at least three points the agent-wise objectives $D_{\mathrm{KL}}(q_1 \| q_2)$ and $D_{\mathrm{KL}}(q_2 \| q_1)$ have unequal mixed partial derivatives unless $q_1 = q_2$. On a binary alphabet these derivatives are $-1/(x_2(1 - x_2))$ and $-1/(x_1(1 - x_1))$ in the success probabilities, which agree only on $x_1 = x_2$ and $x_1 = 1 - x_2$. In both cases the objectives admit no exact potential on any open set. Damped oscillation on directed cycles is the characteristic behavior of non-reciprocal interactions [85]. The reciprocal potential excludes it by construction, and an exact inheritance of a Fisher gradient flow cannot create it, since differentiating an exact intertwining by a submersion at an equilibrium shows that the parent spectrum is contained in the real spectrum of the fine linearization.

4.7 The reciprocal potential is not a global free energy

Proposition 4.24 (No global free-energy identification). Hold the rows of eq. (4.4) fixed and positive. Let all state and model alphabets be finite, with the positivity hypotheses of theorem 4.6, so that $\mathcal{E}$ is finite on the interior of the configuration space.

(a) Suppose agent i attends to a source $j \neq i$ through an identity channel, with $\mathsf{K}_i = \mathsf{K}_j$ of at least two points. There is no normalized joint $\mathbb{P}$ over finitely many finite-valued variables, record value o, rule $x \mapsto Q(x)$ assigning recognition laws, and constants $a > 0$, b with $\mathcal{E}(x) = a\mathcal{F}(Q(x); o) + b$ on the interior.

(b) Suppose the channels are fixed and agent i attends to a source $j \neq i$ through a belief channel $\mathcal{K}_{ij}$ that is not constant in its input, that is, whose rows are not all equal; an identity channel is an instance. There is no normalized joint $\mathbb{P}$ and record value o such that, with the product recognition $Q = \bigotimes_k \zeta_k$ whose factors are the agent laws themselves, $\mathcal{E} = a\mathcal{F}(Q; o) + b$ with $a > 0$ on an open set of interior configurations.

(c) Conversely, suppose every channel is fixed and is either an identity self channel or constant in its input, $\mathcal{K}^q_{ij}(v, \cdot) = C^q_{ij}$ and $\mathcal{K}^s_{il}(w, \cdot) = C^s_{il}$, and suppose the coefficients $1 + \sum_{j\neq i} \beta_{ij}$ and $1 + \sum_{l\neq i} \gamma_{il}$ take one common value c over all agents and both sectors. Then $\mathcal{E} = c\mathcal{F}(\bigotimes_k \zeta_k; o) + b$ on the whole interior, for the normalized product model $\bigotimes_i \Pi_i$ with $\Pi_i \propto \exp(\Phi_i/c)$, $\Phi_i = \log R_i + \sum_{j\neq i} \beta_{ij} \log C^q_{ij} + \sum_{l\neq i} \gamma_{il} \log C^s_{il}$, and a trivial record.

Proof. For (a), every finite value of $\mathcal{F}(Q; o) = D_{\mathrm{KL}}(Q\|\Pi_o) - \log Z_o$ satisfies $D_{\mathrm{KL}}(Q\|\Pi_o) = \sum Q \log Q - \sum Q \log \Pi_o \leq \max_{\operatorname{supp}\Pi_o}(-\log \Pi_o)$, a finite bound independent of Q. Every term of eq. (4.4) is nonnegative, so $\mathcal{E} \geq \beta_{ij} D_{\mathrm{KL}}(q_i\|q_j)$. Let q_i tend to the point mass at u and q_j to the point mass at $v \neq u$ through interior laws. Then $\mathcal{E}$ is finite along the path and tends to infinity, which no bounded function can match.

For (b), $\mathcal{F}(Q; o)$ is a sum of single-factor terms $\int \zeta_k \log \zeta_k$ and the integral of a fixed function against $\bigotimes_k \zeta_k$, which is affine in each factor. Hence the third derivative in a mass-preserving direction w of q_i and twice in a mass-preserving direction v of q_j vanishes identically for $i \neq j$. For $\mathcal{E}$ the only terms containing both q_i and q_j come from the edges between i and j; with fixed channels the edge $i \to j$ contributes nothing to this derivative, since its second derivative in q_j does not depend on q_i, and the edge $j \to i$ contributes

$$\beta_{ij} \sum_u w(u) \frac{(v\mathcal{K}_{ij})(u)^2}{(q_j\mathcal{K}_{ij})(u)^2}. \tag{4.29}$$

Because the rows of $\mathcal{K}_{ij}$ are not all equal, some mass-preserving v has $\phi = v\mathcal{K}_{ij} \neq 0$, and ϕ sums to zero, so it takes both signs. If the ratio $\phi/(q_j\mathcal{K}_{ij})$ had constant absolute value at every point $q_j + tv$ of an interval of t, then $|\phi(u)| = c(t)((q_j\mathcal{K}_{ij})(u) + t\phi(u))$ for all u; comparing a point with $\phi > 0$ and a point with $\phi < 0$ determines t uniquely, a contradiction. Hence every open set contains configurations at which $u \mapsto (\phi/(q_j\mathcal{K}_{ij}))(u)^2$ is nonconstant, and $w = e_a - e_b$ for two points with different values makes eq. (4.29) nonzero. An affine identity on an open set would equate the two derivatives there. For an identity channel on a two-point alphabet, with $w = v = (v_1, -v_1)$, eq. (4.29) equals $\beta_{ij} v_1^3 [q_j(1)^{-2} - q_j(2)^{-2}]$.

For (c), an identity self channel contributes nothing, and a constant channel gives $\beta_{ij} D_{\mathrm{KL}}(q_i \| C_{ij}^q) = -\beta_{ij} H(q_i) - \beta_{ij} \mathbb{E}_{q_i} \log C_{ij}^q$, and likewise in the model sector. With $D_{\mathrm{KL}}(q_i \otimes s_i \| R_i) = -H(q_i) - H(s_i) - \mathbb{E}_{\zeta_i} \log R_i$, the hypothesis on the coefficients gives $\mathcal{E} = \sum_i [-cH(\zeta_i) - \mathbb{E}_{\zeta_i} \Phi_i] + \mathrm{const} = c \sum_i D_{\mathrm{KL}}(\zeta_i \| \Pi_i) + \mathrm{const}$, and $\sum_i D_{\mathrm{KL}}(\zeta_i \| \Pi_i)$ is the free energy of the product recognition in the product model with a trivial record. □

Part (c) is the degenerate exception to (b): such channels transmit no information about the source and exert no source force. The proposition is narrower than the statement that the reciprocal potential is not the free energy of any fixed normalized model whose recognition laws are the agent laws, and that broader statement is false.

Remark 4.25 (Scope of the obstruction). Each hypothesis of proposition 4.24 is used. First, the finiteness of the model in (a) is essential. Let Θ be a countable latent variable with prior $2^{-(n+1)}$, take the model that is uniform on all agent contents and independent of Θ, and let the recognition rule be the product of the agent laws with an auxiliary factor $Q''(x) = (1-t)\pi_\Theta + t\delta_{n^*}$ whose divergence from the prior is tuned to $\mathcal{E}(x)$ minus the agent terms plus a constant; since $\mathcal{E}$ is bounded below and $D_{\mathrm{KL}}(Q'' \| \pi_\Theta)$ ranges over $[0, (n^*+1)\log 2]$, this reproduces $\mathcal{E}$ plus a constant on the whole interior, with the agent laws as recognition factors. Second, the product restriction in (b) is essential off finite alphabets. On atomless spaces, let $Q(x)$ be a Gaussian-copula coupling of the agent laws ζ_i whose total correlation equals $\mathcal{E}(x) - \mathcal{E}_0(x) \geq 0$, which an equicorrelation parameter in $[0, 1)$ attains. The chain rule gives $D_{\mathrm{KL}}(Q \| \bigotimes_i R_i) = \sum_i D_{\mathrm{KL}}(\zeta_i \| R_i) + \mathrm{TC}(Q)$, and total correlation is invariant under measure isomorphisms of the marginals, so $\mathcal{F}(Q(x); o) = \mathcal{E}(x)$ for the fixed model $\bigotimes_i R_i$ with a trivial record, while the marginals of $Q(x)$ are the agent laws. On finite alphabets the total correlation of a coupling is bounded, which is why (a) survives there. Third, the input dependence in (b) is essential, by (c). Fourth, the Gibbs-reduced potential is outside (a): when every receiver has an identity self source with positive prior, $\mathcal{E}_{\mathrm{red}} \leq \mathcal{E}_0 - T_q \sum_i \log \pi_{ii}^q - T_s \sum_i \log \pi_{ii}^s$ is bounded on finite alphabets. Whether (b) extends to $\mathcal{E}_{\mathrm{red}}$ with fixed channels, to law-dependent channels, and to non-finite spaces in a declared chart is not established here. Finally, the obstruction has no content in the mean directions. On the slice of unit-variance Gaussian beliefs with identity channels and fixed rows, $\beta_{ij} D_{\mathrm{KL}}(\mathcal{N}(\mu_i, 1) \| \mathcal{N}(\mu_j, 1)) = \beta_{ij}(\mu_i - \mu_j)^2/2$ differs by a constant from the expected negative log likelihood of a Gaussian comparison record of precision β_{ij} registered at zero, so $\mathcal{E}$ equals a mean-field free energy plus a constant there. The obstruction lives in the variance and shape directions.

The obstruction concerns models in which the agent laws are recognition laws. The per-agent picture evades it because each $\mathcal{F}_i$ is a fixed model only with its peers' laws held as data. Two further identifications place the agent laws elsewhere. In the first, agent i receives n independent messages from a source $J \sim \pi_i^q$ with law ν_{iJ}^q and registers whether their empirical type equals q_i, a type with denominator n. The method of types gives the registration probability $|T_{q_i}| \exp\{-n[H(q_i) + D_{\mathrm{KL}}(q_i \| \nu_{ij}^q)]\}$ given $J = j$ [46, Theorem 11.1.2], so the posterior of J given registration is the Gibbs row eq. (4.5) with $T_q = 1/n$. There the laws are parameters of message laws. In the second, a prior proportional to $e^{-\mathcal{E}/\tau}$ against the Fisher volume on finite positive tables is normalizable because $\mathcal{E} \geq 0$ and that volume is finite; records then add a likelihood, and the mode follows the Fisher flow of $\mathcal{E} - \tau \log \ell$. There the laws are latent variables of a hierarchical model. Neither construction makes q_i or s_i a recognition law.

The first identification is exact on a type lattice. Fix $n \geq 1$, set $T_q = T_s = T = 1/n$, and let $\mathcal{X}_n$ be a finite set of configurations $x = (q_i, s_i)_i$ on finite alphabets in which every q_i, s_i, and $q_i \otimes s_i$ is a type with denominator n and every compared law $\nu^q_{ij}(x), \nu^s_{ij}(x)$ and reference joint R_i is positive; law-dependent channels are allowed. In the registered-agreement model, X is uniform on $\mathcal{X}_n$, the labels $J^q_i \sim \pi^q_i$ and $J^s_i \sim \pi^s_i$ are independent, agent i receives conditionally independent batches $Y^0_i \sim R_i^{\otimes n}$, $Y^q_i \sim \nu^q_{iJ^q_i}(X)^{\otimes n}$, and $Y^s_i \sim \nu^s_{iJ^s_i}(X)^{\otimes n}$, and it registers the binary records that the empirical types of these batches equal $q_i \otimes s_i$, q_i, and s_i. Write o for the event that every record is registered, $|T_p|$ for the size of the type class of p, and $c_n(p) = H(p) - n^{-1} \log |T_p|$. When both sectors are nontrivial, a product $q_i \otimes s_i$ of interior laws is an n-type only if n divides every product of the numerators of q_i and s_i, which is impossible for prime n; interior configurations require composite n, for example marginals that are m-types with $n = m^2$.

Theorem 4.26 (Registered agreement). In the registered-agreement model the following hold.

(i) $\mathbb{P}(o^q_i \mid x, J^q_i = j) = |T_{q_i}| \exp\{-n[H(q_i) + D^q_{ij}(x)]\}$, and likewise in the model sector and for the reference batch with $D_{\mathrm{KL}}(q_i \otimes s_i \| R_i)$.

(ii) The posterior of J^q_i given x and o is the Gibbs row $\beta^*_i(x)$ of eq. (4.5), and likewise for J^s_i.

(iii) For recognition $\delta_x \otimes \bigotimes_i \beta_i \otimes \bigotimes_i \gamma_i$ of (X, J), with the batches integrated out, the free energy satisfies $T\mathcal{F} = \mathcal{E}(x, \beta, \gamma) + C_n(x) + T \log |\mathcal{X}_n|$, where $C_n(x)$ is the sum of c_n over the registered targets and $0 \leq c_n(p) \leq |\operatorname{supp} p| \log(n+1)/n$.

(iv) $\mathcal{E}_{\mathrm{red}}(x) = -T \log \mathbb{P}(o \mid x) - C_n(x)$, and $\mathcal{E}_{\mathrm{red}}(x) + C_n(x) + T \log |\mathcal{X}_n| \geq -T \log \mathbb{P}(o)$.

Proof. A sequence of type p has probability $\exp\{-n[H(p) + D_{\mathrm{KL}}(p\|\nu)]\}$ under $\nu^{\otimes n}$ [46, Theorem 11.1.2]; summing over the type class gives (i). In the posterior of a label the factor $|T_{q_i}| e^{-nH(q_i)}$ is common to all sources and cancels, which gives (ii). For (iii), $D_{\mathrm{KL}}(\delta_x \| \mathrm{Unif}) = \log |\mathcal{X}_n|$, the label terms are $D_{\mathrm{KL}}(\beta_i \| \pi^q_i)$ and $D_{\mathrm{KL}}(\gamma_i \| \pi^s_i)$, and by (i) each expected negative log likelihood is n times a divergence of eq. (4.4) plus nc_n of its target; multiplying by $T = 1/n$ gives the identity. The bounds on c_n are those on the size of a type class [46, Theorem 11.1.3]. For (iv), $-T \log \mathbb{P}(o \mid x)$ equals $\mathcal{E}_0(x) + C_n(x) - T \sum_i \log Z^q_i - T \sum_i \log Z^s_i$ by (i), and the free energy at the optimal rows bounds the negative log evidence. □

In this model the row temperature is the reciprocal number of exchanged messages, the Gibbs rows are exact posteriors of which source was heard, and the reduced potential is, up to the receiver-only term C_n, a negative log evidence at a point recognition of the configuration. The identification needs equal temperatures and one batch size; unequal temperatures give a weighted composite likelihood in the sense of Bissiri et al. [24]. A nonuniform configuration prior ρ adds $-T \log \rho(x)$. On general standard Borel spaces, registration of an empirical law in a neighborhood of the target gives the divergence as a large-deviation rate through Sanov's theorem [49]; the double limit that would make (iii) exact there is not established here. Resampling n messages from a table p has covariance $(\operatorname{diag} p - pp^T)/n$, which is T times the inverse of the Fisher metric eq. (4.14), so the message count would set the temperature of a stochastic version of the flow, up to a factor fixed by the time unit and the noise convention (one resampling round per unit time in a Langevin equation with this covariance gives $1/(2n)$); that version and its metastability are open.

At frozen sources the same device needs no type lattice.

Corollary 4.27 (Frozen-source episodes). Hold the compared laws ν_{ij} of agent i fixed, let $T = 1/N$ with N a positive integer, and consider the normalized model $J \sim \pi_i$, $X^{1:N} \mid J \sim \nu_{iJ}^{\otimes N}$. For the tied-copy recognition $\beta_i \otimes q_i^{\otimes N}$,

$$T D_{\mathrm{KL}}\big(\beta_i \otimes q_i^{\otimes N} \| \pi_i(dJ)\nu_{iJ}^{\otimes N}\big) = \sum_j \beta_{ij} D_{ij} + T D_{\mathrm{KL}}(\beta_i \| \pi_i), \tag{4.30}$$

and for a sequence of type q_i the posterior of J is the Gibbs row at temperature T. Replicating the reference joint in the same way represents the local term, so at $T_q = T_s = 1/N$ the terms of one agent at frozen sources are T times the record-free free energy of a fixed normalized model, and records attached to the copies add a likelihood with an evidence for each frozen episode.

Proof. The chain rule gives $D_{\mathrm{KL}}(\beta_i \| \pi_i) + \sum_j \beta_{ij} D_{\mathrm{KL}}(q_i^{\otimes N} \| \nu_{ij}^{\otimes N})$, and relative entropy is additive over independent copies. The posterior statement is (ii) of theorem 4.26 for one row. □

Freezing the sources removes their source-role forces, so a sequence of such episodes belongs to the per-agent class of section 4.6, not to the reciprocal flow. It is a different instance from the mixture-source model: each episode minimizes agent i's own terms of eq. (4.4), with the additive local term and rows at temperature $1/N$, not $\mathcal{F}_i$, in which the evaluator competes as a source with weight β_{ii}.

Definition 4.28 (Record-interrupted reciprocal dynamics). Fix a Borel set $\mathsf{S}_{\mathrm{adm}}$ of admitted configurations, the reciprocal Fisher flow (Φ_t) on it, declared record times $0 < t_1 < t_2 < \cdots$ with $t_k \to \infty$, record kernels $R_k(x, do)$ evaluated at the pre-record state, and measurable assimilation maps $A : \mathsf{S}_{\mathrm{adm}} \times \mathsf{O} \to \mathsf{S}_{\mathrm{adm}}$, for example the conditioning updates of chapter 5; experiments may be chosen from the history by a declared policy. Assume $(t, x) \mapsto \Phi_t(x)$ is jointly Borel on its domain and that A maps admitted states into admitted states. The record-interrupted dynamics is the adaptive history of definition A.3 with stage kernels

$$J_k(x; dx', do) = R_k(\Phi_{t_k - t_{k-1}}(x), do)\, \delta_{A(\Phi_{t_k - t_{k-1}}(x), o)}(dx'), \tag{4.31}$$

with $t_0 = 0$, right-continuous paths that follow the flow between records, and the history stopped at the first record time beyond the maximal existence time of the flow. Its finite history laws exist and its infinite-history law is unique by the Ionescu-Tulcea theorem.

On finite positive tables joint measurability holds because the flow is continuous; on bounded log-density charts it is part of the hypothesis. The invariance of $\mathsf{S}_{\mathrm{adm}}$ is a real condition: conditioning on a record whose likelihood vanishes on part of an alphabet sends an interior table to the boundary, where the flow of theorem 4.6 is not defined. With deterministic record times the process is a piecewise-deterministic process in the sense of Davis [55] for the time-augmented state, the declared times acting as forced jumps. The reciprocal potential decreases along each flow interval but need not decrease at an assimilation, since conditioning on a record can move an agent away from the minimizer of eq. (4.4). In the free-energy pictures the same record is a likelihood term, and the objective it enters is fixed between records.

Relative to a fixed normalized model, the reciprocal potential therefore gives up three things: an evidence for model comparison, a posterior that the dynamics targets (its stationary points are

compromises among references and peers), and a learning criterion for references, evaluators, priors, and temperatures. Monotonicity across records needs a precise statement. In the free-energy pictures the objective itself changes at each record, by $\mathbb{E}_Q[-\log \ell(o_{k+1} \mid \cdot)]$ at the current recognition, which is nonnegative for discrete records. What is monotone is its optimized value $-\log P(o_{1:k})$, nondecreasing in k for discrete records though not in general for densities, together with each free energy along its own descent between records. No functional is monotone along the composite of descent intervals and records in any of the three pictures. The stationary set of the reciprocal potential does not depend on records, but the stationary point that the flow reaches can. When the potential has several stable stationary points, an assimilation can move the state into another basin, and this selection persists after the flow relaxes, without any declared record dependence. Any other persistent record dependence must be declared.

Definition 4.29 (Evidence-augmented reciprocal potential). For registered records $o = (o_i)_i$ with declared densities of the record kernels ℓ_i of definition 1.11, set

$$\mathcal{E}_o = \mathcal{E} + \sum_i \iint -\log \ell_i(o_i \mid k, m) q_i(dk) s_i(dm). \tag{4.32}$$

The added term is the record term of eq. (4.27), so $\mathcal{E}_o$ is the reciprocal potential with the per-agent record likelihoods kept in the objective. Its stationary points depend on the records. The constructions of chapter 7 to chapter 13 are proved for $\mathcal{E}$ with records entering only as assimilation events, and they are not claimed for $\mathcal{E}_o$, except for the base-only broadcast evidence term of remark 11.10.

For fixed channels the first question one would ask about the three pictures has a short negative answer.

Proposition 4.30 (Per-agent dynamics is not mean-field coordinate descent). On finite positive tables, hold the model laws and rows fixed and let every belief channel be fixed. Suppose that on an open set of interior belief configurations the per-agent belief force of each agent i equals, up to constants, the block force $\log q_i - \bar{e}_i$ of one mean-field global free energy as in proposition 4.16. Then every peer channel $\mathcal{K}_{ij}$ with $\beta_{ij} > 0$ is constant in its input.

Proof. The per-agent force contains $-\beta_{ij} \log(q_j \mathcal{K}_{ij})$ as its only dependence on q_j, while $\bar{e}_i$ is affine in q_j. Equality up to constants on an open set makes $\log(q_j \mathcal{K}_{ij})(k) - \log(q_j \mathcal{K}_{ij})(k')$ affine in q_j there, for all k, k'. Along a line $q_j + tv$ with v mass-preserving, the ratio of the two affine functions $(q_j \mathcal{K}_{ij})(k) + t(v\mathcal{K}_{ij})(k)$ and $(q_j \mathcal{K}_{ij})(k') + t(v\mathcal{K}_{ij})(k')$ is then the exponential of an affine function of t on an interval, which forces the ratio to be constant, so $v\mathcal{K}_{ij}$ is proportional to $q_j \mathcal{K}_{ij}$. Summing over k, $v\mathcal{K}_{ij}$ sums to zero and $q_j \mathcal{K}_{ij}$ to one, so $v\mathcal{K}_{ij} = 0$ for every mass-preserving v, and all rows of $\mathcal{K}_{ij}$ coincide. □

The per-agent dynamics is therefore coordinate descent of a global free energy on an open set only when every attended peer transmits no information about its source. Stationary points can still coincide on special families. For two agents with unit-variance Gaussian beliefs, references at a and b, and mutual edges, the stationary gaps of the reciprocal potential, of the mixture-source per-agent free energies, and of a global model with one agreement record are each proportional to

$a - b$, so they agree under a map between the weights; with one directed edge the reciprocal flow moves the source while the per-agent source stays at its reference, so no such map exists.

Open problem 4.31 (Relating the three objectives). For law-dependent channels, characterize the configurations and channels for which the per-agent belief forces equal the block forces of one mean-field global free energy on an open set, so that the per-agent dynamics is coordinate descent of a global free energy as in proposition 4.16; for fixed channels proposition 4.30 gives the answer. Determine the families, such as linear Gaussian comparisons with fixed covariances, on which the reciprocal, per-agent and global objectives have the same stationary points, with the map between their configurations and parameters stated explicitly. Decide whether a species in the sense of definition 1.13, with matched evaluators and information, admits such a common joint. For each construction of chapter 7 to chapter 13, determine whether its conclusions hold under the per-agent dynamics or under a global free energy.

Chapter 5

Prediction and Selective Communication

A model law is useful through the records it predicts. Communication can preserve those predictions while discarding other distinctions, and a receiver can combine a message with information already present in its model. This chapter gives exact identities for both operations. The retained prediction is always named; agreement about one output is not automatically agreement about every future interaction.

5.1 Model presentations and their evaluators

Let $G_i : \mathsf{M}_i \rightsquigarrow \mathsf{K}_i$ be a normalized local model and let $H_i : \mathsf{K}_i \rightsquigarrow \mathsf{O}_i$ be a normalized record kernel. Their composition

$$K_i^{\mathrm{rec}}(m, A) = \int H_i(k, A) G_i(m, dk), \qquad p_i^{\mathrm{rec}} = s_i K_i^{\mathrm{rec}}, \tag{5.1}$$

gives the model's predictive record law. An evaluator may also take a preparation, accessible history, or chosen interaction as input; these are held fixed when two predictions are compared. More general normalized evaluators need not factor through G_i and H_i. The model prediction p_i^{rec} pushes the model belief through the evaluator, so it predicts a record of freshly drawn content. The belief prediction $\zeta_i H_i$ instead uses the agent's current conditional $q_i(dk \mid m)$; the two agree when $q_i(dk \mid m) = G_i(m, dk)$, in particular before any record, and differ after one.

Suppose a model link $\Theta_{ij}^s : \mathsf{M}_j \to \mathsf{M}_i$ is a bimeasurable bijection. Exact compatibility of the two evaluators, after identifying their record alphabets, means

$$K_i^{\mathrm{rec}}(\Theta_{ij}^s m, A) = K_j^{\mathrm{rec}}(m, A). \tag{5.2}$$

Consequently, $s_i = (\Theta_{ij}^s)_{\#} s_j$ implies equal predictive record laws. The compatibility is additional information: two equal laws on a one-point model space can predict opposite deterministic bits through different evaluators.

Theorem 5.1 (Exact predictive information identity). Let s, t be probability laws on a standard Borel model space and let K be a common normalized record kernel. Assume $D_{\mathrm{KL}}(s\|t) < \infty$. Set $\mathbb{P}_s(dm, do) = s(dm)K(m, do)$ and $\mathbb{P}_t(dm, do) = t(dm)K(m, do)$, and denote their conditional model laws given o by s^o, t^o. Then

$$D_{\mathrm{KL}}(s\|t) = D_{\mathrm{KL}}(sK\|tK) + \int D_{\mathrm{KL}}(s^o\|t^o)(sK)(do). \tag{5.3}$$

Proof. The common kernel cancels in the joint Radon–Nikodym derivative, giving $D_{\mathrm{KL}}(\mathbb{P}_s\|\mathbb{P}_t) = D_{\mathrm{KL}}(s\|t)$. Disintegrating the same joint laws by their record marginal factors their density ratio into the record ratio and the conditional model ratio. The logarithm is their sum. Integrating the first term gives $D_{\mathrm{KL}}(sK\|tK)$, and integrating the second first over m gives the displayed conditional term. Finiteness makes the decomposition unambiguous. □

Theorem 5.1 is the chain rule for relative entropy [46, 124], written for a joint law built from a common kernel. Without the finiteness assumption the identity still holds in $[0,\infty]$, both sides being infinite together [95]. With a deterministic kernel $K(m,\cdot) = \delta_{T(m)}$ it decomposes a divergence into the divergence of the pushforwards and the expected divergence of the conditional laws given T, the form used in corollary 5.3.

The second term is the model distinction that the record leaves unresolved. Its vanishing is an exact sufficiency condition: when $D_{\mathrm{KL}}(s\|t)$ is finite, $D_{\mathrm{KL}}(sK\|tK) = D_{\mathrm{KL}}(s\|t)$ exactly when the Bayesian inverse of K computed at t returns s from sK. This equality case of the data-processing inequality goes back to Kullback and Leibler [124] and is the classical case of Petz's recovery theorem [161]. Finiteness is needed here: for $s = (1/2, 1/2, 0)$, $t = (0, 1/2, 1/2)$ and a kernel that keeps the first point and merges the last two, both divergences are infinite but no version of the Bayesian inverse recovers s. A constant evaluator can make all predictions agree while retaining no model distinction at all. Useful agreement therefore requires a specified predictive task as well as a comparison of laws. For a common history kernel, the same identity holds with the full finite record history in place of o.

If the model link depends on the current laws, theorem 5.1 applies to the resulting fixed comparison at that configuration, provided eq. (5.2) holds there. Preservation of compatibility during an update is a separate condition on the dynamics and the evaluator.

5.2 Conditional prediction within an agent

An evaluator can relate a cue to a different result, so that a record of one changes predictions for the other. On standard Borel presentation, cue, result, and record spaces, choose normalized kernels $\mu_m(du)$, $K_m(u,dv)$, and $\mathcal{L}(u,do)$, with a proper presentation reference $r(dm)$. The joint

$$P(dm,du,dv,do) = r(dm)\mu_m(du)K_m(u,dv)\mathcal{L}(u,do) \tag{5.4}$$

is an instance of definition 1.11, with represented contents (u,v) and evaluator $G(m,d(u,v)) = \mu_m(du)K_m(u,dv)$. For a record likelihood $l(o \mid u)$ with finite positive evidence, the posterior has the correlated form $w^o(dm)\Pi^o_m(du,dv)$ of eq. (B.3). The result prediction averages the component-conditional results using the posterior model belief. The presentation selects a mechanism; its posterior probability is distinct from the parameters defining that mechanism.

The local free energy retains this dependence. By proposition B.2, a candidate $s(dm)q_m(dk)$ satisfies

$$\mathcal{F} = D_{\mathrm{KL}}(s\|w^o) + \int D_{\mathrm{KL}}(q_m\|\Pi^o_m)s(dm) - \log Z(o). \tag{5.5}$$

Exact conditioning minimizes this functional. Requiring one content law independent of the presentation changes the optimization, as in remark 4.1. Thus the general joint agent law supports a concrete local inference operation beyond the product-family flows.

For an additive Gaussian cue record $O = X + \epsilon$, $\epsilon \sim \mathcal{N}(0, C)$, the fixed-model response is

$$D_\xi \mathbb{E}[Y \mid O = \xi] = \mathrm{Cov}(Y, X \mid O = \xi) C^{-1}. \tag{5.6}$$

Proposition B.4 states the moment and differentiation hypotheses and proves the identity. The cross-covariance permits a negative or rotating response. This is conditional prediction through a statistical relation, while the Fisher tensor measures distinguishability of the posterior law. Gaussian mixture regression gives an explicit realization [35].

Returning only a Gaussian with the result moments is a further operation: q_G minimizes $q \mapsto D_{\mathrm{KL}}(p^o_Y \| q)$ over Gaussian laws, not in general the reverse divergence of variational free energy. The exact loss of a product carrier consisting of model weights and that Gaussian is

$$D_{\mathrm{KL}}(\Pi^o_{M,Y} \| w^o \otimes q_G) = I_{\Pi^o}(M; Y) + D_{\mathrm{KL}}(p^o_Y \| q_G), \tag{5.7}$$

whenever these terms are finite. The chain rule proves the split; proposition B.5 supplies the Gaussian projection proof. This local moment-matching operation is familiar from assumed-density filtering [146]. A later Gaussian observation can distinguish the exact mixture from its carrier (proposition B.6), so the retained interface must name which predictions it preserves. Appendix B gives the self-contained construction, its frame laws, and an exact sequential filtering case.

5.3 Communication between conditional predictors

A message sampled from a peer's current prediction depends on that peer's private evidence. Appendix C supplies a two-agent realization with a common preparation law for represented states $Z_i = (A_i, X_i, Y_i)$, private records D_i, known encoders, and a reliably shared transcript $H_{1:t}$. The local marginals are the cue/result models above, and recognition retains model/content dependence. A fixed known schedule, alternating in the running example, chooses the speaker, which samples a fresh result from its conditional prediction. The receiver integrates the resulting message likelihood over the unknown peer record conditional on its own represented state and available information. The speaker's own random output leaves its private posterior unchanged; an incoming message can change its next outgoing distribution.

The common posterior has an exact reduced representation. If $W(d, dz) = P(Z \in dz \mid D = d)$ is the fixed preparation conditional, with $D = (D_0, D_1)$, then

$$\Gamma_t(dz, dd) = W(d, dz)\lambda_t(dd), \qquad \lambda_t = P(D \in \cdot \mid H_{1:t}). \tag{5.8}$$

Every communication likelihood at a fixed history depends on the prepared variables only through D, so it reweights λ_t and leaves W unchanged. Each message factor reads only its speaker's private record, so the history likelihood is a product of one function of D_0 and one of D_1, and λ_t is a product tilt of λ_0 whose cross-agent dependence stays that of the preparation. Each agent conditions λ_t on its own private value and reconstructs its local joint law with W. The retained state

also includes the known policy, the time index with the fixed schedule, and the persistent transcript whenever past-record queries are admitted. This is statistical sufficiency within the declared family, without moment projection or a minimality claim. The appendix gives its normalized finite-history law, correlated local and reduced free-energy identities, and a binary two-way exchange whose expanded posterior is reconstructed from the fixed preparation and two changing private-record evidence ratios.

The same construction has exact finite-dimensional Gaussian closure for one presentation per agent with affine Gaussian kernels. Multiple presentations can produce, after one message with a nonzero received value, a receiver cue posterior and a common private-record posterior outside every finite Gaussian-mixture family. In a preparation enlarged by a registered public parity record, a separate pair of reachable histories has identical local joint beliefs but different next receiver updates, showing why local dependence alone need not preserve cross-agent evidence. These conclusions concern a supplied common-model protocol. Its fixed-history free energy does not identify the three population objectives, and its fixed preparation supplies no new measurements during communication. The common/private information organization is classical [152], common-prior dialogue goes back to Aumann [8] and Geanakoplos and Polemarchakis [87], and communication by fresh posterior samples with exact Bayesian receivers is studied, with its long-run agreement, by Deshpande et al. [62]; the appendix derives the specified finite-history transmission and retention directly.

5.4 What the receiver reconstructs

A sending map $T_j : E_j \to \mathsf{H}$ induces the message law $\mu_j = (T_j)_{\#} q_j$. Different laws can have the same message law. Messages travel from source to receiver. The reciprocal flow of chapter 4 also moves each source by the likelihood ratio of its receivers, so a communication model that realizes that flow must include this reverse message; the per-agent dynamics of definition 4.21 needs only the forward one. For linear maps $T_j : E_j \to \mathsf{H}$ and $O_i : \mathsf{H} \to E_i$,

$$\operatorname{rank}(O_i T_j) \le \operatorname{rank}(T_j). \tag{5.9}$$

Writing a message into a larger space cannot by itself recover distinctions discarded by T_j. A richer reconstruction uses the receiver's conditional model.

For a received value h and a declared likelihood $\ell_i(h \mid k)$, Bayesian assimilation is

$$q_i^+(dk) = \frac{\ell_i(h \mid k) q_i^-(dk)}{\int \ell_i(h \mid k') q_i^-(dk')}, \tag{5.10}$$

when the denominator is finite and positive. An unmeasured coordinate can change because it is conditionally related to the measured one in q_i^-. The observation selects among possibilities already related by the receiver's model. Under the reciprocal flow such an update is an assimilation event in the sense of definition 4.28; its effect persists in the objective only in the free-energy pictures or in the evidence-augmented potential eq. (4.32).

Proposition 5.2 (Exact conditional reconstruction). Let $T : E \to \mathsf{H}$ be measurable on standard Borel spaces. Let q, p be laws with $q \ll p$, and set $\mu = T_{\#} q$. Choose a regular conditional law

$p(dk \mid h)$ for p given $T = h$. Define

$$\widehat{q}(dk) = \int p(dk \mid h)\mu(dh). \tag{5.11}$$

Then $T_{\#}\widehat{q} = \mu$ and, in $[0, \infty]$,

$$D_{\mathrm{KL}}(q\|\widehat{q}) = \int D_{\mathrm{KL}}\big(q(\cdot \mid h)\|p(\cdot \mid h)\big)\mu(dh). \tag{5.12}$$

In particular, $\widehat{q} = q$ exactly when the two conditional laws agree for μ-almost every h.

Proof. Absolute continuity gives $\mu \ll T_{\#}p$, so the conditional version of p is defined at every h needed up to a μ-null set. That conditional is supported on $T^{-1}(h)$ almost surely. Hence its mixture has pushforward μ. The laws q and $\widehat{q}$ have the same h-marginal and respective conditional laws $q(\cdot \mid h)$ and $p(\cdot \mid h)$. Factoring their density ratio by h proves eq. (5.12). Nonnegativity and the equality condition of relative entropy give the last claim. □

The reconstruction eq. (5.11) is Jeffrey's rule of conditioning the prior p on the marginal μ, in the analysis of Diaconis and Zabell [63]. It preserves the communicated marginal and supplies the discarded conditional from p. It makes precise the sense in which a receiver contributes structure that the message alone does not specify. The proposition assumes a common content presentation; a link or a separately declared correspondence is needed before applying it across different fibers.

Corollary 5.3 (Pythagorean identity and information projection). Under the hypotheses of proposition 5.2, in $[0, \infty]$,

$$D_{\mathrm{KL}}(q\|p) = D_{\mathrm{KL}}(q\|\widehat{q}) + D_{\mathrm{KL}}(\widehat{q}\|p), \qquad D_{\mathrm{KL}}(\widehat{q}\|p) = D_{\mathrm{KL}}(\mu\|T_{\#}p). \tag{5.13}$$

Among all laws r with $T_{\#}r = \mu$ and $D_{\mathrm{KL}}(\mu\|T_{\#}p) < \infty$, the reconstruction $\widehat{q}$ is the unique minimizer of $D_{\mathrm{KL}}(r\|p)$.

Proof. Apply the deterministic-kernel case of the chain rule to (q, p), to $(\widehat{q}, p)$, and to (r, p). The pushforwards are μ and $T_{\#}p$ in each case. The conditional term vanishes for $\widehat{q}$, equals the right side of eq. (5.12) for q, and is nonnegative for r, with equality only when r has the conditional laws of p, that is, when $r = \widehat{q}$. □

This is the Pythagorean identity for information projections [48], and the minimization property is the characterization of Jeffrey's rule by minimum relative entropy [63]. What proposition 5.2 adds to these sources is its use as a receiver mechanism: the loss term measures exactly what a receiver's prior conditional fails to restore.

Proposition 5.4 (Common recovery by likelihood descent). Let $\tau : \mathsf{X} \to \mathsf{Z}$ be a measurable map between standard Borel spaces. Let P_0 be a probability law and let $P_\alpha = L_\alpha P_0$ be a family of probability laws dominated by it. Choose the reference conditional $R_0(z, dx) = P_0(dx \mid \tau = z)$, and write $\bar{L}_\alpha(z) = \mathbb{E}_{P_0}[L_\alpha \mid \tau = z]$. Then

$$(\tau_{\#}P_\alpha)R_0(dx) = \bar{L}_\alpha(\tau(x))P_0(dx). \tag{5.14}$$

Consequently this one kernel recovers every P_α exactly when each L_α has a $\sigma(\tau)$-measurable version, that is, exactly when τ is sufficient for the family $\{P_0\} \cup \{P_\alpha\}$. No finite-divergence assumption is needed.

Proof. For each measurable $A \subseteq \mathsf{Z}$, $P_\alpha(\tau^{-1}A) = \int_A \bar{L}_\alpha(z)(\tau_\# P_0)(dz)$, by the defining property of conditional expectation. For a measurable $B \subseteq \mathsf{X}$, disintegration therefore gives

$$\begin{aligned}((\tau_\# P_\alpha)R_0)(B) &= \int \bar{L}_\alpha(z) R_0(z, B)(\tau_\# P_0)(dz) \\ &= \int_B \bar{L}_\alpha(\tau(x)) P_0(dx).\end{aligned} \tag{5.15}$$

Equality with P_α is equivalent to equality of their Radon–Nikodym densities, $L_\alpha = \bar{L}_\alpha \circ \tau$ almost surely. Conditional expectation fixes exactly the integrable $\sigma(\tau)$-measurable functions, proving both directions. □

A posterior-specific conditional always reconstructs that posterior from its own marginal. It need not reconstruct the reference or other posteriors. The common kernel in proposition 5.4 makes the stronger requirement explicit. Regular conditional laws and their almost-sure uniqueness are understood in the standard Borel sense [114]. The equivalence in proposition 5.4 is the Halmos–Savage factorization criterion for a family dominated by one of its members [100]; the common reference conditional is the kernel that defines sufficiency, and the statement is an instance of abstract sufficiency in the Markov category of standard Borel spaces [84]. What is used below is the explicit density eq. (5.14) of the reconstruction when descent fails.

Corollary 5.5 (Recovery without reading the exterior). Let P_0 now be a joint reference law of (X, E) on standard Borel spaces, and let $Z = \tau(X)$. Retain (Z, E) but reconstruct X using only $R_0(z, dx) = P_0(dx \mid Z = z)$, leaving E unchanged. This reconstructs P_0 exactly if and only if X and E are conditionally independent given Z. Under that condition it also reconstructs a dominated family $P_\alpha = L_\alpha P_0$ exactly if and only if each likelihood ratio descends to (Z, E).

Proof. The lifted reference has conditional law $R_0(z, dx)$ given (z, e). Because R_0 is supported on $\tau^{-1}(z)$ almost surely, its joint law with (Z, E) agrees with the true law exactly when $P_0(dx \mid Z, E) = P_0(dx \mid Z)$ almost surely. This is the stated conditional independence. In that case the exterior-preserving lift is the reference conditional for the map $(x, e) \mapsto (\tau(x), e)$. Apply proposition 5.4 to that map. □

Sufficiency together with conditional independence of the retained exterior given the statistic is the adequacy of a subfield in the sense of Skibinsky [181]. Descent of the likelihood ratios alone does not suffice. Let P_0 be uniform on $\{(0, 0), (1, 1)\}$ for (X, E) and let τ be constant. The family $\{P_0\}$ has a trivially descending likelihood ratio, but the lift draws X uniformly and independently of E, so it does not reconstruct P_0; here X and E are dependent given Z.

5.5 A predictive coding criterion

Lossy source coding makes fidelity part of the problem definition [178]. Decoder side information is a realized variable correlated with the source, as in Wyner–Ziv coding [200]; knowing a prior without such a realization does not supply that variable. A few noiseless real numbers also do not specify a finite-bit channel. A communication rate requires a declared coding, precision, or noise model.

For a predictive criterion, let $P(X, Y, Z)$ be a declared preparation law. The source X may be represented content or a complete fine configuration; the choice is fixed in each application. Let Z be accessible receiver information and let Y be a response to be preserved. An encoder $\mathcal{T}(du \mid X)$ gives the conditional independence

$$U - X - (Y, Z). \tag{5.16}$$

The decoder predicts Y from (U, Z). This is the setting of an information bottleneck, with the relevant variable explicitly named [186].

Theorem 5.6 (Predictive distortion and the optimal decoder). On standard Borel spaces, assume the displayed information quantities are finite and eq. (5.16) holds. For any predictive decoder $\Delta(dy \mid u, z)$ with finite expected relative entropy,

$$\begin{aligned} &\mathbb{E} D_{\mathrm{KL}}\big(P(Y \mid X, Z) \| \Delta(Y \mid U, Z)\big) \\ &\quad = I(Y; X \mid U, Z) + \mathbb{E} D_{\mathrm{KL}}\big(P(Y \mid U, Z) \| \Delta(Y \mid U, Z)\big). \end{aligned} \tag{5.17}$$

The optimal decoder is $P(dy \mid u, z)$. Its distortion is

$$\mathcal{D}(\mathcal{T}) = I(Y; X \mid U, Z) = I(Y; X \mid Z) - I(Y; U \mid Z). \tag{5.18}$$

Proof. Conditional independence gives $P(Y \mid X, U, Z) = P(Y \mid X, Z)$. In the logarithmic ratio defining the left side of eq. (5.17), insert the intermediate conditional law $P(Y \mid U, Z)$. The first resulting expectation is the definition of $I(Y; X \mid U, Z)$. In the second, averaging over X gives the conditional law $P(Y \mid U, Z)$, yielding the remaining divergence. The second term vanishes at the stated decoder and is nonnegative for every decoder. Finally, expand $I(Y; X, U \mid Z)$ in both orders and use $I(Y; U \mid X, Z) = 0$. □

The decomposition eq. (5.17) is the logarithmic-loss identity of source coding with soft reconstructions, in which the optimal reconstruction is a conditional law and the distortion is a conditional mutual information [45].

The functional $I(X; U \mid Z) + \lambda \mathcal{D}(\mathcal{T})$, for $\lambda \geq 0$, is one possible criterion for selecting a representation. Its first term is an information cost, not automatically the length of an individual message. An operational coding theorem requires its own source and block-coding assumptions. Transmitting a sample from $(T_j)_{\#} q_j$ and transmitting that whole probability law are also different protocols.

For parent formation, choose X to include the fine configuration and choose Y to be a future boundary response over a specified horizon and interaction. Zero distortion means that U and Z are sufficient for this target under this preparation. A subsequent interaction may read information that this target ignored. Exact inheritance therefore requires the dynamical condition developed in chapter 6, not only predictive sufficiency for one chosen response.

Chapter 6

Inherited Parents and Recursive Formation

A collection becomes a parent when its retained state supports inherited interaction and evolution. A probability law obtained by compression is a first part of this construction. It does not determine how that law will change, what the parent can communicate, or which information its children must retain. We give exact conditions for these further steps, two sufficient mechanisms that make them hold, and a finite drive–response example in which exact lumping works through two levels. The example supplies its reductions; formation of the reductions remains a separate question.

Most exactness criteria below are instances of classical mathematics, and we attribute them where they are stated: factor maps of dynamical systems, exact lumping of differential equations, strong lumpability and Markov functions of Markov chains, Riemannian submersions, equivariant reduction, and conditional independence. What this chapter adds is their assembly into one notion of an inherited parent with declared protocol and preparation classes, the counterexamples showing which natural retentions fail, and the restrictions that these criteria place on the formation constructions of later chapters.

6.1 Two objects that a parent must retain

Definition 6.1 (Sample compression and configuration retention). Let $x \in \mathsf{S}$ be a complete agent configuration and let Q_x be its declared law on a standard Borel space Y of represented contents. A normalized sample channel $C : \mathsf{Y} \rightsquigarrow \mathsf{Z}$ gives the retained law

$$Q_x^c(A) = \int C(y, A) Q_x(dy). \tag{6.1}$$

A configuration retention is a measurable map $R : \mathsf{S} \to \mathsf{S}^c$. Its value includes the parent laws and the additional data needed for its declared interactions and update. Proposition 6.13 identifies the smallest such data for a declared interface.

The distinction applies to both sectors. A parent carrying q_I and s_I also needs the evaluator that makes s_I predictive. Boundary channels, source rows, source marks, and memory can be further components of $R(x)$. Equal marginal laws do not imply equality of these components. When a sample channel reads random links or source variables, those variables belong to the joint input law of the channel. Substitution of a mean link or a selected source is a different retention.

A law of represented contents and a probability law over complete configurations are different objects. Coupled deterministic evolution of law-valued agents requires no probability distribution over x. If such a distribution is used below, its preparation or transition mechanism is stated. All

constructions admit general probability laws. A Gaussian parent is one possible family, subject to its own closure conditions.

Throughout the chapter, $\circ$ composes maps and juxtaposition composes kernels in the order in which they act: for kernels $K : \mathsf{X} \rightsquigarrow \mathsf{O}$ and $D : \mathsf{O} \rightsquigarrow \mathsf{O}'$, $(KD)(x, A) = \int D(o, A)K(x, do)$. A measurable map used in a kernel product stands for its deterministic kernel. With this convention every square below is a commutative diagram of Markov kernels between standard Borel spaces, and the composition results are the pasting of such diagrams in the Markov category of these spaces [84].

6.2 Exact deterministic inheritance

Lemma 6.2 (Factorization through a Borel surjection). Let $\mathsf{S}, \mathsf{S}^c, \mathsf{Y}$ be standard Borel spaces, let $R : \mathsf{S} \to \mathsf{S}^c$ be a Borel surjection, and let $h : \mathsf{S} \to \mathsf{Y}$ be Borel and constant on every fiber of R. There is a unique Borel map $h^c : \mathsf{S}^c \to \mathsf{Y}$ with $h = h^c \circ R$.

Proof. Constancy on fibers and surjectivity define h^c uniquely. Its graph is the image of S under the Borel map $x \mapsto (R(x), h(x))$ into the standard Borel space $\mathsf{S}^c \times \mathsf{Y}$, hence an analytic set. A function between standard Borel spaces with analytic graph is Borel [117, Thm. 14.12]. □

The lemma is used below for maps into spaces of probability laws, which are standard Borel when they carry the σ-algebra generated by the evaluations $p \mapsto p(A)$ [117].

Proposition 6.3 (An update descends precisely when it respects retained states). Let S, S^c be standard Borel spaces, let $R : \mathsf{S} \to \mathsf{S}^c$ be a Borel surjection, and let $F : \mathsf{S} \to \mathsf{S}$ be Borel. There is a Borel parent update F^c satisfying

$$R \circ F = F^c \circ R, \tag{6.2}$$

if and only if

$$R(x) = R(x') \quad \Longrightarrow \quad R(F(x)) = R(F(x')). \tag{6.3}$$

The parent update is unique. The same conclusion holds on arbitrary measurable spaces when R has a measurable right inverse.

Proof. Equation (6.2) implies the implication in eq. (6.3). Conversely, the implication says that $h = R \circ F$ is constant on the fibers of R, and lemma 6.2 gives a unique Borel F^c with $R \circ F = F^c \circ R$. With a measurable right inverse ι on arbitrary measurable spaces, the formula $F^c = R \circ F \circ \iota$ gives measurability directly, and surjectivity gives uniqueness. □

This is the factor-map (semiconjugacy) criterion of dynamical systems. The analytic-graph argument matters because a Borel surjection need not have a Borel right inverse; the uniformization theorem supplies only universally measurable ones [117, Sec. 18.A]. Repeated substitution gives $R \circ F^n = (F^c)^n \circ R$ for every nonnegative integer n. This is an exact statement about the chosen update, including its order and retained parameters. Replacing that update by descent of a newly defined parent objective requires another argument.

For a differentiable version, suppose R is continuously differentiable and the fine and parent vector fields B, B^c have unique local solutions. Their flows satisfy $R \circ \Phi_t = \Phi^c_t \circ R$ on their common

existence interval if

$$DR_x B(x) = B^c(R(x)). \tag{6.4}$$

Indeed, the chain rule makes $R(\Phi_t(x))$ a parent solution with initial value $R(x)$, and uniqueness identifies it. Conversely, differentiation of the flow identity at zero gives eq. (6.4). For a moving map R_t, the left side becomes $\partial_t R_t(x) + DR_t(x)B_t(x)$. Membership changes therefore need an explicit handoff map; a differential identity on one fixed partition does not specify them. Condition eq. (6.4) is the exact-lumping condition for differential equations, introduced for linear kinetics by Wei and Kuo [197], for nonlinear kinetics by Li and Rabitz [137], and for flows on manifolds by Horstmeyer and Atay [110]; flow intertwining has also been proposed as a criterion of macroscopic emergence [15]. The differentiable statement needs R to be continuously differentiable between manifolds on which both flows are defined. A retention into spaces of laws that is differentiable only weakly, with a parent law of motion given by a weak continuity equation, is not covered, and its intertwining must be proved directly.

Proposition 6.4 (Projectable vector fields). Let $R : \mathsf{S} \to \mathsf{S}^c$ be a C^{k+1} surjective submersion between finite-dimensional manifolds, $k \geq 1$, whose fibers are connected, and let B be a C^k vector field on S. A C^k parent field B^c satisfying eq. (6.4) exists if and only if, in every submersion chart $R(a, b) = a$ with $B = (B_A, B_V)$,

$$\partial_b B_A(a, b) = 0. \tag{6.5}$$

For a linear surjection $R = M$ on $\mathbb{R}^n$ this is the condition $M(DB(x)v) = 0$ for all x and all $v \in \ker M$.

Proof. In a submersion chart, $DR_x B(x) = B_A(a, b)$. Condition eq. (6.5) says that this map is locally constant along each fiber; a locally constant function on a connected fiber is constant, so $x \mapsto DR_x B(x)$ is constant on fibers, and conversely constancy on fibers gives eq. (6.5). A submersion has smooth local sections σ through every point, and $B^c = (DR \cdot B) \circ \sigma$ is then well defined and C^k. Since $k \geq 1$, both fields have unique local solutions. In the linear case the fibers are affine, hence connected, and the derivative of MB along $v \in \ker M$ is $M(DB(x)v)$. □

Remark 6.5 (Why inheritance is tested on a set of configurations). Dynamical inheritance checked only along one realized trajectory carries no information. If $x(t)$ is a smooth solution on $[0, T]$ and R is smooth with $z = R \circ x$ an embedded arc, the tangent field of the arc extends to a smooth field B^c on a neighborhood of the arc, and $R(\Phi_t x_0) = \Phi^c_t(Rx_0)$ holds with unique parent solutions. The same holds on a prepared family on which R is injective. The fiber test eq. (6.3) constrains a retention only over a set containing distinct configurations with the same retained value. For this reason the inheritance identities are imposed below on a declared invariant set Ω_I of post-handoff states (definition 6.18), and the formation standard requires an admitted class that is open in the chart topology of the dynamics or relatively open in a declared invariant family that is not a finite union of orbits.

Proposition 6.6 (Gradient descent through a horizontally conformal submersion). Let $R : (\mathsf{S}, g) \to (\mathsf{S}^c, g_c)$ be a smooth surjective submersion between finite-dimensional Riemannian manifolds, with horizontal spaces $H_x = (\ker DR_x)^\perp$. Assume there is a smooth positive function d such that

$$g_c(DR_x u, DR_x v) = d(x)^2 g(u, v), \qquad u, v \in H_x. \tag{6.6}$$

Let E, E_c be smooth potentials and a a positive function. Then

$$DR_x[-\operatorname{grad}_g E(x)] = a(x)[-\operatorname{grad}_{g_c} E_c](R(x)) \tag{6.7}$$

holds at x if and only if

$$d(x)^2 dE_x(v) = a(x)(dE_c)_{R(x)}(DR_x v) \qquad \text{for every } v \in H_x. \tag{6.8}$$

In particular, if $E = \chi \circ E_c \circ R$ with $\chi' > 0$, then eq. (6.7) holds with $a(x) = \chi'(E_c(R(x)))d(x)^2$, and a Riemannian submersion with $E = E_c \circ R$ gives same-time inheritance.

Proof. Let $w = \operatorname{grad}_g E(x)$ and let w^H be its horizontal part. Since $DR_x w = DR_x w^H$, for any coarse tangent z with horizontal lift v,

$$g_c(DR_x w, z) = d(x)^2 g(w^H, v) = d(x)^2 g(w, v) = d(x)^2 dE_x(v). \tag{6.9}$$

Nondegeneracy of g_c shows that $DR_x w = a(x) \operatorname{grad}_{g_c} E_c(R(x))$ exactly when $d(x)^2 dE_x(v) = a(x)(dE_c)_{R(x)}(z)$ for every z, which is eq. (6.8) because horizontal lifting is a bijection $T_{R(x)}\mathsf{S}^c \to H_x$. If $E = \chi \circ E_c \circ R$, then $dE_x(v) = \chi'(E_c(R(x)))(dE_c)_{R(x)}(DR_x v)$ for every v. When $a = 1$, eq. (6.4) applies. □

This is the gradient pushforward for Riemannian submersions [154] and horizontally conformal submersions [11]. The potential hypothesis $E = \chi \circ E_c \circ R$ is much stronger than the criterion. It forces dE to vanish on $\ker DR$, so the fine gradient is horizontal and every fine trajectory is the horizontal lift of its projection: for a product metric and $R(a, b) = a$, the discarded coordinate does not move. Criterion eq. (6.8) constrains only horizontal derivatives and leaves the discarded dynamics free. On Euclidean $\mathbb{R}^2$ with $R(x_1, x_2) = (x_1 + x_2)/\sqrt{2}$, the potential

$$E = \tfrac{1}{2}x_1^2 + \tfrac{1}{2}x_2^2 + W(x_1 - x_2), \qquad E_c(z) = \tfrac{1}{2}z^2, \tag{6.10}$$

with any smooth W, satisfies eq. (6.8) with $a = 1$, so the center of mass descends at the same time. With $s = x_1 - x_2$ one has $E = \frac{1}{2}z^2 + \frac{1}{4}s^2 + W(s)$, so $E \neq \chi \circ E_c \circ R$ unless $W(s) = -\frac{1}{4}s^2 + \text{const}$, and the separation obeys $\dot{s} = -s - 2W'(s)$. The metric hypothesis is necessary in a uniform sense: if eq. (6.7) holds with $a \equiv 1$ for every potential of the form $E_c \circ R$, then $DR_x \circ g^{-1} \circ DR_x^* = g_c^{-1}$ at $R(x)$, because $DR(\operatorname{grad}_g(E_c \circ R)) = (DR \circ g^{-1} \circ DR^*)(dE_c)$ must equal $g_c^{-1} dE_c$ for every covector, and this identity says that R is a Riemannian submersion. For a single potential the metric hypothesis is not necessary.

For $a > 0$, a fine trajectory projects onto the same oriented coarse gradient orbit after the time change $d\tau/dt = a(x(t))$. An autonomous field in the original time additionally requires the product $a(x) \operatorname{grad}_{g_c} E_c(R(x))$ to be constant on retention fibers. Factorization $a = \bar{a} \circ R$ suffices; dependence on discarded state need not satisfy this requirement. The proposition is one sufficient route to inheritance. Fisher contraction and preservation of the collective connection in chapter 3 are different conditions.

Fiber minimization alone does not supply the criterion. For $R(u, v) = v$ on Euclidean $\mathbb{R}^2$, let

$$E(u, v) = \tfrac{1}{2}(u - v)^2 + \tfrac{1}{2}v^2, \qquad E_c(v) = \inf_u E(u, v) = \tfrac{1}{2}v^2. \tag{6.11}$$

Equality $E = E_c \circ R$ holds only on $u = v$. The horizontal direction is ∂_v, and $dE(\partial_v) - dE_c(DR(\partial_v)) = v - u$, so the same-time criterion fails off the diagonal; the pushed field $u - 2v$ is not a function of v. Since $dE(\partial_u) = u - v$ does not vanish on any open set, there is also no identity $E = \chi(E_c \circ R)$ on an open set.

Proposition 6.7 (Equivariant descent). Let a Lie group G act smoothly, freely, and properly on a finite-dimensional manifold S, and let $\pi : \mathsf{S} \to \mathsf{S}/G$ be the quotient map.

(i) If a smooth vector field B is G-equivariant, there is a unique smooth field B^c on S/G with $D\pi \cdot B = B^c \circ \pi$, and the flows intertwine. Every G-invariant measurable response kernel or channel on S factors through π.

(ii) If $B = -\operatorname{grad}_g E$ with g and E both G-invariant, then S/G carries a unique metric g_c making π a Riemannian submersion, $E = E_c \circ \pi$ for a smooth E_c, and $B^c = -\operatorname{grad}_{g_c} E_c$.

Proof. By the quotient manifold theorem [132, Thm. 21.10], S/G is a smooth manifold and π a surjective submersion whose fibers are the orbits. Since $\pi \circ \phi_h = \pi$ for the action ϕ_h, $D\pi_{hx} D\phi_h = D\pi_x$, and equivariance gives $D\pi_{hx} B(hx) = D\pi_{hx} D\phi_h B(x) = D\pi_x B(x)$. Thus $D\pi \cdot B$ is constant on fibers, and $B^c = (D\pi \cdot B) \circ \sigma$ for local smooth sections σ is well defined and smooth. Invariant kernels are constant on orbits and descend by proposition 6.3 applied to the kernel-valued map. For (ii), invariance of E makes dE vanish on orbit tangents, which for a free action span $\ker D\pi$. Isometries carry horizontal spaces to horizontal spaces, so $g_c(D\pi u, D\pi v) = g(u, v)$ for horizontal u, v is well defined [154]. Proposition 6.6 with $d = 1$ and $\chi = \mathrm{id}$ gives the gradient identity. □

Equivariant descent holds on the whole configuration space, so the fiber test needs no restriction to an admitted class. It forgets only orbit data. Without freeness the quotient is stratified by orbit type and the statement holds on each stratum. On infinite-dimensional law charts, (ii) requires in addition a closed complement to the orbit tangent in the Fisher metric and a quotient Banach manifold, which must be verified for the chart in use.

Proposition 6.8 (Coordinate projections are exact only for isolated blocks). Let $\mathsf{S} = \mathsf{S}_A \times \mathsf{S}_B$ be a product of connected manifolds with metric $g = g_A(a) \oplus g_B(b)$, and let $E \in C^2$. The projection $R(a, b) = a$ intertwines $-\operatorname{grad}_g E$ with an autonomous field if and only if $\partial_a \partial_b E \equiv 0$, that is, $E(a, b) = E_A(a) + E_B(b)$. The discarded block is then autonomous as well.

Proof. The retained velocity is $\dot{a} = -g_A(a)^{-1} \partial_a E(a, b)$. Since the fibers $\{a\} \times \mathsf{S}_B$ are connected and $g_A(a)$ is invertible, it is independent of b exactly when $\partial_b \partial_a E = 0$, that is, when $\partial_a E(a, b) = \phi(a)$. Integrating along paths in the connected factors gives $E = E_A(a) + E_B(b)$, and then $\dot{b} = -g_B(b)^{-1} \partial_b E_B(b)$. □

The Fisher flow of chapter 4 with product recognition has agent-local metric blocks and one potential, and a source appears in the cost of every receiver that attends to it. Discarding whole agents or whole sectors is therefore exact in that dynamics only for energetically isolated blocks, with no peer cost in either direction between the retained and the discarded agents. Exact compression of reciprocally coupled agents can come from symmetry (proposition 6.7) or from horizontal energy structure (proposition 6.6), as in the center-of-mass example. These are sufficient mechanisms, not a classification; the general criterion is projectability (proposition 6.4).

Proposition 6.9 (Exact inheritance of a gradient flow creates no oscillatory mode at images of critical points). Let $B = -\operatorname{grad}_g E$ with $E \in C^2$, and let x^* be a critical point of E. Then $L = DB(x^*) = -g(x^*)^{-1} \operatorname{Hess} E(x^*)$ is self-adjoint for $g(x^*)$ and has real spectrum. If a C^2 submersion R intertwines B with a C^1 field B^c near x^* in the sense of eq. (6.4), then $R(x^*)$ is an equilibrium of B^c and $\operatorname{spec} DB^c(R(x^*)) \subseteq \operatorname{spec} L \subset \mathbb{R}$. If z^* is an equilibrium of B^c and E restricted to the fiber $R^{-1}(z^*)$ attains its infimum, for example on a compact fiber, then the fiber contains a critical point of E and the conclusion holds at z^*.

Proof. For the last statement, $B^c(z^*) = 0$ and eq. (6.4) make $B(x)$ vertical at every $x \in R^{-1}(z^*)$, so dE_x vanishes on the horizontal space $(\ker DR_x)^\perp$. At a minimizer of E on the fiber, dE_x also vanishes on the tangent space $\ker DR_x$ of the fiber, so the minimizer is a critical point. For the main statement: At a critical point the Hessian is a symmetric bilinear form, so L is $g(x^*)$-self-adjoint. Evaluating eq. (6.4) at x^* gives $B^c(R(x^*)) = 0$. Differentiating it at x^*, the term containing the second derivative of R is multiplied by $B(x^*) = 0$, leaving $DR \circ L = L^c \circ DR$ with $L^c = DB^c(R(x^*))$. Hence $\ker DR$ is L-invariant, and since DR is surjective, L^c is conjugate through DR to the map that L induces on $T_{x^*}\mathsf{S}/\ker DR$. The characteristic polynomial of L is the product of those of its restriction and of the induced quotient map, so every eigenvalue of L^c is an eigenvalue of L. □

An exactly inherited parent of a Fisher gradient flow therefore has no underdamped linear mode at the image of a critical point. The critical-point hypothesis cannot be dropped. Let one agent have law $\mathcal{N}(z, I_2)$ on $\mathbb{R}^2$, with Euclidean Fisher metric, and another the Poisson law of mean e^{-y}, with Fisher metric $e^{-y}dy^2$, and take $E = \frac{\mu}{2}|z|^2 + e^{-y} \geq 0$ with $\mu > 0$. Then E has no critical point, the flow is $\dot{z} = -\mu z$, $\dot{y} = 1$, and the retention $w = \operatorname{Rot}(\kappa y)z$, which reads the first agent in a frame turned by the second, is exact with parent field $\dot{w} = (-\mu I + \kappa J)w$, J the rotation generator, whose eigenvalues $-\mu \pm i\kappa$ are not real. Oscillatory collective modes therefore require nonreciprocal fields, explicit driving, declared second-order kinetics, or parent equilibria over fibers along which the fine state drifts without reaching a critical point.

Proposition 6.10 (Loss of dynamical roles under product recognition). Take $Q_x = \operatorname{Ber}(p_1) \otimes \operatorname{Ber}(p_2)$, with $x = (p_1, p_2) \in [0,1]^2$, and the coupled deterministic law

$$\dot{p}_1 = -p_1 + \lambda(p_2 - p_1), \qquad \dot{p}_2 = -2p_2 + \lambda(p_1 - p_2), \qquad \lambda \geq 0. \tag{6.12}$$

Retain only the law of $K_1 + K_2$. No autonomous update of that retained law reproduces every trajectory of eq. (6.12).

Proof. The linear system has a unique global solution. Its vector field does not point outward on any face of the square: at $p_1 = 0$, $\dot{p}_1 = \lambda p_2 \geq 0$, at $p_1 = 1$, $\dot{p}_1 \leq -1$, and similarly for p_2. The square is therefore forward invariant; the field is tangent to a face when $\lambda = 0$ and vanishes at the corner $(0,0)$. The configurations $(0.8, 0.2)$ and $(0.2, 0.8)$ both give probabilities $(0.16, 0.68, 0.16)$ for the values $0, 1, 2$. However,

$$\frac{d}{dt}\mathbb{E}(K_1 + K_2) = -(p_1 + 2p_2), \tag{6.13}$$

equals -1.2 and -1.8 respectively. Their retained laws therefore have different derivatives, and no autonomous update of the retained law can reproduce both trajectories. □

This obstruction occurs while recognition remains a product. The missing datum is which probability belongs to which dynamical role. Correlated recognition is neither a prerequisite for this counterexample nor its automatic remedy. Coupled mean-field dynamics are admissible; the proposed retention must still pass its own inheritance test.

6.3 Exact stochastic inheritance

Proposition 6.11 (Finite strong lumpability). Let T be a transition matrix on a finite configuration space S and let $R : \mathsf{S} \to \mathsf{S}^c$ be surjective. A single parent transition matrix reproduces the retained process for every initial law if and only if, for every $z' \in \mathsf{S}^c$,

$$\sum_{y:R(y)=z'} T(x,y) = \sum_{y:R(y)=z'} T(x',y) \quad \text{whenever } R(x) = R(x'). \tag{6.14}$$

In that case the common sum defines $T^c(R(x), z')$ uniquely.

Proof. The common sums are nonnegative and sum to one, so they define a transition matrix. Conditional on any retained history ending at z, the hidden configuration lies in $R^{-1}(z)$. Every state in that fiber gives the same next retained distribution $T^c(z,\cdot)$; averaging over its conditional law leaves this distribution unchanged. Induction gives the parent path law from the pushed initial law. Conversely, start the fine process at each point mass δ_x. A common parent kernel must give the same one-step distribution whenever these initial states have the same image. This is eq. (6.14). □

This is the finite strong-lumpability criterion [118]; its general state-space form is Dynkin's criterion for a function of a Markov process [68]. On standard Borel spaces the same argument applies whenever $x \mapsto T(x, R^{-1}(\cdot))$ is constant on fibers: the analytic-graph argument of proposition 6.3, applied to this map into the standard Borel space of probability laws on S^c, makes the parent kernel measurable. Closure for every initial law is stronger than closure for one chosen preparation. The latter is weak lumpability [118], and the question whether a function of a Markov chain is itself Markov goes back to Burke and Rosenblatt [34]; Buchholz [32] compares these notions with exact lumpability.

An equilibrium conditional average of fine transitions defines a normalized candidate parent kernel. It does not prove eq. (6.14): the hidden conditional law during a transient can differ from the equilibrium conditional. Likewise, agreement of invariant distributions does not establish agreement of histories.

Proposition 6.12 (Intertwined handoffs). Let T be a transition matrix on a finite space S, let $R : \mathsf{S} \to \mathsf{S}^c$ be surjective, and let Φ_R be its deterministic kernel. Let Λ be a stochastic matrix from S^c to S with $\Lambda(z, R^{-1}\{z\}) = 1$ for every z, and suppose

$$\Lambda T = T^c \Lambda \tag{6.15}$$

for some matrix T^c. Then $T^c = \Lambda T \Phi_R$ is a transition matrix. For every law ν on S^c, under the initial law $\nu\Lambda$ the retained process $Z_n = R(X_n)$ is Markov with initial law ν and kernel T^c, and on every

retained history of positive probability the conditional law of X_n is $\Lambda(Z_n, \cdot)$. For a response kernel K_a on S, the predictive response given the retained history is $(\Lambda K_a)(Z_n, \cdot)$.

Proof. The support condition gives $\Lambda\Phi_R = I$, so multiplying eq. (6.15) on the right by Φ_R yields $T^c = \Lambda T \Phi_R$, a product of stochastic matrices. Let P_z be the diagonal selector of $R^{-1}\{z\}$. The support condition also gives $\Lambda(w, \cdot)P_z = \delta_{wz}\Lambda(z, \cdot)$. Let f_n be the row vector whose x entry is the probability of the history $(z_0, \dots, z_n)$ jointly with $X_n = x$. Then $f_0 = \nu\Lambda P_{z_0} = \nu(z_0)\Lambda(z_0, \cdot)$. If $f_n = c_n\Lambda(z_n, \cdot)$, then

$$f_{n+1} = f_n T P_{z_{n+1}} = c_n (T^c\Lambda)(z_n, \cdot)P_{z_{n+1}} = c_n T^c(z_n, z_{n+1})\Lambda(z_{n+1}, \cdot). \tag{6.16}$$

By induction $c_n = \nu(z_0)\prod_{k<n} T^c(z_k, z_{k+1})$ is the probability of the history and $\Lambda(z_n, \cdot)$ is the normalized conditional law. The predictive response follows by integrating K_a against it. □

This is the Markov-function theorem of Rogers and Pitman [170], who also give the continuous-time and general state-space versions. It gives a sufficient condition for a handoff "from a declared conditional law" to yield a memoryless parent, namely the kernel Λ and an initial law of the form $\nu\Lambda$, and it identifies the filter as $\Lambda(Z_n, \cdot)$. It is not necessary: a strongly lumpable chain gives a memoryless parent from every initial law, and on $\{1, 2, 3\}$ with rows $(0.6, 0, 0.4)$, $(0, 0.6, 0.4)$, $(0.5, 0, 0.5)$, $R(1) = R(2) = a$, $R(3) = b$, the chain is strongly lumpable while among $\Lambda(a, \cdot) = (\lambda, 1-\lambda, 0)$, $\Lambda(b, \cdot) = \delta_3$ the identity eq. (6.15) holds only at $\lambda = 1$. Intertwining does not require strong lumpability. On $\mathsf{S} = \{1, 2, 3\}$ with $R(1) = R(2) = a$, $R(3) = b$, rows $(0.3, 0.5, 0.2)$, $(0.4, 0.2, 0.4)$, $(0.25, 0.25, 0.5)$, and $\Lambda(a, \cdot) = (\frac{1}{2}, \frac{1}{2}, 0)$, $\Lambda(b, \cdot) = \delta_3$, condition eq. (6.14) fails ($0.2 \neq 0.4$) while eq. (6.15) holds with $T^c = \left(\begin{smallmatrix} 7/10 & 3/10 \\ 1/2 & 1/2 \end{smallmatrix}\right)$. From the initial law δ_1, which is not of the form $\nu\Lambda$, the retained path laws of length five differ from those of T^c by $1/10$ in total variation.

Proposition 6.13 (The coarsest exact parent containing an interface). Let $F : \mathsf{S} \to \mathsf{S}$ and an interface $Q : \mathsf{S} \to \mathsf{Y}$ be measurable. The interface collects what the parent must reproduce: its laws, the responses of a declared protocol class, and the channels and evaluators it reads. Define

$$\pi(x) = (Q(F^n x))_{n \geq 0} \in \mathsf{Y}^{\mathbb{N}}. \tag{6.17}$$

Then π satisfies eq. (6.3), with parent update the left shift on $\pi(\mathsf{S})$, and Q is its zeroth coordinate. If a retention R satisfies eq. (6.3) and $Q = Q^c \circ R$ for some Q^c, then $R(x) = R(x')$ implies $\pi(x) = \pi(x')$. Thus π is, up to relabeling of its values, the unique coarsest exact retention through which the interface factors. For a forward-complete continuous flow on a Polish space and a continuous Q into a Polish space Y, replace F^n by Φ_t, $t \geq 0$, and take values in the Polish space $C([0, \infty), \mathsf{Y})$. If B and Q are real analytic, $\pi(x) = \pi(x')$ exactly when $L_B^k Q(x) = L_B^k Q(x')$ for every $k \geq 0$. The image $\pi(\mathsf{S})$ is in general an analytic set, and π is a configuration retention in the sense of definition 6.1, onto a standard Borel parent space, when $\pi(\mathsf{S})$ is Borel; this holds, for example, when S is countable or π is injective on a Borel transversal of its fibers.

Proof. The identity $\pi(F(x)) = (Q(F^{n+1}x))_{n \geq 0}$ is the shift of $\pi(x)$, so equal values have equal successors. If R satisfies eq. (6.3), induction gives $R(F^n x) = R(F^n x')$ whenever $R(x) = R(x')$, and applying Q^c gives $\pi(x) = \pi(x')$. In the analytic case $t \mapsto Q(\Phi_t x)$ is real analytic on its interval

of existence and its derivatives at zero are the Lie derivatives $L_B^k Q(x)$, so the identity theorem gives the stated equivalence. □

The analytic case is the observability construction of Hermann and Krener [106]. For finite Markov chains the corresponding object is computed by partition refinement.

Proposition 6.14 (The coarsest strongly lumpable refinement with responses). Let T be a transition matrix on a finite space S, let $\mathcal{B}_0$ be the partition of S into level sets of an interface Q, and let finitely many protocols a have finite retained response alphabets O_a^c and joint kernels $J_a(x, \{o\} \times \{y\}) = \Pr(D_a(O_a) = o, X' = y \mid X = x)$. Define $\mathcal{B}_{k+1}$ by splitting each block of $\mathcal{B}_k$ according to the vector of masses $J_a(x, \{o\} \times B)$ over all protocols a, values $o \in \mathsf{O}_a^c$, and blocks $B \in \mathcal{B}_k$. The sequence stabilizes at a partition $\mathcal{B}_*$, which is the unique coarsest partition refining $\mathcal{B}_0$ that satisfies eq. (6.14) and on which every joint kernel of retained response and next block factors through the block.

Proof. Each step refines the previous partition, so the sequence stabilizes after finitely many steps. At the fixed point, states in one block have equal joint masses into every pair of a response value and a block; summing over o gives eq. (6.14). Let $\mathcal{B}$ be any partition with these properties refining $\mathcal{B}_0$, and suppose it refines $\mathcal{B}_k$. Each block of $\mathcal{B}_k$ is a union of blocks of $\mathcal{B}$, so two states in one block of $\mathcal{B}$ have equal joint masses into every pair (o, B) with $B \in \mathcal{B}_k$ and lie in one block of $\mathcal{B}_{k+1}$. By induction $\mathcal{B}$ refines $\mathcal{B}_*$. □

With no protocols this is probabilistic bisimulation [127], and with them it is bisimulation with outputs; an efficient algorithm is given by Derisavi et al. [61]. The joint kernels are needed: the response example after eq. (6.18) has a strongly lumpable coordinate partition whose joint response kernel does not factor, and splitting by the marginal response laws alone would return it. The two propositions make the phrase “the additional data needed” in definition 6.1 a theorem for deterministic updates and for finite stochastic updates with preparation class equal to all initial laws: given the update and the interface, the canonical parent is determined. For a restricted preparation class no coarsest exact parent is stated here. A larger interface gives a finer canonical parent, so the protocol class decides how much a parent can forget. Unrestricted identity-channel probes make it injective on exposed child laws (section 10.3, proposition 11.15). Every exact retention whose interface contains the declared protocols refines the canonical parent, so noninjectivity of a handoff retention implies noninjectivity of the canonical parent but not conversely. The compression clause (F4) of the formation standard concerns the handoff retention itself and the child-level probe class of definition 6.17.

6.4 Boundary responses and composition

A parent must inherit what it does to the rest of the population. Fix a declared boundary protocol a, including the preparation and receiving conditions. Let $K_a : \mathsf{S} \rightsquigarrow \mathsf{O}_a$ be its normalized response kernel and let $D_a : \mathsf{O}_a \rightsquigarrow \mathsf{O}_a^c$ retain the response of interest. An inherited kernel K_a^c satisfies

$$(K_a D_a)(x, A) = K_a^c(R(x), A). \tag{6.18}$$

On standard Borel spaces, such a kernel exists exactly when $x \mapsto (K_a D_a)(x, \cdot)$ is constant on each fiber of R; the analytic-graph argument of proposition 6.3 gives measurability, and substitution proves necessity. When several outputs are used jointly, this condition applies to their joint kernel. Preserving each output marginal alone need not preserve their dependence.

The same applies to a response and the next retained state. For a stochastic update, response inheritance concerns the joint kernel $x \mapsto \mathrm{law}(D_a(O_a), R(X'))$ on $\mathsf{O}_a^c \times \mathsf{S}^c$, which must factor through R. Separate descent of the response kernel and of the retained transition does not imply it. Let $x = (u, v) \in \{0,1\}^2$, $R(x) = u$, draw a fresh fair bit ξ, set $u' = u \oplus \xi$, draw v' freshly, and respond with $O = \xi \oplus v$. Both the response and the next retained state are uniform from every configuration, but at $(0,0)$ the joint law of (O, u') is uniform on $\{(0,0),(1,1)\}$ and at $(0,1)$ it is uniform on $\{(1,0),(0,1)\}$. If the randomness of the response is conditionally independent of the transition randomness given x, separate descent suffices. For adaptive protocols, in which a receiver chooses its next action from past outputs, the joint condition is required for each admitted action.

Proposition 6.15 (Exact elimination through a conditional separator). Let Z, U, V have a declared joint reference law $\omega(dz, du, dv) = \omega_Z(dz)\omega_{UV}(du, dv \mid z)$ on standard Borel spaces. Denote its conditional marginals by ω_U, ω_V. For bounded nonnegative likelihood factors $L_U(u, z), L_V(v, z)$, jointly measurable, define

$$F_U(z) = \int L_U(u, z)\omega_U(du \mid z), \qquad F_V(z) = \int L_V(v, z)\omega_V(dv \mid z). \tag{6.19}$$

The identity

$$\int L_U(u, z)L_V(v, z)\omega_{UV}(du, dv \mid z) = F_U(z)F_V(z) \tag{6.20}$$

holds ω_Z-almost everywhere for every such pair precisely when U and V are conditionally independent given Z. Under that condition, a nonnegative factor $L_Z(z)$ with finite positive total evidence gives the unnormalized retained measure

$$m_Z(dz) = L_Z(z)F_U(z)F_V(z)\omega_Z(dz). \tag{6.21}$$

Proof. Conditional independence factors ω_{UV} as $\omega_U \otimes \omega_V$ almost surely. Tonelli's theorem then gives eqs. (6.20) and (6.21). Conversely, take indicator factors from countable generating families closed under finite intersections for the two standard Borel spaces. Outside one ω_Z-null set, the conditional joint and product laws agree on every generating rectangle. Uniqueness of probability measures extends this equality to the product sigma algebra, proving conditional independence. □

The shared factor L_Z appears once. If the hidden blocks are not conditionally independent, their joint conditional factor must be retained. This elimination statement concerns the declared reference and likelihood; it does not replace the law-dependent interaction potential by a fixed joint model. It is the separator factorization of Dawid [56] that underlies junction-tree propagation [129].

The proposition presupposes the joint law. A gluing statement holds for two blocks: laws of (Z, U) and (Z, V) with a common Z marginal have exactly one joint law with U and V conditionally independent given Z, namely the conditional product. Elimination of $k \geq 3$ blocks needs mutual conditional independence, not pairwise independence: with U, V independent fair bits, $W = U \oplus V$,

and Z constant, every pair is independent, but $\mathbb{E}[1_{U=1}1_{V=1}1_{W=1}] = 0 \neq 1/8 = F_U F_V F_W$. Gluing over a family of overlapping separators requires a junction tree. Edge laws on a four-cycle forcing $A = B$, $B = C$, $C = D$, and $D \neq A$ with uniform single-site marginals are pairwise consistent but have no joint law; existence of a joint law for every consistent family characterizes the decomposable (running-intersection) hypergraphs [194].

For a child channel $\mathcal{K}_{ij} : \mathsf{Y}_j \rightsquigarrow \mathsf{Y}_i$ and retentions $C_j : \mathsf{Y}_j \rightsquigarrow \mathsf{Z}_J$ and $C_i : \mathsf{Y}_i \rightsquigarrow \mathsf{Z}_I$, the corresponding square is

$$\mathcal{K}_{ij} C_i = C_j \mathcal{K}^c_{IJ}, \qquad \mathcal{K}^c_{IJ} : \mathsf{Z}_J \rightsquigarrow \mathsf{Z}_I. \tag{6.22}$$

The right side means first retain the sender and then apply the parent channel. A parent channel exists exactly when $\mathcal{K}_{ij} C_i$ is a garbling of C_j, that is, when it lies in the set $C_j \mathcal{M}$ with $\mathcal{M}$ ranging over Markov kernels $\mathsf{Z}_J \rightsquigarrow \mathsf{Z}_I$; for finite alphabets this is a linear feasibility problem. For a deterministic retention C_j it reduces to constancy of $\mathcal{K}_{ij} C_i$ on the fibers of C_j: the retained sender must determine every distinction visible to the retained receiver. For a stochastic C_j the condition is stronger than any linear solvability. If C_j is the binary symmetric channel with flip probability $1/4$ and $\mathcal{K}_{ij} = C_i = \mathrm{id}$, the only linear solution is $C_j^{-1} = \left(\begin{smallmatrix} 3/2 & -1/2 \\ -1/2 & 3/2 \end{smallmatrix}\right)$, which is not a Markov kernel, so no parent channel exists. For a law-dependent channel $\mathcal{K}_{ij}[x]$, the square must hold at each configuration and the map $x \mapsto \mathcal{K}_{ij}[x] C_i$ must also factor through the configuration retention. In a deterministic linear realization the square reads $C_i \circ M_{ij} = M^c_{IJ} \circ C_j$; the kernel-inclusion criterion and its proof are given in the supplement.

Proposition 6.16 (Inheritance composes). Suppose $R_{10} : \mathsf{S}_0 \to \mathsf{S}_1$ and $R_{21} : \mathsf{S}_1 \to \mathsf{S}_2$ intertwine deterministic updates at successive levels. Then $R_{20} = R_{21} \circ R_{10}$ intertwines the first and third updates. The analogous assertion holds for finite lumped Markov transitions. Let $K^\ell_a : \mathsf{S}_\ell \rightsquigarrow \mathsf{O}_\ell$ be boundary kernels, with response retentions $D_{10} : \mathsf{O}_0 \rightsquigarrow \mathsf{O}_1$ and $D_{21} : \mathsf{O}_1 \rightsquigarrow \mathsf{O}_2$. If

$$K^0_a D_{10} = R_{10} K^1_a, \qquad K^1_a D_{21} = R_{21} K^2_a, \tag{6.23}$$

then $K^0_a D_{10} D_{21} = R_{10} R_{21} K^2_a$. The same pasting composes the joint kernels J^ℓ_a of retained response and next retained state of clause (B) below: $J^0_a (D_{10} \otimes R_{10}) = R_{10} J^1_a$ and $J^1_a (D_{21} \otimes R_{21}) = R_{21} J^2_a$ give $J^0_a (D_{10} D_{21} \otimes R_{10} R_{21}) = R_{10} R_{21} J^2_a$. Intertwinings compose: if $\Lambda_{10} T^0 = T^1 \Lambda_{10}$ and $\Lambda_{21} T^1 = T^2 \Lambda_{21}$ with supports on the respective fibers, then $(\Lambda_{21}\Lambda_{10}) T^0 = T^2 (\Lambda_{21}\Lambda_{10})$, and $\Lambda_{21}\Lambda_{10}(z, \cdot)$ is supported on the fiber of $R_{21} \circ R_{10}$ over z. Canonical parents compose along nested interfaces: if $Q_2 = Q_{21} \circ Q_1$, the canonical parent of proposition 6.13 for Q_2 is the canonical parent, for the interface Q_{21} applied to the zeroth coordinate, of the shift on $\pi_{Q_1}(\mathsf{S})$, composed with π_{Q_1}.

Proof. For deterministic updates, $R_{21} \circ R_{10} \circ F^0 = R_{21} \circ F^1 \circ R_{10} = F^2 \circ R_{21} \circ R_{10}$. For Markov transitions and their deterministic retention kernels, in kernel order, $T^0 R_{10} R_{21} = R_{10} T^1 R_{21} = R_{10} R_{21} T^2$, and the kernel $R_{10} R_{21}$ is the deterministic kernel of $R_{21} \circ R_{10}$. For boundary kernels and joint kernels, associate the successive integrals and substitute eq. (6.23) or the joint identities. All integrands for events are nonnegative, so interchanging the integration order is valid. For intertwinings, $\Lambda_{21}\Lambda_{10} T^0 = \Lambda_{21} T^1 \Lambda_{10} = T^2 \Lambda_{21}\Lambda_{10}$, and the support statement follows because $\Lambda_{21}(z, \cdot)$ charges only $R_{21}^{-1}\{z\}$ and each $\Lambda_{10}(z', \cdot)$ charges only $R_{10}^{-1}\{z'\}$. For canonical parents, both maps identify x, x' exactly when $Q_2(F^n x) = Q_2(F^n x')$ for every $n \geq 0$. □

When changes of gauge presentation act on the retained spaces, the retentions and boundary kernels must transform compatibly with those actions. Re-expressing a commuting square then gives the same square in the new coordinates. Neither a parent frame chosen for this purpose nor the square itself determines a base connection. If the parent is also required to retain the collective Fisher connection, the additional condition in proposition 3.18 must be checked.

Definition 6.17 (Declared protocol and probe classes). A construction declares, as part of its constitutive data, two rules defined for every block I of every partition of its agents at every level, singletons included.

(A) The parent protocol rule assigns to I a class A_I of boundary protocols. Each $a \in A_I$ consists of its preparation and receiving conditions, its admitted receiver actions, its normalized response kernel K_a, and its response retention D_a, and it addresses I only through channels and records that read the block as a whole; it does not name a member.

(P) The child-level probe rule assigns to I a class B_I of probes that some agent, channel, record kernel, or declared external querier of the construction can apply to individual members of I, or to sets of them, before a handoff of I. It contains every protocol of A_J for every block J merged into I at that handoff, applied before the handoff. A probe b has a response law $\mathrm{law}_x(b)$ at each configuration x.

Both rules are equivariant under relabeling of the agents and are functions of the index set of the block and of the constitutive data alone. They do not depend on the retention, on which blocks are selected, or on the realized trajectory. This is the content of the requirement that the classes be fixed before the retention: a class defined from a retention, such as the protocols whose responses factor through R_I or the responses computed from a retained law, is not a declared class. The probes of B_I are elements of the observable algebra of chapter 1 at the child level, so a distinction that no $b \in B_I$ separates is gauge relative to B_I in the sense of that chapter. A protocol whose response retention is constant imposes nothing in (B) below; a rule consisting of such protocols is a declared rule and is reported as such.

Definition 6.18 (Inherited parent). An inherited parent of a block I selected at a handoff consists of the parent protocol class A_I of definition 6.17; for stochastic dynamics, a preparation class $\mathcal{P}_I$ of initial laws that contains every post-handoff law the handoff produces from the admitted class, together with any kernel Λ used to define it; a Borel invariant set Ω_I that contains the reachable post-handoff set $\Omega_I^{\mathrm{reach}}$, the forward orbits of the post-handoff configurations arising from the admitted class with the block I; and a Borel retention $R_I : \Omega_I \to \mathsf{S}_I^c$ onto a standard Borel parent space. It must satisfy five conditions.

(L) Laws. The parent belief and model laws and the parent evaluator are functions of R_I on Ω_I.

(D) Dynamics. For a deterministic update, eq. (6.3) holds for all $x, x' \in \Omega_I$. For a flow, $R_I \circ \Phi_t = \Phi_t^c \circ R_I$ on Ω_I for every $t \geq 0$ in the forward existence interval, with a parent flow Φ^c that is well posed, with unique solutions, on $R_I(\Omega_I)$ in a declared chart. For stochastic dynamics, a single parent kernel reproduces the retained process for every initial law in $\mathcal{P}_I$: strong lumpability when $\mathcal{P}_I$ contains all laws, and, for example, the intertwining of proposition 6.12 when $\mathcal{P}_I = \{\nu\Lambda\}$.

(B) Responses. For every $a \in A_I$ and every admitted receiver action u of an adaptive protocol, the joint kernel of the retained response and the next retained state factors through R_I on Ω_I. When $\mathcal{P}_I = \{\nu\Lambda\}$ is restricted, this is replaced by output intertwining: with $T_{a,u,o}(x, dy) = \Pr(D_a(O) \in o, X' \in dy \mid X = x)$ for measurable response sets o, require $\Lambda T_{a,u,o} = T^c_{a,u,o}\Lambda$ with $T^c_{a,u,o} = \Lambda T_{a,u,o}\Phi_{R_I}$.

(C) Channels. Every channel and evaluator read by a protocol in A_I or by the parent update factors through R_I, including its dependence on the configuration.

(G) Geometry, when declared. On an admitted regular family R_I restricts to a smooth G-equivariant map independent of the context, and the fine minimizing class is a coarse minimizing class, that is, eq. (3.45) holds; when equality of the full affine classes is required, also $\ker J = \ker \bar{J}$ (proposition 3.18). The accompanying contraction of the minimal tensor, with its exact defect, is corollary 3.19.

This is the sense of inheritance in clause (F3) of definition 6.36. An empty protocol class makes (B) vacuous, and an unrestricted one forces the retention to be injective on exposed child laws. Dynamical, response, and connection inheritance are separate conditions. Output intertwining reduces to the factorization in (B) when $\mathcal{P}_I$ contains all laws, and the induction in the proof of proposition 6.12 goes through with TP_z replaced by $T_{a,u,o}P_z$, so it gives the predictive joint law of responses and retained states and keeps the filter equal to $\Lambda(Z_n, \cdot)$ after every observed response. The plain factorization in (B) would fail for every response of a chain that is intertwined but not strongly lumpable, such as the example after proposition 6.12, because its second marginal is $x \mapsto T(x, R_I^{-1}(\cdot))$.

This notion differs from a Markov-blanket hierarchy [83, 121, 156]. A Markov blanket is a conditional-independence property of a declared joint law over the states of a system. A parent here is a retention that intertwines the given dynamics and the declared responses; no joint law over configurations is assumed or derived, and conditional independence enters only where a separator is eliminated as in proposition 6.15.

Definition 6.19 (Type closure). Fix a constitutive class $\mathcal{C}$ of agent dynamics, for example Fisher gradient flows of reciprocal law potentials with normalized, possibly law-dependent channels and evaluators. An inherited parent is type-closed for $\mathcal{C}$ if its descended update, channels, and responses again belong to $\mathcal{C}$, with the parent laws as agent laws and their Fisher metric as the metric.

Type closure is what allows a result proved for primitive agents to be applied to parents without change, and hence by induction at every level. For the gradient class, proposition 6.6 gives a sufficient test: the retention is horizontally conformal from the fine Fisher metric to the Fisher metric of the parent laws, and the fine potential satisfies eq. (6.8) with $a \equiv 1$ and a reciprocal potential of the parent laws. With a fiber-constant $a = \bar{a} \circ R \neq 1$ the parent is a gradient flow for $g_c/\bar{a}$, not for the parent Fisher metric. Conformality is not necessary. The descended field B^c is a g_c-gradient exactly when the one-form $g_c(B^c, \cdot)$ is exact, which on a simply connected chart is its closedness; type closure then requires in addition that the potential belong to the declared class.

Open problem 6.20 (Type-closed parents). Determine which of the exact parents of the later chapters are type-closed for the class of Fisher gradient flows of reciprocal law potentials. The

additive common-precision Wishart parent of chapter 12 is answered by theorem 12.5, which identifies the descended field with the unit-Fisher flow of the parent reciprocal potential by direct computation. The conformality route does not reach it: on the common-precision family the horizontal space of the sum retention forces the conformal factor one in the precision directions, and the shape direction would then need $\psi_2'(A_I)\sum_{i\in I} 1/\psi_2'(\alpha_i) = 1$, where $\psi_2'(\nu) = \psi'(\nu) + \psi'(\nu - \frac{1}{2})$ is the shape Fisher information of the bivariate Wishart law; for a block of at least two agents the sum is not sufficient for the shapes, and the product is strictly below one (it equals 0.480 for two shapes equal to 1). The remaining parents of chapter 8 to chapter 15 are open.

6.5 A finite two-level construction

Let a complete configuration be $x = (a, b, c, r) \in \{0, 1\}^4$. The coordinates are labels of internal configurations, with r an internal binary register and $\rho(x) = r$ its readout. To attach actual agent laws, choose

$$q_a = \mathrm{Ber}\left(\frac{1 + 2a}{4}\right), \qquad s_a = \mathrm{Ber}\left(\frac{2 + a}{4}\right), \qquad \zeta_a = q_a \otimes s_a. \tag{6.24}$$

Thus both sectors vary, with tied parameters in this supplied family; b, c are additional internal marks. A model presentation $m \in \{0, 1\}$ has evaluator

$$K(m, \{1\}) = \frac{1}{4} + \frac{m}{2}. \tag{6.25}$$

The corresponding predictive law is $\nu_a = s_a K = \mathrm{Ber}(1/2 + a/8)$, which distinguishes the two values of a.

At each step draw independent bits $\xi \sim \mathrm{Ber}(1/4)$, $\eta \sim \mathrm{Ber}(1/3)$, and $\upsilon \sim \mathrm{Ber}(1/5)$, independently of preceding draws, and set

$$a' = a \oplus \xi, \qquad b' = b \oplus a \oplus \eta, \qquad c' = c \oplus b \oplus \upsilon. \tag{6.26}$$

Independently of these bit draws, conditional on the current configuration, give r' law ν_a. This supplies a transition of an internal register; the evaluator alone did not force this choice. The symbol $\oplus$ denotes addition modulo two. The dynamics are coupled: a affects the next b, and b affects the next c.

Retain $R_{10}(a, b, c, r) = (a, b, r)$ and then $R_{21}(a, b, r) = (a, r)$. For $f_p(0) = 1 - p$ and $f_p(1) = p$, the first parent kernel is

$$\begin{aligned} T^1((a, b, r), (a', b', r')) = {} & f_{1/4}(a' \oplus a)\nu_a(\{r'\}) \\ & \cdot f_{1/3}(b' \oplus b \oplus a). \end{aligned} \tag{6.27}$$

Summing the fine transition over c' gives this expression, independently of c. Summing it over b' gives $T^2((a, r), (a', r')) = f_{1/4}(a' \oplus a)\nu_a(\{r'\})$, independently of b. Both reductions satisfy proposition 6.11. Their configuration spaces have sizes sixteen, eight, and four, and both parents retain the laws in eq. (6.24), their evaluator, and the register. Its readout descends by the identity on record values. The first parent additionally retains b.

Both parents reproduce the entire finite record history as well as the retained configuration history. Indeed, write the fine path probability as its initial mass times its transition factors. Summing over

hidden coordinates gives the parent path law by proposition 6.11; the record at each step is already a coordinate of that path. The next record has probability $1/2 + a/8$ of being one, so the inherited response is nonconstant. No independence of the initial coordinates is needed.

This construction has nontrivial reductions and coupled dynamics, but its retentions were specified in advance. It proves exact two-level inheritance, without proving that the dynamics choose either reduction. It is exact lumping of a drive–response chain: the fine configuration carries one law pair and internal marks, so neither level groups two or more agents, and each retained block is autonomous because the update of a reads only a and the update of (a, b) reads only (a, b). For any coordinate projection, eq. (6.14) is precisely such autonomy of the retained block in law. In the reciprocal gradient dynamics of chapter 4, proposition 6.8 shows that this one-way mechanism is unavailable between energetically coupled agents.

6.6 Averaged laws under smooth peer dynamics

The arithmetic mixture of child laws is a normalized candidate parent law, the linear opinion pool [88], which minimizes the average divergence $\sum_j w_j D_{\mathrm{KL}}(p_j \| q)$ [1]. It need not determine either its own derivative or its boundary responses. The following example establishes both failures within the smooth peer-KL family of chapter 4, with identical agents, a coupled evaluator, and equal Fisher mobilities. It concerns this specified retention; it is not an obstruction to every possible parent state.

Definition 6.21 (A coupled binary model with law-dependent channels). Use counting measure on $k, m \in \{-1, 1\}$ and the strictly positive laws

$$q_x(k) = \frac{1 + kx}{2}, \qquad s_y(m) = \frac{1 + my}{2}, \qquad x, y \in (-1, 1). \tag{6.28}$$

Choose $J = 1/2$, reference $r(m) = 1/2$, and normalized evaluator $G_J(k \mid m) = e^{Jkm}/(2 \cosh J)$. For the declared product recognition family, the local variational term is exactly

$$\begin{aligned} E_0(x, y) &= \sum_{k,m} q_x(k) s_y(m) \log \frac{q_x(k) s_y(m)}{r(m) G_J(k \mid m)} \\ &= \psi(x) + \psi(y) - Jxy + \log \cosh J, \\ \psi(x) &= \frac{1 + x}{2} \log(1 + x) + \frac{1 - x}{2} \log(1 - x). \end{aligned} \tag{6.29}$$

All expectations are finite. The mixed derivative is $-J$, so changing either sector changes the force on the other.

In either sector, let a sender with parameter x use the normalized kernel

$$K_x(\ell \mid k) = \frac{1 + \vartheta(x) k \ell}{2}, \qquad \vartheta(x) = \frac{1 + x^2}{4}, \qquad q_x K_x = q_{f(x)}, \quad f(x) = \frac{x + x^3}{4}. \tag{6.30}$$

Every entry exceeds $1/4$. The channel transmits the sampled sender bit, and its transmission coefficient ϑ depends nontrivially on the sender law. For passive frame covariance, present each alphabet as the atoms $\{\pm u_i\}$ of a one-dimensional fiber, with $u_i \neq 0$, and define $\mathcal{K}_{ij}(u_j k, \{u_i \ell\}) = K_{x_j}(\ell \mid k)$. Under $g_i \in \mathrm{GL}(1, \mathbb{R})$, push the laws forward and replace u_i by $g_i u_i$. The reference

parameters remain fixed, and

$$\mathcal{K}'_{ij}(g_j v, g_i A) = \mathcal{K}_{ij}(v, A). \tag{6.31}$$

The model sector, with its own representation of the same group, gives the same construction for s; pushing G_J through its model and belief frames preserves its normalization. These frames are per-agent coordinates on each agent's fiber; the identity is a passive re-expression and constrains nothing in the dynamics. For two identical agents, set

$$\begin{aligned} d(x,z) &= D_{\mathrm{KL}}(q_x \| q_z), \\ P(x_1,x_2) &= d(x_1, f(x_2)) + d(x_2, f(x_1)), \\ E &= \sum_{i=1}^{2} E_0(x_i,y_i) + P(x_1,x_2) + P(y_1,y_2). \end{aligned} \tag{6.32}$$

Each source row contains only the other agent, so its normalized weight is one and its row-relative-entropy term vanishes. This is a specialization of the declared peer potential. The local term is the displayed KL expectation; no further identification of the entire law-dependent potential with a fixed-joint-law ELBO is assumed. The product Fisher metric and its negative gradient flow are

$$g = \sum_i \left(\frac{dx_i^2}{1-x_i^2} + \frac{dy_i^2}{1-y_i^2} \right), \qquad \dot{x}_i = -(1-x_i^2)\partial_{x_i} E, \quad \dot{y}_i = -(1-y_i^2)\partial_{y_i} E. \tag{6.33}$$

Both mobilities are one. The vector field is smooth on $(-1,1)^4$.

Proposition 6.22 (The mixture retention fails the dynamical inheritance test). For the flow in eq. (6.33), the retention

$$R(x_1,y_1,x_2,y_2) = \left(\frac{q_{x_1} + q_{x_2}}{2}, \frac{s_{y_1} + s_{y_2}}{2} \right) = (q_{\bar{x}}, s_{\bar{y}}) \tag{6.34}$$

does not support an autonomous parent vector field on the whole interior. The same failure holds if the averaged joint law $\frac{1}{2}\sum_i q_{x_i} \otimes s_{y_i}$ is also retained.

Proof. For $0 < |a| < 1$ and $0 < |c| < 1$, take $A = (0, c, 0, c)$ and $B = (a, c, -a, c)$ in the coordinate order of eq. (6.34). Both retain (q_0, s_c) and the same averaged joint law $q_0 \otimes s_c$. These are equal full binary probability laws, not merely equal estimates of their means. The identities $f(-x) = -f(x)$ and $d(-x,-z) = d(x,z)$ imply

$$P(x_1,x_2) = P(x_2,x_1) = P(-x_1,-x_2), \qquad \partial_1 P(a,-a) = -\partial_2 P(a,-a), \quad \nabla P(0,0) = 0. \tag{6.35}$$

At B the two Fisher prefactors are equal. Consequently the peer forces and the two atanh terms cancel in $\dot{\bar{x}}$, giving

$$\dot{\bar{x}}(A) = Jc, \qquad \dot{\bar{x}}(B) = Jc(1-a^2). \tag{6.36}$$

Both model-sector derivatives can also be written exactly. Define

$$h(c) = 2\,\mathrm{atanh}\, c - \mathrm{atanh}\, f(c) + f'(c)\frac{f(c)-c}{1-f(c)^2}. \tag{6.37}$$

Indeed, direct differentiation gives $\partial_x d(x,z) = \operatorname{atanh} x - \operatorname{atanh} z$ and $\partial_z d(x,z) = (z-x)/(1-z^2)$. Thus

$$\begin{aligned}(\dot{y}_1, \dot{y}_2)(A) &= -(1-c^2)(h(c), h(c)),\\ (\dot{y}_1, \dot{y}_2)(B) &= -(1-c^2)(h(c) - Ja, h(c) + Ja),\\ \dot{\bar{y}}(A) = \dot{\bar{y}}(B) &= -(1-c^2)h(c).\end{aligned} \tag{6.38}$$

The retained belief derivatives nevertheless differ by $-Jca^2 \neq 0$. For $a = 1/2, c = 1/4$, they are $1/8$ and $3/32$. Hence $DR_x B(x)$ is not constant on a retention fiber, contradicting the necessary condition eq. (6.4). The identical retained joint laws establish the additional assertion. □

The parameter a may be arbitrarily small. A neighborhood of the synchronized configurations therefore does not make this retention exact. Unlike proposition 6.10, the example uses identical dynamical roles.

Proposition 6.23 (The same retention loses a boundary response). Choose a child uniformly, draw its belief bit, and transmit that bit through eq. (6.30) to an external receiver. This declared protocol has a normalized binary response law with mean $b(x_1, x_2) = (f(x_1) + f(x_2))/2$. For any m, a with $|m| + |a| < 1$, compare belief parameters (m, m) and $(m + a, m - a)$, keeping both model parameters equal to c. The retained marginal and averaged joint laws agree, but

$$b(m + a, m - a) - b(m, m) = \frac{3}{4} m a^2. \tag{6.39}$$

For $ma \neq 0$, their full binary response laws differ.

Proof. Expand $f(x) = (x + x^3)/4$ and use $((m + a)^3 + (m - a)^3)/2 = m^3 + 3ma^2$. At $m = a = 1/4$, the two response means are $20/256$ and $17/256$. Thus the response kernel is not constant on a retention fiber, violating eq. (6.18). □

Proposition 6.24 (Interior synchronization cannot first occur at finite time). For eq. (6.33), the diagonal $\Delta = \{x_1 = x_2,\ y_1 = y_2\}$ is invariant. A solution starting outside Δ cannot enter it at a finite time while remaining in the interior.

Proof. Agent exchange preserves the potential and metric, hence the vector field. On its fixed set Δ, the corresponding component derivatives agree, which proves invariance. The smooth field has unique local solutions in both time directions. If an interior solution meets Δ, uniqueness from that meeting point identifies it locally, backward as well as forward, with the solution in Δ. Along the original existence interval, the set of times at which it lies in Δ is therefore open; it is also closed by continuity. Connectedness of that interval makes it the entire interval, contradicting the initial condition. □

The argument is the invariance of fixed-point subspaces of equivariant vector fields [94] combined with backward uniqueness.

This result permits asymptotic synchronization, state-dependent threshold selection, and exact quotients that do not require synchronization. It does not prove convergence. An alternative retained state or a restricted invariant family must be tested against both inheritance conditions; the calculations above do not exclude such constructions.

6.7 Memory and formation

When a proposed retention fails, one exact option is to retain more of the configuration. For a declared Markov process, with transition kernels that may depend on time, another is to retain the conditional law of the current configuration given the accessible retained history. That conditional law determines future predictions. The Markov hypothesis is needed: for independent fair bits X_0, X_1, $X_{n+1} = X_{n-1}$ for $n \geq 1$, and $R = \mathrm{id}$, the histories $(0,0)$ and $(1,0)$ have the same terminal conditional law δ_0 of X_1 but different next records. In a finite model the conditional law is a probability vector with an exact recursive update; on a countable retained alphabet the same formulas hold with measures in place of vectors at every history of positive probability, and on an uncountable one the update is a jointly measurable disintegration defined almost surely (supplement). Its attainable values need not form a finite set. Introducing such a filter presupposes a process and a preparation law; it is not compulsory for deterministic coupled agents.

The conditional law is not a configuration retention in the sense of definition 6.1. It is a function of the retained history, and it evolves as a Markov process on the simplex of laws on S, an information state [7]. In the finite case, with transition matrix T and the diagonal selector P_z of $R^{-1}\{z\}$ as in the proof of proposition 6.12, it moves from $\widehat{p}$ to $\widehat{p}TP_z/(\widehat{p}TP_z\mathbf{1})$ with probability $\widehat{p}TP_z\mathbf{1}$. It compresses only when noise or nonobservability prevents the history from determining the fine state. For deterministic dynamics observed through R, the filter is supported on the image under F^n of the initial configurations consistent with the history. That consistent set decreases to the observability class of proposition 6.13 with interface R, but its image need not shrink. On a finite space, once the consistent set is a point the filter is the point mass at the fine state; on infinite spaces the filter need not concentrate even when the observability class is a point, as for the doubling map $F(x) = 2x \bmod 1$ with $R(x) = 1_{\{x \geq 1/2\}}$ and a uniform prior, where the filter on X_n is uniform on an interval of length $1/2$ at every n. The intertwined handoffs of proposition 6.12 are the case in which the filter is a function of the current retained value.

Theorem 6.25 (Information lost by a retained history). Let $\mathbb{P}$ be a probability law on $\mathsf{Z}_0 \times \cdots \times \mathsf{Z}_{n-1}$, where each record space is standard Borel and the horizon n is finite. Set $H_t = (Z_0, \ldots, Z_{t-1})$, with H_0 a point. For measurable maps into standard Borel statistic spaces, let $M_t = f_t(H_t)$. Write $P_t^M(\cdot \mid m)$ for a regular conditional law of Z_t given $M_t = m$. Let $\widetilde{T}_t$ be any normalized kernel from the statistic space to Z_t, and let $\widetilde{\mathbb{P}}$ be the causal path law whose next-record kernel is $\widetilde{T}_t(f_t(h), \cdot)$ at history h. Then, in $[0, \infty]$,

$$\begin{aligned} D_{\mathrm{KL}}(\mathbb{P} \| \widetilde{\mathbb{P}}) = &\sum_{t=0}^{n-1} I_{\mathbb{P}}(Z_t; H_t \mid M_t) \\ &+ \sum_{t=0}^{n-1} \mathbb{E}_{\mathbb{P}} D_{\mathrm{KL}}\big(P_t^M(\cdot \mid M_t) \| \widetilde{T}_t(M_t, \cdot)\big). \end{aligned} \tag{6.40}$$

The first sum is the minimum path divergence among these predictors, attained by choosing $\widetilde{T}_t = P_t^M$. Its t-th term vanishes exactly when the true next-record law has a version depending on H_t only through M_t.

Proof. Let $P_t(\cdot \mid h)$ be the true conditional law given $H_t = h$. The relative-entropy chain rule expresses the path divergence as

$$\sum_{t=0}^{n-1} \mathbb{E}_{\mathbb{P}} D_{\mathrm{KL}}\big(P_t(\cdot \mid H_t) \| \widetilde{T}_t(M_t, \cdot)\big). \tag{6.41}$$

To decompose its t-th summand, compare the true joint law of (H_t, Z_t) with the law obtained by keeping its H_t marginal and drawing Z_t from $\widetilde{T}_t(M_t, \cdot)$. Disintegrate these two laws by (M_t, Z_t). The divergence of their marginals is $\mathbb{E}_{\mathbb{P}} D_{\mathrm{KL}}(P_t^M(\cdot \mid M_t) \| \widetilde{T}_t(M_t, \cdot))$. In the second joint law the conditional history given (M_t, Z_t) is the true conditional history given M_t alone. The conditional divergence term is consequently $I_{\mathbb{P}}(Z_t; H_t \mid M_t)$.

The chain rule on standard Borel spaces is an equality of nonnegative extended quantities. A singular marginal or conditional contributes infinite divergence; no subtraction of infinite terms is used. Applying it to each summand of eq. (6.41) gives eq. (6.40). Choosing the true statistic-conditional rows makes every term in the second sum zero. Conditional mutual information vanishes exactly when the corresponding conditional joint law is the product of its conditional marginals. Since M_t is a function of H_t, this is equivalent to $P_t(\cdot \mid H_t) = P_t^M(\cdot \mid M_t)$ almost surely. □

The finite-window case is the Markov approximation of a record process [95, Thm. 7.3.1]; the theorem permits any measurable history statistic. Its two terms separate forgetting information from predicting incorrectly with the information kept. Vanishing of the first sum is predictive sufficiency of the statistic [177]. For a statistic of a coarse process it implies, by data processing, the dynamical-independence criterion of Barnett and Seth [15], whose target is the next coarse value rather than the next record; the converse fails, as for records $Z_t = (A_t, B)$ with independent fair bits A_t, one hidden fair bit B, and coarse value A_t. A statistic alone need not have a compact recursive realization. To update from the retained state and new record only, one additionally needs $M_{t+1} = U_t(M_t, Z_t)$, or a sufficient enlargement of M_t; this is the determinism of causal states in Shalizi and Crutchfield [177]. The theorem concerns one preparation and protocol. Sufficiency for every admitted intervention requires the corresponding conditional-law identity for each such protocol.

A handoff initializes the parent from the actual fine state or its declared conditional law, including the source marks and memory read by future interactions. Proposition 6.12 gives the class of declared conditional laws for which the parent is memoryless. When evidence is shared across levels, its likelihood belongs to the joint model once. Reusing a child's record through a complete same-episode parent posterior requires the appropriate conditional or cavity accounting. A subsequent predictive prior is a different operation.

No formed parent in chapter 7 to chapter 15 acts as a source for its children; the supplied hierarchy of chapter 20 declares top-down factors instead. A parent law can enter a child's cost as a law-dependent source, but the exact reverse channel exerts no force: if ζ_I is the pushforward of $\bigotimes_{i \in I} \zeta_i$ under a retained statistic and $\mathcal{K}_{iI}$ is the regular conditional law of child i given that statistic, then $\zeta_I \mathcal{K}_{iI} = \zeta_i$, the marginal of a disintegration. Downward action therefore needs a declared reverse channel that differs from the exact conditional, and its inheritance must be tested with the moving-map form of eq. (6.4); no such construction is given here.

6.8 Statistical partitions and their limits

An independence partition is a property of a declared joint law. A tuple of individual agent laws does not itself specify a cross-agent joint law; taking their product would impose independence in the input. The following constructions apply when such a joint law has been supplied.

Proposition 6.26 (The finest independent partition). Let V be a finite nonempty index set and let $\mathbb{P}$ be a probability law on $\prod_{i\in V} \mathsf{Y}_i$, with standard Borel coordinate spaces. Call a partition π independent when

$$\mathbb{P} = \bigotimes_{B\in\pi} \mathbb{P}_B, \tag{6.42}$$

where $\mathbb{P}_B$ is its marginal on block B. There is a unique finest independent partition $\pi_*(\mathbb{P})$. Relabeling the coordinates relabels this partition, and separate bimeasurable changes of their presentations preserve it.

Proof. The one-block partition is independent. Suppose π and σ are independent. Marginalizing the σ factorization to a block $B \in \pi$ gives

$$\mathbb{P}_B = \bigotimes_{\substack{D\in\sigma \\ B\cap D\neq\varnothing}} \mathbb{P}_{B\cap D}. \tag{6.43}$$

Substituting these marginals into the π factorization proves independence of the common refinement consisting of the nonempty intersections $B \cap D$. There are finitely many partitions of V. Refining all independent ones therefore gives an independent partition finer than every one of them, which is necessarily unique. Relabelings and coordinatewise bimeasurable maps commute with taking marginals and product laws. Applying the inverse transformation gives the reverse implication, so the set of independent partitions and its unique finest member transform as stated. □

This statistical target is the finest-independent clustering problem [174]. Its existence does not require Gaussian laws or densities, but does not identify statistical independence with agency. In particular, it makes no statement about autonomous dynamics or future boundary responses.

Corollary 6.27 (Conditional independence partitions). Let the same coordinate tuple have a joint law with an already retained standard Borel variable C. Define

$$\mathcal{I}_C(\pi) = \int D_{\mathrm{KL}}\left(\mathbb{P}(dy_V \mid c) \,\middle\|\, \bigotimes_{B\in\pi} \mathbb{P}_B(dy_B \mid c)\right) \mathbb{P}_C(dc). \tag{6.44}$$

There is a unique finest partition for which $\mathcal{I}_C(\pi) = 0$.

Proof. Zero divergence means equality of the conditional joint and product laws for $\mathbb{P}_C$-almost every c. There are finitely many partitions. Outside the union of their relevant null sets, the common-refinement argument in proposition 6.26 applies pointwise to every conditionally independent partition. Their common refinement is therefore conditionally independent and uniquely finest. □

Choosing $C = Y_V$ makes every partition conditionally independent while retaining the whole tuple. The corollary does not choose the context that a useful parent should retain.

Proposition 6.28 (Weak coupling can destroy every independent split). For any even $N \geq 4$, let $y_i \in \{-1, 1\}$ and define the strictly positive finite law

$$\mathbb{P}_\epsilon(y) = Z_\epsilon^{-1} \exp\left(J \sum_{b=1}^{N/2} y_{2b-1} y_{2b} + \epsilon \sum_{b=1}^{N/2-1} y_{2b} y_{2b+1} \right), \qquad J > 0, \quad \epsilon \geq 0, \tag{6.45}$$

where Z_ϵ is the sum of the numerator over the finite alphabet. Then $\pi_*(\mathbb{P}_0)$ consists of the pairs $\{2b-1, 2b\}$, while $\pi_*(\mathbb{P}_\epsilon) = \{V\}$ for every $\epsilon > 0$, although $\mathbb{P}_\epsilon \to \mathbb{P}_0$ in total variation.

Proof. At $\epsilon = 0$, distinct pairs are independent and each pair is dependent: its odds ratio is $e^{4J} \neq 1$. Thus the finest partition consists of those pairs. For $\epsilon > 0$ the interaction path is connected. Any nontrivial partition cuts an edge. Fix all coordinates except the endpoints of one cut edge, whose coupling is $J_e \in \{J, \epsilon\}$. Positivity makes every such conditional law well defined. Its odds ratio is $e^{4J_e} \neq 1$, since all terms involving just one endpoint cancel in that ratio. Factorization across the proposed partition would make these endpoints conditionally independent after the other coordinates were fixed, forcing ratio one. No nontrivial independent partition exists. Continuity of each of the finitely many probabilities gives the convergence in total variation. □

Exact independence is therefore not a continuous soft-membership criterion.

Proposition 6.29 (Symmetry constrains a deterministic hard selector). Let all coordinate spaces equal Y, and suppose a deterministic partition selector Π reads only a law on Y^V. Assume relabeling naturality: $\Pi(\sigma_\# \mathbb{P}) = \sigma \Pi(\mathbb{P})$ for every permutation σ of V. Let $H_\mathbb{P} = \{\sigma : \sigma_\# \mathbb{P} = \mathbb{P}\}$ be the stabilizer of the law. Then $\Pi(\mathbb{P})$ is invariant under $H_\mathbb{P}$. If $H_\mathbb{P}$ is transitive, the blocks of $\Pi(\mathbb{P})$ form a block system of $H_\mathbb{P}$; if it is primitive, $\Pi(\mathbb{P})$ is either the singleton or the one-block partition. In particular this holds when $\mathbb{P}$ is invariant under all permutations.

Proof. For $\sigma \in H_\mathbb{P}$, naturality gives $\Pi(\mathbb{P}) = \Pi(\sigma_\# \mathbb{P}) = \sigma \Pi(\mathbb{P})$. An invariant partition of a transitive group is by definition a block system, and a transitive group is primitive when its only block systems are the two trivial partitions [65]. The full symmetric group is primitive. For $|V| \leq 2$ only the trivial partitions exist. For $|V| \geq 3$, if an invariant partition had a block B with $2 \leq |B| < |V|$, choose distinct $i, j \in B$ and $k \notin B$. The transposition exchanging j and k sends B to a distinct block that still contains i, which is impossible for a partition. □

For a law on a four-cycle invariant under its dihedral group, the only nontrivial block system is the diagonal pairing $\{\{1, 3\}, \{2, 4\}\}$, so a relabeling-natural law-only selector cannot output adjacent pairs; adjacent pairing needs asymmetric preparation data. A distribution over partitions, soft memberships, asymmetric realized histories, or additional geometric inputs are outside this proposition. An information-based hierarchy can also be retained before choosing a cut [40]. None of these possibilities supplies the required dynamics without a further construction. The symmetry result does not exclude symmetry breaking from asymmetric initial conditions.

The four results of this section need a cross-agent joint law, and the dynamics of chapter 4 evolves a tuple of agent laws with product recognition. The product of the agent laws has the singleton partition as its finest independent partition. The flow itself produces a joint law only as the pushforward $(\Phi_t)_\# \mu_0$ of a preparation ensemble μ_0, a law over configurations that describes the ensemble rather

than one population; for a Dirac preparation it is a product of point masses whatever the coupling. For a nondegenerate ensemble coupling makes it dependent across interacting agents: for $\dot{x} = Ax$ with $A = \left(\begin{smallmatrix} -1-\lambda & \lambda \\ \lambda & -1-\lambda \end{smallmatrix}\right)$ and $x(0) \sim \mathcal{N}(0, I)$, $\mathrm{Cov}(x_1(t), x_2(t)) = \frac{1}{2}(e^{-2t} - e^{-2(1+2\lambda)t}) > 0$ for $\lambda, t > 0$. The theory also supplies content-level cross-agent joint laws: the posterior Π_o of the global model of section 4.5 and the joint private-record posterior of appendix C. Both are admissible inputs for proposition 6.26 and proposition 6.29, and their finest partitions are determined by the declared model or protocol rather than by the flow of agent laws. These results are therefore constraints on admissible selectors rather than formation mechanisms; the selectors of the later chapters are declared thresholds for this reason.

6.9 Architectures whose selection cannot see the interaction

Two exact statements constrain the membership order that some architectures can produce. They concern the full flow; whether a construction then fails the interaction-necessity clause (F5) of the formation standard depends in addition on its null flows and on the class of selectors it declares, and the remarks after each proposition say when it does.

Proposition 6.30 (Scalar additive closure fixes the membership order). Let scalar parameters θ_i in an open interval J evolve by $\dot{\theta}_i = G(\theta_i, g(t))$, where g is a state common to all agents and G is continuous in its first argument. Suppose that for every pair of agents the sum retention $\theta_i + \theta_j$ intertwines with the flow, that is, $G(x, g) + G(y, g)$ depends on $x, y \in \mathrm{J}$ only through $x + y$. Then $G(\theta, g) = a(g)\theta + c(g)$, and every difference $\theta_i - \theta_j$ and every difference of count-weighted block centroids equals its initial value times the common factor $m(t) = \exp \int_0^t a(g(\tau))d\tau$. Consider a selector that joins two current blocks when their centroid distance first reaches a fixed threshold from above, and suppose m is nonincreasing and the initial centroid distances have no ties. Then the sequence of selected partitions is an initial segment of a sequence determined by the initial data alone; the common state g determines only the event times and the length of the segment.

Proof. For fixed g put $\phi = G(\cdot, g)$. If $x + y = x' + y'$ then $\phi(x) + \phi(y) = \phi(x') + \phi(y')$; taking $x' = y' = (x + y)/2 \in \mathrm{J}$ gives Jensen's equation $\phi((x + y)/2) = (\phi(x) + \phi(y))/2$, whose continuous solutions are affine. Hence $\frac{d}{dt}(\theta_i - \theta_j) = a(g)(\theta_i - \theta_j)$, and a count-weighted centroid obeys the same affine law, so centroid differences are also multiplied by $m(t)$. A pair of current blocks with initial centroid distance D is at distance $Dm(t)$ and first reaches the threshold δ when $m(t) = \delta/D$. Since m is common and nonincreasing, events occur in increasing order of D. After a join at level D, a pair involving the new block with initial centroid distance $D' \le D$ is already at or below the threshold and, m being nonincreasing, never reaches it from above; a pair with $D' > D$ joins later in order of D'. All of these comparisons involve only initial distances. Whether the event at level D occurs depends on whether m reaches δ/D, which is where g enters. □

The same Jensen argument appears as proposition 12.4 for the Wishart family. The proposition says nothing about the null flows, whose force need not be affine or share the factor m. Under scalar additive closure the full-flow membership order is fixed by the initial data; interaction necessity fails in addition when the null flow is scalar additive with a common factor m^0 and the infima of m and m^0 select the same segment, for example when both tend to zero and histories are compared on $[0, \infty)$. When the null factor is $m^0 \equiv 1$, or the two infima lie between different event levels δ/D_k, the

partition sequences differ on open sets although the full-flow order is a dendrogram of the initial data; the calibration example after the formation standard is of this kind.

Anisotropic linear laws escape this proposition. For vector parameters with $\dot\theta_i = A(g)\theta_i + c(g)$, differences evolve by a common propagator that need not preserve the ordering of distances: with $A = \mathrm{diag}(-1,-3)$, initial differences $(1,0)$ and $(0,1.5)$ become approximately $(0.37,0)$ and $(0,0.075)$ at $t=1$, reversing which pair is closer. Within the hypotheses of proposition 6.31 below this cannot happen: on finite tables the centered log-ratios are the natural-parameter differences, and they obey a common propagator equal to -2 times the identity whatever the dimension and the coupling. Whether a smooth reciprocal potential outside that class realizes an anisotropic, interaction-dependent propagator with exact additive closure is open.

Proposition 6.31 (Shared targets give a common clock). Work on finite positive tables as in theorem 4.6, with a common belief alphabet K. Suppose every evaluator is the same model-independent law, $G_i(m,\cdot) = G_0$, and every belief channel from a source $j \neq i$ is $\mathcal{K}^q_{ij}[\mathbf{q}](v,\cdot) = C[M]$ for all inputs v, where C is a twice continuously differentiable normalized law of the additive moment $M = \sum_i \mathbb{E}_{q_i}\mathsf{T}$ of a fixed statistic $\mathsf{T} : \mathsf{K} \to \mathbb{R}^p$. Suppose either (a) the belief source sets exclude self, or (b) they include self with the self channel also equal to $C[M]$. Then along the Fisher flow of eq. (4.4) (with dynamic rows) or of its reduced potential eq. (4.6),

$$\frac{d}{dt}\log\frac{q_i(k)}{q_j(k)} = -2\log\frac{q_i(k)}{q_j(k)} + c_{ij}(t), \tag{6.46}$$

with c_{ij} independent of k. In case (a) the two nulls of definition 6.33 coincide, and along them the coefficient -2 is replaced by -1 with the common target G_0. In case (b) the self costs are nonzero and the nulls differ: along the label-preserving null the coefficient of $\log q_i$ in the force is $1+\beta^0_{ii}$, with the agent- and time-dependent self weight of eq. (6.48), and along the renormalized null it is 2 with a forcing that depends on the agent through the frozen peer moment; neither has a common clock in general.

Proof. With a model-independent evaluator the local term splits as $D_{\mathrm{KL}}(s_i\|r_i) + D_{\mathrm{KL}}(q_i\|G_0)$, whose belief derivative is $\log q_i - \log G_0$ plus a constant. In both cases every cost in the row of i is $D_{\mathrm{KL}}(q_i\|C[M])$ and contributes $\beta_{ij}(\log q_i - \log C[M])$ as receiver, and the row sums to one. As a source, q_i enters every cost only through M, so it contributes $-\sum_l\sum_j \beta_{lj}\sum_u \frac{q_l(u)}{C[M](u)}\frac{\partial C[M](u)}{\partial M}\cdot\mathsf{T}(v)$, the sum running over the source sets, which is the same function of v for every i; with Gibbs rows the envelope identity eq. (4.8) gives the same expression with β^*. Hence $h_i = 2\log q_i + V_t$ with V_t common to all agents, up to constants in v. The flow eq. (4.15) gives $\frac{d}{dt}\log q_i = -(h_i - \mathbb{E}_{q_i}h_i)$, and subtracting the equations for i and j proves eq. (6.46). In case (a) deleting the peer costs leaves no row term under either null, so $h_i = \log q_i - \log G_0$. In case (b) the surviving self cost is $D_{\mathrm{KL}}(q_i\|C[M^0_i])$ with $M^0_i = \mathbb{E}_{q_i}\mathsf{T} + M^0_{-i}$ and the initial peer moment M^0_{-i}; it enters with weight β^0_{ii} or one, and M^0_{-i} differs between agents. □

Proposition 12.3 is the instance of case (b) on the Wishart family, where only the shape coordinates keep a common clock. In case (a) the proof gives more than the identity.

Corollary 6.32 (The full flow is a common tilt of the null flow at doubled time). In case (a) of proposition 6.31, let q^0_i be the null flow from the same initial laws. There are positive functions ω_t on

K, the same for every agent, such that

$$q_i(t) = \frac{\omega_t q_i^0(2t)}{\mathbb{E}_{q_i^0(2t)}\omega_t} \qquad \text{for every } i \text{ and } t \geq 0. \tag{6.47}$$

Proof. From $h_i = 2\log q_i + V_t$, $\log q_i(t) = e^{-2t}\log q_i(0) + W_t + c_i(t)$ with $W_t = -\int_0^t e^{-2(t-s)}V_s ds$ common and c_i constant on K. Along the null, $\log q_i^0(\tau) = e^{-\tau}\log q_i(0) + (1-e^{-\tau})\log G_0 + c_i'(\tau)$. At $\tau = 2t$ the difference of the two logarithms is $W_t - (1-e^{-2t})\log G_0$ up to constants on K, which does not depend on i; take ω_t its exponential. □

A selector that is invariant under common tilts and under monotone changes of time, such as one that reads only pairwise log-ratios modulo constants, therefore produces along the full flow the partition sequence it produces along the null flow, on every horizon, and fails (F5) in this class. A selector that is not tilt-invariant, such as a threshold on the divergence between two agent laws, can give different partition sequences although every relative coordinate evolves as along the null flow. This is one reason why the formation standard fixes a selector family before the flows are analyzed and asks for a selector-free observable. Interaction-sensitive selection under the reciprocal potential, read by tilt-invariant selectors, therefore requires that at least one hypothesis of proposition 6.31 fail: heterogeneous or model-dependent evaluators, self sources with nonzero self costs as in case (b), input- or source-dependent channels, law dependence beyond one additive moment (a shared target that is the normalized geometric mean of all agent laws is the same law for every receiver and is not of this form), or receiver-dependent targets such as selective channels with nonconstant Gibbs rows. Heterogeneous evaluators alone can give (F5) with a membership order supplied by the references, as in the calibration example below.

6.10 The formation standard

The thesis of the theory is that interactions between agents can produce further agents. A definition of formation must therefore exclude three ways in which the selection and inheritance clauses can be met without any such production. A parent that retains every child state compresses nothing: four agents with laws $\mathcal{N}(x_i, 1)$ under symmetric peer-KL consensus, a threshold selector, and the identity retention select two pairs and then their union with exact inheritance at every stage. Agents that do not interact can be grouped by any rule that reads their initial data and a common relaxation clock. A second stage that forms from every handoff whatever the first stage produced is a sequence of supplied scales rather than recursion. The standard below makes each exclusion a clause, and it fixes before the dynamics every object that a construction could otherwise choose after seeing its own flows: the controls, the topology of the admitted class, the protocol and probe classes, the selector and its constants, the horizon, and the counterfactual class.

Definition 6.33 (Null flows and the frozen-environment control). Let a construction declare its objective and its flow Φ, with references, evaluators, source sets and priors, temperatures, records, a selector family, and an admitted class of initial configurations. A cost of receiver i is a peer cost when it belongs to an edge $j \to i$ with $j \neq i$ whose compared law ν_{ij} depends, on the admitted class, on the law of some agent other than i. A cost carrying a peer label whose compared law

depends on no other agent's law, such as one with a constant-output channel, is a self term and is kept by every control.

(N1) The label-preserving null Φ^0_{lp} deletes every peer cost, that is, sets it to zero, while the peer labels, the source priors, the temperatures, and all other data are unchanged. Every law-dependent channel and evaluator that remains is evaluated with the laws of the other agents held at their initial values. With Gibbs rows, a receiver with self prior a and self cost D_{ii} then has self weight

$$\beta^0_{ii} = \frac{a e^{-D_{ii}/T}}{a e^{-D_{ii}/T} + 1 - a}, \tag{6.48}$$

which is below one whenever $a < 1$, and the same form holds in the model sector.

(N2) The renormalized null Φ^0_{rn} is the same, except that the labels of the deleted edges are removed from the source sets. A receiver whose source set contains itself then has self weight one, and a receiver with no remaining source has no row term.

(N3) The frozen-environment control Φ^{fr} deletes nothing. In every cost, row, channel, and evaluator that enters the force on agent i, each law of an agent $j \neq i$ is held at its initial value.

The two nulls coincide when no receiver has a self source whose cost is not identically zero. In the per-agent picture the same rules apply to each $\mathcal{F}_i$ of proposition 4.20, for the dynamics of definition 4.21 and for its Gibbs-reduced version in which the rows are minimized exactly at unit temperature at every state; the self costs are $D_{\mathrm{KL}}(s_i \| r_i)$ and $\int D_{\mathrm{KL}}(q_i \| G_i(m,\cdot)) s_i(dm)$, and (N1) keeps the echo labels. For the global free energy of section 4.5, the null marginalizes every record whose likelihood factor depends on the contents of two or more agents, replaces P_0 by the product of its single-agent marginals, and keeps single-agent factors and records, with latent source labels at their priors; the frozen-environment control holds Q_{-i} at its initial value in eq. (4.20). No construction of this manuscript states a selection result for the global free energy, and no further convention is developed for it here.

Each control is the flow of the modified objective in the chart or invariant family in which the construction declares Φ, and the construction verifies for it the Lipschitz and well-posedness hypotheses it verifies for Φ (for example those of proposition 4.14). The control must exist, with unique solutions, on the admitted class up to the declared horizon; where it does not, (F5) is undefined there and the admitted class is restricted. A construction whose objective lies outside these classes, such as the nonsmooth interaction of chapter 7, declares its controls by the same rules and states their well-posedness.

The selector is a declared measurable function of the configuration of agent laws, the accessible records, and the declared interfaces, built from constitutive data and the same function at every stage. Rows and costs enter it only through the original constitutive formulas as functions of the laws, and it is evaluated in the same way along Φ and along every control: a selector that reads rows reads, along a control, the rows that the original costs give at the control state, not the rows of the control's own objective. After a handoff along a control, the control remains the fine flow, and the selector is evaluated on the block statistics it reads, computed from the fine states; inheritance is not required along a control, and where retained parent laws are not defined along it the construction extends the selector to fine states by a declared formula.

The membership history $H(x)$ of an initial configuration x is the sequence of selected partitions with their event times on the declared horizon $[0, \tau]$, or on the maximal forward existence interval when no finite horizon is declared. The partition sequence $P(x)$ is $H(x)$ with its event times removed, two sequences being identified when one is obtained from the other by exchanging consecutive events whose changes involve disjoint sets of agents. The histories H^0_{lp}, H^0_{rn}, H^{fr} and the sequences P^0_{lp}, P^0_{rn}, P^{fr} are produced by the same selector along the controls, each on its own maximal forward existence interval.

Lemma 6.34 (Agents that do not interact fail interaction necessity). If no cost, row, channel, or evaluator that enters the force on any agent depends on the law of another agent on the admitted class, then $\Phi^0_{\mathrm{lp}} = \Phi^0_{\mathrm{rn}} = \Phi^{\mathrm{fr}} = \Phi$ there, and $P = P^0_{\mathrm{lp}} = P^0_{\mathrm{rn}}$.

Proof. By the definition there is no peer cost, so the nulls delete nothing, and holding the laws of other agents fixed changes no term of any force. The selector is the same function along all four flows. □

The rule that peer costs are those with a compared law depending on another agent is needed for this lemma. Deleting every cost with a peer label would remove, for three Bernoulli agents with a common evaluator and a constant-output channel from each peer, a term $D_{\mathrm{KL}}(q_i \| C)$ that couples no agents but moves every agent's own target, and a relabeling-equivariant selector reading the distance to the evaluator then gives different partition sequences along the full flow and that deletion on an open set. The frozen-environment control answers a different question from the nulls: whether the selection already follows from each agent's response to the initial laws of the others, without feedback.

Definition 6.35 (Counterfactual handoff class). For a construction with two selection stages, let the first stage hand off at a time that is deterministic or a stopping time, with its selected blocks and retentions, and let the second-stage event and its block be measured from the handoff. A counterfactual class $\mathcal{C}$ is declared with the constitutive data, before the second stage is analyzed. It contains every actual first-stage handoff state reachable from the admitted class and the counterfactual handoff states obtained by applying the same retentions, at the same handoff times, to the states of trajectories from a declared class $\mathcal{U}$ of initial configurations containing the admitted class, along which the first-stage event has not occurred. It excludes states outside the domain of the selector, such as parents already within a threshold that the selector requires to be reached from above. It carries the relative topology of the parent configuration space and, for stochastic dynamics, the declared preparation law of the counterfactual handoffs, under which they have probability bounded below.

Definition 6.36 (Inherited recursive formation). An inherited recursive formation is a construction satisfying all of the following clauses.

(F1) Admitted class. The initial configurations form a nonempty set of initially unpartitioned configurations that is open in the chart topology of the declared dynamics: the product of open simplices for finite tables, the bounded log-density chart of definition 4.13, a declared Banach chart, or the parameter manifold of a declared restricted-family flow, which is then stated. This is the full grade. In the family grade the set is instead relatively open in a declared

invariant family defined by stated constitutive constraints and not contained in finitely many orbits, and its codimension is reported. The grade is always reported.

(F2) Selection. The dynamics selects nontrivial blocks with a selector from a declared family S_c, $c \in \mathsf{C}$, that satisfies the readout convention of definition 6.33, with a stated persistence criterion: an invariant block relation, persistence on a finite window, or a declared path event for stochastic dynamics. The family and the set C of its constants are fixed before the flows are analyzed, and every real constant ranges over a set with nonempty interior. Each handoff is a measurable function of the state or of accessible history.

(F3) Inheritance. Each selected block carries an inherited parent (definition 6.18) for the classes of definition 6.17.

(F4) Compression. For each handoff there are $b \in B_I$ and $x \neq x'$ in the reachable post-handoff set $\Omega_I^{\mathrm{reach}}$ with $R_I(x) = R_I(x')$ and $\mathrm{law}_x(b) \neq \mathrm{law}_{x'}(b)$. Equivalently, R_I is noninjective on $\Omega_I^{\mathrm{reach}}$ and erases a distinction that is not gauge relative to B_I.

(F5) Interaction necessity. Against a null Φ^0 of definition 6.33 and for a constant $c \in \mathsf{C}$, there is a nonempty open subset U of the admitted class such that, for every $x \in U$, $P(x) \neq P^0(x)$ when both histories are taken on $[0, \infty)$, and $P(x)$ on the declared finite horizon $[0, \tau]$ is not an initial segment of $P^0(x)$ when a finite horizon is declared. The clause holds when this is true against both nulls and for every $c \in \mathsf{C}$.

(F6) Recursion. (F6a) The parents select a block containing at least two first-stage parents under the inherited interactions, by the first-stage selector family with the same constants and the same constitutive data, without resetting their source data. (F6b) For the declared counterfactual class $\mathcal{C}$, the second-stage event occurs after the actual first-stage handoffs and does not occur from a set of counterfactual handoffs with nonempty interior in $\mathcal{C}$; for stochastic dynamics, the probability of the second-stage event by the declared windows after the actual handoffs exceeds its value after the counterfactual handoffs by a margin bounded below.

A construction satisfying (F1), (F2), (F3), and (F6a) and shown to fail at least one of (F4), (F5), and (F6b) is a two-stage selection with exact inheritance; a construction with only one selection stage that satisfies (F1) to (F3) is a single-stage selection with exact inheritance. A clause that is neither established nor shown to fail leaves the classification undetermined, and the construction is then not shown to be a recursive formation. The clauses that fail or remain undetermined are always stated.

Clause (F5) is a control experiment: it asks whether the grouping would occur with the interaction switched off. Its verdict names the null. When the two nulls give different verdicts it is reported as split, and the clause fails; a pass against the label-preserving null alone can come from the reweighting eq. (6.48) of a nonzero self cost, which carries no information about the interaction. The verdict reports the subset of C on which it holds, a selector-free discrepancy between Φ and the nulls where one exists (a limit point, a sign of a public coordinate, a contraction factor), and the comparison of P with P^{fr}. A difference only in event times, or only in the order of merges of disjoint blocks, does not count, and the initial-segment comparison on a finite horizon excludes a pure change of clock: under the common clock of proposition 6.31 the full flow realizes on $[0, \tau]$ the null sequence up to the null time 2τ, which a comparison of both histories on $[0, \tau]$ would count

as a difference. Persistence can mean an invariant block relation in deterministic dynamics or a specified path event in stochastic dynamics, and a finite-window criterion is stated as such.

Clause (F4) excludes the identity retention described above, and it is always reported with two attributes: whether the separation by b is persistent or tends to zero along the post-handoff flow, and whether the fibers of R_I are orbits of a symmetry of the fine field and of the protocols of A_I. Under (F3) the retained dynamics and every admitted parent response are blind to the erased distinction, while (F4) asks that a child-level probe declared before the handoff see it. The exact routes of this chapter, symmetry quotients and horizontal energy structure, produce such compressions; no mechanism is given here by which an interaction creates an exact symmetry or decoupling on an open admitted class, and a graded notion of inheritance is not developed.

In (F6), a counterfactual class that contained parents already within the threshold of a from-above selector would make (F6b) hold for every such construction, since those parents never unite; the domain exclusion in definition 6.35 removes this case. If the second-stage event occurs from every state of $\mathcal{C}$ outside a set with empty interior, or for stochastic dynamics with probability tending to one uniformly over $\mathcal{C}$, the stages form a supplied-scale sequence; a proof of the second stage that uses only hypotheses on the handoff state that also hold for counterfactual handoffs in $\mathcal{C}$ establishes such a sequence. A margin in the other direction, where the first stage delays or suppresses the second, is first-stage dependence and not recursion.

Two grades refine (F5). The strong grade holds when (F5) holds at every point of the admitted class, for every $c \in \mathsf{C}$, against both nulls. For the amplifying grade a construction declares a metric d on agent laws, for example the Hellinger metric, and for a partition π of a configuration y puts $\varrho(y, \pi)$ equal to the largest within-block d-diameter divided by the smallest d-distance between blocks. It is amplifying with factor $\lambda < 1$ if, at every event time t_k of Φ with partition π_k and for both nulls, $\varrho(\Phi_{t_k} x, \pi_k) \le \lambda \varrho(\Phi^0_{t_k} x, \pi_k)$ on an open set of the admitted class. Both ratios are evaluated on the partition of Φ, so the grade is defined when Φ^0 has no event; when within-block diameters vanish at the events of Φ the grade holds trivially and is reported as degenerate.

The standard does not require that a parent be type-closed in the sense of definition 6.19, that it act on its children through a term of the fine field absent before the handoff, or that it hold a generative model or lower a free energy. Two clauses of (F6) therefore establish two stages with inherited dynamics, not iteration to every depth or top-down efficacy; whether these become clauses is part of open problem 6.38.

Each construction of the later chapters is measured against this standard with the following report: its objective, dynamics, and chart; the grade of (F1); the selector family, its constant set, and the persistence criterion; the rules of definition 6.17; the witness and the attributes of (F4); the verdict of (F5) against each null over C, with the selector-free observable and the frozen-environment comparison; (F6a); the counterfactual class and the verdict of (F6b); the grades; and the resulting classification.

Proposition 6.37 (An anchored linear consensus is a two-stage selection that fails (F6b)). Take four agents with belief laws $\mathcal{N}(x_i, 1)$, $x \in \mathbb{R}^4$, a trivial model sector, evaluators $\mathcal{N}(c_i, 1)$ with $c = (0, 0.6, 1.3, 1.95)$, identity channels including the self channel, and fixed rows of weight $1/4$ on

each of the four sources. The reciprocal potential is

$$\mathcal{E} = \sum_i \tfrac{1}{2}(x_i - c_i)^2 + \tfrac{1}{4}\sum_i \sum_{j\neq i} \tfrac{1}{2}(x_i - x_j)^2, \tag{6.49}$$

and the declared dynamics is its restricted-family flow (definition 4.10) $\dot{x} = -\nabla\mathcal{E}$, with unit Fisher metric. The selector family joins two current blocks when the distance of their count-weighted centroids reaches δ from above, $\delta \in \mathsf{C} = [0.45, 0.55]$; the retention of a block is its centroid and count, all other coordinates kept; A_I consists of the average of one fresh sample from each member, and B_I adds one fresh sample from a named member through the identity channel the population uses. Then, on a neighborhood U of c and for every $\delta \in \mathsf{C}$:

(i) From $x(0) = c$ every difference of coordinates or centroids equals its initial value times $m(t) = (1 + 2e^{-3t})/3$. The full flow selects $\{1, 2\}$, then $\{3, 4\}$, then $\{1, 2, 3, 4\}$, at the times with $m(t_k) = \delta/D_k$, $D_k = 0.6, 0.65, 1.325$; every relation persists. The two nulls coincide and give $x(t) = c + e^{-t}(x(0) - c)$, which selects nothing. The frozen-environment control gives the factor $(1 + 4e^{-5t/2})/5$ from $x(0) = c$ and the same partition sequence as the full flow.

(ii) (F1) holds in the full grade for the declared restricted flow, (F2), (F3), and (F6a) hold, (F4) holds with a transient separation that is not of symmetry type, (F5) holds in the strong grade against both nulls with selector-free observable the contraction of every difference by the factor $1/3$, and $P^{\mathrm{fr}} = P$.

(iii) For every counterfactual class of definition 6.35 the union of the two parents occurs from every counterfactual handoff, so (F6b) fails and the construction is a two-stage selection with exact inheritance.

Proof. The flow is $\dot{x}_i = -(x_i - c_i) - \frac{1}{2}\sum_{j\neq i}(x_i - x_j)$, the source role contributing the same term as the receiver role. Since $\sum_{j\neq i}(x_i - x_j) = 4x_i - \sum_j x_j$, every difference obeys $\dot{\Delta} = -3\Delta + (c_i - c_j)$, so it relaxes at rate three to one third of the corresponding anchor difference, and from $x(0) = c$ it equals $(c_i - c_j)m(t)$; count-weighted centroids are averages and obey the same law. Since m decreases from 1 to $1/3$, a pair of current blocks with initial centroid distance D reaches δ exactly when $m = \delta/D$, which happens for $D/3 < \delta < D$. The initial distances are 0.6 for $\{1, 2\}$, 0.65 for $\{3, 4\}$, at least 0.7 for the other pairs, and after the joins 1.0 and 1.65 from $\{1, 2\}$ to the singletons and 1.325 between the two parents. For $\delta \in \mathsf{C}$ the order is as stated, the last event requires $\delta > 1.325/3 = 53/120$, and m decreasing makes every joined relation invariant. Along both nulls the self cost vanishes and the rows are fixed, so deleting the peer costs, with or without their labels, leaves $\dot{x} = -(x - c)$; its differences move monotonically from their initial values to the anchor differences, which are at least $0.6 > \delta$. Along the frozen-environment control, $\dot{x}_i = -\frac{5}{2}x_i + c_i + \frac{1}{2}\sum_{j\neq i} x_j(0)$, and from $x(0) = c$ the differences equal $(c_i - c_j)(1 + 4e^{-5t/2})/5$, a decreasing factor with limit $1/5 < 0.45/1.325$, which gives the same sequence. Every crossing is transverse, since the factors have negative derivative, and at every event the other current distances exceed δ strictly, uniformly for δ in the compact set C; continuous dependence of the linear flows on initial data therefore gives the same three sequences on a neighborhood U.

For (ii), U is open in $\mathbb{R}^4$, the chart of the declared flow, and is initially unpartitioned. Each centroid evolves by the averaged law, which reads only centroids and counts, and the force of a block I on an outsider k is $\frac{1}{2}|I|(x_k - \bar{x}_I)$; hence (L), (D), (B), and (C) of definition 6.18 hold with the parent

law $\mathcal{N}(\bar{x}_I, 1/|I|)$ and a linear, globally well-posed parent flow. The reachable post-handoff set is four-dimensional and its centroid image three-dimensional, so it contains distinct configurations with the same centroid and different members, and the named-member probe has the laws $\mathcal{N}(x_i, 1) \neq \mathcal{N}(x_i', 1)$ there; since every within-block difference relaxes to a value independent of the handoff state, the separation decays like e^{-3t}, and translations along the fibers are not symmetries of the fine field, since they do not commute with that relaxation. The full flow has events at every point of U for every $\delta \in \mathsf{C}$ and the nulls have none, which is (F5) in the strong grade. The parent centroid distance relaxes at rate three to $53/120 < \delta$, so the parents unite under the descended field with the first-stage selector, which is (F6a).

For (iii), every state of $\mathcal{C}$ is in the domain of the selector, so its parent distance exceeds δ, and the parent distance obeys the same law whatever the fine state inside the blocks; it decreases to $53/120$ and reaches δ from above. □

The calibration separates the clauses. With a different evaluator for each agent the anchors supply the membership order: the full-flow order is the centroid dendrogram of the anchors, as in proposition 6.30, but the null factor is one, so (F5) holds against both nulls, even in the strong grade. The frozen-environment control shows that the selection already follows from each agent's response to the initial laws of the others. What fails is (F6b): the second stage forms from every handoff whatever the first stage produced. Reading (F6b) with a class that contained parents within δ would have reversed this verdict. The example concerns a declared restricted-family flow; the full Fisher flow of eq. (4.4) does not preserve the Gaussian family (proposition 4.11).

The finite construction of this chapter settles inheritance for a supplied hierarchy and makes no selection claim, and chapter 11 gives exact inheritance through a supplied hierarchy without addressing (F2), (F5), or (F6). Each later chapter states its own verdict by the report above; the following summary records what the repaired clauses already decide. Chapter 7 gives explicit fusion events under a nonsmooth interaction. Its fused children are identical, so (F4) fails, and whatever its (F5) verdict for its declared selector it is at most a two-stage selection with exact inheritance. The erasure construction of chapter 9 has one selection stage. Under the original smooth potential, chapter 12 selects two parents and then their union on a prepared common-precision family, with a partition sequence that does not depend on the coupling, a separate control stated there. Its self costs are nonzero, so the two nulls differ: against the label-preserving null the agent-dependent self weights eq. (6.48) change the membership history on a nonempty open set already at zero coupling, while against the renormalized null the partition sequence is reproduced. Its (F5) verdict is therefore split and the clause fails; since the union forms from every counterfactual handoff in the domain of its selector, (F6b) fails as well. Chapter 13 gives a six-agent construction on a full open law neighborhood with native preferential selection; its theorem establishes neither (F5) nor (F6b), integrations of the limit field indicate that the null flow forms the same blocks, and it is not shown to be a recursive formation. The per-agent constructions of chapter 14 and chapter 15 are one-stage results whose erased phase, respectively common rotation, is gauge relative to every algebra declared there, so their retentions fail (F4) as declared here. All the smooth constructions supply their compression architectures.

Open problem 6.38 (A recursive formation). The normalized per-agent construction of chapter 14 supplies a complementary one-stage result: grouping with conflicting records under its declared

selector constant, reported there over the constants for which it holds, with a parent that preserves its declared port dynamics, finite adaptive histories, and nonflat geometry on a finite window. That parent quotients a hidden phase that is gauge relative to every algebra the chapter declares and is distinguished only by a child-level law diagnostic outside its probe rule, so it fails (F4). The phase interface is predeclared and the parent retains primitive ports. Joining already canonical parents loses no new coordinate, so neither a second compressive handoff nor (F6b) follows.

Construct a population dynamics under the reciprocal potential, the per-agent free energies, or a global free energy that satisfies (F1) to (F6), preferably in the strong and amplifying grades, and derive rather than declare its selector, in the sense that the selector family has no threshold, window, or other constant beyond those of the constitutive data. Determine also whether type closure and an action of the parent on its children should be required, and construct a formation that has them.

Chapter 7

Finite Fusion with Exact Statistical Inheritance

This chapter constructs a population whose dynamics makes agents coincide at finite times and whose coincident blocks carry exact inherited laws, evaluator, metric, and responses. The dynamics is a nonsmooth modification of the reciprocal law potential of definition 4.2, made by four constitutive changes stated in section 7.3: each peer divergence is replaced by the square root of twice a divergence, the receiver's law is restored to unit log variance before it is compared, the peer term carries a multiplicity-weighted coefficient, and source rows are fixed. The laws are restricted to a lognormal family with fixed variances. The potential is not, on any open set of means that meets a coincidence hyperplane $\{\mu_i^\alpha = \mu_j^\alpha\}$, the free energy of a normalized model whose log density is integrable against these laws: in logarithmic coordinates such a free energy is, up to a constant, the convolution of the negative log density with a fixed Gaussian kernel, hence real analytic in the means, whereas the potential has a corner on every coincidence hyperplane. Its conclusions do not transfer to the smooth potential or to the per-agent and global free energies of chapter 4.

Pairwise fusion penalties occur in convex clustering [108], and suitable nonsmooth consensus protocols produce finite-time agreement [44]. On the restricted family the square-root interaction is exactly the absolute-value kernel. The population is a weighted particle solution of the two-dimensional aggregation equation with kernel $\lambda\|x\|_1$ in a quadratic external potential; on the common-offset family of theorem 7.5, after a rescaling of positions and a change of time, each sector is the one-dimensional aggregation equation with kernel $\lambda|z|$, whose sticky collisions and finite-time total collapse are known [30, 37]. What the construction adds is the inheritance bookkeeping: represented laws, model evaluator, transport, parent metric, and boundary kernels descend exactly through each collision. Its dynamical outcome is a finite coalescence tree that ends in global consensus for every initial datum (proposition 7.3); persistent heterogeneous organization does not occur. On the common-offset family the tree is a linkage dendrogram of the initial data that does not depend on the coupling strength (proposition 7.6). Section 7.7 reports the construction against the formation standard of definition 6.36.

7.1 A law family closed under geometric averaging

Work over one context, with state and model spaces $\mathsf{K} = \mathsf{M} = (0, \infty)$. Write $\mathrm{LN}(\mu, \sigma^2)$ for the law of e^Z when $Z \sim N(\mu, \sigma^2)$. An agent of integer multiplicity $n \geq 1$ carries the product law

$$q_n = \mathrm{LN}(\mu^q, 1/n), \qquad s_n = \mathrm{LN}(\mu^s, 1/n), \qquad \zeta_n = q_n \otimes s_n. \tag{7.1}$$

Multiplicity is retained state information. The variances are fixed within each type; only the two log means evolve. This is a constraint on the dynamics, not an invariant family of a larger flow. If the common log variance v were released, the local divergence of proposition 7.1 below would have derivative $n(a-1)$ in v at $v = 1/n$, and the peer comparison between restored messages would have a nonzero source-variance derivative whenever the means differ. The Fisher flow with free variances therefore leaves the family, and releasing the variance destroys the encoding of multiplicity in the law. Laws are first written in a common reference coordinate. A fixed positive frame $U_i^q = U_i^s =: U_i \in \mathrm{GL}^+(1)$ represents both sectors locally by multiplication. Frames do not evolve in this example.

Choose constants $a > c > 0$ and define the normalized reference and evaluator

$$r_n = \mathrm{LN}\left(0, \frac{a}{n(a^2-c^2)}\right), \qquad G_n(m, dk) = \mathrm{LN}\left(\frac{c}{a}\log m, \frac{1}{an}\right)(dk). \tag{7.2}$$

The two occurrences of a lognormal expression in this equation denote ordinary probability laws and kernels. In logarithmic coordinates the joint reference $r_n(dm)G_n(m, dk)$ has precision

$$nA, \qquad A = \begin{pmatrix} a & -c \\ -c & a \end{pmatrix}. \tag{7.3}$$

Its eigenvalues $n(a-c)$ and $n(a+c)$ are positive. The predictive law is

$$s_n G_n = \mathrm{LN}\left(\frac{c}{a}\mu^s, \frac{(c/a)^2 + 1/a}{n}\right). \tag{7.4}$$

Thus the evaluator depends on the represented model value. Physical references and evaluators are obtained by pushing both coordinates through the common local frame.

Proposition 7.1 (Local energy and metric). For every $n \geq 1$ and every pair of real means, let

$$C = a - 1 - \frac{1}{2}\log(a^2 - c^2), \qquad \mathcal{U}(\mu^q, \mu^s) = \frac{a}{2}\big((\mu^q)^2 + (\mu^s)^2\big) - c\mu^q\mu^s. \tag{7.5}$$

Then

$$E_{0,n} = D_{\mathrm{KL}}(q_n \otimes s_n \| r_n G_n) = C + n\mathcal{U}, \qquad g_n = n\big((d\mu^q)^2 + (d\mu^s)^2\big). \tag{7.6}$$

Proof. The logarithm is a measurable bijection, so it preserves KL divergence. The recognition law in log coordinates has covariance I_2/n, while the reference has covariance A^{-1}/n. The Gaussian KL formula gives

$$E_{0,n} = \frac{1}{2}\left[\mathrm{tr}\, A - 2 - \log\det A + n(\mu^q, \mu^s)A(\mu^q, \mu^s)^{\mathsf{T}}\right], \tag{7.7}$$

which is the stated expression. The score for either mean is $n(Z - \mu)$, whose squared expectation is n. The product law has zero cross score expectation, giving the displayed Fisher metric. □

For independent synchronized children of multiplicities n_j, set $n_I = \sum_{j\in I} n_j$. Their retained reference samples are

$$k_I = \prod_{j\in I} k_j^{n_j/n_I}, \qquad m_I = \prod_{j\in I} m_j^{n_j/n_I}. \tag{7.8}$$

Weighted log averages have variance $\sum_j (n_j/n_I)^2/n_j = 1/n_I$. The same calculation for the joint reference gives $r_{n_I} G_{n_I}$. Conditional log observation noise has variance $\sum_j (n_j/n_I)^2/(an_j) = 1/(an_I)$, so the evaluator itself is inherited. Physical samples are first returned through U_j^{-1} and the retained sample is expressed in a chosen parent frame U_I.

7.2 Messages and state-dependent scalar links

Define a normalized message kernel by

$$B_n(k, dw) = \mathrm{LN}(\log k, 1 - 1/n)(dw), \tag{7.9}$$

interpreting $B_1(k, dw) = \delta_k(dw)$. It restores unit log variance: $q_n B_n = \mathrm{LN}(\mu^q, 1)$ and $s_n B_n = \mathrm{LN}(\mu^s, 1)$. These messages give all multiplicities a common comparison family. The kernel has a statistical meaning. For n independent unit children with log samples $N(\mu, 1)$, the mean log sample is sufficient for μ and has the law of the parent's log sample; the conditional law of one child's log sample given that mean is $N(\bar{Z}, 1 - 1/n)$, which does not depend on μ. Thus B_n is the conditional law of one child sample given the parent sufficient statistic [9], and this parameter independence is why fresh messages descend exactly.

Model prediction also commutes with this interface. Put $h_G = (c/a)^2 + 1/a$ and define $B_n^{h_G}(k, dw) = \mathrm{LN}(\log k, h_G(1 - 1/n))(dw)$, again using the identity at $n = 1$. Both routes in the kernel identity

$$G_n B_n^{h_G} = B_n G_1 \tag{7.10}$$

have conditional log mean $(c/a) \log m$ and variance $h_G - (c/a)^2/n$. Integrating against s_n gives

$$(s_n B_n) G_1 = (s_n G_n) B_n^{h_G} = \mathrm{LN}\left(\frac{c}{a}\mu^s, h_G\right). \tag{7.11}$$

Thus a parent can emit the original predictive message by either normalized route.

For each sector $\alpha \in \{q, s\}$, choose a fixed $0 < \chi_\alpha < 1$ and set

$$L_{ij}^\alpha = \chi_\alpha(\mu_i^\alpha - \mu_j^\alpha), \qquad \Theta_{ij}^\alpha = U_i \exp(L_{ij}^\alpha) U_j^{-1}. \tag{7.12}$$

Under a local change of frame $U_i \mapsto h_i U_i$, the physical transport changes to $h_i \Theta_{ij}^\alpha h_j^{-1}$. The reference means and L_{ij}^α remain unchanged. These links are pure gauge. With $V_i^\alpha = U_i \exp(\chi_\alpha \mu_i^\alpha)$ one has $\Theta_{ij}^\alpha = V_i^\alpha (V_j^\alpha)^{-1}$, so every cycle product is trivial and $\Theta_{ji}^\alpha = (\Theta_{ij}^\alpha)^{-1}$: the transport is a law-dependent change of frame of the trivial connection.

Transport of the reference message from j shifts its log mean to $\mu_j^\alpha + \chi_\alpha(\mu_i^\alpha - \mu_j^\alpha)$. Consequently the directed comparison cost between restored messages

$$d_{ij}^\alpha = \frac{\sqrt{2 D_{\mathrm{KL}}\big(\mathrm{LN}(\mu_i^\alpha, 1) \| (\exp L_{ij}^\alpha)_{\#} \mathrm{LN}(\mu_j^\alpha, 1)\big)}}{1 - \chi_\alpha} = |\mu_i^\alpha - \mu_j^\alpha| \tag{7.13}$$

is gauge invariant. The normalized cost does not depend on χ_α, so the potential and the flow below are the same for every $\chi_\alpha \in (0, 1)$, including the limit of identity transport. The transport therefore

carries no gauge content beyond passive covariance in this example; a transport whose parameter survives the normalization would need, for example, a noncommuting generator on a vector log space. The equality includes the dependence of the link on both endpoint laws. For unequal means, differentiating its log-mean residual gives

$$\partial_{\mu_i^\alpha}\left[\mu_i^\alpha - \mu_j^\alpha - L_{ij}^\alpha\right] = 1 - \chi_\alpha. \tag{7.14}$$

This factor cancels the denominator in eq. (7.13). Holding the link fixed while differentiating would replace the coupling λ below by $\lambda/(1-\chi_\alpha)$.

7.3 The continuous subgradient dynamics

For multiplicities n_i with total $N = \sum_i n_i$, choose $\lambda > 0$ and the potential

$$\mathcal{F}(\boldsymbol{\mu}) = \sum_i n_i\left[C + \mathcal{U}(\mu_i^q, \mu_i^s)\right] + \lambda \sum_{i<j} n_i n_j\left(|\mu_i^q - \mu_j^q| + |\mu_i^s - \mu_j^s|\right). \tag{7.15}$$

All contacts are fixed and present. If source rows are recorded, take $\beta_{ij} = \gamma_{ij} = n_j/N$, including self entries. Summing child entries gives exactly the same parent row. Self comparison costs vanish, while their row mass records the corresponding contact multiplicity. There is no Gibbs row adaptation. Relative to definition 4.2, the local term is $\sum_i E_{0,n_i}$ up to the constant $\sum_i (n_i - 1)C$. The peer term $\sum_j \beta_{ij} D_{ij}^\alpha$ of agent i is modified in four constitutive ways. The divergence is replaced by its normalized square root d_{ij}^α of eq. (7.13); the receiver's law q_i is replaced by its restored message $q_i B_{n_i} = \mathrm{LN}(\mu_i^\alpha, 1)$ before the comparison; the term is multiplied by $\frac{1}{2}\lambda N n_i$, so that it becomes $\frac{1}{2}\lambda N n_i \sum_j \beta_{ij} d_{ij}^\alpha$; and the rows are fixed. The restoration matters for $n_i \geq 2$: with the unrestored receiver law $\mathrm{LN}(\mu_i^\alpha, 1/n_i)$ one has $2D_{ij}^\alpha = (1/n_i - 1 + \log n_i) + (1-\chi_\alpha)^2(\mu_i^\alpha - \mu_j^\alpha)^2$, whose constant is positive, so its square root would be smooth at coincidence and would depend on χ_α, and parents of multiplicity at least two could not stick. The local term has the stated KL origin; the flow below is a declared gradient flow, not inference in a normalized model, and no fixed sample-model ELBO identity is asserted for the entire potential.

With $W = \mathrm{diag}(n_1, n_1, n_2, n_2, \ldots)$, define the flow on $\mathbb{R}^{2r}$, for r current agents, by

$$W\dot{\boldsymbol{\mu}} \in -\partial\mathcal{F}(\boldsymbol{\mu}), \tag{7.16}$$

where $\partial\mathcal{F}$ is the subdifferential of $\mathcal{F}$ as a function of the whole configuration. Solutions are absolutely continuous in the external time $t \geq 0$ and satisfy this inclusion almost everywhere. There are no boundaries or stochastic driving terms. Both law sectors have the same basic mobility; the factor $1/n_i$ comes from their inherited Fisher metric. In coordinates,

$$\dot{\mu}_i^q = -a\mu_i^q + c\mu_i^s - \lambda \sum_{j\neq i} n_j \epsilon_{ij}^q, \tag{7.17}$$

$$\dot{\mu}_i^s = -a\mu_i^s + c\mu_i^q - \lambda \sum_{j\neq i} n_j \epsilon_{ij}^s, \tag{7.18}$$

where $\epsilon_{ij}^{\alpha} = -\epsilon_{ji}^{\alpha}$ belongs to $\mathrm{Sign}(\mu_i^{\alpha} - \mu_j^{\alpha})$ and $\mathrm{Sign}(0) = [-1, 1]$. Every pair uses one antisymmetric selection; letting each agent choose its own element of $\mathrm{Sign}(0)$, as a product of partial subdifferentials would, admits spurious solutions, for example two coincident unit agents that move together under any constant selection, and is not the Filippov regularization. The multiplicity cancels from an agent's own velocity, which depends only on the multiplicities of the others. Each agent performs overdamped planar motion, in the coordinates (μ_i^q, μ_i^s), in a harmonic trap under the pair potential $n_i n_j \|x_i - x_j\|_1$, so a multiplicity acts as a friction weight and as an interaction charge; it is not an inertia.

Lemma 7.2 (Global well-posedness, contraction, and persistence). For every multiplicity vector and every initial datum, eq. (7.16) has a unique absolutely continuous solution on $[0, \infty)$. Two solutions with the same multiplicities obey

$$\|\boldsymbol{\mu}(t) - \widetilde{\boldsymbol{\mu}}(t)\|_W \le e^{-(a-c)t} \|\boldsymbol{\mu}(0) - \widetilde{\boldsymbol{\mu}}(0)\|_W. \tag{7.19}$$

Synchronized unit-multiplicity children remain synchronized in both sectors.

Proof. The potential $\mathcal{F}$ is convex and finite on $\mathbb{R}^{2r}$, so $W^{-1}\partial\mathcal{F}$ is a maximal monotone operator in the W inner product. The theory of maximal monotone operators gives a unique global absolutely continuous solution for every initial datum, whose right derivative is the element of $-W^{-1}\partial\mathcal{F}(\boldsymbol{\mu})$ of minimal norm in the W norm [31]; the Euclidean minimal element would in general not stick coincident agents of unequal multiplicities. The quadratic part is $(a-c)$-strongly convex in the W metric and the absolute-value part is convex. For subgradients $\xi, \widetilde{\xi}$ at two configurations, monotonicity gives

$$\langle \xi - \widetilde{\xi}, \boldsymbol{\mu} - \widetilde{\boldsymbol{\mu}} \rangle \ge (a-c) \|\boldsymbol{\mu} - \widetilde{\boldsymbol{\mu}}\|_W^2. \tag{7.20}$$

Substitution from eq. (7.16) and integration proves eq. (7.19). Swapping two equal unit-multiplicity children preserves the initial configuration, potential, and metric. Uniqueness therefore preserves their equality. This argument concerns the full two-sector system, not a restricted scalar equation. □

Because the subdifferential of a convex function is its Clarke generalized gradient, eq. (7.16) coincides with the Filippov regularization of the almost-everywhere gradient field $-W^{-1}\nabla\mathcal{F}$ [43, 75, 155], so no ambiguity of solution concept arises. Backward uniqueness fails: the flow is a contraction semigroup and not a group, which is what permits finite-time sticking.

Proposition 7.3 (Complete synchronization in finite time). Fix any multiplicity vector and any initial datum. Put $R_0 = \|\boldsymbol{\mu}(0)\|_W$, $\kappa = a - c$, and let $D^{\alpha} = \max_i \mu_i^{\alpha} - \min_i \mu_i^{\alpha}$ be the spread of sector α. Then $\max_{i,\alpha} |\mu_i^{\alpha}(t)| \le e^{-\kappa t} R_0$, and there is a time

$$T_* \le \frac{1}{\kappa} \log\left(1 + \frac{\kappa \max_{\alpha} D^{\alpha}(0)}{\lambda N}\right) \tag{7.21}$$

such that all agents have equal means in both sectors for every $t \ge T_*$. Every solution therefore produces a finite sequence of synchronization events, at most $r - 1$ of them, ending in a single block.

Proof. The zero configuration is stationary, because $0 \in \partial\mathcal{F}(0)$. Equation (7.19) with this stationary solution gives $\|\boldsymbol{\mu}(t)\|_W \le e^{-\kappa t}R_0$, and $|\mu_i^\alpha| \le \|\boldsymbol{\mu}\|_W$ because $n_i \ge 1$. Fix a sector α and write α' for the other sector. The spread D^α is absolutely continuous. At almost every t all coordinates and both extrema are differentiable. At such a time every maximizer i has $\dot\mu_i^\alpha$ equal to the derivative of the maximum, because the maximum minus μ_i^α is nonnegative and vanishes at t. Let T_+ be the set of maximizers, with multiplicity n_+, and average the coordinate equation over T_+ with weights n_i/n_+. Antisymmetry of ϵ_{ij}^α cancels the internal terms, and $\epsilon_{ij}^\alpha = 1$ for $j \notin T_+$. Hence

$$\frac{d}{dt}\max_i \mu_i^\alpha = -a\max_i \mu_i^\alpha + c\overline{\mu^{\alpha'}}_{T_+} - \lambda(N - n_+), \tag{7.22}$$

where $\overline{\mu^{\alpha'}}_{T_+}$ is the weighted average over T_+. The same argument for the minimizing set T_-, with multiplicity n_-, gives the opposite sign of the last term. While $D^\alpha > 0$ the sets T_+ and T_- are disjoint, so $n_+ + n_- \le N$, and both weighted averages of $\mu^{\alpha'}$ lie in the range of the α' means. Therefore

$$\dot D^\alpha \le -aD^\alpha + cD^{\alpha'} - \lambda N \quad \text{almost everywhere on } \{D^\alpha > 0\}. \tag{7.23}$$

Let $M = \max(D^q, D^s)$. At almost every t where $M > 0$, the derivative of M equals $\dot D^\alpha$ for a sector α with $D^\alpha = M$ (where $D^q = D^s$ the two derivatives agree almost everywhere), so $\dot M \le -\kappa M - \lambda N$ almost everywhere on $\{M > 0\}$. Hence $e^{\kappa t}M(t) \le M(0) - \lambda N(e^{\kappa t} - 1)/\kappa$ as long as M stays positive, so M vanishes by the time eq. (7.21). Agents that coincide in both sectors remain coincident: expanding integer multiplicities into unit children and applying lemma 7.2 proves this, so M stays zero afterward. The number of distinct agents therefore decreases at each event. □

The bound depends only on the initial spreads; it does not involve R_0 or c, and it is attained by two blocks on the common-offset family below. The common-offset family studied next provides explicit event times; it is not needed for finite-time synchronization. The following estimate controls a single pair off that family.

Lemma 7.4 (Isolated pairs synchronize). Let $i \ne j$ and let $\mathcal{T}$ be a time interval on which, in each sector α, no third agent has a mean in the closed interval between μ_i^α and μ_j^α. Put $S = |\mu_i^q - \mu_j^q| + |\mu_i^s - \mu_j^s|$. Then, for $s < t$ in $\mathcal{T}$,

$$S(t) \le \max\{0, S(s) - \lambda(n_i + n_j)(t - s)\}. \tag{7.24}$$

Proof. For a third agent k, the hypothesis gives $\epsilon_{ik}^\alpha = \epsilon_{jk}^\alpha$, so its terms cancel from the equation for $\Delta^\alpha = \mu_i^\alpha - \mu_j^\alpha$, leaving $\dot\Delta^\alpha = -a\Delta^\alpha + c\Delta^{\alpha'} - \lambda(n_i + n_j)\epsilon_{ij}^\alpha$. Where $\Delta^\alpha \ne 0$, $\frac{d}{dt}|\Delta^\alpha| \le -a|\Delta^\alpha| + c|\Delta^{\alpha'}| - \lambda(n_i + n_j)$; where $\Delta^\alpha = 0$ on a set of positive measure, the derivative of $|\Delta^\alpha|$ vanishes almost everywhere. Adding the two sectors gives $\dot S \le -\kappa S - 2\lambda(n_i + n_j)$ when both differences are nonzero, and $\dot S \le -\lambda(n_i + n_j)$ when exactly one is nonzero. Hence $\dot S \le -\lambda(n_i + n_j)$ almost everywhere on $\{S > 0\}$, which proves eq. (7.24). □

7.4 An invariant class with explicit global solutions

Theorem 7.5 (Finite sticking on a common-offset family). Let all initial multiplicities be one and suppose

$$\mu_i^q(0) = u_i^0 + v_0, \qquad \mu_i^s(0) = u_i^0 - v_0, \tag{7.25}$$

where the u_i^0 are distinct real numbers and $v_0 \neq 0$. The unique global solution of the full inclusion for these initial data has, in both sectors, the form

$$\mu_i^q = u_i + v, \qquad \mu_i^s = u_i - v, \qquad v(t) = v_0 e^{-(a+c)t}. \tag{7.26}$$

Blocks formed by equal u_i persist and follow the multiplicity-weighted version of the same equation.

Proof. Set $\kappa = a - c > 0$. Equal sector orderings, with the same antisymmetric selection $\epsilon_{ij} = \epsilon_{ij}^q = \epsilon_{ij}^s$ in both sectors, reduce eq. (7.18) to

$$\dot{u}_i = -\kappa u_i - \lambda \sum_{j \neq i} n_j \epsilon_{ij}, \qquad \epsilon_{ij} = -\epsilon_{ji} \in \operatorname{Sign}(u_i - u_j), \qquad \dot{v} = -(a+c)v. \tag{7.27}$$

Introduce the increasing auxiliary time and rescaled positions

$$\tau(t) = \frac{e^{\kappa t} - 1}{\kappa}, \qquad u_i(t) = e^{-\kappa t} z_i(\tau(t)). \tag{7.28}$$

Since $\dot{\tau} = e^{\kappa t} > 0$, direct differentiation gives

$$z_i' = -\lambda \sum_{j \neq i} n_j \epsilon_{ij}, \qquad \epsilon_{ij} = -\epsilon_{ji} \in \operatorname{Sign}(z_i - z_j). \tag{7.29}$$

This is the sticky-particle solution of the one-dimensional aggregation equation with kernel $\lambda|z|$ on weighted atoms [30, 37]. An ordered block has constant velocity $\lambda(n_{\text{right}} - n_{\text{left}})$ between collisions, where n_{right} and n_{left} are the total multiplicities to its right and left. Adjacent blocks approach with relative speed $\lambda(n_I + n_J) > 0$. At their collision, join their multiplicities, retain the common position, and use the new velocity $\lambda(n_{\text{right}} - n_{\text{left}})$. This equals the multiplicity-weighted average of their velocities just before collision. Choosing zero internal subgradients verifies the original inclusion for every constituent child after collision. Each event reduces the number of blocks, so at most $N - 1$ events occur, and the resulting piecewise-linear curve exists for all $\tau \geq 0$. Reversing eq. (7.28) constructs a global full-system solution. Lemma 7.2 makes it the solution. □

The class eq. (7.25) has a common difference between its two sector means. This is an initial-data restriction, not an imposed constraint force. It is an invariant affine subspace of codimension $r - 1$ in $\mathbb{R}^{2r}$. The two laws of each agent remain distinct at every finite time because $v(t) \neq 0$. The passage to (z, τ) is a rescaling of positions by the common positive factor $e^{\kappa t}$ combined with a monotone change of time; it preserves coincidence, and τ is not a geometric or operational clock.

The merge order on this family has a closed form. For a set G of consecutive sorted agents, write n_G for its multiplicity and c_G for the multiplicity-weighted mean of the initial offsets u_i^0, $i \in G$. The normalized-centroid linkage of the initial offsets starts from the singletons and repeatedly merges

the pair of adjacent current blocks $G < H$ that minimizes

$$\varrho_{GH} = \frac{c_H - c_G}{n_G + n_H}. \tag{7.30}$$

Proposition 7.6 (The coalescence tree is a coupling-independent linkage of the initial data). In theorem 7.5, allowing arbitrary initial multiplicities, the membership history consists exactly of the merges of the normalized-centroid linkage of the initial offsets, in its order, and the k-th merge, of height ϱ_k, occurs at

$$\tau_k = \frac{\varrho_k}{\lambda}, \qquad t_k = \frac{1}{\kappa} \log\left(1 + \frac{\kappa \varrho_k}{\lambda}\right). \tag{7.31}$$

The heights are nondecreasing. Consequently the partition sequence is the same for every $\lambda > 0$ and every $a > c > 0$; only the event times depend on these constants.

Proof. Work in the z frame of eq. (7.29), where the order of the agents is preserved and contact is sticky. Let G be a current block at auxiliary time τ. No agent outside G has touched a member of G before τ, since by stickiness it would then belong to G. Hence, at every earlier time, the signs between members of G and outsiders are single valued and constant, the internal terms cancel by antisymmetry, and $\frac{d}{d\tau} \sum_{i \in G} n_i z_i = \lambda n_G (n_{\mathrm{right}}(G) - n_{\mathrm{left}}(G))$ from $\tau = 0$. For adjacent current blocks $G < H$ one has $n_{\mathrm{left}}(H) = n_{\mathrm{left}}(G) + n_G$ and $n_{\mathrm{right}}(G) = n_{\mathrm{right}}(H) + n_H$, so the distance of their weighted centroids, which is their gap, equals $(c_H - c_G) - \lambda(n_G + n_H)\tau$. They therefore meet at $\tau = \varrho_{GH}/\lambda$ unless an earlier event involves one of them, and the next event is a merge of the adjacent current pair with the smallest ϱ_{GH}. Induction on the number of blocks proves the first claim. When G and H merge into K at τ_k, the gap between K and a neighbor L is still nonnegative at τ_k, so $\varrho_{KL} \geq \lambda \tau_k = \varrho_k$, which gives monotonicity. Inverting eq. (7.28) gives t_k. The heights are computed from the initial offsets and multiplicities alone. □

7.5 Two fusion stages without a supplied partition

Theorem 7.7 (Four agents fuse into two parents and then one). In theorem 7.5, take four agents with the unordered initial set

$$\{u_i^0\} = \{-3d, -2d, 2d, 3d\}, \qquad d > 0. \tag{7.32}$$

Two size-two blocks form at t_1 and remain distinct until $t_2 > t_1$, where they fuse under their inherited equations, with

$$t_1 = \frac{1}{\kappa} \log\left(1 + \frac{\kappa d}{2\lambda}\right), \qquad t_2 = \frac{1}{\kappa} \log\left(1 + \frac{5\kappa d}{4\lambda}\right). \tag{7.33}$$

Proof. The adjacent heights eq. (7.30) of the singletons are $d/2$, $2d$, and $d/2$, so the two outer pairs merge first, at height $d/2$. The two resulting blocks have centroids $\mp 5d/2$ and multiplicity two, so their height is $5d/4$. Proposition 7.6 gives eq. (7.33). Equivalently, in sorted order the initial z velocities are $(3\lambda, \lambda, -\lambda, -3\lambda)$, the outer gaps close at $\tau_1 = d/(2\lambda)$ at the parent positions $\mp 3d/2$, and the parents, with velocities $\pm 2\lambda$, meet after another $3d/(4\lambda)$. □

Exact reflection symmetry is unnecessary. For four sorted initial positions, let their adjacent gaps be $g_1, g_2, g_3 > 0$, and put $H = \max(g_1, g_3)$. The two outer pairs form first and then fuse if and only if

$$\Gamma = g_2 + \frac{g_1 + g_3}{2} - 2H > 0. \tag{7.34}$$

Suppose $g_1 \le g_3$; the other case is the reflected calculation. Before the first outer collision the middle gap decreases at speed 2λ, so at $g_1/(2\lambda)$ it equals $g_2 - g_1$; afterward it decreases at speed 3λ until the other outer pair collides at $g_3/(2\lambda)$, where the same linear expression has the value Γ. If $\Gamma > 0$, then $g_2 > H$, the middle gap stays positive until both outer pairs have formed by $\tau_* = H/(2\lambda)$, and the two parent gap then decreases at speed 4λ, so the final time is

$$\tau_{\mathrm{final}} = \tau_* + \frac{\Gamma}{4\lambda} = \frac{2g_2 + g_1 + g_3}{8\lambda}. \tag{7.35}$$

If $\Gamma \le 0$, the middle gap closes no later than the second outer collision, and the pattern fails; at $\Gamma = 0$ all four agents merge in one event, as three current blocks if $g_1 \neq g_3$ and as four singletons if $g_1 = g_3$. In the language of proposition 7.6, $\Gamma > 0$ is the statement that the normalized-centroid linkage merges the two outer pairs before either joins the middle. Condition eq. (7.34) is strict and defines an open set relative to the ordered common-offset family. Taking all label permutations gives an admitted class with no assigned labels for blocks. The initial laws are unequal across agents, and the interaction rule contains no partition. Membership is selected by attained equality.

Corollary 7.8 (The two-stage pattern is open in the full space). Let $\bar{x} \in \mathbb{R}^8$ be a four-agent configuration of unit multiplicities in the common-offset class satisfying eq. (7.34), and let $\mathcal{O}_1$ and $\mathcal{O}_2$ be its two outer pairs. There is $\varepsilon > 0$ such that every initial configuration within distance ε of $\bar{x}$ has the following membership history: $\mathcal{O}_1$ and $\mathcal{O}_2$ synchronize in both sectors, in some order and before any other synchronization, and the two resulting parents then synchronize. The set of initial data with this history therefore contains an open subset of $\mathbb{R}^8$ that contains the open-gap set. Explicitly, with the gaps ordered so that $g_1 \le g_3$ and t_{f} the final time of $\bar{x}$, every $\varepsilon < \Gamma e^{-\kappa t_{\mathrm{f}}}/16$ suffices, and no radius larger than $(2\sqrt{2}/5)\Gamma$ does.

Proof. Let $\bar{\boldsymbol{\mu}}$ be the solution from $\bar{x}$, let t_* be its second outer collision time and $t_{\mathrm{f}} > t_*$ its final time, and choose $\eta > 0$ with $t_* + \eta < t_{\mathrm{f}}$. On $[0, t_* + \eta]$ every mean of $\mathcal{O}_1$ lies below every mean of $\mathcal{O}_2$ in each sector, with a margin $\sigma_0 > 0$. By eq. (7.19), a solution from a configuration within ε of $\bar{x}$ stays within ε of $\bar{\boldsymbol{\mu}}$ in every coordinate. If $2\varepsilon < \sigma_0$, both pairs are isolated in the sense of lemma 7.4 on $[0, t_* + \eta]$, and no agent of $\mathcal{O}_1$ meets an agent of $\mathcal{O}_2$ in either sector there. At the collision time of $\mathcal{O}_1$ along $\bar{\boldsymbol{\mu}}$, the perturbed pair has $S \le 4\varepsilon$, so eq. (7.24) synchronizes it within $2\varepsilon/\lambda$; the same holds for $\mathcal{O}_2$. If also $2\varepsilon/\lambda \le \eta$, both pairs synchronize before $t_* + \eta$ and no other event occurs before then. Lemma 7.2 keeps each pair synchronized, and proposition 7.3 synchronizes the two parents at a finite later time.

For the explicit radius, choose η by $\tau(t_* + \eta) = \tau_* + \Gamma/(8\lambda)$, which is below τ_{final}. Along $\bar{\boldsymbol{\mu}}$ the gap between the two pairs is $e^{-\kappa t}$ times the z gap, a product of two decreasing positive factors, and at $\tau_* + \Gamma/(8\lambda)$ the z gap is $\Gamma/2$; hence $\sigma_0 \ge e^{-\kappa t_{\mathrm{f}}}\Gamma/2$. Since $\eta = \kappa^{-1}\log\bigl(1 + \kappa\Gamma/(8\lambda(1 + \kappa\tau_*))\bigr)$ and $\log(1 + y) \ge y/(1 + y)$, one has $\eta \ge \Gamma/(8\lambda(1 + \kappa\tau_{\mathrm{final}})) = \Gamma e^{-\kappa t_{\mathrm{f}}}/(8\lambda)$. Both requirements hold when $\varepsilon < \Gamma e^{-\kappa t_{\mathrm{f}}}/16$. Conversely, lowering the third sorted offset by δ in both sectors changes

(g_2, g_3) to $(g_2 - \delta, g_3 + \delta)$ and Γ to $\Gamma - 5\delta/2$, which vanishes at $\delta = 2\Gamma/5$; that configuration lies at distance $(2\sqrt{2}/5)\Gamma$ from $\bar{x}$ and has a four-agent merger. □

The admissible radius is therefore of order Γ, and the open set shrinks toward the boundary $\{\Gamma = 0\}$ of the open-gap set.

Corollary 7.9 (Coalescence of any finite depth). Place $N = 2^k$ unit agents in the common-offset class with sorted initial positions u^0, and assign to the gap between sorted agents p and $p+1$ the level of the smallest dyadic block of indices containing both. If, for each level $\ell < k$,

$$2^\ell \max\{\text{gaps of level } \ell\} < \min\{\text{gaps of level greater than } \ell\}, \tag{7.36}$$

then all dyadic blocks of size 2^ℓ form before any event that joins two of them, for $\ell = 1, \dots, k$, and each stage is inherited exactly. The condition is strict, so the k-stage history holds on an open subset of the common-offset class.

Proof. In the z frame, adjacent blocks with multiplicities n_I, n_J approach at speed $\lambda(n_I + n_J) \geq 2\lambda$, so a gap of initial size g between two sorted agents closes by the auxiliary time $g/(2\lambda)$. By eq. (7.36) at the lower levels, the maximal gap of level ℓ exceeds every gap of lower level, so every gap of level at most ℓ has closed by $\max\{\text{gaps of level } \ell\}/(2\lambda)$. Before the first closure of a gap of level greater than ℓ, every current block lies inside a dyadic block of size 2^ℓ, so adjacent blocks approach at speed at most $\lambda 2^{\ell+1}$, and no gap of level greater than ℓ closes before $\min\{\text{gaps of level greater than } \ell\}/(\lambda 2^{\ell+1})$, which by eq. (7.36) is later. Induction on ℓ proves the ordering. Event times are $t = \kappa^{-1}\log(1 + \kappa\tau)$, and inheritance at every stage is the handoff identity of the next section. □

The condition $\frac{N}{2}\max\{\text{gaps of level } \ell\} < \min\{\text{gaps of level } \ell + 1\}$ for every ℓ implies eq. (7.36), because $N/2 \geq 2^\ell$ and the inequalities chain across levels. Every such hierarchy ends in one block, by proposition 7.3.

7.6 The inherited handoff and its scope

At a collision, restrict a block I to its common means. Summing its local energies and pulling back its metric gives

$$\sum_{i \in I} n_i(C + \mathcal{U}) = n_I(C + \mathcal{U}) = E_{0,n_I} + (n_I - 1)C, \qquad g_I = n_I\big((d\mu_I^q)^2 + (d\mu_I^s)^2\big). \tag{7.37}$$

Every external pair term sums to $n_I n_J |\mu_I^\alpha - \mu_J^\alpha|$. Zero internal subgradients therefore lift each solution of the inherited equation to a fine solution. Uniqueness identifies that lift with the original trajectory. This also proves persistence for inherited multiplicities by expanding a multiplicity into synchronized unit children. With R retaining block means, multiplicities, and frames, the corresponding parameter flows satisfy the following identity. For a fixed formed partition, keep both component index sets fixed even when further components coincide; a later merger is a separate retention composed with R. Then

$$R \circ \Phi_t = \Phi_t^c \circ R \tag{7.38}$$

on every synchronized stratum reached by the construction, for all future times. Weighted geometric averaging is associative, so later collisions give the same retained laws as direct averaging. The state is continuous at a collision; the velocity changes through the inherited multiplicity sum.

The averaged dynamics alone does not single out synchronized blocks. Call a partition separated if, in each sector and for every pair of blocks, all means of one block lie strictly below all means of the other. Then every external sign is single valued and constant, the internal terms cancel by antisymmetry, and the weighted block averages obey the coarse equation until two blocks interleave, whether or not the blocks are synchronized. What holds only on synchronized strata is forward invariance of the retained structure and inheritance of the declared responses: for unsynchronized children the log samples given the retained log mean have a law that depends on the discarded deviations, so fresh child messages and the joint reconstruction below do not descend. The selected partition is therefore distinguished by response inheritance and invariance, not by the averaged dynamics.

Along every solution the convex chain rule for absolutely continuous curves [31] gives

$$\frac{d\mathcal{F}}{dt} = -\sum_i n_i\big((\dot{\mu}_i^q)^2 + (\dot{\mu}_i^s)^2\big) \quad \text{almost everywhere,} \tag{7.39}$$

including intervals on which pairs slide together in one sector only. The objective is continuous at collisions when the constants above are retained. By proposition 7.3, every solution converges to the stationary configuration with every mean zero, after complete synchronization at a finite time.

Remark 7.10 (A discrete update that keeps exact sticking). The natural time discretization with step $\Delta t > 0$ is the proximal step

$$\boldsymbol{\mu}^{k+1} = \operatorname*{argmin}_x \left\{ \mathcal{F}(x) + \frac{1}{2\Delta t}\|x - \boldsymbol{\mu}^k\|_W^2 \right\}, \tag{7.40}$$

whose minimizer is unique because the objective is strongly convex. Its optimality condition $W(\boldsymbol{\mu}^k - \boldsymbol{\mu}^{k+1})/\Delta t \in \partial\mathcal{F}(\boldsymbol{\mu}^{k+1})$ and strong monotonicity give the discrete contraction $\|\boldsymbol{\mu}^{k+1} - \widetilde{\boldsymbol{\mu}}^{k+1}\|_W \le (1+\kappa\Delta t)^{-1}\|\boldsymbol{\mu}^k - \widetilde{\boldsymbol{\mu}}^k\|_W$, and comparing the objective at its minimizer with its value at $\boldsymbol{\mu}^k$ gives $\mathcal{F}(\boldsymbol{\mu}^{k+1}) + (2\Delta t)^{-1}\|\boldsymbol{\mu}^{k+1} - \boldsymbol{\mu}^k\|_W^2 \le \mathcal{F}(\boldsymbol{\mu}^k)$. Uniqueness and the exchange symmetry of equal-multiplicity agents keep synchronized blocks synchronized, and the step creates exact ties: for two unit agents under the pair term alone, the step returns equal means exactly when the gap before the step is at most $2\lambda\Delta t$. The iterates converge to the solution of eq. (7.16) uniformly on bounded time intervals as $\Delta t \to 0$ [47]. An explicit Euler step with sign selections keeps ties that are already present but does not create exact ties generically; it oscillates around coincidence, so equality-based membership is only meaningful for the proximal scheme or for an event-driven integration of eq. (7.29).

Sample retention is genuinely noninjective: different positive sample vectors have the same geometric mean. For n synchronized unit children, their log samples conditional on the retained log mean $\bar{Z}$ have law

$$(Z_1, \dots, Z_n) \mid \bar{Z} \sim N\left(\bar{Z}\mathbf{1}, I_n - \frac{\mathbf{1}\mathbf{1}^{\mathsf{T}}}{n}\right). \tag{7.41}$$

The residual is independent of the mean: their joint Gaussian cross covariance is zero. Exponentiation gives a normalized joint reconstruction kernel. Its one-child marginal is eq. (7.9). Independent residuals for all children would be incorrect because the off-diagonal conditional covariances are $-1/n$. Generative joint samples similarly use the covariance of $(\log m, \log k)$ under $s_1 G_1$; their residual kernel is determined by that covariance, not by the recognition product law. The local KL residual $(n-1)C$ records the corresponding state-independent information loss under the reference comparison.

Remark 7.11 (Sufficiency beyond the lognormal family). The kernel eq. (7.41) is the conditional law of the children given their sufficient statistic, and it is free of the parameter for that reason. For an exponential family with sufficient statistic T, the same holds for independent children of equal parameter: the parent law is the law of $\sum_j T(X_j)$, the Fisher metric of the children restricted to equal parameters is multiplied by the multiplicity, and the parameter-free conditional law supplies both the message kernel and the joint reconstruction. Under fixed support and the usual regularity, a sufficient statistic of fixed dimension for every number of children exists only for exponential families, by the Pitman–Koopman–Darmois theorem, so this is the scope of the remark. For Gamma children of shape k, for example, one child given the sum S of n children is S times a $\mathrm{Beta}(k,(n-1)k)$ variable. Two ingredients are specific to Gaussian location families with fixed covariances. First, by the chain rule for relative entropy, $\sum_{i\in I} E_{0,n_i} - E_{0,n_I}$ equals the expected KL divergence between the parameter-free recognition conditional of the children given T and the conditional of the reference given T; for Gaussian location families both conditionals are Gaussian with the same mean and state-independent covariances, which gives the constant $(n_I-1)C$ above, and the residual vanishes when the joint reference belongs to an exponential family for which T is sufficient. In general it depends on the state, for example for Bernoulli children with an evaluator proportional to e^{Jkm}, and the restricted energy is then the parent's own KL plus a state-dependent term, so the parent is not an agent of the children's type in the sense of the local energy. Second, the square-root divergence between restored messages is the convex function $|\Delta\mu|$; on a general regular family it equals the Fisher distance only to leading order near the diagonal. Global uniqueness would then need semiconvexity of the square-root divergence along Fisher geodesics on a forward-invariant compact set; this fails near the boundary of the Bernoulli family, its constant grows with λ, and semiconvexity gives uniqueness but not the contraction used in proposition 7.3.

Parameter retention has a narrower meaning. On an exactly synchronized stratum, child means are redundant copies; retaining their common value does not identify different nonsynchronized configurations. Away from that stratum, the weighted average closes the dynamics only on separated partitions, as explained above. Equation (7.38) makes no claim about such hidden parameter fibers.

The admitted boundary protocol draws fresh messages through B_n, fresh predictive messages through eq. (7.11), or complete fresh represented samples through the joint reconstruction. To type an accessible stored readout, augment the complete population configuration by $o \in \mathsf{O} = (0,\infty)$ and set $\rho(\boldsymbol{\mu}, \boldsymbol{n}, o) = o$. At declared epochs, the transition Γ_{rd} advances the deterministic parameter flow and overwrites o using the selected normalized scalar output kernel. Between epochs, o is fixed, and it never enters the mean equation. Retention copies this accessible o exactly; it does not claim access to cached private residual samples. For vector readouts the same definition uses the

corresponding finite product space and joint kernel. Equal parameter trajectories and conditional output kernels imply equal joint accessible readout laws for every finite sequence of such queries, by multiplying successive conditional kernels. Reusing persistent latent samples or introducing record feedback requires another state specification.

7.7 Status against the formation standard

We report the construction by the scheme following definition 6.36. The objective is the nonsmooth modification eq. (7.15) of the reciprocal law potential; the dynamics is the subdifferential inclusion eq. (7.16), a declared restricted-family flow on the parameter space $\mathbb{R}^{2r}$ of log means, well posed by lemma 7.2. (F1) holds in the full grade: the admitted class is the open subset of $\mathbb{R}^8$ given by corollary 7.8, with all label permutations, and its trace on the common-offset family, an invariant family of codimension three, is open there. The selector is the partition of the current state into agents with equal means in both sectors. It has no real constant, so the declared family is a single selector; it reads laws only, it is a Borel function of the state, and its persistence criterion is the invariant block relation of lemma 7.2; (F2) holds. The parent protocol rule A_I consists of fresh messages through B_{n_I}, fresh predictive messages through eq. (7.11), complete fresh represented samples through the joint reconstruction, and the stored readout, each addressing the block as a whole. The child-level probe rule B_I consists of these protocols applied to individual members and the peer channels applied to each member, with response laws the restored messages of that member. On synchronized strata the laws, evaluator, metric, rows, contacts, and every protocol of A_I are inherited exactly, and the parent flow is the multiplicity-weighted inclusion, well posed by lemma 7.2, so (F3) holds; the geometric clause is not declared, since the links are pure gauge.

Clause (F4) fails. The reachable post-handoff set consists of configurations whose block members are identical copies, so the retention of common means and multiplicity is injective there, and no probe of B_I has a pair of distinct configurations to separate. Child frames are fixed passive data. The geometric mean is noninjective on sample vectors, but samples are not configurations, and the conditional law of the child samples given the retained sample is the parameter-free kernel eq. (7.41).

For the controls of definition 6.33, the peer costs are the pair terms of eq. (7.15), whose compared laws depend on another agent's law, and the self costs vanish identically. Deleting the peer costs, with or without their labels, gives the same flow, the case $\lambda = 0$, so the label-preserving and renormalized nulls coincide. Each agent then follows the linear equation $\dot{\mu}_i = -A\mu_i$, which is globally well posed and invertible, so distinct agents never coincide and P^0 contains no event. By proposition 7.3, every configuration with two distinct agents has at least one event along Φ, so $P \neq P^0$ at every point of the admitted class. For the declared selector (F5) therefore holds against both nulls, in the strong grade. This verdict measures the exactness of the selector rather than which blocks the interaction forms. No backward-unique null flow produces an attained equality, so every finite-time sticking interaction read by an exact-equality selector passes in this way. On the common-offset family the partition sequence along Φ is the normalized-centroid linkage of the initial offsets and is the same for every $\lambda > 0$ (proposition 7.6). Along both nulls every difference of means is $e^{-\kappa t}$ times its initial value in both sectors, so a selector that joins two adjacent current groups when the normalized centroid distance eq. (7.30) of their current offsets reaches δ from

above gives, for every δ below the smallest linkage height, the same linkage sequence along the nulls; along Φ it gives that sequence as well, since by the proof of proposition 7.6 the threshold is reached for the pair of height ϱ at $t = \kappa^{-1}\log\big((\kappa\varrho+\lambda)/(\kappa\delta+\lambda)\big)$, an increasing function of ϱ, before the corresponding coincidence. For such a threshold family, with constants in an interval below the smallest linkage height, $P = P^0$ on the family-grade admitted class, and (F5) fails there. The interaction determines the event times and the exactness of coincidence, not which blocks form. The selector-free discrepancy between Φ and the nulls is the finite-time attainment of equality; both converge to the zero configuration. The frozen-environment control decouples the agents into single-agent subgradient flows of strongly convex functions, well posed by the argument of lemma 7.2; its partition sequence is not determined here. Whether the partition sequence depends on λ off the common-offset family is open.

Clause (F6a) holds: by theorem 7.7 and corollary 7.8, the two first-stage parents select their union under the inherited interaction, with the same selector and constitutive data, no reset of their data, and no retuned coupling. Clause (F6b) fails for every counterfactual class of definition 6.35: by proposition 7.3 applied to multiplicities two, every configuration of two first-stage parents, whatever their means, fuses at a finite time, so the second-stage event occurs from every counterfactual handoff, and the stages form a supplied-scale sequence ordered by the initial gaps of eq. (7.34). The amplifying grade holds only in the degenerate sense of definition 6.36, since within-block diameters vanish at every event of Φ.

The construction satisfies (F1), (F2), (F3), and (F6a), and (F4) and (F6b) are shown to fail, so it is a two-stage selection with exact inheritance and not a recursive formation. Its (F5) verdict holds only through the exact-equality selector; its coalescence tree is a coupling-independent linkage of the initial data, and the tree always ends in one block.

Remark 7.12 (Smooth peer KL dynamics and persistent parents). The finite mergers above depend on the nonsmooth interaction. Under the smooth reciprocal potential with a locally Lipschitz vector field, an interior trajectory cannot first reach an invariant synchronization manifold at finite time (proposition 6.24), so smooth constructions need a different selection criterion; the threshold selectors of chapters 12 and 13 are such criteria, and the remaining problem is open problem 6.38. Persistent parents are excluded here by proposition 7.3. A contact range, for example $\lambda_{ij} = \lambda\mathbf{1}\{|\mu_i - \mu_j| < \varrho_c\}$ as in bounded-confidence dynamics [105], or a multiwell local reference would let separated blocks stop interacting. Both destroy the convexity on which uniqueness rests, and well-posedness and inheritance for such variants have not been established. Persistence does not by itself require giving up convexity: agent-specific single-well references $n_i\mathcal{U}(\mu_i - x_i)$ keep the potential $(a-c)$-strongly convex, so lemma 7.2 and eq. (7.19) hold verbatim, and the unique minimizer can have several blocks. Under any strongly convex potential, however, every solution converges to that minimizer whatever the initial datum, so the limiting partition is a function of the declared data alone and a final-stage counterfactual dependence (F6b) needs several minima. For such references, finite-time arrival at the limiting partition, and the behavior of blocks before it, are not established here; synchronized blocks need not persist before arrival, since agent-specific references break the exchange symmetry used in lemma 7.2.

Chapter 8

Sufficient States for Smooth KL Evolution

This chapter works with the reciprocal law potential of definition 4.2 and its unit-mobility Fisher flow; its conclusions concern that potential, not the per-agent or global free energies of chapter 4. The question is which information a coarse description of an agent must keep when an exact reduction is wanted while agents remain different. We first identify the conditional information that enters a retained force, then construct an exact state completion made of general probability laws for one architecture: fixed independent private reference factors, and receiver reconstruction channels that discard the sender's private coordinate. These channels are additional constitutive data within the potential, not a consequence of an arbitrary law-dependent transport.

In this architecture the private sector is passive. Each private conditional relaxes along an explicit exponential arc toward its reference, independently of every public law, row, and other agent, and it acts on the public flow only as open-loop forcing through its conditional KL profile. The completion is therefore a solvable baseline. Its compression is the equimeasurability quotient of each private log density under its declared reference: it forgets which private point carries a log-density value, so it identifies two conditionals exactly when their log densities are equimeasurable under the reference. On a finite alphabet this requires two distinct sets of atoms of equal mass, which is weaker than a symmetry of the reference. The completion is not minimal for the public flow (proposition 8.11), and the per-agent completion fails once a channel transmits the sender's private data (remark 8.13). The construction selects no group of agents, so clauses (F2), (F5), and (F6) of definition 6.36 are not addressed; its private sector evolves identically under both nulls and under the frozen-environment control of definition 6.33, because the conditional equations contain no peer term.

8.1 The force that survives marginalization

Let Q be a positive law on a standard Borel space, let C be a measurable statistic, and write $\mu = C_{\#}Q$. Suppose a differentiable functional has variational derivative h_Q and Fisher velocity

$$\dot{Q} = -Q(h_Q - \mathbb{E}_Q h_Q). \tag{8.1}$$

All identities below are immediate for finite positive tables. On general spaces they need a dominated differentiable path, square-integrable scores, and integrability that permits the displayed conditional expectations and differentiations. Statements about flows on general spaces are made under the hypotheses of hypothesis 8.4, which give unique solutions in bounded log-density charts (proposition 8.8); they do not assert existence on every space of probability measures.

Proposition 8.1 (Conditional mean force). The retained velocity is

$$\dot{\mu} = -\mu \left(\mathbb{E}_Q[h_Q \mid C] - \mathbb{E}_Q h_Q \right). \tag{8.2}$$

Consequently, an autonomous velocity for μ exists on an admitted class precisely when the centered conditional mean on the right is determined by μ on that class. If, in addition, the admitted class is invariant under the fine flow and the resulting retained equation $\dot{\mu} = V(\mu)$ has unique solutions, then two fine trajectories in the class with the same initial marginal have the same marginal at all common times.

Proof. For every bounded measurable f, integrate $f(C)$ against eq. (8.1) and condition on C. This gives the weak form of eq. (8.2). Two laws with the same marginal have the same marginal velocity if and only if their centered conditional forces agree almost everywhere under that marginal. Invariance keeps both fine trajectories in the class, so each marginal solves $\dot{\mu} = V(\mu)$, and uniqueness identifies them. □

Equation eq. (8.2) is the class-mean aggregation of a replicator equation over types, familiar from the Price equation of selection [165]. For $Q(dz, dv) = \mu(dz)\kappa_z(dv)$ and a fixed reference $P(dz, dv) = \nu(dz)\pi_z(dv)$, the local KL force illustrates the missing information:

$$\mathbb{E}_Q[\log(dQ/dP) \mid z] = \log \frac{d\mu}{d\nu}(z) + H(z), \qquad H(z) = D_{\mathrm{KL}}(\kappa_z \| \pi_z). \tag{8.3}$$

Thus a scalar total conditional loss does not usually determine the retained force. Its dependence on z matters. The Fisher metric makes the same separation: if a score is $f(z, v) = f_1(z) + f_2(z, v)$, with $\mathbb{E}_\mu f_1 = 0$ and $\mathbb{E}_{\kappa_z} f_2 = 0$, then

$$g_Q(fQ, fQ) = \int f_1(z)^2 \mu(dz) + \int \left[\int f_2(z, v)^2 \kappa_z(dv) \right] \mu(dz). \tag{8.4}$$

The cross term vanishes by conditional centering. This calculation is consistent with the statistical behavior of Fisher information under sufficient statistics [9]. Dynamical sufficiency asks a different question: it preserves the specified evolution and responses, and need not preserve inference about every discarded parameter.

Corollary 8.2 (Pushforward as a Fisher submersion). On a regular positive class, the differential of $Q \mapsto C_\# Q$ sends the tangent fQ to $\mathbb{E}_Q[f \mid C]\mu$. Its kernel consists of the scores with zero conditional mean given C, and it is isometric on scores that are functions of C. If the functional has the form $\mathcal{E} = \mathcal{E}_c \circ C_\#$, the retained marginal follows the Fisher gradient flow of $\mathcal{E}_c$ at the same time.

Proof. For bounded g, $\int g(C) f dQ = \int g \mathbb{E}_Q[f \mid C] d\mu$, which identifies the pushforward tangent. With $f_1 = \mathbb{E}_Q[f \mid C]$ and $f_2 = f - f_1$, eq. (8.4) gives the kernel and the isometry. If $\mathcal{E} = \mathcal{E}_c \circ C_\#$, the chain rule gives $h_Q = h_c \circ C$ up to a constant, so eq. (8.2) becomes $\dot{\mu} = -\mu(h_c - \mathbb{E}_\mu h_c)$. □

This is the equality case of the monotonicity of Fisher information under statistics [10]; the content of the present chapter is the case in which h_Q is not a function of C.

8.2 Conditional reconstruction within the original potential

For each agent take $\mathsf{K}_i = \mathsf{Z}_i \times \mathsf{V}_i$ and $\mathsf{M}_i = \mathsf{W}_i \times \mathsf{U}_i$, all standard Borel. The coordinates z, w are public in the sense that the channels below read them; the coordinates v, u are private in the sense that no channel reads the sender's value of them, and the declared probes read them only through the marks of section 8.4. Use public densities against fixed reference probabilities ρ_i on Z_i and ϱ_i on W_i, so conditional versions share common full-measure public sets. Disintegrate

$$q_i(dz, dv) = \mu_i(dz)\kappa_{i,z}(dv), \qquad s_i(dw, du) = \sigma_i(dw)\varkappa_{i,w}(du). \tag{8.5}$$

Neither conditional need be independent of its retained coordinate. Choose fixed private reference probabilities A_i^q, A_i^s and a fixed normalized public reference

$$R_i^c(dz, dw) = r_i^c(dw)G_i^c(w, dz), \qquad R_i = R_i^c \otimes A_i^q \otimes A_i^s, \tag{8.6}$$

with the product reordered into state and model coordinates. The evaluator G_i^c can couple μ_i and σ_i. Independence of the private reference factors is an assumption of this construction, not a restriction of the general theory.

Let $B_{ij}^q[\boldsymbol{\mu}] : \mathsf{Z}_j \rightsquigarrow \mathsf{Z}_i$ be normalized channels, differentiable in the laws in the sense required by hypothesis 8.4 on general spaces. Define the full channels by

$$\mathcal{K}_{ij}^q[\mathbf{q}]((z_j, v_j), d(z_i, v_i)) = B_{ij}^q[\boldsymbol{\mu}](z_j, dz_i)\kappa_{i,z_i}(dv_i). \tag{8.7}$$

The model channel uses $B_{ij}^s[\boldsymbol{\sigma}]$ and $\varkappa_{i,w_i}$ in the same way. Source sets may exclude self edges; if self comparisons are included, use the primitive identity convention separately. Public channels may be informative and nonconstant functions of the retained laws. For example, on the public space $\{-1, 1\}$, the channel $B_\mu(z, \{z'\}) = (1 + \vartheta(\mu)zz')/2$, with $\vartheta(\mu) = (1 + (\mathbb{E}_\mu Z)^2)/4$, is strictly positive, transmits source information, and varies with the law. Public/private maps and reference factors are declared interface data; carrying them through a bimeasurable change of presentation preserves these constructions.

Proposition 8.3 (Exact cancellation of private peer information). Where the divergences are defined,

$$\begin{aligned} q_j\mathcal{K}_{ij}^q &= (\mu_j B_{ij}^q)\kappa_i, \\ D_{\mathrm{KL}}(q_i \| q_j\mathcal{K}_{ij}^q) &= D_{\mathrm{KL}}(\mu_i \| \mu_j B_{ij}^q). \end{aligned} \tag{8.8}$$

This is an identity on the full admitted class, including correlated q_i. Its derivative includes the sender, receiver, and channel derivatives.

Proof. Integration over the source conditional gives the first identity. The compared laws have the same receiver conditional $\kappa_{i,z}$, so their density ratio depends only on z. Integrating its logarithm proves the second identity. Differentiating the resulting functional identity on any regular path proves the derivative statement. Holding κ_i fixed in the denominator leaves the force at the frozen state unchanged, because the derivative of the divergence in the receiver conditional of the compared law vanishes there; it violates the identity away from that state, changes the second variation, and, if the snapshot is held while the laws move, changes the force. □

Let $\mathcal{E}_c$ denote exactly eq. (4.4) applied to $\mu_i, \sigma_i, R_i^c, B_{ij}^q, B_{ij}^s$, with the same rows, source priors, and temperatures. Define

$$\begin{aligned} \ell_{i,z}^q &= \log(d\kappa_{i,z}/dA_i^q), \quad H_i^q(z) = \int \ell_{i,z}^q d\kappa_{i,z}, \\ \ell_{i,w}^s &= \log(d\varkappa_{i,w}/dA_i^s), \quad H_i^s(w) = \int \ell_{i,w}^s d\varkappa_{i,w}. \end{aligned} \tag{8.9}$$

On finite public alphabets with strictly positive r_i^c and G_i^c, twice continuously differentiable channel entries, and positive compared laws on the interior, (H2) and (H3) below hold, and it suffices that each private log density be bounded. On general public spaces we use the following hypotheses.

Hypothesis 8.4 (Regularity on general spaces). Use the bounded log-density chart for every public law: $\mu_i = e^{u_i}\rho_i/\mathbb{E}_{\rho_i} e^{u_i}$ with u_i in $L_0^\infty(\rho_i) = \{u \in L^\infty(\rho_i) : \mathbb{E}_{\rho_i} u = 0\}$, the same for σ_i against ϱ_i, and log-odds coordinates for interior rows.

(H1) The initial private log densities are bounded uniformly in the public coordinate: at time zero,

$$\operatorname*{ess\,sup}_{(z,v)} |\ell_{i,z}^q(v)| \le M < \infty \tag{8.10}$$

with respect to $\rho_i \otimes A_i^q$, and likewise in the model sector.

(H2) The public reference has bounded log densities: $G_i^c(w, dz) = g_i^c(w, z)\rho_i(dz)$ and $r_i^c = \tilde{r}_i \varrho_i$, with $\log g_i^c \in L^\infty(\varrho_i \otimes \rho_i)$ and $\log \tilde{r}_i \in L^\infty(\varrho_i)$.

(H3) On bounded subsets of the product of the charts, the public first variations $h_i^q = \delta\mathcal{E}_c/\delta\mu_i$ and $h_i^s = \delta\mathcal{E}_c/\delta\sigma_i$, including every interaction role and channel derivative, are Lipschitz into $L^\infty(\rho_i)$ and $L^\infty(\varrho_i)$, and the costs D_{ij}^q, D_{ij}^s are Lipschitz. Identity channels between agents sharing a public reference space satisfy this, as do channels with kernel densities $b_{ij}[\boldsymbol{\mu}]$ whose logarithms are bounded on bounded subsets, provided $\boldsymbol{\mu} \mapsto b_{ij}[\boldsymbol{\mu}]$ is continuously differentiable into $L^\infty(\rho_j \otimes \rho_i)$ with a derivative represented by bounded kernels, $Db_{ij}[\boldsymbol{\mu}](\delta\mu_k) = \int \beta_{ijk}[\boldsymbol{\mu}](z', \cdot, \cdot)\delta\mu_k(dz')$ with $\beta_{ijk} \in L^\infty(\rho_k \otimes \rho_j \otimes \rho_i)$ locally Lipschitz in $\boldsymbol{\mu}$. Channels that depend on the laws through moments $\int g d\mu_k$ qualify when g is bounded.

Positivity of the public densities alone does not suffice: a normalized, strictly positive, law-dependent channel whose first variation contains an unbounded function of z gives a public force outside L^∞, and the chart vector field is then undefined. Continuous differentiability into L^∞ alone does not exclude this: on $(0, 1]$ with Lebesgue reference, the kernel density $1 + \varsigma(y)\cos(2\pi z')\cos(2\pi z)$, with $y = \int \log(1/z)\mu(dz)$ and a smooth $|\varsigma| \le 1/2$, is a smooth function of the chart coordinate into L^∞ because $\log(1/z)$ is integrable, but the first variation of a peer cost through it contains $\varsigma'(y)\log(1/z)$ times a factor that is nonzero in general. A bound on each $\ell_{i,z}$ separately does not suffice either: a family of bounded conditionals whose bounds grow as z approaches a boundary point can make the public log density unbounded for every positive time. The class (H1) excludes conditionals whose density vanishes on a set of positive reference measure or is unbounded, such as Gaussian conditionals against a Gaussian reference; remark 8.12 treats those separately.

Theorem 8.5 (Exact marginal and conditional dynamics). For the architecture above the original energy is

$$\mathcal{E} = \mathcal{E}_c + \sum_i \int H_i^q(z)\mu_i(dz) + \sum_i \int H_i^s(w)\sigma_i(dw). \tag{8.11}$$

Write $h_i^q = \delta\mathcal{E}_c/\delta\mu_i$ and $h_i^s = \delta\mathcal{E}_c/\delta\sigma_i$, including every interaction role. The full unit-mobility Fisher dynamics gives

$$\begin{aligned} \dot{\mu}_i &= -\mu_i \left[h_i^q + H_i^q - \mathbb{E}_{\mu_i}(h_i^q + H_i^q)\right], \\ \partial_t \ell_{i,z}^q &= H_i^q(z) - \ell_{i,z}^q. \end{aligned} \tag{8.12}$$

The equations for σ_i, ℓ_i^s are obtained by exchanging sectors. Source rows evolve by their original Fisher equations with the public costs, or may be exactly optimized as in proposition 4.3.

Proof. Apply the relative-entropy chain rule to the local reference in eq. (8.6) and use eq. (8.8). This proves eq. (8.11). Up to a constant independent of (z, v), the full derivative with respect to q_i is $h_i^q(z) + \ell_{i,z}^q(v)$. This can be checked directly from the log-density ratio in the local KL; every peer contribution is a functional of μ alone. Conditional averaging gives the first equation of eq. (8.12). Subtract its logarithmic derivative from the full logarithmic derivative of $q_i = \mu_i\kappa_i$. The public score and global centering cancel, leaving the second equation. The private terms contain no source rows, so their variations do not change the row equations. □

The two row choices share stationary points but are different flows. At frozen costs the Fisher row equation relaxes log-ratio contrasts at the rate T_q (eq. (8.28)), so exactly optimized rows are the limit of infinite row mobility, and the two choices differ most in their transients when T_q is small.

The private equation in eq. (8.12) contains no public law, no row, and no other agent. Each private conditional is solved in closed form, independently of the public dynamics, by eq. (8.19) below: the exponential arc from $\kappa_{i,z}(0)$ to A_i^q. The public flow sees the private sector only through the function $H_i^q(z,t)$, which is fixed by the initial conditionals. The completion constructed next is therefore the public flow of $\mathcal{E}_c$ driven by the precomputable time-dependent potential $\int H_i^q(\cdot,t)d\mu_i$. Private information never influences an interaction in this architecture. The dependence of H_i^q on z is the only private contribution to the retained belief velocity, and it cannot generally be discarded.

8.3 An exact law-valued completion

Definition 8.6 (Conditional log-density spectrum). For a private reference A and conditional density κ_z, define

$$\xi_z = (\ell_z)_{\#}A, \qquad \ell_z = \log(d\kappa_z/dA). \tag{8.13}$$

This probability law records how much reference mass carries each log-density value. It forgets which private point carries that value. It obeys

$$\int e^r \xi_z(dr) = 1, \qquad H(z) = \int r e^r \xi_z(dr). \tag{8.14}$$

The spectrum is the law under A of the log-likelihood ratio of the dichotomy (κ_z, A). For mutually absolutely continuous pairs, equal spectra are equivalent to Blackwell equivalence of the two dichotomies [27, 189]. Two conditionals have the same spectrum exactly when their log densities are equimeasurable under A. The compression is therefore a property of the declared reference, not of the dynamics. On a finite private alphabet $\xi_z(\{r\}) = A(\{\ell_z = r\})$, so if no two distinct sets of atoms of A have equal mass, equal spectra force equal level sets and hence equal conditionals: the retention is injective. With $A = (0.6, 0.4)$, for example, the conditionals $(3/4, 1/4)$ and $(1/4, 3/4)$

have the different spectra with atoms $\log(5/4), \log(5/8)$ and $\log(5/12), \log(15/8)$ at masses $0.6, 0.4$. Conversely, if two distinct sets of atoms have equal mass, their set differences S, S' are disjoint, nonempty, and of equal mass, and exchanging two distinct density values placed on S and S' gives distinct conditionals with equal spectra. Noninjectivity therefore requires a reference with equal-mass sets of atoms, such as a uniform finite reference, or a reference with a nonzero atomless part, in which disjoint sets of equal positive mass always exist. The log-Laplace transform of the spectrum is

$$\Lambda_z(b) = \log \int e^{br} \xi_z(dr) = (b-1) D_b(\kappa_z \| A), \tag{8.15}$$

where D_b is the Rényi divergence of order b [192]. For every κ_z mutually absolutely continuous with A and $0 < b < 1$, Hölder's inequality gives $\int e^{br} \xi_z(dr) = \int (d\kappa_z/dA)^b dA \in (0, 1]$, and this function is analytic in the strip $0 < \operatorname{Re} b < 1$. A law is determined by its moment generating function on any nondegenerate interval on which it is finite, so Λ_z on any subinterval of $(0, 1)$ determines ξ_z; (H1) is not needed for this.

Theorem 8.7 (Closed conditional-spectrum evolution). For theorem 8.5, the velocity of the state consisting of all μ_i, σ_i, conditional spectra ξ_i^q, ξ_i^s, and source rows is a function of that state. Each spectrum satisfies, in weak form,

$$\frac{d}{dt} \int f(r) \xi_z(dr) = \int f'(r)(H(z) - r) \xi_z(dr), \tag{8.16}$$

for continuously differentiable test functions on the initial range $I = [m_0, M_0]$ of the log density, which contains the range at every later time. For a time interval t starting at zero, its exact solution is

$$\begin{aligned} a &= e^{-t}, & Z_z(a) &= \int e^{ar} \xi_z^0(dr), \\ \xi_z^t &= (r \mapsto ar - \log Z_z(a))_{\#} \xi_z^0, \end{aligned} \tag{8.17}$$

and it is the only narrowly continuous curve of probability laws supported in I that solves eq. (8.16) with $H(z)$ given by eq. (8.14) from the same initial law with $\int e^r \xi_z^0(dr) = 1$. The entropy profile obeys

$$\dot{H}(z) = -\operatorname{Var}_{e^r \xi_z}(r). \tag{8.18}$$

Proof. The reference A is fixed. Differentiate $\int f(\ell_z(v)) A(dv)$ and substitute the conditional equation in eq. (8.12); this proves eq. (8.16). Direct differentiation shows that

$$\frac{d\kappa_z(t)}{dA}(v) = \frac{\exp(a\ell_z^0(v))}{Z_z(a)} \tag{8.19}$$

solves the normalized conditional equation; this is the explicit solution of the Fisher–Rao gradient flow of a relative entropy, the exponential arc toward the reference [41]. Pushing A through its logarithm gives eq. (8.17). For the range, Hölder's inequality gives $Z_z(a) \le 1$, and $e^{ar} \ge e^r e^{-(1-a)M_0}$ for $r \le M_0$ gives $Z_z(a) \ge e^{-(1-a)M_0}$; since $m_0 \le 0 \le M_0$ by normalization, $a\ell_z^0 - \log Z_z(a)$ takes values in $[m_0, M_0]$, and the width of its range is $a(M_0 - m_0)$.

For uniqueness, let ξ_t be a curve in the stated class and put $H(t) = \int r e^r \xi_t(dr)$, which is continuous. Then ξ_t solves the linear continuity equation with the velocity field $H(t) - r$, which is Lipschitz in r,

and continuity equations with such fields have unique narrowly continuous solutions [5, Sec. 8.1]. Hence $\xi_t = (X_t)_{\#}\xi_0$ with $X_t(r) = e^{-t}r + c(t)$ and $c(t) = \int_0^t e^{-(t-s)}H(s)ds$. Taking $f(r) = e^r$ in eq. (8.16) gives $\dot{N} = H(N-1)$ for $N = \int e^r \xi_t(dr)$, so $N \equiv 1$, which forces $e^{c(t)}Z(a) = 1$ and gives eq. (8.17).

Setting $f(r) = re^r$ in the weak equation gives $\dot{H} = H(H+1) - \int (r^2+r)e^r \xi_z(dr)$, which is eq. (8.18). The probability $e^r \xi_z(dr)$ is normalized by eq. (8.14). All private contributions to eq. (8.12) are now functions of the retained spectra. The remaining scores and row equations depend only on retained public data, so the retained velocity is a function of the retained state. □

That the retained state determines its own future additionally requires uniqueness of the public flow forced by $H_i(z,t)$. For finite public alphabets with twice continuously differentiable channels and the positivity hypotheses of theorem 4.6, its Picard–Lindelöf argument applies to this forced system. On general public spaces the following proposition supplies it.

Proposition 8.8 (Well-posedness and exact lift on general spaces). Assume hypothesis 8.4. Then for every initial state in the charts:

(i) each private log density keeps its values in its initial essential range, $0 \le H_i^q(z,t) \le H_i^q(z,0) \le M$, and $|\partial_t H_i^q(z,t)| \le M^2$ uniformly in z, and likewise in the model sector;

(ii) the retained system of public laws, rows, and spectra has a unique maximal solution on an interval $[0, T^*)$, with public laws and rows continuously differentiable into the charts and spectra given by eq. (8.17);

(iii) the fine Fisher system has a unique maximal continuously differentiable solution in the charts $L_0^\infty(\rho_i \otimes A_i^q)$ and $L_0^\infty(\varrho_i \otimes A_i^s)$ on the same interval, and it equals the lift $q_i(t) = \mu_i(t)\kappa_i(t)$ with $\kappa_i(t)$ given by eq. (8.19), and likewise in the model sector;

(iv) if R sends a fine state to its public laws, spectra, and rows, then $R \circ \Phi_t = \Phi_t^c \circ R$ on $[0, T^*)$, where Φ^c is the solution map of (ii).

Proof. (i) The range statement is in the proof of theorem 8.7, and it holds uniformly in z by (H1). The function H is a relative entropy, hence nonnegative, and it is nonincreasing by eq. (8.18). A law on an interval of width at most $2M$ has variance at most M^2.

(ii) In the chart the public equation reads $\dot{u}_i = -(\phi_i - \mathbb{E}_{\rho_i}\phi_i)$ with $\phi_i = h_i^q + H_i^q(\cdot,t)$, because then $\partial_t \log(d\mu_i/d\rho_i) = -(\phi_i - \mathbb{E}_{\mu_i}\phi_i)$. This is the chart flow of definition 4.13 for $\mathcal{E}_c$, with the time-dependent term $H_i^q(\cdot,t)$ added to the first variation. By (i), $t \mapsto H_i^q(\cdot,t)$ is Lipschitz into $L^\infty(\rho_i)$, and (H2) and (H3) make h_i^q Lipschitz on bounded sets; the row block is a finite-dimensional equation whose costs are Lipschitz on bounded sets, and Gibbs rows are smooth functions of the costs. The proof of proposition 4.14, applied to the system augmented by the time coordinate, gives a unique maximal solution, with the alternative that it leaves every bounded set as $t \to T^*$ if $T^* < \infty$.

(iii) In the fine chart, the first variation of the energy with respect to q_i is $\varphi_i(z,v) = h_i^q(z) + \ell_{i,z}(v)$ up to a constant, by the proof of theorem 8.5. The public marginal map $l \mapsto \log \int e^{l(\cdot,v)} A_i^q(dv)$ is smooth from $L^\infty(\rho_i \otimes A_i^q)$ to $L^\infty(\rho_i)$, so the fine first variation is Lipschitz on bounded sets, and proposition 4.14 gives uniqueness of fine solutions. For the lift, $\partial_t \log q_i = -(h_i^q + H_i^q - \mathbb{E}_{\mu_i}(h_i^q + H_i^q)) + (H_i^q - \ell_i) = -(\varphi_i - \mathbb{E}_{q_i}\varphi_i)$, because $\mathbb{E}_{q_i}\varphi_i = \mathbb{E}_{\mu_i}h_i^q + \mathbb{E}_{\mu_i}H_i^q$. The lift stays bounded on compact subintervals of $[0, T^*)$ by (i), so it is the fine solution there. Conversely, the projection of a

fine solution solves the retained system, so a fine solution bounded beyond T^* would contradict the alternative in (ii).

(iv) Two fine states with the same retained value have the same public laws, rows, and spectra, hence the same forcing $H(t)$ and, by (ii), the same retained solution; by (iii) their projections are retained solutions. □

Under these hypotheses the retained state is therefore autonomous on the common maximal interval. For law-dependent public channels, global existence, $T^* = \infty$, requires further estimates on the public sector, which are not given here. For fixed public channels it holds.

Corollary 8.9 (Global existence for fixed public channels). Assume hypothesis 8.4 and, in addition, that every public channel B^q_{ij}, B^s_{ij} is fixed and either an identity between agents sharing a public reference or a kernel whose density lies in $[c, C_K]$ with $0 < c \le C_K < \infty$, and that the rows are fixed or follow their Fisher equations. Then $T^* = \infty$ in proposition 8.8.

Proof. The public system is the chart flow of $\mathcal{E}_c$ with the first variation of each public law increased by $H^q_i(\cdot, t)$ or $H^s_i(\cdot, t)$, which lies in $[0, M]$ by proposition 8.8(i). Run the proof of proposition 4.15 for this system. In the lower bound the first variation increases by at most M and its mean does not decrease, so the constant A of eq. (4.18) is replaced by $A + M$. In the upper bound $\mathcal{E}_c(t)$ is at most the full energy eq. (8.11) at time t, which is nonincreasing along the fine flow, and the forcing adds at most M to the mean of the first variation, so the bound holds with B computed from the full initial energy and increased by M. The public log densities therefore stay bounded on every finite interval, and the continuation alternative of proposition 8.8(ii) gives $T^* = \infty$; by (iii) the fine solution is global as well. □

This is a configuration retention. Its spectra can be infinite-dimensional, and there is no claim that they give a fixed finite number of parameters. If represented as a sample statistic, the map $v \mapsto \ell_z(v)$ depends on the current law; its motion is exactly the term differentiated in eq. (8.16). It must not be frozen while the law changes. The inherited spectral evolution is not a Fisher–Rao gradient flow on probability laws over (z, r). Such a flow $\dot{Q} = -Q(h - \mathbb{E}_Q h)$ keeps Q_t mutually absolutely continuous with Q_0, whereas the retained conditional $e^r \xi^t_z(dr)$ is supported on $a \operatorname{supp} \xi^0_z - \log Z_z(a)$, which moves unless $\kappa_z = A$. The spectral evolution is a continuity equation in r with velocity $H(z) - r$.

Corollary 8.10 (Different internal conditionals with the original public flow). Suppose that, for each agent and each sector separately, the initial spectrum is independent of the public coordinate. Then this property persists, and the public laws and source rows solve the original coarse KL/Fisher system generated by $\mathcal{E}_c$ on the common maximal interval.

Proof. The same initial spectrum gives the same solution of eq. (8.17) at every public coordinate. Consequently $H^q_i(z)$ is constant in z, and $H^s_i(w)$ is constant in w. Each constant disappears under its own marginal centering in eq. (8.12). □

Equal spectra do not require equal conditionals when the reference is uniform. With a private reference uniform on two points, the conditionals $(3/4, 1/4)$ and $(1/4, 3/4)$ share

$$\xi = \tfrac{1}{2}\delta_{\log(3/2)} + \tfrac{1}{2}\delta_{\log(1/2)}. \tag{8.20}$$

A public bit may select either conditional, producing correlation between public and private variables. Another law with the same public marginal may use the first conditional at both public values and be a product. When the original public flow from the common initial public state is unique, as it is for this finite public alphabet by theorem 4.6 and on general spaces under hypothesis 8.4, both have the same public trajectory in this architecture, although their internal laws differ. Different agents may have different public laws and different spectra. No equality between agents is imposed.

This class is invariant, but it is not reached for the first time at finite interior time. For any $a > 0$, the spectral update in eq. (8.17) is invertible on its image. Indeed, if ξ^t is given, the initial log value is recovered by the map

$$y \longmapsto \frac{y}{a} - \log \int e^{r/a} \xi^t(dr). \tag{8.21}$$

Substituting eq. (8.17) and using $\int e^r \xi^0(dr) = 1$ proves the assertion. Equal spectra at a finite time therefore had to be equal initially. The same inversion applied to eq. (8.19) proves that distinct conditionals cannot first become equal at finite time. Neither fact forbids an autonomous quotient that keeps the unequal spectra.

Proposition 8.11 (A coarser completion for the public flow). In the setting of proposition 8.8, the public laws and rows are determined, on the common maximal interval, by their initial values together with the classes of the maps $z \mapsto \Lambda^q_{i,z}|_{(0,1]}$ and $w \mapsto \Lambda^s_{i,w}|_{(0,1]}$ of eq. (8.15) modulo functions of b alone. These classes evolve autonomously by

$$\Lambda^t_z(b) = \Lambda^0_z(ab) - b\Lambda^0_z(a), \qquad a = e^{-t}. \tag{8.22}$$

Conversely, two private configurations with the same initial public laws and rows whose public trajectories agree on some interval $[0, \varepsilon)$ have the same classes, so the class is exactly the private information that the public flow needs and reveals. The class is strictly coarser than the spectrum: if $\mathsf{V} = \mathsf{V}_1 \times \mathsf{V}_2$ with a product reference, the conditionals $\kappa^{(1)}_z \otimes \psi$ and $\kappa^{(1)}_z \otimes \psi'$ with $\psi \neq \psi'$ fixed have different spectra at every z whenever ψ and ψ' have different spectra, but the same class.

Proof. Substituting eq. (8.17) into the definition gives eq. (8.22). If $\Lambda^0_z = \widetilde{\Lambda}^0_z + k(b)$ with k independent of z, then eq. (8.22) adds $k(ab) - bk(a)$, again independent of z, so the flow acts on classes. The forcing is $H(z,t) = \partial_b \Lambda^t_z(1)$, which the class determines up to a function of t alone, and such a function disappears under marginal centering in eq. (8.12). Uniqueness in proposition 8.8 gives the conclusion. For the example, the log-Laplace transform of a product conditional against a product reference is the sum of the transforms of the factors, so the two maps differ by the z-independent function $\Lambda_\psi - \Lambda_{\psi'}$, while the spectra are the convolutions of a common factor with the different spectra of ψ and ψ'.

For the converse, equal public trajectories have equal public velocities, so $H(\cdot,t) - \widetilde{H}(\cdot,t) = k(t)$ almost everywhere for each t in the interval, with one null set for all t because both sides are continuous in t. With $\Delta_z = \Lambda^0_z - \widetilde{\Lambda}^0_z$, $a = e^{-t}$, and $H(z,t) = \partial_b \Lambda^t_z(1) = a^2 \frac{d}{da}\big(\Lambda^0_z(a)/a\big)$, this reads $\frac{d}{da}(\Delta_z(a)/a) = k(t(a))/a^2$ on $(e^{-\varepsilon}, 1]$. Since $\Delta_z(1) = 0$, $\Delta_z(a) = aK(a)$ there with K independent of z. Each Λ^0_z is real analytic on $(0,1)$, so $\Delta_z - \Delta_{z_0}$ vanishes on $(0,1]$, and the two classes coincide. □

Beyond the class, the full spectrum is needed for responses that read the conditional spectra themselves, such as f-divergence readouts $\int f(e^r)\xi_z(dr)$, and its marked version below for records that read private data. Corollary 8.10 is the case in which the classes are trivial.

Remark 8.12 (Exponential families and finite-parameter private closure). Hölder's inequality gives $0 < Z_z(a) \le 1$ for every $\kappa_z(0)$ mutually absolutely continuous with A, so eq. (8.19) defines the private solution without (H1). If $\kappa_z(0) \propto \exp(\theta_z \cdot T)A$ for a statistic T with log-partition function ψ, then $\kappa_z(t) \propto \exp(e^{-t}\theta_z \cdot T)A$: the natural parameters contract linearly, and $H(z,t) = a\theta_z \cdot \nabla\psi(a\theta_z) - \psi(a\theta_z)$ with $a = e^{-t}$. The private sector then closes on finitely many parameters per public value, for instance on Gaussian conditionals relative to a Gaussian reference. Such families fall outside (H1) when T is unbounded, and the public part then needs its own justification; for the public equation alone, a bound on $\operatorname{ess\,sup}_z H(z,0)$ suffices for the forcing to be bounded, because H is nonincreasing in t. Nothing here makes the public laws finite-dimensional.

Remark 8.13 (Which channels permit the completion). Receiver reconstruction is not what produces closure. If every channel in a row is replaced by $B^q_{ij}(z_j, dz_i)A^q_i(dv_i)$, a fixed reconstruction independent of state and source, the chain rule gives $D^q_{ij} = D_{\mathrm{KL}}(\mu_i \| \mu_j B^q_{ij}) + \int H^q_i d\mu_i$. The added term is the same for every source, so it cancels from Gibbs rows and from the row velocities, the private energy doubles, and the conditional equation becomes $\partial_t \ell^q_{i,z} = 2(H^q_i(z) - \ell^q_{i,z})$; the spectrum still closes. What matters is that the channel discards the sender's private coordinate. A channel that transmits it fails. Take a trivial public space, $\mathsf{V} = \{1,2\}$ with uniform A, two agents attending to each other through the identity channel, and $\kappa_1 = (3/4, 1/4)$. With $\kappa_2 = \kappa_1$ the peer costs vanish, whereas with $\kappa_2 = (1/4, 3/4)$, which has the same spectrum, each peer cost is $\frac{1}{2}\log 3$ and the peer forces move κ_1 toward κ_2. The spectral velocity of agent 1 then differs between two configurations with the same per-agent spectra, so those spectra are not a closed state. For identity private channels on a common (V, A), every peer integrand is a function of the vector $(\ell_k(v))_k$ at one private point, which suggests the joint law of that vector under A as the retained state. For identity public and private channels with base-only references this joint law is closed (proposition 11.6); its closure with law-dependent public channels, and its compression beyond equimeasurable rearrangements of the private value vector, are not established here.

8.4 Boundary records and feedback

Public-only response kernels descend immediately: the channels B^q_{ij}, B^s_{ij} and evaluator G^c_i are determined by the retained public laws. A private-sensitive response needs more information. In eq. (8.20), let the probabilities of record 1 at the two private points be $(3/4, 1/4)$. The two conditionals yield record probabilities $5/8$ and $3/8$. After record 1, their private posteriors are $(9/10, 1/10)$ and $(1/2, 1/2)$. Thus an unmarked spectrum can fail both the boundary test and the next-update test.

There is an exact completion for a specified family of records. Fix a measurable mark $T : \mathsf{V} \to \mathsf{D}$, with D standard Borel, through which every allowed private readout factors. Retain

$$\eta_z = (\ell_z, T)_{\#}A. \tag{8.23}$$

The correlation between the log-density value and the mark is retained. For a normalized record kernel with strictly positive density $L(o \mid z, d)$, jointly Borel in (o, z, d), assume the following integrals are finite and nonzero and each updated law stays in the admitted regular domain. A bounded log likelihood is a sufficient local condition for the conditional log-density bounds. Put

$$\begin{aligned} N_z(o) &= \int e^r L(o \mid z, d)\eta_z(dr, dd), \\ N(o) &= \int N_z(o)\mu(dz). \end{aligned} \tag{8.24}$$

Proposition 8.14 (Exact marked feedback). The record density is $N(o)$. Exact conditioning of this sector gives the closed updates

$$\begin{aligned} \mu^+(dz) &= \frac{N_z(o)}{N(o)}\mu(dz), \\ \eta_z^+ &= \big((r, d) \mapsto (r + \log L(o \mid z, d) - \log N_z(o), d)\big)_{\#}\eta_z. \end{aligned} \tag{8.25}$$

Between records, eqs. (8.16) and (8.17) transport the first coordinate of η_z and leave its mark fixed. Hence, for the stage kernels eq. (4.31) of the record-interrupted dynamics with these records and updates, the retention R_{ret} to the public laws, marked spectra, and rows satisfies $(R_{\mathrm{ret}} \times \mathrm{id})_{\#} J_k(x; \cdot) = J_k^c(R_{\mathrm{ret}}(x); \cdot)$ for a retained stage kernel J_k^c, at every state whose flow interval lies inside the maximal interval of proposition 8.8, so the marked state determines every finite adaptive history law of these records, feedback updates, and original smooth-flow intervals up to the stopping time of definition A.3.

Proof. Since $d\kappa_z = e^{\ell_z} dA$, integration of the likelihood gives eq. (8.24). Bayes' rule gives the displayed marginal update and $d\kappa_z^+/dA = e^{\ell_z} L(o \mid z, T)/N_z(o)$. Taking the logarithm and pushing A gives the marked update. The fine reference remains fixed. The conditional equation after the update is therefore the same equation with this new conditional law. Inductively, the inherited flow gives the next retained state, the inherited record kernel gives the same next-record law, and the inherited update gives the same posterior retained state, which is the stated identity of stage kernels. Actions selected from retained histories by Borel policies in the sense of definition A.3 preserve this induction, because each stage kernel then depends on the past only through retained quantities. The history laws are defined when the flow map $(t, x) \mapsto \Phi_t(x)$ is jointly Borel on its domain, which definition 4.28 requires and which is part of the hypothesis on these charts. Any reference or likelihood parameters changed by records must also be retained and updated by a rule that depends only on this state. □

The composite of flow intervals and these updates is the record-interrupted dynamics of definition 4.28, with record kernel $N(o)$ at the pre-record state and the update eq. (8.25), stopped at the maximal existence time. The potential contains no record term and is not monotone across records. Records act only at their times: no rule changing the references is declared here, and between records each conditional relaxes toward its fixed reference by eq. (8.19), so the effect of a record on the private sector decays.

The update is sectorwise. A product of separate belief and model likelihoods preserves the recognition restriction $q_i \otimes s_i$. A likelihood jointly coupling the two sectors need not preserve that product; exact joint conditioning would require a joint recognition state, or a separately declared projection. Marked feedback can also destroy equality of unmarked spectra across public coordinates. The augmented marked state still closes, but the simpler conclusion of corollary 8.10 then need not hold.

For an explicit retained sample law, use

$$Q^c(dz, dr, dd) = \mu(dz)e^r\eta_z(dr, dd). \tag{8.26}$$

It is normalized and reproduces every declared marked record. It is the pushforward under the current, law-dependent statistic $(z, v) \mapsto (z, \ell_z(v), T(v))$. Different assignments of density values to private points within a mark can give the same retained state. If the permitted records distinguish every private point, the mark may have to retain that point, eliminating this compression. The boundary protocol therefore fixes what can be forgotten. Predictive-state constructions likewise make sufficiency relative to future observations [177]; the formulas here specify the state and update for this particular KL architecture.

8.5 What is still required for formation

Positive source temperature does not give exact isolation by itself. For optimized rows, every allowed finite-cost edge has positive weight. For a dynamic row, set

$$\theta_{jk} = \log \frac{\beta_j/\pi_j}{\beta_k/\pi_k}. \tag{8.27}$$

The Fisher row equation from eq. (4.4) gives

$$\dot{\theta}_{jk} = -(D_j - D_k) - T_q\theta_{jk}. \tag{8.28}$$

Multiplying by $e^{T_q t}$ and integrating shows that finite cost differences keep every such ratio finite on a regular finite interval. Initially positive weights therefore stay positive there. Uniformly bounded cost differences also bound these ratios for all time and prevent asymptotic disappearance of an allowed weight. A formed collective may nevertheless keep nonzero boundary interactions; isolation is not required for inheritance.

The construction above compresses private conditional information of each agent separately, and only up to the equimeasurability quotient of its declared reference. It does not combine several distinct public agent laws into one smaller collective state. Retaining their complete labeled tuple and calling it a parent would leave that aggregation problem untouched; whether a retention of such a tuple compresses is decided by clause (F4) of definition 6.36, which requires a noninjective retention whose fibers some admitted child-level probe distinguishes. Nor does recognizing a persistent source-preference group establish a compatible collective retention. The distinction between a coarse observable and an autonomous macroscopic evolution is also expressed by flow intertwining in Barnett and Seth [15].

Remark 8.15 (Aggregation with conditional information retained). The construction of this chapter selects no group and involves no grouping of agents, and its private sector is unchanged by deleting every peer cost. A recursive formation that uses the conditional-spectrum state as part of its retention is an instance of open problem 6.38. For identity public and private channels with base-only references, private conditionals do feed back into public interactions, and the joint conditional spectrum is an exact completion (proposition 11.6); it compresses only through equimeasurable rearrangements of the private value vector and keeps every public child state. What remains open is a coupled architecture under the reciprocal potential, with law-dependent public channels or rows that read private data, whose exact completion compresses beyond such rearrangements or forgets distinct public child states.

Chapter 9

State-Controlled Erasure and Persistent Source Preferences

This chapter concerns the reciprocal law potential of definition 4.2, in its exact Gibbs-reduced form eq. (4.6); nothing in it is a statement about the per-agent or global free-energy pictures. Inside that potential there is a conditional first-stage construction. A prescribed channel can lose its dependence on some internal laws when a dynamically evolving gate coordinate enters a flat part of that channel. The preferred sources of the remaining agents can change, and the retained dynamics can then become exactly autonomous. The mechanism is an engineered constitutive choice, and both of its conclusions follow from a declared ordering of scales. It supplies neither a general explanation of sufficient interfaces nor a second formation stage. Measured against the formation standard definition 6.36, it is a single-stage selection with exact inheritance by a decoupled-factor projection, whose interaction-necessity clause is not determined by the results proved here; section 9.5 records the clause-by-clause status and the controls. The general laws of the preceding chapters remain the framework; the existence example below uses an explicitly invariant family of positive finite laws.

9.1 A homogeneous population with internal gate coordinates

Write $b_x(k) = (1 + kx)/2$ on $\{-1, 1\}$, set $d(x, z) = D_{\mathrm{KL}}(b_x \| b_z)$, and put $d_0(x) = d(x, 0)$. Four agents carry

$$q_i = b_{x_i} \otimes b_{z_i} \otimes b_{h_i}, \qquad s_i = b_{y_i} \otimes b_{w_i} \otimes b_{\widehat{h}_i}. \tag{9.1}$$

The first coordinate is called public, the second private, and the third is a gate coordinate. These words refer to a declared observable algebra: the peer channel below transmits samples of the public coordinate, it reads the private coordinate of a source only through that source's mean multiplied by the gate, and the admitted response class of proposition 9.4 reads public coordinates only. Their unit Fisher metric has diagonal entries $1/(1 - u^2)$ for each mean u. The product preparation within each sector is an assumption. It is invariant under the full table dynamics specified below, rather than a projection imposed after each step.

Take $J = \log 3$ and $h_* = -1/2$. Use the proper reference $r = b_0 \otimes b_0 \otimes b_{h_*}$ and normalized evaluator

$$G_J(k \mid m) = \frac{e^{Jkm}}{2 \cosh J}, \tag{9.2}$$

on the public coordinates, with independent private output b_0 and gate output b_{h_*}. The local term is exactly

$$\begin{aligned} E_{0,i} = d_0(x_i) + d_0(y_i) - Jx_iy_i + \log\cosh J \\ + d_0(z_i) + d_0(w_i) + d(h_i, h_*) + d(\widehat{h}_i, h_*). \end{aligned} \tag{9.3}$$

All agents have the same reference and evaluator. Every nonself agent is an allowed source, with uniform prior and fixed temperature $T > 0$. We use the exact Gibbs-reduced version of the reciprocal potential. The bits are reference coordinates: a change of presentation carries the reference laws, channels, and readout maps with it, so the KL and Fisher covariance statements apply.

Define $\chi(r) = 0$ for $r \le 0$, $\chi(r) = 1$ for $r \ge 1$, and, for $0 < r < 1$,

$$\chi(r) = \frac{e^{-1/r}}{e^{-1/r} + e^{-1/(1-r)}}. \tag{9.4}$$

This increasing function is smooth, all its derivatives match at the endpoints, and $M_\chi = \max|\chi'| < \infty$. Fix $h_0 = 1/4$ and a gate width $\Delta > 0$, and set

$$H_q = \frac{1}{4}\sum_i h_i, \quad H_s = \frac{1}{4}\sum_i \widehat{h}_i, \quad \alpha(H) = \chi((H - h_0)/\Delta). \tag{9.5}$$

The gate is a nonlocal function of internal laws. Its threshold and flat region are constitutive data. There is no external time schedule; the schedule is internal, set by the initial gate positions and by h_*, as lemma 9.1 makes precise.

Choose gains $0 < \varepsilon, \varrho < 1$. On a belief edge $j \to i$, let the public output kernel be $(1-\varepsilon)b_{x_i} + \varepsilon\delta_{k_{\mathrm{pub}}}$, and independently draw private output from $b_{\alpha(H_q)\varrho z_j}$ and gate output from b_{h_*}. Its compared law is

$$\nu_{ij}^q = b_{(1-\varepsilon)x_i + \varepsilon x_j} \otimes b_{\alpha(H_q)\varrho z_j} \otimes b_{h_*}. \tag{9.6}$$

This normalized channel is strictly positive on the interior and depends smoothly on the current marginal means. It transmits public samples, mixed with the receiver's own public law, and uses the source's private mean rather than its private sample. At $\alpha = 0$ every private cost therefore equals $d_0(z_i)$ by construction. The model channel exchanges (x, z, h) with $(y, w, \widehat{h})$. Define

$$D_\varepsilon(x, v) = d(x, (1-\varepsilon)x + \varepsilon v). \tag{9.7}$$

The reciprocal peer cost becomes

$$D_{ij}^q = D_\varepsilon(x_i, x_j) + d(z_i, \alpha(H_q)\varrho z_j) + d(h_i, h_*). \tag{9.8}$$

No fusion cost has been added. Because the public kernel mixes in the receiver's own law, every public peer cost and force is of order ε^2.

The definitions extend to arbitrary positive tables on each three-bit space by taking the indicated marginal means. At a product table, the derivative of the receiver entropy is additive in its coordinates. Each compared law is a product, and every channel derivative is a function of a single

coordinate because the channel parameters depend on marginal means. The local evaluator also gives additive scores within each sector. The full Fisher score is consequently a sum of coordinate scores on eq. (9.1). Differentiating a product shows that the full table flow of theorem 4.6 preserves this family and induces the stated Bernoulli Fisher metric. This argument includes incoming and channel derivatives.

9.2 Finite entry into an exactly erasing region

Lemma 9.1 (Permanent gate closure under the reciprocal flow). For sufficiently small $\Delta > 0$ and then sufficiently small $\varrho > 0$ relative to Δ, there is an open class within eq. (9.1), with

$$|z_i|, |w_i| < \sigma_{\max} = 3/4, \qquad h_i, \widehat{h}_i \in (h_0 + \Delta/2, h_0 + 3\Delta/4), \tag{9.9}$$

and public means unrestricted, on which, for as long as the solution exists, both gates reach zero by the time $3\Delta/(2m_K)$ defined below and remain zero afterward. The gate means follow an autonomous relaxation toward h_* perturbed by a term of size at most $B\varrho M_\chi/\Delta$. On the admitted class of theorem 9.2 the solution exists for all time.

Proof. The exact envelope derivative eq. (4.8) gives

$$\dot{h}_i = -(1 - h_i^2) \left[2(\operatorname{atanh} h_i - \operatorname{atanh} h_*) + \frac{\alpha'(H_q)}{4} \sum_{a,j} \beta^*_{aj} \partial_\alpha d(z_a, \alpha\varrho z_j) \right]. \tag{9.10}$$

The factor two comes from the local term and the receiver's source-independent gate cost. For $|z_a|, |z_j| \le \sigma_{\max}$ and $\varrho \le 1/2$,

$$\partial_\alpha d(z_a, \alpha\varrho z_j) = \varrho z_j \frac{\alpha\varrho z_j - z_a}{1 - \alpha^2\varrho^2 z_j^2}. \tag{9.11}$$

Its absolute value is bounded by $B\varrho$, where one may take $B = 3\sigma_{\max}^2/[2(1 - \sigma_{\max}^2/4)]$. The four row sums equal one, so the correction to the gate force is bounded by $B\varrho M_\chi/\Delta$.

On $K = [1/8, 1/2]$, define the positive constants

$$\begin{aligned} m_K &= \min_{h \in K} 2(1 - h^2)(\operatorname{atanh} h - \operatorname{atanh} h_*), \\ M_K &= \max_{h \in K} 2(1 - h^2)(\operatorname{atanh} h - \operatorname{atanh} h_*), \qquad V_K = M_K + m_K/2. \end{aligned} \tag{9.12}$$

Choose $B\varrho M_\chi/\Delta \le m_K/2$. While its coordinates stay in K, each gate coordinate has $\dot{h}_i \le -m_K/2$ and $|\dot{h}_i| \le V_K$. Also choose Δ so that the initial interval lies in K and

$$\frac{3V_K\Delta}{2m_K} < h_0 - 1/8. \tag{9.13}$$

The sector mean reaches h_0 by time $3\Delta/(2m_K)$, before any coordinate can leave K. On $H_q = h_0$, both α and α' vanish. Thereafter

$$\dot{h}_i = -2(1 - h_i^2)(\operatorname{atanh} h_i - \operatorname{atanh} h_*). \tag{9.14}$$

All these coordinates start above h_* and decrease toward it. Their mean therefore cannot leave the closed-gate region. The model-sector proof is identical.

It remains to justify the private bounds used in the argument. At $\varrho = 0$ the private field is $\dot{z} = -2(1 - z^2)\,\mathrm{atanh}\, z$, strictly inward at $z = \pm\sigma_{\max}$. Private derivatives of eq. (9.8) converge uniformly to this field as $\varrho \to 0$, uniformly over normalized source rows and $0 \le \alpha \le 1$. The incoming terms are of order ϱ; differentiating a private mean does not differentiate α. Compactness therefore preserves the strict inward signs for sufficiently small ϱ. The same holds for w. This closes the interval argument. The private and gate means stay in compact subsets of $(-1, 1)$. On the admitted class of theorem 9.2 so do the public means, by the box invariance proved there, and the solution therefore continues for all time on that class. □

Let τ be the later of the two gate-closure times. The construction uses a smooth but nonanalytic channel: its exactly flat region makes finite entry into exact erasure possible. Decreasing a small nonzero analytic gate would not establish this conclusion. The gate mean is an internal clock: its motion is monotone and decoupled from the other coordinates up to a perturbation that the parameter choice makes subordinate.

9.3 A change of preferred sources without synchronization

Theorem 9.2 (Changing and persistent source preferences). For sufficiently small $\varepsilon > 0$, admit two public states in $(l, u)^2$ and two in $(-u, -l)^2$, with $l = 0.49$, $u = 0.51$, including every assignment of these signs to the four labels. These boxes are forward invariant. After τ, each agent's unique preferred source in both sectors is the other agent in its box. There are initial private states for which the preferred pairs are instead crossed between these boxes before closure. The public states within each final pair can remain distinct at every finite time.

Proof. For the local public flow, the inequalities

$$Jl - \mathrm{atanh}\, l > 0, \qquad \mathrm{atanh}\, u - Ju > 0, \tag{9.15}$$

give strict inward signs on every face of the two boxes. They follow by differentiating $Jv - \mathrm{atanh}\, v$ near its zero $v = 1/2$. Set $c = (1 - \varepsilon)x + \varepsilon v$. Direct differentiation gives

$$\begin{aligned} \partial_1 D_\varepsilon(x, v) &= \mathrm{atanh}\, x - \mathrm{atanh}\, c + (1 - \varepsilon)\frac{c - x}{1 - c^2}, \\ \partial_2 D_\varepsilon(x, v) &= \varepsilon \frac{c - x}{1 - c^2}. \end{aligned} \tag{9.16}$$

Both are uniformly $O(\varepsilon^2)$ on $[-u, u]^2$; the first-order terms cancel in the first line. The full public force has one outgoing row and three incoming sources, each weighted by a number in $[0, 1]$. A sufficiently small ε preserves every strict inward margin, independently of the private state and the gate.

After gate closure, the private and gate costs in a row do not depend on its source, so Gibbs ranking uses D_ε alone. Taylor's theorem in the first argument of d gives

$$\frac{1}{2}(x-c)^2 \le d(x,c) \le \frac{(x-c)^2}{2(1-u^2)}. \tag{9.17}$$

A same-box source has separation at most $u-l$, whereas an opposite-box source has separation at least $2l$. Since $(u-l)^2/(1-u^2) < (2l)^2$, the same-box partner is the unique preferred source throughout the invariant region. Cross-pair weights remain positive.

For an explicit initial reorganization, choose private means $\pm\sigma$, with $0 < \sigma < \sigma_{\max}$, in both sectors, assigning each same-private-sign pair one positive and one negative public member. Initially $\alpha \ge 1/2$. The private advantage of the same-sign partner over an opposite-sign source is

$$d(\sigma, -\alpha\varrho\sigma) - d(\sigma, \alpha\varrho\sigma) = 2\sigma\,\mathrm{atanh}(\alpha\varrho\sigma) \ge \varrho\sigma^2. \tag{9.18}$$

The public cost range is at most $2u^2\varepsilon^2/(1-u^2)$. Choosing this smaller than $\varrho\sigma^2$ makes the initial preferred pairs the crossed pairs. Strict inequalities persist under small perturbations of the initial private values. Thus the final persistent grouping is not an unchanged initial source assignment.

Choose initial public states within a box separated by a fixed positive distance. Their speeds are uniformly bounded, whereas $\tau \le 3\Delta/(2m_K)$. Taking Δ sufficiently small keeps them distinct through τ. Afterward the public field is autonomous, smooth, and permutation equivariant. Its equality diagonal for a pair is invariant. Backward uniqueness precludes first entry into that diagonal at a finite time. This proves finite-time distinctness. □

The preference change is a switch between two prescribed scales: before closure the private advantage $\varrho\sigma^2$ dominates the public cost range of order ε^2, and after closure only the public costs remain. The parameter choices are compatible: first choose a sufficiently small Δ, then ϱ sufficiently small relative to Δ, and finally ε sufficiently small that ε^2 is small relative to ϱ and the fixed public inward margins. All means stay in compact subsets of $(-1,1)$, giving global continuation in this admitted family. The two final groups are the two wells of the bistable single-agent energy $d_0(x) + d_0(y) - Jxy$, whose minima are $x = y = \pm 1/2$, and box membership is the supplied initial well.

The temperature enters this ordering. The proofs above use only that row weights lie in $[0,1]$, so $T > 0$ may be fixed before the gains are chosen. At fixed T the source-dependent part of every row cost is $O(\varrho + \varepsilon^2)$, so every Gibbs row is within $O((\varrho+\varepsilon^2)/T)$ of uniform, and a preferred source is then the largest entry of a nearly uniform row. Quantitatively, eq. (9.17) gives after closure

$$\beta^*_{i,\mathrm{same}} \ge \frac{1}{1+2e^{-\Delta_\varepsilon/T}}, \qquad \Delta_\varepsilon = \varepsilon^2\left[2l^2 - \frac{(u-l)^2}{2(1-u^2)}\right], \tag{9.19}$$

and at the initial time of the explicit crossed preparation, with $\alpha \ge 1/2$, the crossed partner has weight at least

$$\left[1 + 2\exp\left(-\frac{\varrho\sigma^2 - 2u^2\varepsilon^2/(1-u^2)}{T}\right)\right]^{-1}. \tag{9.20}$$

Near one-hot selection therefore requires T of order ε^2 or smaller. The next result holds uniformly in T, so it covers that regime.

The heterogeneity of the final pairs is transient, as the next result shows.

Proposition 9.3 (Each preferred pair converges to a common state). Fix the parameters of theorem 9.2. There is $\varepsilon_0 > 0$ such that for every $T > 0$ and $0 < \varepsilon < \varepsilon_0$ the public system after τ has exactly one equilibrium in the closed boxes $Q = ([l, u]^2)^2 \times ([-u, -l]^2)^2$, ordered by box. At this equilibrium the two members of each box have equal public states, the two boxes carry opposite states $\pm(x_\varepsilon, y_\varepsilon)$, and $(x_\varepsilon, y_\varepsilon) \to (1/2, 1/2)$ as $\varepsilon \to 0$. Every admitted trajectory converges to it exponentially. In particular the in-pair public separation tends to zero, and each retained empirical measure η_I of eq. (9.25) converges to a Dirac mass.

Proof. After τ the public coordinates follow the unit Fisher gradient flow of the public part $\mathcal{E}_{\mathrm{pub}}$ of eq. (9.22), so

$$\frac{d\mathcal{E}_{\mathrm{pub}}}{dt} = -\sum_i (1 - x_i^2)(\partial_{x_i}\mathcal{E}_{\mathrm{pub}})^2 - \sum_i (1 - y_i^2)(\partial_{y_i}\mathcal{E}_{\mathrm{pub}})^2 \le 0, \tag{9.21}$$

with equality exactly at equilibria. The set Q is compact and forward invariant by theorem 9.2. By the invariance principle [128], the ω-limit set of each trajectory is a nonempty connected set of equilibria in Q.

At $\varepsilon = 0$ the public field decouples into single-agent fields. In the positive box their zeros solve $\operatorname{atanh} x = Jy$ and $\operatorname{atanh} y = Jx$. With the increasing map $g(t) = \operatorname{atanh}(t)/J$ this reads $x = g(y)$ and $y = g(x)$. If $x > g(x)$, then $y = g(x) < x$ and $x = g(y) < g(x)$, a contradiction; the case $x < g(x)$ is symmetric. Hence $\operatorname{atanh} x = Jx$, and since $\operatorname{atanh}(x)/x$ increases from one and $J > 1$, the only solution in $(0, 1)$ is $x = 1/2$. The Jacobian of $(\operatorname{atanh} x - Jy, \operatorname{atanh} y - Jx)$ there has diagonal entries $4/3$ and off-diagonal entries $-J$, and $16/9 - J^2 > 0$, so the zero is nondegenerate. The negative box is the mirror image. Thus at $\varepsilon = 0$ the field on Q has exactly one zero, and it is nondegenerate and interior.

The peer cost D_ε is a smooth function of (x, v, ε) that vanishes to second order in ε, so its derivatives in (x, v) up to second order are bounded by $C_1\varepsilon^2$ on Q. For one Gibbs row, $|d\beta_j| \le (2/T) \max_k |dD_k| \beta_j(1 - \beta_j)$. On Q, eq. (9.17) and eq. (9.19) give $1 - \beta_j \le 2e^{-\Delta_\varepsilon/T}$ for the same-box source and $\beta_j \le e^{-\Delta_\varepsilon/T}$ for every other source, so $\beta_j(1 - \beta_j) \le 2e^{-\Delta_\varepsilon/T}$ with $\Delta_\varepsilon = c_2\varepsilon^2$, $c_2 > 0$. Hence $|d\beta_j| \le 4C_1(\varepsilon^2/T)e^{-c_2\varepsilon^2/T} \le 4C_1/(ec_2)$, uniformly in $T > 0$ and ε. The peer part of the public field is a sum of rows times first derivatives of D_ε, so its $C^1(Q)$ norm is $O(\varepsilon^2)$ uniformly in T, and the public field converges to the decoupled field in $C^1(Q)$ as $\varepsilon \to 0$, uniformly in T. The decoupled field is bounded away from zero on Q outside any neighborhood of its zero, and the implicit function theorem gives a unique nondegenerate zero inside a small neighborhood. This proves uniqueness and the limit $(1/2, 1/2)$.

The post-closure public system has identical constitutive data for all agents and uniform nonself priors, so it is equivariant under permutations of the agents. It is also equivariant under the global reversal $(x, y) \mapsto (-x, -y)$, since d_0 is even, Jxy is invariant, and $d(-x, -c) = d(x, c)$. The transposition of two box-mates, and the composition of global reversal with the exchange of the two boxes, map Q to itself and therefore fix the unique equilibrium. This gives equal box-mates and opposite boxes.

Every ω-limit set is therefore this single equilibrium. It attracts every interior point of Q, so it is a local minimum of $\mathcal{E}_{\mathrm{pub}}$: a nondegenerate critical point of a gradient flow that is not a minimum has a stable set of positive codimension. The linearized gradient flow at a nondegenerate minimum has negative spectrum, which gives exponential convergence. □

Within the regime of proposition 9.3, persistent heterogeneous pairs would require heterogeneous constitutive data. With identical agents each box carries a single attracting equilibrium, and proposition 9.3 forces its members together. If agent i instead carried its own coupling $J_i = \log 3 + \eta_i$, the single-agent equilibria near $1/2$ would separate to first order by $|\eta_i - \eta_k|/(8/3 - 2\log 3)$ per coordinate, while the inward margins of eq. (9.15) survive for small $|\eta_i|$; box-mates would then be expected to converge to distinct states. We do not prove this variant here. A gate driven by a public statistic, such as the polarization $\frac{1}{4}\sum_i d_0(x_i)$, would make erasure depend on public relaxation rather than on a dedicated clock; its entry estimate is also not established.

9.4 The exact inherited state

For $t \geq \tau$ the reduced potential separates exactly:

$$\begin{aligned}\mathcal{E}_{\mathrm{red}} &= \mathcal{E}_{\mathrm{pub}}(\mathbf{x}, \mathbf{y}) \\ &\quad + 2\sum_i [d_0(z_i) + d_0(w_i) + d(h_i, h_*) + d(\widehat{h}_i, h_*)].\end{aligned} \tag{9.22}$$

Here $\mathcal{E}_{\mathrm{pub}}$ is the coupled local KL and Gibbs-reduced reciprocal peer potential with costs D_ε. The product Fisher metric separates as well. After closure the state space is the product of eight public means and sixteen private and gate means, the flow is a product flow, and the second factor relaxes autonomously to $z = w = 0$ and $h = \widehat{h} = h_*$. Discarding that factor is projection onto a factor of a product flow, the most elementary exact lumping [110], and it gives an exact autonomous public system without fitting another potential or mobility.

This independence is generally absent before gate closure. To see that this is an actual change of the vector field, expand one reduced row with $\alpha > 0$ fixed. If $A_{ij} = (x_j - x_i)^2/[2(1 - x_i^2)]$, its first mixed public-private term is

$$\frac{\alpha\varrho\varepsilon^2}{T} z_i \operatorname{Cov}_{j\neq i}(A_{ij}, z_j). \tag{9.23}$$

This follows by expanding $D_\varepsilon = \varepsilon^2 A + O(\varepsilon^3)$, $d(z_i, \alpha\varrho z_j) = d_0(z_i) - \alpha\varrho z_i z_j + O(\varrho^2)$, and the logarithm of the row normalization in powers of the costs divided by T; the expansion is valid at fixed T as the gains tend to zero. The covariance is under the uniform source prior. It is not identically zero even within the crossed-preference preparation. Fix $v = (-b, -a, a, b)$ with $l < a < b < u$, keep $z = \sigma(1, -1, 1, -1)$ fixed, and vary only public $x_i = r v_i$ near $r = 1$. The summed coefficient is

$$\frac{2\sigma^2 r^2}{9}\left[\frac{a^2 - 3b^2 + 6ab}{1 - r^2 b^2} + \frac{b^2 - 3a^2 + 6ab}{1 - r^2 a^2}\right]. \tag{9.24}$$

Both numerator factors are positive on the stated intervals, so this expression has positive derivative in $r > 0$. Two fixed admissible values of σ therefore give different public directional gradients at leading order $\varrho\varepsilon^2$. The differentiated remainder is $O(\varrho\varepsilon^3 + \varrho^2\varepsilon^2)$ on these compact sets. For

sufficiently small positive gains, equal public data thus have different public derivatives while the gate is open. This expansion establishes nonidentity of the preclosure force; eq. (9.22) is an exact postclosure identity. For T of order ε^2 or smaller the cumulant expansion is unavailable, and nonautonomy before closure would need a direct comparison of the forces under two private assignments.

For each selected pair I, retain its size and the empirical measure

$$\eta_I = \frac{1}{|I|}\sum_{i\in I}\delta_{(x_i,y_i)}, \qquad \mathsf{M} = \sum_I |I|\eta_I. \tag{9.25}$$

This is a deterministic configuration statistic, not an assumed probability distribution over uncertain population configurations. Let $V[v;\mathsf{M}]$ be the public Fisher vector field at the public state $v=(x,y)$, computed from $\mathcal{E}_{\mathrm{pub}}$. It is determined by v and the counting measure M: source sums use $\mathsf{M}-\delta_v$, and the derivative includes both receiver and incoming source terms from eq. (9.16). The normalized Gibbs denominator is computed from the same counting measure. Permutation equivariance makes this definition independent of labels, including multiplicities.

Proposition 9.4 (Exact parent evolution and public responses). The inherited parent equations are

$$\frac{d}{dt}\int f(v)\eta_I(dv) = \int \nabla f(v)\cdot V[v;\mathsf{M}]\eta_I(dv), \tag{9.26}$$

for every continuously differentiable test function. Among curves of realizable finite-atom states, that is, empirical measures of solutions of the public system, two curves with equal initial retained measures coincide at all later times. The retained measures also determine every declared fresh public response whose source selection depends only on public states, including joint responses with a shared sampled child.

Proof. Differentiate the finite sum in eq. (9.25) using the exact public equations. This gives eq. (9.26). Equal retained parent measures admit enumerations that differ only by within-pair permutations. Uniqueness and permutation equivariance of the public system give identical evolved measures, proving the uniqueness assertion among realizable curves. For a fresh response, select the child or children using the declared normalized public kernel, then integrate their public sample laws against the retained empirical measures. Both the selection probabilities and sample laws are determined by the retained states. For shared-source outputs, use one shared draw rather than multiplying separate marginals; the same calculation preserves their joint law. □

The proposition does not assert uniqueness for eq. (9.26) among general measure-valued curves; that would need a Lipschitz stability estimate for V in (v,M). A single η_I has no autonomous dynamics, since $V[v;\mathsf{M}]$ depends on the counting measure of both pairs; only the pair of parents is closed. The descent also holds for every partition of the four agents, so inheritance does not distinguish the selected pairs. In the sense of definition 6.18, the retention of either pair is therefore the pair (η_I,η_J) of both retained measures, which intertwines the public flow; a retention of η_I alone is controlled projectable with η_J as exterior state, in the sense of definition 10.5.

The parent can carry belief and model marginals $q_I=\int b_x\eta_I(d(x,y))$ and $s_I=\int b_y\eta_I(d(x,y))$, with normalized public evaluator G_J. These two marginals alone are insufficient: η_I retains the paired

public roles and receiver-specific routing information. The inherited equations are eq. (9.26), not a Fisher flow of the reciprocal potential for the two mixture marginals. The parent is therefore not an agent of the children's type: no law pair with an evaluator and a reciprocal KL interaction carries its dynamics, and the construction cannot be iterated as it stands. A recursion would need a retention whose image is again a law pair with an inherited reciprocal interaction.

The compression here discards the private and gate laws, which no admitted response reads after closure: the admitted family has twenty-four continuous means, and the retained empirical state keeps eight public means, up to within-pair permutations. It is a decoupled-factor projection followed by a finite label quotient, and it keeps every public child state. Cached private readouts, feedback that changes the erased channels, and protocols that later inspect discarded private coordinates are outside the response class. The reference, gate, initial sign-box preparation, and boundary restrictions are all supplied assumptions. The dynamics select changing source preferences and a persistent grouping within them, but do not discover the erasing architecture.

9.5 Status against the formation standard

We report the construction by the scheme following definition 6.36. The objective is the exact Gibbs-reduced reciprocal potential eq. (4.6), and the dynamics is its Fisher flow on finite positive tables, restricted to the invariant product family eq. (9.1). (F1) holds in the family grade: the admitted class of lemma 9.1 and theorem 9.2 is relatively open in that family, which has twenty-four means against fifty-six table coordinates, so its codimension is thirty-two; openness in the full table chart is not shown. The selector is declared as follows. At each time each agent's preferred source in each sector is the source of smallest reciprocal row cost eq. (9.8), computed from the original formulas at the current state, all minimizers being kept at a tie; the selected partition consists of the connected components of the graph that joins every agent to its preferred sources in either sector. The family has no real constant. The persistence criterion is the invariant block relation of the box pairs after τ, and the handoff is the closure time τ, a function of the state; (F2) holds. The parent protocol rule A_I consists of the fresh public responses whose source selection depends only on public states, including joint responses with a shared sampled child; the child-level probe rule B_I consists of the peer channels of eq. (9.6) applied to each member, whose public output reads the member's public mean and whose private output reads its private mean through the gate, together with the protocols of A_J for the merged singletons. Inheritance, (F3), holds for A_I through proposition 9.4, with the retention of each pair taken to be the pair of retained measures and with the qualification that the parent state is an empirical measure rather than a law pair.

Compression, (F4), holds in the letter of the clause through the label quotient only. The private and gate coordinates that the retention discards provide no witness: on the reachable post-handoff set the gates are closed, every gated private output has law b_0, and no probe of B_I reads those coordinates. Within a pair, however, the empirical measure forgets which member carries which public state, and the reachable set contains, with a configuration whose box-mates are distinct, the configuration in which their public states are exchanged; the peer channel applied to one of them separates the two. The witness is transient, since box-mates converge by proposition 9.3, and it is of symmetry type, since the fibers are orbits of within-pair permutations, which are symmetries of

the fine field and of the protocols of A_I. Recursion, (F6), fails, since there is a single stage and the two final boxes stay separated.

For the controls of definition 6.33, every cost eq. (9.8) is a peer cost, because its compared law reads the source's public and private means, and the source sets exclude the receiver, so the label-preserving and renormalized nulls coincide. Deleting the peer costs leaves the local terms. The public coordinates then follow the single-agent flows, which keep the two sign boxes invariant by eq. (9.15); the private means relax monotonically toward zero without changing sign; and the gate means follow eq. (9.14) with the factor two replaced by one, so the gates still close, at about half the speed. The selector reads the original costs at the null state. Both flows start from the same state, so their first selected partitions agree; on the explicit crossed preparation it is the partition into crossed pairs. After closure along each flow, the private costs of a row do not depend on the source and the public states lie in the boxes, so by eq. (9.17) both flows end with the partition into box pairs. Between these, the four agents need not change their preferred sources at the same instant, and the intermediate partitions, which depend on the order of those changes, are not determined by the results proved here. Clause (F5) is therefore neither established nor shown to fail: P and P^0 have the same first and last partitions, and their equality is not proved. On the explicit preparation the first and last partitions are fixed by the initial private and public sign patterns, which both flows preserve; the grades are not assessed. No selector-free discrepancy beyond the closure times is established. The frozen-environment control holds the other agents' laws, and with them the gate statistic of each agent's own force, at their initial values; its partition sequence is not determined here. The construction is a single-stage selection with exact inheritance by a decoupled-factor projection, with (F5) undetermined; it is not a two-stage selection and not a recursive formation.

Chapter 10

Public Compression and Inherited Responses

This chapter concerns the reciprocal law potential of definition 4.2, in its exact Gibbs-reduced form eq. (4.6), with identity channels; it makes no claim about the per-agent or global free-energy pictures. A nonlinear projection of the reciprocal Fisher flow can discard continuous public differences without an erasure channel or finite-time equality. The projection is the asymptotic phase of the approach to a stable common reference, so it compresses the decaying part of the transient. This autonomous parent nevertheless fails to predict ordinary child sampling, and it is the only parent of its kind. A separate identifiability result explains why an unrestricted Gibbs interface retains every exposed child law, and how a fixed channel changes what it retains. The block is supplied and nothing is selected, so these are retention results rather than formation results; section 10.5 records their status against the formation standard definition 6.36.

10.1 An asymptotic collective phase near a stable reference

Let A and B be probability measures on standard Borel spaces K and M. Suppose there are measurable functions $\varphi : \mathsf{K} \to \{-1, 1\}$ and $\psi : \mathsf{M} \to \{-1, 1\}$ with zero reference means. For example, either space may be a balanced two-point factor times an arbitrary probability space. The agent laws below need not factor over these coordinates. Fix $|J| < 1$ and the normalized reference

$$R(dk, dm) = \frac{e^{J\varphi(k)\psi(m)}}{\cosh J} A(dk)B(dm). \tag{10.1}$$

Its model marginal is B, and its evaluator is $G(m, dk) = e^{J\varphi(k)\psi(m)}A(dk)/\cosh J$. Thus

$$D_{\mathrm{KL}}(q_i \otimes s_i \| R) = D_{\mathrm{KL}}(q_i \| A) + D_{\mathrm{KL}}(s_i \| B) - J\mathbb{E}_{q_i}\varphi\mathbb{E}_{s_i}\psi + \log\cosh J. \tag{10.2}$$

For $J \neq 0$ the belief and model laws interact through the evaluator.

Take $N \geq 2$ identical agents, identity peer channels, all other agents as sources, uniform nonself source priors, and fixed positive temperatures. Use the exact Gibbs-reduced reciprocal potential of eq. (4.6), with unit Fisher mobility. No source or channel derivative is frozen. Introduce the Banach spaces

$$\begin{aligned} \mathsf{E}_q &= \{u \in L^\infty(A) : \mathbb{E}_A u = 0\}, \quad q_u = \frac{e^u}{\mathbb{E}_A e^u} A, \\ \mathsf{E}_s &= \{v \in L^\infty(B) : \mathbb{E}_B v = 0\}, \quad s_v = \frac{e^v}{\mathbb{E}_B e^v} B, \end{aligned} \tag{10.3}$$

and $\mathsf{E} = \mathsf{E}_q \times \mathsf{E}_s$, with the maximum norm. These are full bounded log-density charts, rather than a finite-parameter family, and the flow is the chart flow of definition 4.13 for the reduced potential, which has no row coordinates. Let $h_i^q \in L^\infty(A)$ be the first variation of the reduced potential with respect to q_i, given by eq. (4.8) and eq. (4.10), and similarly h_i^s. The Fisher-Rao flow $\partial_t \log q_i = -(h_i^q - \mathbb{E}_{q_i} h_i^q)$ becomes, after centering against A,

$$\dot{u}_i = -(h_i^q - \mathbb{E}_A h_i^q), \qquad \dot{v}_i = -(h_i^s - \mathbb{E}_B h_i^s). \tag{10.4}$$

By eq. (4.8) the first variation of the reduced potential is that of eq. (4.4) at the Gibbs rows, which are smooth functions of the costs. The channels are identities between agents sharing A and B, and $\log(dR/d(A \otimes B)) = J\varphi\psi - \log\cosh J$ is bounded, so the sufficient condition of proposition 4.14 holds for the costs, and the Gibbs weights add only smooth bounded factors. On every bounded ball of E^N the right side is in fact a smooth map into E^N: exponentiation is analytic in the Banach algebra L^∞, the normalizers and compared densities are bounded away from zero, and the KL costs, Gibbs sums, and their first variations are finite combinations of integrals of such functions and of logarithms of positive bounded functions. Proposition 4.14 therefore gives a unique maximal solution for every initial point, and along it the reduced potential decreases at the rate $\sum_i [\mathrm{Var}_{q_i}(h_i^q) + \mathrm{Var}_{s_i}(h_i^s)]$. Set $z_i = (u_i, v_i)$, $\bar{z} = N^{-1}\sum_i z_i$, and $\delta_i = z_i - \bar{z}$. Write Φ_t for the resulting fine flow in E^N. The statements below are made for each fixed initial configuration; no measurability of Φ_t with respect to a sigma-algebra on E^N is used, so the norm topology of the nonseparable space L^∞ raises no measurability question here.

On the diagonal $z_1 = \cdots = z_N = z$, every peer force vanishes. The diagonal vector field is

$$f(u, v) = \big(-u + J\varphi \mathbb{E}_{s_v}\psi, -v + J\psi \mathbb{E}_{q_u}\varphi\big). \tag{10.5}$$

Denote its flow by ϕ_t. Its derivative is $Df(u,v)[h,k] = (-h + J\varphi\,\mathrm{Cov}_{s_v}(\psi, k), -k + J\psi\,\mathrm{Cov}_{q_u}(\varphi, h))$. If $\psi = \pm 1$ has mean m under s_v, then $|\mathrm{Cov}_{s_v}(\psi, k)| \le \|k\|\mathbb{E}_{s_v}|\psi - m| = (1 - m^2)\|k\| \le \|k\|$, and likewise for φ. Hence f is globally Lipschitz with constant

$$L = 1 + |J|, \tag{10.6}$$

has at most linear growth, and has a bounded second derivative. Its flow is defined for all $z \in \mathsf{E}$ and all positive and negative finite times. In particular, it can be used below without extending a locally defined diagonal dynamics.

Theorem 10.1 (An asymptotic-phase projection under the reciprocal KL flow). For the preceding data there is a forward-invariant neighborhood U of the common reference configuration in E^N on which

$$\Pi(x) = \lim_{t\to\infty} \phi_{-t}\big(\overline{\Phi_t x}\big), \tag{10.7}$$

exists in E and is continuously differentiable. It satisfies

$$\Pi(z, \dots, z) = z, \qquad \Pi(\Phi_t x) = \phi_t(\Pi x), \qquad D\Pi_{(z,\dots,z)}(h_1, \dots, h_N) = \frac{1}{N}\sum_i h_i. \tag{10.8}$$

After shrinking U if necessary, Π is a submersion with nontrivial fibers parametrized locally by $N-1$ public log-density differences. The parent laws q_{u^c}, s_{v^c}, where $(u^c, v^c) = \Pi(x)$, follow eq. (10.5) exactly. If two fine child states (q_i, s_i) are initially different, they remain different at every finite time in this neighborhood. Every trajectory in U converges to the common reference.

Proof. By the preceding paragraph the reduced Fisher vector field V of eq. (10.4) is smooth on bounded log-density balls.

At the origin every peer KL cost has zero first variation, so the Gibbs rows equal $1/(N-1)$ and their first variations vanish. The receiver and sender derivatives each contribute one copy of the complete-graph difference operator. Therefore, with $\kappa_N = 2N/(N-1)$,

$$\begin{aligned}(DV(0)h)_i &= \mathcal{A}h_i - \kappa_N(h_i - \bar{h}),\\ \mathcal{A}(u,v) &= \big(-u + J\varphi\mathbb{E}_B[\psi v], -v + J\psi\mathbb{E}_A[\varphi u]\big).\end{aligned} \tag{10.9}$$

The off-diagonal operator in $\mathcal{A}$ has norm at most $|J|$. Hence its common and difference semigroups have bounds $e^{-(1-|J|)t}$ and $e^{-(1+\kappa_N-|J|)t}$ in the corresponding maximum norms. In the adapted norm $\max(\|\bar{z}\|, \max_i \|\delta_i\|)$ both linear semigroups are contractions with constant one; in the plain maximum norm on E^N the full linear semigroup has a constant up to three, which is why the adapted norm is used below.

Here are the local estimates needed for the limit, including its regularity. Choose $\mu_0 > 0$ and ω strictly below these two decay rates, respectively, with

$$\omega > L = 1 + |J|. \tag{10.10}$$

This is possible because $1 + |J| < 1 + \kappa_N - |J|$ whenever $|J| < N/(N-1)$, which holds since $|J| < 1$; and μ_0 exists because $|J| < 1$. In the adapted norm the nonlinear part of V is $o(r)$ on a ball of radius r, so for small r the one-sided Lipschitz constant of the full field on the ball stays below $-\mu_0$ and the ball is forward invariant. Taylor expansion and the variation-of-constants formula on such a ball give

$$\|\Phi_t x\| + \|D\Phi_t(x)\| \le Ce^{-\mu_0 t}, \qquad \|\delta(t)\| + \|D_x\delta(t)\| \le Ce^{-\omega t}. \tag{10.11}$$

The constants are uniform on a smaller neighborhood. To detail the second estimate, the diagonal is invariant, being the fixed-point subspace of the permutation action [93], so the difference equation is linear in δ with coefficient $\mathcal{A} - \kappa_N I + O(\|x\|)$. Its differentiated equation has the same stable homogeneous part and an additional forcing bounded by $C\|\delta(t)\|\|D\Phi_t(x)\|$. Variation of constants bounds that forcing by $e^{-\omega t}$ times a convergent integral of $e^{-\mu_0 t}$. Small losses in the two decay exponents absorb the coefficient perturbations. The first estimate follows in the same way from the strictly stable full linearization, after choosing the neighborhood small enough that its nonlinear derivative perturbation is less than the chosen decay margins. It also shows that every trajectory in U converges to the reference.

Permutation equivariance implies that the average derivative of V in a zero-sum direction vanishes at every diagonal point. Thus the exact average equation has the form

$$\dot{\bar{z}} = f(\bar{z}) + g(\bar{z}, \delta), \qquad g = O(\|\delta\|^2), \quad D_{\bar{z}}g = O(\|\delta\|^2), \quad D_\delta g = O(\|\delta\|). \tag{10.12}$$

It follows from eq. (10.11) that both $g(\Phi_t x)$ and its derivative with respect to x are bounded by $Ce^{-2\omega t}$.

By eq. (10.6) and the bounded second derivative of f, the complete flow obeys

$$\|D\phi_{-t}\| \le e^{Lt}, \qquad \|D^2\phi_{-t}\| \le Ce^{2Lt}, \qquad t \ge 0. \tag{10.13}$$

Differentiation in time gives the convergent integral

$$\Pi(x) = \bar{z}(0) + \int_0^\infty D\phi_{-t}\big|_{\bar{z}(t)} g(\bar{z}(t), \delta(t))dt. \tag{10.14}$$

The integrand is bounded by $Ce^{-(2\omega-L)t}$. Its derivative in x is the sum of a term with $D\phi_{-t}D_x g$ and one with $D^2\phi_{-t}[D_x\bar{z}, g]$. They are bounded by integrable multiples of $e^{-(2\omega-L)t}$ and $e^{-(2\omega-2L)t}$. Uniform convergence proves continuous differentiability.

On the diagonal the residual and its first derivative vanish, proving the retraction and derivative assertions. Shifting the limit in eq. (10.7) by a finite time proves the intertwining identity. The averaging derivative has the diagonal embedding as a bounded right inverse. The Banach implicit function theorem therefore gives the claimed local fibers, including continuously varying heterogeneous preparations. Finally, if two complete child states first became equal at a finite time, exchanging their labels would produce the same configuration at that time. Permutation equivariance and local uniqueness backward to the initial time would imply that they were initially equal, a contradiction. □

The limit does not depend on which average is used. Any smooth permutation-invariant retraction ρ of E^N onto the diagonal has the averaging map as its derivative at diagonal points, so $\rho(x) - \bar{z} = O(\|\delta\|^2)$, and $\|\phi_{-t}(\rho(\Phi_t x)) - \phi_{-t}(\overline{\Phi_t x})\| \le Ce^{-(2\omega-L)t} \to 0$. The same estimate gives a computable approximation. With $\Pi_{t_0}(x) = \phi_{-t_0}(\overline{\Phi_{t_0} x})$, eq. (10.14) gives

$$\|\Pi(x) - \Pi_{t_0}(x)\| \le \frac{C}{2\omega - L} e^{-(2\omega-L)t_0}, \tag{10.15}$$

so in a finite-dimensional subfamily the phase parent is evaluated, up to an exponentially small error, by one forward and one backward integration.

Corollary 10.2 (Uniqueness of the phase parent and extension to the basin). Let Π' be a map defined on U, Lipschitz on a neighborhood of the common reference, equal to the identity on the diagonal, and satisfying $\Pi' \circ \Phi_t = \phi_t \circ \Pi'$. Then $\Pi' = \Pi$. Moreover, for every configuration x whose trajectory enters U at some time $t_0 \ge 0$, the value $\widetilde{\Pi}(x) = \phi_{-t_0}(\Pi(\Phi_{t_0} x))$ does not depend on the choice of t_0, and $\widetilde{\Pi}$ is a continuously differentiable intertwining extension of Π to the basin of U; it is a submersion wherever $\Phi_{t_0} x$ lies in the neighborhood on which Π is.

Proof. Both maps are Lipschitz near the reference and equal the identity on the diagonal, so $\|\Pi(y) - \bar{y}\|$ and $\|\Pi'(y) - \bar{y}\|$ are bounded by $C\|\delta(y)\|$ there. Trajectories in U enter that neighborhood and stay in it. Hence $\|\phi_t(\Pi x) - \phi_t(\Pi' x)\| = \|\Pi(\Phi_t x) - \Pi'(\Phi_t x)\| \le Ce^{-\omega t}$ for large t, and eq. (10.13) gives $\|\Pi x - \Pi' x\| \le Ce^{-(\omega-L)t}$, which tends to zero by eq. (10.10). Independence of t_0 follows from the intertwining identity. Near x the extension is the composition $\phi_{-t_0} \circ \Pi \circ \Phi_{t_0}$ for one fixed t_0,

because U is open and Φ_{t_0} is continuous; Φ_{t_0} and ϕ_{-t_0} are continuously differentiable with invertible derivatives, which proves the remaining assertions. □

The parent is a virtual collective state, generally different from either the arithmetic or the geometric pool [1, 88]. The restricted potential $ND_{\mathrm{KL}}(q^c \otimes s^c \| R)$ and metric Ng_{Fisher} give eq. (10.5). Fine states remain unchanged by the construction. They approach the common reference, so the theorem supplies neither persistent nonzero separation nor selected membership, and every fiber with its parent converges to one point.

The map Π is the asymptotic phase of the stable common reference relative to its invariant diagonal: it is the projection along the strong-stable foliation of the diagonal, whose existence is governed by the rate condition eq. (10.10) [73]. For periodic orbits this is the asymptotic phase of Guckenheimer [97]. For a stable equilibrium with a spectral gap its fibers are the joint level sets of the Koopman eigenfunctions of the diagonal spectrum, the fibrations of Mauroy et al. [145]; they are isostables in the sense of that work only when the diagonal is one-dimensional, as in the binary example below. The general theorems are stated for compact normally attracting invariant manifolds in finite dimensions [71] or for compact invariant manifolds of semiflows on Banach spaces [18]; the diagonal here is infinite dimensional near the reference, so no neighborhood of the reference in it is compact, and the proof above is given directly in the chart rather than imported. The construction is an exact dynamical lumping [110]. What the reciprocal interaction contributes is the spectral gap: the difference rate exceeds the common rate by $\kappa_N = 2N/(N-1)$, which comes from the receiver and sender roles of the peer costs and does not depend on the temperatures at linear order, because row variations vanish at equality. The gap gives uniqueness of the phase among Lipschitz intertwining retractions and strong-stable fibers; it is not needed for the existence of some exact intertwining retraction, as the null control of section 10.5 shows.

10.2 An ordinary response that the phase does not preserve

The failure of boundary sufficiency can be seen inside the preceding example. Take two binary agents, $q_i = s_i = \mathrm{Ber}_{\pm 1}(x_i)$, $0 < J < 1$, and set $x_1 = m + a$, $x_2 = m - a$. This is an invariant subfamily. The exact single-agent diagonal equation is

$$f_b(x) = -(1 - x^2)(\operatorname{atanh} x - Jx). \tag{10.16}$$

Both source rows have one entry. Including the incoming peer derivative gives

$$\dot{x}_i = -(1 - x_i^2)[2 \operatorname{atanh} x_i - \operatorname{atanh} x_j - Jx_i] - (x_i - x_j). \tag{10.17}$$

Expansion at the reference state yields

$$\begin{aligned} \dot{m} &= f_b(m) + (6 - 3J)ma^2 + O((|m| + |a|)^5), \\ \dot{a} &= -(5 - J)a + O((|m| + |a|)^3). \end{aligned} \tag{10.18}$$

Let $\bar{p}$ be the mean parameter of the phase law. By the remark after theorem 10.1, Π may be computed with the arithmetic mean m of the mean parameters, so eq. (10.14) reads $\bar{p} = m + \int_0^\infty D\phi^b_{-t}|_{m(t)} g_b(m(t), a(t))dt$, with ϕ^b the flow of f_b and $g_b = \dot{m} - f_b(m)$. Child exchange

makes g_b even in a, simultaneous bit reversal makes it odd under $(m, a) \mapsto (-m, -a)$, and it vanishes at $a = 0$; hence $g_b = (6 - 3J)ma^2 + O(|m|a^2\varsigma^2)$ with $\varsigma = |m| + |a|$. Along the flow, $m(t) = e^{-(1-J)t}m(1 + O(\varsigma^2))$, $a(t) = e^{-(5-J)t}a(1 + O(\varsigma^2))$, and $D\phi^b_{-t}|_{m(t)} = e^{(1-J)t}(1 + O(\varsigma^2))$, uniformly in $t \geq 0$, because the corrections to the linear rates are integrals of $O(m(s)^2 + a(s)^2)$ along exponentially decaying orbits. Integration gives

$$\bar{p} = m + \frac{3(2 - J)}{2(5 - J)}ma^2 + O(|m|a^2\varsigma^2). \tag{10.19}$$

The coefficient is $(6 - 3J)/[2(5 - J)]$, the time integral of $e^{-2(5-J)t}$ multiplied by the leading residual. Equivalently, substituting $\bar{p} = m + cma^2$ into the invariance equation $D\bar{p}(V) = f_b(\bar{p})$ gives $(6-3J)-2(5-J)c = 0$. Carried to the next order, with $\dot{m} - f_b(m) = ma^2[6-3J+\frac{16}{3}m^2+2a^2]+O(\varsigma^7)$ and $\dot{a} = -a[5 - J + (3J - 2)m^2 + (J - 2)a^2] + O(\varsigma^5)$, the same equation gives

$$\bar{p} = m + \frac{3(2 - J)}{2(5 - J)}ma^2 + \frac{9J^2 - 32J + 52}{6(5 - J)(3 - J)}m^3a^2 + \frac{15J^2 - 64J + 80}{8(5 - J)^2}ma^4 + O(|m|a^2\varsigma^4), \tag{10.20}$$

with coefficients bounded uniformly for $J \in [0, 1)$; the divisors of the invariance equation at these orders have modulus at least $2(5 - J)$, so there are no small divisors. Along $m = a > 0$ sufficiently small, $\bar{p} \neq m$. The heterogeneous state and the diagonal state $(\bar{p}, \bar{p})$ have exactly the same phase parent, whereas choosing a child uniformly and sampling its public bit gives means m and $\bar{p}$, respectively. This response is not constant on the phase fibers. Replacing it by sampling the phase law would specify a new interface. By corollary 10.2, no Lipschitz intertwining parent that is a single-agent law and restricts to the identity on equal children can preserve this response.

The defect is of second order in the child differences. Let $\mathcal{R}$ be a permutation-invariant continuously differentiable response with $\mathcal{R}(z, \ldots, z) = \bar{\mathcal{R}}(z)$, such as uniform child sampling. Permutation invariance gives $D\mathcal{R}_{(z,\ldots,z)}(h) = D\bar{\mathcal{R}}(z)\big(N^{-1}\sum_i h_i\big) = D(\bar{\mathcal{R}} \circ \Pi)_{(z,\ldots,z)}(h)$, so $\mathcal{R} - \bar{\mathcal{R}} \circ \Pi$ vanishes together with its first derivative on the diagonal. Where both are twice differentiable it is $O(\|\delta\|^2)$, decaying like $e^{-2\omega t}$ along the flow while the parent signal decays like $e^{-(1-|J|)t}$; eq. (10.19) shows that the second order is attained.

10.3 What unrestricted Gibbs probes retain

Consider a block of exposed belief laws $q_j = f_j\lambda$ against a probability reference λ, with f_j bounded above and away from zero. Fix positive source weights w_j and a temperature $T > 0$. An identity-channel receiver with law $r\lambda$ sees the block normalizer

$$Z(r) = \sum_j w_j \exp[-D_{\mathrm{KL}}(r\lambda\|q_j)/T]. \tag{10.21}$$

The probability of selecting this block against a known outside source determines $Z(r)$, so the normalizer is an observable routing response without revealing a child label.

Proposition 10.3 (Unrestricted Gibbs response identifies the exposed child laws). If two finite source collections have equal eq. (10.21) for every positive receiver density in an L^∞ neighborhood

of 1, then their weighted empirical measures of densities are equal:

$$\sum_j w_j \delta_{f_j} = \sum_k w'_k \delta_{f'_k}. \tag{10.22}$$

Repeated identical laws may be combined by adding their weights; distinct exposed laws cannot be forgotten by a retention preserving this entire response class.

Proof. Multiplication by the known receiver-entropy factor gives

$$e^{\int r \log r d\lambda/T} Z(r) = \sum_j w_j e^{\int r \log f_j d\lambda/T}. \tag{10.23}$$

List the distinct normalized densities in the union of the two collections. No two of their logarithms differ by a constant. There is therefore a bounded mean-zero function h for which the numbers $v_j = \int h \log f_j d\lambda$ are all different: each equality excludes one proper hyperplane in the finite-dimensional span of centered log-density differences, and finitely many such hyperplanes cannot cover that span. Put $r = 1 + th$ for sufficiently small $|t|$. Subtracting the two expressions in eq. (10.23) gives $\sum_j c_j e^{tv_j/T} = 0$. Differentiation through orders zero to one less than the number of distinct laws gives an invertible Vandermonde system. Every c_j vanishes. Its positive factor $e^{\int \log f_j d\lambda/T}$ leaves exactly equality of the total weights on each density. □

The proof uses a single probe line $r = 1 + th$, with h outside a finite union of hyperplanes fixed by the children. Any receiver family that contains such a line therefore identifies the children, and a restricted family fails only on children whose centered log-density differences are annihilated by every probe direction. The Vandermonde nodes v_j/T approach one another as the children approach one another, so near the diagonal of theorem 10.1 the resolving power of the probe decays with the child differences.

In particular, a single ordinary law with one positive weight reproduces all these responses only if every exposed child law already equals it. The proposition concerns identity comparison channels and unrestricted receiver probes. It is not an impossibility result for arbitrary channels, restricted interfaces, or isolated dynamical factors such as theorem 10.1. For a fixed channel its scope is exactly the following.

Corollary 10.4 (Gibbs probes through a fixed channel). Let every comparison pass through one fixed Markov kernel $\mathcal{K}$, so that the compared laws are $\nu_j = q_j\mathcal{K}$, and suppose their densities against λ are bounded above and away from zero. Then the block normalizer $\sum_j w_j \exp[-D_{\mathrm{KL}}(r\lambda\|\nu_j)/T]$, for all positive receiver densities near 1, determines $\sum_j w_j \delta_{d\nu_j/d\lambda}$ and nothing more. Two collections with equal channelled laws give identical responses, so exact compression of this interface is possible precisely on the fibers of $q \mapsto q\mathcal{K}$ restricted to the exposed laws.

Proof. The normalizer depends on the collection only through the weighted channelled densities; apply proposition 10.3 to them. □

For a deterministic coarse-graining channel, the pushforward by a measurable map c, the incoming force also descends to the channelled laws. The incoming Fisher force of one Gibbs receiver with law r and weight β on source j is $\partial_t \log q_j = \beta(\mathcal{K}(r/\nu_j) - 1)$, and for $\mathcal{K}$ given by c this

gives $\dot{\nu}_j = \beta(r - \nu_j)$, closed on the image. For a nondeterministic kernel two sources with equal channelled laws generally have different channelled velocities, so even the incoming force does not descend. The within-fiber conditional laws of a deterministic channel are hidden conditionals of the type treated in chapter 8; whether a deterministic comparison channel combined with that mechanism yields an exact parent for the full channelled Gibbs interface is not established here.

For a restricted affine receiver family $r_\theta = r_0 + \sum_a \theta_a h_a$, the identity-channel response instead identifies

$$\mu = \sum_j w_j e^{\iota_j/T}\delta_{v_j}, \qquad \iota_j = \int r_0 \log f_j d\lambda, \qquad (v_j)_a = \int h_a \log f_j d\lambda. \tag{10.24}$$

Equal slopes can be combined by adding the tilted weights. This possible boundary compression still requires a dynamical check. The incoming Fisher force from one Gibbs receiver is $\dot{f}_j = \alpha_j(r_\theta - f_j)$, and consequently

$$\dot{\iota}_j = \alpha_j \left(\int \frac{r_0 r_\theta}{f_j} d\lambda - 1 \right), \qquad (\dot{v}_j)_a = \alpha_j \int \frac{h_a r_\theta}{f_j} d\lambda. \tag{10.25}$$

These reciprocal moments are additional information unless the admitted family supplies a closure identity. For an explicit witness, let λ be uniform on three points, take the extreme restricted family $\{r_0\}$ with $r_0 = 1$, and compare the children $(1/4, 1/4, 1/2)$ and $(c, c, 1-2c)$ with $c = (1+\sqrt{5})/8$. Their intercepts are equal, since $3c \cdot 3c \cdot 3(1-2c) = 27/32$, but with $\alpha_j = 1$ their intercept velocities are $1/9$ and $(5\sqrt{5} - 10)/9$. The force $r_\theta - f_j$ is a mixture-geodesic term; it leaves every finite-dimensional exponential family in general, which is why these statements are made on full density charts.

10.4 The remaining recursive requirement

The phase construction retains one supplied block. A recursive use of any retention needs two further properties, which we state as a definition.

Definition 10.5 (Controlled projectability and descent of joining). Let a block I have fine state space X_I and a unique flow generated by a vector field $V_I(x_I; e)$ that depends on the state e of the agents outside I in a space $\mathsf{X}_{\mathrm{ext}}$. A retention $R_I : \mathsf{X}_I \to \mathsf{Y}_I$, together with an external retention $R_{\mathrm{ext}} : \mathsf{X}_{\mathrm{ext}} \to \mathsf{Y}_{\mathrm{ext}}$, is controlled projectable if there is a vector field $\bar{V}_I(y; \bar{e})$ on Y_I, parametrized by $\bar{e} \in \mathsf{Y}_{\mathrm{ext}}$, with

$$DR_I(x_I)V_I(x_I; e) = \bar{V}_I(R_I x_I; R_{\mathrm{ext}} e), \tag{10.26}$$

for all x_I and e, and if every declared boundary kernel of the block, including the memory of a reused child, factors through R_I. Within a family of blocks closed under joining, let $\mathcal{J}_{I,J} : \mathsf{X}_I \times \mathsf{X}_J \to \mathsf{X}_{I\cup J}$ identify the fine state of a union of disjoint blocks. Joining descends if there is a map $\bar{\mathcal{J}}_{I,J} : \mathsf{Y}_I \times \mathsf{Y}_J \to \mathsf{Y}_{I\cup J}$ with

$$R_{I\cup J} \circ \mathcal{J}_{I,J} = \bar{\mathcal{J}}_{I,J} \circ (R_I \times R_J). \tag{10.27}$$

Condition eq. (10.26) is the relatedness of control systems under a surjection, used in hierarchical control [159], with the external state as the control; for an empty exterior it is the intertwining identity of eq. (10.8). Condition eq. (10.27) says that the parent of a union is computed from the parents of its parts, which is what clause (F6) of definition 6.36 needs from a second stage that starts from first-stage parents. For the phase projection the block is the whole population, so only

the intertwining case has been verified; for proper sub-blocks, where the external state enters V_I, neither condition has been checked, and the sampling response of section 10.2 already fails to factor through Π. At the common reference with uniform nonself rows, theorem 10.1 cannot be applied to a proper sub-block I with $N \geq 3$: the set on which the members of I are equal is invariant, but every zero-sum mode carries the operator $\mathcal{A} - \kappa_N$ of eq. (10.9), so the intra-block differences normal to that set decay at rates between $1 + \kappa_N - |J|$ and $1 + \kappa_N + |J|$, and the differences between the block and the exterior, which are tangent to it, decay at the same rates; the rate condition, normal contraction stronger than every tangential contraction, fails for every J. The erasure parent of proposition 9.4 satisfies both conditions for its proper blocks, with R_{ext} the other pair's retained measure and joining given by $\eta_{I\cup J} = (|I|\eta_I + |J|\eta_J)/(|I| + |J|)$, but it discards no public information, so the two conditions are compatible with exact inheritance and say nothing by themselves about compression.

10.5 Status against the formation standard

The objective is the Gibbs-reduced reciprocal potential with identity channels, and the dynamics is its chart flow on E^N. The phase projection is a retention of a supplied block, the whole population, and the dynamics selects nothing, so clauses (F2), (F5), and (F6) of definition 6.36 are not addressed. The child-level probe rule B_I consists of the identity peer channels applied to each member, with response law that member's law, and of uniform child sampling. The retention is non-injective on the forward-invariant neighborhood, and a member's channel separates two configurations in one fiber that differ in that member's law, so the separation required by (F4) exists; it is transient, since every fiber converges to one point, and it is not of symmetry type, since the fibers are continuous families of heterogeneous configurations rather than orbits of a symmetry. For (F3), every declared parent protocol rule A_I that contains uniform child sampling fails, because that response is not constant on the fibers of Π and, by corollary 10.2, no Lipschitz intertwining single-law parent preserves it. A rule consisting of responses computed from the phase law is defined from the retention and is not a declared rule in the sense of definition 6.17. With the empty rule, (B) is vacuous and (L), (D), and (C) hold for Π.

The source sets exclude the receiver, so the two nulls of definition 6.33 coincide. With every peer cost deleted, each agent follows the diagonal field eq. (10.5) independently, the linearized difference and common modes have the same spectrum $\mathcal{A}$, and the rate condition eq. (10.10) fails, since the difference rate $1 - |J|$ is smaller than L. The null flow nevertheless admits exact intertwining retractions onto the diagonal. On the binary subfamily, the function $\mathfrak{h}(x) = x \exp \int_0^x [-(1-J)/f_b(s) - 1/s]ds$ is analytic near zero, with $\mathfrak{h}'(0) = 1$ and $\mathfrak{h}(\phi_t^b x) = e^{-(1-J)t}\mathfrak{h}(x)$, and for every $c \geq 0$ the map $\Pi_c^0(x_1, x_2) = \mathfrak{h}^{-1}\big((\mathfrak{h}(x_1) + \mathfrak{h}(x_2))/2 + c|\mathfrak{h}(x_1) - \mathfrak{h}(x_2)|\big)$ is a Lipschitz retraction onto the diagonal that intertwines the null flow with ϕ_t^b, by homogeneity of degree one, and has one-dimensional fibers; Π_0^0 is analytic. These retractions are not unique in the Lipschitz class, and their fibers are not strong-stable leaves: two configurations in one fiber approach each other only at the common rate. What the peer costs produce is therefore the spectral gap κ_N, which makes the Lipschitz intertwining parent unique (corollary 10.2) and makes the discarded differences decay faster than the parent signal, not the existence of an exact compressing coordinate. No grouping is selected, and the results of this chapter are retention and identifiability statements rather than formation.

Chapter 11

Boundary Closure and Recursive Composition

This chapter concerns the reciprocal law potential of definition 4.2 and its Fisher flow, with identity peer channels, all-source uniform priors, and a coupled evaluator. It supplies a binary hierarchy together with an invariant family of agent laws, and proves that the family carries an exact hierarchy of public compressions. Each level preserves fresh uniform-descendant responses, their subsequent evolution, and a declared class of record assimilations. The chapter makes no selection claim. Membership and the hierarchy are supplied, so it is an inheritance construction without selection: measured against definition 6.36, (F1) holds in the family grade for the declared invariant family, (F3) and (F4) hold at every join of the supplied hierarchy, and (F2), (F5), and (F6) are not addressed. The source sets exclude self, so the two nulls of definition 6.33 coincide. That null deletes every peer cost, is equivariant under the same symmetry group, preserves the same family, descriptors, and descended interface, and has the profile equation $\dot{f} = -f(\log f + V^q - c_q)$, along which every bit contrast decays at rate one. The exact hierarchy is therefore a property of the prepared family and its symmetry, not of the interaction, which contributes the class costs D_k, the incoming term, and a faster decay of the discarded contrast. Within this construction the gauge structure supplies the orbits that a parent forgets; it does no dynamical work. The mechanism is an equivariant symmetry reduction (section 11.3). The public partition data that the hierarchy discards are invisible to the fine energy, the Fisher metric, and the admitted records, and near the partition-independent laws their contrast decays exponentially under the same flow (proposition 11.13).

11.1 A general-law family on a supplied tree

Let A and B be probability references on standard Borel spaces with coordinates r and w. Choose a positive joint density $\rho(r, w)$ against $A \otimes B$, normalized to one, with bounded logarithm. Its disintegration supplies a proper model reference and normalized evaluator as in eq. (4.1). It may couple its arguments; for example, balanced signs $\phi(r), \psi(w)$ give $\rho = \exp(J\phi\psi)/\cosh J$.

Fix a binary tree of depth $d \geq 1$ with $N = 2^d$ leaves. Each leaf is a primitive agent. For every internal tree node v, append to the belief sample space an independent public coordinate θ_v with law λ_v, and append a separate set of such coordinates to the model sample space. The references extend by these product laws. Each node carries a measurable sign $\sigma_v(r, \theta_v) \in \{-1, 1\}$. The profile equations of section 11.2 and the descent computations of section 11.4 use only one property of these signs: conditional on the base, they are independent and fair. The symmetry interpretation of section 11.3 uses more, namely node laws that admit reference-preserving sign reversals and maps acting transitively within each sign cell; the two realizations below have this property, while a node law with atoms of unequal mass inside a sign cell does not. Two realizations are used. In

the coin realization, λ_v is uniform on $\{-1,1\}$ and $\sigma_v(r,\theta_v)=\theta_v s_v(r)$ with a measurable orientation $s_v(r)\in\{-1,1\}$. In the circle realization, λ_v is Haar measure on a circle, and a measurable phase function $\alpha_v(r)$ defines

$$\sigma_v(r,\theta_v)=\operatorname{sign}\cos(\theta_v-\alpha_v(r)). \tag{11.1}$$

The sign on its null zero set is immaterial. The atomless circle realization is needed only where a continuum of balanced partitions is used, namely for the continuous phase families and the total-variation and Hellinger statements of section 11.5. In both realizations the coordinates θ_v are original public sample coordinates. Different phase functions or orientations change the laws in those same coordinates; they are held fixed along a trajectory.

Label the left and right edges by $\epsilon=+1,-1$. A leaf i defines a signed path

$$u_j(i,\theta)=\epsilon_j(i)\sigma_{v_{j-1}(i)}(r,\theta_{v_{j-1}(i)}),\qquad j=1,\dots,d. \tag{11.2}$$

Let U_d be the uniform law on $\{-1,1\}^d$. Choose any positive bounded profile $f(r,u)$, with bounded logarithm and $\int f\,dA\,dU_d=1$, and set

$$q_i(r,\theta)=f(r,u(i,\theta)),\qquad f_0(r)=\mathbb{E}_{U_d}f(r,u). \tag{11.3}$$

These are densities against A times all the node laws. Each leaf has the same base marginal f_0A, but its full public law can differ from every other leaf law. Define model densities from an independent profile $g(w,z)$ and model coordinates in the same way. The base spaces and the profile functions are unrestricted within the stated regularity domain; no Gaussian or finite-parameter family is imposed.

All physical peer channels are identities, every other agent is an allowed source, and the original source priors are uniform. If two leaves share exactly k initial edges, there are $n_k=2^{d-k-1}$ possible sources in that class for each receiver. Give each such source weight b_k/n_k, where b is a probability vector on $k=0,\dots,d-1$. The belief and model vectors b^q,b^s evolve separately. Uniform original rows are included by taking

$$\pi_k=\frac{n_k}{N-1},\qquad b^q_k=b^s_k=\pi_k. \tag{11.4}$$

The fine system is therefore the reciprocal potential eq. (4.4) with identity channels, the evaluator determined by ρ, and uniform all-source priors. The tree enters only the prepared relation among the agent laws and row classes. The fine potential and its uniform priors contain no additional preference for a chosen pair, and they are invariant under every permutation of the agents.

11.2 The exact inherited profile equations

For each k, define an operator P_k on bit profiles: keep the first k bits, reverse bit $k+1$, and average all later bits. It is a Markov operator, self-adjoint for U_d. It describes an average over a class of original source labels; it is not a replacement physical communication channel. Define

$$D_k(f)=\int f(r,u)[\log f(r,u)-(P_k\log f)(r,u)]\,dA(r)\,dU_d(u). \tag{11.5}$$

The Fisher flow of theorem 4.6 is proved on finite tables. The fine system here lives on general sample spaces. Its well-posedness is the identity-channel case of proposition 4.14, whose sufficient conditions (identity channels between agents sharing a reference, evaluators with bounded log densities) it meets; the lemma below restates that case in unnormalized log-density coordinates and records the explicit first variation used later.

Lemma 11.1 (Fine identity-channel flow in bounded log-density charts). Let (X, λ) and (Y, μ) be standard Borel probability spaces whose coordinates include the base coordinates r and w, with base marginals A and B. Consider finitely many agents with normalized belief densities $q_i = e^{\ell_i}$ against λ and model densities $s_i = e^{m_i}$ against μ, where $\ell_i \in L^\infty(\lambda)$ and $m_i \in L^\infty(\mu)$. Suppose the peer channels are identities on a fixed directed edge set with nonempty source sets and positive priors, $T_q, T_s > 0$, and each local term is $D_{\mathrm{KL}}(q_i \otimes s_i \| R_i)$, where R_i has a density $\rho_i(r, w)$ against $\lambda \otimes \mu$ that depends only on the base coordinates and has bounded logarithm. Up to additive constants, the first variation of eq. (4.4) with respect to q_i is

$$\begin{aligned} h_i^q &= \log q_i + V_i^q + \sum_{j \in \mathcal{N}_i^q} \beta_{ij}(\log q_i - \log q_j) - \sum_{l : i \in \mathcal{N}_l^q} \beta_{li} \frac{q_l}{q_i}, \\ V_i^q(r) &= -\int s_{i,0}(w) \log \rho_i(r, w) dB(w), \end{aligned} \tag{11.6}$$

where $s_{i,0}$ is the base marginal density of s_i; the model variation h_i^s follows by exchanging the sectors. Then the system

$$\dot{\ell}_i = -(h_i^q - \mathbb{E}_{q_i} h_i^q), \qquad \dot{m}_i = -(h_i^s - \mathbb{E}_{s_i} h_i^s), \tag{11.7}$$

together with the row equations of eq. (4.15), has for every initial condition with interior rows a unique maximal solution in $L^\infty(\lambda)^N \times L^\infty(\mu)^N$ times the row coordinates, with interior rows; if its existence time is finite, the solution leaves every bounded set of the chart with rows bounded away from the boundary. Along it the laws stay normalized, $\dot{q}_i = -q_i(h_i^q - \mathbb{E}_{q_i} h_i^q)$, the right side is the Fisher gradient of $\mathcal{E}$ for the metric $\int (\delta q)^2 / q$ in each block, and $\dot{\mathcal{E}} = -\sum \mathrm{Var} \le 0$ as in eq. (4.16).

Proof. For identity channels, eq. (4.10) gives the receiver variation $\log q_i - \log q_j + 1$ and the source variation $-q_i/q_j$ of $D_{\mathrm{KL}}(q_i \| q_j)$. The local term has variation $\log q_i + 1 + V_i^q$, because ρ_i reads only the base coordinates. This gives eq. (11.6). Define the field off the normalized set through $q_i = e^{\ell_i} / \int e^{\ell_i} d\lambda$. On a ball of radius L in the sup norm, every density lies between e^{-2L} and e^{2L}. The maps $\ell \mapsto \log q_i$, $\ell \mapsto q_l/q_i$, the base marginals $s_{i,0}$ as maps into $L^1(B)$, the linear map $s_{i,0} \mapsto V_i^q$ from $L^1(B)$ into $L^\infty(A)$ with norm at most $\| \log \rho_i \|_\infty$, the costs D_{ij}, and the expectations $\mathbb{E}_{q_i} h_i^q$ are therefore locally Lipschitz, and the row field is smooth on the interior of the simplices. Finite sums and products of bounded locally Lipschitz maps are locally Lipschitz, so the Picard–Lindelöf theorem in the Banach space gives a unique maximal solution and the continuation alternative, as in proposition 4.14: a field that is Lipschitz on bounded sets is bounded there, so a solution confined to such a set on a finite interval is Cauchy at its end and extends. Since $\mathbb{E}_{q_i} \dot{\ell}_i = 0$, the integral of e^{ℓ_i} is constant, so the normalized set is invariant and $\dot{q}_i = q_i \dot{\ell}_i$. For a tangent vector v with $\int v d\lambda = 0$, the metric pairing of $q_i(h_i^q - \mathbb{E}_{q_i} h_i^q)$ with v is $\int h_i^q v d\lambda$, which identifies the Fisher gradient exactly as in the proof of theorem 4.6. The chain rule in the chart gives the dissipation identity. □

Theorem 11.2 (Exact coupled hierarchy dynamics). Consider the reciprocal potential eq. (4.4) with the data of section 11.1, in either realization of the signs, and its Fisher flow in the chart of lemma 11.1, including source-row evolution. The family eq. (11.3), with its model analogue and class-constant rows, is invariant under this flow. Its inherited profile energy and metric are the original restrictions divided by their common factor N. The complete profile equations retain the incoming source derivatives, and their solutions exist for all $t \geq 0$.

Proof. For any two leaves in class k, the first k path bits agree, bit $k+1$ is opposite, and later bits lie on disjoint branches. Integrating their independent signs gives exactly $D_{\mathrm{KL}}(q_i \| q_j) = D_k(f)$. Every leaf path has uniform bit law. Each local product law therefore has the same base marginals f_0, g_0 in its evaluator term. Direct substitution gives $\mathcal{E} = NF$, with

$$\begin{aligned} F = &\int f \log f dA dU_d + \int g \log g dB dU_d - \int f_0(r) g_0(w) \log \rho(r,w) dA dB \\ &+ \sum_{k=0}^{d-1} b_k^q D_k(f) + T_q D_{\mathrm{KL}}(b^q \| \pi) + \sum_{k=0}^{d-1} b_k^s D_k(g) + T_s D_{\mathrm{KL}}(b^s \| \pi). \end{aligned} \tag{11.8}$$

The metric restricts to N times

$$\mathcal{G}^c = \int \frac{(\delta f)^2}{f} dA dU_d + \int \frac{(\delta g)^2}{g} dB dU_d + \sum_{a=q,s} \sum_k \frac{(\delta b_k^a)^2}{b_k^a}. \tag{11.9}$$

Thus no new mobility is introduced.

Self-adjointness gives the exact derivative

$$\delta D_k(f) = \int \delta f \left[\log f + 1 - P_k \log f - \frac{P_k f}{f} \right] dA dU_d. \tag{11.10}$$

The last ratio is the incoming source contribution. Define

$$V^q(r) = -\int g_0(w) \log \rho(r,w) dB(w), \qquad V^s(w) = -\int f_0(r) \log \rho(r,w) dA(r). \tag{11.11}$$

Up to an irrelevant additive constant, the belief force and evolution are

$$\begin{aligned} h^q &= 2 \log f + V^q - \sum_k b_k^q \left[P_k \log f + \frac{P_k f}{f} \right], \\ c_q &= \int f h^q dA dU_d, \qquad \dot{f} = -f(h^q - c_q). \end{aligned} \tag{11.12}$$

Exchange the sectors for $\dot{g}$. With $D_k^q = D_k(f)$ and $D_k^s = D_k(g)$, the rows obey

$$\dot{b}_k^a = -b_k^a \left[D_k^a + T_a \log \frac{b_k^a}{\pi_k} - \sum_l b_l^a \left(D_l^a + T_a \log \frac{b_l^a}{\pi_l} \right) \right]. \tag{11.13}$$

To establish full-flow invariance, fix a receiver and a public sample point. The leaves in its class-k source subtree run through every suffix bit string once, regardless of the physical signs at that

point. Their averages of $\log q_j$ and q_j are therefore $P_k \log f$ and $P_k f$. Incoming weights have the same class values. The fine force eq. (11.6) at that leaf is precisely eq. (11.12) evaluated at its path bits. All leaf normalizations agree because every path-bit vector is uniform under the reference. Source costs also agree within their classes, preserving the row form. Hence the full field is tangent to the profile family, and the restricted gradient is its actual evolution. The profile equations are locally Lipschitz in the bounded-profile chart by the argument of lemma 11.1, so they have a unique maximal solution. Its lift solves the fine equation by the preceding computation, and by the uniqueness in lemma 11.1 it is the fine solution. Global existence is lemma 11.3. □

In the coin realization with a finite base, the fine laws are positive tables on the base times $\{-1,1\}^{N-1}$, and theorem 4.6 applies to the fine system directly; the theorem is then a statement about finite tables that can be checked by exact computation.

Lemma 11.3 (Global bounded-log existence of the profile flow). Let $F(0)$ be the initial profile energy eq. (11.8) and $C_K = F(0) + 3\|\log\rho\|_\infty + 2$. Along the solution of theorem 11.2,

$$\min\left(\operatorname{ess\,inf}\log f(0), -C_K\right) \le \log f(t) \le \max\left(\operatorname{ess\,sup}\log f(0), C_K\right), \tag{11.14}$$

with the same bound for $\log g$, and $\log(b_k^a(t)/\pi_k) \ge \min(\log(b_k^a(0)/\pi_k), -C_a/T_a)$, where C_a is twice the bound eq. (11.14) on the absolute value of the corresponding log profile. The solution therefore exists for all $t \ge 0$.

Proof. Since $\sum_k b_k^q = 1$ and P_k preserves U_d, the normalizer in eq. (11.12) is $c_q = \int f \log f dA dU_d + \sum_k b_k^q D_k(f) + \int f_0 V^q dA - 1$. Every term of F except the evaluator term is nonnegative, the evaluator term is at least $-\|\log\rho\|_\infty$, and F decreases along the flow. Hence the first two terms of c_q lie between 0 and $F(0) + \|\log\rho\|_\infty$, and $|V^q| \le \|\log\rho\|_\infty$ gives $|c_q| \le F(0) + 2\|\log\rho\|_\infty + 1$. Given the continuous scalar inputs $c_q(t)$, $b^q(t)$ and the base function $V^q(t,\cdot)$, the profile equation is, for each base point, a finite system of ordinary differential equations for $l = \log f(r,\cdot) \in \mathbb{R}^{2^d}$:

$$\dot{l} = -2l - V^q + \sum_k b_k^q \left[P_k l + \frac{P_k e^l}{e^l}\right] + c_q.$$

At a bit string where l attains its maximum M_r, one has $P_k l \le M_r$ and $P_k e^l \le e^{M_r}$, so the upper Dini derivative of M_r is at most $-M_r + C_K$. At a minimizing bit string, $P_k l \ge m_r$ and $P_k e^l \ge e^{m_r}$, so the lower Dini derivative of the minimum m_r is at least $-m_r - C_K$. Comparison gives eq. (11.14) uniformly in r. For the rows, $y_k = \log(b_k/\pi_k)$ satisfies $\dot{y}_k = -D_k - T y_k + \sum_l b_l (D_l + T y_l)$, and the last sum is nonnegative because $D_l \ge 0$ and $\sum_l b_l y_l = D_{\mathrm{KL}}(b\|\pi) \ge 0$. Since $D_k \le \sup l - \inf l \le C$, we obtain $\dot{y}_k \ge -C - T y_k$. The model sector is identical. The solution thus stays in a bounded set of the chart with rows uniformly interior, and the continuation alternative of lemma 11.1 extends it to $[0,\infty)$. This is the profile form of the maximum-principle argument of proposition 4.15, which gives global existence of the fine flow under the same hypotheses. □

The same retentions apply to the original Gibbs-reduced flow. Replace each row by $b_k^{a,*} = \pi_k \exp(-D_k^a/T_a) / \sum_l \pi_l \exp(-D_l^a/T_a)$ and use the matter metric. The envelope identity in eq. (4.8) gives the displayed profile force with these weights, including its incoming term. All symmetry and response arguments below remain unchanged. The Gibbs-reduced flow and the flow with evolving

rows are different flows. At frozen costs the row log-odds in eq. (11.13) relax at rate T_a, so their transients differ most when T_a is small; the retention statements hold for each of them.

The inherited cost is still a relative entropy. Let

$$J_k(dr, du, dv) = A(dr)f(r, u)U_d(du)P_k(u, dv), \tag{11.15}$$

and let J_k^{rev} exchange u, v. Reversibility gives $D_k(f) = D_{\mathrm{KL}}(J_k \| J_k^{\mathrm{rev}})$. Since J_k and J_k^{rev} are linear in f, joint convexity of relative entropy makes each D_k convex in f. Equivalently, $D_k(f) = \frac{1}{2}\iiint (f(r,u) - f(r,v))(\log f(r,u) - \log f(r,v))A(dr)U_d(du)P_k(u,dv)$ is the Dirichlet form of the reversible kernel P_k evaluated at $(f, \log f)$, the entropy-production functional of modified logarithmic Sobolev inequalities [29]. The class cost differs from the divergence to the averaged source law by a Jensen term:

$$D_k(f) - D_{\mathrm{KL}}(fA \otimes U_d \| (P_k f)A \otimes U_d) = \int f[\log(P_k f) - P_k \log f] dA dU_d \geq 0. \tag{11.16}$$

Because f is positive and $P_k(u, \cdot)$ is uniform on its support, the right side vanishes exactly when $f(r, \cdot)$ does not depend on bits $k+2, \ldots, d$ for A-almost every r. It vanishes identically for $k = d-1$, where P_{d-1} is the deterministic reversal of the last bit, and is positive for $k < d-1$ whenever f depends on a later bit. Replacing the exact inherited cost by the averaged-source divergence would discard this term and change the dynamics.

11.3 The hierarchy as an equivariant symmetry reduction

The inheritance statements below are instances of classical facts about equivariant gradient flows. Let $\mathcal{S}$ be the group of pairs $\omega = (\Phi, \varpi)$ in which $\Phi = (\Phi^q, \Phi^s)$ consists of fibered bimeasurable maps of the belief and model sample spaces, $\Phi^q(r, \theta) = (r, \varphi_r(\theta))$ and similarly for Φ^s, that preserve the references, and ϖ is a permutation of the agents that preserves the edge set, the priors, and the evaluators. The group acts on configurations by $(\omega \cdot q)_i = q_{\varpi(i)} \circ \Phi^q$, $(\omega \cdot s)_i = s_{\varpi(i)} \circ \Phi^s$, and $(\omega \cdot \beta)_{ij} = \beta_{\varpi(i)\varpi(j)}$, with the same rule for model rows. The product is $\omega_1\omega_2 = (\Phi_2 \circ \Phi_1, \varpi_2 \circ \varpi_1)$, componentwise in each sector; with this law the formulas define a left action, $\omega_1 \cdot (\omega_2 \cdot x) = (\omega_1\omega_2) \cdot x$. The finite-dimensional statement proposition 6.7 does not apply here: $\mathcal{S}$ is not a Lie group, it acts on a Banach chart, and the sets used below are fixed-point sets of actions that are not free, so no quotient manifold or quotient metric is asserted.

Lemma 11.4 (Equivariant reduction). In the setting of lemma 11.1, every $\omega \in \mathcal{S}$ acts on the chart by a linear isometry of the sup norm, preserves $\mathcal{E}$ and the Fisher metric, and commutes with the flow: if $x(t)$ solves the flow from x_0, then $\omega \cdot x(t)$ solves it from $\omega \cdot x_0$. Consequently the following hold. (i) For every subgroup $H \leq \mathcal{S}$, the fixed-point set $\mathrm{Fix}(H)$ is invariant under the flow, and on it the flow is the Fisher gradient flow of the restriction of $\mathcal{E}$ for the restricted metric. (ii) Every $\omega \in \mathcal{S}$ maps $\mathrm{Fix}(H)$ onto $\mathrm{Fix}(\omega H \omega^{-1})$ and conjugates the restricted flows. A subgroup $K \leq \mathcal{S}$ that maps $\mathrm{Fix}(H)$ into itself, for example a subgroup of the normalizer of H, acts on $\mathrm{Fix}(H)$ commuting with the flow, and the flow descends to the orbit space $\mathrm{Fix}(H)/K$. More generally, if X is a flow-invariant union of fixed-point sets, the flow descends to X modulo the equivalence relation induced on X by $\mathcal{S}$. (iii) A response kernel Q from configurations to laws on an output space with

$Q(\omega \cdot x) = Q(x)$ for $\omega \in K$ descends to $\mathrm{Fix}(H)/K$. If an assimilation map A_o of definition 4.28 satisfies $A_o(\omega \cdot x) = \omega \cdot A_o(x)$ for $\omega \in K$, and the law of its record is K-invariant, then every finite adaptive history law of the record-interrupted dynamics descends as well.

Proof. A reference-preserving bimeasurable Φ^q preserves relative entropy, so $D_{\mathrm{KL}}(q_{\varpi(i)} \circ \Phi^q \| q_{\varpi(j)} \circ \Phi^q) = D_{\mathrm{KL}}(q_{\varpi(i)} \| q_{\varpi(j)})$, and ϖ preserves the edge set and priors, so the reindexed peer sums agree. The entropy part of each local term is preserved, and because Φ^q is fibered and reference-preserving, $q \circ \Phi^q$ has the same base marginal as q. The evaluator term reads only base marginals, and ϖ preserves the evaluators, so $\mathcal{E}(\omega \cdot x) = \mathcal{E}(x)$. The metric is preserved since $\int (\delta q \circ \Phi^q)^2/(q \circ \Phi^q) = \int (\delta q)^2/q$. On log densities the action is $\ell_i \mapsto \ell_{\varpi(i)} \circ \Phi^q$, a linear isometry of L^∞. Direct substitution in eq. (11.6), including the incoming term, gives $h_i^q(\omega \cdot x) = h_{\varpi(i)}^q(x) \circ \Phi^q$, so the field is equivariant, and uniqueness in lemma 11.1 gives the commutation. For (i), if $x_0 \in \mathrm{Fix}(H)$ and $\eta \in H$, then $\eta \cdot x(t)$ is a solution from x_0 and equals $x(t)$. At a point of $\mathrm{Fix}(H)$ the field is H-fixed, hence tangent to $\mathrm{Fix}(H)$, and its metric pairing with every tangent vector v of $\mathrm{Fix}(H)$ equals the derivative of the restricted energy along v; it is therefore the restricted gradient. Statement (ii) follows from $\omega \cdot \mathrm{Fix}(H) = \mathrm{Fix}(\omega H \omega^{-1})$ and the commutation: if $y_0 = \omega \cdot x_0$ with $\omega \in K$, then $y(t) = \omega \cdot x(t)$, so orbits are carried to orbits; the same commutation applies to any two points of X related by an element of $\mathcal{S}$. For (iii), $Q(y(t)) = Q(x(t))$, and an equivariant assimilation keeps related states related while assigning the same law to the record. Induction over the record times, with the flow between them, gives the history statement. □

Statement (i) is the flow form of the principle of symmetric criticality [158], and the invariance of fixed-point subspaces of equivariant vector fields is standard in equivariant bifurcation theory and dynamics [74, 94]. The lemma adds nothing beyond these classical facts; it fixes the form in which they are used here. The following proposition identifies the profile family as a fixed-point set.

Proposition 11.5 (The profile family is a fixed-point set). Fix a realization of the signs. Let $H_\alpha \leq \mathcal{S}$ consist of the pairs (Φ, ϖ) such that ϖ is an automorphism of the rooted binary tree acting on the leaves, and $u(\varpi(i), \Phi^q(\xi)) = u(i, \xi)$ for every leaf i and almost every belief sample point ξ, with the analogous condition on Φ^s. Then $\mathrm{Fix}(H_\alpha)$ consists exactly of the configurations of the form eq. (11.3) in both sectors, with class-constant rows $\beta_{ij} = b_k^q/n_k$ and $\gamma_{ij} = b_k^s/n_k$ for pairs in class k.

Proof. If $q_i = f(r, u(i, \xi))$ and $(\Phi, \varpi) \in H_\alpha$, then $q_{\varpi(i)}(\Phi^q \xi) = f(r, u(\varpi(i), \Phi^q \xi)) = f(r, u(i, \xi)) = q_i(\xi)$, and class-constant rows are preserved because tree automorphisms preserve the common-prefix length. Conversely, let a belief configuration be fixed by H_α. Elements (Φ, id) whose Φ^q preserves every sign belong to H_α. In the coin realization the signs determine the node coordinates given r. In the circle realization, choose Φ^q to act on each cell $\{\sigma = \mathrm{s}\}$, in arc coordinates measured from $\alpha_v(r)$, by rotations of the arcs through fractions c_v of their length such that 1 and the c_v are linearly independent over the rationals. This map is ergodic on each cell for the normalized Haar product by the Kronecker–Weyl criterion [195, Chapter 1]; rational independence of the c_v alone does not suffice, since $c_1 = \sqrt{2} - 1$ and $c_2 = 2 - \sqrt{2}$ leave the character $e^{2\pi i(x_1 + x_2)}$ invariant, so by Fubini's theorem each invariant q_i is almost everywhere a function $F_i(r, \sigma)$ of the base and the node signs.

For an internal node w, let ϖ_w exchange the two subtrees below w by their mirror map, and let Φ_w^q reverse σ_w (a half-period shift of θ_w, or a coin flip) and exchange the coordinates of mirror nodes

in the two subtrees, adjusted by their phases or orientations so that mirror signs are carried to mirror signs. Following the edge labels along each path shows that $(\Phi_w, \varpi_w) \in H_\alpha$. For a leaf i not below w, $\varpi_w(i) = i$, so F_i is invariant under reversing σ_w and permuting the signs strictly below w. Induction from the lowest internal nodes upward shows that F_i does not depend on the sign of any node that is not an ancestor of i. The ancestral signs of i determine $u(i, \xi)$ bijectively, so $q_i = f_i(r, u(i, \xi))$. For $(\Phi, \varpi) \in H_\alpha$, fixedness gives $f_{\varpi(i)}(r, u(i, \xi)) = f_i(r, u(i, \xi))$, and $u(i, \cdot)$ is uniform given r, so $f_{\varpi(i)} = f_i$. The node elements generate a group that acts transitively on the leaves and on the pairs of leaves with a given common-prefix length. Hence $f_i = f$ for all i, and the rows are class constant. The model sector is identical. □

Two further subgroups give the quotient. For a measurable map $r \mapsto \tau(r)$ into the automorphisms of bit strings, let Φ_τ shift θ_v by a half period exactly when $\tau(r)$ reverses the bit that follows the original signed prefix of v. This map is fibered, bimeasurable, and reference-preserving, since each shift depends only on r and on ancestor coordinates, and $u(i, \Phi_\tau \xi) = \tau(r)u(i, \xi)$ for every leaf. Hence (Φ_τ, id) maps the family with profile f to the family with profile $f(r, \tau(r)u)$. Phase rotations $\theta_v \mapsto \theta_v + \alpha_v(r) - \alpha'_v(r)$ map $\mathrm{Fix}(H_\alpha)$ onto $\mathrm{Fix}(H_{\alpha'})$ with the profile unchanged, and orientation changes act in the same way in the coin realization. These maps do not form one subgroup acting on a single $\mathrm{Fix}(H_\alpha)$, since phase rotations change α and Φ_τ depends on it. The union $X = \bigcup_\alpha \mathrm{Fix}(H_\alpha)$ is flow invariant, and by the general form of lemma 11.4 (ii) the flow descends to X modulo the relation $(\alpha, f) \sim (\alpha', f(r, \tau(r)u))$, which is the unordered profile tree of section 11.4.

The data discarded by this quotient, namely the phases or orientations and the base-dependent ordering of branches, are orbit data of a group of exact symmetries of the fine energy and the Fisher metric. Relative to the observable algebra generated by the fine energy and metric, the fresh uniform-descendant responses, and the admitted base-only broadcast records, they are gauge: no agent, channel, record, or energy term of this construction reads them. They are resolved only by the protocols excluded in section 11.4, namely reused children, shared labels, and child-specific or unrestricted identity-channel probes. Symmetry does not supply the explicit class costs D_k or their pair-law form, the incoming term of eq. (11.12), the identification eq. (11.18) of a fresh response with a subtree mean, the contrast decay of proposition 11.13, or the observability obstruction of section 11.6; these are computed directly.

The root-level statement holds for every identity-channel configuration, not only on the profile family.

Proposition 11.6 (Descent of the joint conditional spectrum). In the setting of lemma 11.1, let μ^q be the law of $(r, q_1(\xi), \ldots, q_N(\xi))$ for $\xi \sim \lambda$, and μ^s the law of $(w, s_1, \ldots, s_N)$ under μ. For any edge set, priors, and initial rows, two configurations with equal μ^q, μ^s, and rows have solutions with equal $\mu^q(t)$, $\mu^s(t)$, and rows throughout their common existence interval. Response kernels that read only base coordinates of a fresh sample from a fixed agent, and assimilations by a base-only likelihood applied to every agent, descend to these spectra.

Proof. By eq. (11.6), the force h_i^q at ξ is a function of r, of the value vector $(q_1(\xi), \ldots, q_N(\xi))$, and of quantities determined by the spectra and rows: V_i^q reads the base marginals $s_{i,0}(w) = \mathbb{E}[s_i \mid w]$, and the expectations $\mathbb{E}_{q_i} h_i^q$ and the costs $D_{ij} = \int q_i \log(q_i/q_j) d\lambda$ are integrals against μ^q. Fix the solution $x(t)$ from the first configuration, and let $\Psi_t(r, \cdot)$ be the flow of the finite-dimensional equations $\dot{v}_i = -v_i(h_i^q(r, v; t) - c_i(t))$ with the scalar inputs of $x(t)$, and similarly in the model sector.

Two routine identifications make this precise. A continuously differentiable curve in L^∞ has a jointly measurable version that is absolutely continuous in t for almost every ξ and satisfies the pointwise equation, so $x(t)$ is the pointwise flow almost everywhere; conversely, since Ψ_t is locally Lipschitz in v uniformly on the common essential range of the values, which is bounded and bounded away from zero, $\xi \mapsto \Psi_t(r, v(\xi))$ is a continuously differentiable curve in L^∞. Applying Ψ_t pointwise to the second configuration gives a path whose spectra are the push-forwards of the common initial spectra, hence equal to those of $x(t)$. Its scalar inputs therefore coincide with those of $x(t)$, so it solves the fine flow, and by uniqueness it is the solution from the second configuration. The rows read only the costs, which agree. A base-only response reads only (r, μ^q). A base-only assimilation sends the value vector at r to $(L(o \mid r)v_i/Z_i)_i$ with $Z_i = \int L(o \mid r)q_{i,0}(r)dA(r)$; these factors depend on the agent unless the base marginals agree, but they are determined by μ^q, so the update acts on the spectra. □

On the profile family, the conditional law of the value vector given r is uniform on the orbit of the labeled vector $(f(r, u(i)))_i$ under the tree automorphisms, so μ^q and the unordered profile tree determine each other. The root case of theorem 11.9 is therefore an instance of proposition 11.6. What the hierarchy adds is that responses at intermediate nodes, which read physical subtree membership, also descend; there the relevant descriptors are the conditional spectra of subtrees given ancestor coordinates. This is the identity-channel analogue of the conditional spectrum of chapter 8.

Remark 11.7 (Compression requires the symmetry). Suppose the leaves of the two root branches carry different profiles f^L, f^R, a configuration outside $\mathrm{Fix}(H_\alpha)$. With $f_1^X(r, u_1) = \mathbb{E}f^X(r, u_1, \cdot)$, the fresh root response density is

$$\tfrac{1}{2}\left[f_1^L(r, \sigma_0) + f_1^R(r, -\sigma_0)\right], \tag{11.17}$$

which depends on the root sign, and hence on the root phase, unless $f_1^L(r, s) + f_1^R(r, -s)$ does not depend on s. Discarding the join phase under the fresh interface therefore needs this conjugacy condition, which holds automatically on the symmetric family. The two-block family is invariant: it is $\mathrm{Fix}(H'_\alpha)$ for the subgroup $H'_\alpha \leq H_\alpha$ of elements whose permutation preserves the two root branches, by the argument of proposition 11.5 without the root node element, so lemma 11.4 (i) applies. When the flow preserves the conjugacy condition, so that phase compression at a heterogeneous join holds along the whole trajectory and not only at one time, is not established here.

Remark 11.8 (Other symmetric hierarchies). The proofs use, at each node, a group of permutations of the children that acts 2-transitively on them, for example the full symmetric group, which gives an iterated wreath product of symmetric groups on the leaves, together with public coordinates that realize at each node a uniformly random element of that group, fair given the base and ancestor coordinates. Transitivity alone does not suffice: with cyclic arrangements of three children the two sibling costs of a receiver differ in general, so class-constant rows are not preserved. Rooted trees with larger branching, realized by uniform random child permutations, class-constant priors with an edge set that is a union of common-prefix classes, and phases that depend measurably on ancestor coordinates fall under lemma 11.4 in the same way. The operator

P_k is then replaced by the average that keeps the first k digits, replaces digit $k+1$ by a uniformly chosen different digit, and averages the rest. We do not write out these cases.

11.4 Recursive retention and the response process

At each base point, arrange the values of $f(r,u)$ in their binary profile tree. Retain this tree with unordered children at every internal node. Its leaves still carry density values, and its nested grouping is retained. Flattening it into a multiset would lose which values share a prefix and could change D_k.

For a physical subtree rooted at depth l, apply this retention only to its descendant values, keeping the dependence on the base and physical ancestor coordinates. This defines its parent descriptor. Include the corresponding model descriptor, the shared row-class data, primitive counts, and the remaining tree connections. At a lowest pair, the unordered values p_+, p_- can equivalently be written as a mean $(p_+ + p_-)/2$ and a squared contrast $((p_+ - p_-)/(p_+ + p_-))^2$. A first parent therefore retains more than its mixture law, but does not retain the two full child laws on their original node coordinates.

Theorem 11.9 (Recursive boundary inheritance on the supplied hierarchy). Under the hypotheses of theorem 11.2, these parent descriptors inherit the joint controlled dynamics at every level of the same fine system. Joining two sibling descriptors gives their unordered pair. The join commutes with evolution and with fresh uniform-descendant response kernels. Record likelihoods that depend only on the base and are applied to every agent, used as the assimilation events of definition 4.28, preserve these statements under arbitrary finite adaptive sequences, whenever the updates remain in the regular domain.

Proof. By proposition 11.5, the family is $\mathrm{Fix}(H_\alpha)$, and the maps (Φ_τ, id) and the phase rotations of section 11.3 act on it as stated there. At a fixed physical point, changing an internal sign exchanges the two branches below that node. Thus the map from physical leaf labels to their bit strings is an automorphism of the rooted binary tree. Forgetting its realization removes the phase at the joined node and leaves exactly the unordered profile subtree. As a direct check of lemma 11.4(ii), tree automorphisms preserve common-prefix length, U_d, and every P_k. They consequently preserve eqs. (11.8) and (11.9) and commute with eqs. (11.12) and (11.13), also when they depend measurably on the base, since the equations have no derivatives in the base coordinate, the local evaluator reads only f_0, g_0, and the global integrals are unchanged. The quotient flow is therefore independent of the chosen ordered representatives.

The quotient admits Borel representatives. Order $\mathbb{R}^{2^m}$ lexicographically. At each internal node, from the lowest level upward, place first the child subtree whose current vector of values is lexicographically smaller. Each step is a finite composition of Borel comparisons and coordinate permutations, and applying it at each base point to the measurable map $r \mapsto f(r,\cdot)$ gives a measurable canonical profile that depends only on the unordered tree. On a finite base this also gives measurability of the descended updates, since the chart is then finite dimensional and the profile flow is continuous; on a general base the record statements below assume, as definition 4.28 does, that the flow is jointly Borel in time and state.

For partial retentions, the same argument applies to the product of automorphism groups of the retained subtrees. The complement supplies the retained sibling and ancestor data needed for their interactions; the joint parent network closes. Thus this is controlled closure for complements in the stated prepared family, rather than autonomy under every external law. Combining sibling trees and then forgetting their ordering is the same operation as retaining their union directly. Repeated compatible joins give the same descriptor, and joins of disjoint subtrees commute. These are representation changes inside one fixed N-agent interaction graph. They do not add or delete fine edges, and do not authorize an arbitrary reassociation of the leaves into another hierarchy.

For a response at a node of depth l, choose a descendant freshly and uniformly, draw its full original public sample, and conceal its label. At every physical sample point the descendant labels run through every suffix bit string. The response density is exactly

$$f_l(r, u_1, \dots, u_l) = \mathbb{E}_{u_{l+1},\dots,u_d} f(r, u_1, \dots, u_d). \tag{11.18}$$

It is the mean of the retained subtree values, with all unused node factors left in the original output space, and it is invariant under the automorphisms acting below the node, as lemma 11.4(iii) requires. At the root it is $f_0(r)$. Choosing a sibling parent uniformly and then a descendant uniformly gives exactly the same full sample law. The model sector follows the same rule with a separate fresh label.

Fix a normalized positive record kernel $L_e^q(do \mid r)$ with bounded positive density on the admitted update domain, jointly Borel in (o, r), applied as a readout to the original samples of every agent. Every child has base marginal $f_0 A$, so a common record has the same component normalization

$$Z_o = \int f_0(r) L_e^q(o \mid r) dA(r). \tag{11.19}$$

Conditioning every mixture component preserves uniform labels and gives

$$f^+(r, u) = \frac{L_e^q(o \mid r) f(r, u)}{Z_o}. \tag{11.20}$$

This multiplies every value in a profile tree at the given base point by the same factor. It commutes with every element of $\mathcal{S}$ whose maps are fibered, and in particular with all retentions and joins, and its record law depends only on f_0. The model update is identical in form, and rows remain unchanged at the record update. Equality of retained states gives equality of the next response law and the retained update. Induction, using the inherited continuous flow between records, proves equality of every finite history law under policies choosing the next admitted experiment from the observed history; the policies are Borel kernels from finite histories to admitted experiments and waiting times, and the history laws are those of definition A.3. At a representation join, the already observed history is carried forward as controller state; no past record is erased. □

The records enter as assimilation events in the sense of definition 4.28. The potential is unchanged by them, and after an update the force eq. (11.12) depends on the record only through the updated profile. The admitted records are those whose likelihoods are invariant under the compression group and that are applied to every agent, which here means base-only and broadcast likelihoods; every admitted record history is a functional of the root trajectory $f_0(t)$. A likelihood that reads a

public coordinate, for example $L(o \mid r, \sigma_0)$ at the root node, multiplies the profiles of the two root branches by $L(o \mid r, u_1)$ and $L(o \mid r, -u_1)$ respectively, and the next fresh root response eq. (11.17) then depends on the root phase. A base-only record delivered to one subtree produces the same asymmetry. Records reading ancestor path bits would require retaining the ancestor phases and give a different descriptor.

Remark 11.10 (A persistent base-only evidence term). Add to the potential a fixed evidence term $\sum_i \int q_i(d\xi)[-\log L(o \mid r)]$ for a recorded base-only likelihood with bounded logarithm, and analogously in the model sector. This evidence-augmented potential is not the reciprocal potential of definition 4.2. Every leaf has base marginal $f_0 A$, so the term equals $N \int f_0(-\log L(o \mid r))dA$, and its force is the base-only function $-\log L(o \mid r)$, which enters h^q exactly as V^q does. The term is invariant under $\mathcal{S}$, so theorems 11.2 and 11.9 and lemma 11.3 hold with V^q replaced by $V^q - \log L(o \mid r)$ and $\|\log \rho\|_\infty$ increased by $\|\log L\|_\infty$, where $\|\log L\|_\infty$ is the sum of the bounds in the two sectors when both carry evidence. The added term is the case of eq. (4.32) with $o_i = o$ for every agent and record kernels that read only the base coordinates, $\ell_i(o \mid k, m) = L^q(o \mid r(k))L^s(o \mid w(m))$. The stationary laws then depend on the accumulated evidence, while the compression is preserved.

The fresh-sampling condition matters. For a pair $q_\pm = p(1 \pm a\sigma)$, two samples ξ, ξ' from one reused child have joint density

$$p(\xi)p(\xi')[1 + a(\xi)a(\xi')\sigma(\xi)\sigma(\xi')], \tag{11.21}$$

which can reveal the discarded phase. A shared child label for a joint belief/model response can likewise expose a relative contrast. Unrestricted external identity-channel probes can distinguish these laws, as in chapter 10. These protocols are outside the stated inherited interface.

Remark 11.11 (The root interface does not see base-uniform shapes). Let $f = f_0(r)F(u)$ with $\mathbb{E}_{U_d} F = 1$. Since $\sum_k b_k^q = 1$, eq. (11.12) gives $h^q = \log f_0 + V^q + \beta(u)$ with $\beta = 2\log F - \sum_k b_k^q (P_k \log F + P_k F/F)$, so $\partial_t \log F = -(\beta - \mathbb{E}_{U_d}[F\beta])$ is independent of r, the product form is invariant, and $\partial_t f_0 = -f_0[(\log f_0 + V^q) - \int f_0(\log f_0 + V^q)dA]$ involves neither F nor the rows; the shape enters only the row equations, through $D_k(f) = D_k(F)$. The model sector is the same, V^q and V^s read only base marginals, a base-only broadcast record multiplies f_0 by $L(o \mid r)/Z_o$ and leaves F unchanged, and its law depends on f_0 alone. Two base-uniform shapes F, F' in different tree-automorphism orbits therefore give identical histories of the root responses under every admitted adaptive record policy. The unordered profile tree is not minimal for the root interface alone.

Open problem 11.12 (Minimality of the unordered profile tree). Decide whether the unordered profile tree is minimal for the fresh-response interface of the hierarchy: whether the response histories of all nodes, under admitted adaptive record policies, separate the base-dependent tree-automorphism orbits of profiles, at least generically. By remark 11.11 the root histories alone do not, so a minimal root retention identifies every base-uniform shape with the flat profile. Proposition 11.15 proves local injectivity for a different two-agent binary model; for $d = 1$ and a two-point base, the analogous Jacobian computation for the root response and its time derivatives is a natural first case.

11.5 Public distinctions discarded at every depth

Choose, only as an initial witness,

$$f(r,u) = f_0(r)\prod_{j=1}^{d}[1 + a_j(r)u_j], \qquad 0 < a_j(r) < 1, \tag{11.22}$$

with regular bounds and $\int f_0 dA = 1$. The evolution subsequently uses the unrestricted positive profile f, and is not assumed to preserve this product. Before joining at depth l, the two child-parent response densities differ by their factor $1 \pm a_{l+1}\sigma_v$. Rotating α_v changes these full public laws, while their joined unordered profile tree stays fixed. Strict differences persist on a nonzero time interval by continuity. The same construction works at every depth and in both sectors. Under the same flow, however, these differences decay near the partition-independent profiles.

Proposition 11.13 (Decay of public contrast). Under the profile flow of theorem 11.2, the following hold in each sector. (i) The set of profiles that do not depend on u is invariant, and on it the rows converge to the class priors eq. (11.4). (ii) Along a solution in this set, write a nearby profile as $f = f_0(1 + \varphi)$ with $\mathbb{E}_{U_d}\varphi = 0$. To first order in φ,

$$\dot{\varphi} = -\left(3 - 2\sum_k b_k^q P_k\right)\varphi. \tag{11.23}$$

For the Walsh function $\chi_S(u) = \prod_{j\in S} u_j$ with largest index m, the rate is $\lambda_m = 3 + 2b_{m-1}^q - 2\sum_{k\geq m} b_k^q \geq 1$. At the class priors every rate equals $3 + 2/(N-1)$. (iii) Among profiles with a fixed base marginal, the belief part $\int f \log f dA dU_d + \sum_k b_k^q D_k(f)$ of the profile energy is minimized exactly by the profile that does not depend on u.

Proof. If f does not depend on u, then $P_k \log f = \log f$ and $P_k f/f = 1$, so h^q does not depend on u and the set is invariant. There $D_k = 0$, and eq. (11.13) is the Fisher flow of $T_a D_{\mathrm{KL}}(b^a\|\pi)$, whose log-odds relative to π decay at rate T_a. For (ii), $\log f = \log f_0 + \varphi + O(\varphi^2)$, $P_k \log f = \log f_0 + P_k\varphi + O(\varphi^2)$, and $P_k f/f = 1 + P_k\varphi - \varphi + O(\varphi^2)$. The u-dependent part of h^q is therefore $(3 - 2\sum_k b_k P_k)\varphi$, which has zero U_d-mean, while D_k is of second order and the rows and V^q are unchanged at first order. Writing $\dot{f} = \dot{f}_0(1+\varphi) + f_0\dot{\varphi}$ and taking the U_d-mean of $\dot{f} = -f(h^q - c_q)$ gives eq. (11.23). On χ_S, $P_k\chi_S = 0$ when $m > k+1$, $P_k\chi_S = -\chi_S$ when $m = k+1$, and $P_k\chi_S = \chi_S$ when $m \leq k$, which gives λ_m; since $\sum_k b_k = 1$, $\lambda_m \geq 1$. At the class priors, $\pi_{m-1} - \sum_{k\geq m}\pi_k = [2^{d-m} - (2^{d-m} - 1)]/(N-1) = 1/(N-1)$. For (iii), $\int f \log f dA dU_d = \int f_0 \log f_0 dA + \int f_0 D_{\mathrm{KL}}(f(r,\cdot)/f_0(r)\|U_d) dA$, the conditional term vanishes exactly at the u-independent profile, and each D_k is nonnegative and vanishes there. □

The common rate $3 + 2/(N-1)$ equals $1 + \kappa_N$ with $\kappa_N = 2N/(N-1)$, the rate of the complete-graph difference mode in eq. (10.9) at $J = 0$. The proposition is local. Two global statements follow from the lemmas above.

Corollary 11.14 (No finite-time loss of contrast, and global decay on a finite base). Under the profile flow of theorem 11.2, a profile that depends on u at $t = 0$ depends on u at every finite

time. If the base spaces are finite, every solution has bit contrast tending to zero in both sectors and rows tending to the class priors eq. (11.4).

Proof. The set of u-independent profiles is a closed linear subspace of the log chart, invariant by proposition 11.13 (i). The field is locally Lipschitz, so solutions are unique backward as well as forward, and if a solution entered the subspace at a time t^*, the solution inside the subspace through that point would coincide with it on $[0, t^*]$; this is the argument of proposition 6.24. On a finite base the chart is finite dimensional, and by lemma 11.3 the solution stays in a compact subset of the interior. The profile energy decreases at the rate of the summed Fisher variances, which vanish exactly at equilibria, so by LaSalle's invariance principle the ω-limit set consists of equilibria. At an equilibrium $h^q = c_q$, so f is a critical point, and since $\int f \log f$ and each D_k are convex and the evaluator term is linear in f, a minimizer, of $f \mapsto F(f, g, b)$ on normalized profiles. Averaging f over u at fixed r leaves f_0 and the evaluator term unchanged and, by proposition 11.13 (iii), lowers the rest strictly unless f is already u-independent. Every equilibrium therefore has u-independent profiles in both sectors, hence $D_k = 0$ and rows equal to the class priors. □

The distinctions discarded at every depth are therefore the orbit data of lemma 11.4. Their effects through the excluded protocols decay near the partition-independent profiles, and on a finite base along every solution; on a general base global decay is not established, because bounded sets of L^∞ are not compact.

In the circle realization these losses are continuous in total variation. For a phase shift δ with $|\delta| \le \pi$, the sign disagreement has Haar mass $|\delta|/\pi$. If the two branch response densities at a node are p_+, p_-, then, including the base and ancestor integrations, the total variation distance induced in one child-parent response is

$$\frac{|\delta|}{2\pi} \int \mathbb{E}_{U_l} |p_+(r, U_l) - p_-(r, U_l)| dA(r). \tag{11.24}$$

For $\pi < |\delta| \le 2\pi$, the factor $|\delta|$ is replaced by $2\pi - |\delta|$. With the squared Hellinger distance $H^2 = \frac{1}{2} \int (\sqrt{p} - \sqrt{p'})^2$, the same computation gives

$$H^2 = \frac{|\delta|}{2\pi} \int \mathbb{E}_{U_l} \left(\sqrt{p_+} - \sqrt{p_-}\right)^2 dA, \qquad |\delta| \le \pi. \tag{11.25}$$

The Fisher–Rao distance between the two laws is $2 \arccos(1 - H^2)$, which lies between $2\sqrt{2H^2}$ and $\pi\sqrt{2H^2}$. Hence every phase path along which $p_+ \neq p_-$ on a set of positive measure is Lipschitz in total variation and Hölder of order one half in the Fisher–Rao distance, and it is not rectifiable in that distance: the chord sum over n equal steps of a fixed phase interval grows like $\sqrt{n}$. These paths have no finite Fisher tangent and are not continuous in the bounded-log-density norm. In the coin realization, orientation changes are discrete.

Each phase can be an arbitrary measurable function of the base. The quotient therefore discards spatial patterns in the original public laws, beyond a permutation of primitive agents. It retains the amplitude profile tree, whose size can grow with the number of descendants; no fixed-size parent representation is asserted.

The sample-coordinate density is piecewise constant in the node coordinates, while the law-valued Fisher evolution remains smooth in time on the regular profile domain. The result does not give a smooth dimension reduction on an open law chart. It gives exact recursive public compression on a prepared invariant class, and that compression is a symmetry reduction. The general base laws and the coupled evaluator enter only through the base marginals f_0, g_0, which every level retains; the compressed data are orthogonal to them. The tree and the class are supplied, and no endogenous formation time has been derived.

The mechanism relies on complementary child laws whose equal mixture is flat in the public coordinate. Equal mixtures of Gaussian children do not have this property: with ϕ_d the standard normal density, $\frac{1}{2}[\phi_d(y-v)+\phi_d(y+v)] = \phi_d(y)e^{-|v|^2/2}\cosh(y\cdot v)$ depends on the direction of v unless $v=0$. A continuous phase of the kind used here therefore has no analogue for Gaussian child laws with a common isotropic covariance.

The reference coordinates r, w in this construction belong to represented sample spaces. They are not identified with the contextual base $\mathcal{C}$. Applying chapter 3 would additionally require an admitted smooth context-parameterized family and compatible connections. The measurable retention proved here does not supply the collective-connection condition in proposition 3.18.

11.6 Why ordinary response completion can remove all compression

The preceding restrictions cannot generally be removed by adding a few response moments to the phase parent. Observation derivatives can already recover a whole fine configuration, as in the local observability method of Hermann and Krener [106]. The following calculation uses only the inverse function theorem.

Proposition 11.15 (Fresh responses locally identify a binary pair). Consider two agents under the reciprocal potential with identity peer channels, binary state and model spaces, the uniform model reference, and the coupled binary evaluator $G_J(\kappa \mid \kappa') = e^{J\kappa\kappa'}/(2\cosh J)$ for any real J; this includes the range $|J|<1$ used in chapter 10. There is an open heterogeneous region where any retention preserving both fresh belief/model mixture responses and their future evolution is locally injective.

Proof. Write $q_i = p_{x_i}$, $s_i = p_{y_i}$, where $p_x(\kappa) = (1+\kappa x)/2$. The exact original equations, including incoming derivatives, are

$$\dot{x}_i = -(1-x_i^2)[2\,\operatorname{atanh} x_i - \operatorname{atanh} x_j - Jy_i] - (x_i - x_j), \tag{11.26}$$

and the exchanged equation for y; a nonuniform model reference would add its own term to the model equation. Set $x_{1,2} = m \pm a$, $y_{1,2} = n \pm e$. Fresh mixture responses are p_m, p_n. Their velocities have the convergent local expansions

$$\begin{aligned} F_x &= -(1-m^2)(\operatorname{atanh} m - Jn) + (6m - Jn)a^2 - 2Jmae + O(5), \\ F_y &= -(1-n^2)(\operatorname{atanh} n - Jm) + (6n - Jm)e^2 - 2Jnae + O(5). \end{aligned} \tag{11.27}$$

Here $O(5)$ denotes terms of total degree at least five. To check the coefficient, the uncoupled part of F_x is

$$-\tfrac{1}{2}\sum_{\pm}[1-(m\pm a)^2]\,\mathrm{atanh}(m\pm a)+2ma[\mathrm{atanh}(m+a)-\mathrm{atanh}(m-a)], \tag{11.28}$$

whose expansion is $-m+2m^3/3+6ma^2+O(5)$.

For $\mathcal{H}=(m,n,F_x,F_y)$, its Jacobian determinant at $m=n=\varepsilon c_1$, $a=e=\varepsilon c_2$, $c_1c_2\neq 0$, is

$$\det D\mathcal{H}=4\varepsilon^4c_1^2c_2^2(6-3J)(6-J)+O(\varepsilon^6). \tag{11.29}$$

Indeed, the lower-right block is $2\varepsilon^2c_1c_2$ times the matrix with diagonal $6-2J$ and off-diagonal $-J$, plus $O(\varepsilon^4)$. Its leading coefficient vanishes only at $J=2$ and $J=6$. At a general base point $(m,n,a,e)=\varepsilon(c_m,c_n,c_a,c_e)$ the same block gives the leading coefficient $\varepsilon^4Q(c;J)$ with $Q=[2(6c_m-Jc_n)c_a-2Jc_mc_e][2(6c_n-Jc_m)c_e-2Jc_nc_a]-4J^2c_mc_nc_ac_e$, which equals -32 at $J=2$, $c=(1,0,1,1)$ and -36 at $J=6$, $c=(1,0,1,1/2)$. For every real J there is therefore a base point at which the determinant is nonzero for sufficiently small nonzero ε, and it remains nonzero nearby. If two states in that neighborhood have the same inherited parent, their response curves agree. Equality at time zero and of their first derivatives gives equality of $\mathcal{H}$, hence equality of the states by local invertibility. No differentiability assumption on the retention is needed. □

This is a local obstruction in one original model. It neither removes the global child-exchange ambiguity nor excludes the selective protocols and invariant classes constructed above.

Chapter 12

Two-Stage Selection with Exact Inheritance on a Wishart Family

This chapter studies the reciprocal law potential of definition 4.2, with its exact Gibbs rows (proposition 4.3) and unit-Fisher dynamics, on an invariant family of matrix Wishart laws. It does not concern the global or per-agent free energies of chapter 4, and its conclusions are not transferred to them. Four initially singleton agents are grouped into two parents and then into their union by a declared threshold rule applied to their evolving laws. Every parent inherits the original-time dynamics, a declared joint response interface, and the collective connection and contextual tensor of its children. Every join discards a continuous distinction between model laws.

Measured against the formation standard definition 6.36, the construction is a two-stage selection with exact inheritance, not a recursive formation. The selector reads belief shapes, and the shape law $\dot{\alpha}_i = -2(\alpha_i - \alpha_0)$ contains neither the coupling, nor the precisions, nor the model sector, nor any other agent. Proposition 12.7 shows that the membership history is the count-weighted centroid dendrogram of the initial belief shapes, read on a clock whose origin is set by the threshold, that it is the same for every coupling strength, and that the renormalized null and the frozen-environment control of definition 6.33 reproduce it, event times included, on the entire admitted class. The self costs of this construction are nonzero, so the label-preserving null is a different flow: its self weights eq. (6.48) make the shape clocks agent dependent already at zero coupling (proposition 12.8), and integrations give a different partition sequence on an open set. The verdict of interaction necessity (F5) is therefore split between the two nulls, and the clause fails; the difference against the label-preserving null is present where no agent interacts and carries no information about the interaction. The second stage forms from every counterfactual handoff in the domain of the selector, so the counterfactual clause (F6b) fails as well, whatever the null. Proposition 12.3 identifies the structural cause, which every architecture with receiver-independent targets and additive moment channels shares for selectors that read relative log-likelihood ratios. The discarded model-shape allocation is dynamically inert, and the inherited contextual geometry is prepared in the initial section, is present at zero coupling, and decays exponentially (proposition 12.12). What the construction establishes is exact inheritance: original-time parent dynamics with the complete channel derivative, extensive response histories under adaptive protocols, and preservation of the collective connection and contextual tensor through both joins.

The architecture is declared. Children have matrix Wishart laws with common precision in each sector, product recognition, a constant model-independent evaluator, and comparison channels depending on the population moment. Individual shapes are constant over the contextual base. Responses use fresh independent samples and return extensive sums. Membership is selected by a declared threshold rather than by a threshold derived from the potential. The preparation class is

open within this invariant family. Each additive quotient is available initially, and inheritance holds for every partition at every time, so inheritance does not distinguish the selected partition; the rule selects when a quotient becomes the operative description. No law is reset at a join.

12.1 Matrix laws and the declared architecture

Let both represented spaces be the cone S_2^{++} of real symmetric positive-definite two-by-two matrices. A random matrix X or Y is represented content, distinct from a contextual coordinate. Write P, Q for precision matrices, to distinguish them from the principal bundle P. For $\nu > 1/2$ and $\mathsf{P} \succ 0$, define

$$W_2(\nu, \mathsf{P})(dX) = \frac{\det(\mathsf{P})^\nu}{\Gamma_2(\nu)} \det(X)^{\nu-3/2} e^{-\operatorname{tr}(\mathsf{P}X)} dX, \qquad \Gamma_2(\nu) = \sqrt{\pi}\Gamma(\nu)\Gamma(\nu - \tfrac{1}{2}). \tag{12.1}$$

Here $dX = dx_{11}dx_{12}dx_{22}$. This is the central Wishart law with real shape. Its normalizer is the multivariate gamma function, and congruence by B has Jacobian $\det(B)^3$ on this cone [149, Section 2.1 and Chapter 3]. The range $\nu > 1/2$ of real shapes for which eq. (12.1) is a probability density is the absolutely continuous part of Gindikin's set $\{0, \frac{1}{2}\} \cup (\frac{1}{2}, \infty)$, whose two isolated points give singular laws [90]; existence questions for the noncentral case are discussed by Letac and Massam [135]. No Gaussian representation or integer number of degrees of freedom is required. The proof below is self-contained.

Lemma 12.1 (Normalization and addition). The density in eq. (12.1) is normalized. Its mean is $\nu\mathsf{P}^{-1}$. Independent variables with laws $W_2(\nu, \mathsf{P})$ and $W_2(\mu, \mathsf{P})$ have sum law $W_2(\nu + \mu, \mathsf{P})$.

Proof. At $\mathsf{P} = I_2$, set $x_{11} = u > 0$, $x_{12} = \sqrt{u}z$, and $x_{22} = v + z^2$, with $v > 0$ and $z \in \mathbb{R}$. The determinant is uv, the trace is $u + v + z^2$, and the Jacobian is $\sqrt{u}$. The integral is $\Gamma(\nu)\Gamma(\nu - \frac{1}{2})\sqrt{\pi}$. Substitution $X = \mathsf{P}^{-1/2} Z \mathsf{P}^{-1/2}$ gives the general normalizer: congruence by a positive matrix B has Jacobian $\det(B)^3$ on this cone. Differentiating the normalizing integral with respect to P gives the stated mean.

The convolution density at $S \succ 0$ contains the integral

$$e^{-\operatorname{tr}(\mathsf{P}S)} \int_{0<X<S} \det(X)^{\nu-3/2} \det(S - X)^{\mu-3/2} dX. \tag{12.2}$$

The substitution $X = S^{1/2} U S^{1/2}$ makes it a constant times $e^{-\operatorname{tr}(\mathsf{P}S)} \det(S)^{\nu+\mu-3/2}$. The constant is positive and finite by Tonelli's theorem and normalization of the convolution. Normalizing the resulting density determines it and proves the addition formula. □

Initially take $N = 4$ agents with

$$q_i = W_2(\alpha_i, \mathsf{P}), \qquad s_i = W_2(\eta_i, \mathsf{Q}), \qquad \zeta_i = q_i \otimes s_i, \tag{12.3}$$

where every shape exceeds $1/2$. The model shape η_i is unrelated to a source-row weight. Fix $\alpha_0, \eta_0 > 1/2$ and $\kappa > 0$. The local data are the proper reference $r_i = W_2(\eta_0, I_2)$ and the normalized

constant evaluator $G_i(m, dX) = W_2(\alpha_0, I_2)(dX)$. For any admitted finite-moment configuration set

$$M_q = \sum_i \mathbb{E}_{q_i} X, \qquad \Lambda_q = I_2 + \frac{\kappa}{N} M_q, \qquad \mathcal{K}^q_{ij}[\mathbf{q}](k, dX) = W_2(\alpha_0, \Lambda_q)(dX). \tag{12.4}$$

The model sector uses $M_s = \sum_i \mathbb{E}_{s_i} Y$, $\Lambda_s = I_2 + \kappa M_s/N$, and target $W_2(\eta_0, \Lambda_s)$. Every source, including self, is allowed with prior $\pi_{ij} = 1/N$ and positive original temperatures. Each channel ignores its particular source sample k but reads the current laws through the total moment. It therefore depends on the whole configuration, not only on the two endpoint laws, as eq. (4.3) allows. The channel is a continuous function of M_q on the finite-moment configurations, so it is jointly measurable in the configuration and the input. Independent draws across receiver children specify the joint channel experiment used when their outputs are added.

These primitive self channels are nonidentity channels by constitutive choice. With identity self channels, definition 4.2 would give $D^q_{ii} = 0$ while the peer costs are positive away from agreement, so the Gibbs rows would depend on each receiver's peer cost and would differ from the priors. The construction requires the nonidentity choice.

All source costs in a receiver row agree, so the exact Gibbs row equals its prior. For the same reason, promoting the source priors to state variables of the same potential would leave them fixed. The Gibbs-reduced reciprocal potential is $\mathcal{E}_q + \mathcal{E}_s$, where

$$\mathcal{E}_q = \sum_i \left[D_{\mathrm{KL}}(q_i \| W_2(\alpha_0, I_2)) + D_{\mathrm{KL}}(q_i \| W_2(\alpha_0, \Lambda_q)) \right]. \tag{12.5}$$

The two sectors have separate population coupling. The evaluator is constant in its model input, and this assumption is load-bearing. By eq. (4.2), a model-dependent evaluator $G_i(m, \cdot) = W_2(\alpha_0, \Gamma(m))$ would add the term $\mathrm{tr}(\mathbb{E}_{s_i}[\Gamma(m)]X)$ to the belief force. Because the model laws differ across agents, the precisions would then cease to be common, and lemma 12.1 would no longer apply to the parents. The sum-pushforward of independent child evaluator outputs is a kernel from the retained model sum only because the evaluator ignores the model sample; for unequal precisions $\Gamma(m_i)$ the sum is not even a Wishart law. With the constant evaluator the belief and model sectors are uncoupled, and each sector is coupled across agents only through its own population moment. Outside the Wishart family the displayed functional requires finite relative entropies and moments; the differential also requires integrable log densities and log determinants, and differentiation under the integrals along the admitted paths.

12.2 The complete smooth Fisher flow

Proposition 12.2 (Invariant family and original-time evolution under the reciprocal potential). With the unit Fisher metric and all channel derivatives included, eq. (12.3) is invariant under eq. (12.5). In the belief sector,

$$C_q = \Lambda_q + \frac{\kappa}{N}(M_q - N\alpha_0\Lambda_q^{-1}) = I_2 + \frac{2\kappa}{N} M_q - \kappa\alpha_0\Lambda_q^{-1}, \tag{12.6}$$

$$\dot{\alpha}_i = -2(\alpha_i - \alpha_0), \qquad \dot{\mathsf{P}} = -2\mathsf{P} + I_2 + C_q. \tag{12.7}$$

The model equations exchange $\alpha, \alpha_0, \mathsf{P}, M_q, C_q$ with $\eta, \eta_0, \mathsf{Q}, M_s, C_s$. These solutions are global on the stated family, and every regular solution, in the sense defined in the proof, with these initial laws coincides with them. Distinct initial shapes remain distinct at every finite time.

Proof. Put $c_0 = \alpha_0 - 3/2$. Expanding the normalized densities gives

$$\begin{aligned}\mathcal{E}_q = 2\sum_i \int q_i \log q_i dX - 2c_0 \sum_i \mathbb{E}_{q_i} \log\det X + \operatorname{tr} M_q + \operatorname{tr}(\Lambda_q M_q) \\ - N\alpha_0 \log\det \Lambda_q + 2N \log\Gamma_2(\alpha_0).\end{aligned} \tag{12.8}$$

Since $d\Lambda_q = (\kappa/N) dM_q$, the differential of the last moment-dependent pair is $\operatorname{tr}(C_q dM_q)$. Thus the full variational force, up to a sample-independent scalar, is

$$h_i(X) = 2\log q_i(X) - 2c_0 \log\det X + \operatorname{tr}[(I_2 + C_q)X]. \tag{12.9}$$

The direct sender variation is zero because a mass-zero variation traversing a constant-in-input kernel gives zero. The channel variation is not zero: it is precisely the second term in the first expression for C_q. The unrestricted density equation is $\dot{q}_i = -q_i(h_i - \mathbb{E}_{q_i} h_i)$. Comparing the coefficients of $\log\det X$ and X in this equation with the derivative of eq. (12.1) gives eq. (12.7); normalization fixes the remaining scalar. This proves tangency of the full density field.

For regular Wishart data with possibly different precisions the same calculation gives $\dot{\mathsf{P}}_i = -2\mathsf{P}_i + I_2 + C_q$. Consequently

$$\mathsf{P}_i(t) - \mathsf{P}_j(t) = e^{-2t}(\mathsf{P}_i(0) - \mathsf{P}_j(0)). \tag{12.10}$$

Initially common precision remains common. On that family put $A = \sum_i \alpha_i$ and $\bar{\alpha} = A/N$. Then $M_q = A\mathsf{P}^{-1}$ and $\bar{\alpha}(t) = \alpha_0 + e^{-2t}(\bar{\alpha}(0) - \alpha_0)$. The precision field commutes with P, so its initial eigenvectors remain fixed and each positive eigenvalue solves

$$\dot{p} = f(p, \bar{\alpha}(t)) = -2p + 2 + \frac{2\kappa\bar{\alpha}(t)}{p} - \frac{\kappa\alpha_0 p}{p + \kappa\bar{\alpha}(t)}. \tag{12.11}$$

Let $m_- = \min(\bar{\alpha}(0), \alpha_0)$ and $m_+ = \max(\bar{\alpha}(0), \alpha_0)$. For $\varepsilon = \min(1/2, m_-/\alpha_0)$, $f > 0$ when $0 < p \le \varepsilon$. For $L = 2 + \kappa m_+$, $f \le -2$ when $p \ge L$. Every eigenvalue therefore stays between $\min(p(0), \varepsilon) > 0$ and $\max(p(0), L) < \infty$. Shapes stay between their initial values and their reference shape, all strictly above $1/2$. The smooth finite-dimensional field consequently has a unique global solution on this invariant family.

Slightly enlarged compact parameter neighborhoods dominate polynomial moments and logarithmic determinant derivatives in eq. (12.1). They justify finite KL, square-integrable scores, and the differentiations above.

Uniqueness is stated for the following class. A regular solution on $[0, \tau)$ is a family of Lebesgue densities $q_i(t, X) > 0$ on S_2^{++} such that each $q_i(t)$ is a probability density with $\mathbb{E}|X|$, $\mathbb{E}|\log\det X|$, and $\mathbb{E}|\log q_i|$ finite, $t \mapsto M_q(t)$ is locally integrable, and for almost every X the map $t \mapsto \log q_i(t, X)$ is absolutely continuous with $\partial_t \log q_i = -h_i + \mathbb{E}_{q_i} h_i$ for almost every t, where h_i is eq. (12.9) and $t \mapsto \mathbb{E}_{q_i} h_i$ is locally integrable. Put $C(t) = C_q(M_q(t))$, which is locally integrable because $\Lambda_q^{-1} \preceq I_2$.

For almost every X the equation is scalar and linear, so

$$\log q_i(t,X) = e^{-2t}\log q_i(0,X) + (1-e^{-2t})c_0 \log\det X - \operatorname{tr}(\widehat{C}(t)X) + k_i(t),$$

with $\widehat{C}(t) = \int_0^t e^{-2(t-u)}(I_2 + C(u))du$ and k_i a function of t alone. With Wishart initial laws this is $\det(X)^{a_i(t)-3/2}e^{-\operatorname{tr}(\Pi_i(t)X)}$ up to normalization, with $a_i(t) = \alpha_0 + e^{-2t}(\alpha_i(0)-\alpha_0) > 1/2$ and $\Pi_i(t) = e^{-2t}\mathsf{P}_i(0) + \widehat{C}(t)$. Normalizability forces $\Pi_i(t) \succ 0$: if $\Pi_i(t)$ had an eigenvalue $\pi_2 \le 0$, then after a rotation to $\operatorname{diag}(\pi_1,\pi_2)$ the chart of lemma 12.1 gives a factor $\int_0^\infty v^{a_i-3/2}e^{-\pi_2 v}dv = \infty$. Hence $q_i(t) = W_2(a_i(t),\Pi_i(t))$, so $M_q(t) = \sum_j a_j(t)\Pi_j(t)^{-1}$ and C are continuous, Π_i is continuously differentiable, and (a,Π) solves the closed parameter system above, whose field is smooth on $\{a > 1/2,\ \Pi \succ 0\}$. Uniqueness for that system identifies it with the solution constructed above. Bounded log-density charts in the sense of definition 4.13 do not contain this family, since $\log\det X$ and $\operatorname{tr}(\mathsf{P}X)$ are unbounded, so proposition 4.14 is not used; the chapter works on the finite-dimensional invariant family together with this uniqueness statement. Finally, shape differences equal their initial differences times e^{-2t}. □

The log-density equation has the form $\partial_t \log q_i = -2\log q_i - V_t + c_i(t)$, with V_t common to all agents. For a fixed target this is the Fisher–Rao gradient flow of a relative entropy, whose explicit solution interpolates geometrically between the initial law and the target [41, Appendix B]; the rate is two here because two relative-entropy terms contain q_i. In the present construction the target moves through M_q, and the Wishart family carries the solution.

The precision flow is coupled across agents. For a symmetric moment variation H,

$$DC_q[H] = \frac{2\kappa}{N}H + \frac{\kappa^2\alpha_0}{N}\Lambda_q^{-1}H\Lambda_q^{-1}. \tag{12.12}$$

Its trace pairing with nonzero H is strictly positive. Varying another agent's shape produces $H = \mathsf{P}^{-1}d\alpha_j$ and changes the coefficient of the linear statistic X in the force on q_i, hence the precision velocity. It does not change any shape velocity: in eq. (12.9) the coupling enters only the coefficient of X, while the coefficient of $\log\det X$ after substitution of $\log q_i$ is $2(\alpha_i - \alpha_0)$ for every $\kappa \ge 0$. The coupling comes from the declared population moment; the source rows remain at their priors and do not select preferred neighbors.

12.3 A common relaxation clock

The independence of the shape law from the interaction is an instance of a general identity for receiver-independent targets.

Proposition 12.3 (Shared targets impose a common relaxation clock). Consider the reciprocal potential for agents with common state and model spaces. Suppose every agent has the same model reference r, the same evaluator $G(m,dk) = g(dk)$, constant in m, and that every belief channel, for every receiver, source, and input, equals one normalized law $\nu^q[M_q]$ depending smoothly on $M_q = \sum_i \mathbb{E}_{q_i}T$ for a fixed measurable statistic $T : \mathsf{K} \to \mathbb{R}^d$; the model sector is analogous. Let all source priors and temperatures be positive. Along any solution of the unit-Fisher density equation of the Gibbs-reduced potential in the sense of the regular class of the proof of proposition 12.2, with T in place of X and with differentiation under the integrals assumed, every

pair of agents satisfies

$$\partial_t \log \frac{q_i}{q_j} = -2 \log \frac{q_i}{q_j} + c_{ij}(t), \tag{12.13}$$

with c_{ij} independent of the sample. Hence $\log(q_i/q_j)(t) = e^{-2t} \log(q_i/q_j)(0)$ up to a sample-independent function of t, for every target map ν^q, in particular for every coupling strength and for a constant target. If the laws lie in an exponential family with natural parameter θ and affinely independent sufficient statistic, then $\theta_i(t) - \theta_j(t) = e^{-2t}(\theta_i(0) - \theta_j(0))$.

Proof. All costs in row i equal $D_i = D_{\mathrm{KL}}(q_i \| \nu^q[M_q])$, so by proposition 4.3 the Gibbs row equals the prior and $-T_q \log Z_i^q = D_i$. By eq. (4.2) with a constant evaluator, the part of the reduced potential that depends on the belief laws is $\sum_l [D_{\mathrm{KL}}(q_l \| g) + D_{\mathrm{KL}}(q_l \| \nu^q[M_q])]$. By eq. (4.10), a variation of q_i contributes $\log(q_i/g) + \log(q_i/\nu^q[M_q])$ through the two terms in which q_i is the receiver, and $\langle \Psi_t, T \rangle$ through M_q, where $\Psi_t = -\sum_l \mathbb{E}_{q_l} \partial_M \log \nu^q[M]$ at $M = M_q$. Since M_q is additive and every agent enters it through the same statistic, Ψ_t is the same for every i. Thus $h_i = 2 \log q_i + V_t$ up to a constant, with $V_t = -\log g - \log \nu^q[M_q] + \langle \Psi_t, T \rangle$ common to all agents. The density equation gives $\partial_t \log q_i = -2 \log q_i - V_t + \mathbb{E}_{q_i} h_i$, and subtraction gives eq. (12.13). Integrating the linear equation proves the second statement. In an exponential family, $\log(q_i/q_j) = \langle \theta_i - \theta_j, S \rangle - \psi(\theta_i) + \psi(\theta_j)$, and affine independence of S identifies the coefficients. □

For the Wishart family the natural parameter is $(\alpha - 3/2, -\mathrm{P})$, so proposition 12.3 contains both the shape law and eq. (12.10). On finite positive tables the same identity is case (b) of proposition 6.31, which also records what happens along the nulls of definition 6.33: with self in every source set the self costs are nonzero, the two nulls differ, and neither keeps a common clock in general. In the present family only the shape coordinates of the renormalized null and of the frozen-environment control keep the clock, because the frozen moment enters only the coefficient of X (proposition 12.7). Any relabeling-equivariant selector that reads only centered log-likelihood ratios between agents, for example natural-parameter differences, sees along the full flow a rescaled copy of the initial data on a clock shared by every interaction in this class, so its partition sequence does not depend on the coupling. The conclusion is relative to that selector class. The cost of agent i satisfies, at common precision,

$$D_i - D_j = F(\alpha_i) - F(\alpha_j) + (\alpha_i - \alpha_j)\tau(t), \qquad D_i = D_{\mathrm{KL}}(q_i \| W_2(\alpha_0, \Lambda_q)), \tag{12.14}$$

with $F(a) = (a - \alpha_0)\psi_2(a) - \log \Gamma_2(a)$, $\psi_2 = (\log \Gamma_2)'$, and $\tau = \mathrm{tr}(\Lambda_q \mathsf{P}^{-1}) - 2$, which depends on κ through P and Λ_q. A threshold rule on $|D_i - D_j|$ can therefore be coupling sensitive although every target is receiver independent. This is one reason the formation standard fixes the selector family before the flows are analyzed. Whether exact sum closure on an open fiber of child allocations is compatible with receiver-selective rows is not settled here.

A second obstruction concerns exact additive closure itself.

Proposition 12.4 (Scalar sum closure forces a common clock). Let scalar parameters θ_i in an open interval Θ evolve by $\dot{\theta}_i = G(\theta_i, g(t))$, where $g(t)$ is a state common to all agents and G is continuous in its first argument. Suppose that for every value of g the sum $G(x, g) + G(y, g)$ depends on $(x, y) \in \Theta^2$ only through $x + y$, as exact intertwining of sum retention for every two-element block requires. Then $G(\theta, g) = a(g)\theta + c(g)$, and every difference of count-weighted centroids of disjoint

blocks is multiplied by the common factor $\exp \int_0^t a(g(u))du$. The ordering of all centroid distances is therefore that of the initial data at every time.

Proof. Fix g and write $G(\theta)$. For $x, y \in \Theta$ the midpoint $m = (x+y)/2$ lies in Θ and $x + y = m + m$, so $G(x) + G(y) = 2G(m)$. Choose $u < v$ in Θ, and let ℓ be the affine function agreeing with G at u and v. Then $G - \ell$ satisfies the same midpoint equation and vanishes at u and v, hence at every dyadic point of $[u, v]$, hence on $[u, v]$ by continuity. Since u, v are arbitrary, G is affine on Θ. A count-weighted centroid $\theta_I = n_I^{-1} \sum_{i \in I} \theta_i$ then obeys $\dot{\theta}_I = a\theta_I + c$, so centroid differences obey a linear homogeneous equation with the common coefficient $a(g(t))$. □

The Wishart shapes are the instance $a = -2$, $c = 2\alpha_0$, in which even a is independent of the interaction. The same Jensen argument, with its consequence for threshold selectors, is proposition 6.30. Vector parameters with a population-dependent generator that is not a multiple of the identity escape proposition 12.4, because a common anisotropic propagator can change which of two differences is smaller. Within the hypotheses of proposition 12.3 this escape is closed: natural-parameter differences obey a common propagator equal to -2 times the identity, whatever the dimension and the coupling. An anisotropic propagator therefore needs receiver- or source-dependent targets, unequal mobilities, or a selector that reads more than natural-parameter differences; a smooth reciprocal-potential instance with exact closure is not constructed here.

12.4 Parent laws and exact joining

Theorem 12.5 (Additive parents with the original Fisher dynamics). For any disjoint current partition, retain

$$n_I = |I|, \qquad A_I = \sum_{i \in I} \alpha_i, \qquad B_I = \sum_{i \in I} \eta_i, \qquad q_I = W_2(A_I, \mathsf{P}), \quad s_I = W_2(B_I, \mathsf{Q}). \tag{12.15}$$

The parent evaluator, reference, and comparison channels are the corresponding sum-pushforwards of independent child outputs. Their shapes are $n_I\alpha_0, n_I\eta_0$, and their source priors are $\pi^c_{IJ} = n_J/N$. With the primitive $N = \sum_I n_I$ kept fixed, their ordinary unit-Fisher dynamics is

$$\dot{A}_I = -2(A_I - n_I\alpha_0), \qquad \dot{B}_I = -2(B_I - n_I\eta_0), \tag{12.16}$$

together with the same precision equations eq. (12.7). Joining disjoint blocks adds (n, A, B) and keeps the common precisions. It commutes with laws, channels, and evolution, and is associative.

Proof. Use the fixed sample map $(X_i, Y_i)_{i \in I} \mapsto (\sum_i X_i, \sum_i Y_i)$. Lemma 12.1 gives eq. (12.15). The same map on the independent reference or channel outputs gives $W_2(n_I\alpha_0, I_2)$ or $W_2(n_I\alpha_0, \Lambda_q)$, and the model exchange. The parent evaluator is constant in its model input, just as the fine evaluator is; its descent through the sum map uses this constancy. Source normalization gives $\pi^c_{IJ} = n_J/N$ by summing primitive sender weights. Its row is again the exact Gibbs prior.

Total moments satisfy $M_q = \sum_I \mathbb{E}_{q_I} X_I$ and $M_s = \sum_I \mathbb{E}_{s_I} Y_I$. The parent reciprocal energy in the belief sector is

$$\mathcal{E}^c_q = \sum_I \left[D_{\mathrm{KL}}(q_I \| W_2(n_I\alpha_0, I_2)) + D_{\mathrm{KL}}(q_I \| W_2(n_I\alpha_0, \Lambda_q)) \right]. \tag{12.17}$$

Its Λ_q terms are still $\operatorname{tr}(\Lambda_q M_q) - N\alpha_0 \log\det\Lambda_q$ because $\sum_I n_I = N$. Its complete moment derivative is therefore the same C_q. The density-force calculation in eq. (12.9), with reference shape $n_I\alpha_0$, gives eq. (12.16) and the original precision equation. This is the projected fine vector field at the same time, with no mobility correction.

Adding two parents preserves total moments, counts, and forcing. Shape equations add, and sample addition gives the joined law and normalized channel. Associativity follows from addition of the three retained scalars and of the sample matrices. □

The theorem holds for every partition at every time, so exact inheritance does not single out the partitions that the selector of the next section produces; selection and inheritance are logically independent here.

The parent is type closed in the sense of definition 6.19: its descended field is the unit-Fisher flow of a reciprocal potential of the parent laws, with normalized law-dependent channels and a constant evaluator. A shorter route to proposition 12.2 and theorem 12.5 uses natural coordinates. In $x = (\nu, \mathsf{P})$ the law $W_2(x)$ has log density $\nu \log\det X - \operatorname{tr}(\mathsf{P}X) - F(x)$ against $\det(X)^{-3/2}dX$, with $F(\nu, \mathsf{P}) = \log\Gamma_2(\nu) - \nu\log\det\mathsf{P}$. Its Fisher matrix is the Hessian of F, and $D_{\mathrm{KL}}(W_2(x)\|W_2(y))$ is the Bregman divergence of F [4], so the natural gradient of $x \mapsto D_{\mathrm{KL}}(W_2(x)\|W_2(y))$ is $x - y$. The natural gradient of a function Φ of the mean $m = \mathbb{E}X = \nu\mathsf{P}^{-1}$ with $d\Phi = \operatorname{tr}(G\,dm)$ is $(0, -G)$. Summing these for the reference, for the target at fixed Λ_q, and for the moment dependence with $G = (\kappa/N)(M_q - N\alpha_0\Lambda_q^{-1})$ gives eq. (12.7) at every state with individual precisions; the same sum for the parents, with targets of shape $n_I\alpha_0$, gives eq. (12.16). The retention is not horizontally conformal for the Fisher metrics, so proposition 6.6 does not give this descent (open problem 6.20).

The final single parent still has its inherited nonidentity internal-source channel. Replacing it by an identity self-comparison would delete the second KL term and change its flow. A parent source label is the pushforward of one primitive source-label draw; a batch exposing separate primitive labels would require their joint mark law. Those labels are not exposed by the interface below.

Exact evolution does not assert equality of fine and parent energies. For example, at $\mathsf{P} = I_2$ replace two shapes α_0, α_0 by $\alpha_0 + \epsilon, \alpha_0 - \epsilon$. The total moment and parent state are unchanged, but the fine energy increases by $2D_{\mathrm{KL}}(W_2(\alpha_0 + \epsilon, I_2)\|W_2(\alpha_0, I_2)) + 2D_{\mathrm{KL}}(W_2(\alpha_0 - \epsilon, I_2)\|W_2(\alpha_0, I_2)) > 0$. The vector-field calculation, rather than energy factorization, proves inheritance.

12.5 Selected membership

Definition 12.6 (Threshold membership rule). Fix a threshold $\delta > 0$ and begin with singletons whose initial belief shapes are pairwise farther than δ apart. For a current block read its mean belief-shape deviation

$$b_I = A_I/n_I - \alpha_0, \qquad \dot{b}_I = -2b_I, \qquad b_{I\cup J} = \frac{n_I b_I + n_J b_J}{n_I + n_J}. \tag{12.18}$$

Join two current blocks when their distance $|b_I - b_J|$ first reaches δ from above. Simultaneous threshold edges join their connected components. Recompute distances after each event.

This rule reads only retained laws and counts, is invariant under relabeling, and contains no supplied membership tree. The threshold specifies a persistence criterion; it is additional constitutive data.

The separation hypothesis belongs to the domain of the rule: two shapes initially closer than δ never reach δ from above and are never joined to each other, even when they are nearest neighbors. Because every score is explicit, the event times below are computed in closed form, and no discretization or event-location scheme is involved. For the report against the formation standard the selector family is this rule with $\delta \in \mathsf{C} = [9/100, 11/100]$, declared with the constitutive data before the flows are analyzed, and the admitted class consists of the preparations of theorem 12.9 whose initial gaps all exceed $11/100$.

Proposition 12.7 (Coupling-independent centroid dendrogram and the controls that reproduce it). On the invariant family of proposition 12.2, let $\kappa \geq 0$, all precisions, and all model data be arbitrary.

(i) Every shape obeys $\alpha_i(t) = \alpha_0 + e^{-2t}(\alpha_i(0) - \alpha_0)$, so every block score obeys $b_I(t) = e^{-2t} b_I(0)$, where $b_I(0)$ is the count-weighted centroid of the initial deviations $\alpha_i(0) - \alpha_0$, $i \in I$.

(ii) Let $N \geq 2$ initial belief shapes be pairwise farther than δ apart, and suppose that at every stage of the agglomeration below the closest pair of current centroids is unique. Then definition 12.6 makes $N-1$ binary joins and coincides with count-weighted centroid agglomerative clustering of the initial shapes: at stage k the two current blocks whose initial centroids are closest, at distance D_k, join at time $t_k = \frac{1}{2}\log(D_k/\delta)$. The heights D_k increase strictly; at each event every other current distance exceeds δ; and the scores of the two blocks joined at t_k differ by $\delta e^{-2(t-t_k)} < \delta$ at every later time. The membership tree depends only on the initial belief shapes. It is independent of κ, of the precisions, of the model sector, and of δ, which shifts every event time by $-\frac{1}{2}\log\delta$.

(iii) Coupling-deformation control. For the one-parameter family of constructions indexed by $\kappa \geq 0$, whose member at $\kappa = 0$ has channel targets equal to the reference and hence no law-dependent channel, the membership history $H(x)$, event times included, is the same for every $\kappa \geq 0$ at every point x of the admitted class. This control is not one of the controls of definition 6.33.

(iv) Renormalized null. The null Φ^0_{rn} of definition 6.33, in which the peer costs and their labels are deleted, the self row has weight one, and each self channel is evaluated with the peer laws held at their initial values, exists globally on the Wishart family with individual precisions and has the same shape trajectories as Φ. Hence $H^0_{\mathrm{rn}}(x) = H(x)$, event times included, and $P^0_{\mathrm{rn}}(x) = P(x)$ at every point x of the admitted class, for every $\kappa \geq 0$ and every δ. This null is not the flow at $\kappa = 0$: it keeps κ inside each self channel, so its precisions differ and need not stay common.

(v) Frozen-environment control. The control Φ^{fr} of definition 6.33 exists globally on the same family and has the same shape trajectories, so $H^{\mathrm{fr}}(x) = H(x)$ and $P^{\mathrm{fr}}(x) = P(x)$ at every point of the admitted class.

After a handoff every control remains the fine flow, and the selector reads the block shape sums A_I computed from the fine shapes. Along Φ^0_{rn} and Φ^{fr} with $\kappa > 0$ the precisions are individual, the block laws are not Wishart, and theorem 12.5 and corollary 12.10 are not claimed along them.

Proof. (i) Substituting $\log q_i = (\alpha_i - \frac{3}{2})\log\det X - \mathrm{tr}(\mathsf{P}X) + \mathrm{const}$ into eq. (12.9) gives $h_i = 2(\alpha_i - \alpha_0)\log\det X + \mathrm{tr}[(I_2 + C_q - 2\mathsf{P})X] + \mathrm{const}$. The coupling, the precision, and the other agents

enter only the coefficient of X; matching the coefficient of $\log\det X$ gives the shape law for every $\kappa \geq 0$, and at $\kappa = 0$ the channel target equals the reference. Block scores are count-weighted averages of agent scores, and averaging commutes with the common factor e^{-2t}.

(ii) By (i), every current distance is De^{-2t} with D a distance between initial centroids, so a pair reaches δ at $\frac{1}{2}\log(D/\delta)$, and the first pair to reach it is the pair with the smallest D. Since all initial distances exceed δ, every pair reaches δ from above. Suppose I and J are the closest current pair, at distance D, and let K be any other current block. The merged centroid lies strictly between the centroids of I and J. The centroid of K does not lie between them, since otherwise it would be closer than D to one of them. If it lies beyond the centroid of J, its distance to the merged centroid is its distance to J plus $n_I D/(n_I + n_J)$, which exceeds D; the other side is symmetric. Thus every distance after the join exceeds D, which proves that heights increase strictly, that centroids remain distinct, and that at the event time every other current distance is above δ, so there are no cascades and every later pair again reaches δ from above. The two joined blocks keep their scores along the fine flow, so their difference is $\delta e^{-2(t-t_k)}$. This is the agglomeration that joins the closest pair of current centroids and replaces it by its count-weighted centroid. The heights D_k are functions of the initial shapes alone.

(iii) is (i) and (ii), whose conclusions do not depend on κ.

(iv) The renormalized null deletes the peer costs with their labels and evaluates each self channel with the peer laws held at their initial values. Its belief potential is $\sum_i [D_{\mathrm{KL}}(q_i \| W_2(\alpha_0, I_2)) + D_{\mathrm{KL}}(q_i \| W_2(\alpha_0, \Lambda_i^0))]$ with $\Lambda_i^0 = I_2 + (\kappa/N)(\mathbb{E}_{q_i} X + M_{-i}^0)$, where M_{-i}^0 is the initial moment of the other agents. The calculation of proposition 12.2 with this single-agent moment gives the same shape law and $\dot{\mathsf{P}}_i = -2\mathsf{P}_i + I_2 + C_i^0$, where $C_i^0 = \Lambda_i^0 + (\kappa/N)(\alpha_i \mathsf{P}_i^{-1} - \alpha_0 (\Lambda_i^0)^{-1})$. For a unit eigenvector v of the smallest eigenvalue p of P_i, $\Lambda_i^0 \succeq I_2$ gives $v^T \dot{\mathsf{P}}_i v \geq -2p + 2 - \kappa\alpha_0/N + 2\kappa\alpha_i/(Np)$, which is positive for small p because α_i stays above $\min(\alpha_i(0), \alpha_0) > 1/2$. With the smallest eigenvalue bounded below, the largest eigenvalue p satisfies $\dot{p} \leq -2p + K$ for a constant K on each finite interval. The precisions therefore remain in compact subsets of the cone, and the solution is global and unique as in proposition 12.2. The selector reads only belief shapes, whose trajectories are the same under both flows, also after every handoff, so the membership histories agree, event times included.

(v) Along Φ^{fr} nothing is deleted, every row stays at its prior because all costs in a row agree, and in every term that enters the force on q_i the other laws take their initial values. The force is $h_i = 2\log q_i - 2c_0 \log\det X + \mathrm{tr}[(I_2 + C_i^{\mathrm{fr}})X]$ with $C_i^{\mathrm{fr}} = \Lambda_i^0 + (\kappa/N)(\alpha_i \mathsf{P}_i^{-1} + M_{-i}^0 - N\alpha_0 (\Lambda_i^0)^{-1})$, so the shape law is that of (i) and $\dot{\mathsf{P}}_i = -2\mathsf{P}_i + I_2 + C_i^{\mathrm{fr}}$. At the smallest eigenvalue p of P_i, $v^T \dot{\mathsf{P}}_i v \geq -2p + 2 - \kappa\alpha_0 + \kappa\alpha_i/(Np)$, which is positive for small p when $\kappa > 0$, and the case $\kappa = 0$ is the flow $\dot{\mathsf{P}}_i = -2\mathsf{P}_i + 2I_2$; the largest eigenvalue is bounded as in (iv). The solution is global, and the selector sees the shape trajectories of (i). □

The label-preserving null of definition 6.33 keeps the peer labels and priors, so with Gibbs rows the self row of receiver i has the weight eq. (6.48), which is below one.

Proposition 12.8 (The label-preserving null has agent-dependent clocks). Along Φ_{lp}^0, with self prior $a = 1/N$ and belief temperature T_q, put $D_{ii} = D_{\mathrm{KL}}(q_i \| W_2(\alpha_0, \Lambda_i^0))$, with Λ_i^0 and C_i^0 as in the proof of proposition 12.7, and let $\beta_i = ae^{-D_{ii}/T_q}/(ae^{-D_{ii}/T_q} + 1 - a) \in (0, 1/N]$. The Wishart family

with individual precisions is invariant, with

$$\dot{\alpha}_i = -(1+\beta_i)(\alpha_i - \alpha_0), \qquad \dot{\mathsf{P}}_i = -(1+\beta_i)\mathsf{P}_i + I_2 + \beta_i C_i^0, \tag{12.19}$$

and the model analog. For $\kappa\alpha_0 \le N^2$ the solution exists for all $t \ge 0$ and is unique on this family. At $\kappa = 0$ and $\mathsf{P}_i(0) = I_2$ the precisions stay at I_2 and each deviation $b_i = \alpha_i - \alpha_0$ obeys the autonomous equation

$$\dot{b} = -(1+\beta(b))b, \qquad \beta(b) = \frac{1}{1+(N-1)e^{F_0(b)/T_q}}, \qquad F_0(b) = D_{\mathrm{KL}}(W_2(\alpha_0+b, I_2)\|W_2(\alpha_0, I_2)), \tag{12.20}$$

so that agents farther from α_0 relax more slowly, and the clocks differ between agents whenever their shapes differ in distance from α_0.

Proof. With peer costs set to zero and exact Gibbs rows, the row term of receiver i is $-T_q \log(ae^{-D_{ii}/T_q} + 1 - a)$, whose first variation in q_i is β_i times that of D_{ii}. The first variation of D_{ii} is $\log q_i - \log W_2(\alpha_0, \Lambda_i^0)$ plus the channel term $\mathrm{tr}[(\kappa/N)(\alpha_i \mathsf{P}_i^{-1} - \alpha_0(\Lambda_i^0)^{-1})X]$, so the force is $(1+\beta_i)\log q_i - (1+\beta_i)c_0 \log\det X + \mathrm{tr}[(I_2 + \beta_i C_i^0)X]$ up to a constant, and matching coefficients as in proposition 12.2 gives eq. (12.19). The closed form $D_{\mathrm{KL}}(W_2(a,\mathsf{P})\|W_2(b,\mathsf{Q})) = (a-b)\psi_2(a) - \log\Gamma_2(a) + \log\Gamma_2(b) + b\log\det(\mathsf{P}\mathsf{Q}^{-1}) + a(\mathrm{tr}(\mathsf{Q}\mathsf{P}^{-1}) - 2)$ makes β_i a smooth function of the state. Shapes move monotonically from $\alpha_i(0)$ toward α_0. At the smallest eigenvalue p of P_i with unit eigenvector v, $\Lambda_i^0 \succeq I_2$ gives $v^T C_i^0 v \ge 1 - \kappa\alpha_0/N$, so, since $0 < \beta_i \le 1/N$, $v^T \dot{\mathsf{P}}_i v \ge -2p + 1 + \beta_i(1 - \kappa\alpha_0/N) \ge -2p + 1 + 1/N - \kappa\alpha_0/N^2$ when $\kappa\alpha_0 \ge N$ and $\ge -2p+1$ otherwise. For $\kappa\alpha_0 \le N^2$ both bounds are positive for small p. With the smallest eigenvalue bounded below, C_i^0 is bounded and the largest eigenvalue satisfies $\dot{p} \le -p + K$. The parameter field is smooth on the invariant region, which gives global existence and uniqueness. At $\kappa = 0$, $\Lambda_i^0 = C_i^0 = I_2$ and $\dot{\mathsf{P}}_i = (1+\beta_i)(I_2 - \mathsf{P}_i)$, so $\mathsf{P}_i \equiv I_2$ from I_2, and then $D_{ii} = F_0(b_i)$, which is zero at $b = 0$ and increases with $|b|$ because F_0 is a relative entropy in the shape of an exponential family with convex log partition. Hence β decreases in $|b|$. □

As $T_q \to \infty$ every β_i tends to $1/N$, the clocks become the common rate $1 + 1/N$, and the partition sequence of the full flow returns. At $T_q = 1$ the agent-dependent clocks change the partition sequence. For $\alpha_0 = 1$, $\kappa = 0$, $\mathsf{P}(0) = I_2$, and initial shapes $(1.2807, 1.5057, 1.8665, 2.3057)$, which satisfy eq. (12.21) for every $\delta \in \mathsf{C}$, the full flow at $\delta = 1/10$ forms $\{1,2\}$, $\{3,4\}$, and their union at $t = 0.406$, 0.740, 0.968, while integration of eq. (12.19) with the selector applied to fine shape sums forms $\{1,2\}$, $\{1,2,3\}$, and the union at $t = 0.689$, 1.340, 1.776. These sequences differ even modulo exchange of disjoint merges. At each null event every other block distance exceeds δ by at least 3.5×10^{-3}, and every crossing is transverse, at five values of δ spanning C with $\kappa = 0$; the null sequence is the same at $\kappa = 0.05$ and 0.1, and the same two sequences were obtained under twelve random perturbations of size 4×10^{-3} in the shapes and 10^{-2} in $\mathsf{P}(0)$, with $\kappa \in \{0, 0.01\}$ and $T_q \in [0.9, 1.1]$. By continuous dependence of both flows on the initial data, $P_{\mathrm{lp}}^0 \ne P$ then holds on a nonempty open subset of the admitted class, already at zero coupling. This conclusion rests on floating-point integration with these margins, not on a validated enclosure. At $\kappa = T_q = 1$, $\mathsf{P}(0) = I_2$, and the example shapes below, the label-preserving null forms the same blocks as the full flow, at the later times 0.650, 1.111, 2.058. Where $\kappa\alpha_0 > N^2$ the existence of this null is not shown, and (F5) against it is undefined there.

Centroid clustering is one of the pair-group strategies of numerical taxonomy [183], unified with the other agglomerative strategies by Lance and Williams [126]. In one dimension it has no reversals, which is the monotonicity in (ii). The rule is not single-linkage clustering. For initial deviations $0, 1, 2.05, 3.5$, single linkage joins the third point to the first pair before the last point, whereas the rule forms the two pairs first, at heights 1 and 1.45, and their union at 2.275. The union time in eq. (12.23) below already carries the centroid distance $d_2+(d_1+d_3)/2$ rather than the single-linkage distance d_2. The construction also differs from bounded-confidence dynamics [105], in which the interaction determines which agents cluster. What it adds to centroid clustering is exact inheritance of laws, dynamics, responses, and geometry at each join.

Theorem 12.9 (Two selected stages on the four-agent family). Let $u_1 < u_2 < u_3 < u_4$ be the sorted initial belief shapes, with consecutive gaps d_1, d_2, d_3, and suppose

$$\delta < \min(d_1, d_3) < d_2, \qquad \max(d_1, d_3) < d_2 + \tfrac{1}{2}\min(d_1, d_3). \tag{12.21}$$

All four shapes exceed $1/2$, while model shapes vary independently above $1/2$. Under definition 12.6, the two lower and the two upper shapes are joined into parents at

$$t_L = \tfrac{1}{2}\log(d_1/\delta), \qquad t_R = \tfrac{1}{2}\log(d_3/\delta). \tag{12.22}$$

Their union is joined at

$$t_2 = \tfrac{1}{2}\log\left(\frac{d_2 + (d_1 + d_3)/2}{\delta}\right) > \max(t_L, t_R). \tag{12.23}$$

There are no intervening cross joins or cascades. Both first parents coexist on $[\max(t_L, t_R), t_2)$. Each joined child-block distance remains below threshold afterward, and all fine belief laws remain distinct at every finite time. When strict inequalities replace ties, eq. (12.21) is also necessary for this membership history. Each join forgets a continuous model-shape distinction even if all membership labels and event times are retained.

Proof. Apply proposition 12.7. Suppose $d_1 \le d_3$. Since $d_1 < d_2$, the lower pair is the closest initial pair. After it joins, the distance of its centroid to u_3 is $d_2 + d_1/2$, which exceeds d_3 by eq. (12.21), so the upper pair joins next. The distance between the two parent centroids is $d_2 + (d_1 + d_3)/2$, giving eq. (12.23). The opposite order follows by exchanging upper and lower pairs. If $d_1 = d_3$, exactly the two disjoint outer edges reach threshold together, and the component rule forms the same pairs. Conversely, if $d_2 < \min(d_1, d_3)$ the middle pair joins first, and if $\max(d_1, d_3) > d_2 + \frac{1}{2}\min(d_1, d_3)$ the first parent absorbs the nearer middle shape before the other pair joins.

The persistence statement is part of proposition 12.7; it follows from the equations rather than from storing a membership flag. Primitive shape differences never vanish at finite time. The two first-parent laws also remain distinct, because their mean belief shapes remain different.

For loss at a first join, fix all belief shapes and perturb two initial model shapes in that pair by $(\eta_i, \eta_j) \mapsto (\eta_i + \epsilon, \eta_j - \epsilon)$ for sufficiently small ϵ. The parent model sum, both population moments, all subsequent precision evolution, and every membership event and time are unchanged. Individual model laws change continuously. Distinct fixed belief shapes distinguish the labels, so this change is not a permutation. At the second join, transfer model shape between the two first-parent sums

while keeping their union sum fixed. This gives a further continuous lost distinction and still leaves the belief-only selection schedule unchanged. □

The discarded allocation is a dynamical spectator in both descriptions. Individual model shapes obey $\dot{\eta}_i = -2(\eta_i - \eta_0)$, the model precision depends on the model shapes only through their total, the belief sector never reads the model laws because the evaluator is constant, and the selector reads only belief shapes. The witness therefore establishes noninjectivity of the retention, not loss of a coordinate to which any agent, channel, or selector responds. The belief allocation that the selector reads is not lost given the event record: at a join the two child scores differ by exactly δ, so the parent sum, the child counts, the join time, δ, and α_0 determine the child belief shapes up to exchange. The loss witness accordingly uses the model-shape allocation. Already observed sample records remain with the controller; they are not erased or asserted to have the same law across different initial preparations. Loss concerns the current parent state and the future interface specified next.

The hierarchy is transient. All shapes converge to α_0, η_0, and every precision eigenvalue converges to the unique positive zero of $f(\cdot, \alpha_0)$ or of its model analog, so every agent tends to one common law. The children are distinct at every finite time, and the consensus is asymptotic rather than reached in finite time.

The preparation class is nonempty and relatively open. For example take $\alpha_0 = \eta_0 = \kappa = 1$, $\delta = 1/10$, $(u_1, u_2, u_3, u_4) = (1, 6/5, 9/5, 21/10)$, and model shapes in a neighborhood of $(4/5, 11/10, 7/5, 17/10)$. The gaps are $(1/5, 3/5, 3/10)$, and the events occur at $\frac{1}{2}\log 2$, $\frac{1}{2}\log 3$, and $\frac{1}{2}\log(17/2)$. The two first parents coexist for a time $\frac{1}{2}\log(17/6)$. A sufficiently small neighborhood and every relabeling obey eq. (12.21). Openness is within the common-precision family, not among all positive density tuples. The smooth statistical retention applies on each fixed membership branch; no global smoothness across switching boundaries is required.

Against definition 6.36 the report is as follows. The objective is the reciprocal potential, and the dynamics is the unit-Fisher flow of proposition 12.2 on the parameter manifold of the common-precision Wishart family; no chart of the full density flow is declared, since bounded log-density charts do not contain the family. (F1) holds in the family grade: the admitted class is relatively open in the invariant family defined by Wishart laws with common precision in each sector, which has dimension 14, infinite codimension among positive densities, and is not contained in finitely many orbits. (F2) holds for the selector family $\delta \in \mathsf{C} = [9/100, 11/100]$, which reads only retained shapes and counts and so satisfies the readout convention, with an invariant block relation as persistence criterion. The parent protocol rule assigns to a block I the fresh block-sum queries of the response section below, read through whole current blocks; the probe rule adds, before a handoff of I, the protocols of the blocks merged into it, which before a first join are fresh samples of single children and before the second join fresh sums over each first parent. (F3) holds for these classes by theorem 12.5, corollary 12.10, and theorem 12.11. (F4) holds: the model-shape transfer in the proof of theorem 12.9 gives two reachable post-handoff configurations with the same retained state, separated by the fresh model sample of one child, and at the second join by the fresh model sum of one first parent. The separation is transient, since every model shape relaxes to η_0 at rate two, and it is not of symmetry type, since that relaxation contracts every fiber displacement by e^{-2t}; the separated coordinate is moreover dynamically inert. (F5) fails against the renormalized null at every

point and for every $\delta \in \mathsf{C}$ (proposition 12.7 (iv)), while against the label-preserving null the partition sequence differs on an open set already at zero coupling (proposition 12.8 and the integrations after it). The verdict is split, and by definition 6.36 the clause fails. The selector-free discrepancy is the shape contraction rate, which is two along Φ, Φ^0_{rn}, and Φ^{fr} and the agent-dependent $1+\beta_i$ along Φ^0_{lp} at every coupling, including zero; $P^{\mathrm{fr}} = P$, and the coupling-deformation control of proposition 12.7 (iii) shows no dependence on κ at all. (F6a) holds: the parents select their union under the inherited dynamics with the same selector, constants, and data, without a reset. For (F6b) declare $\mathcal{U}$ to be the whole common-precision family with shapes above $1/2$, and let $\mathcal{C}$ be built by definition 6.35 at the time the second first parent forms. Every state of $\mathcal{C}$ lies in the domain of the selector, so its two parent scores differ by more than δ, and that difference contracts by e^{-2t} whatever the fine allocation; the union forms from every state of $\mathcal{C}$, the stages form a supplied-scale sequence, and (F6b) fails, against either null. The strong grade fails with (F5); the amplifying grade is not evaluated. The construction is a two-stage selection with exact inheritance along a prepared contraction ordering, with (F5) and (F6b) failing. Since (F6b) fails independently, this classification does not depend on counting a split (F5) verdict as a failure; the (F5) verdict itself does.

12.6 Exact joint response histories

The response interface is noninvasive. At each query, draw fresh samples independently across children and sectors from $\bigotimes_i (q_i \otimes s_i)$. For the current disjoint partition Π, form the entire vector

$$Z_\Pi = ((X_I, Y_I))_{I\in\Pi}, \qquad X_I = \sum_{i\in I} X_i, \quad Y_I = \sum_{i\in I} Y_i. \tag{12.24}$$

An action may choose any normalized readout kernel on this vector. It can report overlapping unions of whole current blocks using the shared batch, so no independence between such overlapping reports is assumed. Query times and readouts may depend measurably on carried record history and retained state. Recording an output does not reset a belief or model law.

Corollary 12.10 (Adaptive extensive-response inheritance). For every admitted preparation and every finite sequence of finite query times selected by the preceding protocol, the fine and parent descriptions have the same joint law of records, actions, query times, and retained states. This includes the membership events of theorem 12.9.

Proof. Independence across disjoint blocks and lemma 12.1 give the conditional batch kernel

$$K_\Pi = \bigotimes_{I\in\Pi} [W_2(A_I, \mathsf{P}) \otimes W_2(B_I, \mathsf{Q})]. \tag{12.25}$$

Initialize the retained process from the same fine configuration. Condition on the retained state and carried history at a query. The same policy chooses the next waiting time and readout in both descriptions. Theorem 12.5 gives equal retained states after waiting, and definition 12.6 gives the same intervening events and addition handoffs. Freshness makes the next conditional batch law exactly eq. (12.25); composition with the same normalized readout gives the same next-record kernel. Induction proves equality of every finite joint history. Independent external randomization can be conditioned on in the same proof. □

The result is a corollary of two facts: the interface exposes only block sums, whose law depends on the fine allocation only through A_I and B_I, and the retained state is a deterministic function of the preparation and time. Records never feed back into laws, so adaptivity selects only query times and readouts, and the induction carries no further content. Every decision functional of the joint record law, such as the expected cost of an adaptive policy or the entropy of the record history, takes the same value in both descriptions; information gain about the discarded allocation does not transfer, because the parent has no such coordinate. The informative extension, in which records drive updates of the laws, is outside the declared interface.

The corollary compares two descriptions of each given preparation. A final aggregate is not claimed to reconstruct previously discarded records or the distribution of earlier records from a different preparation. Queries cutting through a current block need additional intersection data. For example, same-batch sums over primitive subsets C, D have

$$\operatorname{Cov}(\operatorname{tr}(UX_C), \operatorname{tr}(VX_D)) = A_{C\cap D} \operatorname{tr}(\mathsf{P}^{-1}U\mathsf{P}^{-1}V), \tag{12.26}$$

which can expose a discarded shape sum. Repeated hidden-child sampling, child-specific Bayesian updates, and interventions changing the reference potential are outside the declared interface.

12.7 Inherited collective geometry

Use the trivial principal bundle $\mathcal{C}\times SO(2)$ and the common left sample action $X \mapsto gXg^T, Y \mapsto gYg^T$. It sends precisions to $g\mathsf{P}g^T, g\mathsf{Q}g^T$. The identity-precision references are invariant under this action. One rotation acting on belief and model samples simultaneously is a declared choice of collective group. Because the evaluator is constant, the constitutive data are also invariant under independent rotations of the two sectors, and nothing in the potential ties them. Use the collective product-law Fisher metric with unit weights, the common comparison frame, and the convention $D^A p = dp + J_p A$ of chapter 3. Let $\mathcal{C} = (a_-, a_+) \times (-\chi_0, \chi_0)$, with $0 < a_- < a_+$, and prepare the contextual section by

$$\mathsf{P}(0; a, \chi) = R_\chi \operatorname{diag}(e^a, e^{-a}) R_\chi^T, \qquad \mathsf{Q}(0) = \operatorname{diag}(e^{\tilde{a}}, e^{-\tilde{a}}), \qquad \tilde{a} > 0. \tag{12.27}$$

Here R_χ is plane rotation by χ. All individual shape parameters are constant over $\mathcal{C}$. Consequently the selection schedule is also constant over this contextual chart. The flow acts pointwise in the context, and no vector field, row, channel, or record depends on the connection: the connection below is a diagnostic of the evolved sections, and its curvature is the image under the flow of the prepared contextual dependence in eq. (12.27).

Theorem 12.11 (Exact connection and contextual-tensor inheritance for a prepared contextual family). For every block, before and after either join, sum retention preserves its collective minimizing connection class and minimized contextual tensor for the common $SO(2)$. For eq. (12.27), that tensor has rank two and the connection has nonzero curvature at every finite time, for every $\kappa \geq 0$.

Proof. Differentiating eq. (12.1) at fixed shape gives the precision score and Fisher metric

$$\ell_U(X) = \alpha \operatorname{tr}(\mathsf{P}^{-1}U) - \operatorname{tr}(UX), \qquad g^F(U, V) = \alpha \operatorname{tr}(\mathsf{P}^{-1}U\mathsf{P}^{-1}V). \tag{12.28}$$

The second identity follows by differentiating the normalizing integral twice. For common precision variations (U, V), the summed fine score is

$$A_I \operatorname{tr}(\mathsf{P}^{-1}U) - \operatorname{tr}(UX_I) + B_I \operatorname{tr}(\mathsf{Q}^{-1}V) - \operatorname{tr}(VY_I). \tag{12.29}$$

It equals the parent score and is measurable from the retained sums. Both contextual and common rotation-orbit directions have this form. Their Fisher pairings and all mixed pairings are therefore preserved. The infinitesimal stabilizers agree as well: in each description they are the rotations commuting with both precisions. Thus the normal equations for the minimizing connection have the same solutions modulo the same isotropy, and the residual tensors agree. This also verifies the mixed-loss criterion in proposition 3.18. The sample addition channel is fixed, so it contributes no moving-channel derivative.

For explicit curvature, let $p_\pm(t, a)$ be the two eigenvalues of $\mathsf{P}(t; a, \chi)$, and let $\tilde{p}_\pm(t)$ be those of $\mathsf{Q}(t)$. The scalar field in eq. (12.11) has

$$f_p = -2 - \frac{2\kappa\bar{\alpha}}{p^2} - \frac{\kappa^2\alpha_0\bar{\alpha}}{(p + \kappa\bar{\alpha})^2} < 0. \tag{12.30}$$

Its solution map $\Phi_t(p_0)$ satisfies $\partial_{p_0}\Phi_t = \exp(\int_0^t f_p du) > 0$ for finite t. Hence $p_+ = \Phi_t(e^a) > p_- = \Phi_t(e^{-a}) > 0$, $\partial_a p_+ > 0$, and $\partial_a p_- < 0$. The model eigenvalues also remain distinct and have no a dependence. Define the rotational inertias

$$\mathcal{I}_1 = 2A_I \frac{(p_+ - p_-)^2}{p_+ p_-}, \qquad \mathcal{I}_2 = 2B_I \frac{(\tilde{p}_+ - \tilde{p}_-)^2}{\tilde{p}_+ \tilde{p}_-} > 0. \tag{12.31}$$

Writing $r = p_+/p_- > 1$ gives $\mathcal{I}_1 = 2A_I(r + r^{-1} - 2)$ and $\partial_a \mathcal{I}_1 > 0$. Radial precision variations are diagonal in an eigenframe, while orbit variations are off-diagonal, so their Fisher pairing vanishes. The collective normal equation then gives

$$A^F = -\frac{\mathcal{I}_1}{\mathcal{I}_1 + \mathcal{I}_2} d\chi, \qquad h^{\min} = g_{aa} da^2 + \frac{\mathcal{I}_1 \mathcal{I}_2}{\mathcal{I}_1 + \mathcal{I}_2} d\chi^2, \tag{12.32}$$
$$g_{aa} = A_I \left[\left(\frac{\partial_a p_+}{p_+} \right)^2 + \left(\frac{\partial_a p_-}{p_-} \right)^2 \right] > 0.$$

Identifying the one-dimensional Lie algebra with its rotation generator, the curvature is

$$F^F = dA^F = -\frac{(\partial_a \mathcal{I}_1)\mathcal{I}_2}{(\mathcal{I}_1 + \mathcal{I}_2)^2} da \wedge d\chi \neq 0. \tag{12.33}$$

These computations hold for every retained block and its fine constituents with the same totals. None of them uses $\kappa > 0$. Under a passive local rotation by $\vartheta(c)$, the angular momentum one-form gains $(\mathcal{I}_1 + \mathcal{I}_2)d\vartheta$, so $A^{F\prime} = A^F - d\vartheta$. The tensor is invariant, with exactly the affine connection law of the general construction. □

The minimizing connection is the mechanical connection of a two-component rotor with locked inertia $\mathcal{I}_1 + \mathcal{I}_2$ [182], and $h^{\min}$ is the pullback of the quotient metric of the corresponding Riemannian submersion [154]. Its curvature measures how the ratio of belief to model rotational inertia varies

along the context. Two features delimit it. First, the curvature belongs to the tied belief and model connection. For the independent group $SO(2)_q \times SO(2)_s$, which the constitutive data also admit, the Fisher cost of a connection (ξ_q, ξ_s) is $\mathcal{I}_1(d\chi + \xi_q)^2 + \mathcal{I}_2\xi_s^2 + g_{aa}da^2$, minimized by $\xi_q = -d\chi$ and $\xi_s = 0$; that connection is flat and the residual tensor $g_{aa}da^2$ has rank one. Second, the geometry is prepared rather than produced: it is present at $t = 0$ and at $\kappa = 0$, where the agents do not interact, and the next proposition shows that the flow erases it. At finite times $p_+ > p_-$, so the infinitesimal isotropy is zero and the constant-rank hypothesis of chapter 3 holds; agreement of the children's laws does not affect it, because the collective action is common.

Proposition 12.12 (Decay of the inherited contextual geometry). Let p_* be the unique positive zero of $f(\cdot, \alpha_0)$ in eq. (12.11), and put $\lambda = -f_p(p_*, \alpha_0)$, so that $\lambda > 2$ for $\kappa > 0$ and $\lambda = 2$ for $\kappa = 0$. For every context, $\mathsf{P}(t) \to p_* I_2$, and $e^{\lambda t}(p_+ - p_-)$, $e^{\lambda t}\partial_a p_+$, and $e^{\lambda t}\partial_a p_-$ converge to nonzero limits. Consequently $e^{2\lambda t}g_{aa}$ and $e^{2\lambda t}\mathcal{I}_1$ converge to positive limits, the minimized tensor tends to zero, and the limit laws are isotropic, with stabilizer all of $SO(2)$. For $\alpha_0 = \kappa = 1$, $p_* = \sqrt{2}$ and $2\lambda = 12 - 4\sqrt{2}$.

Proof. Since $f_p < 0$ and $f(p, \alpha_0)$ tends to $+\infty$ as $p \to 0$ and to $-\infty$ as $p \to \infty$, the zero p_* is unique. On the compact invariant region of the proof of proposition 12.2, f and f_p are Lipschitz. With $y = p - p_*$, write $\dot{y} = [f(p, \bar{\alpha}) - f(p_*, \bar{\alpha})] + [f(p_*, \bar{\alpha}) - f(p_*, \alpha_0)]$. Since $f_p \le -2$, the first bracket has the sign of $-y$ and modulus at least $2|y|$, and the second is bounded by a constant times $|\bar{\alpha} - \alpha_0| = O(e^{-2t})$. Hence $|y(t)| \le (|y(0)| + Kt)e^{-2t}$, uniformly for initial values in a compact set. Then $f_p(p(u), \bar{\alpha}(u)) + \lambda$ is integrable on $[0, \infty)$, so $\partial_{p_0}\Phi_t = e^{-\lambda t}(c(p_0) + o(1))$ with $c(p_0) > 0$ continuous, uniformly on compact sets of p_0. The identities $\partial_a p_\pm = \pm e^{\pm a}\partial_{p_0}\Phi_t(e^{\pm a})$ and $p_+ - p_- = \int_{e^{-a}}^{e^a} \partial_{p_0}\Phi_t(p_0)dp_0$ give the stated limits, and the formulas for g_{aa} and $\mathcal{I}_1$, with $A_I \to n_I\alpha_0$ and $p_\pm \to p_*$, give the rest. For $\alpha_0 = \kappa = 1$, substitution verifies $f(\sqrt{2}, 1) = 0$ and $f_p(\sqrt{2}, 1) = -6 + 2\sqrt{2}$. □

For the example constants and the block of the first two agents at $a = 3/10$, g_{aa} equals 4.40, 2.69×10^{-3}, and 5.20×10^{-6} at $t = 0, 1, 2$; by the root join at $t_2 \approx 1.07$ it is below 10^{-3} of its initial value. The model inertia decays at the rate given by the model analog of λ. Because A^F is a ratio of two vanishing inertias, the limit of the finite-time curvature depends on whether the two rates agree: for $\alpha_0 = \eta_0 = \kappa = 1$, model log-eigenvalue $\tilde{a} = 3/10$, and the example model shapes it tends to approximately -1.81, a value that depends on $\tilde{a}$ (about -1.55 at $\tilde{a} = 0.2$ and -1.27 at $\tilde{a} = 0.5$), while for $\alpha_0 = 1, \eta_0 = 2$ the connection tends to the flat value $-d\chi$ and the curvature to zero. In neither case is the limit a property of the limit laws, at which the connection is undetermined.

Remark 12.13 (Persistent contextual geometry from anisotropic references). Replace the belief evaluator by $W_2(\alpha_0, \Gamma(c))$ with $\Gamma(a, \chi) = R_\chi \operatorname{diag}(e^a, e^{-a})R_\chi^T$, the belief channel target by $W_2(\alpha_0, \Lambda_q)$ with $\Lambda_q = \Gamma + \kappa M_q/N$, and the model reference by $W_2(\eta_0, \operatorname{diag}(e^{\tilde{a}}, e^{-\tilde{a}}))$, with the analogous model channel. The calculation of proposition 12.2 gives $C_q = \Lambda_q + (\kappa/N)(M_q - N\alpha_0\Lambda_q^{-1})$ and $\dot{\mathsf{P}} = -2\mathsf{P} + \Gamma + C_q$, with the same shape law. Precisions remain common, evaluators and targets still add under sums, and theorem 12.5, corollary 12.10, proposition 12.7, and the selection schedule are unchanged. With the section $\mathsf{P}(0; a, \chi) = \Gamma(a, \chi)$, the eigenvalues converge to the zeros of $-2p + 2\gamma + 2\kappa\alpha_0/p - \kappa\alpha_0 p/(\gamma p + \kappa\alpha_0)$ with $\gamma = e^{\pm a}$, which differ and depend on a; the limit tensor and curvature are computed from these fixed points. For the example constants at $a = \tilde{a} = 3/10$ the block tensor tends to $\operatorname{diag}(1.168, 0.199)$ and the curvature to approximately -1.689, so rank two and nonzero curvature persist at infinite time. Both sectors need anisotropic local data;

with an isotropic model reference the model sector isotropizes and the curvature tends to zero. The local data are then contextual sections that transform under the common action, so the statement above that the references are invariant no longer holds; the curvature remains a property of the declared contextual data, not of the interaction.

The context-constant shape assumption is needed only for the tensor. On every smooth section of the common-precision family the fine and retained Fisher costs of a belief-sector tangent vector $v = (\delta\alpha, U)$ differ by $\Delta(v, v) = \sum_{i\in I} \psi_2'(\alpha_i)\delta\alpha_i^2 - \psi_2'(A_I)(\sum_{i\in I} \delta\alpha_i)^2$, plus the model analog, because the precision block of the Wishart Fisher matrix is linear in the shape and the mixed block adds over children. This vanishes on every vector without a shape component, in particular on the orbit directions, so by proposition 3.18 the minimizing connection class is inherited for every such section, while by corollary 3.19 the minimized tensors differ by Δ evaluated on the contextual derivatives of the shapes, which is positive unless every individual shape is context constant. For the tensor the assumption is substantive. If $d\eta_1 = 1, d\eta_2 = -1$ while precisions and all other shapes are fixed, the parent model sum has zero derivative, whereas the fine score is the centered difference of two log determinants with positive variance. It is orthogonal to rotation scores because determinant is conjugation invariant. Its fine minimized length is positive and its retained length is zero. Such a contextual direction violates exact tensor inheritance.

Open problem 12.14 (Approximate inheritance off the common-precision family). The common-precision family has positive codimension, and exact sum laws fail off it. By eq. (12.10) and the shape law, the ratio of precision spread to shape spread is constant in time, so at a join the precision mismatch is δ times a preparation constant. Under moment matching, the first-order term of the log-Laplace transform of a sum of Wishart laws with nearby precisions cancels, so the sum law deviates from the Wishart family at second order in the precision spread. Prove that on an open set of Wishart preparations with unequal precisions the moment-matched parent inherits the law, the dynamics, and the extensive responses up to a stated relative-entropy defect at each join that decays afterward, and determine its order.

Chapter 13

Preferential Selection in Two Stages from Open Law Classes

This chapter concerns the reciprocal law potential of definition 4.2 with exact Gibbs rows (proposition 4.3); no statement below is transferred to the per-agent or global free energies of chapter 4. The common-precision construction selects an exact additive description within a prepared family. The construction given here instead admits initial laws ranging over a full open neighborhood. Native source rows, read by a fixed threshold selector, select three pairs and then a pair of their parents. The selection reads only a transmitted bit. The compression is an orientation symmetry of a fixed selective channel; it exists before the selected events and descends for every partition at every time. What the events change is the resolution of a declared external response interface, and the information discarded at a handoff is exactly the information that this interface stops resolving there. The compression group is a gauge group of the population in the sense of chapter 1; within this construction the gauge structure supplies the orbits that a parent forgets and does no dynamical work. Measured against the formation standard definition 6.36, the main theorem establishes selection, exact inheritance, and compression on an open class over a finite window. It establishes neither interaction necessity (F5) nor the counterfactual clause (F6b). In the regime of its proof, integrations of the limit field indicate that the nulls of definition 6.33 form the same blocks in the same stages (remark 13.6); this is not proved. Every block relation is transient, and the selector later merges the exterior pair (corollary 13.10). The construction is therefore not shown to be a recursive formation, and it is not shown to be a two-stage selection with exact inheritance either, since no clause is shown to fail.

13.1 A homogeneous selective-channel model

Every primitive belief and model space is $\mathsf{K} = \mathsf{M} = \{-1, 1\} \times S^1$. Use the probability measure

$$d\nu = \frac{1}{2} \sum_{z=\pm 1} \delta_z \otimes \frac{d\theta}{2\pi}. \tag{13.1}$$

All densities below are relative to this measure. The $2N$ normalized densities q_i, s_i vary independently over positive densities of class C^r in the angle, with $r \geq 3$ and logarithms bounded in C^r; this is the chart of the chapter. Internal bit-angle correlations are unrestricted. Recognition is the declared product $q_i \otimes s_i$. Take uniform proper references and evaluators, $r_i = G_i = 1$, equal unit Fisher mobilities, a fixed temperature $T > 0$, and all-source priors $\pi_{ij} = 1/N$, including self.

Use the same fixed normalized comparison channel in both sectors and on every edge, self edges included:

$$\mathcal{K}((z,\theta),d(z',\theta')) = \delta_z(dz')\frac{d\theta'}{2\pi}. \tag{13.2}$$

It transmits the bit and replaces the angle by an independent uniform draw. There is no channel switch at a selected event. In the terminology of chapter 8, the bit is the public coordinate, meaning the coordinate that peer channels transmit, and the angle is a private coordinate with Haar private reference. The word public is used only in this sense below. The channel is the S^1 twirl of the angular coordinate, which averages a law over all rotations of its angle; it is the map that describes communication between parties that share no angular reference frame [17].

If

$$\mu_i(z) = \int q_i(z,\theta)\frac{d\theta}{2\pi}, \qquad \eta_i(w) = \int s_i(w,\psi)\frac{d\psi}{2\pi}, \tag{13.3}$$

then $q_j\mathcal{K}$ has density $\mu_j(z)$, and the model output has density $\eta_j(w)$. The chain rule gives

$$D_{\mathrm{KL}}(q_i\|\mu_j\otimes\mathrm{Haar}) = H_i^q + D_{\mathrm{KL}}(\mu_i\|\mu_j), \qquad H_i^q = D_{\mathrm{KL}}(q_i\|\mu_i\otimes\mathrm{Haar}). \tag{13.4}$$

The receiver term H_i^q cancels from the Gibbs normalization. Thus

$$\beta_{ij} = \frac{\exp[-D_{\mathrm{KL}}(\mu_i\|\mu_j)/T]}{\sum_k \exp[-D_{\mathrm{KL}}(\mu_i\|\mu_k)/T]}, \tag{13.5}$$

and γ is the same expression in η. All entries are positive. The cancellation uses the selective self channel. With an identity self channel one would have $D_{ii}^q = 0$ while every peer cost retains H_i^q, so each peer weight would carry the factor $e^{-H_i^q/T}$ relative to self; at the small temperatures used below this suppresses every peer weight unless H_i^q is of order T. The construction therefore requires the selective self channel, which is a constitutive choice. The angular term remains in the reduced energy:

$$\mathcal{E}_{\mathrm{red}} = \sum_i \big[D_{\mathrm{KL}}(q_i\|1) + D_{\mathrm{KL}}(s_i\|1)\big] - T\sum_i \log\left(\frac{1}{N}\sum_j e^{-D_{ij}^q/T}\right) - T\sum_i \log\left(\frac{1}{N}\sum_j e^{-D_{ij}^s/T}\right), \tag{13.6}$$

where D_{ij}^q is the left side of eq. (13.4).

Proposition 13.1 (Complete density force). On the positive chart, the exact unit-Fisher field is

$$\begin{aligned} h_i^q(z,\theta) &= 2\log q_i(z,\theta) - \sum_j \beta_{ij}\log\mu_j(z) - \sum_k \beta_{ki}\frac{\mu_k(z)}{\mu_i(z)}, \\ \partial_t q_i &= -q_i\big(h_i^q - \mathbb{E}_{q_i}h_i^q\big), \end{aligned} \tag{13.7}$$

with the exchanged formula for s,η,γ. Every sum includes self.

Proof. The local KL contributes $\log q_i$ modulo a constant. Differentiating a reduced row by its envelope identity gives its weighted divergence derivatives. The normalized receiver row contributes another $\log q_i - \sum_j \beta_{ij}\log\mu_j$. Varying a source marginal in every incoming row gives $-\sum_k \beta_{ki}\mu_k/\mu_i$.

The fixed channel has no law derivative. Fisher inversion on normalized densities subtracts the q_i expectation and multiplies by $-q_i$.

For a self term the incoming ratio is one, but $D^q_{ii} = H^q_i$ need not vanish. Its constant incoming contribution and its angular entropy are both included. Logarithms and marginal ratios are smooth on a neighborhood bounded away from zero. Bounded integration, multiplication, and finite Gibbs normalization preserve C^r. The field contains no angular derivative, so it is locally Lipschitz both for the C^r norm of log-densities and for their supremum norm. This is the Lipschitz hypothesis of proposition 4.14 for the Gibbs-reduced potential (its sufficient condition on fixed kernels does not apply, since the selective channel has no density bounded below), which gives a unique maximal interior flow on the chart with the continuation alternative and the dissipation identity. Its centering preserves normalization. □

13.2 Two stages selected by native routing

For a current partition $\mathcal{P}$, retain the actual block rows

$$B^q_{IJ} = \frac{1}{|I|}\sum_{i\in I}\sum_{j\in J}\beta_{ij}, \qquad B^s_{IJ} = \frac{1}{|I|}\sum_{i\in I}\sum_{j\in J}\gamma_{ij}. \tag{13.8}$$

Self mass remains in each normalized row. Its experiment chooses a receiver uniformly in I, draws its native source index, and reports the source block. If blocks are joined, finite summation gives

$$B^a_{CD} = \frac{1}{|C|}\sum_{I\subset C}|I|\sum_{J\subset D}B^a_{IJ}, \qquad a\in\{q,s\}. \tag{13.9}$$

This is associative aggregation with inherited primitive counts.

Definition 13.2 (Preferential pair rule). Fix $\varepsilon = 1/10$ and $\delta = 1/20$. Distinct current blocks I, J qualify when, in both sectors and directions,

$$B^a_{IJ} > \varepsilon, \qquad B^a_{IJ} - \sum_{K\in\mathcal{P}\setminus\{I,J\}} B^a_{IK} > \delta. \tag{13.10}$$

The exterior excludes self and the proposed partner. A partner need not beat the self loop. Merge qualifying disjoint pairs at their first closed-threshold contact when the strict inequalities hold immediately afterward: the event time is the first time at which the closed inequalities hold and the strict inequalities hold on a right neighborhood. In every event proved below the relevant crossing is transverse, so this time is the unique zero of a block-row difference with positive time derivative. A block cannot have two qualifying partners: each would have to exceed the other by more than δ. The rule is therefore unambiguous at simultaneous disjoint events and respects relabeling. Persistence below means strict contrast over the stated positive time windows.

The routing calculation first uses binary densities $\mu_i(z) = 1 + zx_i$, $\eta_i(w) = 1 + wy_i$, with $(x, y) \in (-1, 1)^{2N}$. Define $d(u, v) = D_{\mathrm{KL}}(\mu_u\|\mu_v)$. Its derivatives are

$$\partial_u d(u,v) = \operatorname{atanh} u - \operatorname{atanh} v, \qquad \partial_v d(u,v) = \frac{v-u}{1-v^2}. \tag{13.11}$$

The exact binary field is

$$\begin{aligned}\dot{x}_i = (1-x_i^2)\left[-\operatorname{atanh} x_i - \sum_j \beta_{ij}(\operatorname{atanh} x_i - \operatorname{atanh} x_j)\right] \\ + \sum_k \beta_{ki}(x_k - x_i).\end{aligned} \tag{13.12}$$

The model equation is identical and independent. Self summands vanish in this binary equation, consistently with $d(x_i, x_i) = 0$.

Lemma 13.3 (Interior binary evolution). Every binary initial configuration has a unique global interior solution, and all its means tend to zero.

Proof. At a maximum of $\vartheta_i = \operatorname{atanh} x_i$, both peer sums in the equation for $\dot{\vartheta}_i$ are nonpositive; at a minimum they are nonnegative. The finite maximum and minimum satisfy the corresponding one-sided differential inequalities. Scalar comparison yields

$$e^{-t} \min_i \vartheta_i(0) \le \vartheta_i(t) \le e^{-t} \max_i \vartheta_i(0). \tag{13.13}$$

The same holds for the model logits. The solution stays in a compact interior box, where the smooth field extends it past any proposed finite maximal time. The bounds imply convergence. This does not by itself establish the order of selection events. □

The perturbation steps below all use one elementary statement. It compares a reference trajectory with a perturbed one and tracks both the trajectory and the observables that define events, since the observables themselves change with each perturbation.

Lemma 13.4 (Persistence of strict windows). Let X solve $\dot{X} = \Phi(X)$ on $[0, L]$ in a finite-dimensional space with the supremum norm, and let K_ρ be the tube of closed balls of radius ρ about $X(t)$. On K_ρ suppose that a possibly time-dependent field Ψ satisfies $|\Psi - \Phi| \le e$, that Φ is C-Lipschitz on each slice, and that $|\Phi|, |\Psi| \le M$. Let O be a C^1 observable with $|\nabla O|_1 \le G$ and ∇O Lipschitz with constant L_1, and let $\widetilde{O}$ satisfy $|\widetilde{O} - O| \le \eta_0$ and $|\nabla \widetilde{O} - \nabla O|_1 \le \eta_1$ on K_ρ. If $\dot{Y} = \Psi(Y)$, $|Y(0) - X(0)| \le \delta_0$, and

$$\Delta = \delta_0 e^{CL} + e\frac{e^{CL} - 1}{C} < \rho, \tag{13.14}$$

then $Y(t) \in K_\rho$ on $[0, L]$, and

$$\begin{aligned}|\widetilde{O}(Y) - O(X)| &\le \eta_0 + G\Delta, \\ \left|\tfrac{d}{dt}\widetilde{O}(Y) - \tfrac{d}{dt}O(X)\right| &\le \eta_1 M + L_1 \Delta M + G(e + C\Delta).\end{aligned} \tag{13.15}$$

Suppose $|O(X) - \theta_0| \ge m$ outside an interval I, $O(X) - \theta_0$ has opposite signs at the ends of I, and $\frac{d}{dt}O(X) \ge s > 0$ on I. If the two bounds in eq. (13.15) are below m and s respectively, then $\widetilde{O}(Y) - \theta_0$ has exactly one zero in $[0, L]$, it lies in I, and it is transverse; strict windows of the reference persist.

Proof. While Y is in the tube, $|Y - X|' \le |\Psi(Y) - \Phi(Y)| + |\Phi(Y) - \Phi(X)| \le e + C|Y - X|$ because each slice is convex. Gronwall's inequality bounds $|Y - X|$ by $\Delta < \rho$, so the first exit time cannot occur. The first observable bound follows from $|\widetilde{O}(Y) - O(Y)| + |O(Y) - O(X)|$ and the mean value theorem on the convex slice. For the second, write $\frac{d}{dt}\widetilde{O}(Y) = \nabla\widetilde{O}(Y) \cdot \Psi(Y)$ and $\frac{d}{dt}O(X) = \nabla O(X) \cdot \Phi(X)$, and split the difference into the four terms $(\nabla\widetilde{O} - \nabla O)(Y) \cdot \Psi(Y)$, $(\nabla O(Y) - \nabla O(X)) \cdot \Psi(Y)$, $\nabla O(X) \cdot (\Psi - \Phi)(Y)$, and $\nabla O(X) \cdot (\Phi(Y) - \Phi(X))$. Outside I the perturbed observable keeps the reference sign; inside I it is strictly increasing and changes sign once. □

Proposition 13.5 (Two-stage preferential routing). For $N = 6$, some fixed $T > 0$ and a nonempty relabeling-invariant open set in $(-1, 1)^{12}$ have the following behavior. Every initial off-diagonal native weight is below $1/10$. Three preferential pairs form and coexist for a positive interval. Two of their parents subsequently join while the third remains an exterior with strictly smaller routing masses. The first pairs and the selected parent pair satisfy eq. (13.10) over positive windows. All events occur at finite interior times under eq. (13.12), with no parameter change.

Proof. Choose $T = h^2$ and write $x_i = hu_i$. The function $h^{-2}d(hu, hv)$ is even and analytic in h on $h \max(|u|, |v|) < 1$, with

$$h^{-2}d(hu, hv) = \frac{(u - v)^2}{2} + \frac{h^2}{12}(u - v)^2(u^2 + 2uv + 3v^2) + O(h^4). \tag{13.16}$$

Dividing eq. (13.12) by h, without changing time, gives a field and block rows that are analytic in (h^2, u) on $\{h|u|_\infty < 1\}$. At $h = 0$ the field is

$$\dot{u}_i = -u_i + \sum_j (w_{ij} + w_{ji})(u_j - u_i), \qquad w_{ij} = \frac{e^{-(u_i - u_j)^2/2}}{\sum_k e^{-(u_i - u_k)^2/2}}. \tag{13.17}$$

The self summand one bounds every denominator away from zero. Limit trajectories remain bounded on finite intervals by the same extrema comparison. The limit eq. (13.17) is the Euclidean gradient flow of the Gaussian log-sum-exp energy $\sum_i u_i^2/2 - \sum_i \log \sum_k e^{-(u_i - u_k)^2/2}$. Its pair formation is the clustering mechanism of Gaussian mean shift and its blurring variant [36, 42] and of self-attention particle dynamics [89]; here it is obtained from the original KL source rows at low temperature. The rest of the proof constructs a reference trajectory of eq. (13.17) with strict windows and transfers them three times by lemma 13.4: from decoupled clusters to the six-agent limit field, from the limit field to the exact scaled field, and from symmetric to open initial data.

For four agents first take the reflection-invariant configuration

$$u = (-a - b, -a + b, a - b, a + b), \qquad a(0) = A, \quad b(0) = 2. \tag{13.18}$$

Here a is half the distance between the two pair centers and b is the internal half-gap. Set $w = e^{-2b^2}$, $v_0 = e^{-2a^2}$, $v_\pm = e^{-2(a \pm b)^2}$, $Z_\pm = 1 + w + v_0 + v_\pm$, and

$$c_p = w(Z_+^{-1} + Z_-^{-1}), \qquad c_0 = v_0(Z_+^{-1} + Z_-^{-1}). \tag{13.19}$$

Substitution into the full field gives

$$\begin{aligned}\dot{a} &= -a - 2ac_0 - 2\left[\frac{(a+b)v_+}{Z_+} + \frac{(a-b)v_-}{Z_-}\right],\\ \dot{b} &= -b - 2bc_p - 2\left[\frac{(a+b)v_+}{Z_+} - \frac{(a-b)v_-}{Z_-}\right].\end{aligned} \tag{13.20}$$

Both unequal row normalizers are retained.

For $a \geq 4$ and $0 \leq b \leq 2$, put $c = 1 + w + v_0 \geq 1$. The function $g_c(r) = re^{-2r^2}/(c + e^{-2r^2})$ satisfies

$$|g_c'(r)| \leq (1 + 4r^2)e^{-2r^2}. \tag{13.21}$$

This bound decreases for $r \geq 2$, so $|g_c(a+b) - g_c(a-b)| \leq 34be^{-8}$. Since $0 \leq c_p \leq 1$, $2ac_0 \leq 4ae^{-32}$, and $(a+b)v_+ + (a-b)v_- \leq 2av_- \leq 2ae^{-8}$, eq. (13.20) implies

$$-(3 + 68e^{-8})b \leq \dot{b} \leq -(1 - 68e^{-8})b, \qquad -(1 + 4e^{-32} + 4e^{-8})a \leq \dot{a} \leq -a. \tag{13.22}$$

Here $1 - 68e^{-8} > 0.977$ and $1 + 4e^{-32} + 4e^{-8} < 1.0014$. The positive coordinates cannot first reach their invariant zero axes at finite time.

Take $A \geq 34$. The first time τ with $b = 1/2$ satisfies $\tau \leq \log 4/0.977 < 1.42$, and $a(t) \geq Ae^{-1.0014 \cdot 1.42} > 0.239A \geq 8$ through this interval. Initial peer weights are at most e^{-8} and cross weights much smaller. A directed within-pair weight is $p_\pm = w/Z_\pm$. While $b \geq 1/2$, $\dot{w} \geq w/2$, whereas

$$|\dot{v}_0 + \dot{v}_\pm| \leq 8a^2e^{-2a^2} + 4(a+2)(2a+8)e^{-2(a-2)^2} < 1/100 \tag{13.23}$$

for $a \geq 8$. Both positive terms decrease on that range. Differentiating $p_\pm$ therefore gives $\dot{p}_\pm > 0$. At τ,

$$p_\pm \geq \frac{e^{-1/2}}{1 + e^{-1/2} + 2e^{-8}} > 1/4. \tag{13.24}$$

Each peer threshold is crossed transversely, with exterior mass less than $2e^{-8}$. The contrast is strict at entry.

Until the first time t_A with $a = 4$, the pair weights remain above $1/4$ and every cross weight is at most e^{-8}. An unwanted parent source column has at most two such entries, excluding a premature join even if one pair has formed before the other. Moreover,

$$\frac{\log(A/4)}{1.0014} \leq t_A \leq \log(A/4), \qquad b(t_A) \leq 2(A/4)^{-0.975}. \tag{13.25}$$

Thus the pairs coexist while their residual internal gaps become arbitrarily small.

Starting at $a = 4$, compare with the invariant trajectory $b = 0$. Its exact inherited cross mass and primitive peer weight are

$$B = \frac{e^{-2a^2}}{1 + e^{-2a^2}}, \qquad p = \frac{1 - B}{2}, \qquad \dot{a} = -a(1 + 4B). \tag{13.26}$$

Here $\dot{B} = 4a^2e^{-2a^2}(1+4B)/(1+e^{-2a^2})^2 > 0$. The parent threshold is crossed at $a = \sqrt{\log 9/2}$. Stop this comparison when $B = 1/5$, at a finite time $S \le \log(4/\sqrt{\log 4/2})$. Throughout it, $p \ge 2/5$ and the primitive exterior mass is at most $1/5$, so first-pair contrast is at least $1/5$. The parent crossing has a positive derivative. By lemma 13.4, with $\Psi = \Phi$ given by eq. (13.20) and the block rows as observables, these compact margins and the transverse crossing persist when the initial half-gap at $a = 4$ is below some $\varepsilon_b > 0$. This ε_b does not depend on A, since the comparison starts at $a = 4$. Choose one finite $A \ge 34$ with $2(A/4)^{-0.975} < \varepsilon_b$ using eq. (13.25), and fix $L = t_A + S$. The parent event then occurs between $\log(A/4)/1.0014$ and $\log(A/4) + S$: the second stage happens when the supplied core separation has contracted to order one.

A third pair makes the second event have a genuine exterior. Add initial coordinates $R-2, R+2$. In the decoupled comparison the two-agent interaction is antisymmetric, so their center is Re^{-t}, and the half-gap b_c obeys

$$\dot{b}_c = -b_c(1+4p_c), \qquad p_c = \frac{e^{-2b_c^2}}{1+e^{-2b_c^2}}, \qquad b_c(0) = 2. \tag{13.27}$$

Its threshold is transverse, and by $\log 4$ its peer weight exceeds $1/4$.

Let $X(t)$ be the decoupled reference, the four-agent core trajectory together with this two-agent trajectory, and compare it in the original coordinates with the six-agent limit field. On the radius-one supremum-norm tube about X, the core coordinates satisfy $|u_i| \le A+3$, because the extrema comparison keeps the reference core in $[-A-2, A+2]$, and the pair coordinates lie within 3 of Re^{-t}. Every core-pair separation is therefore at least $D_R = Re^{-L} - A - 6$. Let $\varphi(D) = e^{-D^2/2}$. For an agent in one cluster, the six-agent row differs from the decoupled row through the changed normalizer, $w_{ij} - \widetilde{w}_{ij} = -\widetilde{w}_{ij}\epsilon_i/Z_i$ with ϵ_i the cross-cluster numerator mass, in both the outgoing and incoming directions, and through the incoming and outgoing cross forces. Using $\widetilde{w}|x| \le e^{-1/2}$ and $|x|\varphi(|x|) \le D\varphi(D)$ for $|x| \ge D \ge 1$, these contributions give

$$|\Psi - \Phi|_\infty \le 8\big(e^{-1/2} + D_R\big)\varphi(D_R). \tag{13.28}$$

The same quotient-rule bookkeeping bounds the differences of the Jacobians and of the block-row gradients by absolute multiples of $(1+D_R^2)\varphi(D_R)$; for the core-parent row, $|B^{(6)} - B^{(4)}| \le 2\varphi(D_R)$. None of these multiples depends on A or R. The field bound M of lemma 13.4 does: on the tube $M \le R + 3 + 10e^{-1/2}$, so the terms $\eta_1 M$ and $L_1 \Delta M$ of eq. (13.15) grow linearly in R. Since $D_R = Re^{-L} - A - 6$, every error term is of order $(D_R + A)e^L(1+D_R^2)\varphi(D_R)$ and still tends to zero as R grows. The input of lemma 13.4 is the Lipschitz constant of the decoupled reference field, not a bound on Jacobian differences. The decoupled field is the limit field on each component. Its interaction depends only on differences, and $w|x|$ and wx^2 are bounded, so its Lipschitz constant on the tube is bounded independently of A and R. Choosing R sufficiently large and finite, lemma 13.4 preserves every window and crossing of the reference and keeps every aggregated mass between the core and the third pair below 10^{-3} through L. At the core-parent crossing the contrast is then greater than $0.099 > 1/20$; the third pair cannot join either core block or their union.

Fix this R and let X now denote the six-agent limit trajectory on $[0, L]$. On the radius-$1/2$ tube about it, $|u|_\infty \le R + 3$. For $h(R+3) < 1$, analyticity in (h^2, u) makes the exact scaled field and its block rows converge to the limit field and its rows in C^1, uniformly on this compact tube, as

$h \to 0$. Lemma 13.4 with $\delta_0 = 0$ gives one sufficiently small fixed $h > 0$ for which the exact scaled binary field has the same strict windows and transverse crossings. Choose equal sector trajectories only as a reference, then perturb all twelve initial coordinates independently. Lemma 13.4 with $\Psi = \Phi$ preserves the windows and the positive crossing derivatives for sufficiently small initial perturbations. Each pair's event is the latest of its four directional-sector crossings, before the next separated window. The means can be initially distinct; their invariant equality diagonals and uniqueness preclude a first finite collision. Finally take the union of all population permutations of this open neighborhood. The constitutive data contain no block labels; the initial configuration chooses the eventual blocks through its scale hierarchy. □

Remark 13.6 (Null-flow control). Both nulls of definition 6.33 delete every peer cost; the label-preserving null keeps the peer labels and priors, and the renormalized null removes them. On the binary family every self cost vanishes, so the two nulls coincide there and give the decoupled field $\dot{x}_i = -(1 - x_i^2)\,\operatorname{atanh} x_i$, that is $\operatorname{atanh} x_i(t) = e^{-t} \operatorname{atanh} x_i(0)$, with scaled limit $u(t) = e^{-t}u(0)$. For correlated laws the angular self cost H_i^q is nonzero and the nulls differ: the angular conditionals relax at rate two along the renormalized null and at the agent-dependent rate $1 + \beta_{ii}^0$, with the self weight eq. (6.48), along the label-preserving null, so the bit forcing eq. (13.30) differs. By the readout convention of definition 6.33, the selector reads along a null the native rows eq. (13.5) computed from the original costs at the null state. On the binary family its membership history is then a function of the initial positions alone: the rows at time t are a Gaussian kernel of bandwidth e^t on $u(0)$. Integrations of eq. (13.17) and of this null limit from reference configurations of the form used in the proof, at $A = 8$, 64, and 4096, with the third pair at $R = 200$, 2000, and 4×10^5 rather than at the far larger R of the proof, produce the same three pairs, the same core-parent join with the same exterior, and the same order of events; the pair events move from $t \approx 0.589$ to 0.647, and at $A = 4096$ the parent join moves from $t \approx 8.214$ to 8.271. These are integrations of the limit field, not of the exact scaled field at the fixed h of the proof, and they are not part of any statement proved here. The near-simultaneous first-stage pair events are merges of disjoint agent sets, so a reordering of them between the full and null flows does not change the partition sequence as compared by definition 6.33.

What is proved is one-sided: while $a \geq 4$ and $0 < b \leq 2$ the internal half-gap contracts along the full field at a rate above one, against exactly one along the null limit. Indeed, by eq. (13.20), $-\dot{b}/b - 1 = 2c_p + 2[g_c(a+b) - g_c(a-b)]/b$ with g_c as above. For $b \geq 1/2$, $a - b \geq b$ and the decrease of re^{-2r^2} on $[1/2, \infty)$ give $bw \geq (a-b)v_-$, hence $-\dot{b}/b - 1 \geq 2w/Z_+ > 0$ after dropping the positive v_+ term. For $b \leq 1$, $a - b \geq 3$, the mean-value bound $|g_c(a+b) - g_c(a-b)| \leq 2b(1 + 4(a-b)^2)e^{-2(a-b)^2}$, and $c_p \geq 2w/(1 + w + 2e^{-18})$ with $w \geq e^{-2}$ give $-\dot{b}/b - 1 > 0.47$. The peer forces therefore tighten the selected pairs. Neither interaction necessity (F5) nor the amplifying grade of definition 6.36 is established by proposition 13.5. A proof of the null comparison would run the explicit null solution through the same windows and crossing estimates by lemma 13.4; it is not carried out here.

Remark 13.7 (Size of the constants). The witness is nonconstructive. With $A = 4096$ the core reference has $t_A \approx 6.93$ and $L \approx 8.36$, and a third pair separated by D_R of order ten requires $R \approx 1.8 \times 10^7$. Admissibility $h|u|_\infty < 1$ alone then forces T below about 3×10^{-15}. The tube step as written is more restrictive: for a far agent the scaled field is $-h^{-1}(1 - h^2u^2)\operatorname{atanh}(hu) = -u + \frac{2}{3}h^2u^3 + O(h^4u^5)$, so the uniform field error on the tube contains the common-mode term

$\frac{2}{3}h^2R^3$, and keeping the far pair within a radius-$1/2$ tube needs $T = h^2$ below about 10^{-21} before any Gronwall amplification. Existence is unaffected, since h is only required to be sufficiently small. A comparison relative to each cluster's exact center would remove the common-mode error and need only hR small; we do not carry it out. These values of A, t_A, L, and R are illustrative: the proof takes some finite $A \geq 34$ with $2(A/4)^{-0.975} < \varepsilon_b$, where ε_b is only known to be positive, and $A = 4096$ is certified once $\varepsilon_b \geq 2.3 \times 10^{-3}$ is shown. The temperature bounds above are computed before the Gronwall factor of lemma 13.4; including it makes the temperature implied by the written route far smaller. The early phase above works from $A \geq 34$; a numerical interval enclosure of the late comparison indicates that $\varepsilon_b = 1$ suffices, which would make $A = 34$ admissible if the horizon of the late comparison stops where a reaches $\sqrt{\log 2}$, but that enclosure is not part of the proof. The temperature of the theorem below is not quantified.

13.3 A full open domain of correlated laws

The binary result embeds exactly into eq. (13.7). The embedding extends to every correlated law, because the conditional angular laws evolve autonomously.

Proposition 13.8 (Exact reduction of the bit-circle flow). Write $q_i(z,\theta) = \mu_i(z) f_i(\theta \mid z)$ and $s_i(w,\psi) = \eta_i(w) g_i(\psi \mid w)$ for positive laws with bounded log-densities. Along eq. (13.7),

$$f_i(\theta \mid z,t) = \frac{f_i(\theta \mid z,0)^{e^{-2t}}}{\int f_i(\psi \mid z,0)^{e^{-2t}} d\psi/(2\pi)}, \tag{13.29}$$

for each z separately, and the bit means obey eq. (13.12) plus the exogenous forcing

$$-\frac{1}{2}(1-x_i^2)\big[c_i(1,t) - c_i(-1,t)\big], \qquad c_i(z,t) = 2D_{\mathrm{KL}}\big(f_i(\cdot \mid z,t) \big\| \mathrm{Haar}\big), \tag{13.30}$$

with the exchanged statements in the model sector. The solution exists for all $t \geq 0$, and all bit and model means tend to zero. Rows, block rows, and the bit forcing are invariant under independent rotations of every belief and every model angle.

Proof. In eq. (13.7), $2\log q_i = 2\log \mu_i + 2\log f_i$, so $h_i^q = 2\log f_i(\theta \mid z) + \phi_i(z)$, where ϕ_i collects all terms depending on the bit alone. Integrating $\partial_t q_i = -q_i(h_i^q - \mathbb{E}_{q_i} h_i^q)$ over the angle gives $\partial_t \log \mu_i(z) = -(\mathbb{E}[h_i^q \mid z] - \mathbb{E}_{q_i} h_i^q)$. Subtracting, $\partial_t \log f_i = -2(\log f_i - \mathbb{E}[\log f_i \mid z])$, a separate Fisher-Rao relaxation toward Haar measure for each z, whose solution is eq. (13.29). Since $\mathbb{E}[h_i^q \mid z] = c_i(z,t) + \phi_i(z)$, the bit part reproduces the binary field and adds $-\frac{1}{2}\mu_i(1)\mu_i(-1)[c_i(1,t) - c_i(-1,t)]$ to $\dot{x}_i$, with $\mu_i(1)\mu_i(-1) = 1 - x_i^2$. In logit coordinates the forcing is bounded by $\sup_z c_i(z,0)$, because the relative entropy decreases along eq. (13.29); the extremum comparison of lemma 13.3 with this bounded forcing keeps the logits bounded on every finite interval, and since $c_i(z,t) \to 0$ it gives convergence to zero. The explicit solution satisfies eq. (13.7) and therefore coincides with its unique local solution. Rows depend only on bit marginals, and c_i is rotation invariant. □

The conditional relaxation eq. (13.29) is the explicit solution of the Fisher-Rao gradient flow of a relative entropy [41], and the reduction is the Haar-reference analogue of theorem 8.5, with relaxation rate two. The receiver term $H_i^q = D_{\mathrm{KL}}(q_i \| \mu_i \otimes \mathrm{Haar})$ is the μ_i-average of $c_i(z,t)/2$. For

product laws $c_i(1,t) = c_i(-1,t)$ and the forcing vanishes. In particular, for arbitrary positive angular profiles, the product reference family

$$q_i(z,\theta) = \mu_i(z) f_i(\theta), \qquad s_i(w,\psi) = \eta_i(w) g_i(\psi) \tag{13.31}$$

is invariant, and its angular factors obey

$$\partial_t f_i = -2 f_i(\log f_i - \mathbb{E}_{f_i} \log f_i), \qquad f_i(\theta,t) = \frac{f_i(\theta,0)^{e^{-2t}}}{\int f_i(\psi,0)^{e^{-2t}} d\psi/(2\pi)}. \tag{13.32}$$

A von Mises factor $p_\kappa(\theta - a) = e^{\kappa\cos(\theta-a)}/I_0(\kappa)$ keeps its phase and has concentration κe^{-2t}.

Proposition 13.9 (Open-law transfer). The routing behavior of proposition 13.5 persists on a nonempty relabeling-invariant set of independently variable positive belief and model densities on $\{-1,1\} \times S^1$, including correlated bit-angle laws. The set is open for uniform convergence of log-densities, and hence in the C^r chart.

Proof. Let Θ bound the oscillation of every log-conditional $\log f_i(\cdot \mid z)$. For $\ell = \log f$ and $\rho = e^{-2t}$ write $\mathcal{R}_\rho(\ell)$ for the relative entropy to Haar measure of the law proportional to $e^{\rho\ell}$. Its derivative in a direction ς is $\rho^2 \operatorname{Cov}(\ell,\varsigma)$ under that law, which is at most $\rho^2\Theta\|\varsigma\|_\infty$ in absolute value along the segment between the two log-conditionals. Hence

$$|c_i(1,t) - c_i(-1,t)| \le 2e^{-4t}\Theta \big\|\log f_i(\cdot \mid 1,0) - \log f_i(\cdot \mid -1,0)\big\|_\infty. \tag{13.33}$$

In the scaled coordinates of proposition 13.5, the forcing eq. (13.30) is therefore a time-dependent field perturbation of size at most $e^{-4t}\Theta\epsilon_c/h$, where ϵ_c is the largest of these conditional differences over agents and sectors. Take the binary open set of proposition 13.5, shrunk so that its windows survive a field error $e_0 > 0$ by lemma 13.4. Every law whose bit and model means lie in it and whose conditionals satisfy $\Theta\epsilon_c < he_0$ has the routing behavior. This condition is open for uniform convergence of log-densities and contains every product law eq. (13.31); rows and block rows are the binary ones by proposition 13.8. Admissible bit-angle correlations are thus of order h. □

Corollary 13.10 (Transient relations and a forced final merge). For every initial law of proposition 13.9, every native row tends to $1/N$. Every first-stage contrast tends to $1/6 - 4/6 < 0$ and the core-parent contrast against the third pair tends to $0 < \delta$, so no block relation used in proposition 13.5 is forward invariant. After the core-parent join the partition has two blocks, the exterior sum in eq. (13.10) is empty, and the remaining block rows tend to $2/6$ and $4/6$. Under definition 13.2 every such trajectory therefore reaches the one-block partition at a finite time after L, with an empty exterior. Under the retention below, this last join discards one further relative orientation, since $\mathsf{H}_{I\cup J} \times \mathsf{H}_K \subsetneq \mathsf{H}_{\{1,\dots,6\}}$.

Proof. All means tend to zero by proposition 13.8, so every divergence in eq. (13.5) tends to zero. With two blocks the contrast condition reduces to $B^a > \varepsilon$ in both directions and sectors. The cross block rows are below 10^{-3} at the core-parent join and tend to limits above ε, so there is a finite time after which every strict inequality holds, and the event occurs no later than that time. □

The persistence in this chapter is therefore finite-window by construction. The integrations of remark 13.6 show the same final merge along the null limit flow.

13.4 Orientation compression and inherited dynamics

For an angle a, let $(\tau_a f)(z,\theta) = f(z,\theta-a)$. An active rotation of agent i applies τ_{a_i} to its belief and model laws in fixed sample coordinates. It changes the laws, unlike the passive changes of presentation of chapter 2. The energy, metric, rows, and eq. (13.7) are invariant or equivariant under independent active rotations of every belief and every model angle, a group $(S^1)^{2N}$. Below we use the diagonal subgroup $(S^1)^N$ that rotates both laws of one agent by the same angle. Whether an active rotation is detectable is a statement about a declared observable algebra (definition 13.14): no agent, channel, row, or evaluator of the population resolves it, and only the external response protocol of the next section can.

For a block I discard the subgroup

$$\mathsf{H}_I = \{(a_i)_{i\in I} : \sum_{i\in I} a_i = 0 \pmod{2\pi}\}. \tag{13.34}$$

Its dimension is $|I|-1$. A quotient alone does not give the required contextual statistical realization, so retain a whole local contextual section with fixed anchor c_*. Suppose $u_i(c_*) = \mathbb{E}_{q_i(c_*)} e^{i\theta} \neq 0$, choose compatible local phase lifts, and define

$$\vartheta_i = \arg u_i(c_*), \qquad \phi_I = \frac{1}{|I|}\sum_{i\in I}\vartheta_i, \qquad a_i = \phi_I - \vartheta_i. \tag{13.35}$$

This normalization is a first-Fourier-mode slice [33] in the method of slices for equivariant evolution equations [22, 172], a cross-section in the sense of moving frames [72]; it is singular where an anchor moment vanishes. Apply a_i to the entire section, independently of evaluation context:

$$\widetilde{q}_i(c) = \tau_{a_i} q_i(c), \qquad \widetilde{s}_i(c) = \tau_{a_i} s_i(c), \qquad q_I = \bigotimes_{i\in I}\widetilde{q}_i, \quad s_I = \bigotimes_{i\in I}\widetilde{s}_i. \tag{13.36}$$

These are actual normalized parent laws. Their factors are recovered by marginalization. The relative phase profiles $\arg u_i(c) - \arg u_i(c_*)$ remain retained; only constant relative offsets are discarded. On this chart the fibers are precisely the constant H_I orbits. Choosing the offsets separately at each context would instead remove contextual phase variation itself and change the contextual geometry; the anchored choice avoids this. The model offsets are rotated by the belief-anchored angles. Independent model offsets are also symmetries and could be discarded as well; we do not do so.

The parent retains every child law up to one constant rotation per child with zero sum, together with its primitive ports. Its state size grows with the number of children, and all public bit marginals are retained. In the sense of the closing discussion of chapter 8, it does not combine distinct public child laws into a smaller collective state.

Parent references and evaluators are products of their normalized primitive data. For each primitive port $i \in I, j \in J$, project a source-parent sample onto j, apply eq. (13.2), and compare with factor i.

This is a normalized inherited channel. A full-parent version appends independent recipient factors on the other coordinates:

$$\mathcal{L}^{ij}_{IJ}(\boldsymbol{k}_J, d\boldsymbol{k}'_I) = \mathcal{K}(k_j, dk'_i) \bigotimes_{\ell \in I\setminus\{i\}} \widetilde{q}_\ell(dk'_\ell). \tag{13.37}$$

Consequently $D_{\mathrm{KL}}(q_I \| q_J \mathcal{L}^{ij}_{IJ}) = D^q_{ij}$ by cancellation of the matching normalized factors. If this representation is differentiated, its recipient-fill channel derivatives must be retained. It represents the original primitive sums and rows, not one newly fitted parent KL cost.

At a join of I, J, rotate their factors further by $\phi_{I\cup J} - \phi_I$ and $\phi_{I\cup J} - \phi_J$, where

$$\phi_{I\cup J} = \frac{|I|\phi_I + |J|\phi_J}{|I| + |J|}. \tag{13.38}$$

Each primitive total rotation becomes $\phi_{I\cup J} - \vartheta_i$, so direct and sequential retention agree. Primitive counts are inherited. Since $\mathsf{H}_I \times \mathsf{H}_J \subsetneq \mathsf{H}_{I\cup J}$, this join discards one further continuous relative block orientation. Native row trajectories and all selected event times are unchanged by these rotations, so recording the times does not recover the discarded offsets.

Remark 13.11 (Global representatives). For $n = |I| \geq 2$ no map $(S^1)^n \to S^1$ is both invariant under relabeling and equivariant under a common rotation. If v is the rotation by $2\pi/n$ and $x = (e, v, \ldots, v^{n-1})$, then vx is a cyclic relabeling of x, so such a map would give $\phi(x) = \phi(vx) = v\phi(x)$. The equal-phase anchor eq. (13.35) therefore exists only on local phase charts, with or without continuity. Dropping relabeling symmetry removes the obstruction: rotating every agent $j \geq 2$ of I to anchor phase zero and the first agent by the sum of their phases is a continuous H_I-canonical representative on the whole domain where the anchor moments are nonzero. Rows, events, response laws, and collective costs are H_I invariant, so they do not depend on which representative is used. The common rotation meets H_I in $\mathbb{Z}_n$; on parent orbits the common group acts through $S^1/\mathbb{Z}_n$, and the parent sees the holonomy of the common connection only modulo $\mathbb{Z}_n$.

Proposition 13.12 (Original-time parent flow). On the regular local chart, up to the first time an anchor moment of a block vanishes, the retained network of sections has an autonomous flow intertwining the original dynamics at the same time. The regular chart is not flow invariant: anchor moments tend to zero and can vanish at finite times, where the slice phase jumps by π; the quotient flow on H_I orbits exists for all times.

Proof. Equivariance and uniqueness imply that two sections related by a constant H_I rotation remain so for their common existence interval. This defines the quotient flow, the projectability condition underlying exact lumpability [110]. An explicit representative equation is also available. Let

$$\omega_i = \operatorname{Im} \frac{\int e^{i\theta} V^q_i(c_*) d\nu}{u_i(c_*)}, \qquad \overline{\omega}_I = |I|^{-1} \sum_{i\in I} \omega_i. \tag{13.39}$$

This ratio is invariant under rotation and is computable from retained factors. Since $\dot{a}_i = \overline{\omega}_I - \omega_i$ and $\partial_a \tau_a f = -\partial_\theta \tau_a f$,

$$\partial_t \widetilde{q}_i = V^q_i(\widetilde{x}) - (\overline{\omega}_I - \omega_i)\partial_\theta \widetilde{q}_i, \tag{13.40}$$

with the same coefficient for $\widetilde{s}_i$. This is the in-slice equation of the freezing method [22]. The product rule gives the parent-law derivative. Every term is retained. Successive correction coefficients telescope by eq. (13.38). For finite C^r regularity the displayed derivative is in C^{r-1}; existence and uniqueness follow from the fine flow and equivariance, without assuming rotations are differentiable as maps $C^r \to C^r$. □

The Fisher metric of this canonical product slice need not be the kinetic quotient metric. For two location families with inertias $I_1, I_2 > 0$ and retained total phase F, minimizing lift norm subject to $\dot{\vartheta}_1 + \dot{\vartheta}_2 = \dot{F}$ gives

$$ds^2_{\text{quotient}} = \frac{dF^2}{I_1^{-1} + I_2^{-1}}, \qquad ds^2_{\text{slice}} = \frac{I_1 + I_2}{4} dF^2. \tag{13.41}$$

They differ unless the inertias agree. The first is the quotient metric of a Riemannian submersion [154]; the second is the metric induced on the section. The exact parent update is eq. (13.40), not an uncorrected unit-Fisher gradient of a new single parent KL. Its dissipation is that of the quotient metric, by eq. (13.42) below.

Proposition 13.13 (The parent flow and the quotient metric). Let V be the fine field of eq. (13.7) and let $\overline{g}$ be the quotient metric on H_I orbits, $\overline{g}(\pi_* v, \pi_* v) = \min_w g(v + w, v + w)$ over orbit directions w, where g is the unit Fisher metric. At every law of the regular chart, V is Fisher-orthogonal to the H_I orbit directions, the moving term in eq. (13.40) is the orbit component of the slice velocity, and the descended potential $\overline{\mathcal{E}}$ satisfies

$$\frac{d}{dt}\overline{\mathcal{E}} = -\overline{g}(\pi_* V, \pi_* V). \tag{13.42}$$

On the finite-dimensional invariant family of products of binary marginals and von Mises angular factors with positive concentrations, the H_I action is free, proper, and Fisher-isometric, and the parent flow is the gradient flow of $\overline{\mathcal{E}}$ for $\overline{g}$ at the original time.

Proof. The potential is C^1 for the supremum norm of positive densities, and for a density of class C^1 in the angle the orbit curve $a \mapsto \tau_a q$ is C^1 in that norm with tangent $-\partial_\theta q$. Invariance gives $d\mathcal{E}(J\xi) = 0$ for every orbit direction $J\xi$, hence $g(V, J\xi) = -d\mathcal{E}(J\xi) = 0$. The slice velocity in eq. (13.40) is V plus an orbit direction. Since V is orthogonal to orbits, $\overline{g}(\pi_* V, \pi_* V) = g(V, V)$, and $\frac{d}{dt}\overline{\mathcal{E}} = d\mathcal{E}(V) = -g(V, V)$. The family of the last statement is invariant by eq. (13.32); the full field is tangent to it, so it equals the Fisher gradient of the restricted potential for the induced metric. A rotation fixing a von Mises factor with positive concentration is trivial, the torus is compact, and rotations preserve the Fisher metric and the potential. The orbit map is then a Riemannian submersion for $\overline{g}$, and proposition 6.6 with $d = 1$ and $\chi = \mathrm{id}$ gives same-time inheritance of the gradient flow. □

On the finite-r chart a rotation is not differentiable as a map $C^r \to C^r$, so no quotient Banach manifold is asserted there; the pointwise dissipation identity eq. (13.42) is what holds on the whole chart. On the von Mises family the phases are constant along the flow, so the moving term vanishes there and the last statement does not exercise it; on that family the restricted flow is equally the gradient flow for the slice-induced metric, and only the pointwise dissipation identity distinguishes the quotient metric.

13.5 Responses and retained histories

Declare one recursive response rule before any selection. A query to current block I draws independent constituent samples $(Z_i, \Theta_i) \sim q_i$ and returns

$$\left((Z_i)_{i\in I},\ \sum_{i\in I} \Theta_i \pmod{2\pi} \right). \tag{13.43}$$

The model sector has its corresponding independent joint response. Native source draws, transmitted bit samples, and joint measurable functions of whole-block outputs are also admitted. A policy may depend on past records, but cannot split a compressed block to reveal an individual absolute angle. Policies are Borel kernels from finite histories to admitted queries and waiting times, record densities are jointly Borel in the outcome and the state, and the history laws are those of definition A.3. Queries are observational and do not reset the law state. The querier is not an agent of the population and does not act on its dynamics; it reads angles in a fixed angular frame that no agent holds, since every channel twirls the angle.

Definition 13.14 (Declared observable algebra). For a current partition $\mathcal{P}$, let $\mathcal{A}_{\mathrm{pop}}$ be the algebra of bounded measurable functions of the configuration generated by the quantities the population reads: rows, channel output laws, evaluators, and costs. The vector field is equivariant, so the population's evolution of these quantities does not depend on the orbit representative. Let $\mathcal{A}_{\mathcal{P}}$ be generated by $\mathcal{A}_{\mathrm{pop}}$ and by the probabilities of finite adaptive response histories under eq. (13.43) for $\mathcal{P}$. A distinction between two configurations is resolved by an algebra if some element takes different values on them.

Every element of $\mathcal{A}_{\mathrm{pop}}$ is invariant under $(S^1)^{2N}$, by proposition 13.8 and the invariance of the potential. Thus no active rotation is resolved within the population, and the words gauge and physical have no content here without the response protocol. The kernel in eq. (13.43) is a normalized pushforward of a product law. Since $\sum_i a_i = 0$, it is exactly the same pushforward of the parent law in eq. (13.36). Its first angular moment is $\prod_i u_i$, generally nonzero, so an aggregate angle is resolved by $\mathcal{A}_{\mathcal{P}}$. Joining concatenates bit vectors and adds angle sums. Thus the same experiment composes at each level.

Equal retained states have equal waiting-time evolution and equal next-record joint kernels. Conditioning successively on the realized history proves equality of every finite adaptive joint-history law. Prior records are carried forward. A finite collection of sampled angular records does not deterministically identify the erased offsets: all the densities are positive, leaving continuously many compatible configurations for the same finite history. An exact record of constituent laws would instead invalidate the loss assertion. An individual absolute-angle query also fails descent immediately because its first moment changes under eq. (13.34). These experiments are outside the declared interface after the handoff.

Remark 13.15 (How much is lost). The loss is non-identifiability, not absence of information. Before a merge every agent is its own block, so eq. (13.43) returns samples of its angular marginal. For an angular marginal f, the Fisher information about its phase carried by one sample is the rotation inertia $\mathbb{E}_f[(\partial_\theta \log f)^2]$, which is $\mathcal{I}(\kappa)$ below for a von Mises factor. By equivariance the discarded offsets are constant group elements along the flow, so n_i pre-merge records of agent

i carry information about them equal to the sum of the rotation inertias $\mathbb{E}_{q_i(t)}[(\partial_\theta \log q_i)^2]$ at their record times, and the maximum-likelihood location estimate from them has error of order the inverse square root of that sum; for von Mises factors, whose phases are constant, the circular mean attains this order. This information is finite for every finite record set and grows linearly with the number of records. A uniform loss statement needs a protocol that bounds the number of pre-merge angle queries or discards records at merges; the present protocol does neither.

13.6 Collective geometry on actual parent laws

Take the common collective group to be S^1, acting on every belief and model angle simultaneously. This tie is a declared representation. The constitutive data are also invariant under independent belief and model rotations; for the untied group $S^1_q \times S^1_s$ the minimizing connection below is $A_q = -d\chi$, $A_s = 0$, which is flat, and the minimized tensor has rank one. The curvature computed below belongs to the tied representation. No row, channel, record, or vector field of this construction reads the connection; A^F is a diagnostic of the evolved sections.

The common rotation commutes with the independent active rotations. Use the original sum Fisher metric and convention $D^A p = dp + J_p A$. At fixed time the constants a_i in eq. (13.35) do not vary with evaluation context. Hence $d\widetilde{q_i} = \tau_{a_i} dq_i$ and similarly for s_i. A product score is a sum of centered independent factor scores, so all cross terms vanish. Rotation preserves the remaining integrals. For every contextual direction X and scalar gauge velocity ξ this proves

$$\overline{g}(dRpX + \overline{J}\xi, dRpX + \overline{J}\xi) = g(dpX + J\xi, dpX + J\xi). \tag{13.44}$$

Equality holds before minimization. The orbit inertia, contextual momentum, minimizing connection class, and minimized tensor are therefore identical. On a regular chart their curvature agrees. This is a calculation from normalized laws, rather than assigning the fine tensor to retained metadata. It is consistent with fixed-statistic Fisher sufficiency results [9], but the present retention is section valued.

Under a passive common rotation $\chi_0(c)$, each anchor phase shifts by $\chi_0(c_*)$, leaving every a_i unchanged. Parent sections transform by the same $\chi_0(c)$, and $A' = A - d\chi_0$ gives the usual covariant derivative. The local Lie-algebra stabilizer is preserved. Globally the total phase has $|I|$ roots; a common rotation $2\pi/|I|$ lies in the discarded group (remark 13.11). The construction stays on compatible local charts.

A nonflat family accompanies the routing trajectory. Define

$$p_\kappa(\theta) = \frac{e^{\kappa\cos\theta}}{I_0(\kappa)}, \qquad \mathcal{I}(\kappa) = \kappa\frac{I_1(\kappa)}{I_0(\kappa)}. \tag{13.45}$$

Here I_0 normalizes Haar measure and $I_1/I_0 = \mathbb{E}_{p_\kappa}\cos\theta$ [142]. Integrating the derivative of $\sin\theta p_\kappa(\theta)$ around the circle gives rotation inertia $\mathbb{E}_{p_\kappa}[\kappa^2\sin^2\theta] = \mathcal{I}(\kappa)$. For $\kappa > 0$ this mean cosine is positive, and its derivative is $\mathrm{Var}_{p_\kappa}(\cos\theta) > 0$, so $\mathcal{I}, \mathcal{I}' > 0$.

Use contexts (κ, χ), belief factors $p_\kappa(\theta - \chi)$, and model factors $p_\lambda(\psi)$ with fixed $\lambda > 0$. Binary coordinates are initially context independent. Put $\rho_t = e^{-2t}$, $a_t = \mathcal{I}(\rho_t\kappa)$ and $b_t = \mathcal{I}(\rho_t\lambda)$. Parity makes concentration and rotation scores orthogonal. For a block of primitive count n the common-

gauge cost in its angular direction is $n[a_t(d\chi+\xi)^2+b_t\xi^2]$. Thus

$$A_t^F=-\frac{a_t}{a_t+b_t}d\chi,\qquad F_t^F=-\frac{b_t\rho_t\mathcal{I}'(\rho_t\kappa)}{(a_t+b_t)^2}d\kappa\wedge d\chi, \tag{13.46}$$

and

$$h_{\chi\chi}=n\frac{a_tb_t}{a_t+b_t}>0,\qquad h_{\kappa\kappa}=n\rho_t^2\operatorname{Var}_{p_{\rho_t\kappa}}(\cos\theta)>0,\qquad h_{\kappa\chi}=0. \tag{13.47}$$

Curvature is nonzero at every finite time. As $t\to\infty$, $\mathcal{I}(v)\sim v^2/2$ and $\mathcal{I}'(v)\sim v$ give

$$F_t^F\longrightarrow-\frac{2\lambda^2\kappa}{(\kappa^2+\lambda^2)^2}d\kappa\wedge d\chi,\qquad e^{4t}h_{\chi\chi}\longrightarrow\frac{n\kappa^2\lambda^2}{2(\kappa^2+\lambda^2)},\qquad e^{4t}h_{\kappa\kappa}\longrightarrow\frac{n}{2}. \tag{13.48}$$

On the reference family the curvature coefficient is therefore bounded away from zero uniformly in $t\geq0$ on compact context patches with $\kappa>0$, while the tensor decays like e^{-4t}; the infinite-time degeneration is a loss of rank, not of curvature. The curvature is already present at $t=0$ and does not depend on the selected partition. The retention preserves it because eq. (13.44) holds for every partition at every time (proposition 13.16). This geometry is supplied by the declared contextual family; it is not produced by the selection.

To extend this geometry through every nearby law value without encoding its erased orientation, fix $c_*=(\kappa_*,0)$. For arbitrary nearby initial $x=(q_i,s_i)_i$, let $\vartheta_i(x)=\arg\mathbb{E}_{q_i}e^{i\theta}$ and assign

$$q_{i,x}(\kappa,\chi)=\tau_\chi\left[\frac{e^{(\kappa-\kappa_*)\cos(\theta-\vartheta_i(x))}q_i(z,\theta)}{\int e^{(\kappa-\kappa_*)\cos(\theta-\vartheta_i(x))}q_i(z,\theta)d\nu}\right],\qquad s_{i,x}(\kappa,\chi)=s_i. \tag{13.49}$$

This normalized section passes through x at the anchor and reproduces the preceding family at the product reference. An active rotation shifts both the density and its centering phase, so the entire section rotates by that same constant. Thus its retained profiles do not secretly store the discarded offsets.

On a compact context patch with $\kappa>0$ and a finite time horizon, the reference densities, moments, inertia, tensor eigenvalues, and absolute curvature coefficient have positive lower bounds. Smooth dependence of the fine field and its contextual variational equations preserves these bounds on a sufficiently small full neighborhood of independent initial laws. This step uses that the fine field is real analytic on the bounded log-density chart, so that the flow map is twice continuously differentiable with derivatives continuous in the initial point uniformly on the horizon [64]; that equivariance transports angular derivatives, $\partial_\theta\Phi_t(q)=D\Phi_t(q)\partial_\theta q$; and that the curvature $F=-b\partial_\kappa a/(a+b)^2$, with a and b the belief and model rotation inertias, needs only the jets $\partial_\kappa q$, $\partial_\theta q$, and $\partial_\kappa\partial_\theta q$. The neighborhood is therefore open in the C^1 chart, hence in the C^r chart; uniform closeness of log-densities, the topology of proposition 13.9, does not suffice for this step. The lift chooses the contextual jets on which geometry is asserted; it imposes no product or common-parameter restriction on their anchor values.

Select membership at the fixed anchor and hand off the whole section on that common time slice. This convention matters. If each context instead chooses its own birth time $\tau(c)$, the derivative is

$$d_c[p_{\tau(c)}(c)]X=(d_cp_tX)_{t=\tau(c)}+V(p_{\tau(c)}(c))X\tau. \tag{13.50}$$

That distinct comparison includes the additional score and is not implied by eq. (13.44).

13.7 The combined existence statement

The inheritance statements above never use the selection. This is the content of the following proposition, which makes the division of labor in the main theorem explicit.

Proposition 13.16 (Symmetry reduction for every partition). Let $\mathcal{P}$ be any partition of the agents, let $t_0 \geq 0$, and suppose every anchor moment of the blocks of $\mathcal{P}$ is nonzero on an interval $[t_0, t_1)$; the statements below hold on that interval. Put $\mathsf{H}_{\mathcal{P}} = \prod_{I \in \mathcal{P}} \mathsf{H}_I$. Then every row, block row, selector input, and event time is invariant under $(S^1)^{2N}$, hence under $\mathsf{H}_{\mathcal{P}}$; the anchored retention of $\mathcal{P}$ at t_0 has the original-time flow of proposition 13.12; every finite adaptive history of eq. (13.43) for the current partition $\mathcal{P}$ descends; and the collective minimizing costs satisfy eq. (13.44).

Proof. The first statement is proposition 13.8. The proofs of proposition 13.12, of response descent, and of eq. (13.44) use only equivariance under H_I, the anchor, and the product form of the parent laws, and none uses an event time or a selected block. □

The selected partition is therefore not distinguished by inheritance. Native routing contributes a law-determined and relabeling-natural choice of partition and of times. The response rule contributes the coarsening that makes the discarded directions unresolvable. Before an event the current partition is finer: a singleton query returns Θ_i, whose first moment changes under a rotation in $\mathsf{H}_{\{i,j\}}$, and a block query returns a sum whose first moment changes under a rotation in $\mathsf{H}_{I \cup J}$ that is not in $\mathsf{H}_I \times \mathsf{H}_J$. After the event these responses are no longer admitted. The distinction discarded at a handoff is exactly the one the declared interface stops resolving at that handoff; the handoff loss is created by the response interface, not by the selection.

Theorem 13.17 (Open-law selection in two stages with exact inheritance). Consider the reciprocal law potential definition 4.2 with the constitutive data of this chapter: six agents, bit-circle sample spaces, the selective channel eq. (13.2) on every edge including self, uniform references and evaluators, uniform all-source priors, unit Fisher mobilities, and one fixed temperature. There exist $T > 0$, a finite horizon, and a nonempty relabeling-invariant set of independently variable positive belief and model laws, open in the C^r chart, on which the following hold over that horizon and on compatible local phase charts.

(a) Selection. Definition 13.2 selects three pairs, which coexist over a positive window, and then joins two of their parents while the third pair is a nonempty exterior with strictly smaller routing masses. Rows and events depend only on the bit marginals.

(b) Interface coarsening. At each event the response interface eq. (13.43) passes to the coarser partition. A continuous distinction between laws, along the newly discarded subgroup, is resolved before the event by admitted queries and after it by no admitted query; no element of $\mathcal{A}_{\mathrm{pop}}$ resolves it at any time. Records acquired before the event carry finite information about it (remark 13.15).

(c) Exact inheritance. For the selected partitions, the anchored section-valued parent laws, inherited port channels, original-time parent flow, and every finite adaptive history of eq. (13.43) descend exactly.

(d) Geometry. On the declared contextual family eq. (13.49), the handoffs preserve the tied collective minimizing connection and a rank-two tensor with nonzero curvature.

The temperature is not quantified (remark 13.7). The theorem does not establish interaction necessity or the counterfactual clause, and the block relations it selects are transient (remark 13.6 and corollary 13.10).

Proof. Use proposition 13.5 to fix finite A, R, h and its horizon, then proposition 13.9 to obtain a full positive-law neighborhood. Choose the angular profiles with nonzero first moments and the equivariant sections of eq. (13.49). Shrink this neighborhood in the C^r chart to preserve the geometric bounds, phase chart, and all routing windows. Intersection remains nonempty because the same product trajectory satisfies them with strict compact margins.

The primitive laws are distinct and stay distinct at finite regular times by uniqueness on the invariant equality diagonals. The selector and inherited rows are exactly definition 13.2 and eq. (13.9), and (a) follows. The first joins discard their H_I directions; the later join discards an additional relative parent direction. Small active rotations stay within the open neighborhood, and equivariance of the assigned sections preserves these fibers after all retained profile data are included. Rows and event times are invariant along them. Statement (b) is the paragraph after proposition 13.16, together with the invariance of $\mathcal{A}_{\mathrm{pop}}$ and positivity of the anchor moments. Parent normalization, port channels, recursive composition, original-time flow, and history descent in (c) follow from eqs. (13.36) and (13.38) and proposition 13.12 and proposition 13.16. Equation (13.44) preserves the actual minimizing costs, and eqs. (13.46) and (13.47) establish their nontrivial geometry throughout the same finite interval, proving (d). □

Against definition 6.36 the report is as follows. The objective is the reciprocal potential with exact Gibbs rows, and the dynamics is its unit-Fisher flow in the bounded log-density chart, with proposition 4.14 supplying well-posedness. (F1) holds in the full grade: the admitted class is open in the chart, both for uniform convergence of log-densities and in the C^r topology, and for the geometric statement in the C^1 topology. Selection is proved for the selector definition 13.2 with its two constants fixed at $\varepsilon = 1/10$ and $\delta = 1/20$, which reads native rows and so satisfies the readout convention, with persistence on finite windows and the finite horizon of theorem 13.17. Clause (F2) asks in addition for a declared constant set with nonempty interior; none is declared, the theorem is not proved over such a set, and (F2) in that form is undetermined. The parent protocol rule assigns to a block the responses eq. (13.43) with native source draws and transmitted bits; the probe rule adds, before a handoff, the protocols of the merged blocks, which before a first join are singleton queries returning individual angles. (F3) holds for these classes by (c). (F4) holds: the newly discarded subgroup gives distinct reachable configurations with the same retained state, separated by a pre-handoff singleton or block query, and the separation is of symmetry type, since the fibers are orbits of the active rotations, and transient, since every angular law relaxes to Haar measure. (F5) is not established against either null. The two nulls coincide on the binary family and differ on correlated laws; integrations of the limit field indicate that both form the same blocks in the same stages, and the frozen-environment control is not computed. (F6a) holds: the parent pair consists of first-stage parents under inherited rows and the same selector, without reset or retuning. (F6b) is not established: no counterfactual class is declared, the second stage depends on the first only through the partner exclusion of definition 13.2, and it occurs when the supplied core separation

has contracted to order one. Neither grade is established. The construction is not shown to be a recursive formation, and since no clause is shown to fail, the classification is undetermined, with (F2) over a constant set, (F5), and (F6b) open. The symmetry enabling compression is fixed in the selective architecture, and the parent is not a fixed-size ordinary KL agent. The result gives neither a preferred threshold nor generic selection for arbitrary channels or all initial laws.

Remark 13.18 (Finite depth by collapsed clusters). In eq. (13.17), two clusters of m coincident agents at $\pm a$ have inherited cross mass $B = v/(1+v)$ with $v = e^{-2a^2}$, and $\dot{a} = -a(1+4B)$, independently of m: each member's row has m unit entries and m entries v. The late comparison in the proof of proposition 13.5 therefore applies to collapsed clusters of every size. An induction over geometric scales would plausibly give any finite number of selected stages on a finite horizon, at a temperature that decreases with the depth. The uniformity of the tube estimates in the depth and the effect of the self mass on the contrast at large depth have not been checked, and we do not claim the result.

Open problem 13.19 (Interaction-necessary selection in the bit-circle architecture). Prove the null comparison of remark 13.6 for the exact field, which would show that (F5) fails on the admitted class. Prove a version of theorem 13.17 on an open set where neither null reproduces the partition sequence, so that (F5) holds; integrations of the limit field at moderate separations suggest that such sets exist. Separately, find an architecture with a forward-invariant set on which a block relation holds and a later join of first-stage parents occurs; a compression symmetry produced by the dynamics rather than fixed by the channel, for example through an angle-transmitting channel of small concentration that makes the compression approximate on the selected blocks with a computable defect; a loss statement in which the information about discarded coordinates given all admitted records stays bounded; and explicit or validated witness constants.

13.8 Two tests of stronger inheritance claims

The first test makes the identity-channel response obstruction quantitative. For two binary source means $x_1 \neq x_2$ and fixed positive weights, an exterior receiver mean r determines the block normalizer. With $a_j = \operatorname{atanh} x_j$, $v_j = a_j/T$, and $\psi(r) = \frac{1+r}{2}\log(1+r) + \frac{1-r}{2}\log(1-r)$, removing its receiver entropy factor gives

$$F(r) = e^{\psi(r)/T} Z_I(r) = \sum_{j=1}^{2} b_j e^{v_j r}, \qquad b_j > 0. \tag{13.51}$$

At an interior r_0, set $m_k = F^{(k)}(r_0)$ and $c_j = b_j e^{v_j r_0}$. Then

$$H = m_0 m_2 - m_1^2 = c_1 c_2 (v_1 - v_2)^2 > 0. \tag{13.52}$$

The two scaled logits $v_j = a_j/T$ are the roots of

$$z^2 - \frac{m_0 m_3 - m_1 m_2}{H} z + \frac{m_1 m_3 - m_2^2}{H} = 0. \tag{13.53}$$

This is Prony's method for a two-term exponential sum [57]. Thus equality of response functions on any open receiver interval identifies the unordered source laws. Small within-pair KL makes this inverse ill-conditioned, not exactly redundant. This is a finite reconstruction within the unrestricted Gibbs-interface result; it does not obstruct the selective angular interface above.

The second test concerns propagation of Fisher sufficiency. Let a fixed statistic retain Z from positive densities $p_t(c; z, y) = \mu_t(c; z)\varkappa_t(c; y \mid z)$. Write P_t for conditional expectation given Z, let X be a contextual direction, and suppose the complete centered force is $\partial_t \log p_t = -(h_t - \mathbb{E}h_t)$. The hidden contextual score is $a_X = X \log \varkappa_t$. Conditional normalization gives

$$\partial_t a_X = -(I - P_t)Xh_t + \mathrm{Cov}_{\varkappa_t}(h_t, a_X). \tag{13.54}$$

If $a_X(0) = 0$, the Fisher loss therefore starts as

$$g_t(dp_tX, dp_tX) - \overline{g}_t(dRp_tX, dRp_tX) = t^2\mathbb{E}_0[((I - P_0)Xh_0)^2] + o(t^2). \tag{13.55}$$

This follows by differentiating the conditional score and expanding its squared norm; on finite positive alphabets all operations are finite sums.

For two identity-coupled laws $q_i(z, y) = \mu_i(z)\varkappa_i(y \mid z)$ on a finite alphabet, written as probabilities with respect to counting measure, with local references r_i, the complete forces are $h_1 = 2\log q_1 - \log r_1 - \log q_2 - q_2/q_1$ and its exchanged expression. Initially vary only μ_2 with tangent v, keeping both conditionals fixed. Conditional centering of the incoming ratios gives the coefficient

$$\sum_z v_z^2 \left[\frac{\chi^2(\varkappa_2 \| \varkappa_1)}{\mu_1(z)} + \frac{\mu_1(z)^2}{\mu_2(z)^3}\chi^2(\varkappa_1 \| \varkappa_2) \right]. \tag{13.56}$$

Here the conditional divergences are evaluated at z. Taking $\mu_1 = \mu_2 = (1/2, 1/2)$, $v = (1, -1)$, and $\varkappa_1(y = 1 \mid z) = 1/4$, $\varkappa_2(y = 1 \mid z) = 3/4$ for both values of z gives $32/3$. Initial score sufficiency is lost immediately. This example uses a trivial collective group; with a nontrivial action one must test the actual minimizing residual rather than infer optimized loss from an arbitrary raw score. The anchored retention avoids this failure through the exact equality of its full minimizing costs.

Chapter 14

Per-Agent Grouping with Persistent Records

This chapter concerns the per-agent objective, the normalized per-agent model of definition 4.19 under its Gibbs-reduced dynamics, rather than the reciprocal potential used in chapter 7 to chapter 13. Each agent changes its own recognition laws while treating the communicated laws of its peers as current model data. Records remain in its free energy. The first construction uses a four-state channel and exhibits a grouping on a finite window that neither null of definition 6.33 selects. A second construction gives persistent grouping against a conflicting record. Its verdict under interaction necessity depends on the selector constant: it holds for every same-sign ratio threshold in $(1/8, 1/3)$ and fails for every threshold in $(1/3, 1)$, and the frozen-environment control reproduces the selected sequence. A third construction adds an angular sector and a parent that quotients a hidden phase, with exact original-time dynamics, admitted response histories, and nonflat collective geometry on a declared family. That phase is gauge relative to every algebra the chapter declares, so the parent fails compression (F4). Its interface is supplied before selection, and no second compressive handoff is established. The angular construction is a single-stage selection with exact inheritance, the other two have no compressive parent, and none is a recursive formation.

14.1 What changes under an own-objective derivative

All source priors below are positive and normalized. Source rows are minimized exactly at unit temperature at every state: the dynamics is the Gibbs-reduced per-agent dynamics of definition 4.21, not its finite-mobility row evolution. The two share equilibria but not transients, and no transfer of stability between the two row conventions is proved for the non-gradient per-agent field. On the circle spaces below the dynamics is the chart version of the same definition, whose well-posedness is checked where it is used. At a fixed configuration let $D^q_{ii} = \int D_{\mathrm{KL}}(q_i \| G_i(m,\cdot)) s_i(dm)$ and $D^s_{ii} = D_{\mathrm{KL}}(s_i \| r_i)$, with the nonself costs defined by the communicated laws. These self costs are the evaluator and reference divergences of the per-agent model, the costs written D_{ii} in eq. (6.48); they are not the identity-channel self divergences of the reciprocal potential, which vanish. Then

$$\beta_{ij} = \frac{\pi^q_{ij} e^{-D^q_{ij}}}{\sum_k \pi^q_{ik} e^{-D^q_{ik}}}, \qquad \gamma_{ij} = \frac{\pi^s_{ij} e^{-D^s_{ij}}}{\sum_k \pi^s_{ik} e^{-D^s_{ik}}}. \tag{14.1}$$

The envelope derivative is the row-weighted derivative of the costs. It does not differentiate the optimized row again. Each law follows its own agent's Fisher field, so derivatives of another agent's free energy are absent. The probability-coordinate form is the replicator equation associated with

the Shahshahani metric [176]; the construction here concerns its particular normalized sources, records, and exact inheritance.

Every peer cost below has a compared law that depends on the sender's law, so definition 6.33 deletes all of them. The label-preserving null (N1) keeps their labels and priors. If $a = \pi^q_{ii}$, its content self row is the instance

$$\beta^0_{ii} = \frac{a e^{-D^q_{ii}}}{a e^{-D^q_{ii}} + 1 - a} \tag{14.2}$$

of eq. (6.48). The model row has the same form. The renormalized null (N2) removes the deleted labels, so both self weights equal one. Since every self cost here is nonzero, the two nulls differ, and each (F5) verdict below is stated against both. The frozen-environment control (N3) holds every peer law at its initial value in every cost, row, and channel. Every example uses the same records, selector, and initial state along the full flow and along all three controls, and the selector reads public laws only, so the readout convention of definition 6.33 is met trivially.

14.1.1 A conditional four-state channel

Let $k = (a, x)$ and $m = (b, u)$ range over $\{-1, 1\}^2$ with counting measure, and let q_i, s_i be arbitrary positive normalized tables. Fix $J > 1$ and $0 < \alpha, \kappa < 1$. Take

$$r(b, u) = \frac{1}{4}, \qquad G((b, u), (a, x)) = \frac{e^{Jab}}{4 \cosh J}. \tag{14.3}$$

For a positive table p, define $\mu_p(x) = \sum_a p(a, x)$ and $L_p(b \mid x) = p(b, x)/\mu_p(x)$. The nonself kernel is

$$\begin{aligned} W_b(y \mid x) &= \frac{1 + \kappa b x y}{2}, \\ K[p]((a, x), (b, y)) &= \big[\alpha \mathbf{1}_{\{a=b\}} + (1 - \alpha) L_p(b \mid x)\big] W_b(y \mid x). \end{aligned} \tag{14.4}$$

Each row sums to one and is positive. The first operation has determinant α on each conditional two-state block, and the second has determinant κb on each content block, so $K[p]$ is invertible. Nevertheless its compared output simplifies exactly:

$$(pK[p])(b, y) = \sum_x p(b, x) W_b(y \mid x) =: (pW)(b, y), \qquad D(pK[p])[v] = vW. \tag{14.5}$$

The first identity follows by summing over a; its derivative includes the channel derivative. No loss of channel rank or frozen derivative is used. In the per-agent dynamics an own-block variation holds the sender fixed, and the receiver sees the sender only through $pK[p] = pW$, so the law-dependent part of $K[p]$ is inert here: every result below holds verbatim with the fixed channel W. The channel derivative would enter only the source-role forces of the reciprocal potential, which the per-agent dynamics omits.

Let the homogeneous self priors be $\rho_q, \rho_s \in (0, 1)$ and divide each remaining prior equally over the $N - 1$ peers. Write $m_i = \mathbb{E}_{q_i} a$, $n_i = \mathbb{E}_{s_i} b$, $\nu^q_j = q_j W$, and $\nu^s_j = s_j W$. One retained record $(o_q, o_s) \in \{-1, 1\}^2$ has normalized likelihood

$$\ell(o_q, o_s \mid a, x, b, u) = \frac{e^{h_q o_q a}}{2 \cosh h_q} \frac{e^{h_s o_s b}}{2 \cosh h_s}, \qquad h_q, h_s > 0. \tag{14.6}$$

For finitely many retained records put $H_i^q = \sum_t h_{q,t} o_{q,t}$ and $H_i^s = \sum_t h_{s,t} o_{s,t}$. Their negative expected log likelihood is a constant minus $H_i^q m_i + H_i^s n_i$. The ordered record word is retained whenever it remains an accessible response.

Proposition 14.1 (Transferred own-block and null fields). With product recognition $q_i \otimes s_i$, the normalized per-agent model with sources eqs. (14.3) and (14.4) has the original-time equations

$$\dot{q}_i = -q_i(h_i^q - \mathbb{E}_{q_i} h_i^q), \qquad \dot{s}_i = -s_i(h_i^s - \mathbb{E}_{s_i} h_i^s), \tag{14.7}$$

where, modulo constants,

$$\begin{aligned} h_i^q(a,x) &= \log q_i(a,x) - \sum_{j \neq i} \beta_{ij} \log \nu_j^q(a,x) - \beta_{ii} J n_i a - H_i^q a, \\ h_i^s(b,u) &= \log s_i(b,u) - \sum_{j \neq i} \gamma_{ij} \log \nu_j^s(b,u) - \beta_{ii} J m_i b - H_i^s b. \end{aligned} \tag{14.8}$$

Writing $b_i^0 = \beta_{ii}^0$ and $g_i^0 = \gamma_{ii}^0$, the null potentials are

$$h_i^{q,0} = b_i^0(\log q_i - J n_i a) - H_i^q a, \qquad h_i^{s,0} = g_i^0 \log s_i - b_i^0 J m_i b - H_i^s b. \tag{14.9}$$

Proof. Integrating the record, content source, presentation source, and source priors in the local model proves normalization. Its content self cost is $\sum q_i \log q_i - J m_i n_i + \log(4 \cosh J)$, and its presentation self cost is $\sum s_i \log s_i + \log 4$. Every peer is fixed in the own-block variation. Differentiating the row-weighted costs and the persistent record term gives eq. (14.8); the coefficients of each full-law entropy sum to one. In particular both derivatives of the evaluator interaction carry β_{ii}, not γ_{ii}. After deletion only the self entropy coefficients remain, giving eq. (14.9). Centering these first variations in the Fisher metric gives eq. (14.7). □

On the synchronized family $q_i(a,x) = (1 + ma)/4$, $s_i(b,u) = (1 + nb)/4$, every peer cost and normalized peer tangent vanishes. Full and null fields coincide there. With no records, stationarity requires

$$\operatorname{atanh} m = Jn, \qquad \gamma_{ii} \operatorname{atanh} n = \beta_{ii} J m. \tag{14.10}$$

Thus a symmetric synchronized equilibrium $m = n = r_*$ with $\operatorname{atanh} r_* = J r_*$ additionally requires equality of the two self rows. Synchronized bistability cannot by itself establish interaction necessity.

Proposition 14.2 (Interaction-dependent selection on a full open table class). For $N = 2$, $J = 3/2$, $\rho_s = 1/2$, and $\rho_q = \cosh J/(1+\cosh J)$, there are positive record strengths and a nonempty full open set of initial positive tables for which a permutation-covariant conditional-sign selector selects the pair for a nonzero time window, while neither the label-preserving nor the renormalized null selects the pair at any time.

Proof. At the record-free uniform table the self rows equal $1/2$ and their first variations vanish. The sign linearization is

$$\dot{m}_i = -m_i + \frac{1}{2} m_j + \frac{3}{4} n_i, \qquad \dot{n}_i = -n_i + \frac{1}{2} n_j + \frac{3}{4} m_i, \qquad j \neq i. \tag{14.11}$$

Its common sign eigenvalue is $1/4$ and the matching difference eigenvalue is $-3/4$. Other sign eigenvalues are $-5/4, -9/4$, and the content modes have eigenvalues $-1 \pm \kappa/2$, hence are stable. Start with $m = n = \eta(3, -1)$ and uniform conditional content. The relevant linear solution is

$$m_i(t) = n_i(t) = \eta\big[e^{t/4} + (a_i - 1)e^{-3t/4}\big], \qquad (a_1, a_2) = (3, -1). \tag{14.12}$$

At $t_* = \log 4$ its values are $\eta(3/\sqrt{2}, 1/\sqrt{2})$. The analytic field is its linearization plus a quadratic remainder. Variation of constants on a fixed finite interval gives an $O(\eta^2)$ state and velocity error for sufficiently small $\eta > 0$. Now fix such an η and add small records with $H_1^q = H_1^s = h > 0$ and $H_2^q = H_2^s = -h$. Continuous dependence preserves the strict positive crossing and a subsequent common interval, also under arbitrary small positive-table perturbations.

For the null, let $c_i^q(x) = \mathbb{E}[a \mid x]$ and $c_i^s(u) = \mathbb{E}[b \mid u]$. Conditional Fisher centering gives

$$\begin{aligned} \dot{c}_i^q(x) &= (1 - (c_i^q(x))^2)\{b_i^0[Jn_i - \operatorname{atanh} c_i^q(x)] + H_i^q\}, \\ \dot{c}_i^s(u) &= (1 - (c_i^s(u))^2)\{b_i^0 Jm_i - g_i^0 \operatorname{atanh} c_i^s(u) + H_i^s\}. \end{aligned} \tag{14.13}$$

At a first zero of a conditional initially aligned with its record, all other signed conditionals are nonnegative and the signed derivative is strictly positive. The two opposite cones are therefore invariant. All null log ratios have linear growth bounds because the self rows lie in $(0, 1]$ and $|m_i|, |n_i| \le 1$, so this conclusion holds for every finite time. The renormalized null is eq. (14.13) with $b_i^0 = g_i^0 = 1$, and the same first-zero argument applies to it.

Declare the selector family S_c: the pair is selected when all eight conditional means exceed c, or all are below $-c$. For every $c \in (0, \eta/(8\sqrt{2})]$ the full flow meets this condition on the common interval; neither null does for any $c \ge 0$, since its two agents stay in opposite cones. Initially there is no selected pair for any $c \ge 0$. The condition is strict on a full open neighborhood of the chosen initial tables and is preserved by simultaneous relabeling of agents and records. The partition sequences differ after deleting their event times, against both nulls and for every such c. □

The open set is a small neighborhood of a near-uniform unstable equilibrium, of size of order η, and the constant set $(0, \eta/(8\sqrt{2})]$ scales with the initial amplitude. Numerical integration of the full flow from the base point indicates that the pair, once selected, persists and that the state converges to a hyperbolic sink inside the selection region; this is not proved, and the theorem asserts only a finite window. The frozen-environment control is not analyzed here. On the product-content family through the base point, a numerical integration at $\eta = 0.02$ and record strength 2×10^{-4} shows the frozen control also selecting the pair, on a window beginning near $t = 0.7$, and losing it near $t = 3.6$, while the full flow keeps it. Within the theorem's finite window the selection therefore does not distinguish the full flow from the response of each agent to the initial laws of the other. The construction has no compressive retention.

14.2 A persistent group against conflicting evidence

A simpler local evaluator yields indefinite persistence with the initial records held fixed. Take three agents with positive Bernoulli belief and presentation laws of log odds l_i, n_i, uniform self reference and evaluator, and identity peer channels. Set $\pi_{ii} = a = 1/5$ and $\pi_{ij} = p = 2/5$ for $j \ne i$. Their

common binary record likelihood, relative to counting measure, is

$$\ell_\epsilon(o \mid z) = \frac{e^{\epsilon o z/2}}{2\cosh(\epsilon/2)}, \qquad o, z \in \{-1, 1\}, \quad \epsilon > 0. \tag{14.14}$$

Condition on $o = (1, 1, -1)$ and retain it in the objective. The species are identical; the different observations are data.

For a constant $c \in (0, 1)$, the selector S_c connects i, j when l_i, l_j have the same nonzero sign and

$$\frac{\min(|l_i|, |l_j|)}{\max(|l_i|, |l_j|)} > c, \tag{14.15}$$

and takes connected components. It reads public laws only, treats labels symmetrically, and is the same function along every flow compared below.

Theorem 14.3 (Persistent grouping for selector constants below one third). Let C be a compact subset of $(1/8, 1/3)$ and $u^{\mathrm{in}} = (4, 1/2, -1)$. There are $\epsilon_0 > 0$ and $r > 0$ such that for every $0 < \epsilon < \epsilon_0$ the following holds for all initial belief log odds $l(0) \in \epsilon B(u^{\mathrm{in}}, r)$ and all positive initial presentation laws. Under every S_c with $c \in \mathsf{C}$ the configuration starts with three singleton blocks, and the per-agent flow selects the three-agent block at a finite time after which the block persists for all time, whereas neither the label-preserving nor the renormalized null selects that block at any time.

Proof. The exact log-odds equations are

$$\dot{l}_i = -l_i + \sum_{j \neq i} \beta_{ij} l_j + \epsilon o_i, \qquad \dot{n}_i = -n_i + \sum_{j \neq i} \gamma_{ij} n_j. \tag{14.16}$$

For a Bernoulli law, the Fisher inverse cancels $dl/dq(+1)$, so the peer KL derivative contributes $l_i - l_j$ and the log-likelihood ratio contributes ϵo_i. The rows use the corresponding Bernoulli costs. The field is analytic with at most linear growth. The presentation sector is retained but does not enter the evaluator.

Put $l = \epsilon u$. Then exactly

$$\dot{u} = Mu + o + R_\epsilon(u), \qquad M = -\frac{7}{5} I + \frac{2}{5} \mathbf{1}\mathbf{1}^\top, \qquad u_* = -M^{-1} o = \left(\frac{15}{7}, \frac{15}{7}, \frac{5}{7}\right), \tag{14.17}$$

with $R_\epsilon(u)_i = \sum_{j \neq i} (\beta_{ij} - \pi_{ij}) u_j$. The matrix M is symmetric with eigenvalues $-1/5, -7/5, -7/5$. The log-partition function $\log(2\cosh(l/2))$ of the Bernoulli family has second derivative at most $1/4$, so every Bernoulli divergence obeys $D_{\mathrm{KL}} \leq (l - l')^2/8$. Let $\rho = 13/4$ and let B be the closed ball of radius ρ about u_*. On B we have $|u_i| < 6$, so every cost lies in $[0, 18\epsilon^2]$, each ratio β_{ij}/π_{ij} lies in $[e^{-18\epsilon^2}, e^{18\epsilon^2}]$, and

$$|R_\epsilon(u)| \leq \bar{R}_\epsilon := \frac{24\sqrt{3}}{5}\left(e^{18\epsilon^2} - 1\right) \qquad (u \in \mathsf{B}). \tag{14.18}$$

For $e = u - u_*$, symmetry of M gives $\frac{d}{dt}|e| \leq -|e|/5 + |R_\epsilon(u)|$, so B is forward invariant whenever $\bar{R}_\epsilon < \rho/5$, in particular for $\epsilon \leq 1/20$. Since $|u^{\mathrm{in}} - u_*| = \sqrt{1781}/14 < 3.02$, the initial set lies in B for $r \leq 1/5$. Let $\bar{u}$ solve the limit equation $\dot{\bar{u}} = M\bar{u} + o$ with $\bar{u}(0) = u(0)$. Then $\frac{d}{dt}(u - \bar{u}) =$

$M(u-\bar{u})+R_\epsilon(u)$ and $\|e^{Mt}\| = e^{-t/5}$ give

$$|u(t)-\bar{u}(t)| \le 5\bar{R}_\epsilon \qquad \text{for all } t \ge 0. \tag{14.19}$$

This bound is uniform in time, so persistence needs no separate attracting-ball argument.

From $u(0) = u^{\mathrm{in}}$ the limit solution is $\bar{u}(t) = u_* - \frac{1}{2}e^{-t/5}\mathbf{1} + \frac{1}{14}e^{-7t/5}(33,-16,-17)$. For an edge between agents with $u_i \ge u_j > 0$ the selector margin is $u_j - cu_i$, and along this solution

$$\begin{aligned}
\bar{u}_2 - c\bar{u}_1 &= \tfrac{15}{7}(1-c) - \tfrac{1-c}{2}e^{-t/5} - \tfrac{16+33c}{14}e^{-7t/5},\\
\bar{u}_3 - c\bar{u}_2 &= \tfrac{5-15c}{7} - \tfrac{1-c}{2}e^{-t/5} - \tfrac{17-16c}{14}e^{-7t/5},\\
\bar{u}_3 - c\bar{u}_1 &= \tfrac{5-15c}{7} - \tfrac{1-c}{2}e^{-t/5} - \tfrac{17+33c}{14}e^{-7t/5},
\end{aligned} \tag{14.20}$$

while $\bar{u}_1 - \bar{u}_2 = \frac{7}{2}e^{-7t/5} > 0$, $\bar{u}_2 - \bar{u}_3 > 10/7$, and $\bar{u}_2 \ge 1/2$. For $c \in (0,1)$ every time-dependent coefficient is negative, so each margin increases strictly in time. For small r these signs persist uniformly over the initial set and over c in a compact subset of $(0,1)$, and the limit margins move by $O(r)$. The margins are $\sqrt{2}$-Lipschitz in u, so by eq. (14.19) the exact margins differ from those of $\bar{u}$ by at most $5\sqrt{2}\bar{R}_\epsilon$.

Initially $u_2 - cu_1 = 1/2 - 4c + O(r) < 0$ and $u_3 < 0$, so all blocks are singletons for $c > 1/8$ and small r. For $c \in \mathsf{C}$ the limits $\frac{15}{7}(1-c)$ and $(5-15c)/7$ of the margins are bounded below by a positive constant m_{C}. Choose $\epsilon_0 \le 1/20$ with $5\sqrt{2}\bar{R}_\epsilon < m_{\mathsf{C}}/4$ and then T such that every margin of $\bar{u}$ exceeds $m_{\mathsf{C}}/2$ for $t \ge T$, uniformly over the initial set and C. For $t \ge T$ the exact edges $\{1,2\}$ and $\{2,3\}$ are present in both orientations of each pair, so the full block persists. The equilibrium ratio $1/3$ and the initial ratio $1/8$ are the two quantities that bound the admissible constants.

The label-preserving null is

$$\dot{l}_i^0 = -\beta_{ii}^0(l_i^0)l_i^0 + \epsilon o_i, \qquad \beta_{ii}^0 = \frac{ae^{-D_{\mathrm{KL}}(q_i\|q_0)}}{ae^{-D_{\mathrm{KL}}(q_i\|q_0)} + 2p}, \tag{14.21}$$

and the renormalized null is $\dot{l}_i^0 = -l_i^0 + \epsilon o_i$. In both, the derivative at $l_i = 0$ has the strict sign of o_i. The first two initial odds remain positive and the third remains negative for all time. Thus the full block never appears under any S_c, irrespective of event timing. Independent presentation-law perturbations do not change the conclusion. □

Proposition 14.4 (The verdict depends on the selector constant). Let C' be a compact subset of $(1/3,1)$ and C as in theorem 14.3. For sufficiently small ϵ and r and the same initial set, the full flow and both nulls produce the same partition sequence $1|2|3,\ 12|3$ under every S_c with $c \in \mathsf{C}'$. Under every S_c with $c \in \mathsf{C}$, the full flow and the frozen-environment control both produce $1|2|3,\ 12|3,\ 123$.

Proof. For $c \in \mathsf{C}'$ the limit $(5-15c)/7$ is negative and the last two margins of eq. (14.20) lie below it for all $t \ge 0$; by eq. (14.19) the exact margins $u_3 - cu_1$ and $u_3 - cu_2$ stay negative once $5\sqrt{2}\bar{R}_\epsilon < (15c-5)/14$. If $u_3 > 0$ then $u_3 < cu_j < u_j$, so the ratio of agent 3 with either peer is below c, and agent 3 is never connected. The first margin increases strictly to $\frac{15}{7}(1-c) > 0$ with a positive derivative at its unique zero, so for small ϵ the exact margin has exactly one zero, and the opposite-orientation margin $u_1 - cu_2$ stays positive. The full sequence is therefore $1|2|3,\ 12|3$.

Along the renormalized null, $l_i = \epsilon o_i + (l_i(0) - \epsilon o_i)e^{-t}$, and the ratio of agents 1 and 2 increases strictly to one. Along the label-preserving null put $r_i = o_i l_i/\epsilon$, so $\dot r_i = f(r_i)$ with $f(r) = 1 - \beta^0(r)r$ for one function β^0. Since $D_{\mathrm{KL}}(q_i\|q_0) \le \epsilon^2 r^2/8$, its r-derivative is at most $\epsilon^2 r/4$, and $|d\beta^0/dD_{\mathrm{KL}}| \le 1/4$, the interval $1/4 \le r \le 12$ is forward invariant for $\epsilon \le 1/100$, it contains the initial values, and on it $f(r)/r$ and f are strictly decreasing. Hence agents 1 and 2 keep their order $r_2 < r_1$, $\frac{d}{dt}\log(r_2/r_1) = f(r_2)/r_2 - f(r_1)/r_1 > 0$, and both converge to the unique zero of f. In both nulls the ratio of agents 1 and 2 crosses c exactly once and agent 3 stays negative, giving $1|2|3,\ 12|3$.

The frozen-environment control is $\dot u_i = -u_i + \sum_{j\neq i} \beta^{\mathrm{fr}}_{ij} u_j(0) + o_i$ with rows computed from the own current law and the initial peer laws, so $\beta^{\mathrm{fr}}_{ij} = \pi_{ij} + O(\epsilon^2)$ and its solution stays within $O(\epsilon^2)$ of the limit $u_i(t) = w_i + (u_i(0) - w_i)e^{-t}$, $w_i = o_i + \frac{2}{5}\sum_{j\neq i} u_j(0)$, uniformly in time. From u^{in}, $w = (4/5, 11/5, 4/5)$, and with $x = e^{-t}$ the margins $u_2 - cu_1$, $u_3 - cu_1$, $u_3 - cu_2$ are affine in x and vanish at $x_{12} = 2(11-4c)/(17+32c)$, $x_{13} = 4(1-c)/(9+16c)$, $x_{23} = 2(4-11c)/(18-17c)$, each crossing once, while the opposite-orientation margins stay positive for $c < 4/11$. For $c \in (1/8, 1/3)$ one has $x_{12} > 58/83 > 4/9 \ge \max(x_{13}, x_{23})$, so the sequence is $1|2|3,\ 12|3,\ 123$. For the full flow, at $e^{-t/5} = 5/6$ the limit solution has $\bar u_3 < 0$ and $\bar u_2 - c\bar u_1 > 0$ for every $c \le 1/3$, so the pair $\{1,2\}$ forms before agent 3 turns positive, and by the proof of theorem 14.3 agent 3 then joins once and stays. All crossings are transversal, so the sequences persist for small ϵ and r. □

The (F5) report for this construction under definition 6.36 is therefore the following. Its objective is the per-agent free energy under the Gibbs-reduced dynamics on positive Bernoulli tables, a full-grade admitted class. Against both nulls (F5) holds for every compact constant set inside $(1/8, 1/3)$, at every point of the initial set of theorem 14.3, and fails for every constant set that meets $(1/3, 1)$; the two endpoints are the initial ratio and the equilibrium ratio, which are properties of the flow rather than of the constitutive data. The selector-free discrepancy is the sign of agent 3's public odds: it converges to approximately $5\epsilon/7 > 0$, against its own record, under the full flow, and stays negative under both nulls, near -5ϵ and $-\epsilon$. The frozen-environment control reproduces the full partition sequence on the same constant set and also reverses agent 3's sign, with limit near $4\epsilon/5$ from u^{in}: the selection already follows from each agent's response to the initial laws of the others, and feedback among the agents is not needed for it. For $c = 1/4$ the limit solution meets the last selector margin at $t = -5\log y_* \approx 3.83$, where $y_* \approx 0.465$ is the root in $(0,1)$ of $\frac{5}{28} - \frac{3}{8}y - \frac{13}{14}y^7$. In the limit $\epsilon \to 0$, eq. (14.17) is continuous-time averaging [58] with each agent anchored to its fixed record, the model of Friedkin and Johnsen [79], whose equilibrium is the same linear solve; its stationary points are logarithmic pools of private-signal likelihoods, as in log-linear non-Bayesian social learning [111]. What is new here is its derivation from normalized per-agent free energies with exact Gibbs rows, the uniform-in-time bound eq. (14.19) for the exact field, and the inheritance of the selected block below. The construction discards no child law.

14.3 A fixed-model angular extension

Let $\mathsf{K} = \{-1,1\} \times S^1 \times S^1$, with uniform binary reference and independent Haar references H on both circles. Let $\mathsf{M} = \{-1,1\}$ with counting measure. Write $e(\phi) = (\cos\phi, \sin\phi)$ and $h(A) = \log I_0(|A|)$,

where $I_0(r) = \int e^{r\cos\phi} H(d\phi)$. The invariant belief family is

$$q_i(z,\phi,\psi) = \frac{\exp[l_i z/2 + A_i \cdot e(\phi) + zC_i \cdot e(\phi) + b_i \cos(\psi - \theta_i)]}{Z_i}, \tag{14.22}$$

where $A_i, C_i \in \mathbb{R}^2$, $b_i > 0$, $\theta_i \in S^1$, and

$$Z_i = I_0(b_i)\frac{e^{l_i/2} I_0(|A_i + C_i|) + e^{-l_i/2} I_0(|A_i - C_i|)}{2}. \tag{14.23}$$

The Cartesian hidden parameter $b_i e(\theta_i)$ is regular at zero; the phase chart used for compression is restricted to $b_i > 0$. Retain the independent Bernoulli presentation laws and priors of theorem 14.3. Fix $J = \epsilon^3$ and the normalized self evaluator

$$G_i(m; z, \phi, \psi) = \frac{e^{Jz\cos\phi}}{I_0(J)}. \tag{14.24}$$

It is independent of m. Each nonself channel copies z and resamples both circles from Haar measure. Thus its output is $\mu_j \otimes H \otimes H$, where μ_j is the public binary marginal. Its density is relative to the uniform binary reference, not counting measure:

$$\mu_i(z) = \frac{e^{L_i z/2}}{\cosh(L_i/2)}, \qquad L_i = l_i + h(A_i + C_i) - h(A_i - C_i). \tag{14.25}$$

The record space is a bit o with counting measure and an angle ω with Haar measure, with likelihood

$$\ell_\epsilon(o, \omega \mid z, \phi, \psi, m) = \frac{e^{\epsilon o z/2}}{2\cosh(\epsilon/2)} \frac{e^{\epsilon\cos(\omega - \phi)}}{I_0(\epsilon)}. \tag{14.26}$$

Integrating each circle and summing the record bit proves all normalizations. The channels, evaluator, and likelihood are fixed during an own-block variation. In particular this example does not reconstruct the receiver's changing conditional law inside its generative model.

Proposition 14.5 (Exact angular field). The full own-law Fisher field preserves eq. (14.22) and is

$$\begin{aligned} \dot{l}_i &= -l_i + \sum_{j \neq i} \beta_{ij} L_j + \epsilon o_i, \quad \dot{A}_i = -A_i + \epsilon e(\omega_i), \\ \dot{C}_i &= -C_i + \beta_{ii}\epsilon^3 e_1, \qquad \dot{b}_i = -b_i, \qquad \dot{\theta}_i = 0. \end{aligned} \tag{14.27}$$

The presentation equation remains the second equation of eq. (14.16). If $m_i = \tanh(L_i/2)$ and $t_i = \mathbb{E}_{q_i}[z\cos\phi]$, the unnormalized content row weights are

$$w_{ii} = \frac{a e^{Jt_i}}{I_0(J)}, \qquad w_{ij} = \frac{p e^{L_j m_i/2}}{\cosh(L_j/2)}, \qquad \beta_{ij} = \frac{w_{ij}}{\sum_r w_{ir}}. \tag{14.28}$$

Proof. Put $S_i = \mathbb{E}_{q_i} \log q_i$. The costs are $D_{ii} = S_i - Jt_i + \log I_0(J)$ and $D_{ij} = S_i - L_j m_i/2 + \log\cosh(L_j/2)$. The common entropy cancels in the Gibbs normalization, giving eq. (14.28). The centered negative own derivative is the logarithmic replicator velocity. It lies in the span of $z, e(\phi), ze(\phi), e(\psi)$ and constants. Matching coefficients gives eq. (14.27). The covariance of

these seven sufficient statistics is positive definite: a linear combination constant under a positive density must be constant under the reference, and its Fourier modes and sign parity force every nonconstant coefficient to vanish. This gives tangency: the chart field of definition 4.21, evaluated on the family, is the family velocity. Invariance needs uniqueness of that chart flow. Take the bounded log-density chart of definition 4.13 on K relative to $\lambda = \frac{1}{2}(\delta_{-1} + \delta_1) \otimes H \otimes H$, with log densities $f_i \in L_0^\infty(\lambda)$. The evaluator eq. (14.24) and the likelihood eq. (14.26) have bounded log densities. The peer output has density $E_\lambda[e^{f_j} \mid Z = z]/E_\lambda e^{f_j}$; since $f \mapsto \log E_\lambda e^f$ and its conditional version are 1-Lipschitz in the supremum norm, its logarithm is a 2-Lipschitz function of f_j. The Gibbs rows are smooth functions of bounded costs. The sufficient conditions stated after definition 4.21 therefore hold, the chart flow has unique maximal solutions, and the family curve, which solves it, is that solution. The family is invariant, and eq. (14.27) is the exact Fisher velocity on it rather than a Euclidean update or a projected approximation. The same argument applies to the null and frozen-environment fields. □

The marked sector participates in the public force. At $A = \alpha e_1$ and $C = ce_1$ with $c > 0$, $\partial_\alpha L = h_1'(\alpha + c) - h_1'(\alpha - c) > 0$, since $h_1''(r) = \mathrm{Var}_r(\cos\phi) > 0$. Starting with $C = 0$ generates positive c at positive time. Also $\partial_{L_j}\beta_{ij} = \beta_{ij}(1 - \beta_{ij})(m_i - m_j)/2$. Thus angular data can change source competition and the receiver's force. The separate hidden phase does not enter those quantities.

Theorem 14.6 (Grouping survives the angular extension). Let C be a compact subset of $(1/8, 1/3)$. There is $\epsilon_0 > 0$ such that for every $0 < \epsilon < \epsilon_0$, every selector S_c of eq. (14.15) with $c \in \mathsf{C}$, applied to the actual public odds L_i, selects a persistent full block from a nonempty relatively open initial set in eq. (14.22), with records $o = (1, 1, -1)$ held fixed. Neither the label-preserving nor the renormalized null selects that block at any time. Initial marked and hidden concentrations can lie in a fixed small positive compact patch independent of ϵ, and the conclusion is uniform over the initial angular record values.

Proof. The null equations are

$$\begin{aligned} \dot{l}_i^0 &= -\beta_{ii}^0 l_i^0 + \epsilon o_i, \qquad \dot{A}_i^0 = -\beta_{ii}^0 A_i^0 + \epsilon e(\omega_i), \\ \dot{C}_i^0 &= \beta_{ii}^0(\epsilon^3 e_1 - C_i^0), \quad \dot{b}_i^0 = -\beta_{ii}^0 b_i^0, \qquad \dot{\theta}_i^0 = 0, \end{aligned} \tag{14.29}$$

with eq. (14.2) evaluated at the complete self cost. Its angular entropy cannot be canceled after deleting the nonself costs.

For $\epsilon \le 1/100$, consider

$$\frac{1}{4} \le \frac{o_i l_i}{\epsilon} \le 12, \qquad |A_i| \le \frac{1}{10}, \qquad |C_i| \le 2\epsilon^3, \qquad 0 \le b_i \le \frac{1}{10}. \tag{14.30}$$

In natural parameters $\eta = (l/2, A, C, be(\theta))$ the sufficient-statistic vector has squared norm four, so the log-partition Hessian has operator norm at most four. Taylor's integral remainder for the exponential-family divergence, relative to $\eta_G = (0, 0, \epsilon^3 e_1, 0)$, gives

$$D_{ii} \le 2|\eta - \eta_G|^2 \le 72\epsilon^2 + \frac{1}{25} + 18\epsilon^6 < \log 2, \qquad \frac{1}{9} \le \beta_{ii}^0 \le \frac{1}{5}. \tag{14.31}$$

For $r_i = o_i l_i/\epsilon$, $\dot r_i = 1 - \beta^0_{ii} r_i$ points inward at both endpoints. The outward derivatives of the A and C norms are at most $\epsilon - 1/90 < 0$ and $-\beta^0_{ii}\epsilon^3 < 0$ on their respective boundaries, while b decreases. Thus eq. (14.30) is forward invariant. Since $\nabla h(0) = 0$ and $\|\nabla^2 h\| \le 1$,

$$|L_i - l_i| \le 2|C_i|(|A_i| + |C_i|) \le \frac{2}{5}\epsilon^3 + 8\epsilon^6 < \frac{\epsilon}{8}. \tag{14.32}$$

Consequently $o_i L_i \ge \epsilon/8$ for all finite null times.

For the full flow put $u = l/\epsilon$ and $c = C/\epsilon^3$, leaving A unscaled. On bounded sets, $L_i/\epsilon = u_i + O_{C^1}(\epsilon^2)$ and $m_i, t_i = O(\epsilon)$. Equation eq. (14.28) implies $\beta_{ij} = \pi_{ij} + O_{C^1}(\epsilon^2)$ even for fixed nonzero initial angular amplitudes. Hence

$$\dot u = Mu + o + O_{C^1}(\epsilon^2), \qquad \dot c_i = -c_i + a e_1 + O_{C^1}(\epsilon^2), \qquad \dot A_i = -A_i + \epsilon e(\omega_i). \tag{14.33}$$

At $\epsilon = 0$ this is a stable affine field with equilibrium $(u_*, c_i = a e_1, A_i = 0)$ and symmetric negative-definite derivative. The implicit function theorem supplies a nearby equilibrium with $u = u_* + O(\epsilon^2)$, $c_i = a e_1 + O(\epsilon^2)$, and $A_i = \epsilon e(\omega_i)$. On a sufficiently small common ball its symmetric derivative is at most $-I/10$. After recentering, the derivative of squared distance is at most minus one fifth of that distance squared. This proves an invariant attracting neighborhood with all public signs positive and all pair ratios above $\max \mathsf{C}$.

Choose $u(0)$ near $(4, 1/2, -1)$, $|C_i(0)| < \epsilon^3$, and $|A_i(0)|, b_i(0) < 1/20$ with positive hidden amplitudes. The limiting affine flow enters the attraction neighborhood after a fixed finite time, uniformly over compact initial marked patches. Finite-time continuous dependence gives the same entry for small positive ϵ and a relatively open initial set. Initially $L_i = l_i + O(\epsilon^3)$, so the strict singleton margins persist. The angular records range over a compact torus and enter only through bounded constant forcings of size ϵ; the same entrance time and smallness threshold work uniformly over their values. The label-preserving null retains opposite signs by eq. (14.32). The renormalized null is eq. (14.29) with $\beta^0_{ii} = 1$; along it $o_i l_i/\epsilon \ge \min(o_i l_i(0)/\epsilon, 1) \ge 1/4$, $|A_i| \le \max(|A_i(0)|, \epsilon)$ and $|C_i| \le \epsilon^3$, so eq. (14.32) holds there as well.

Finally A, C, b have finite-interval bounds directly from their equations, $|L - l| \le 2|C|$, and both l and the presentation log odds have at most linear growth. Solutions therefore continue for every finite time. The subsequent group persistence uses the invariant attraction neighborhood, not indefinite extrapolation of finite-time continuity. □

The smallness of the marked and hidden amplitudes is used only by the null estimate eq. (14.31). In the full flow the marked amplitudes enter the scaled public equation only through $|L_i/\epsilon - u_i| \le 2\epsilon^2(|A_i| + |C_i|)$, and the hidden concentrations do not enter it at all. The scaled public equation differs from eq. (14.17) by a remainder of order ϵ^2 that is bounded uniformly on the invariant ball, so the uniform-in-time comparison eq. (14.19) and the proof of proposition 14.4 apply with L in place of l. The (F5) report after proposition 14.4 therefore holds for this theorem, with the same constant sets, the same frozen-environment comparison and the same sign reversal of agent 3's public odds.

14.4 A hidden-phase quotient with inherited records

Fix the operational protocol before selection. It admits fresh public and marked-coordinate queries on the primitive ports, normalized generative or recognition-predictive record transactions with the likelihood family eq. (14.26), and adaptive choices of ports, strengths and waiting times from the retained history. These two kinds of predictive law are declared separately and need not agree. A prediction is not automatically assimilated. An evidence transaction appends its actual outcome and adds its likelihood to the persistent objective. Every finite continuation has finite record coefficients. Hidden absolute-angle queries are excluded at all stages. An external source-label readout does not change the agent's conditioning convention; revealing such a label to the inferring agent would specify another source-row update. Policies are Borel kernels, independent of context, from finite histories and fixed lab data to admitted actions and waiting times, and every record density is jointly Borel in the outcome and the state, as in definition A.3. For the collective geometry below, the expected number of transactions in $[0, t]$ is required to be bounded on compact contextual patches; without this bound a policy can issue unboundedly many context-sensitive fresh queries before a fixed time, and the history experiment is then not differentiable in quadratic mean. In the terms of definition 6.17, the parent protocol rule A_I consists of these queries and transactions addressed to a block through its primitive ports, and the child-level probe rule B_I consists of the same queries applied to individual members, or to sets of them, before the handoff, together with every channel, evaluator and record kernel of the construction. Neither rule depends on the retention or on the selected blocks, and neither contains a query whose response law depends on a hidden phase θ_i.

Proposition 14.7 (Canonical phase parent). On the invariant family eq. (14.22), replace each hidden phase by zero and retain l_i, A_i, C_i, b_i, the presentation laws, primitive port labels, all constitutive data, event information, and the full record history. For a selected block take the product of these normalized canonical laws. This retention is noninjective at every finite time, intertwines the original-time full flow, both null flows and the frozen-environment control, and preserves every admitted finite adaptive joint response-record history, including its dependence with retained next states. The phase it erases is gauge relative to the probe rule B_I and to every observable algebra of the construction, so the retention fails (F4).

Proof. All full and null costs and retained velocities are independent of θ_i, and $\dot{\theta}_i = 0$. The canonical laws therefore obey the exact inherited port equations. Positive initial b_i remains positive at each finite time. The child diagnostic

$$\mathbb{E}_{q_i} e^{i\Psi} = \frac{I_1(b_i)}{I_0(b_i)} e^{i\theta_i} \tag{14.34}$$

separates a phase fiber because $I_1(b)/I_0(b) > 0$ for $b > 0$, so the retention is noninjective after event data and records are included. That diagnostic is a law functional that no agent, channel, evaluator, record kernel or admitted query reads. By definition 1.5 it belongs to no observable algebra and is not an element of B_I. Every cost, row, channel output, retained velocity and admitted response law is independent of θ_i at every time, so no element of B_I separates two points of a fiber, and (F4) of definition 6.36 fails.

The current retained state determines every admitted next response law. Conditional on its actual outcome, the same likelihood adds the same binary and marked-vector coefficients to the retained

objective and the same entry to history. The joint response and retained next-state kernel therefore factors through retention. Deterministic waiting also factors by the exact equations. Induction over the causal product of these kernels proves every finite adaptive joint history. This is the predictive-equivalence requirement familiar from causal-state constructions [177]; for input-driven processes its analogue is the ϵ-transducer [16], and for finite controlled chains it is probabilistic bisimulation [127]. It is proved here directly for the declared controlled protocol. □

The retention is the equivariant quotient of proposition 6.7 by the hidden torus acting by $\theta_i \mapsto \theta_i + g_i$ on the stratum $b_i > 0$, with the quotient identified with the canonical section $\theta = 0$; that action is free and proper and is a symmetry of the full field, of both nulls, of the Fisher metric and of every admitted kernel. The history induction follows the pattern of lemma 11.4. What is new is the assembly of these general results for the per-agent model with persistent records. The two attributes required with (F4) are these: the fibers are orbits of that symmetry, and the separation they carry fades, since $b_i(t) = e^{-t} b_i(0)$ along the full flow, so that the excluded diagnostic has modulus $I_1(b_i)/I_0(b_i) \sim b_i/2$ and the phase Fisher information $b_i I_1(b_i)/I_0(b_i)$ decays like e^{-2t}; along the nulls the rate is $\beta^0_{ii} \in [1/9, 1/5]$ or one.

The parent carries primitive ports and their directed own-objective field. No claim is made that it is a fixed-size agent with one new source row, or the Fisher gradient of the sum of its children's free energies. Such a sum would add source-role derivatives. Normalized inherited channels act by projecting to the selected primitive source port, applying the existing public channel, and appending normalized canonical factors on other output coordinates. No new parent force is fitted.

The diagnostic eq. (14.34) lies outside A_I and outside B_I. If a hidden-angle query is admitted in A_I, the quotient fails (B). A rule that admitted such a query in B_I for singletons and withdrew it after the handoff would make (F4) hold, at the price of deleting a probe at the handoff; the chapter declares no such rule, and no agent of the construction reads the hidden angle. A persistent hidden-angle likelihood proportional to $e^{\lambda \cos(v-\psi)}$ also changes the hidden amplitude and phase:

$$\dot{b} = -b + \lambda \cos(v - \theta), \qquad \dot{\theta} = \frac{\lambda}{b} \sin(v - \theta) \tag{14.35}$$

in the full fixed-model example. Two states with the same retained b, v can then have different future b velocities. Finite-record uncertainty about θ does not remove this dynamical obstruction.

14.5 Collective geometry of the same parent

The marked comparison group is a common S^1 acting on ϕ within the selected block. Under a frame change g, transport ϕ and the record angle by addition of g, rotate A_i, C_i and the constitutive evaluator direction e_1 by the corresponding plane rotation, and leave z, b, ψ, θ unchanged. The presentation sector carries no marked orbit. This is passive covariance of the declared comparison data. Rotating only the beliefs while holding the numerical evaluator and record directions fixed is generally not a symmetry of the energy. The hidden phase rotations act outside the marked comparison group. They are symmetries of every constitutive datum and admitted kernel, and the distinctions they relate are gauge relative to every algebra declared in this chapter.

To include records in the actual statistical experiment, fix one finite initial episode and let $\Pi_c(dh)$ be its normalized generative predictive law at contextual point c. Conditional on the prepared configuration the agents' record kernels may be sampled independently; this defines an operational experiment and does not posit a global latent generative model. Let $Q^h_{c,t}$ be the product of fresh recognition and Gibbs-label laws conditional on the retained history h. The comparison law is

$$\mathcal{P}_{c,t}(dh, dy) = \Pi_c(dh)Q^h_{c,t}(dy), \qquad g^{\text{joint}} = g^{\text{history}} + \mathbb{E}_H[g^{\text{conditional}}]. \tag{14.36}$$

The metric identity is the chain rule for Fisher information [201], compare proposition 3.2: the conditional score has mean zero. It presupposes that $c \mapsto \mathcal{P}_{c,t}$ is differentiable in quadratic mean (eq. (3.1)), which is where the bound on the expected number of transactions in the protocol is used. For the grouping witness one may condition on the positive-probability public pattern $(+,+,-)$, keeping all angular records random and including the conditioning normalizer in the history score. Source-label scores remain part of Q^h.

Proposition 14.8 (Exact collective cost inheritance). On every regular contextual section, the canonical parent of proposition 14.7 preserves the actual collective minimizing connection class and its curvature for eq. (14.36). It preserves the minimized contextual Fisher tensor exactly when $\sum_i \mathcal{I}(b_i)(d\theta_i)^2 = 0$, with $\mathcal{I}(b) = bI_1(b)/I_0(b)$, in particular when every discarded θ_i is constant in context.

Proof. At each context the product shift $\psi_i \mapsto \psi_i - \theta_i$ is constant in time and history. It sends the fine joint law to the actual canonical parent law with unit Haar Jacobian and commutes with the marked action. The fine contextual score is the parent score at corresponding samples plus $\sum_i (X\theta_i) b_i \sin(\psi_i - \theta_i)$. Under $\mathcal{P}$ each ψ_i is independent of the marked and public coordinates, of the history and of the source labels, because no likelihood, channel output or row reads it; and $\mathbb{E}\sin(\psi - \theta) = \mathbb{E}[(\cos(\psi-\theta) - \mathbb{E}\cos(\psi-\theta))\sin(\psi-\theta)] = 0$. The added term is therefore orthogonal to every retained score and to the marked orbit score s_{orb}, and for every tangent X and comparison velocity ξ

$$\mathcal{C}_{\text{fine}}(X,\xi) = \mathbb{E}_{\mathcal{P}}(s_X + \xi s_{\text{orb}})^2 = \mathcal{C}_{\text{parent}}(X,\xi) + \sum_i \mathcal{I}(b_i)(X\theta_i)^2. \tag{14.37}$$

The added term does not depend on ξ, so the minimizer class modulo isotropy, hence the connection and its curvature, agree, and the minima differ by $\sum_i \mathcal{I}(b_i)(X\theta_i)^2$. The parent metric is computed from its normalized law rather than assigned by copying a tensor. □

This is proposition 3.18 with a vanishing horizontal defect, and the tensor gap is the exact defect of corollary 3.19. The common marked connection is a diagnostic of these evolved sections; it does not enter the present dynamical equations.

Theorem 14.9 (A common nonflat grouping window). The initial set in theorem 14.6 admits a compact rank-two contextual patch and a fixed finite post-selection window on which the actual collective connection has nonzero curvature, its minimized tensor has rank two, and the canonical parent has exactly the same quantities.

Proof. Choose fixed small positive $a_- < a_+$, d, and b_i, independent of ϵ. Use contextual coordinates (a, χ) with $A_1(0) = ae(\chi)$, $a \in [a_-, a_+]$, $A_2(0) = de_1$, and $C_i(0) = 0$. Other selected marked factors

may have fixed positive concentrations. Keep hidden phases constant over context and take the public initial odds from theorem 14.6. Choose a finite T_+ containing its common entrance time and a positive subsequent interval, with no additional assimilation during this window.

At $\epsilon = 0$, the episode likelihood is uniform, public laws are uniform, and the full Gibbs rows equal the fixed priors. The history score and the score of the conditioning normalizer vanish identically in this witness, at every ϵ: the generative predictive kernel of the initial episode reads the configuration only through the peers' initial public odds, which equal $l_j(0)$ because $C_i(0) = 0$ and are independent of context. Label scores vanish at the limiting parameter. The solution is $A_i(t) = e^{-t}A_i(0)$, $C_i(t) = 0$, $l_i(t) = 0$, and $b_i(t) = e^{-t}b_i(0)$. Put

$$I(x) = x\frac{I_1(x)}{I_0(x)}, \qquad V(x) = \mathrm{Var}_{v_x}(\cos\phi), \qquad B(t) = \sum_{i\neq 1} I(e^{-t}|A_i(0)|) > 0. \tag{14.38}$$

Here $v_x = e^{x\cos\phi}/I_0(x)$. Differentiating its partition function gives $I'(x) = I_1(x)/I_0(x) + xV(x) > 0$ for $x > 0$. Integration of the derivative of $\sin\phi e^{x\cos\phi}$ around the circle gives the phase Fisher identity $x^2\mathbb{E}\sin^2\phi = I(x)$. Minimizing the sum of marked phase-score costs gives

$$\begin{aligned}
\mathcal{A}_{0,t} &= -\frac{I(e^{-t}a)}{I(e^{-t}a) + B(t)}d\chi,\\
\mathcal{F}_{0,t} &= -\frac{B(t)e^{-t}I'(e^{-t}a)}{[I(e^{-t}a) + B(t)]^2}da\wedge d\chi,\\
h^{0,t}_{aa} &= e^{-2t}V(e^{-t}a), \qquad h^{0,t}_{\chi\chi} = \frac{I(e^{-t}a)B(t)}{I(e^{-t}a) + B(t)}, \qquad h^{0,t}_{a\chi} = 0.
\end{aligned} \tag{14.39}$$

The locked inertia, both tensor eigenvalues, and the magnitude of the curvature coefficient have positive minima on the compact contextual patch and $0 \le t \le T_+$.

The exact densities, episode kernels, and ODE coefficients are analytic on a common compact finite-time tube. The field is real-analytic jointly in the state and in ϵ, and the initial section is analytic in context and ϵ, so the flow and its first two contextual derivatives, which solve $\dot{X}_c = DV(X)X_c$ and $\dot{X}_{cc} = DV(X)X_{cc} + D^2V(X)[X_c, X_c]$, depend continuously on ϵ uniformly on the compact tube [103, Chapter V]; they converge to the displayed limit as $\epsilon \to 0$. Compact record spaces and positive bounded densities justify differentiating the Fisher integrals. Conditioning on the public pattern is regular because its limiting probability is positive. Thus the locked inertia K_ϵ, momentum j_ϵ, and unminimized contextual tensor B_ϵ converge with the derivatives required in $\mathcal{A}_\epsilon = -K_\epsilon^{-1}j_\epsilon$, $h_\epsilon = B_\epsilon - j_\epsilon \otimes j_\epsilon/K_\epsilon$, and $\mathcal{F}_\epsilon = d\mathcal{A}_\epsilon$. The positive limiting margins persist for sufficiently small positive ϵ, uniformly in the conditional integrands over the compact episode space. The grouping threshold can be imposed simultaneously by taking the smaller admissible ϵ_0. Equation eq. (14.37) gives exact parent equality. □

The fixed positive initial marked amplitudes are needed in this finite-window argument: they keep the limiting inertia nonzero at the same finite entrance time as grouping. No numerical bound on ϵ_0 is asserted. The group persists indefinitely with its initial record episode fixed, while nonflatness is asserted on the declared finite window. Later admitted record continuations preserve quotient closure but may alter the grouping or geometry.

The curvature is prepared rather than produced by the grouping. Context enters only through $A_1(0) = ae(\chi)$, the marked equation $\dot{A}_i = -A_i + \epsilon e(\omega_i)$ has no peer term, and eq. (14.39) already holds at $t = 0$, at $\epsilon = 0$ with no coupling and no records, and along the null flows with the same limit. The window is a choice of interval containing the entrance time. At a fixed positive ϵ the evaluator-driven C_i gives the locked inertia a positive floor of order ϵ^6 while the contextual momentum decays like e^{-t}, so the curvature of the evolved section is expected to tend to zero; numerical evaluations of a reduced model indicate this, and the finite window is then necessary rather than a limitation of the proof. Neither statement is proved here.

14.6 Report against the formation standard

The angular construction concerns the per-agent objective, with the Gibbs-reduced per-agent dynamics in the bounded log-density chart on $\{-1, 1\} \times S^1 \times S^1$. Its admitted class is relatively open in the declared invariant family eq. (14.22) together with the Bernoulli presentation laws, a finite-dimensional family of infinite codimension in the chart that is not contained in finitely many orbits, so (F1) holds in the family grade; the Bernoulli construction of theorem 14.3 has the full grade. The selector family is S_c of eq. (14.15), with persistence as an invariant block relation while the initial records are held fixed. Its constant set is not derived from the constitutive data, and the verdicts below are reported for each choice. The protocol and probe rules A_I and B_I are those declared before proposition 14.7, and the parent of that proposition satisfies (F3). Compression (F4) fails: the retention is noninjective, but the phase it erases is gauge relative to B_I; the fibers are orbits of the hidden torus symmetry and the excluded separation fades. Interaction necessity (F5) holds against both nulls for every compact constant set inside $(1/8, 1/3)$ and fails for every constant set that meets $(1/3, 1)$ (proposition 14.4). The selector-free discrepancy is the reversal of agent 3's public sign against its record, and the frozen-environment control reproduces both the partition sequence and that reversal. There is one selection stage: no block of first-stage parents is selected, so (F6a) fails and no counterfactual class is declared. With the admitted class equal to the initial set of the theorem and the constant set inside $(1/8, 1/3)$, the strong grade holds because the proof gives the conclusion at every point; the amplifying grade is not examined. The construction is therefore a single-stage selection with exact inheritance, and it is not a recursive formation. The four-state construction of proposition 14.2 has full-grade (F1), finite-window persistence, (F5) against both nulls for constants in $(0, \eta/(8\sqrt{2})]$, and no compressive retention.

The selector and the phase-insensitive interface are constitutive choices. The phase quotient is available before any group is selected and also under both nulls, and coupling the marked sector to public communication does not make the erased phase invisible by interaction. Joining two already canonical parent products is associative and preserves their inherited port fields and histories. It erases no new phase, so it supplies neither a second noninjective handoff nor the counterfactual dependence required by (F6b). The next obligation is an interface that the interaction selects and that some declared probe resolves before the handoff, followed by a further compressive inherited selection that depends on the first handoff. The present conditional theorem does not close open problem 6.38.

Chapter 15

Collective Compression and Conditional Local Parents

This chapter concerns the per-agent objective under the Gibbs-reduced dynamics, as in chapter 14. The phase parent of proposition 14.7 erases a distinction already invisible on each primitive port. A collective alternative transmits the second angular law between peers and admits relative-angle responses. Each proper block is then exposed by a response across its boundary, while a retention of the whole connected population can erase one common orientation. That orientation is gauge relative to every algebra declared here, so the whole-block retention fails compression (F4) of definition 6.36, and the construction is a single-stage selection with exact inheritance, not a recursive formation. This chapter constructs that quotient and a local realization with finite reference memory. A finite preparation generator gives a second realization. A causal local phase memory removes the reference evaluation when the initial collective direction is coherent across context. That preparation condition is explicit; it is not generated by the recurrence. The unrestricted local-parent question remains inconclusive, with obstructions for specified memory classes rather than every possible mechanism.

15.1 A transmitted angle

There are three agents with the normalized per-agent model of definition 4.19. Relative to a uniform sign and two normalized Haar circles, write

$$q_i(z,\phi,\psi) = Z_i^{-1}\exp\{l_i z/2 + A_i\cdot e_\phi + zC_i\cdot e_\phi + B_i\cdot e_\psi\}, \tag{15.1}$$
$$Z_i = I_0(|B_i|)\frac{e^{l_i/2}I_0(|A_i+C_i|)+e^{-l_i/2}I_0(|A_i-C_i|)}{2}.$$

Here $e_\theta=(\cos\theta,\sin\theta)$ and $A_i,C_i,B_i\in\mathbb{R}^2$. The presentation probability is

$$s_i(w) = \frac{e^{n_i w/2}}{2\cosh(n_i/2)}, \qquad w\in\{-1,1\}. \tag{15.2}$$

The self evaluator is $G_i = e^{Jz\cos\phi}/I_0(J)$, with $J=\epsilon^3$, and the self presentation reference is uniform. Source priors in both sectors are $1/5$ for self and $2/5$ for each peer. Rows are reduced exactly at unit temperature, and the dynamics is the chart version of the Gibbs-reduced per-agent dynamics of definition 4.21 on $\{-1,1\}\times S^1\times S^1$.

The nonself content channel copies z, ψ and resamples ϕ from Haar measure. Define $h_0(B) = \log I_0(|B|)$ and $M(B) = \nabla h_0(B)$. The transmitted density is

$$\nu_j(z, \phi, \psi) = \frac{e^{L_j z/2}}{\cosh(L_j/2)} \frac{e^{B_j \cdot e_\psi}}{I_0(|B_j|)}, \qquad L_j = l_j + h_0(A_j + C_j) - h_0(A_j - C_j). \tag{15.3}$$

The sign reference is uniform, so its density differs by a factor of two from its probability mass. Presentation peers use the identity channel. Every source remains normalized.

Evidence uses the positive normalized likelihood

$$\ell_{r,\lambda}(o, \omega \mid z, \phi, \psi, w) = \frac{e^{roz/2}}{2\cosh(r/2)} \frac{e^{\lambda \cos(\omega - \phi)}}{I_0(\lambda)}, \tag{15.4}$$

relative to sign counting measure and angular Haar measure. Every admitted finite-strength evidence transaction appends its actual outcome and leaves its likelihood in the objective. Write $H_i = \sum ro$ and $\Lambda_i = \sum \lambda e_\omega$ for its persistent coefficients, retaining the entire ordered word as well. A passive marked frame change rotates the local embeddings of these fixed lab labels and the evaluator direction. There is one marked structure group $G = S^1$ with trivial representation on the presentation sector. The common rotation of every ψ_i is a separate action, outside the marked structure group. It is a symmetry of every constitutive datum: the self evaluator and the likelihood do not read ψ, the peer channels copy it, and the rows read only $B_j \cdot M(B_i)$ and $|B_j|$. The distinction it relates is therefore gauge relative to every observable algebra declared in this chapter (definition 1.5). Only the child-level law diagnostic $\mathbb{E}e^{\mathrm{i}\Psi_i}$, which no agent, channel, record or admitted query reads, resolves it.

Proposition 15.1 (The exact transmitted-angle field). On the positive family eq. (15.1), put $m_i = \tanh(L_i/2)$ and $t_i = \mathbb{E}_{q_i}[Z \cos \Phi]$. The content Gibbs weights and own-block Fisher equations are

$$\begin{aligned} w_{ii} &= \frac{e^{Jt_i}}{5I_0(J)}, \qquad w_{ij} = \frac{2e^{L_j m_i/2 + B_j \cdot M(B_i)}}{5\cosh(L_j/2) I_0(|B_j|)}, \\ \beta_{ij} &= w_{ij} / \sum_r w_{ir}, \qquad j \neq i, \end{aligned} \tag{15.5}$$

$$\begin{aligned} \dot{l}_i &= -l_i + \sum_{j \neq i} \beta_{ij} L_j + H_i, \quad \dot{A}_i = -A_i + \Lambda_i, \\ \dot{C}_i &= -C_i + \beta_{ii} J e_1, \qquad \dot{B}_i = -B_i + \sum_{j \neq i} \beta_{ij} B_j, \\ \dot{n}_i &= -n_i + \sum_{j \neq i} \gamma_{ij} n_j. \end{aligned} \tag{15.6}$$

Here γ is the original presentation Gibbs row. Every equation uses the original inference time.

Proof. Let $S_i = \mathbb{E}_{q_i} \log q_i$. The self cost is $S_i - Jt_i + \log I_0(J)$ and the peer cost is $S_i - L_j m_i/2 + \log\cosh(L_j/2) - B_j \cdot M(B_i) + h_0(B_j)$. Their common entropy cancels from the content row, giving eq. (15.5). The Gibbs envelope removes derivatives of the optimizing row from the own-objective first variation. The centered logarithmic Fisher velocity is a linear combination of $z, e_\phi, ze_\phi, e_\psi$. Matching these coefficients gives eq. (15.6). Their covariance is positive definite: a linear combination

constant under a positive density has zero nonconstant Fourier and sign coefficients. This gives tangency of the chart field to the family. The evaluator and likelihood have bounded log densities, the logarithm of the peer output density $E_\lambda[e^{f_j} \mid Z, \Psi]/E_\lambda e^{f_j}$ is a 2-Lipschitz function of the sender's log density f_j in the supremum norm, and the Gibbs rows are smooth in bounded costs, so the sufficient conditions stated after definition 4.21 hold and the chart flow has unique solutions. The family is therefore invariant, and eq. (15.6) is its exact Fisher velocity, without a Euclidean approximation or source-role forces from other agents' objectives. □

With the initial record pattern $(+,+,-)$ held fixed, this model has the grouping of theorem 14.6, for the selector family S_c of eq. (14.15) applied to the public odds L_i and the same constant sets. Throughout, $0 < \epsilon \le 1/100$ is below the threshold of that theorem, $|A_i(0)| \le 1/20$, $|C_i(0)| \le \epsilon^3$ and $\delta \max_i |b_i| \le 1/20$, so that the null box below contains the initial set. To see why the transmitted angle does not invalidate the argument, take $B_i = \delta b_i$ with b_i in a compact pointed cone and nonzero. Relative to the common term, the extra peer log weight is $-D_{\mathrm{KL}}(\mathrm{vM}(B_i)\|\mathrm{vM}(B_j))$ and the self log weight gains $-D_{\mathrm{KL}}(\mathrm{vM}(B_i)\|H)$, so the new term in the row logarithm is $O_{C^1}(\delta^2)$ on bounded scaled charts. With $u = l/\epsilon$ and C/ϵ^3 scaled, the public limiting field is

$$\dot u = \left(-\frac{7}{5}I + \frac{2}{5}\mathbf{1}\mathbf{1}^\top\right) u + (1,1,-1)^\top + O_{C^1}(\epsilon^2+\delta^2), \qquad u_* = (15/7, 15/7, 5/7). \tag{15.7}$$

The remaining scaled blocks have a symmetric negative-definite limiting derivative. For a compact constant set $\mathsf{C} \subset (1/8, 1/3)$, a sufficiently small attraction ball retains positive public signs and pairwise magnitude ratios above $\max \mathsf{C}$, since the smallest equilibrium ratio is $1/3$. Initial u near $(4, 1/2, -1)$ has strict singleton margins. Finite-time dependence followed by the attracting-ball estimate gives a common entrance time T and persistence with that initial episode fixed.

The label-preserving null deletes nonself costs while retaining their source labels, priors and row entropy. Its surviving self weight is not one. On $|l_i| \le 12\epsilon$, $|A_i| \le 1/10$, $|C_i| \le 2\epsilon^3$, $|B_i| \le 1/10$, the self KL bound of eq. (14.31) still gives $1/9 \le \beta^0_{ii} \le 1/5$. Its equations include $\dot l^0_i = -\beta^0_{ii} l^0_i + \epsilon o_i$ and $\dot B^0_i = -\beta^0_{ii} B^0_i$. The signed strip $1/4 \le o_i l^0_i/\epsilon \le 12$ is invariant, while $|L_i - l_i| < \epsilon/8$. The renormalized null has self weight one and the same equations with $\beta^0_{ii} = 1$, and the same bounds hold along it. Thus the third public sign remains negative under both nulls and the full block is never selected. For constants in $(1/8, 1/3)$, (F5) therefore holds against both nulls. For constants in $(1/3, 1)$ the proof of proposition 14.4 applies with a remainder of order $\epsilon^2 + \delta^2$: the full flow and both nulls select only the pair $\{1, 2\}$, and (F5) fails. For constants in $(1/8, 1/3)$ the frozen-environment control reproduces the full partition sequence, and under every constant the full flow, but neither null, reverses agent 3's public sign against its record. Sample the same state graph at fixed census times $0, T$, retaining both partition labels and times. Strict uniform margins make those outputs constant on the chosen contextual patch. No continuously reported parameter-dependent crossing time is inserted into a differentiable statistical experiment.

15.2 A collective rather than primitive quotient

Declare the protocol before the handoff. It admits the public, marked-lab and presentation queries, original normalized evidence transactions and source-label readouts, and relative-angle queries on the complete graph. A joint relative query draws each involved primitive angle once and

returns all requested differences. Shared vertices share samples; triangle outputs obey their cycle relation. Every transaction draws fresh primitive samples, and no sample is reused by a later transaction. A diagnostic appends history without becoming an assimilated likelihood. Policies are Borel kernels, independent of context, from the complete history, fixed lab data and optional independent randomization to admitted actions and waiting times, and every record density is jointly Borel in the outcome and the state (definition A.3). For the collective geometry the expected number of transactions in $[0,t]$ is bounded on compact contextual patches; without this bound the history experiment can fail to be differentiable in quadratic mean (eq. (3.1)). No absolute ψ response is admitted at any stage. In the terms of definition 6.17, A_I consists of these queries and transactions addressed to a block through channels that read it as a whole, and B_I of the same queries applied to individual members, or to sets of them, before the handoff, together with every channel, evaluator and record kernel; neither depends on the retention.

For $\zeta_i = [I_1(|B_i|)/I_0(|B_i|)]e^{\mathrm{i}\arg B_i}$, the pair moment is

$$c_{ij} = \mathbb{E}e^{\mathrm{i}(\Psi_i - \Psi_j)} = \zeta_i\overline{\zeta_j}, \qquad |\zeta_i| = \sqrt{\frac{|c_{ij}||c_{ik}|}{|c_{jk}|}}. \tag{15.8}$$

The function $I_1(b)/I_0(b)$ has strictly positive derivative $\mathrm{Var}_b(\cos\Psi)$ for $b>0$. Thus the three pair laws identify the tuple up to one common rotation. With an exterior nonzero ζ_j fixed, $c_{ij}/\overline{\zeta_j}$ identifies every interior vector. Public and marked law queries identify l, A, C, and presentation queries identify n. The same erasure cannot therefore be performed independently on a proper block with its boundary retained. This is identification from exact response laws, not estimation from finitely many samples. The common symmetry preexists grouping; no interaction-created invisibility is asserted.

The whole-block retention erases only the common ψ rotation, and every element of B_I, the joint relative queries included, is invariant under it at every time. That distinction is gauge relative to B_I, so the retention fails (F4) of definition 6.36. Its fibers are orbits of a symmetry of every constitutive datum and admitted kernel, and the separation they carry fades by the bound after eq. (15.9). What the relative queries establish is that no proper block can be compressed with its boundary retained.

15.3 Finite reference memory and closed local jets

Let $x(c) = (l_i, A_i, C_i, B_i, n_i)_i \in \mathbb{R}^{24}$ be a C^2 section on a compact contextual patch, and fix c_0 before retention. At every context the same realized lab history is applied to eq. (15.6). Suppose the initial $B_i(c)$ lie in a common open pointed cone, with its rotational union admitted. For a vector v positive on that cone,

$$v \cdot B_i(c,t) \geq e^{-t} v \cdot B_i(c,0) > 0. \tag{15.9}$$

This follows by variation of constants in the cooperative B equation. Finite evidence words preserve it. The other natural parameters have finite-interval bounds from eq. (15.6), so every finite admitted branch remains regular. The same equation gives an upper bound. Since $\sum_{j\neq i}\beta_{ij} = 1-\beta_{ii}$, an

agent attaining $\max_j |B_j|$ has $\frac{d}{dt}|B_i| \le -\beta_{ii}|B_i|$, so

$$\max_i |B_i(t)| \le \exp\Big(-\int_0^t \min_i \beta_{ii}\Big) \max_i |B_i(0)| \tag{15.10}$$

on every branch, because no admitted likelihood reads ψ. The transmitted sector therefore fades, and with it the relative moments, the anchor S and the information that the common rotation carries: the quotient below erases an orientation of a sector that becomes informationless as $t \to \infty$, and its noninjectivity, like the nonflatness below, is significant only on finite windows.

Write $x_\circ = x(c_0)$, $S = \sum_i B_i(x_\circ)$ and

$$U(S) = \frac{1}{|S|}\begin{pmatrix} S_x & S_y \\ -S_y & S_x \end{pmatrix}, \qquad y = \mathsf{R}_U x(c), \qquad \widehat{x}_\circ = \mathsf{R}_U x_\circ, \tag{15.11}$$

where R_U rotates only the B blocks. This smooth formula uses no angle branch. A whole-section retention applies this one U at every c and removes precisely one context-constant common angle. The local realization below retains only the data required at c and the reference.

Theorem 15.2 (Conditional evaluation-local realization). Retain $j_c^2 y$, the full finite native tuple $\widehat{x}_\circ$, all primitive ports, constitutive data, complete ordered history, census records and persistent likelihood coefficients. If record embeddings vary with context, retain their known two-jets. Give the past-law experiment its actual compatible initialization. These data have an autonomous original-time evolution and determine the exact local two-jet and every admitted future joint response/update kernel at c. They preserve the actual collective Fisher cost on the declared regular experiment. Initialization uses reference-context data and is not a rule from the germ at c alone.

Proof. Let $V(x, c, h)$ denote eq. (15.6), let $V_S = \sum_i V_{B_i}(\widehat{x}_\circ, c_0, h)$, and let $\mathcal{J}$ act by $J_2 = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ on each B block and by zero elsewhere. Set

$$\Omega(\widehat{x}_\circ, h) = \frac{\det(S, V_S)}{|S|^2}, \qquad W(y, c, h; \widehat{x}_\circ) = V(y, c, h) - \Omega \mathcal{J} y. \tag{15.12}$$

Equivariance and $\dot{U} = -\Omega J_2 U$ give

$$\dot{y} = W(y, c, h; \widehat{x}_\circ), \qquad \dot{\widehat{x}}_\circ = V(\widehat{x}_\circ, c_0, h) - \Omega \mathcal{J} \widehat{x}_\circ. \tag{15.13}$$

The canonical anchor stays on $S_y = 0, S_x > 0$. The subtraction is the derivative of the retention, not a separately fitted parent force. This is the exact vector-field intertwining criterion of eq. (6.4), an instance of exact lumping [110]. The anchor $U(S)$ is a slice with phase condition $S_y = 0$ in the method of slices [33, 172], and eq. (15.13) is its in-slice equation with the phase condition evaluated at the reference context, as in eq. (13.35) and eq. (13.40); the cone estimate keeps the flow away from the slice border $S = 0$. What is new here is a common rotation exposed against every proper block, the per-agent field, and evaluation-local jets.

At fixed t, h and prepared section, the reference is held fixed under evaluation differentiation in c. For contextual indices r, s,

$$\begin{aligned} \dot{y}_r &= W_y y_r + W_r, \\ \dot{y}_{rs} &= W_y y_{rs} + W_{yy}[y_r, y_s] + W_{yr} y_s + W_{ys} y_r + W_{rs}. \end{aligned} \tag{15.14}$$

The explicit partials hold $y, \widehat{x}_\circ, h$ fixed. Every state dependence of the Gibbs rows is differentiated. Since the native field reads no neighboring contexts or contextual derivatives, no third jet enters. Evidence adds the same saved likelihood at both evaluations without resetting either law. Diagnostics change only history. For a two-dimensional base the local law jet has $24(1+2+3) = 144$ coordinates, plus 24 reference coordinates before constraints. The full ordered history remains unbounded in length; this is not a fixed-memory claim.

The actual parent laws are the normalized products of eq. (15.1) in the y coordinates, together with their original presentation and Gibbs-label factors. The reference is deterministic internal memory, not an extra independent sample or agent. The common ψ rotation preserves every joint relative response, every other response and its actual next retained state. Equal histories therefore supply the same next action and wait law. Induction over the joint transaction kernels proves all admitted finite adaptive histories, including response/next-state dependence.

For an unconditional history/current experiment, retain an initial local two-jet and initial reference tuple in one canonical frame as a replay seed, together with the full protocol and chronological word. Applying the original normalized kernels defines its actual history law Π_c. A current tuple and one observed word alone do not determine that past probability. Conditioning on the initial public pattern retains its positive probability normalizer. No distribution over law configurations is introduced.

Finally, for fixed t, h, rotating the fresh transmitted-angle samples by $U(S(x_\circ^h(t)))$ is independent of the evaluation coordinate c. The joint map $T(h, Y) = (h, T_h Y)$ preserves Haar measure and commutes with the marked action. Stored marked outcomes are fixed lab labels; their local direction embeddings transform. Hence, for

$$\mathcal{P}_{c,t}(dh, dY) = \Pi_c(dh) Q_{c,t}^h(dY), \qquad \overline{\mathcal{P}}_{c,t} = T_* \mathcal{P}_{c,t}, \tag{15.15}$$

the contextual score and marked orbit score are carried isometrically. For every X, ξ,

$$\overline{\mathcal{C}}(X, \xi) = \mathbb{E}_{\overline{\mathcal{P}}}(\overline{s}_X + \overline{\ell}_\xi)^2 = \mathbb{E}_{\mathcal{P}}(s_X + \ell_\xi)^2 = \mathcal{C}(X, \xi). \tag{15.16}$$

This proves actual-law cost inheritance before minimization, including history and label scores. It gives the same minimizing affine class, isotropy and minimized tensor. On a locally free patch their connection and curvature agree. □

The retained reference contains no inverse rotation: $U(R_\alpha S) R_\alpha = U(S)$, so simultaneous rotation changes none of the canonical local, reference or seed data. It still changes the excluded child absolute-angle diagnostic. The reference retains the phase difference between c and c_0, not the discarded global angle. For $B_i(c) = b_i e_{\chi(c)}$, the actual canonical density contains $b_i e_{\chi(c) - \chi(c_0)}$, and

its ψ-phase information is

$$\sum_i b_i \frac{I_1(b_i)}{I_0(b_i)} (d\chi)^2 > 0. \tag{15.17}$$

Pointwise canonicalization would delete this term. Copying it into a stored tensor would not restore it to a normalized parent law.

A local initial two-jet has an actual law realization: use its degree-two natural-parameter polynomial near the evaluation point and then evolve that preparation by the exact pointwise field, without truncating the evolved density. Its two-jet agrees with the original there throughout the common finite regular interval by eq. (15.14). This supplies actual law germs and their connection curvature, but does not reconstruct the unknown original section away from that point. Compatible initial data over all contexts remain a field. Marked overlap maps preserve this compatibility because the ψ canonicalization commutes with them.

For a common nonflat window take $A_1(c,0) = ae_\chi$, $A_2(c,0) = de_1$, with $a \in [a_-, a_+]$, $0 < a_- < a_+ < 1/20$ and $0 < d < 1/20$ independent of ϵ, δ, and small noncollinear cone vectors $B_i = \delta b_i$ with $\delta \max_i |b_i| \le 1/20$. Keep the initial episode fixed. At the uncanonicalized $\epsilon = \delta = 0$ comparison, put $a_t = e^{-t}a$ and $\mathcal{D}(t) = \sum_{i\neq 1} \mathcal{I}(e^{-t}|A_i(0)|) > 0$, where $\mathcal{I}(x) = xI_1(x)/I_0(x)$; $\mathcal{D}(t)$ is the marked inertia written $B(t)$ in eq. (14.38). The same minimization as in eq. (14.39) gives

$$\begin{aligned}
\mathcal{A}_{0,t} &= -\frac{\mathcal{I}(a_t)}{\mathcal{I}(a_t)+\mathcal{D}(t)} d\chi, \\
\mathcal{F}_{0,t} &= -\frac{\mathcal{D}(t) e^{-t} \mathcal{I}'(a_t)}{[\mathcal{I}(a_t)+\mathcal{D}(t)]^2} da \wedge d\chi, \\
h^{0,t}_{aa} &= e^{-2t} \operatorname{Var}_{a_t}(\cos\Phi) > 0, \qquad h^{0,t}_{\chi\chi} = \frac{\mathcal{I}(a_t)\mathcal{D}(t)}{\mathcal{I}(a_t)+\mathcal{D}(t)} > 0.
\end{aligned} \tag{15.18}$$

The cross coefficient vanishes and $\mathcal{I}'(a_t) > 0$. As in chapter 14, this curvature is prepared in the initial section, is present at $t = 0$ and at zero coupling, and has the same limit along the null flows. Compact finite-window margins persist for sufficiently small positive ϵ, δ: the field is real-analytic in the state and in (ϵ, δ), so the exact finite-time flow and its two-jet depend continuously on those parameters uniformly on a common positive tube [103, Chapter V], and so do the normalized history kernels and Fisher integrals. Apply continuity to the fine laws, since the canonical anchor is singular at $\delta = 0$, and apply eq. (15.16) only at positive δ. This gives the same nonflat rank-two window containing T. Later records preserve inheritance, but need not preserve this grouping or nonflatness.

15.4 Handoff frames and finite preparation generators

Fix the census handoff T and freeze $U_T = U(S(x_\circ(T)))$ once. Then $z(c,t) = \mathsf{R}_{U_T} x(c,t)$ obeys $\dot z = V(z,c,h)$ for $t \ge T$, with no moving-frame term. Its actual costs still satisfy eq. (15.16). Future local evolution needs no live reference, but its initialization is reference-coherent memory, not an instantaneous quotient of $x(c,t)$ alone. Recovering the continuously reanchored representative later requires the reference evolution or its accumulated phase.

Proposition 15.3 (Conditional finite preparation-and-history realization). Suppose the prepared section has a known finite C^2 generator $x_0(c) = f(c; p)$, with $p \in P \subset \mathbb{R}^m$, equivariant under the common ψ rotation and with nonzero reference anchor. Retain its canonical coefficients p_*, the full chronological history and original time. Then the current contextual laws and their two-jets have an exact whole-patch realization using these data, with all the preceding operational and geometric inheritance properties.

Proof. Let $\mathcal{E}_{t,h}(c)$ be the chronological composition of the unique native flows and original fixed-record coefficient appends. Evaluate

$$x_t^h(c; p_*) = \mathcal{E}_{t,h}(c) f(c; p_*). \tag{15.19}$$

Between transactions $\dot{p}_* = 0$, $\dot{h} = 0$ and $\dot{t} = 1$; cached local values satisfy the native field and eq. (15.14). At a transaction append its actual outcome and likelihood, if it is evidence. The original normalized response/update kernels evaluated on eq. (15.19) generate its full path law. Physical equivariance removes exactly the common seed orientation; it is not stored in the canonical seed. Evaluate the same generator at c_0 if a moving representative is wanted. The sample rotation is again independent of evaluation c, proving eq. (15.16). No projection onto a finite basis of evolved sections is used. □

For a nonempty family, take the marked patch used in eq. (15.18), constant initial l_i/ϵ near $(4, 1/2, -1)$, $C_i = 0$, small fixed presentation laws and three context-constant noncollinear b_i in the cone, with their rotational union. These are finite preparation coefficients, relatively open in their parameter manifold. They are not an open set of arbitrary C^2 functions. The same persistent grouping and nonflat window apply. The full word remains retained, and evaluation at a new context may require replay of its entire native history. Finite seed dimension gives neither constant evaluation cost nor bounded record memory.

15.5 A causal local phase memory

The reference frame can be replaced by an accumulated local phase on a restricted preparation family. Keep the same field V, common ψ action R, generator $\mathcal{J}$ and cone domain. At the actual initialization time t_* define, using only the value at the evaluation context,

$$S_{\mathrm{loc}}(x) = \sum_i B_i(x), \qquad \bar{x} = \mathsf{R}_{U(S_{\mathrm{loc}}(x))} x. \tag{15.20}$$

The sum of the B blocks of $\bar{x}$ lies on the positive first axis. Unlike the reference tuple, this slice uses no value at c_0. Define its local angular velocity by

$$\omega_{\mathrm{loc}}(\bar{x}, c, h) = \frac{\det(S_{\mathrm{loc}}(\bar{x}), \sum_i V_{B_i}(\bar{x}, c, h))}{|S_{\mathrm{loc}}(\bar{x})|^2}. \tag{15.21}$$

It uses the actual native Gibbs rows, including their dependence on the local public and marked laws. It is not a function of the sum alone.

Proposition 15.4 (Closed causal reconstruction in the original time). Starting at t_* with $\vartheta = 0$, evolve

$$\dot{\bar{x}} = W_{\mathrm{loc}}(\bar{x}, c, h) := V(\bar{x}, c, h) - \omega_{\mathrm{loc}} \mathcal{J} \bar{x}, \qquad \dot{\vartheta} = \omega_{\mathrm{loc}}. \tag{15.22}$$

The actual parent laws are the normalized native laws at

$$\widetilde{x} = \mathsf{R}_{R_\vartheta} \bar{x}. \tag{15.23}$$

They satisfy $\dot{\widetilde{x}} = V(\widetilde{x}, c, h)$ on every regular finite branch. Their local two-jet, together with that of ϑ, closes at order two. Every original finite adaptive joint response/update history, persistent likelihood and census record is preserved, with a compatible actual preparation and past-law initialization.

Proof. The cone estimate eq. (15.9) keeps S_{loc} nonzero. Differentiating its canonicalizing matrix gives $\dot{U} = -\omega_{\mathrm{loc}} J_2 U$. Physical equivariance therefore gives eq. (15.22). Differentiating the actual readout cancels its rotating-coordinate term:

$$\dot{\widetilde{x}} = \mathsf{R}_{R_\vartheta}(\dot{\bar{x}} + \dot{\vartheta} \mathcal{J} \bar{x}) = \mathsf{R}_{R_\vartheta} V(\bar{x}, c, h) = V(\widetilde{x}, c, h). \tag{15.24}$$

Thus no independently fitted force or time change is used. Uniqueness, with the canonical initial value, gives

$$\widetilde{x}(c, t) = \mathsf{R}_{U(S_{\mathrm{loc}}(x(c, t_*)))} x(c, t). \tag{15.25}$$

The memory may equivalently be an $\mathrm{SO}(2)$ variable satisfying $\dot{g} = \omega_{\mathrm{loc}} J_2 g$, $g(t_*) = 1$, so no global angle branch is required. The pair eq. (15.22) is the in-slice equation and the reconstruction equation of the freezing method with phase condition $S_y = 0$ [22, 171], and $(\bar{x}, \vartheta)$ are coordinates of the native solution started from the pointwise-canonicalized preparation, as eq. (15.25) records. The remaining conclusions follow from equivariance of V and invariance of the transaction kernels under a constant rotation.

For explicit contextual derivatives a, b and state derivative D, the slice jets obey the prolongation eq. (15.14) with W_{loc} in place of W. The phase jets, initialized to zero, obey

$$\begin{aligned} \dot{\vartheta}_a &= D\omega_{\mathrm{loc}} \bar{x}_a + (\omega_{\mathrm{loc}})_a, \\ \dot{\vartheta}_{ab} &= D\omega_{\mathrm{loc}} \bar{x}_{ab} + D^2 \omega_{\mathrm{loc}}[\bar{x}_a, \bar{x}_b] \\ &\quad + D(\omega_{\mathrm{loc}})_a \bar{x}_b + D(\omega_{\mathrm{loc}})_b \bar{x}_a + (\omega_{\mathrm{loc}})_{ab}. \end{aligned} \tag{15.26}$$

All explicit partials hold $\bar{x}, h$ fixed. In contrast to the fixed-reference derivative, the contextual derivatives of ω_{loc} enter. The field has contextual order zero, so no third jet is needed. In each ψ block the actual law derivatives are

$$\begin{aligned} \widetilde{B}_{i,a} &= R_\vartheta(\bar{B}_{i,a} + \vartheta_a J_2 \bar{B}_i), \\ \widetilde{B}_{i,ab} &= R_\vartheta\big(\bar{B}_{i,ab} + \vartheta_a J_2 \bar{B}_{i,b} + \vartheta_b J_2 \bar{B}_{i,a} \\ &\qquad + \vartheta_{ab} J_2 \bar{B}_i - \vartheta_a \vartheta_b \bar{B}_i\big). \end{aligned} \tag{15.27}$$

Every row and embedding derivative is included. Actual slice-valued initial families realize compatible initial jets, and their native evolution realizes these law germs.

At each fixed c, eq. (15.25) is a time-independent common ψ rotation. All original transaction kernels are invariant under it, including the entire joint relative response with one sample per involved primitive vertex. Induction on responses, waits and updates proves equality of every admitted adaptive joint history. Evidence appends its actual persistent likelihood without jumping the law; diagnostics append history without assimilation. The laws, slice and phase memory are continuous at these transactions. Both original census records remain retained. A local canonical preparation seed and full chronological word, or their compatible initialized past-law experiment, supply the unconditional history law. They do not introduce a prior over prepared sections or an independently sampled memory factor. □

The recurrence implements a frozen local initial frame, whereas the earlier handoff construction froze a reference frame. Retain its actual accumulated value at a later census T. Initializing $\vartheta = 0$ afresh at T instead changes eq. (15.25) to the frame at T; its compatibility with the original preparation is not automatic. Neither existing records nor elapsed phase memory may be reset. One finite state per context and its two-jet are still distinct from a finite global encoding of an unknown contextual function, and the complete word has unbounded length.

Corollary 15.5 (Actual geometric inheritance under initial coherence). In addition to proposition 15.4, suppose the actual preparation satisfies

$$\frac{S_{\mathrm{loc}}(x(c,t_*))}{|S_{\mathrm{loc}}(x(c,t_*))|} = e_* \in S^1 \qquad \text{for every } c \in \mathcal{C}, \tag{15.28}$$

where e_* is independent of subsequent history and its absolute angle is not retained. The causal parent preserves the actual complete Fisher cost eq. (15.16), hence the minimizing marked connection classes, isotropy, tensor and curvature. It removes one common ψ orientation after all memory is counted. There is a nonempty family satisfying this condition with the grouping of theorem 14.6, under the constant sets reported there, and a positive-rank nonflat window. The condition is sufficient; it is not shown to be necessary.

Proof. The hypothesis is exactly the condition that makes the sample map independent of context, so the conclusion is that of eq. (15.16) and of eq. (14.37). Every local initialization in eq. (15.20) applies the same matrix U_* under eq. (15.28). The map of current samples in eq. (15.25) is consequently independent of c, t, h, preserves Haar measure, leaves history and numerical lab labels fixed, and commutes with the marked action. Differentiating its normalized joint density proves eq. (15.16) for every contextual tangent and marked comparison velocity, including history and public-pattern normalization scores. The parent samples the laws at $\widetilde{x}$; the phase derivatives in eq. (15.27) therefore enter its actual scores. Sampling only the slice laws and attaching a stored tensor would not give this equality.

Take the marked family of eq. (15.18) with context-constant initial $B_i = \delta b_i$ in the pointed cone, allowing noncollinear b_i. It satisfies eq. (15.28); the finite-window continuity argument after eq. (15.18) then supplies positive rank and nonzero marked curvature for small positive ϵ, δ. The weak initial record episode is retained. The equality of actual costs transfers that same window. Later arbitrary evidence preserves exact dynamical and operational inheritance on each regular finite branch, while it need not preserve grouping or nonflatness.

The initial slice, its canonical replay seed, generated phase difference and complete history are invariant under a constant common ψ rotation of the original preparation. Conversely, equal canonical initial fields under eq. (15.28) determine preparations differing by one context-constant rotation. The erased angle is therefore not restored by memory. Relative probes still expose every proper block against its exterior; no independently invisible primitive factor is added. □

Initial coherence is a preparation restriction, not a consequence of causal updating or an open condition among arbitrary contextual sections. For a general preparation write locally $\alpha_0(c) = \arg S_{\mathrm{loc}}(x(c,t_*))$. The actual sample map now depends on c. At corresponding samples its contextual score is

$$\widetilde{s}_X(T_0Y) = s_X(Y) - (X\alpha_0)\ell_\psi(Y), \tag{15.29}$$

where ℓ_ψ is the score of positive common rotation of the B_i. Indeed $\psi_{\mathrm{parent}} = \psi_{\mathrm{fine}} - \alpha_0$ and $\ell_\psi = -\sum_i \partial_{\psi_i} \log q_i$. Thus operational history equality alone has not preserved the actual geometry; lemma 15.6 below gives the exact defect. With $\alpha(c,t) = \arg S_{\mathrm{loc}}(x(c,t))$ locally,

$$d_c\alpha(c,t) = d_c\alpha_0(c) + d_c \int_{t_*}^{t} \omega_{\mathrm{loc}}(c,s)ds. \tag{15.30}$$

Temporal integration supplies the second term, not the initial contextual one-form. This exact ψ-phase derivative is distinct from the marked minimizing connection, which can remain nonflat. No context-relocation action is part of the fixed protocol.

Lemma 15.6 (History decomposition and phase-field parents). For the experiment eq. (15.15) with an admitted protocol, write s_X^Π for the history score and s_X^Q, ℓ_ξ^Q for the conditional contextual and marked orbit scores given the history. Then

$$\mathcal{C}(X,\xi) = \mathbb{E}_\Pi(s_X^\Pi)^2 + \mathbb{E}_\Pi\mathbb{E}_{Q^H}(s_X^Q + \ell_\xi^Q)^2, \tag{15.31}$$

so the minimizing marked velocity is a ratio of history averages, $-\mathbb{E}_\Pi\mathbb{E}_{Q^H}[s_X^Q\ell_1^Q]/\mathbb{E}_\Pi\mathbb{E}_{Q^H}(\ell_1^Q)^2$, and not an average of branchwise minimizers. Let a parent sample the native laws at the configuration obtained by rotating every B_i by $-\Theta^H(c)$, where Θ^H is smooth in c for each history. It has the same minimizing marked connection class and isotropy as the fine experiment, and its minimized tensor satisfies

$$\bar{h}(X,X) - h(X,X) = \mathbb{E}_H\big[(X\Theta^H)^2 I^H - 2(X\Theta^H)b^H(X)\big], \tag{15.32}$$

with $I^H = \sum_i \mathcal{I}(|B_i^H|)$, $b^H(X) = \sum_i (X \arg B_i^H)\mathcal{I}(|B_i^H|)$ and $\mathcal{I}(x) = xI_1(x)/I_0(x)$.

Proof. Conditional scores have conditional mean zero given the history, so their cross terms with s_X^Π vanish, and the history law has no marked orbit score because stored outcomes are fixed lab labels. This gives eq. (15.31). Every admitted kernel is invariant under a common ψ rotation, so the parent has the history law of the fine experiment, as in eq. (15.36). At corresponding samples its conditional contextual score is $s_X^Q - (X\Theta^H)\ell_\psi$. Conditionally on the history each ψ_i is independent of the marked, public and presentation factors and of the source labels, so ℓ_ψ is orthogonal to every marked orbit score, and the von Mises identities give $\mathbb{E}[s_X^Q\ell_\psi \mid H] = b^H(X)$ and $\mathbb{E}[\ell_\psi^2 \mid H] = I^H$. Expanding the square, the part that depends on ξ is unchanged, which proves the statements about the connection class and isotropy, and the minima differ by eq. (15.32). □

The causal parent is the case $\Theta = \alpha_0$, independent of the history. Its connection class is therefore inherited exactly for every preparation, without the coherence hypothesis, and only its minimized tensor can change. The defect eq. (15.32) vanishes at a given time when $X\alpha_0 = 2b(X)/I$, so the identity does not show that eq. (15.28) is necessary at a single time; whether it is necessary on a time interval in the exact model is not established.

Remark 15.7 (The obstructions refer to the declared marked group). Every datum of this model is invariant under the common ψ rotation, so the group $S^1_{\mathrm{marked}} \times S^1_{\psi}$ acting on (ϕ, ψ) is an admissible declaration for proposition 3.9. With it ℓ_ψ is an orbit score. A context-dependent common rotation changes the fine contextual score by a multiple of ℓ_ψ, which the ψ component of the comparison velocity absorbs: the minimized tensor is unchanged, and only the ψ component of the minimizing velocity shifts by the phase gradient. Under the enlarged group pointwise canonicalization preserves the minimized tensor, and the score defect eq. (15.29), the coherence hypothesis and the obstructions below lose their force. The obstructions are therefore statements about the marked group declared in this chapter, which leaves horizontal a phase field that no agent, channel or admitted query resolves; the construction gives no internal reason to prefer the smaller group.

15.6 Scoped obstructions to stronger locality

The reference dependence cannot be removed by merely storing its sum. For $B_1 = B_3 = be_1$, $B_2 = be_2$, $A = C = 0$, $l_1 = l_3 = 0$ and $l_2 = \varpi$, put

$$\begin{gathered} s = \frac{1}{5I_0(J)}, \quad k = \frac{2}{5I_0(b)}, \quad E = e^{bI_1(b)/I_0(b)}, \\ u = \operatorname{sech}(\varpi/2), \qquad N = s + kE + ku. \end{gathered} \tag{15.33}$$

The exact rows give

$$\Omega = \frac{1}{5}\left(\frac{4ku - 2kE}{N} - \frac{2k}{s+2k}\right), \qquad \frac{d\Omega}{du} = \frac{4ks + 6k^2E}{5N^2} > 0. \tag{15.34}$$

Thus $S = (2b, b)$ stays fixed while its angular velocity changes. Also, for $c \neq c_0$, a section can be changed only near c_0 without changing its germ at c, yet its canonical value at c changes. These facts exclude sum-only closure and germ-only computation of this representative, not all alternative parent realizations.

Definition 15.8 (Rich preparation class). Fix a compact contextual patch, a finite record word and the window of eq. (15.18). The rich class $\mathcal{X}_{\mathrm{rich}}$ consists of the smooth sections x_0 of initial native parameters in $\mathbb{R}^{24}$ whose B_i lie in a common open pointed cone, with its rotational union admitted, and that satisfy the strict census and window margins of eq. (15.18) with the parameter ranges stated there. It is open in the C^2 topology within the smooth sections, and it is closed under $x_0 \mapsto x_0 + \tau\chi P$ for supported polynomial bumps χP of every order and sufficiently small τ, and under $x_0 \mapsto \mathsf{R}_\kappa x_0$ for smooth κ of sufficiently small C^2 norm. The coherent family of corollary 15.5, the generator families of proposition 15.3 and the real-analytic sections are smooth but violate one of the two closure properties.

Theorem 15.9 (Finite differential-order obstruction). Consider sections in $\mathcal{X}_{\mathrm{rich}}$ (definition 15.8), which allows independent finite-jet variations. Suppose the parent's actual law is a smooth rule $\bar{p}(c) = F(c, j_c^k x, h)$ of some finite order. Local native-history summaries may be included if they are themselves finite-jet functions on the fixed-history stratum. There is no separately retained phase-bearing field or reference-context memory. If the actual minimizing marked connection classes and minimized tensors agree for every section in that class, then the actual-law rule is independent of positive-order jets. It cannot also erase the common ψ angle while preserving that geometry.

Proof. For $k \geq 1$ fix a k-jet and vary the next jet in a contextual direction $X \neq 0$. Contraction of symmetric $(k+1)$-tensors with X is onto symmetric k-tensors; a supported polynomial primitive realizes each variation. At the point the parent law, orbit map $\bar{J}$, fine first score and fine minimizing class are fixed. The parent tangent has the form $\bar{u}_X(\tau) = \bar{u}_X(0) + \tau w$. Choose one representative $A_f(X)$ of the common minimizing class. Equality of tensors requires

$$\|\bar{u}_X(0) + \bar{J}A_f(X) + \tau w\|_{\bar{g}}^2 = h_f^{\min}(X, X) \tag{15.35}$$

on an interval. Its quadratic coefficient is $\bar{g}(w, w)$, so positivity forces $w = 0$. Descending in derivative order removes all positive-order dependence of the actual law.

For random histories make the argument in the joint L^2 space. Vary the highest initial derivative while holding lower initial jets fixed. The history law and fine first scores stay fixed, and the highest current variation is transported by the invertible fixed-record native-flow derivative on each regular branch. If the parent generates its own history law from its laws at past transactions, its history score also varies affinely, by a history function $w^{\mathrm{hist}}(H)$ of mean zero; it is orthogonal to the conditional score variation w_H in the joint L^2 space, so the quadratic coefficient is $\mathbb{E}|w^{\mathrm{hist}}|^2 + \mathbb{E}_H\|w_H\|^2$, which must vanish, and both terms vanish. A finite basis of initial variations gives derivative independence almost surely on the actual history experiment. Smooth positive kernels give its full-support continuity extension where those hypotheses hold. No conditional equality is inferred merely from an equality of averaged minima.

An order-zero phase-erasing rule has zero derivative along a common rotation of the B_i. The fine score in that direction has squared norm $\sum_i |B_i| I_1(|B_i|)/I_0(|B_i|) > 0$. It is orthogonal to the marked orbit, which acts on ϕ, and the relative-history and row laws are unchanged. Its fine horizontal norm is therefore positive while the parent norm is zero. Inside a nonflat section patch one may instead hold its marked derivatives fixed and vary that ψ-phase derivative; the positive quadratic term gives the same contradiction. □

This theorem applies to the smooth subset of the admitted C^2 class. It does not assume that every C^2 section has arbitrarily high derivatives. Independent jet variation and equality of the connection classes are necessary hypotheses of the proof: tensor equality alone could hide a changing marked-orbit tangent. The result is not an application of fixed-statistic Fisher contraction to a moving law map; the latter has additional derivative terms, as in proposition 3.15. Score sufficiency for a fixed statistic is a separate statement [164]. Finite reference memory and constrained generator families are outside the theorem.

Proposition 15.10 (One regular finite encoding cannot hold arbitrary profiles). On a C^2-open class of sections, no single smooth state of dimension m can reproduce every direct public response law at every c, if retention and response evaluation are C^1 on finite-dimensional smooth section variations and all finite smooth memory is counted in m.

Proof. On one retained-history stratum choose $m+1$ disjoint contextual neighborhoods and points c_j in them. Perturb l_1 by $\sum_j a_j f_j(c)$ with $f_j(c_j) = 1$ and disjoint supports. At a regular finite time these current perturbations have admitted predecessors by local invertibility of the fixed-record flow. Strict census and window margins persist for small coefficients. The Jacobian of the $m+1$ direct sign probabilities is diagonal with positive entries, since $\Pr(Z = 1) = \mathrm{sigmoid}(L_1)$ and $\partial L_1/\partial l_1 = 1$. Its rank is $m+1$. Factoring through an m-dimensional state would bound the rank by m, a contradiction. □

This is a statement about one encoding of an indexed contextual experiment, not the addition of context-relocation actions to the protocol. It excludes neither one initialized local germ, compatible local fields, a finite generator family, nonfinite memory nor irregular encodings. No construction of this chapter uses such an encoding: each keeps a state per context or a generator restricted by hypothesis. The proposition is a dimension count that marks the boundary of those realizations, not an obstruction to any of them.

Theorem 15.11 (Order-zero obstruction for rotation-invariant value-path retentions). On the preparation class $\mathcal{X}_{\mathrm{rich}}$, let a strictly local causal retention read the entire value path $x(c,s)$, $t_* \le s \le t$, all actual records, persistent evidence, censuses and fixed templates. It reads no contextual derivatives, neighboring or reference contexts, or initialized contextual phase field. Require the complete retained state, including every memory and seed, to be invariant under a constant common ψ rotation, and let the actual parent laws depend only on that state and fixed data. Such a retention cannot universally preserve the actual minimized Fisher tensor. There is no bound on its temporal horizon or memory dimension. The conclusion also holds for a regular stochastic retained-state kernel with this invariance and phase-independent randomization.

Proof. For any smooth time-independent $\kappa(c)$, pointwise equivariance gives

$$x^{\kappa,h}(c,t) = \mathsf{R}_{R_{\kappa(c)}} x^h(c,t), \qquad \Pi_c^\kappa = \Pi_c. \tag{15.36}$$

The second identity follows by coupling the original invariant transaction kernels and policy randomness; records and censuses coincide. At each fixed c the full local value paths differ by one constant rotation. Local rotation-invariant retention therefore gives identical parent law families for all κ, even with the complete temporal path available.

Choose $\kappa(c_*) = 0$ and $X\kappa(c_*) = \lambda$. The fine contextual score becomes $s_X + \lambda \ell_\psi$. The score ℓ_ψ is orthogonal to every marked orbit score: conditional on history the ψ factors are independent of the marked factors, have mean-zero scores, and neither the history nor the rows have a common-phase score. Its joint norm is

$$I_\psi := \|\ell_\psi\|^2 = \mathbb{E}_H \sum_i |B_i^h| \frac{I_1(|B_i^h|)}{I_0(|B_i^h|)} > 0. \tag{15.37}$$

The minimizing marked class at c_* is unchanged. With $s_X^{\mathrm{hor}} = s_X + \ell_{A_f(X)}$, the fine minimized cost is

$$h_f^{\kappa}(X,X) = h_f(X,X) + 2\lambda\mathbb{E}[s_X^{\mathrm{hor}}\ell_\psi] + \lambda^2 I_\psi. \tag{15.38}$$

It cannot be constant on an interval of λ, whereas the parent tensor is. Small supported smooth phase variations stay inside the strict common nonflat-window class and preserve the grouping exactly. This is a joint-experiment score identity; no conditional geometric equality is inferred from an averaged one. □

The unbounded horizon and memory dimension add no information on fixed-record branches. There the native field is an ordinary differential equation with piecewise-constant coefficients, and evidence does not jump the state, so the value path $x(c,s)$, $t_* \le s \le t$, is a smooth function of $(c, x(c,t), h)$. The retentions of the theorem are therefore rotation-invariant functions, possibly randomized, of the current value, the records and fixed data, and the theorem is the order-zero case of theorem 15.9 without its regularity hypothesis on the retention; its content is the pointwise-canonicalization defect of eq. (15.17) and eq. (15.29).

Corollary 15.12 (Smooth bounded-jet causal memories). Suppose a finite-dimensional deterministic memory is initialized smoothly from $c, j_c^k x(t_*)$ and the original initial records. Between transactions let $\dot m = \Gamma(m, j_c^k x, c, h)$; let record updates and the actual-law readout also be smooth functions of these data. Assume one finite order k bounds initialization, evolution and readout on every branch, with no reference seed or separately supplied contextual phase field. On $\mathcal{X}_{\mathrm{rich}}$, such a parent cannot erase the common ψ phase and inherit both the original minimizing marked affine classes and minimized tensors.

Proof. At a fixed finite word h, the native k-jet and memory form a closed smooth ordinary differential system. Its memory value is a smooth function of the initial k-jet. The native fixed-record flow is locally invertible on its reached regular image; record appends do not reset the law. Its jet prolongation is locally invertible as well. Hence the actual parent law has the finite-order form $F_{t,h}(c, j_c^k x(t))$. Apply theorem 15.9, including its joint-L^2 history argument. The chronological law remains initialized from the actual preparation; branchwise inversion does not infer an unconditional past law from a current snapshot or add a reverse-time protocol. □

The coherent preparation family of eq. (15.28) does not permit the independent phase-gradient variations used in these obstructions. Its causal construction therefore coexists with them. The unrestricted question remains inconclusive: an admissible causal source of the initial contextual phase relation outside these classes, or an obstruction covering it, is still needed. Genuinely unbounded contextual-order memory, differently specified infinite-dimensional memory and stochastic mechanisms outside theorem 15.11 are not classified by corollary 15.12. Temporal records do not supply an undeclared contextual calibration channel. The common ψ symmetry remains predeclared; interaction-created invisibility and a new second compressive handoff remain open. All three obstructions concern the minimizing geometry of the declared marked group (remark 15.7).

Open problem 15.13 (Causal local parents on arbitrary preparations). Does there exist a retention that is causal and germ-local at each context, whose complete retained state is invariant under a constant common ψ rotation, whose actual parent laws lie in a declared family, and

which for every $x_0 \in \mathcal{X}_{\mathrm{rich}}$ reproduces every admitted fixed-context joint history together with the minimizing marked connection classes and minimized tensors on the window, either jointly or on each record branch as declared? The answer is negative for finite-order rules (theorem 15.9), for rotation-invariant value-path retentions (theorem 15.11) and for bounded-jet causal memories (corollary 15.12). It is positive with one reference read (theorem 15.2 and the frozen handoff frame) and on the restricted coherent and finite-generator classes (corollary 15.5, proposition 15.3). On real-analytic preparations over a connected patch the germ at one context determines the section, so germ locality is no restriction there, and an obstruction for germ memories must use non-analytic variations.

15.7 Report against the formation standard

The construction concerns the per-agent objective under the chart Gibbs-reduced dynamics on $\{-1, 1\} \times S^1 \times S^1$. Its admitted class is relatively open in the declared invariant family eq. (15.1) with Bernoulli presentation laws, so (F1) holds in the family grade. The selector family is that of theorem 14.6, with persistence as an invariant block relation while the initial records are held fixed, and the rules A_I and B_I are those of the protocol of section 15.2. The whole-block retention with reference memory satisfies (F3) by theorem 15.2. Compression (F4) fails: the retention erases only the common ψ rotation, which is gauge relative to B_I; its fibers are orbits of a symmetry of every datum and admitted kernel, and the separation fades by eq. (15.10). Interaction necessity (F5) holds against both nulls for every compact constant set inside $(1/8, 1/3)$ and fails for constant sets that meet $(1/3, 1)$; the frozen-environment control reproduces the partition sequence, and the selector-free discrepancy is the reversal of agent 3's public sign. There is one selection stage, so (F6a) fails and no counterfactual class is declared; the amplifying grade is not examined. The construction is a single-stage selection with exact inheritance, and it is not a recursive formation.

Chapter 16

Contextual Coupling of Statistical Laws

The collective connection of proposition 3.9 acquires a dynamical role when a contextual alignment cost, minimized over auxiliary transports, is added to the reciprocal law potential. This chapter gives two constructions on supplied contextual bases. On a finite graph, a regular branch of auxiliary alignments produces neighboring-context forces and a locally unique Fisher flow; the reduced cost equals the minimized cost only while that branch remains the global minimizer. On a flat torus, algebraic elimination of the spatial connection gives a shape Dirichlet energy and a coupled parabolic and ordinary differential system. The finite and continuum theorems use explicit regular-family hypotheses. Neither is a refinement theorem, and neither changes the formation results of the preceding chapters.

The evolving laws remain general normalized probability measures within declared finite-dimensional product families. Gaussian laws appear only in explicit examples. The selected potential is the reciprocal objective of chapter 4; its full derivative includes every receiver, source, and channel role. The normalized product experiments below define the Fisher metrics and do not turn this objective into the variational free energy of a fixed normalized global model. Of the three objectives of chapter 4, every theorem here concerns the reciprocal potential, apart from the own-block remark after theorem 16.7, which treats per-agent free energies and proves no dissipation identity for them. No membership selection, handoff, or null-flow comparison occurs, so the formation clauses of chapter 6 are not tested. The finite contextual vertices and the continuum coordinates are separate supplied descriptions of the base.

16.1 Finite contextual graphs

16.1.1 Statistical and contextual data

Let V be a finite set of contexts and E a finite set of unoriented contextual edges. Choose one orientation $e : u \to v$ for each edge, positive vertex masses m_v, and positive edge weights w_e. The graph, masses, weights, agent census, and comparisons below are fixed. They are supplied contextual data. A change in the agent census is outside the evolution defined here.

At each context there are n agents with normalized laws on standard Borel spaces. The label i identifies one agent across all contexts: the edge cost below compares agent i at u with agent i at v. This correspondence is a modeling choice, and the cost is not invariant under relabeling the agents at one context only. For the reciprocal potential studied here impose the product restriction

$$\zeta_{iv}(dk, dm) = q_{iv}(dk)s_{iv}(dm). \tag{16.1}$$

This is an admitted family, not an assertion that unrestricted joint Fisher dynamics preserves independence. Each factor belongs to a finite-dimensional regular statistical manifold. Assume differentiability in quadratic mean with centered square-integrable scores and a smooth positive-definite Fisher tensor. Redundant parameters have been removed. No Gaussian assumption or finite sample alphabet is imposed.

Fixed bimeasurable comparisons S_{iv} place the joint laws in compatible target spaces, preserving the factorization when it is used. Write $p_{iv} = (\mathsf{S}_{iv})_{\#}\zeta_{iv}$ and $p_v = (p_{iv})_{i=1}^n \in \mathcal{N}$. The manifold $\mathcal{N}$ is the declared product family, equipped with the metric of the normalized product experiment

$$P_{p_v} = \bigotimes_{i=1}^{n} p_{iv}, \qquad \mathsf{G}_{p_v}(a,b) = \sum_{i=1}^{n} g^F_{p_{iv}}(a_i, b_i), \qquad g^F_p(a,b) = \mathbb{E}_p[\ell_a \ell_b]. \tag{16.2}$$

The equality follows because the product score is the sum of the individual centered scores, so its cross terms vanish. For eq. (16.1), $g^F_{p_{iv}}$ is the transported sum of the belief and model Fisher metrics. The product experiment specifies the metric; it is not a proposed population generative model whose variational free energy is the reciprocal potential. Unit agent weights are chosen to match the local Fisher flow. Other positive constant weights define a different declared metric and therefore different mobilities.

Let a finite-dimensional Lie group G act smoothly on $\mathcal{N}$ by diagonal pushforward through bimeasurable sample maps. One element acts on both sectors, through their respective representations when those are linear. The admitted family is stable under every element of G, since the transports below are minimized over all of G. The action is isometric: a fixed sample bijection sends a score ℓ_a to $\ell_a \circ g^{-1}$, leaving its squared integral unchanged. Define the fundamental tangent

$$J_p\xi = \left.\frac{d}{dt}\right|_{t=0} \exp(t\xi)\cdot p, \qquad \xi \in \mathfrak{g}. \tag{16.3}$$

The notation $g \cdot p$ means pushforward, not multiplication of density values. A nontrivial continuous action on a finite sample alphabet is not assumed.

An agent-coordinate change g_{iv} and a common target change h_v act as

$$\zeta'_{iv} = (g_{iv})_{\#}\zeta_{iv}, \qquad \mathsf{S}'_{iv} = h_v \mathsf{S}_{iv} g_{iv}^{-1}, \qquad p'_v = h_v \cdot p_v. \tag{16.4}$$

Here h_v is a change of target, acting on laws by pushforward; it is the inverse of the reference change h of chapter 17, which sends reference laws to $(h^{-1})_{\#}$. Formulas of the two chapters agree after h is replaced by h^{-1}. These changes are fixed in evolution time. Comparisons that depend on the evolving laws would require their derivatives in both the metric pullback and the energy variation; they are not used here.

16.1.2 The chosen context-local potential

At context v, fix normalized references r_{iv}, normalized evaluator kernels $G_{iv}(m, dk)$, and the normalized joint references

$$R_{iv}(dk, dm) = r_{iv}(dm) G_{iv}(m, dk). \tag{16.5}$$

Choose finite nonempty source sets, positive normalized prior rows π^q_{iv}, π^s_{iv}, and temperatures $T_q, T_s > 0$. Belief rows β_{iv} and model rows γ_{iv} lie in the interiors of their probability simplexes. For normalized, possibly law-dependent, channels at that same context set

$$\nu^q_{ijv} = q_{jv}\mathcal{K}^q_{ijv}[p_v], \qquad D^q_{ijv} = D_{\mathrm{KL}}(q_{iv}\|\nu^q_{ijv}), \tag{16.6}$$

$$\nu^s_{ijv} = s_{jv}\mathcal{K}^s_{ijv}[p_v], \qquad D^s_{ijv} = D_{\mathrm{KL}}(s_{iv}\|\nu^s_{ijv}). \tag{16.7}$$

The local reciprocal law potential is

$$\begin{aligned}\mathcal{E}_v = &\sum_i D_{\mathrm{KL}}(q_{iv}\otimes s_{iv}\|R_{iv}) \\ &+ \sum_i \left[\sum_j \beta_{ijv}D^q_{ijv} + T_q D_{\mathrm{KL}}(\beta_{iv}\|\pi^q_{iv})\right] \\ &+ \sum_i \left[\sum_j \gamma_{ijv}D^s_{ijv} + T_s D_{\mathrm{KL}}(\gamma_{iv}\|\pi^s_{iv})\right].\end{aligned} \tag{16.8}$$

Assume every term is finite and $\mathcal{E}_v$ is C^2 on the chosen law and row chart; this is all that the local theorem below uses. The hypothesis concerns the sum, not its individual terms, and it is not inherited by other objectives assembled from the same terms (see the own-block remark after theorem 16.7). Differentiability in quadratic mean alone does not imply this KL regularity. It is a separate hypothesis about the family, supports, references, and channels. Evaluators, references, and channel rules are fixed; their declared dependence on the evolving laws is differentiated. Links between agents at one context remain independent directed communication data.

This objective is a direct potential on laws. A fixed normalized global generative model with coordinate-descent recognition, and an agent's own normalized model treating peers as external sources, are different choices. The derivative of eq. (16.8) contains all receiver, source, and channel roles. On a differentiable dominated subfamily with justified differentiation under the integrals,

$$dD_{\mathrm{KL}}(a\|b) = \int \log(a/b)da - \int (a/b)db, \tag{16.9}$$

$$d\nu^q_{ijv} = (dq_{jv})\mathcal{K}^q_{ijv} + q_{jv}D\mathcal{K}^q_{ijv}(dp_v). \tag{16.10}$$

The absent $\int da$ term is zero by normalization. Thus differentiating a source law must include the costs of receivers attending to it. The coordinate derivative of eq. (16.8) is used when a density formula is unavailable.

Under eq. (16.4), carry references, evaluators, and channels with the same sample maps. For a sector channel this means

$$\mathcal{K}'_{ijv}(g_{jv}y, A) = \mathcal{K}_{ijv}(y, g_{iv}^{-1}A). \tag{16.11}$$

Relative entropy is then invariant by change of variables. Any law-dependent constitutive rule is required to obey this relation. Passive covariance does not imply active symmetry of arbitrarily fixed references.

16.1.3 Finite transports and their stationarity equation

For $e : u \to v$, let $U_e \in G$ send target coordinates at u to those at v. The other orientation of this geometric edge uses U_e^{-1}. This convention places no inverse constraint on independently attending agent links. Set

$$c_e(p_u, p_v, U_e) = \frac{1}{2} d_{\mathcal{N}}(p_v, U_e \cdot p_u)^2, \qquad \mathcal{C}(p, U) = \sum_{e \in E} w_e c_e(p_u, p_v, U_e). \tag{16.12}$$

Here $d_{\mathcal{N}}$ is the intrinsic Riemannian distance of the admitted family with metric eq. (16.2), not necessarily the distance in the unrestricted space of all laws. We work where the compared pairs have a unique minimizing geodesic and squared distance is smooth. The edge transports transform by the endpoint law

$$U'_e = h_v U_e h_u^{-1}. \tag{16.13}$$

This is the transformation law of the link variables of lattice gauge theory [199], here attached to contextual edges rather than to agent pairs. Indeed, $U'_e \cdot p'_u = h_v \cdot (U_e \cdot p_u)$, so isometry proves invariance of every c_e. Reversing an edge leaves its cost unchanged because $d(p_v, U_e p_u) = d(p_u, U_e^{-1} p_v)$.

Proposition 16.1 (Auxiliary transport equation and endpoint forces). Write $x = p_v$, $y = p_u$, and $z = U_e \cdot y$. At a smooth stationary transport,

$$\mathsf{G}_z(\operatorname{Log}_z x, J_z \xi) = 0 \quad \text{for every } \xi \in \mathfrak{g}. \tag{16.14}$$

Holding U_e fixed, the two endpoint gradients are

$$\operatorname{grad}_x c_e = -\operatorname{Log}_x(U_e \cdot y), \qquad \operatorname{grad}_y c_e = -\operatorname{Log}_y(U_e^{-1} \cdot x). \tag{16.15}$$

Proof. Let $\eta : [0,1] \to \mathcal{N}$ be the minimizing geodesic from x to z, with constant speed. Its energy is $d(x,z)^2/2$. For an endpoint variation, differentiating its energy and integrating the covariant derivative by parts gives $\langle \dot\eta(1), \delta z\rangle - \langle \dot\eta(0), \delta x\rangle$; the interior term vanishes because $\nabla_t \dot\eta = 0$. Since $\dot\eta(0) = \operatorname{Log}_x z$ and $\dot\eta(1) = -\operatorname{Log}_z x$, this gives both distance gradients. For $U_e(t) = \exp(t\xi) U_e$, the target velocity is $J_z\xi$. Setting its derivative to zero proves eq. (16.14). For the source endpoint, pull the target gradient back by the isometry U_e. Isometries carry minimizing geodesics and their initial velocities to those of the transformed endpoints, giving the second formula in eq. (16.15). □

Equation (16.14) is the exact finite connection equation of this construction. It is generally nonlinear in U_e. It selects comparisons between neighboring collective laws. There is no transport kinetic energy, curvature penalty, or independent evolution equation for U_e. Cyclic products of the selected transports can have holonomy; no flatness constraint has been imposed. Finite group elements on edges are not connection one-forms. Discrete connections and their relation to continuous connection data have a separate geometric theory [134]; here the edge cost and its elimination specify the finite model directly.

Proposition 16.2 (Stationary transports give horizontal endpoint forces). Let ϕ be a C^1 function on a neighborhood of (x, z) in $\mathcal{N} \times \mathcal{N}$ with $\phi(g \cdot x, g \cdot z) = \phi(x, z)$ for g near e. If U_e is a

stationary point of $U \mapsto \phi(x, U \cdot y)$ and $z = U_e \cdot y$, then

$$d_x\phi(x,z)[J_x\xi] = 0 \qquad \text{for every } \xi \in \mathfrak{g}. \tag{16.16}$$

For $\phi = \frac{1}{2}d^2_{\mathcal{N}}$ this reads $\mathsf{G}_x(\mathrm{Log}_x z, J_x\xi) = 0$, and the momentum $\mathsf{G}(\dot\eta, J_\eta\xi)$ vanishes along the whole minimizing geodesic η from x to z. The same statement holds at the source endpoint.

Proof. Differentiating the invariance at $g = \exp(t\xi)$, $t = 0$, gives $d_x\phi[J_x\xi] + d_z\phi[J_z\xi] = 0$. Varying U as $\exp(t\xi)U_e$ moves z with velocity $J_z\xi$ at fixed x, so stationarity is $d_z\phi[J_z\xi] = 0$, and eq. (16.16) follows. For half squared distance $d_x\phi = -\mathsf{G}_x(\mathrm{Log}_x z, \cdot)$. The fields $J\xi$ generate isometries and are therefore Killing, so $\mathsf{G}(\dot\eta, J_\eta\xi)$ is constant along the geodesic η; at $t = 1$ it equals $-\mathsf{G}_z(\mathrm{Log}_z x, J_z\xi)$, which vanishes by eq. (16.14). Pulling the problem back by the isometry U_e gives the source statement. □

Thus stationarity of the transport at one endpoint is equivalent to horizontality of the connecting geodesic. The proof uses only invariance of the edge cost, which is also what makes the remark on divergence edge costs after proposition 16.4 possible.

16.1.4 Elimination on a regular minimizing branch

Hypothesis 16.3 (Regular branch). On an open law domain $\mathcal{O} \subset \mathcal{N}^V$, each edge admits a chosen stationary point U_e^* at a reference configuration. Squared distance is smooth on a neighborhood of that point and those endpoints. In group coordinates a_e near U_e^*, the Hessian $\partial^2_{a_e a_e} c_e$ is positive definite. The local branch is continued only while these conditions hold.

Continuation takes place in the regular critical set

$$\mathcal{R} = \big\{(p,U) \in \mathcal{N}^V \times G^E :\ \partial_{U_e} c_e = 0,\ \partial^2_{U_eU_e} c_e > 0,\ \tfrac{1}{2}d^2_{\mathcal{N}} \text{ smooth near } (p_v, U_e \cdot p_u) \text{ for every } e\big\}, \tag{16.17}$$

not in law space. By the implicit-function theorem $\mathcal{R}$ is an embedded submanifold of dimension $\dim \mathcal{N}^V$ whose projection to $\mathcal{N}^V$ is a local diffeomorphism. That projection need not be injective: a branch continued around a closed loop of laws can return to a different transport, so a branch energy is not in general a single-valued function on a law-space domain (an example follows eq. (16.55)). Write $\mathcal{R}_{\mathrm{glob}} \subset \mathcal{R}$ for the configurations at which every U_e is the unique global minimizer of $c_e(p_u, p_v, \cdot)$ over G.

Proposition 16.4 (Local elimination and envelope). After shrinking $\mathcal{O}$, hypothesis 16.3 determines a unique smooth local minimizing transport branch $U_e^*(p_u, p_v)$ through the specified point. Define its reduced energy by

$$\mathcal{C}_*(p) = \mathcal{C}(p, U^*(p)). \tag{16.18}$$

Then

$$d\mathcal{C}_*[\delta p] = \sum_e w_e \left[-\mathsf{G}_{p_v}(\mathrm{Log}_{p_v}(U_e^* p_u), \delta p_v) - \mathsf{G}_{p_u}(\mathrm{Log}_{p_u}((U_e^*)^{-1} p_v), \delta p_u)\right]. \tag{16.19}$$

Proof. The stationarity map $\partial_{a_e} c_e$ has invertible derivative in a_e. The implicit-function theorem gives a smooth critical branch; continuity of its positive-definite Hessian makes it a strict local

minimum. Differentiate the composite energy. Its transport derivative is zero on the branch, so the term containing dU_e^* vanishes. The two remaining derivatives are eq. (16.15). The smooth implicit-function theorem applies because the metric and action are smooth; C^3 data would already suffice for a C^2 reduced energy and a C^1 vector field. □

A local minimum is not automatically the global infimum over G.

Proposition 16.5 (A global minimizing branch near a nondegenerate pair). Assume either that G is compact, or that $\mathcal{N}$ is complete and the action of G on $\mathcal{N}$ is proper. Let (p_u^0, p_v^0) be an endpoint pair at which $U \mapsto d_{\mathcal{N}}(p_v^0, U \cdot p_u^0)$ has a unique global minimizer U^0, with squared distance smooth near $(p_v^0, U^0 \cdot p_u^0)$ and positive group Hessian there. Then there are a neighborhood W of U^0 on which the implicit branch U_e^* of proposition 16.4 is defined and $\Delta > 0$ such that, whenever

$$d_{\mathcal{N}}(p_u, p_u^0) + d_{\mathcal{N}}(p_v, p_v^0) < \Delta/2, \tag{16.20}$$

every global minimizer of $U \mapsto d_{\mathcal{N}}(p_v, U \cdot p_u)$ exists, lies in W, and equals $U_e^*(p_u, p_v)$.

Proof. Write $f(U) = d_{\mathcal{N}}(p_v, U \cdot p_u)$ and $f_0(U) = d_{\mathcal{N}}(p_v^0, U \cdot p_u^0)$. Isometry and the triangle inequality give, simultaneously for all $U \in G$,

$$|f(U) - f_0(U)| \le d_{\mathcal{N}}(p_v, p_v^0) + d_{\mathcal{N}}(p_u, p_u^0). \tag{16.21}$$

By the implicit-function theorem there is an open neighborhood W of U^0 such that, for all endpoint pairs in some neighborhood of (p_u^0, p_v^0), the branch value $U_e^*(p_u, p_v)$ is the only critical point of f^2 in W. The set $S = \{U : f_0(U) \le f_0(U^0) + 1\}$ is compact: for compact G trivially, and otherwise because closed metric balls of the complete finite-dimensional manifold $\mathcal{N}$ are compact by the Hopf–Rinow theorem [133], and properness makes $\{U : U \cdot p_u^0 \in K\}$ compact for compact K. The continuous function f_0 attains its minimum over the compact set $S \setminus W$ (if it is nonempty), and by uniqueness that minimum exceeds $f_0(U^0)$; together with $f_0 > f_0(U^0) + 1$ off S this gives $\Delta \in (0, 1]$ with $f_0 \ge f_0(U^0) + \Delta$ outside W. Shrink Δ so that the displayed condition also places the pair in the neighborhood chosen for W. Under that condition $f \ge f_0 - \Delta/2 \ge f_0(U^0) + \Delta/2 > f(U^0)$ outside W. The sublevel set $\{f \le f(U^0)\}$ is closed, nonempty, and contained in $\{f_0 \le f_0(U^0) + \Delta\} \subset S$, hence compact, so f attains its minimum, and every minimizer lies in W. A minimizer in W is a critical point of f^2 there, hence equals $U_e^*(p_u, p_v)$. □

Under these hypotheses $\mathcal{R}_{\mathrm{glob}}$ is open in $\mathcal{R}$, and on it $\mathcal{C}_*$ is the actual minimized energy. The hypotheses hold near agreement whenever the action at a common law point p_0 is free and either G is compact or the action is proper on a complete $\mathcal{N}$: at (p_0, p_0) only the identity gives zero distance, and its group Hessian $\mathsf{G}_{p_0}(J_{p_0}\cdot, J_{p_0}\cdot)$ is positive definite because freeness makes J_{p_0} injective. Properness covers noncompact frame groups. The affine group acts transitively and isometrically on the Gaussian family $\mathcal{N}(\mu, \Sigma)$ on $\mathbb{R}^K$, which is therefore complete, and $\mathrm{GL}(K)$ acting by pushforward $(\mu, \Sigma) \mapsto (A\mu, A\Sigma A^T)$ is proper on it, because congruence on positive-definite matrices is a transitive action with compact isotropy $O(K)$. A product of such factors, one per agent and sector, is again complete with a proper diagonal action. These statements are local; they say nothing about distant endpoint pairs.

Freeness at agreement is restrictive for the frame groups of interest. For $\mathrm{GL}(K)$ acting by pushforward on Gaussian laws, the stabilizer of (μ, Σ) is $\{A : A\mu = \mu,\ A\Sigma A^T = \Sigma\}$, which is conjugate to the orthogonal group of $\mathbb{R}^{K-1}$, of dimension $(K-1)(K-2)/2$, when $\mu \neq 0$ (and to $O(K)$ when $\mu = 0$). A collective of n identical Gaussian laws has the same stabilizer, and so does a context occupied by a single agent. For $K \geq 3$ the group Hessian at such a configuration is therefore singular, hypothesis 16.3 fails at every intra-context consensus and at every one-agent context, and the near-agreement statement above does not apply there; for $K = 2$ and $\mu \neq 0$ the stabilizer is finite. Generic collectives of $n \geq 2$ distinct laws have trivial infinitesimal stabilizer, so the theorem below applies to free non-consensus configurations, but it cannot follow a trajectory into a consensus. A flow on the stratum of consensus configurations would be a different dynamics unless the local data are actively invariant under the stabilizer; with passively covariant fixed references the local force at a consensus point need not be tangent to that stratum.

Positive-definite group Hessians exclude infinitesimal stabilizers of the endpoints. With isotropy, the cost is invariant under

$$U_e \longmapsto h_v U_e h_u^{-1}, \qquad h_v \in G_{p_v}, \quad h_u \in G_{p_u}, \tag{16.22}$$

so transport representatives can be nonunique. A quotient version requires a specified smooth law stratum and a smooth bundle of minimizing classes, with positive normal Hessian in a smooth transverse slice. It also requires a smooth local representative and constancy of the cost along each minimizing fiber. On that stratum the same chain-rule proof gives a representative-independent differential. The gradient uses the induced Fisher metric, equivalently the Fisher-orthogonal projection of eq. (16.15) onto the stratum tangent. These are additional hypotheses, not consequences of constant orbit rank. The main theorem below uses hypothesis 16.3 on an open chart and does not assert unrestricted passage through singular or competing-minimum configurations.

Under a fixed gauge change, transport the branch neighborhood and the chosen initial minimizer by eq. (16.13). Isometry identifies the two minimization problems. Local uniqueness then gives

$$U_e^{*\prime}(p'_u, p'_v) = h_v U_e^*(p_u, p_v) h_u^{-1}. \tag{16.23}$$

A numerical branch rule defined by a preferred coordinate angle must satisfy this condition; coordinate-dependent branch switching need not.

On $\mathcal{R}_{\mathrm{glob}}$, with a free proper action, $\min_U d_{\mathcal{N}}(p_v, U \cdot p_u)$ is the distance between the orbits of p_v and p_u, which is the distance d_Q of the quotient $Q = \mathcal{N}/G$ with its submersion metric: horizontal lifts of quotient curves preserve length, and projections of curves do not increase it [154]. Hence

$$\mathcal{C}_*(p) = \tfrac{1}{2} \sum_e w_e d_Q([p_u], [p_v])^2 \tag{16.24}$$

is a weighted graph Dirichlet energy of the shape map $V \to Q$, the finite counterpart of the quotient Dirichlet energy of proposition 16.9. In the planar location examples below, this is the Procrustes distance of shape theory [119]. Off $\mathcal{R}_{\mathrm{glob}}$ a branch energy exceeds the orbit distance, and the identity fails.

Remark 16.6 (Edge costs without a closed-form distance). For Gaussian laws whose mean and covariance both vary, no closed form of the Fisher–Rao distance is known in general [162],

so the cost eq. (16.12) and the forces eq. (16.15) require a geodesic boundary-value problem for every edge. The proofs of propositions 16.2, 16.4 and 16.5 and theorem 16.7 use the squared distance only through smoothness, invariance $c_e(h_u \cdot p_u, h_v \cdot p_v, h_v U h_u^{-1}) = c_e(p_u, p_v, U)$, and a positive group Hessian, except that the Lipschitz bound in the proof of proposition 16.5 is specific to the distance. The local statements therefore hold for a declared smooth G-invariant edge cost, such as $D_{\mathrm{KL}}(p_v \| U \cdot p_u)$ or its symmetrization, which are closed form on Gaussian families and invariant under pushforward; the endpoint forces become minus the divergence gradients instead of geodesic logarithms. Relative entropy is asymmetric, so the edge orientation becomes data unless the symmetrized cost is used. Because the second-order expansion of relative entropy at the diagonal is one half of the Fisher metric, such a cost has the same infinitesimal normal equation eq. (16.40) as the geodesic cost. A global version for such a cost needs its own compactness argument in place of the Lipschitz bound. These are statements about a changed model, and the declared cost is a constitutive choice.

16.1.5 The coupled Fisher field and its local solution

Let b_v denote all rows at v. On their interior simplexes use

$$g_\beta^{\mathrm{row}}(a, b) = \sum_j \frac{a_j b_j}{\beta_j}, \qquad \sum_j a_j = \sum_j b_j = 0, \tag{16.25}$$

and the analogous metric for γ. On law and row configurations define

$$\mathbb{G} = \sum_v m_v \left(\mathsf{G}_{p_v} + \sum_i g_{\beta_{iv}}^{\mathrm{row}} + \sum_i g_{\gamma_{iv}}^{\mathrm{row}} \right), \tag{16.26}$$

$$\mathcal{F}_* = \sum_v m_v \mathcal{E}_v(p_v, b_v) + \lambda \mathcal{C}_*(p), \qquad \lambda \geq 0. \tag{16.27}$$

This is a Fisher metric up to a fixed overall scale. To see this, choose a context label with probabilities m_v / M_0, $M_0 = \sum_v m_v$, then conditionally draw the product experiment eq. (16.2) and independent categorical labels with the rows at that context. Its normalized joint law has Fisher metric $\mathbb{G}/M_0$. The overall scale only fixes the evolution time convention. This auxiliary sampling description does not identify $\mathcal{F}_*$ with a free energy.

For either orientation of an incident contextual edge write $U^*_{v \leftarrow u}$ for the transport into v, using its inverse when necessary, and put

$$B_v(p) = \sum_{e=\{u,v\}} w_e \operatorname{Log}_{p_v}(U^*_{v \leftarrow u} \cdot p_u). \tag{16.28}$$

The natural-gradient equation [3] is

$$\boxed{\dot{p}_v = -\operatorname{grad}_{\mathsf{G}} \mathcal{E}_v + \frac{\lambda}{m_v} B_v(p).} \tag{16.29}$$

In a chart θ_v, if $c_v = \partial_{\theta_v}\mathcal{C}_*$ is a covector, this is

$$\dot{\theta}_v^a = -(\mathsf{G}_v^{-1})^{ab}\left(\partial_{\theta_v^b}\mathcal{E}_v + \frac{\lambda}{m_v}c_{v,b}\right). \tag{16.30}$$

All local derivatives include eq. (16.10). By proposition 16.2, applied to each incident edge, $\mathsf{G}_{p_v}(B_v, J_{p_v}\xi) = 0$ for every ξ: the contextual force is horizontal and exerts no torque, so the orbit component of $\dot{p}_v$ is that of the local reciprocal force alone. Define the row costs

$$h^q_{ijv} = D^q_{ijv} + T_q\log(\beta_{ijv}/\pi^q_{ijv}), \qquad \bar{h}^q_{iv} = \sum_j \beta_{ijv}h^q_{ijv}, \tag{16.31}$$

$$h^s_{ijv} = D^s_{ijv} + T_s\log(\gamma_{ijv}/\pi^s_{ijv}), \qquad \bar{h}^s_{iv} = \sum_j \gamma_{ijv}h^s_{ijv}. \tag{16.32}$$

Their equations remain

$$\dot{\beta}_{ijv} = -\beta_{ijv}(h^q_{ijv} - \bar{h}^q_{iv}), \qquad \dot{\gamma}_{ijv} = -\gamma_{ijv}(h^s_{ijv} - \bar{h}^s_{iv}). \tag{16.33}$$

The constant $+T_a$ in an unconstrained derivative cancels under centering. To derive the formula, pair its right side with any mass-zero row tangent using the row metric; the result is the negative directional derivative. The factor m_v cancels between the local energy and its metric. A singleton row has no tangent variables and remains one. The contextual cost has no direct row derivative.

Theorem 16.7 (Finite-context local well-posedness and dissipation). Assume the stated smooth positive Fisher metric, C^2 finite local potentials on an open chart with positive rows, and hypothesis 16.3 for every edge. Through each configuration in a sufficiently small such domain, eqs. (16.29) to (16.33) have a unique local solution, continuously differentiable in the initial configuration. Lifted together with its transports, it extends uniquely to a maximal solution $t \mapsto (p(t), U^*(t), b(t))$ whose transport part stays in the regular critical set eq. (16.17). Along it,

$$\frac{d\mathcal{F}_*}{dt} = -\sum_v m_v\Bigg[\left\|\operatorname{grad}_{\mathsf{G}}\mathcal{E}_v - \frac{\lambda}{m_v}B_v\right\|^2_{\mathsf{G}} + \sum_{ij}\beta_{ijv}(h^q_{ijv} - \bar{h}^q_{iv})^2 + \sum_{ij}\gamma_{ijv}(h^s_{ijv} - \bar{h}^s_{iv})^2\Bigg]. \tag{16.34}$$

Fixed gauge re-expression with transported constitutive data sends this solution to the solution of the re-expressed system. At $\lambda = 0$ the equations recover exactly the restricted-family flows (definition 4.10) of the context-local reciprocal potentials on the declared families. The reduced energy $\mathcal{C}_*$ along the solution is the minimized contextual cost only while $(p(t), U^*(t))$ lies in $\mathcal{R}_{\mathrm{glob}}$.

Proof. Use ordinary coordinates for the law families and the first $r-1$ entries of each interior row simplex. The Fisher matrices and their inverses are smooth and positive definite. Implicit elimination makes the branch energy $\mathcal{C}_*$ smooth, because the metric and action are smooth, and the local potentials are C^2; hence the metric gradient is C^1 and locally Lipschitz. Locally $\mathcal{R}$ is the graph of the smooth branch, so the field and its solutions may be written either in law coordinates or on $\mathcal{R}$. For completeness, choose a closed coordinate ball inside the regular domain,

on which the vector field f is bounded by K and has Lipschitz constant L. On a sufficiently short interval, the map $x(\cdot) \mapsto x_0 + \int_0^{\cdot} f(x(s))ds$ maps the closed continuous-path ball into itself and contracts by a factor at most $Lt < 1$. Its fixed point is a unique differentiable solution. The integral inequality for two solutions and Gronwall's inequality give continuous dependence. Differentiating this integral equation gives the variational equation and the corresponding differentiable dependence. Compatible local solutions on $\mathcal{R}$ concatenate by uniqueness, giving the maximal solution; the transport is continued along the solution, not read off from the current laws, which matters because the projection of $\mathcal{R}$ to law space need not be injective. Interior normalization is built into the coordinates; the displayed row equations also have zero sum.

The envelope identity shows that the full vector field is $-\operatorname{grad}_{\mathbb{G}} \mathcal{F}_*$. Pairing it with $d\mathcal{F}_*$ gives minus its squared metric norm. Substitution of the law and row components gives eq. (16.34). Metric isometry, invariance of the transformed local potential, and eq. (16.23) imply covariance of the gradient. Uniqueness then gives covariance of trajectories. Setting $\lambda = 0$ deletes precisely B_v and leaves eq. (16.33) unchanged. □

The lifted solution can be continued whenever (p, U^*, b) remains in a compact subset of $\mathcal{R}$ times the interior row simplexes, with compactness taken in the transports as well as the laws. A finite endpoint therefore requires leaving every such compact subset: possibilities include a statistical-family boundary, a row approaching zero, unbounded coordinates, loss of KL regularity, a geodesic cut locus, or degeneration of the selected branch. Energy decrease alone excludes none of these in general. Leaving $\mathcal{R}_{\mathrm{glob}}$ while staying in $\mathcal{R}$ is not an endpoint: the solution continues, but it then follows a branch that is no longer the global minimizer, and nothing in the equations detects the crossing. At $\lambda = 0$ total-energy stationarity places no condition on U at all. The law and row ODE is still unique; a separately chosen diagnostic transport does not become dynamically determined. Time-dependent frame changes add their coordinate-velocity term and are not covered by fixed-gauge covariance.

A coordinate-chart boundary alone can be crossed in another regular chart. Row positivity has a sharper criterion: for two entries of the same row,

$$\frac{d}{dt} \log \frac{\beta_j}{\beta_k} = -(D_j - D_k) - T_q \left(\log \frac{\beta_j}{\beta_k} - \log \frac{\pi_j}{\pi_k} \right). \tag{16.35}$$

Variation of constants bounds these log ratios on every finite interval with bounded costs, giving a positive row floor there. Thus a row cannot reach zero in finite time while all its costs remain bounded. This does not prevent the other regular-domain exits.

If the rows are instead minimized instantaneously, finite positive costs give

$$\beta^*_{ijv} = \frac{\pi^q_{ijv} e^{-D^q_{ijv}/T_q}}{Z^q_{iv}}, \qquad Z^q_{iv} = \sum_j \pi^q_{ijv} e^{-D^q_{ijv}/T_q}, \tag{16.36}$$

and analogously for γ^*. Replacing their row energies by $-T_q \log Z^q_{iv}$ and $-T_s \log Z^s_{iv}$ gives a different, reduced law-only flow. Direct differentiation yields $-T_q d \log Z^q_{iv} = \sum_j \beta^*_{ijv} dD^q_{ijv}$; there is no extra row-response term. The theorem applies to that smooth reduced potential, with the row dissipation terms absent. The main result above retains the rows as evolving variables.

The per-agent objective of chapter 4 is a different construction. For own-block per-agent free energies $\mathcal{F}_{iv}$, including their persistent record terms, define $A_{iv} = \mathrm{grad}_{iv} \mathcal{F}_{iv}$ with peers held fixed. If each $\mathcal{F}_{iv}$ is C^2 jointly in all laws and interior rows of its context on the chosen chart, adding the contextual force gives a locally Lipschitz, hence locally well-posed, vector field. That hypothesis is not implied by the regularity of the reciprocal potential assumed above: a C^2 or even smooth sum of costs can have own-block gradients that are not locally Lipschitz. For instance, with $q_i = \mathcal{N}(m_i, 1)$ and law-dependent translation channels $\nu_{12} = \mathcal{N}(m_2 + \sigma, 1)$, $\nu_{21} = \mathcal{N}(m_1 + \sigma, 1)$, $\sigma = |m_1|^{7/4}$, the sum $D_{12} + D_{21} = (m_1 - m_2)^2 + |m_1|^{7/2}$ is C^3, while $\partial_{m_1} D_{12} = (m_1 - m_2 - \sigma)(1 - \sigma')$ contains $\sigma' \propto |m_1|^{3/4}$, which is not Lipschitz at $m_1 = 0$. Even when it is well posed, the own-block field need not be a gradient of $\sum_{iv} \mathcal{F}_{iv} + \lambda \mathcal{C}_*$. If R denotes the omitted source-role and channel derivatives in that sum, its gradient is $A + \lambda \,\mathrm{grad}\, \mathcal{C}_* + R$ (with the chosen metric weights). Along the proposed own-block field the derivative is

$$-\|A + \lambda \,\mathrm{grad}\, \mathcal{C}_*\|^2 - \langle R, A + \lambda \,\mathrm{grad}\, \mathcal{C}_* \rangle. \tag{16.37}$$

The second term has no fixed sign. Thus eq. (16.34) belongs to the reciprocal gradient construction.

16.1.6 Relation to the infinitesimal connection equation

Take a smooth local section $p(c)$ in a common frame, a base tangent X, and a displacement ϵX. Suppose the near-identity minimizing transport branch satisfies $U^*(0) = e$ and is smooth. Put

$$U^*(\epsilon) = \exp[-\epsilon A(X) + O(\epsilon^2)], \qquad p(c + \epsilon X) = p(c) + \epsilon dp(X) + O(\epsilon^2) \tag{16.38}$$

in local charts. The initial geodesic residual, identified to $T_{p(c)}\mathcal{N}$ by any smooth local trivialization, is

$$\mathrm{Log}_{U^*(\epsilon)p(c)}\, p(c + \epsilon X) = \epsilon\big(dp(X) + J_p A(X)\big) + O(\epsilon^2). \tag{16.39}$$

Dividing eq. (16.14) by ϵ and taking its first-order coefficient gives

$$\boxed{\mathfrak{k}_p(A(X), \xi) = -\mathsf{G}_p(dp(X), J_p\xi), \qquad \mathfrak{k}_p(\eta, \xi) = \mathsf{G}_p(J_p\eta, J_p\xi).} \tag{16.40}$$

If J_p is injective, $\mathfrak{k}_p$ is positive definite. At coincident endpoints the group Hessian of c_e is exactly $\mathfrak{k}_p$, so the near-identity branch and its smooth dependence follow from the same implicit-function argument. Differentiating stationarity gives eq. (16.40), and its solution is linear in X. With a kernel, solutions form an affine class modulo $\ker J_p$ under the corresponding constant-rank hypotheses. The residual is the Fisher-orthogonal projection of $dp(X)$ off the orbit directions. This is the mechanical-connection construction expressed with a Fisher metric; the separation of rotation and shape in mechanics provides its classical antecedent [139].

Expanding the finite edge cost gives

$$c_e = \frac{\epsilon^2}{2}\|dp(X) + J_p A(X)\|^2_{\mathsf{G}} + O(\epsilon^3). \tag{16.41}$$

Under a smooth base-dependent frame change, the endpoint law expands as

$$U'(c+\epsilon X,c)=h(c+\epsilon X)U(c+\epsilon X,c)h(c)^{-1}, \tag{16.42}$$

$$A'(X)=\mathrm{Ad}_h\,A(X)-dh(X)h^{-1}. \tag{16.43}$$

These are local consistency statements, not convergence of a refining graph or existence for a continuum equation. The familiar candidate continuum energy

$$\frac{1}{2}\int \gamma^{\mu\nu}\mathsf{G}(D^A_\mu p, D^A_\nu p)\mathrm{dvol}_\gamma \tag{16.44}$$

requires a supplied base metric, measure, functional domain, and boundary conditions. Finite graph weights do not supply those choices or their analytic estimates.

16.1.7 An explicit two-context witness

Use two contexts of unit mass, one edge of unit weight, and two agents. Both belief and model sample spaces are $\mathbb{R}^2$. At each context let

$$q_i=\mathcal{N}(\mu_i,I_2),\quad s_i=\mathcal{N}(\nu_i,I_2),\quad r_i=\mathcal{N}(0,I_2),\quad G_i(m,dk)=\mathcal{N}(m,I_2)(dk). \tag{16.45}$$

The evaluator depends on the model presentation, so both sectors contribute to the local dynamics. Direct integration gives

$$D_{\mathrm{KL}}(q_i\otimes s_i\|r_iG_i)=\frac{1}{2}\|\mu_i-\nu_i\|^2+\frac{1}{2}\|\nu_i\|^2+1. \tag{16.46}$$

Give each agent the other agent as its singleton source in each sector, with an identity channel. Rows and their priors equal one. The two directed belief divergences sum to $\|\mu_1-\mu_2\|^2$, and likewise for models. Thus

$$\mathcal{E}(X)=\sum_{i=1}^{2}\left[\frac{1}{2}\|\mu_i-\nu_i\|^2+\frac{1}{2}\|\nu_i\|^2+1\right]+\|\mu_1-\mu_2\|^2+\|\nu_1-\nu_2\|^2, \tag{16.47}$$

where $X=(\mu_1,\nu_1,\mu_2,\nu_2)\in\mathbb{R}^{2\times4}$. The Fisher metric is the Frobenius inner product on these means. It is the intrinsic metric of the fixed-covariance location family; that family is not totally geodesic in the full Gaussian family, so $d_{\mathcal{N}}$ here is not the Fisher–Rao distance of unrestricted Gaussian laws. A single $SO(2)$ rotation acts on every column, in both sectors. References, evaluator, and channels obey this symmetry.

For a source matrix X and a target matrix Y, write

$$J=\begin{pmatrix}0&-1\\1&0\end{pmatrix},\quad R_\theta=\cos\theta I+\sin\theta J,\quad C=\langle Y,X\rangle_F,\quad S=\langle Y,JX\rangle_F,\quad \rho=\sqrt{C^2+S^2}. \tag{16.48}$$

Then

$$c(X,Y,R_\theta)=\tfrac{1}{2}(\|X\|_F^2+\|Y\|_F^2)-C\cos\theta-S\sin\theta. \tag{16.49}$$

For $\rho > 0$ the unique minimizing group element and reduced energy are

$$R_* = (CI + SJ)/\rho, \qquad c_* = \tfrac{1}{2}(\|X\|_F^2 + \|Y\|_F^2) - \rho. \tag{16.50}$$

Stationarity is $C\sin\theta - S\cos\theta = 0$; the Hessian at this minimum is $\rho > 0$. The two envelope gradients are

$$\nabla_X c_* = X - R_*^T Y, \qquad \nabla_Y c_* = Y - R_* X. \tag{16.51}$$

The formula for R_* is smooth on $\rho > 0$, regardless of any angle-coordinate branch cut. This is the planar case of the orthogonal Procrustes problem [175], and c_* is half the squared unscaled Procrustes distance of shape theory [119]; the singular set $\rho = 0$ is the classical nonuniqueness of the Procrustes rotation.

Choose the ordered columns

$$X = (e_1, e_1, e_2, e_2), \qquad Y = (2e_1, 3e_1, e_2, 2e_2). \tag{16.52}$$

Here $(C, S, \rho) = (8, 0, 8)$, $R_* = I$, and $c_* = 3$. From eq. (16.47),

$$\nabla\mathcal{E}(X) = \begin{pmatrix} 2 & 3 & -2 & -2 \\ -2 & -2 & 2 & 3 \end{pmatrix}. \tag{16.53}$$

Consequently the source-context velocity is

$$\dot{X} = \begin{pmatrix} -2+\lambda & -3+2\lambda & 2 & 2 \\ 2 & 2 & -2 & -3+\lambda \end{pmatrix}. \tag{16.54}$$

Changing only the neighbor's first column from $2e_1$ to $3e_1$ keeps $R_* = I$ and adds exactly λe_1 to the local $\dot{\mu}_1$. This is a genuine contextual force on top of the reciprocal local flow. All laws, reference joints, and channels in the example are normalized. Gaussianity supplies the explicit formula; theorem 16.7 is a general regular-family result.

Free action does not prevent a branch singularity

Consider two Gaussian location factors with means $x_\epsilon = (e_1, (1+\epsilon)e_2)$ and $y = (e_1, -e_2)$ and the same diagonal rotation group. Both endpoints lie in the free-action region near $\epsilon = 0$. Nevertheless $C = -\epsilon$, $S = 0$, and

$$\min_R \tfrac{1}{2}\|x_\epsilon - Ry\|_F^2 = 2 + \epsilon + \tfrac{1}{2}\epsilon^2 - |\epsilon|. \tag{16.55}$$

For $\epsilon < 0$ the minimizer is I, and for $\epsilon > 0$ it is $-I$; at zero every rotation minimizes. The minimized energy is not differentiable at zero. The statistical metric is nonsingular and there is no geodesic cut locus. Thus constant orbit rank, even freeness, cannot replace the transport Hessian and branch hypotheses. The defect is not only nondifferentiability. At the singular configuration the minimized energy is a minimum of the smooth functions $c(\cdot, \cdot, R)$ over a minimizing set that is all of $SO(2)$; by Danskin's theorem its one-sided directional derivatives are minima over that set of the derivatives of $c(\cdot, \cdot, R)$, whose gradients differ from one R to another. There is no single descent

direction, and uniqueness of a gradient flow of the minimized energy from that point is not available. Hypothesis 16.3 is what restores a differentiable reduced energy and uniqueness.

A regular branch can stop being the minimizer

The branch set $\mathcal{R}$ can carry monodromy even where every hypothesis of hypothesis 16.3 holds. Take one agent per context with belief factor $\mathcal{N}(\mu, I_2)$ and centered model factor $\mathcal{N}(0, \Sigma)$, and let $SO(2)$ rotate both sectors. The Fisher metric is Euclidean on μ and $\frac{1}{2}\operatorname{tr}(\Sigma^{-1}\delta\Sigma\Sigma^{-1}\delta\Sigma)$ on Σ. At agreement $p_u = p_v = (\mu, \Sigma)$ the edge cost, as a function of the rotation angle θ, is the sum of $|\mu|^2(1-\cos\theta)$ and a covariance term of period π that vanishes at $\theta = 0, \pi$. For $\mu = 0.3e_1$ and $\Sigma = \operatorname{diag}(4, 1/4)$ the covariance term has second derivative $\frac{1}{2}\operatorname{tr}\big((\Sigma^{-1}[J,\Sigma])^2\big) = 14.0625$ at $\theta \in \{0, \pi\}$, so $\theta = 0$ and $\theta = \pi$ are both nondegenerate local minima, with costs 0 and $2|\mu|^2 = 0.18$ and group Hessians 14.1525 and 13.9725. Pure common frame variation, here rotation by π, therefore has positive cost on the second branch. Rotate Σ_v through an angle α from 0 to π; this is a closed loop of laws, since rotation by π fixes centered covariances. Along it the covariance term is minimized at $\theta = \alpha$, and the mean term perturbs its gradient by at most $|\mu|^2 = 0.09$ against a Hessian at least 13.9, so the continued branch stays within about 0.01 of α; at $\alpha = \pi$ both gradients vanish at $\theta = \pi$, so the branch started at $\theta = 0$ returns at $\theta = \pi$, with energy 0.18 at the same laws where it started with energy 0. The projection of $\mathcal{R}$ to law space is not injective, and a flow that carries a configuration around such a loop leaves $\mathcal{R}_{\mathrm{glob}}$ while all hypotheses remain in force. For $SO(2)$ acting on location laws alone, the cost is $\mathrm{const} - \rho\cos(\theta - \theta_*)$ and has one local minimum whenever $\rho > 0$; competing nondegenerate minima need a sector with second-harmonic dependence, such as a covariance.

16.1.8 What the contextual energy controls

On $\mathcal{R}_{\mathrm{glob}}$, auxiliary elimination penalizes variation of collective law shape after group alignment, by eq. (16.24), and pure common frame variation has zero cost; on other branches it can have positive cost, as in the preceding example. A single family on which G acts transitively has zero minimized contextual cost wherever alignment is possible; collective laws are needed if the common action otherwise removes every difference. The example retains shape mismatch despite optimizing the common rotation.

With the local potential absent, the contextual flow decreases $\mathcal{C}_*$ by its own negative squared gradient. With the local potential present, only the total dissipation eq. (16.34) is asserted: local forces can increase contextual cost, and individual edge costs need not decrease even under the contextual flow. In a finite graph, smoothing means this aligned law-variation penalty and its gradient, not a continuum regularity theorem.

Independent rows have no contextual penalty in this model. Their local equations can react to changing laws, but this supplies no separate spatial regularity estimate for rows. Independent communication links and frames have no new evolution equation here, and by the horizontality of B_v the frame component of the velocity is driven by the local potential alone. Law-dependent channels inherit whatever regularity follows from their declared constitutive maps, without becoming independent smoothed fields. The contextual graph, comparisons, reference data, and transport

branch selection remain supplied; the laws and rows evolve, and the auxiliary transports follow their algebraic minimizers.

The continuum construction in section 16.2 supplies these data and a local theorem on a flat torus. It does not follow by taking a graph-refinement limit of this finite model. Chapter 17 separately constructs contextual displacement transport from regular law, interface, and inter-agent link data. Its comparison metric is a constitutive choice and need not be the law-Fisher metric minimized here. No formation, recursive inheritance, global convergence, or graph-refinement result follows from the local finite-system theorem.

16.2 A periodic continuum base

We now use x for the contextual coordinate and V for the context-local reciprocal potential. These symbols replace the vertex labels and $\mathcal{E}_v$ of the finite construction. The statistical target is again denoted $\mathcal{N}$ and carries the same unit-weight collective Fisher convention. The following hypotheses and proofs specify the continuum problem directly.

16.2.1 Declared geometry and normalized statistical data

Fix an integer $d \geq 1$ and take the contextual base $\mathcal{C}$ to be the flat torus

$$\mathbb{T}^d = \mathbb{R}^d/(2\pi\mathbb{Z})^d, \qquad \gamma_{\mu\nu} = \delta_{\mu\nu}, \qquad \mathrm{d}\mu_C = (2\pi)^{-d}\mathrm{d}x^1 \cdots \mathrm{d}x^d. \tag{16.56}$$

Every field and constitutive datum is periodic. There is no boundary contribution. The base metric, measure, agent census, and contextual dimension are supplied and independent of evolution time. Work in the trivial principal bundle $\mathbb{T}^d \times G$. This choice excludes the additional patching problem for nontrivial base bundles and avoids implicit boundary conditions.

At each x, agent i has normalized belief and model laws q_i and s_i on standard Borel sample spaces. Impose the admitted product restriction

$$\zeta_i(\mathrm{d}k, \mathrm{d}m) = q_i(\mathrm{d}k)s_i(\mathrm{d}m). \tag{16.57}$$

Each factor belongs to a finite-dimensional regular family: it is differentiable in quadratic mean, its scores are centered and square integrable, and its Fisher tensor is smooth and positive definite after redundant parameters are removed. This is a restriction of the evolving statistical model. Unrestricted joint Fisher dynamics need not preserve the restriction. The sample spaces need not be finite, and the general construction makes no Gaussian assumption.

Fixed bimeasurable comparisons identify the agents' product laws with compatible target spaces. These comparisons are independent of x in the chosen trivialization and of the evolving state. Write $p = (p_i)_i \in \mathcal{N}$ for the collective target configuration. Its metric is the Fisher metric of the normalized product experiment

$$P_p = \bigotimes_{i=1}^{n} p_i, \qquad \mathsf{G}_p(U, W) = \sum_i g^F_{p_i}(U_i, W_i), \qquad g^F_p(U, W) = \mathbb{E}_p[\ell_U \ell_W]. \tag{16.58}$$

The identity follows by adding product scores: centered independent scores have zero cross expectation. For eq. (16.57), each summand is the transported sum of belief and model Fisher tensors. Unit agent weights are part of the declared metric. Other weights give other mobilities. This product experiment specifies a normalized statistical metric; it does not make the reciprocal energy below the variational free energy of a fixed population generative model. Fisher gradient descent follows the metric principle of natural gradients [3].

Let a finite-dimensional Lie group G act on $\mathcal{N}$ by smooth diagonal pushforward. One group element acts on both sectors, through their respective sample actions. Fixed sample bijections preserve score integrals, so assume the admitted family is stable and the induced action is isometric. Define

$$J_p\xi = \frac{\mathrm{d}}{\mathrm{d}a}\bigg|_{a=0} \exp(a\xi)\cdot p, \qquad \xi \in \mathfrak{g}. \tag{16.59}$$

No nontrivial continuous action on a finite alphabet is presumed.

Hypothesis 16.8 (Regular quotient and one adapted chart). There is a G-invariant open region $\mathcal{N}_0 \subset \mathcal{N}$ on which the action is free and proper. Its smooth quotient $Q = \mathcal{N}_0/G$ has the submersion metric induced by G. The initial field lies compactly inside one adapted product coordinate chart

$$p = \Psi(u, r), \qquad u \in \mathbb{R}^{n_u}, \quad r \in \mathbb{R}^{n_r}, \quad n_r = \dim G, \tag{16.60}$$

where u coordinates the quotient and r the group fiber. All metric coefficients are smooth on a neighborhood of this chart.

Freeness makes the infinitesimal action injective. A free proper smooth action has a smooth quotient manifold with $\mathcal{N}_0 \to Q$ a principal bundle [132, Theorem 21.10], and the isometric action supplies a positive quotient metric. These are substantive hypotheses. An action with stabilizers, a rank-changing orbit, or an initial image requiring several charts is outside the theorem as stated. The theorem is local in this regular region; it does not infer orbit regularity from normalization alone. Riemannian submersion geometry provides the horizontal/vertical decomposition [154]; rotational shape geometry gives a classical mechanical-connection example [139].

16.2.2 The reciprocal local potential and the field metric

At each x, fix normalized references $r_i(x, \mathrm{d}m)$ and evaluator kernels $G_i(x, m, \mathrm{d}k)$, with normalized joint reference

$$R_i(x, \mathrm{d}k, \mathrm{d}m) = r_i(x, \mathrm{d}m)G_i(x, m, \mathrm{d}k). \tag{16.61}$$

The reference symbol r_i is distinct from the fiber coordinate r in eq. (16.60). Fix finite nonempty source sets, positive normalized prior rows $\pi_i^q(x), \pi_i^s(x)$, and temperatures $T_q, T_s > 0$. The evolving rows β_i, γ_i are in their simplex interiors. Normalized channels may depend on p through fixed smooth constitutive rules:

$$\nu_{ij}^q = q_j \mathcal{K}_{ij}^q[x, p], \qquad D_{ij}^q = D_{\mathrm{KL}}(q_i \| \nu_{ij}^q), \tag{16.62}$$

$$\nu_{ij}^s = s_j \mathcal{K}_{ij}^s[x, p], \qquad D_{ij}^s = D_{\mathrm{KL}}(s_i \| \nu_{ij}^s). \tag{16.63}$$

There is no row dependence in these channel rules. Define

$$\begin{aligned} V(x,p,\beta,\gamma) = &\sum_i D_{\mathrm{KL}}(q_i \otimes s_i \| R_i(x)) \\ &+ \sum_i \left[\sum_j \beta_{ij} D^q_{ij} + T_q D_{\mathrm{KL}}(\beta_i \| \pi^q_i(x)) \right] \\ &+ \sum_i \left[\sum_j \gamma_{ij} D^s_{ij} + T_s D_{\mathrm{KL}}(\gamma_i \| \pi^s_i(x)) \right]. \end{aligned} \tag{16.64}$$

Assume every term is finite and V is smooth jointly in x and the admitted law and row coordinates on the open regular chart of hypothesis 16.8, so that the continuation statement below refers to compact subsets of the domain on which V is smooth. Smooth Fisher geometry alone does not imply smooth relative entropy; support, integrability, and channel regularity are additional hypotheses here. All data are fixed in evolution time. Derivatives of V include every receiver role, source role, and prescribed law dependence of a channel. For example, when a dominated density calculation is justified,

$$\mathrm{d}D_{\mathrm{KL}}(a\|b) = \int \log(a/b)\mathrm{d}a - \int (a/b)\mathrm{d}b, \qquad \mathrm{d}(q_j \mathcal{K}[p]) = (\mathrm{d}q_j)\mathcal{K}[p] + q_j D\mathcal{K}[p](\mathrm{d}p). \tag{16.65}$$

Normalization removes the extra integral of $\mathrm{d}a$. Coordinate differentiation defines the derivative when this density formula is unavailable.

For a row with l entries, use log odds $b_j = \log(\beta_j/\beta_l)$, $1 \le j < l$. Collect all belief and model log odds in b. A singleton row contributes no coordinates. The categorical Fisher matrix in these coordinates is

$$\mathsf{B}_{jk}(b) = \beta_j \delta_{jk} - \beta_j \beta_k, \qquad 1 \le j,k < l, \tag{16.66}$$

with one block per row. It is positive definite at every interior row. Thus the field metric is

$$\mathbb{G}_{p,b}((U,a),(W,c)) = \int_{\mathbb{T}^d} [\mathsf{G}_p(U,W) + a^T \mathsf{B}(b) c] \mathrm{d}\mu_C. \tag{16.67}$$

It is the Fisher metric of the normalized experiment that first samples $x \sim \mu_C$, then independently samples $P_{p(x)}$ and each categorical source row conditional on x. Differentiation under this fixed experiment gives the integral in eq. (16.67). It is not a formal product of uncountably many probability laws.

Arbitrary fixed references need not be actively G-invariant. In adapted coordinates, $V(x,u,r,b)$ may depend on r. A passive re-expression transforms the reference laws, evaluators, and channels together with the laws. That covariance is distinct from a symmetry obtained while leaving the references fixed.

16.2.3 Auxiliary spatial connection and its elimination

For a Lie-algebra-valued periodic one-form $A = A_\mu \mathrm{d}x^\mu$, set

$$v_\mu = D_\mu^A p = \partial_\mu p + J_p A_\mu, \qquad \mathcal{F}[p, b, A] = \int_{\mathbb{T}^d} \left[V(x, p, b) + \frac{\lambda}{2} \sum_\mu \mathsf{G}_p(v_\mu, v_\mu) \right] \mathrm{d}\mu_C, \tag{16.68}$$

where $\lambda > 0$ is fixed. The connection has no kinetic term, curvature energy, or independent evolution equation. Define the positive locked-inertia form

$$\mathfrak{k}_p(\xi, \eta) = \mathsf{G}_p(J_p\xi, J_p\eta). \tag{16.69}$$

At fixed p the integrand is a strictly convex quadratic in each A_μ. Its unique minimizer satisfies

$$\boxed{\mathfrak{k}_p(A_\mu^*, \xi) = -\mathsf{G}_p(\partial_\mu p, J_p\xi) \quad \text{for all } \xi.} \tag{16.70}$$

Let the mechanical connection $\theta_p : T_p\mathcal{N}_0 \to \mathfrak{g}$ be defined by

$$\mathfrak{k}_p(\theta_p U, \xi) = \mathsf{G}_p(U, J_p\xi), \qquad P_H U = U - J_p\theta_p U. \tag{16.71}$$

Then

$$A^* = -p^*\theta, \qquad D_\mu^{A^*} p = P_H \partial_\mu p. \tag{16.72}$$

The notation $p^*\theta$ denotes pullback along the field $p : \mathbb{T}^d \to \mathcal{N}_0$. Smooth positive inertia makes this elimination smooth on the regular domain, unlike a possibly nonunique finite-distance minimizer over group elements.

In the adapted chart the metric has the form

$$\mathsf{G} = h_{ab}(u)\mathrm{d}u^a\mathrm{d}u^b + k_{\alpha\beta}(u, r)(\mathrm{d}r^\alpha + C_a^\alpha(u, r)\mathrm{d}u^a)(\mathrm{d}r^\beta + C_b^\beta(u, r)\mathrm{d}u^b). \tag{16.73}$$

Here h and k are positive definite. Orthogonal complement gives horizontal coordinate vectors $H_a = \partial_{u^a} - C_a^\alpha \partial_{r^\alpha}$. The quotient metric h is independent of r. The coefficient C represents this horizontal distribution in a fiber coordinate chart; it is not itself the Lie-algebra-valued field A^*.

Proposition 16.9 (Reduced energy). Under hypothesis 16.8, auxiliary minimization gives

$$\boxed{\mathcal{F}_*[u, r, b] = \int_{\mathbb{T}^d} \left[V(x, u, r, b) + \frac{\lambda}{2} h_{ab}(u)\partial_\mu u^a \partial_\mu u^b \right] \mathrm{d}\mu_C.} \tag{16.74}$$

Repeated base and coordinate indices are summed. Its contextual term is the Dirichlet energy of the shape map $u : \mathbb{T}^d \to Q$.

Proof. Equation (16.70) makes every residual orthogonal to the group orbit. The horizontal component of $\partial_\mu p$ is $H_a \partial_\mu u^a$. Pairing with eq. (16.73) gives $\mathsf{G}(P_H\partial_\mu p, P_H\partial_\mu p) = h_{ab}\partial_\mu u^a \partial_\mu u^b$. The quadratic reaches its unique minimum there. Substitution proves the formula. □

The unreduced energy eq. (16.68) is a gauged nonlinear sigma model whose gauge field has no kinetic term, and proposition 16.9 is the classical elimination of such an auxiliary field, which leaves

the sigma model on the quotient. This is the mechanism of the CP^{N-1} model [51] (see also open problem 3.12) and, for general cosets, of hidden local symmetry [13]. What the present construction adds is the Fisher metric of normalized product laws as the target metric, the nonbasic reference force derived below, and the coupled frame and row equations.

Pure common frame variation has zero contextual cost. If the group is transitive on the admitted family, there are no shape variables and the reduced contextual term vanishes. The theorem below includes that degenerate case as a local ODE, but actual contextual smoothing requires retained shape directions.

The absence of frame diffusion below is not specific to this quadratic energy. Let $\mathcal{E}_{\mathrm{ctx}}$ be any differentiable functional of law fields with $\mathcal{E}_{\mathrm{ctx}}[g \cdot p] = \mathcal{E}_{\mathrm{ctx}}[p]$ for every smooth $g : \mathbb{T}^d \to G$. Differentiating along $g = \exp(a\xi)$ for an arbitrary field $\xi : \mathbb{T}^d \to \mathfrak{g}$ gives $d\mathcal{E}_{\mathrm{ctx}}[p](J_p\xi) = 0$, so its Fisher gradient for the field metric eq. (16.67) is horizontal at every point. Added to a pointwise potential V, it leaves the vertical part of the velocity equal to the vertical part of $-\operatorname{grad}_{\mathsf{G}} V$, which contains no derivative in x. Hence no locally gauge-invariant contextual energy, of whatever differential order, produces frame diffusion once the connection is eliminated; frame smoothing requires a gauge-variant term or a connection with its own dynamics. Rows are not acted on by G, so a row term such as $\frac{\lambda_b}{2}\int \sum_\mu \partial_\mu b^T \mathsf{B}(b)\partial_\mu b \mathrm{d}\mu_C$ would be admissible; it is a constitutive choice not made here.

16.2.4 Derivation of the coupled Fisher flow

Let Γ^a_{bc} be the Levi-Civita Christoffel symbols of h, and define the quotient tension

$$\tau_h(u)^a = \Delta u^a + \Gamma^a_{bc}(u)\partial_\mu u^b \partial_\mu u^c. \tag{16.75}$$

For a periodic variation δu, integration by parts in the Dirichlet term gives $-\int h_{ab}\tau_h^b \delta u^a \mathrm{d}\mu_C$. Thus

$$\frac{\delta \mathcal{F}_*}{\delta u} = V_u - \lambda h \tau_h, \qquad \frac{\delta \mathcal{F}_*}{\delta r} = V_r, \qquad \frac{\delta \mathcal{F}_*}{\delta b} = V_b. \tag{16.76}$$

The kinetic norm induced by eq. (16.67) is

$$\|\partial_t(u,r,b)\|^2_{\mathbb{G}} = \int_{\mathbb{T}^d} \left[u_t^T h u_t + (r_t + Cu_t)^T k (r_t + Cu_t) + b_t^T \mathsf{B} b_t\right] \mathrm{d}\mu_C. \tag{16.77}$$

Pairing the negative gradient condition with independent coordinate variations yields

$$hu_t + C^T k(r_t + Cu_t) = -V_u + \lambda h \tau_h, \tag{16.78}$$

$$k(r_t + Cu_t) = -V_r, \qquad \mathsf{B} b_t = -V_b. \tag{16.79}$$

Solving this block system gives the coupled equations

$$\boxed{u_t = \lambda \Delta u + \lambda \Gamma(u)(\partial_\mu u, \partial_\mu u) - h^{-1}(V_u - C^T V_r),} \tag{16.80}$$

$$\boxed{r_t = -C(u,r)u_t - k(u,r)^{-1}V_r, \qquad b_t = -\mathsf{B}(b)^{-1}V_b.} \tag{16.81}$$

The $C^T V_r$ term is required when fixed references give a nonbasic potential. Setting it to zero without $V_r = 0$ would change the Fisher gradient.

For the potential eq. (16.64), the row equation has the explicit form

$$H_{ij}^q = D_{ij}^q + T_q \log(\beta_{ij}/\pi_{ij}^q), \qquad \overline{H}_i^q = \sum_j \beta_{ij} H_{ij}^q, \tag{16.82}$$

$$\partial_t \beta_{ij} = -\beta_{ij}(H_{ij}^q - \overline{H}_i^q), \tag{16.83}$$

and analogously for γ. Equivalently, for each nonreference entry of a row of sector $a \in \{q, s\}$,

$$\partial_t b_j = -(D_j - D_l) - T_a \left(b_j - \log \frac{\pi_j}{\pi_l} \right). \tag{16.84}$$

The context penalty has no direct row derivative. Positive normalized rows are reconstructed by softmax from b. Bounded costs on a finite time interval bound all log odds by variation of constants in eq. (16.84), hence give a positive row floor. There is no row Laplacian or separate spatial regularization estimate.

Intrinsic variation and gauge covariance

At fixed A, set $X_\mu(p) = J_p A_\mu(x)$. The derivative of v_μ in a law variation U is

$$\nabla_U v_\mu = \nabla_\mu U + \nabla_U X_\mu. \tag{16.85}$$

Because the group action is isometric, each X_μ is Killing. Its covariant derivative is skew adjoint. Periodic integration by parts therefore gives the negative gradient of the contextual term as

$$\tau_A(p) = \sum_\mu \left[\nabla_\mu v_\mu + \nabla_{v_\mu} X_\mu \right]. \tag{16.86}$$

The second summand accounts for the state dependence of the fundamental vector field. The envelope derivative from varying $A^*(p, \mathrm{d}p)$ vanishes by eq. (16.70). Since the reduced contextual energy depends only on u, its metric gradient is horizontal. Comparing its first variation with eq. (16.76) gives

$$\tau_{A^*}(p) = H_a \tau_h(u)^a, \qquad p_t = -\operatorname{grad}_{\mathsf{G}} V + \lambda \tau_{A^*}(p). \tag{16.87}$$

The full flow need not be horizontal, since $\operatorname{grad} V$ may have a vertical component.

For a smooth frame re-expression $g : \mathbb{T}^d \to G$ fixed in evolution time,

$$p' = g \cdot p, \qquad A' = \mathrm{Ad}_g A - (\mathrm{d}g)g^{-1}, \qquad D^{A'} p' = g_* D^A p. \tag{16.88}$$

Push references and evaluators forward as well, and transform each sector channel by $\mathcal{K}'_{ij}(g_j y, S) = \mathcal{K}_{ij}(y, g_i^{-1} S)$ in its corresponding agent coordinates. Relative entropy and the Fisher norm are unchanged. The auxiliary minimizer transforms by eq. (16.88), by uniqueness, and the gradient equations are covariant. This statement applies wherever both field descriptions admit the regular charts in the theorem. A time-dependent $g(t, x)$ adds $J_{p'}((\partial_t g)g^{-1})$ to the velocity. An associated temporal connection or modified evolution would be required; it is not part of the present system.

The principal second-order operator in law coordinates is a horizontal projection of the Laplacian. It is degenerate in frame directions. Harmonic-map heat flow supplies the shape operator [70], but a

uniformly parabolic theorem for all law coordinates cannot be invoked. The next proof retains the frame equation coupled to u_t.

A check with a fixed reference that breaks active symmetry

For a unit-covariance Gaussian location law with nonzero mean $p \in \mathbb{R}^2$, let $SO(2)$ rotate the mean. In a local chart choose

$$p = uR_{r+\eta u}e_1, \qquad u > 0, \qquad \mathsf{G} = \mathrm{d}u^2 + u^2(\mathrm{d}r + \eta \mathrm{d}u)^2, \tag{16.89}$$

where η is a fixed real constant. A fixed reference with mean ae_1, $a \neq 0$, gives the local KL term

$$V = \tfrac{1}{2}(u^2 + a^2 - 2au\cos\varphi), \qquad \varphi = r + \eta u. \tag{16.90}$$

Here $V_r = au\sin\varphi$ and $V_u = u - a\cos\varphi + a\eta u\sin\varphi$. With $C = \eta$ and $k = u^2$, the coupled equations are

$$u_t = \lambda\Delta u - u + a\cos\varphi, \qquad r_t = -\eta u_t - \frac{a}{u}\sin\varphi. \tag{16.91}$$

At zero coupling they reconstruct the Cartesian Fisher equation $p_t = -(p - ae_1)$. Omitting $C^T V_r$ would incorrectly change the radial velocity by $-a\eta u\sin\varphi$. This local term illustrates why passive coordinate covariance cannot replace the missing active symmetry assumption.

The same example reaches the singular orbit in finite time. Take $a > 0$, $\varphi \equiv \pi$, and spatially constant $u(0) = u_0 > 0$. Then $\sin\varphi = 0$, so $\varphi_t = r_t + \eta u_t = 0$ and φ stays equal to π, while $u_t = -u - a$ gives $u(t) = -a + (u_0 + a)e^{-t}$. The radius vanishes at $t_* = \log(1 + u_0/a)$, every Sobolev norm of u and r stays bounded, and the energy $\frac{1}{2}(u + a)^2$ decreases; the image leaves every compact subset of $\mathcal{N}_0 = \{p \neq 0\}$. The Cartesian law $p(t) = (a - (u_0 + a)e^{-t})e_1$ passes smoothly through zero, so the endpoint is a breakdown of the quotient description, not of the law flow. This is an instance of the image clause of the continuation statement in theorem 16.10.

16.2.5 Local well-posedness in an explicit Sobolev class

Choose an integer $s > d/2 + 1$ and put $m = s - 1 > d/2$; then $m \geq 1$. Every estimate below uses only $m > d/2$: the algebra property and the embedding $H^m \hookrightarrow C^0$, smooth composition in H^m, and $\partial_\mu u \in H^m$ for $u \in H^s$. A higher threshold buys additional statements, such as C^1 control of the law field, that the theorem does not use. All Sobolev spaces below are periodic, vector valued, and use the normalized measure eq. (16.56). For $z = (u, r, b)$ write $CH^j = C([0,T]; H^j(\mathbb{T}^d))$, and likewise for time-integrated spaces.

Theorem 16.10 (Continuum local existence, uniqueness, and continuation). Assume eqs. (16.56) to (16.64) and hypothesis 16.8, with smooth coefficients and finite smooth V on an open chart. Fix $\lambda > 0$. Suppose

$$u_0 \in H^s(\mathbb{T}^d), \qquad r_0, b_0 \in H^m(\mathbb{T}^d), \tag{16.92}$$

and the pointwise image of z_0 is contained in a compact subset of that regular interior chart. Then there is $T > 0$, depending only on λ, upper bounds for the three data norms, and a compact set containing the image of z_0, and a unique solution of eqs. (16.80) to (16.81), remaining in the chart,

with

$$u \in CH^s \cap L^2(0,T;H^{s+1}) \cap H^1(0,T;H^m), \tag{16.93}$$

$$r, b \in CH^m \cap H^1(0,T;H^m). \tag{16.94}$$

In fact $b \in C^1([0,T];H^m)$. The solution depends locally Lipschitz-continuously on the initial data, in these norms, on a common sufficiently short interval near each admissible datum. It extends uniquely while

$$\|u(t)\|_{H^s} + \|r(t)\|_{H^m} + \|b(t)\|_{H^m} \tag{16.95}$$

stays bounded and its image remains in a fixed compact subset of the regular coordinate domain.

The asymmetric regularity is intentional. Only the shape coordinate is assigned the extra spatial derivative. Smooth initial data satisfy these assumptions; for $d = 1$, one admissible choice is $s = 2$, $m = 1$. For frame and row data only in H^m, the reconstructed law map has H^m regularity, eq. (16.123). The theorem asserts neither instantaneous smoothing of the frame coordinate or rows nor a loss of their regularity: whether H^s frame and row data remain in H^s on the existence interval is not addressed by this proof, whose Picard map uses inputs without time derivatives.

Lemma 16.11 (Heat estimate on the declared base). For $0 < T \le 1$, $v_0 \in H^s$, and $f \in L^2(0,T;H^{s-1})$, the equation

$$v_t - \lambda \Delta v = f, \qquad v(0) = v_0, \tag{16.96}$$

has a unique solution with

$$\|v\|_{CH^s} + \|v\|_{L^2H^{s+1}} + \|v_t\|_{L^2H^{s-1}} \le K_\lambda \left(\|v_0\|_{H^s} + \|f\|_{L^2H^{s-1}} \right). \tag{16.97}$$

The constant is independent of $T \le 1$. The same estimate for a difference with zero initial value omits the initial term. Also,

$$\|(e^{\lambda t \Delta} - I)v_0\|_{H^{s-1}} \le \sqrt{\lambda t}\|v_0\|_{H^s}. \tag{16.98}$$

Proof. For Fourier modes $k \in \mathbb{Z}^d$, solve $\partial_t \widehat{v}_k + \lambda |k|^2 \widehat{v}_k = \widehat{f}_k$. For nonzero modes, multiplication by $(1+|k|^2)^s \overline{\widehat{v}_k}$, summation, and Young's inequality give

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\|P_{\neq 0} v\|_{H^s}^2 + \frac{\lambda}{2}\sum_{k \neq 0} |k|^2 (1+|k|^2)^s |\widehat{v}_k|^2 \le \frac{c}{\lambda}\|f\|_{H^{s-1}}^2. \tag{16.99}$$

Indeed, $(1+|k|^2)/|k|^2 \le 2$ for nonzero integer modes, which is the required weight comparison. The dissipative term controls the nonzero-mode H^{s+1} norm. The zero mode satisfies $\widehat{v}_0(t) = \widehat{v}_0(0) + \int_0^t \widehat{f}_0(a)\mathrm{d}a$, so Cauchy–Schwarz controls its remaining norms for $T \le 1$. Finally $v_t = \lambda \Delta v + f$ gives the derivative estimate. Fourier truncations converge by this same bound. Approximation by smooth data gives continuity into H^s, including at zero. The homogeneous estimate proves uniqueness. For eq. (16.98), use $(1-e^{-a})^2 \le a$ for $a \ge 0$ on each Fourier coefficient and sum. □

Proof of theorem 16.10. Write the system as

$$u_t - \lambda\Delta u = F(z), \qquad F(z) = \lambda\Gamma(u)(\partial_\mu u, \partial_\mu u) - h^{-1}(V_u - C^T V_r), \tag{16.100}$$

$$r_t = -C(z)u_t + \Phi(z), \qquad \Phi(z) = -k^{-1}V_r, \tag{16.101}$$

$$b_t = L(z), \qquad L(z) = -\mathsf{B}^{-1}V_b. \tag{16.102}$$

The maps include the fixed smooth x dependence of V and of the prior rows. Since $m > d/2$, $H^m(\mathbb{T}^d)$ is a Banach algebra. For clarity, its product bound follows from Fourier convolution, the weight inequality

$$\langle k\rangle^m \le c_m(\langle l\rangle^m + \langle k-l\rangle^m), \qquad \|\widehat{a}\|_{\ell^1} \le \left(\sum_k \langle k\rangle^{-2m}\right)^{1/2} \|a\|_{H^m}, \tag{16.103}$$

where the displayed sum is finite. Young's convolution inequality gives

$$\|ab\|_{H^m} \le c_m \|a\|_{H^m}\|b\|_{H^m}. \tag{16.104}$$

The associated smooth-composition (Moser) estimate on bounded H^m sets with compact image [185, Chapter 13] follows by the chain rule, Leibniz's formula, Sobolev interpolation for its derivative products, and the identity

$$N(z) - N(\widetilde{z}) = \int_0^1 DN(\widetilde{z} + a(z-\widetilde{z}))(z-\widetilde{z})\mathrm{d}a. \tag{16.105}$$

One can first extend the smooth coefficients to a compact neighborhood of the initial image; their derivatives are bounded there. The small pointwise neighborhoods used below ensure that the needed segments remain in that extension. These estimates apply to smooth periodic x dependence as well.

On a bounded admissible set in $H^s \times H^m \times H^m$, there are constants M_N, L_N for $N = F, C, Q, L$ such that

$$\|N(z)\|_{H^m} \le M_N, \tag{16.106}$$

$$\|N(z) - N(\widetilde{z})\|_{H^m} \le L_N\big(\|u-\widetilde{u}\|_{H^s} + \|r-\widetilde{r}\|_{H^m} + \|b-\widetilde{b}\|_{H^m}\big). \tag{16.107}$$

For F, the gradient product is harmless in this estimate because $\partial_\mu u \in H^m$. No derivative of r or b occurs in F.

Use the complete path space

$$X_T = CH^s \times CH^m \times CH^m \tag{16.108}$$

with fixed initial values. On a closed subset with bounded norms and small CH^m displacement from z_0, define a map $z \mapsto z^+ = (u^+, r^+, b^+)$ by solving

$$u_t^+ - \lambda\Delta u^+ = F(z), \qquad u^+(0) = u_0, \tag{16.109}$$

and then setting

$$r^+(t) = r_0 + \int_0^t [-C(z)u_t^+ + \Phi(z)]\mathrm{d}a, \tag{16.110}$$

$$b^+(t) = b_0 + \int_0^t L(z)\mathrm{d}a. \tag{16.111}$$

The new heat solution's derivative is used. Thus an input path needs no time derivative. Choose the norm bounds larger than the initial norms. Lemma 16.11 and contraction of the homogeneous heat semigroup on H^s give

$$\|u^+\|_{CH^s} \le \|u_0\|_{H^s} + K_\lambda\sqrt{T}M_F, \tag{16.112}$$

$$\|u_t^+\|_{L^2H^m} \le K_\lambda(\|u_0\|_{H^s} + \sqrt{T}M_F) =: A_T. \tag{16.113}$$

The displacement estimates are

$$\|u^+ - u_0\|_{CH^m} \le \sqrt{\lambda T}\|u_0\|_{H^s} + TM_F, \tag{16.114}$$

$$\|r^+ - r_0\|_{CH^m} \le c_m\sqrt{T}M_CA_T + TM_Q, \tag{16.115}$$

$$\|b^+ - b_0\|_{CH^m} \le TM_L. \tag{16.116}$$

The first follows from eq. (16.98) and the Duhamel formula using heat contraction on H^m. Sobolev embedding $H^m \hookrightarrow C^0$ makes a sufficiently small displacement preserve the chart and its positive metric bounds. All right sides tend to zero with T. Hence the map preserves the chosen closed set for sufficiently small T.

For two input paths let $D = \|z - \widetilde{z}\|_{X_T}$. The zero-initial heat estimate gives

$$\|u^+ - \widetilde{u}^+\|_{CH^s} + \|u_t^+ - \widetilde{u}_t^+\|_{L^2H^m} \le K_\lambda L_F\sqrt{T}D. \tag{16.117}$$

Split the fiber integrand difference into $C(z)(u_t^+ - \widetilde{u}_t^+) + (C(z) - C(\widetilde{z}))\widetilde{u}_t^+$ and the Q difference. The algebra bound gives

$$\|r^+ - \widetilde{r}^+\|_{CH^m} \le [c_mM_CK_\lambda L_FT + c_mL_CA_T\sqrt{T} + L_QT]D, \tag{16.118}$$

$$\|b^+ - \widetilde{b}^+\|_{CH^m} \le L_LTD. \tag{16.119}$$

Together with eq. (16.117), this is a contraction for small T. Its fixed point solves the system. Lemma 16.11 supplies eq. (16.93); the products in the fiber equation belong to L^2H^m, giving eq. (16.94). The row right side is continuous into H^m, giving the stronger row regularity.

Retaining initial-data differences in the same estimates proves locally Lipschitz dependence in X_T and the heat regularity norms. Substitution into the fiber equation then proves it also for the H^1H^m norms of r, b. Uniqueness among solutions in the stated class follows from these difference estimates on a sufficiently short common interval; concatenation extends it to their common interval of existence.

Finally, a bound on eq. (16.95) and a fixed compact interior image give uniform coefficient bounds, norm radii, and a positive distance to the chart boundary. The displacement estimates eqs. (16.114)

to (16.116) make the restart time uniform. This uses continuity of the heat semigroup from H^s into H^{s-1} with a uniform rate on bounded sets, rather than a false operator-norm continuity claim on H^s. Restarting at times approaching a finite endpoint extends the solution beyond it. This proves the continuation alternative. □

Dissipation and reconstructed-law regularity

Proposition 16.12 (Exact energy identity). For the solution of theorem 16.10, $\mathcal{F}_*(t)$ is absolutely continuous and, for almost every t,

$$\begin{aligned}\frac{\mathrm{d}\mathcal{F}_*}{\mathrm{d}t} &= -\int_{\mathbb{T}^d}\left[u_t^T h u_t + (r_t + Cu_t)^T k(r_t + Cu_t) + b_t^T \mathsf{B} b_t\right]\mathrm{d}\mu_C \\ &= -\int_{\mathbb{T}^d}\left[u_t^T h u_t + V_r^T k^{-1} V_r + V_b^T \mathsf{B}^{-1} V_b\right]\mathrm{d}\mu_C. \end{aligned} \tag{16.120}$$

Consequently

$$\mathcal{F}_*(t) + \int_0^t \|\partial_a(u,r,b)\|_{\mathbb{G}}^2 \mathrm{d}a = \mathcal{F}_*(0). \tag{16.121}$$

Proof. The regularity $m > d/2$, $m \geq 1$, makes $\mathcal{F}_*$ continuously differentiable on the $H^s \times H^m \times H^m$ chart neighborhood, so the chain rule along $H^1(0,T;\cdot)$ paths applies, with spatial gradients and their time derivatives integrable in the needed norms. Approximation by smooth periodic fields, or direct Sobolev integration by parts, gives

$$\frac{\mathrm{d}\mathcal{F}_*}{\mathrm{d}t} = \int_{\mathbb{T}^d}[(V_u - \lambda h \tau_h)^T u_t + V_r^T r_t + V_b^T b_t]\mathrm{d}\mu_C. \tag{16.122}$$

Every term is integrable in time by eqs. (16.93) to (16.94) and the compact-image coefficient bounds. Insert the metric-gradient block equations preceding eq. (16.80). This gives the first identity; the fiber and row equations give the second. Integration in time proves the assertion, including absolute continuity. □

The reconstructed law parameters satisfy

$$p = \Psi(u,r) \in CH^m \cap H^1(0,T;H^m), \qquad A^* = -p^*\theta \in CH^{m-1} \cap H^1(0,T;H^{m-1}). \tag{16.123}$$

The latter follows by differentiating p once in space and using smooth-composition and product estimates; its time derivative contains $p_t\partial_\mu p$ and $\partial_\mu p_t$, both in the stated class. The cancellation in eq. (16.72) gives the better residual regularity $D^{A^*}p = H_a\partial_\mu u^a \in CH^m$. There is no separate connection evolution hidden in these claims.

Dissipation controls the total energy, not each summand separately. It gives no general bound on the Sobolev norms in eq. (16.95) and no exclusion of a statistical-family boundary, loss of finite smooth KL costs, or approach to a singular orbit. A coordinate-chart exit may be removable by another chart, but such patching is not needed for this local theorem. Positive target curvature is not an obstruction to this local proof. Global harmonic-map results with additional curvature hypotheses [70] cannot be imported as global conclusions for this coupled nonbasic system. Related sigma-flow work studies statistical-manifold energies but leaves its general existence question open [38]; the

present result uses the declared Levi-Civita quotient metric, fixed flat base, and explicit regularity class.

Zero coupling is a separate local ODE theorem

Setting $\lambda = 0$ in the metric-gradient equations removes exactly the contextual term and recovers the pointwise restricted-family flow (definition 4.10) of the chosen reciprocal potential. Smooth composition makes this an ordinary differential equation in the Banach space $(H^m)^{n_u+n_r+n_b}$, locally Lipschitz on bounded compact-image subsets. The integral contraction therefore gives a unique local $C^1_t H^m_x$ solution. The energy identity holds with the contextual term absent. The energy at zero coupling does not select A.

The constants in lemma 16.11 are not uniform as $\lambda \downarrow 0$. Nor is the extra H^s regularity of u automatic for the zero-coupling equation. For example, Euclidean Fisher coordinates of Gaussian location laws with a correlated Gaussian reference can give

$$V(u, r) = \tfrac{1}{2}(u^2 - 2aur + r^2) + \text{constant}, \qquad 0 < a < 1. \tag{16.124}$$

Take the isometric translation action in r, so u is the quotient coordinate. The zero-coupling solution is

$$u(t) = e^{-t}[\cosh(at)u_0 + \sinh(at)r_0]. \tag{16.125}$$

If $u_0 \in H^s$ and $r_0 \in H^m \setminus H^s$, then $u(t) \notin H^s$ for every $t > 0$. This example concerns the regular statistical-gradient class underlying the proof; it is not needed to specialize the multi-agent potential. Algebraic zero-coupling recovery is established, while a uniform singular-limit theorem is not asserted.

16.2.6 An exact two-sector continuum witness

Take $\mathcal{C} = S^1$ with measure $\mathrm{d}x/(2\pi)$ and two agents, with

$$q_i = \mathcal{N}(\mu_i, I_2), \quad s_i = \mathcal{N}(\nu_i, I_2), \quad r_i = \mathcal{N}(0, I_2), \quad G_i(m, \mathrm{d}k) = \mathcal{N}(m, I_2)(\mathrm{d}k). \tag{16.126}$$

Each agent attends to the other through an identity channel in both sectors. These singleton rows have no tangent variables. Direct integration gives

$$D_{\mathrm{KL}}(q_i \otimes s_i \| r_i G_i) = \tfrac{1}{2}\|\mu_i - \nu_i\|^2 + \tfrac{1}{2}\|\nu_i\|^2 + 1. \tag{16.127}$$

Thus both sectors participate in the evaluator term, and both directed peer interactions participate in the reciprocal potential. With columns $P = (\mu_1, \nu_1, \mu_2, \nu_2)$, the Fisher metric is Frobenius and

$$V(P) = 2 + \tfrac{1}{2}\langle P, P\mathsf{H}\rangle_F, \qquad \mathsf{H} = \begin{pmatrix} 3 & -1 & -2 & 0 \\ -1 & 4 & 0 & -2 \\ -2 & 0 & 3 & -1 \\ 0 & -2 & -1 & 4 \end{pmatrix}. \tag{16.128}$$

Equivalently, the peer terms are $\|\mu_1 - \mu_2\|^2 + \|\nu_1 - \nu_2\|^2$. A single $SO(2)$ rotation acts on every column. The action is free for $P \neq 0$, is proper, and preserves these constitutive data. Let

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad R_\varphi = \cos\varphi I + \sin\varphi J, \qquad A_x = \alpha J. \tag{16.129}$$

The connection equation and reduced energy are

$$\alpha_* = -\frac{\langle P_x, JP\rangle_F}{\|P\|_F^2}, \tag{16.130}$$

$$\mathcal{F}_*[P] = \int_{S^1} \left[2 + \tfrac{1}{2}\langle P, P\mathsf{H}\rangle_F + \frac{\lambda}{2}\left(\|P_x\|_F^2 - \frac{\langle P_x, JP\rangle_F^2}{\|P\|_F^2} \right) \right] \mathrm{d}\mu_C. \tag{16.131}$$

For fixed α, $D_x = \partial_x + \alpha J$ is skew adjoint on periodic matrix fields. Its envelope variation therefore gives

$$\boxed{P_t = -P\mathsf{H} + \lambda D_x D_x P, \qquad \alpha = \alpha_*.} \tag{16.132}$$

In expanded form the second-order term is $P_{xx} + \alpha_x JP + 2\alpha JP_x - \alpha^2 P$. Its principal part after eliminating α is $P_{xx} - \langle P_{xx}, JP\rangle_F JP/\|P\|_F^2$, which displays the missing vertical diffusion. The vector field in this particular symmetric example is horizontal: $\mathsf{H} = \mathsf{H}^T$ gives $\langle P\mathsf{H}, JP\rangle_F = 0$, and differentiating $\langle D_x P, JP\rangle_F = 0$ covariantly gives $\langle D_x D_x P, JP\rangle_F = 0$. The same degeneracy matters for computation in these law coordinates: the vertical component of the right side is a cancellation among the terms containing α, and it is not damped, so a discretization that does not preserve this cancellation exactly can grow off the horizontal family. Integration in the adapted chart, or exact computation of the vertical part, avoids this, and $\langle P_t, JP\rangle_F = 0$ is a check.

Set

$$\varsigma = \frac{\sqrt{5}-1}{2}, \qquad \kappa = 5 - \varsigma = \frac{11-\sqrt{5}}{2}, \qquad v = (1, \varsigma, -1, -\varsigma)^T, \qquad E_0 = e_1 v^T. \tag{16.133}$$

The identity $\varsigma^2 + \varsigma = 1$ gives $\mathsf{H}v = \kappa v$ and $w := \|E_0\|_F^2 = 5 - \sqrt{5}$. For any smooth periodic real phase $\varphi(x)$, take

$$P(t,x) = R_{\varphi(x)} a(t,x) E_0, \qquad a(t,x) > 0. \tag{16.134}$$

Direct substitution yields

$$\alpha_* = -\varphi_x, \qquad D_x P = R_\varphi a_x E_0, \qquad D_x D_x P = R_\varphi a_{xx} E_0. \tag{16.135}$$

The full matrix equation reduces exactly to

$$a_t = \lambda a_{xx} - \kappa a, \qquad \varphi_t = 0. \tag{16.136}$$

For $n \geq 1$ and $a_0 > |a_1| > 0$, an exact solution is

$$\boxed{a(t,x) = a_0 e^{-\kappa t} + a_1 e^{-(\kappa + \lambda n^2)t} \cos(nx).} \tag{16.137}$$

It is positive at every finite $t \geq 0$, since $a(t,x) \geq e^{-\kappa t}(a_0 - |a_1|) > 0$. All four means are nonzero, and the collective action remains free at every finite time. The zero-mean singular orbit is approached only as $t \to \infty$ for this solution. With $\varphi(x) = \varepsilon \sin x$, $\varepsilon \neq 0$, the nonzero connection is $A_x = -\varepsilon \cos x J$. It is pure gauge. On a one-dimensional base the curvature vanishes identically, and the gauge-invariant content of A_* is its holonomy around the base circle, a rotation through $\pm \oint \alpha_* \mathrm{d}x$ depending on the orientation convention; here $\oint \alpha_* \mathrm{d}x = -\oint \varphi_x \mathrm{d}x = 0$ for every periodic phase, so the holonomy is trivial.

The amplitude $a = \|P\|_F / \sqrt{w}$ is gauge invariant. Its spatial-mode-to-constant-mode ratio is

$$\frac{a_1}{a_0} e^{-\lambda n^2 t}. \tag{16.138}$$

At zero coupling this ratio is constant; at positive coupling it decays faster for higher spatial frequency. This is an exact contextual effect in shape, with a nontrivial reciprocal local reaction. Initial fields with $n = 1$ and $n = 2$ have the same laws and first derivative at $x = 0$, but

$$P_t^{(2)}(0,0) - P_t^{(1)}(0,0) = -3\lambda a_1 R_{\varphi(0)} E_0 \neq 0. \tag{16.139}$$

Thus the local velocity depends on neighboring contextual variation.

Using the normalized base measure, the exact energy is

$$\mathcal{F}_*(t) = 2 + \frac{w\kappa a_0^2}{2} e^{-2\kappa t} + \frac{w(\kappa + \lambda n^2) a_1^2}{4} e^{-2(\kappa + \lambda n^2)t}. \tag{16.140}$$

Its derivative is

$$\frac{\mathrm{d}\mathcal{F}_*}{\mathrm{d}t} = -w\kappa^2 a_0^2 e^{-2\kappa t} - \frac{w}{2}(\kappa + \lambda n^2)^2 a_1^2 e^{-2(\kappa + \lambda n^2)t} = -\int_{S^1} \|P_t\|_F^2 \mathrm{d}\mu_C. \tag{16.141}$$

The existence for every finite time in this explicit sector is a property of its formula. It does not turn the general local theorem into a global theorem. The witness is the linear radial sector of the system: along the ansatz the shape moves on one ray of the quotient, where the quotient metric is flat and the Christoffel terms vanish; the references are rotation invariant, so $V_r = 0$ and the nonbasic correction is absent; $\varphi_t = 0$, so there is no frame dynamics; the connection is pure gauge; and the singleton source rows have no row dynamics. It therefore tests none of the nonlinear, frame, nonbasic, or row parts of the theorem; the fixed-reference example above exhibits the nonbasic correction, but only in a one-agent chart computation.

16.2.7 Scope and remaining obligations

The theorem establishes local well-posedness on a supplied flat torus and a regular finite-dimensional family of general normalized laws. It retains nonbasic fixed-reference forces. Global continuation requires the stated Sobolev and compact-image bounds. Unrestricted infinite-dimensional law dynamics, changes of orbit type (which the fixed-reference example reaches in finite time), nontrivial base bundles, multiple-chart solutions, physical boundaries, an evolving base metric, and alternative statistical connections require additional analysis.

The construction does not derive its base, agent census, references, or channel data. It proves no graph-refinement limit, formation, or recursive inheritance. Deriving displacement transport from peer links is the separate construction of chapter 17; substituting its selector into this energy would require its full constitutive variation and a new evolution theorem. Independent frames and source rows acquire no separate diffusion; for frames this holds for every locally gauge-invariant contextual energy after elimination. The finite theorem holds on the regular critical set and identifies the reduced energy with the minimized cost only on its global part, and for $\mathrm{GL}(K)$ frames with $K \geq 3$ it excludes intra-context consensus. The exact Gaussian sector illustrates the continuum coupling within these boundaries.

Chapter 17

Base Transport from Contextual Agent Data

A smooth family of agent interactions can determine a connection once its cross-context comparison rule is specified. This chapter constructs such rules from law moments, receiving interfaces, and directed links. A full-rank message matrix gives displacement maps with identity at zero displacement; their first derivative gives the connection, and products along refining paths give its parallel transport. A second construction aligns the contextual link data directly and admits nonzero curvature even when every agent backtrack is the identity.

The comparison data are constitutive choices. The message metric and an augmented link metric need not be the Fisher metric of the joint law in proposition 3.9. The new selectors therefore need not equal that chapter's collective Fisher connection. They supply a conditional construction for open problem D.4; they add no evolution equation or formation result to the contextual-flow extension in chapter 16. The selectors are kinematic: they take supplied fields as input and belong to none of the three objectives of chapter 4.

17.1 Contextual data and their transformation laws

Let $\mathcal{C}$ be a smooth finite-dimensional contextual base, $P \to \mathcal{C}$ a principal matrix-group bundle, and $E = P \times_G V$ an associated vector bundle. Fix one sector temporarily. The explicit construction first uses $G = \mathrm{GL}(K, \mathbb{R})$ and $V = \mathbb{R}^K$. Section 17.7 treats a smaller declared matrix group. A context is denoted c; agent labels i, j are not base coordinates. All derivatives below are contextual derivatives at fixed evolution time.

On a patch choose a reference presentation of E. Agent i displays vectors in a coordinate copy E_i through an invertible map $U_i : V \to E_i$. Its normalized law ν_i may be a belief or model marginal. A passive change of displayed coordinates and a change of common reference act differently:

$$U_i' = g_i U_i, \qquad \nu_i' = (g_i)_{\#}\nu_i, \tag{17.1}$$

$$U_i' = U_i h, \qquad \bar{\nu}_i' = (h^{-1})_{\#}\bar{\nu}_i, \qquad \bar{\nu}_i = (U_i^{-1})_{\#}\nu_i. \tag{17.2}$$

The presentation maps and coordinate changes are C^2 and may vary with context. Reference-independent statements must satisfy both rules.

A directed, invertible interaction at one context is

$$\Theta_{ij} = U_i \bar{\Theta}_{ij} U_j^{-1} : E_j \longrightarrow E_i, \qquad \bar{\Theta}_{ij}' = h^{-1}\bar{\Theta}_{ij} h. \tag{17.3}$$

For example, $\bar{\Theta}_{ij} = \exp(L_{ij})$, where $L_{ij} = O_i T_j$ is a receiving–sending product on V that may depend on the current laws. General normalized channels need not be invertible; the factorization arguments here apply to invertible links only. Neither Θ_{ij} nor its exponential generator is already a displacement map between c and $c + \delta c$.

Our convention for a reference connection is $D^a = d + a$, with

$$a' = h^{-1}ah + h^{-1}dh. \tag{17.4}$$

In agent i's displayed coordinates the same connection is

$$B_i = U_i a U_i^{-1} - dU_i U_i^{-1}, \qquad F_i = U_i F^a U_i^{-1}, \qquad F^a = da + a \wedge a. \tag{17.5}$$

Thus $B_i' = g_i B_i g_i^{-1} - dg_i g_i^{-1}$ under eq. (17.1), and B_i is unchanged by eq. (17.2) together with eq. (17.4). Products of matrix-valued forms include matrix multiplication; $(a \wedge a)(Y, Z) = [a(Y), a(Z)]$.

The issue is to derive a from stated contextual agent data. Merely declaring a_μ and inserting $\exp(-a_\mu \delta c^\mu)$ supplies the answer as additional data. The construction below begins with finite cross-context messages and differentiates their normalized alignment.

17.2 What an agent frame does determine

Proposition 17.1 (A square frame selects flat local transport). Let $Q(c)$ be a C^2 invertible matrix whose columns are declared marked frame vectors, expressed in one reference coordinate system. Put

$$a_Q = -dQQ^{-1}, \qquad T_Q(c', c) = Q(c')Q(c)^{-1}. \tag{17.6}$$

Then $T_Q(c, c) = I$, $D^{a_Q} Q = 0$, $F^{a_Q} = 0$, and $T_Q(c'', c')T_Q(c', c) = T_Q(c'', c)$. If $Q' = h^{-1}Q$, then a_Q obeys eq. (17.4) and $T_Q' = h(c')^{-1} T_Q h(c)$.

Proof. Identity, composition and covariance follow by multiplication. Taylor expansion gives $T_Q(c + \delta c, c) = I - a_Q(\delta c) + O(|\delta c|^2)$. The identity $dQ^{-1} = -Q^{-1}(dQ)Q^{-1}$ gives $da_Q = -a_Q \wedge a_Q$, proving flatness. □

An agent frame can therefore select a connection if declaring that frame parallel is part of the model. When the frame is only a coordinate choice, that declaration changes the model under an arbitrary change of frame unless the connection is transformed with it. In particular $B_i = -dU_i U_i^{-1}$, with U_i as in eq. (17.2), holds the reference connection $a = 0$ fixed. Replacing U_i by $U_i h(c)$ while again setting $a = 0$ changes B_i by $-U_i dh h^{-1} U_i^{-1}$. This rule is not independent of the common reference.

Several individually flat connections can give a curved affine average. They must already be connections on the same bundle. For two such connections a_1, a_2, put $\Delta = a_1 - a_2$ and let λ be a smooth scalar. Then $a = a_2 + \lambda\Delta$ has

$$F^a = d\lambda \wedge \Delta - \lambda(1 - \lambda)\Delta \wedge \Delta. \tag{17.7}$$

Flatness gives $D^{a_2}\Delta = -\Delta \wedge \Delta$, which substituted into $F^{a_2 + \lambda\Delta}$ proves the formula. Weight variation can produce Abelian curvature, and noncommuting disagreements can produce non-Abelian

curvature even at constant weight. The weights and identification of the individual connections remain constitutive inputs. Averaging square frames and applying eq. (17.6) still gives a flat connection.

17.3 Retaining reciprocal and backtrack information

Suppose each backtrack $N_{ij,j} = \Theta_{ji}\Theta_{ij}$ has spectrum disjoint from $(-\infty, 0]$. The pair ij names the interacting pair and the last subscript the agent at which the loop is based, so $N_{ij,j}$ and $N_{ij,i}$ are the two loops of one pair. The following is the split of proposition 2.11, written with these subscripts. Define

$$S_{ij,j} = N_{ij,j}^{1/2}, \qquad \Omega_{ij} = \Theta_{ij} S_{ij,j}^{-1}, \qquad \Theta_{ij} = \Omega_{ij} S_{ij,j}. \tag{17.8}$$

The principal real matrix root is smooth on this spectral domain and commutes with similarity [107]. Since $N_{ij,i} = \Theta_{ij} N_{ij,j} \Theta_{ij}^{-1}$, multiplication gives

$$\Omega_{ji} = \Omega_{ij}^{-1}, \qquad \Omega'_{ij} = g_i \Omega_{ij} g_j^{-1}, \qquad S'_{ij,j} = g_j S_{ij,j} g_j^{-1}. \tag{17.9}$$

This factorization retains the original directed pair exactly. It does not impose inverse relations on the original Θ fields. An interaction loop at one context is also distinct from a loop in the contextual base.

The construction may use raw Θ fields directly, without any square-root branch. Where the branch exists, including both Ω-transformed and Θ-transformed messages makes their different roles explicit. Storing Ω and S retains all directed links, even though their eventual connection need not encode them injectively.

17.4 A full-rank contextual message construction

Fix a finite census and persistent message labels. Their message space is $H = \mathbb{R}^r$, with a stated, context-independent Euclidean inner product. A fixed positive message metric can be absorbed into the columns. Identifying these labels between contexts is an assumption; agent labels or fixed receiving-interface columns give one realization. Section 17.10 treats changing message coordinates.

For a normalized law with a finite first moment, let $\mu_i = \mathbb{E}_{\nu_i}[v] \in E_i$ and $\bar{\mu}_i = U_i^{-1}\mu_i$. The law need not be Gaussian. Smoothness of its moments is an assumption on the law family, not a consequence of normalization. Define

$$R_{ij} = U_i^{-1} \Omega_{ij} U_j, \qquad \bar{\Theta}_{ij} = U_i^{-1} \Theta_{ij} U_j. \tag{17.10}$$

One permissible collective message matrix is a chosen concatenation

$$X(c) = \big[\{\bar{\mu}_i\}_i,\ \{R_{ij}\bar{\mu}_j\}_{(i,j)},\ \{\bar{\Theta}_{ij}\bar{\mu}_j\}_{(i,j)}\big] : H \longrightarrow V. \tag{17.11}$$

Each column has a stated label. Receiving-interface columns $O_i : H_i \to V$, or their link-transformed columns, can replace or supplement the means. Gauge-invariant scalar column weights may depend smoothly on laws and rows; their derivatives are included in dX. Column selection, weights

and message metric specify the rule. The identity self link does not imply that these columns span V.

Under passive agent re-expression every reference column is unchanged. Under reference rechoice it obeys

$$X' = h^{-1}X. \tag{17.12}$$

For example $(h^{-1}R_{ij}h)(h^{-1}\bar{\mu}_j) = h^{-1}R_{ij}\bar{\mu}_j$. The same calculation applies to raw link and interface columns. Thus the construction is reference consistent when its original link and interface rules are reference consistent. This is naturality: the rule is built equivariantly from the laws and from declared data that are re-expressed with the reference. It is not an active symmetry of the model with those data held fixed, which fixed receiving axes or labels generally break.

Hypothesis 17.2 (Regular message region). On an open region $\mathcal{U} \subset \mathcal{C}$, X is C^2 with row rank K. Its columns refer to one fixed labeled message space. Local expressions on overlaps obey eq. (17.12). No spatial connection on E is included in these data.

Put

$$H_X = XX^T, \qquad Q_X = X^T H_X^{-1}, \qquad P_X = Q_X X. \tag{17.13}$$

Here H_X is a positive-definite $K \times K$ Gram matrix, $Q_X : V \to H$ is a right inverse of X, and P_X is the orthogonal projector in message space onto $\operatorname{im} X^T$.

Theorem 17.3 (Displacement links and their connection). Under the regular message assumption, the map

$$C(c', c) = X(c')Q_X(c) \tag{17.14}$$

equals I at $c' = c$, is invertible on a neighborhood of the diagonal, and obeys

$$C'(c', c) = h(c')^{-1}C(c', c)h(c). \tag{17.15}$$

Its first derivative determines a C^1 connection:

$$C(c + \delta c, c) = I - a_c(\delta c) + O(|\delta c|^2), \qquad a = -dXQ_X. \tag{17.16}$$

The derivative is linear in the tangent displacement. The local forms obey eq. (17.4) and define a connection wherever the data have the stated overlap laws. In agent coordinates

$$C_i(c', c) = U_i(c')C(c', c)U_i(c)^{-1} \tag{17.17}$$

has connection eq. (17.5), is reference independent, and transforms by $g_i(c')$ and $g_i(c)^{-1}$ at its two endpoints.

Proof. Full row rank gives $XQ_X = I$. Continuity of the determinant makes C invertible near each diagonal point. Under eq. (17.12),

$$H'_X = h^{-1}H_X h^{-T}, \qquad Q'_X = Q_X h, \qquad P'_X = P_X. \tag{17.18}$$

This proves the endpoint law. Differentiating the first argument of eq. (17.14) gives eq. (17.16); Taylor's theorem gives its remainder locally uniformly on compact subsets. Moreover,

$$-d(h^{-1}X)Q_X h = h^{-1}(-dXQ_X)h + h^{-1}dh. \tag{17.19}$$

The affine law and the bundle transition cocycle glue the local covariant derivatives. Expanding eq. (17.17) gives B_i; substitution proves passive covariance and reference independence. □

Proposition 17.4 (Global message surjections). Let $E \to \mathcal{C}$ be a rank-K vector bundle and $X : \mathcal{C} \times \mathbb{R}^r \to E$ a C^2 fiberwise surjective bundle map whose local forms obey eq. (17.12). Put $W(c) = \operatorname{im} X(c)^T \subset \mathbb{R}^r$. Then the local forms $a = -dXQ_X$ define a global C^1 connection, with covariant derivative

$$\nabla\sigma = Xd(Q_X\sigma). \tag{17.20}$$

It is the connection induced on the moving subspace W by orthogonal projection of the trivial connection of $\mathbb{R}^r$, transported to E by the isomorphism $X|_W$. The maps $C(c', c) = X(c')Q_X(c)$ are defined for all pairs, equal the identity on the diagonal, and are invertible near it. The map $c \mapsto W(c)$ into the Grassmannian of K-planes in $\mathbb{R}^r$ classifies E; in particular $r = K$ forces E to be trivial.

Proof. The affine law follows from the identity in the proof of theorem 17.3, so the local forms glue. Since $XQ_X = I$, $XdQ_X = -dXQ_X$, and therefore $Xd(Q_X\sigma) = d\sigma - dXQ_X\sigma = D^a\sigma$. Because Q_X is an isomorphism onto W with inverse $X|_W$, and X annihilates $W^\perp$, this is $X|_W$ applied to the orthogonal projection onto W of the derivative of the W-valued section $Q_X\sigma$. The statements about C were proved in theorem 17.3. Finally $X|_{W(c)} : W(c) \to E_c$ is an isomorphism depending continuously on c, so E is isomorphic to the pullback of the tautological bundle along $c \mapsto W(c)$; for $r = K$ that pullback is trivial. □

When $r > K$, such a surjection need not trivialize E. For instance, on the tangent bundle of the unit sphere $S^2 \subset \mathbb{R}^3$, the tangential projections of the three coordinate vectors give a surjection with $r = 3$; the induced connection is the Levi-Civita connection, and the Euler number 2 shows that this bundle is nontrivial. A global square frame is the more restrictive case $r = K$.

A receiver's displayed columns $\mu_i, \Omega_{ij}\mu_j, \Theta_{ij}\mu_j$ also transform by left multiplication by g_i. Applying the formula gives a receiver-selected connection. Different receivers' selections need not agree. The collective reference construction eq. (17.11) instead selects one a, displayed by all agents through eq. (17.5).

17.5 Reversal, finite paths, and zero displacement

The cross-map C generally fails both reversal and composition. These failures do not invalidate its first jet as a connection, but matter before calling a finite map parallel transport. Finite maps on links whose ordered products refine to path-ordered transport are the link variables of lattice gauge theory and their continuum limit [199]; a discrete-connection formalism is developed in Leok et al. [134].

Proposition 17.5 (Reciprocal displacement maps with the same derivative). After shrinking the neighborhood of the diagonal, define

$$N(c',c) = C(c,c')C(c',c), \qquad T(c',c) = C(c',c)N(c',c)^{-1/2}. \tag{17.21}$$

Then $T(c,c) = I$, $T(c,c') = T(c',c)^{-1}$, and T obeys eq. (17.15). Furthermore,

$$N(c+\delta c, c) = I + O(|\delta c|^2), \qquad T(c+\delta c, c) = I - a_c(\delta c) + O(|\delta c|^2). \tag{17.22}$$

Proof. At the diagonal $N = I$, so its principal root is smooth nearby. The calculation in eq. (17.9) proves reversal and endpoint covariance. Differentiating $XQ_X = I$ gives $XdQ_X = -dXQ_X$, so the two factors of N have opposite linear terms. Its root is $I + O(|\delta c|^2)$, proving the expansion. □

Proposition 17.6 (Principal angles and the reciprocal map). Let $W(c) = \operatorname{im} X(c)^T$ and let $\theta_1, \dots, \theta_K \in [0, \pi/2]$ be the principal angles between $W(c)$ and $W(c')$. Then the eigenvalues of $N(c',c)$ are $\cos^2\theta_k$. Wherever $C(c',c)$ is invertible, that is, wherever every $\theta_k < \pi/2$, the operator $N(c',c)$ is similar to a positive operator, $T(c',c)$ is defined without further shrinking, and $T(c',c)$ is an isometry from $(E_c, H_X(c)^{-1})$ to $(E_{c'}, H_X(c')^{-1})$. Read through these isometries, $T(c',c)$ is the direct rotation of $W(c)$ onto $W(c')$, which equals parallel transport of the connection eq. (17.20) of the moving subspace along the Grassmannian geodesic from $W(c)$ to $W(c')$.

Proof. Since $Q_X^T Q_X = H_X^{-1}$, the map $\iota_c = Q_X(c)$ is an isometry from $(E_c, H_X(c)^{-1})$ onto $W(c)$, with inverse $X(c)|_{W(c)}$, and $X(c')$ vanishes on $W(c')^\perp$. Hence $C(c',c) = \iota_{c'}^{-1}\Pi'\iota_c$, where Π' is the orthogonal projection onto $W(c')$ restricted to $W(c)$, and similarly $C(c,c') = \iota_c^{-1}\Pi\iota_{c'}$. Thus $N(c',c)$ is conjugate to $\Pi\Pi'$ restricted to $W(c)$, a positive semidefinite self-adjoint operator with eigenvalues $\cos^2\theta_k$ [26]; it is invertible exactly when Π' is injective on $W(c)$. The map $T = CN^{-1/2}$ is then conjugate to the orthogonal polar factor of Π', an isometry of $W(c)$ onto $W(c')$, which is the direct rotation [54]. Choose principal vectors $w_k \in W(c)$ and $w'_k \in W(c')$ with $\langle w_j, w'_k\rangle = \delta_{jk}\cos\theta_k$, and for $\theta_k > 0$ put $n_k = (w'_k - \cos\theta_k w_k)/\sin\theta_k$, an orthonormal family in $W(c)^\perp$. Let Ξ be the skew map with $\Xi w_k = \theta_k n_k$, $\Xi n_k = -\theta_k w_k$, and zero on the remaining complement. The curve $W(t) = \exp(t\Xi)W(c)$ is a Grassmannian geodesic from $W(c)$ to $W(c')$, and the vectors $\exp(t\Xi)w_k = \cos(t\theta_k)w_k + \sin(t\theta_k)n_k$ have derivatives orthogonal to $W(t)$, so they are parallel for the projected connection. At $t = 1$ they equal w'_k, and the polar factor of Π' also sends w_k to w'_k, because $\Pi' w_k = \cos\theta_k w'_k$. □

The Grassmannian geodesic need not be the image of any path in the base, so $T(c',c)$ is exact transport along a path that the base may not contain; this is the precise sense of the remark below that a finite endpoint rule contains additional higher-order data. Because $T(c,c') = T(c',c)^{-1}$, ordered products of T form a symmetric one-step rule, and for C^3 data their local errors cancel at second order, whereas products of C are first-order accurate.

The new $N(c',c)$ is a cross-context reconstruction defect. It differs from the agent backtrack $N_{ij,j}(c)$, retained in the input state through eq. (17.8). Reciprocalizing a displacement map does not delete the backtrack data that determined X.

Theorem 17.7 (Parallel transport from the derived first jet). For a piecewise C^2 path $\gamma : [0,1] \to \mathcal{U}$, the equation

$$\dot{\mathcal{T}}(t) = -a_{\gamma(t)}(\dot{\gamma}(t))\mathcal{T}(t), \qquad \mathcal{T}(0) = I \tag{17.23}$$

has a unique invertible solution. It composes under concatenation, inverts under reversal, and has the endpoint law. Ordered products of C or T along partitions of γ converge to $\mathcal{T}(1)$ as the maximum time step tends to zero.

Proof. On each smooth segment the coefficient in eq. (17.23) is continuous and bounded. Picard iteration on a sufficiently short interval is a contraction, and finitely many intervals cover the path. The solution of $\dot{\mathcal{S}} = \mathcal{S}a(\dot{\gamma})$, $\mathcal{S}(0) = I$, satisfies $\frac{d}{dt}(\mathcal{S}\mathcal{T}) = 0$, proving invertibility. Uniqueness proves concatenation and reversal. Differentiating $h(\gamma(t))^{-1}\mathcal{T}(t)h(\gamma(0))$ verifies its transformed equation.

On each step contained in one smooth segment the exact short propagator and discrete factor equal $I - a(\dot{\gamma})\Delta t + O(\Delta t^2)$ uniformly on the compact path image. Their norms are bounded by $1 + L\Delta t$. Telescoping the product difference bounds the contribution of these steps by $c_0 \exp(2L) \sum_k \Delta t_k^2$, which tends to zero because $\sum_k \Delta t_k^2 \le \max_k \Delta t_k$. A step that straddles one of the finitely many corners has both factors equal to $I + O(\Delta t)$, so these steps add at most a constant times the number of corners times $\max_k \Delta t_k$. This proves convergence for both discrete factors and every partition sequence. Local trivializations can be joined by transition maps; the endpoint laws make the result independent of this description. □

An unrefined $C(c', c)$ or $T(c', c)$ need not equal exact transport along a selected path. Its first jet determines eq. (17.23), while the finite endpoint rule contains additional, nonunique higher-order data.

17.6 Curvature and a curved law-dependent example

Proposition 17.8 (Curvature of the rectangular construction). For eq. (17.16),

$$\begin{aligned} F^a &= dX \wedge dP_X Q_X = dX \wedge (I - P_X) dQ_X \\ &= dX \wedge (I - P_X) dX^T H_X^{-1}, \end{aligned} \tag{17.24}$$

$$dH_X + aH_X + H_X a^T = 0, \qquad \operatorname{tr} a = -\tfrac{1}{2} d \log\det H_X, \qquad \operatorname{tr} F^a = 0. \tag{17.25}$$

The connection preserves the fiber metric H_X^{-1}. When $r = K$, $P_X = I$ and the curvature vanishes. Redundant columns permit nonzero curvature.

Proof. Exterior differentiation of $a = -dXQ_X$ gives $da = dX \wedge dQ_X$. Differentiating $P_X = Q_X X$ and multiplying by Q_X gives $dP_X Q_X = dQ_X + Q_X dX Q_X$. This proves the first equality after adding $a \wedge a$. Differentiating $P_X Q_X = Q_X$ proves the second. In $dQ_X = dX^T H_X^{-1} + X^T dH_X^{-1}$ the second term is killed by $I - P_X$, giving the last expression. Finally $aH_X = -dXX^T$ and $H_X a^T = -XdX^T$, proving metric compatibility. Taking its trace after multiplying by H_X^{-1} gives $2 \operatorname{tr} a = -d \log\det H_X$. The trace of a commutator vanishes, so $\operatorname{tr} F^a = d \operatorname{tr} a = 0$. □

Corollary 17.9 (Orthogonal holonomy of the rectangular selector). Along every path, the transport of eq. (17.23) for eq. (17.16) is an isometry between the fiber metrics H_X^{-1} at its endpoints, so every holonomy at c_0 lies in the orthogonal group of $H_X(c_0)^{-1}$. On a loop along which E is trivialized, its determinant is 1. Each curvature value $F^a(Y,Z)$ is skew for H_X^{-1} and therefore has purely imaginary spectrum. The same holds when a message connection Γ that preserves the message metric is used in eq. (17.39).

Proof. With $M = H_X^{-1}$, eq. (17.25) gives $dM = Ma + a^T M$. For $\dot{\mathcal{T}} = -a(\dot\gamma)\mathcal{T}$, $\frac{d}{dt}(\mathcal{T}^T M \mathcal{T}) = \mathcal{T}^T(-a^T M + dM(\dot\gamma) - Ma)\mathcal{T} = 0$. In a trivialization along the loop, $\det \mathcal{T}(1) = \exp(-\oint \operatorname{tr} a) = \exp(\frac{1}{2}\oint d\log\det H_X) = 1$. Metric compatibility gives $MF^a + (F^a)^T M = 0$, so $M^{1/2}F^a M^{-1/2}$ is skew. A skew Γ in eq. (17.39) gives $aH_X + H_X a^T = -dH_X$ by the same computation. □

Under the structure group $\mathrm{GL}(K)$ the rectangular selector therefore reduces to the H_X^{-1}-orthonormal frames and never produces shear or dilation holonomy; noncompact holonomy requires a selector that does not preserve a fiber metric. The curvature formula also shows that F^a depends on dX only through $(I - P_X)dX^T$, which represents the derivative of $c \mapsto W(c)$; $F^a(c) = 0$ wherever that map has rank below two.

This projector construction is an instance of induced subbundle geometry: by proposition 17.4 it is the pullback along $c \mapsto W(c)$ of the connection that the trivial bundle induces on the tautological subbundle of a Grassmannian, the setting of universal connections [144, 150, 151]. By corollary 17.9 every connection produced by this real rule preserves a fiber metric; whether every metric connection arises for some finite r is the universal-connection question, which is not settled here for this rule. The choice here is how agent laws and directed links supply the columns.

On $\mathbb{R}^2$ with coordinates (x, y) take

$$X = \begin{pmatrix} 1 & 0 & x \\ 0 & 1 & y \end{pmatrix}, \qquad v = \begin{pmatrix} x \\ y \end{pmatrix}, \qquad \chi = 1 + x^2 + y^2. \tag{17.26}$$

It has rank two everywhere. Substitution gives

$$a = -\frac{(dv)v^T}{\chi}, \tag{17.27}$$

$$F^a = \frac{1}{\chi^2}\begin{pmatrix} -xy & 1+x^2 \\ -(1+y^2) & xy \end{pmatrix} dx \wedge dy. \tag{17.28}$$

At the origin $a = 0$ but $F^a_{xy} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \neq 0$. Here $W(c)$ is the plane orthogonal to $(v, -1)$, so $c \mapsto W(c)$ is the inverse central projection onto the unit sphere, and the connection is the Levi-Civita connection of S^2 pulled back by it.

An explicit realization uses normalized non-Gaussian laws and an invertible link. Let a receiver carry receiving columns e_1, e_2. Let a source law be the product Laplace law of unit coordinate

variances on $\mathbb{R}^2$, translated to mean $(x,1)^T$. Choose

$$\bar{\Theta}_{ij} = \begin{pmatrix} 1 & 0 \\ y & 1 \end{pmatrix} = \exp(yE_{21}), \qquad \bar{\Theta}_{ji} = \bar{\Theta}_{ij}^{-1}. \tag{17.29}$$

The linked mean is $(x, 1+xy)^T$. Replacing v in eq. (17.26) by this vector gives a full-rank, link-dependent matrix. Its curvature is the pullback of eq. (17.28) under $(x,y) \mapsto (x, 1+xy)$, nonzero wherever $x \neq 0$ because the Jacobian determinant is x. The labels, receiving axes, column selection, and message metric are declared data, and the law enters only through its mean: any law with mean $(x,1)^T$ gives the same connection. No Gaussian closure or evolving distributional ansatz is used.

Backtracks can affect the gauge class of the connection while the reciprocal agent link stays fixed. Take receiving columns e_1, e_2, a source mean $(1,1)^T$ in reference coordinates, and $\Theta_{ij} = \Theta_{ji} = \mathrm{diag}(e^x, e^y)$. Then $N_{ij,j} = \mathrm{diag}(e^{2x}, e^{2y})$, $S_{ij,j} = \mathrm{diag}(e^x, e^y)$, and $\Omega_{ij} = I$. The raw linked column $(e^x, e^y)^T$ gives the matrix eq. (17.26) with $v = (e^x, e^y)^T$, whose curvature is the pullback of eq. (17.28) by a map with Jacobian determinant $e^{x+y} \neq 0$, hence nonzero everywhere; at the origin it equals $\frac{1}{9}\begin{pmatrix} -1 & 2 \\ -2 & 1 \end{pmatrix} dx \wedge dy$. The reciprocal column $\Omega_{ij}(1,1)^T$ is constant and gives $a = 0$. Curvature spectra are gauge invariant, so the backtrack factor alone changes the gauge class of the selected connection. This proves that a backtrack contribution can survive the selector, without asserting that every backtrack distinction is recoverable from one connection. In rank one the rectangular selection is always pure gauge, $a = -\frac{1}{2} d \log H_X$; for example, a scalar link $\Theta_{ij} = \Theta_{ji} = \vartheta(c) > 0$ with raw column gives $a = -\vartheta d\vartheta/(1+\vartheta^2)$, which differs from the reciprocal selection $a = 0$ only by a gauge transformation.

17.7 A declared structure group and joint sector data

The explicit matrix eq. (17.16) need not lie in an arbitrarily chosen subalgebra $\mathfrak{g} \subset \mathfrak{gl}(K)$. A $\mathrm{GL}(E)$ connection is not automatically a connection on a prescribed G-reduction.

Let $M(c)$ be a C^1 positive-definite fiber metric with $M' = h^T M h$, and let X have full row rank and eq. (17.12), with reference changes $h(c) \in G$. The choice $M = H_X^{-1}$ is available. For $Y \in T_c\mathcal{C}$, minimize over $\xi \in \mathfrak{g}$

$$\mathcal{J}_Y(\xi) = \tfrac{1}{2} \operatorname{tr}\big[M(dX(Y) + \xi X)(dX(Y) + \xi X)^T\big]. \tag{17.30}$$

Proposition 17.10 (A selector in the declared Lie algebra). For a closed matrix group G, eq. (17.30) has a unique minimizer $a(Y) \in \mathfrak{g}$, linear in Y, C^1 in c, and satisfying eq. (17.4). It solves

$$\operatorname{tr}[M(a(Y)X)(\eta X)^T] = -\operatorname{tr}[M dX(Y)(\eta X)^T] \quad (\eta \in \mathfrak{g}). \tag{17.31}$$

For $\mathfrak{g} = \mathfrak{gl}(K)$ the solution is eq. (17.16).

Proof. The Hessian is positive: $\xi X = 0$ implies $\xi = 0$ after multiplying by Q_X. Its matrix in any basis of $\mathfrak{g}$ is invertible, giving existence, uniqueness and linear, smooth dependence. Under reference

rechoice the candidate $\xi' = h^{-1}\xi h + h^{-1}dh(Y)$ belongs to $\mathfrak{g}$ and $dX'(Y) + \xi' X' = h^{-1}(dX(Y) + \xi X)$. The norm is unchanged; uniqueness gives the affine law. In the unrestricted algebra the normal equation is $(dX + aX)X^T = 0$, yielding eq. (17.16). The affine law uses $h^{-1}dh \in \mathfrak{g}$, that is, h valued in G. □

For a proper subalgebra the minimizer depends on M, which is then an additional comparison choice. For $G = \mathrm{SL}(K)$ and $M = H_X^{-1}$ the Hessian form is $\mathrm{tr}(H_X^{-1}\xi H_X \eta^T)$, for which the identity is orthogonal to every traceless η; the minimizer is therefore the traceless part $a + \frac{1}{2K}(d \log\det H_X)I$ of eq. (17.16), with the same curvature, since the removed part is an exact scalar form.

For a proper subalgebra, the induced-metric and explicit curvature formulas of proposition 17.8 need not hold for the constrained minimizer. Finite G-valued transport follows by eq. (17.23) along every path in the regular region with this derived a. Belief and model sectors carry representations ϱ_q, ϱ_s of one group. To select one principal connection from both, sum their costs with a declared relative sector weight, replacing ξX by $d\varrho_a(\xi)X^a$, and minimize over the same $\xi \in \mathfrak{g}$; the weight changes the selected generator. The required nondegeneracy is

$$\{\xi : d\varrho_q(\xi)X^q = 0,\ d\varrho_s(\xi)X^s = 0\} = \{0\}. \tag{17.32}$$

The same proof applies to the summed normal equation. Two unrelated sector connections do not automatically arise from one principal connection. The two interaction link fields can remain distinct; equality of the belief and model links is a separate hypothesis.

17.8 Direct alignment of contextual link data

The rectangular construction is sufficient, but a full-rank vector collection is not necessary when the group action on links is itself identifiable and an invariant metric on the link target exists. For compact groups the second condition is automatic; for noncompact groups it is restrictive, as explained after proposition 17.11. Let G be a finite-dimensional Lie group and let $\mathcal{Z}$ be a finite-dimensional Riemannian target of laws, interfaces and links with a smooth isometric left G-action and metric $\mathcal{H}$. Under reference rechoice the target obeys $Z'(c) = h(c)^{-1} \cdot Z(c)$. Write $Z(c) \in \mathcal{Z}$ and $J_Z\xi = \frac{d}{dt}\big|_0 \exp(t\xi) \cdot Z$. Suppose Z is smooth and J_Z is injective on the region considered.

Proposition 17.11 (Local link alignment). Near each diagonal point there is a unique smooth branch $t(c', c)$ near e of strict local minimizers of

$$g \longmapsto \tfrac{1}{2} d_{\mathcal{H}}(Z(c'), g \cdot Z(c))^2. \tag{17.33}$$

It satisfies $t(c, c) = e$, reversal by inversion, and the endpoint law. Its connection solves

$$\mathcal{H}(J_Z a(Y), J_Z \eta) = -\mathcal{H}(dZ(Y), J_Z \eta). \tag{17.34}$$

Proof. Use squared distance inside a normal neighborhood. At a diagonal point the group gradient vanishes and its Hessian is the positive locked form $\mathcal{H}(J_Z\xi, J_Z\eta)$. The implicit function theorem gives a unique smooth critical branch, whose Hessian remains positive nearby. Expanding $t(c + \delta c, c) = e - a(\delta c) + O(|\delta c|^2)$ gives residual $dZ(Y) + J_Z a(Y)$ and hence eq. (17.34). Isometry makes a reference rechoice carry the selected branch to $h(c')^{-1}t(c', c)h(c)$. Symmetry of distance and

isometry identify the reversed objective at g^{-1} with the original at g; local uniqueness proves inversion. □

This is the mechanical connection of the declared target metric [182], as in the rotation–shape separation of Littlejohn and Reinsch [139]. With $\mathcal{Z}$ the declared law family, $\mathcal{H}$ its product Fisher metric, and h replaced by h^{-1} (chapter 16 writes target changes, this chapter reference changes), $t(c', c)$ is the near-identity branch of hypothesis 16.3 and its first jet is the collective connection of proposition 3.9; the new content here is the general target. For a free action the argument of proposition 3.11 shows that the curvature is the pullback of the curvature of the target's mechanical connection by Z, so it depends only on the derivative of the shape map $c \mapsto [Z(c)]$ and vanishes where that map has rank below two. The statement concerns the branch continuous from the identity. Infinitesimal identifiability does not exclude distant competing minima; eq. (16.55) is an example of a branch singularity under a free action. On overlaps the local identity branches agree after restriction by uniqueness. Its first jet gives path transport by the preceding ODE argument.

For noncompact G the isometric target is a substantive hypothesis, and injectivity of J_Z does not supply it. If z has stabilizer G_z, each element of G_z acts on $T_z\mathcal{Z}$ by an isometry of $\mathcal{H}_z$, so the isotropy representation of G_z must have relatively compact image. For $\mathrm{GL}(2)$ acting by conjugation on one link $\mathrm{diag}(\lambda_1, \lambda_2)$ with real $\lambda_1 \neq \lambda_2$, the element $\mathrm{diag}(t, 1)$ fixes the link and scales the tangent vector E_{12} by t, which no positive inner product allows for all $t > 0$. Even where the action is free, an invariant metric can fail to exist because the action is not proper. For $\mathrm{SL}(2)$ acting by conjugation on pairs, let $D = \mathrm{diag}(2, 1/2)$, $z = (D, I+E_{12})$, $z' = (D, I+E_{21})$, $z_t = \left(D, \left(\begin{smallmatrix} 1+t^2 & 1 \\ t^2 & 1 \end{smallmatrix}\right)\right)$ and $g_t = \mathrm{diag}(t, 1/t)$. Both z and z' have trivial infinitesimal stabilizer, $z_t \to z$ and $g_t z_t g_t^{-1} \to z'$ as $t \to 0$, while $g_t z g_t^{-1} \to (D, I)$, a point with a one-dimensional stabilizer. An invariant metric on an invariant open set of free points containing z and z' would give $d(g_t z, z') \leq d(z, z_t) + d(g_t z_t g_t^{-1}, z') \to 0$, which is impossible. Invariant metrics do exist on invariant open sets where the action is proper [157], but none is canonical in general. For the defining representation of $\mathrm{GL}(K)$, scalar matrices fix every conjugation link (proposition 17.12), so link-only targets are never identifiable and the link term must be dressed by law or interface data, as in the example metric below. The link-only curved example below accordingly uses a compact group.

The target can retain reciprocal link matrices and backtrack factors, both transforming by conjugation, as well as laws and interfaces. Its invariant metric is an additional comparison choice: it is one smooth metric on the augmented target, including any dependence on the law or interface coordinates of that target. For example on a full-rank matrix region the terms

$$\mathrm{tr}(H_X^{-1}\delta X \delta X^T) + \sum_e \mathrm{tr}(H_X^{-1}\delta\bar{\Theta}_e H_X \delta\bar{\Theta}_e^T) \tag{17.35}$$

are positive and invariant under $X \mapsto h^{-1}X$, $\bar{\Theta}_e \mapsto h^{-1}\bar{\Theta}_e h$. Unlike the rectangular selector, the selector of this augmented metric need not preserve H_X^{-1}, so it is not subject to corollary 17.9. If the target retains reciprocal parts of tied links from both sectors, the sector-wise principal roots of eq. (17.8) need not be the images of one element of G when the two representations have different charges; a reciprocal part computed in G and then represented in each sector avoids this. One can include Fisher terms for regular statistical families. An interface or link norm is not automatically the Fisher metric of a normalized statistical experiment.

17.8.1 A curved connection from reciprocal links alone

There is a concrete link-only example for $G = SO(3)$. Identify its Lie algebra with vectors by $\widehat{v}w = v \times w$, and use the product bi-invariant metric on $SO(3)^2$, normalized by $\langle \widehat{v}, \widehat{w} \rangle = v \cdot w$. On $(x, y) \in (0, \pi)^2$, fix $\beta \in (0, \pi)$ and set

$$Q_1 = R_x(x), \qquad Q_2 = R_z(y) R_x(\beta) R_z(-y). \tag{17.36}$$

These can be the two edges of a three-agent star, with the reverse links set to their inverses. All agent backtracks are the identity. Under reference change the pair transforms by simultaneous conjugation.

Let $\mathsf{C}_r = I - \mathrm{Ad}_{Q_r}$ and $V_r = dQ_r Q_r^{-1}$. The right-trivialized infinitesimal conjugation is $\mathsf{C}_r \xi$, so eq. (17.34) reads

$$\mathfrak{k} a = -\sum_{r=1}^{2} \mathsf{C}_r^* V_r, \qquad \mathfrak{k} = \sum_{r=1}^{2} \mathsf{C}_r^* \mathsf{C}_r. \tag{17.37}$$

For a rotation through angle α about unit axis n, $(I - Q)^T (I - Q) = 4 \sin^2(\alpha/2)(I - nn^T)$. Here the axes are e_1 and $n(y) = (\cos y, \sin y, 0)^T$, which are nonparallel on the stated region. Thus $\mathfrak{k}$ is positive definite. Write $w_x = 4 \sin^2(x/2)$ and $w_\beta = 4 \sin^2(\beta/2)$. The identities $V_1 = \widehat{e}_1 dx$ and $V_2 = \mathsf{C}_2 \widehat{e}_3 dy$ imply $\mathsf{C}_1^* V_1 = 0$ and $\mathsf{C}_2^* V_2 = w_\beta \widehat{e}_3 dy$. Since $\mathfrak{k} \widehat{e}_3 = (w_x + w_\beta) \widehat{e}_3$, one obtains

$$a = -\frac{w_\beta}{w_x + w_\beta} \widehat{e}_3 dy, \qquad F^a = \frac{2 w_\beta \sin x}{(w_x + w_\beta)^2} \widehat{e}_3 dx \wedge dy \neq 0. \tag{17.38}$$

The wedge square vanishes because a has only one base component. This proves nonzero curvature from smoothly varying reciprocal agent links and a declared bi-invariant comparison metric, without adding directional generators or a spatial connection as input. The connection and its curvature take values in the line spanned by $\widehat{e}_3$, the common normal of the two axes, so the connection reduces to the rotations about e_3 and its holonomy is Abelian; the example exhibits no non-Abelian contextual holonomy. Normalized rotationally invariant agent laws may be attached with a product metric on laws and links; their orbit terms are zero and do not alter this example. The axes are selected by the links, not by those invariant laws.

17.9 What pointwise links do not determine

Proposition 17.12 (An invisible gauge subgroup). Suppose the only input fields are reference links and their contextual derivatives, with links transforming by simultaneous conjugation. If a nonzero $\zeta \in \mathfrak{g}$ commutes with every admitted link field, then no full connection selector from these inputs alone can obey eq. (17.4) for every smooth reference change $h(c) = \exp(f(c)\zeta)$.

Proof. Every link field is unchanged as a function of c, so all its derivatives are also unchanged. A selector of those data returns the same a. Choose a point c_0 and f with $f(c_0) = 0$ and $df(c_0) \neq 0$. At that point the affine law requires $a' = a + \zeta df$, a contradiction. □

The obstruction is pointwise for selectors that read only values and first derivatives of the links, as the normal equation eq. (17.34) does. If a nonzero ζ commutes with every link value $\bar{\Theta}_e(c_0)$

at one point, take $h = \exp(f\zeta)$ with $f(c_0) = 0$ and $df(c_0) \neq 0$. Then $\bar{\Theta}'_e(c_0) = \bar{\Theta}_e(c_0)$ and $d\bar{\Theta}'_e(c_0) = d\bar{\Theta}_e(c_0) + [\bar{\Theta}_e(c_0), \zeta]df(c_0) = d\bar{\Theta}_e(c_0)$, while the affine law requires $a'(c_0) = a(c_0) + \zeta df(c_0)$. No such selector is covariant at a point of nonzero common infinitesimal stabilizer.

For group-valued links the Lie algebra of the center is always invisible to conjugation. For arbitrary endomorphism-valued links, the relevant ambiguity is their actual common commutant within the represented group; a central group element need not commute with arbitrary maps between inequivalent representation blocks. In the defining $\mathrm{GL}(K)$ representation scalar matrices are invisible to every conjugation link. An Abelian group acting on its own group-valued links is entirely invisible. The $SO(3)$ example avoids this obstruction because the two nonparallel axes have zero common infinitesimal stabilizer.

Laws, labeled vectors, or interfaces can resolve an ambiguity if their combined infinitesimal action is identifiable. Full row rank of X is sufficient in the fundamental representation. It is not implied by many links: centered invariant laws and deficient interfaces can fail it.

Even after identifiability is supplied, several covariant comparison rules can select different connections. Column sets, weights, message metrics, augmented target metrics, the auxiliary fiber metric M of a proper subalgebra, and the relative sector weight affect the selector. Gauge covariance does not choose these data. A finite link collection at one context supplies no contextual derivative; the constructions require a smooth contextual field of such data.

17.10 Message coordinates, dynamics, and remaining scope

The fixed message space assumption has an intrinsic formulation. If X maps a message bundle into E and the message bundle has connection Γ, use

$$a = -(dX - X\Gamma)Q_X. \tag{17.39}$$

For a change $w' = \Lambda(c)w$ of message coordinates, $X' = X\Lambda^{-1}$, $\Gamma' = \Lambda\Gamma\Lambda^{-1} - d\Lambda\Lambda^{-1}$ and $Q'_X = \Lambda Q_X$, provided the message metric is transformed as well. Hence $dX' - X'\Gamma' = (dX - X\Gamma)\Lambda^{-1}$, and eq. (17.39) is unchanged. Fixed labeled coordinates with $\Gamma = 0$ give eq. (17.16). Resetting Γ to zero after a context-dependent message re-expression changes the model. A supplied nonflat message connection can contribute additional curvature beyond eq. (17.24).

The contextual laws, rows, interfaces and links remain supplied fields; no evolution theorem is implied. If the selected connection is inserted into a contextual energy, its complete constitutive dependence must be varied. An envelope simplification requires stationarity for that same energy. Agreement with a law-Fisher selector requires matching its target, action, metric and inputs; the shared affine law alone transfers no Fisher-flow theorem.

Dynamic selection of the comparison rule, preservation or recovery of rank, singular continuation, and inheritance of the link sector remain open. The base, its dimension and topology are supplied; no formation result follows.

Chapter 18

Communication Codes and Learned Alignment

A communication channel can preserve a transmitted marginal while leaving the receiver to supply its conditional interpretation. An invertible dictionary instead aligns complete content spaces. Keeping these objects separate permits a precise account of lost information, learned alignment, and the distinctions that a loop of communicating agents can resolve.

The constructions take place at one context of the principal bundle P of chapter 1. One group G acts on belief and model contents through ϱ_q and ϱ_s. A frame change is one group element acting in both sectors. General-law reconstruction uses standard Borel spaces and normalized kernels. For the matrix calculations, write the belief representation as $V = \mathbb{R}^K$ with $G \subseteq \mathrm{GL}(V)$ and suppress the sector superscript. The model-sector calculation uses its own representation and codes; using one dictionary for both sectors requires the tied-link hypothesis. Opposite attending dictionaries remain independent.

For sender j and receiver i, selective transmission has the form

$$q_j \longrightarrow (T_j)_{\#} q_j \longrightarrow \text{transmission noise} \longrightarrow \text{receiver conditional law} \longrightarrow q_i'. \tag{18.1}$$

This is a law-reconstruction protocol when the message law is supplied. Conditioning on one realized message is a different operation. The general-law information identities below apply to both finite alphabets and continuous contents. Gaussian assumptions enter only the explicit decoder and tangent posterior calculations.

18.1 Codes and dictionaries

Definition 18.1 (Code of a modality). For general laws, a modality is a normalized Markov kernel from a standard Borel content space to a standard Borel message space. A noiseless modality is a measurable map. The linear realization used for identifiability below is more restrictive. A modality c, such as speech, drawing or gesture, has a public message space $H_c = \mathbb{R}^{r_c}$ and a noise covariance R_c, either positive definite or zero for a noiseless code. The message space carries no frame of its own: its coordinates are public conventions. Sender j's code in modality c is a linear map $\bar{T}_j^c : V \to H_c$ of rank $r_c \le K$ on reference coordinates. Its displayed form is $T_j^c = \bar{T}_j^c U_j^{-1}$. A message produced from displayed content y_j is

$$h = T_j^c y_j + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, R_c). \tag{18.2}$$

Under a passive change $U_j \mapsto g_j U_j$ the displayed content becomes $g_j y_j$ and the displayed code becomes $T_j^c g_j^{-1}$, so the message is unchanged. Under a common reference rechoice $U_j \mapsto U_j k$ with $k \in G$, the reference code becomes $\bar{T}_j^c k$ and the reference dictionary becomes $k^{-1}\bar{\theta}_{ij}k$. Re-expressing these constitutive data is the reference-consistency requirement. Public messages therefore do not depend on the sender's frame, and the reference code $\bar{T}_j^c$ is unchanged by passive display changes at fixed reference coordinates. Codes are agent specific. The shared generative model of section 1.4 contains no code, so a shared language is an additional declared hypothesis: members of a species use codes related by one group element in reference coordinates. Whether learned codes come to satisfy this hypothesis is open problem 18.34. The identification results below state explicitly when they assume that the receiver knows the sender's reference code.

Definition 18.2 (Dictionary). Receiver i's dictionary for sender j is

$$\theta_{ij} = U_i \bar{\theta}_{ij} U_j^{-1} \in G, \qquad \bar{\theta}_{ij} \in G. \tag{18.3}$$

This is the group-valued link of definition 2.5 with the reference value taken in all of G. The general dictionary is unrestricted in G. The local parameterization $\bar{\theta}_{ij} = \exp(\omega_{ij})$ with $\omega_{ij} \in \mathfrak{g}$ is used for tangent learning. It can cover less than G: a single real matrix exponential need not cover even $\mathrm{GL}^+(K, \mathbb{R})$ [50]. The global coset and exact-identification results below concern the stated group, and a restriction to the exponential image retains only the solutions lying in that image.

It is owned by the receiver: it is i's hypothesis about how j's internal reference coordinates sit relative to its own. Content with reference coordinates $\bar{y}$ for i corresponds, according to i, to $\bar{\theta}_{ij}^{-1}\bar{y}$ for j. The effective code that i attributes to j is therefore

$$\hat{A}_{ij}^c = \bar{T}_j^c \bar{\theta}_{ij}^{-1}. \tag{18.4}$$

The dictionary θ_{ji} is j's hypothesis about i. It is an independent datum, learned by a different agent, so θ_{ji} need not equal θ_{ij}^{-1}. The product $\bar{\theta}_{ij}\bar{\theta}_{ji}$ is the holonomy of the two-cycle $i \to j \to i$: the composite of what i thinks j means with what j thinks i means. It is the identity exactly when the two agents' translations of each other are mutually inverse; otherwise it measures mutual misunderstanding. Geometric reversal of one particular dictionary is its inverse; the separately learned opposite attending edge is independent.

Remark 18.3 (Why the dictionary is group valued). A model that i holds of j need not in general be a group element, and the group-valued form is a hypothesis with a stated scope. Four considerations support it within a species. First, the two agents' internal spaces are copies of one G-space, so a hypothesis about how the sender reads its coordinates is naturally G-valued; the bundle identification of the two frames is itself the value $\bar{\theta}_{ij} = I$, and a dictionary with nontrivial holonomy is an additional edge field rather than a relation between frames. Partial access is carried by the code, not by the dictionary. Second, one dictionary serves every modality, which is what lets several codes pool their evidence in corollary 18.12. Third, invertibility is used: the consensus theorem theorem 18.17 needs inverses to show that transported beliefs are well defined along different paths, and group-valued holonomy composes invertible links; singular path products can still have trace invariants, but do not define transport by group elements. Fourth, i's knowledge of j is a probability law over G, or over ω_{ij} in a declared exponential chart; uncertainty about

the dictionary is carried by that law. For agents of different kinds, whose internal spaces differ in dimension or representation, an invertible dictionary need not exist. The natural replacement is a linear or Markov map between different spaces, with consistency measured by global sections of a sheaf rather than by holonomy [102].

18.2 Reception through a code

Fix a receiver i, a sender j and one modality, and suppress the indices. In i's reference coordinates let p be i's prior law of the content and let $q = (\bar{\theta})_{\#}\bar{q}_j$ be the sender's belief carried into i's coordinates by the dictionary. Write $A = \hat{A}$ for the effective code.

A conditional law decodes a message under a receiver prior. Reconstructing an entire sender law uses that decoder for every possible message. For a measurable code T and laws q, p on a standard Borel space, choose a regular conditional law $p(dy \mid h)$ of p given $T = h$, and define the reconstruction

$$\hat{q}(dy) = \int p(dy \mid h)(T_{\#}q)(dh). \tag{18.5}$$

It keeps exactly what the code transmits, the message law $T_{\#}q$, and fills every discarded direction with the receiver's own conditional belief.

Corollary 18.4 (Meaning lost through a noiseless code). If $D_{\mathrm{KL}}(q\|p) < \infty$, then

$$D_{\mathrm{KL}}(q\|\hat{q}) = D_{\mathrm{KL}}(q\|p) - D_{\mathrm{KL}}(T_{\#}q\|T_{\#}p). \tag{18.6}$$

This is the conditional-lift construction of proposition 5.2 after transporting the sender into the receiver's reference coordinates. The decisive proof is the chain-rule argument in corollary 5.3: its two nonnegative terms are finite under the hypothesis here, so subtraction gives eq. (18.6). The code is sufficient for the pair precisely when the discarded conditional laws agree.

In words, the meaning lost when a belief is sent through a code and rebuilt by a receiver equals the full disagreement between sender and receiver minus the disagreement that the code makes visible. It vanishes exactly when the code is sufficient for the pair, that is, when the sender's conditional belief given the message already equals the receiver's. The loss is evaluated at the true dictionary, through $q = (\bar{\theta})_{\#}\bar{q}_j$. It is not constant on the coset $\bar{\theta}\,\mathrm{Stab}_G(\bar{T})$ of theorem 18.9 and therefore cannot be estimated from messages, whereas $\hat{q}$ depends on the dictionary only through the effective code.

Corollary 18.5 (Noisy codes). Let a noisy code be a Markov kernel $\mathcal{K}$ from the content space to a standard Borel message space, for example the Gaussian kernel $\mathcal{K}(\bar{y}, \cdot) = \mathcal{N}(A\bar{y}, R)$. Let p^h and q^h be regular versions of the posteriors of p and q given h, and define the reconstruction $\hat{q} = \int p^h(q\mathcal{K})(dh)$. If $D_{\mathrm{KL}}(q\|p) < \infty$, then

$$D_{\mathrm{KL}}(q\|\hat{q}) \le D_{\mathrm{KL}}(q\|p) - D_{\mathrm{KL}}(q\mathcal{K}\|p\mathcal{K}). \tag{18.7}$$

Proof. The chain rule applied to the joint laws of $(\bar{y}, h)$ under $q \otimes \mathcal{K}$ and $p \otimes \mathcal{K}$, which share the kernel, gives $D_{\mathrm{KL}}(q\|p) = D_{\mathrm{KL}}(q\mathcal{K}\|p\mathcal{K}) + \int D_{\mathrm{KL}}(q^h\|p^h)(q\mathcal{K})(dh)$. Since $q = \int q^h(q\mathcal{K})(dh)$, joint convexity of relative entropy bounds $D_{\mathrm{KL}}(q\|\hat{q})$ by the integral. □

The proof uses only the chain rule and joint convexity, so the bound holds for every kernel and every pair of laws with $D_{\mathrm{KL}}(q\|p) < \infty$. For $q = \mathcal{N}(a, \Sigma_q)$, $p = \mathcal{N}(m, \Sigma_p)$, and the Gaussian kernel, $\hat{q}$ has mean $m + OA(a - m)$ and covariance $\Sigma_p - OA\Sigma_p + O(A\Sigma_q A^T + R)O^T$ with O from eq. (18.8).

Proposition 18.6 (Reception of one message). Let $p = \mathcal{N}(m, \Sigma_p)$ with Σ_p positive definite, let A be an $r \times K$ matrix, and let R be positive definite. Given a message h from the model $h = A\bar{y} + \varepsilon$, the posterior of $\bar{y}$ is $\mathcal{N}(m + O(h - Am), \Sigma_p - OA\Sigma_p)$ with

$$O = \Sigma_p A^T (A\Sigma_p A^T + R)^{-1}. \tag{18.8}$$

The posterior precision is $\Sigma_p^{-1} + A^T R^{-1} A$, a correction of rank at most r, and the information carried by one message is

$$I(\bar{y}; h) = \tfrac{1}{2} \log\det\big(I_r + R^{-1} A\Sigma_p A^T\big). \tag{18.9}$$

Proof. The pair $(\bar{y}, h)$ is jointly Gaussian with $\mathrm{Cov}(\bar{y}, h) = \Sigma_p A^T$ and $\mathrm{Cov}(h) = A\Sigma_p A^T + R$, and Gaussian conditioning gives the posterior [23, Section 2.3.1]; this is the update step of the Kalman filter [115]. Since R is positive definite, $A\Sigma_p A^T + R$ is invertible without a rank condition on A. The precision form follows from the Woodbury identity. The mutual information is the difference of the entropies of $\bar{y}$ and $\bar{y}$ given h, $\frac{1}{2}\log\det\Sigma_p - \frac{1}{2}\log\det(\Sigma_p - OA\Sigma_p)$, and the matrix determinant lemma turns this into eq. (18.9). □

The receiving map is therefore derived. It is the gain O, fixed by the receiver's own uncertainty Σ_p and by its model A of the sender's code. In this Gaussian instance one message changes the receiver's precision in at most r directions, however large K is.

This is the Gaussian instance of conditional decoding. That the decoder is a linear map, that the posterior covariance does not depend on the message received, and that one message changes the precision in at most r directions are features of the Gaussian family, not of decoding in general.

Proposition 18.7 (Gaussian reconstruction). Let $q = \mathcal{N}(a, \Sigma_q)$, $p = \mathcal{N}(m, \Sigma_p)$ with positive-definite covariances, and let A have full row rank. Put

$$O_0 = \Sigma_p A^T (A\Sigma_p A^T)^{-1}, \qquad P = O_0 A. \tag{18.10}$$

Then $AO_0 = I_r$, $P^2 = P$, $\ker P = \ker A$, the range of P is the range of $\Sigma_p A^T$, and the reconstruction through the noiseless code A is

$$\hat{q} = \mathcal{N}\big((I - P)m + Pa,\ (I - P)\Sigma_p (I - P)^T + P\Sigma_q P^T\big). \tag{18.11}$$

Proof. The identities for O_0 and P are immediate. Conditioning p on $A\bar{y} = h$ gives $\mathcal{N}(m + O_0(h - Am), \Sigma_p - O_0 A\Sigma_p)$. Mixing over $h \sim \mathcal{N}(Aa, A\Sigma_q A^T)$ gives mean $m + O_0 A(a - m) = (I - P)m + Pa$ and covariance $\Sigma_p - P\Sigma_p + O_0 A\Sigma_q A^T O_0^T$. The last term is $P\Sigma_q P^T$. The matrix $P\Sigma_p = \Sigma_p A^T (A\Sigma_p A^T)^{-1} A\Sigma_p$ is symmetric and satisfies $P\Sigma_p P^T = P\Sigma_p$, so $\Sigma_p - P\Sigma_p = (I - P)\Sigma_p (I - P)^T$. □

The reconstructed belief adopts the sender's belief along the communicated directions and keeps the receiver's own belief along the discarded ones. Its update of the mean,

$$\mu_i' = \mu_i + O_0 \hat{A}_{ij}\big(\bar{\theta}_{ij}\bar{\mu}_j - \mu_i\big), \tag{18.12}$$

is a projected message of the kind computed by attention. The direction in which a communicated coordinate is lifted, the range of $\Sigma_p A^T$, is set by the receiver's own correlations: two receivers with different priors read the same message as meaning different things. The linear lift is the Gaussian instance of a general-law statement: by eq. (18.5) the receiver keeps the transmitted marginal and supplies its own conditional law given the coded statistic, so the meaning attached to a message is fixed by the receiver's conditional structure. The loss eq. (18.6) evaluates in closed form for eq. (18.11).

Remark 18.8 (The link with a code-shaped generator). For matrices O of size $K \times r$ and T of size $r \times K$, $e^{OT} = I + O\varphi_1(TO)T$ with $\varphi_1(z) = (e^z - 1)/z$ [107, Section 1.8], and $\det e^{OT} = e^{\mathrm{tr}(OT)} > 0$. With the noiseless gain of eq. (18.10), $AO_0 = I_r$ gives $e^P = I + (e-1)P$: the invertible link whose generator is the reconstruction projector is a dilation along the communicated directions, not the reconstruction. Reconstruction is the affine use $(I-P)m + Pa$ of the projector; as a map of the sender mean a with fixed receiver mean m, it is noninvertible when $r < K$. For a square full-rank code, $P = I$ and this particular loss disappears. Thus no group element factors through a code of rank $r < K$, and a code-shaped generator OT exponentiates to a link that acts as the identity on $\ker T$. Codes and group elements are nevertheless related directly, through the right action in $\hat{A} = \bar{T}\bar{\theta}^{-1}$ and through the stabilizer $\mathrm{Stab}_G(\bar{T})$ of the next section.

18.3 What messages reveal about the dictionary

A dictionary can be tested only against something both agents refer to. Let a referent have law ρ on V in i's reference coordinates, with finite positive-definite second moment $M = \mathbb{E}_\rho[\bar{y}\bar{y}^T]$. The sender encodes the referent in its own coordinates, and the receiver models the message as

$$h \mid \bar{y} \sim \mathcal{N}\big(\bar{T}\bar{\theta}^{-1}\bar{y}, R\big), \tag{18.13}$$

with R positive definite.

Theorem 18.9 (Identifiability up to the code stabilizer). Assume that the receiver knows the sender's reference code $\bar{T}$, for example through a declared shared language in reference coordinates, so that only the dictionary is unknown. Let $\mathrm{Stab}_G(\bar{T}) = \{g \in G : \bar{T}g = \bar{T}\}$. For parts (i) and (iii), allow any fixed additive-noise law on $\mathbb{R}^r$, independent of the referent; part (ii) assumes the Gaussian noise of eq. (18.13).

(i) Two dictionaries $\bar{\theta}', \bar{\theta}''$ give the same conditional message law for every referent if and only if $\bar{\theta}'^{-1}\bar{\theta}'' \in \mathrm{Stab}_G(\bar{T})$.

(ii) For the perturbation $\bar{\theta}e^\delta$, $\delta \in \mathfrak{g}$, the Fisher information of eq. (18.13), averaged over ρ, is

$$F(\delta,\delta) = \mathrm{tr}\big(R^{-1}\bar{T}\delta C \delta^T \bar{T}^T\big), \qquad C = \bar{\theta}^{-1} M \bar{\theta}^{-T}, \tag{18.14}$$

and its kernel is $\{\delta \in \mathfrak{g} : \bar{T}\delta = 0\}$, the Lie algebra of $\mathrm{Stab}_G(\bar{T})$.

(iii) For $G = \mathrm{GL}(K, \mathbb{R})$ and $\bar{T}$ of rank r, $\mathrm{Stab}_G(\bar{T})$ has dimension $K(K-r)$, and the identifiable part of the dictionary has dimension Kr.

Proof. (i) A probability law on $\mathbb{R}^r$ has no nonzero translation symmetry: its characteristic function is nonzero near the origin, where translation by a would require $e^{\mathrm{i}\langle t,a\rangle} = 1$ and hence $a = 0$. Thus two translates of the same noise law agree exactly when their locations agree, and $\bar{T}\bar{\theta}'^{-1}\bar{y} = \bar{T}\bar{\theta}''^{-1}\bar{y}$ for every $\bar{y}$ means $\bar{T}\bar{\theta}'^{-1} = \bar{T}\bar{\theta}''^{-1}$, that is, $\bar{T}\bar{\theta}'^{-1}\bar{\theta}'' = \bar{T}$. The set $\mathrm{Stab}_G(\bar{T})$ is a subgroup because $\bar{T}g_1g_2 = \bar{T}g_2 = \bar{T}$. (ii) The mean $\bar{T}e^{-\delta}\bar{\theta}^{-1}\bar{y}$ has derivative $-\bar{T}\delta\bar{\theta}^{-1}\bar{y}$ at $\delta = 0$, and the Fisher information of a Gaussian mean with fixed covariance R is $\mathbb{E}_\rho[(\bar{T}\delta\bar{\theta}^{-1}\bar{y})^T R^{-1}(\bar{T}\delta\bar{\theta}^{-1}\bar{y})]$, which is eq. (18.14). Since C and R^{-1} are positive definite, $F(\delta,\delta) = 0$ exactly when $R^{-1/2}\bar{T}\delta C^{1/2} = 0$, that is, when $\bar{T}\delta = 0$. For such δ every power of δ is annihilated by $\bar{T}$, so $\bar{T}e^{t\delta} = \bar{T}$ for all t. (iii) $\bar{T}g = \bar{T}$ means that $g - I$ maps V into $\ker\bar{T}$, which has dimension $K - r$; such g form an open subset of an affine space of dimension $K(K-r)$ containing the identity. □

Messages never test the part of the dictionary that the code discards. Relative to the observable algebra of coded messages that part is gauge in the sense of the observable-algebra convention of chapter 1: no message distinguishes two dictionaries that differ by an element of the stabilizer. In interpretive terms this is an instance of the indeterminacy of translation [168]; the correspondence is not a theorem about that thesis. The known-code hypothesis is essential. Messages depend on the pair $(\bar{T}, \bar{\theta})$ only through the effective code $\hat{A} = \bar{T}\bar{\theta}^{-1}$, and $(\bar{T}g)(\bar{\theta}g)^{-1} = \bar{T}\bar{\theta}^{-1}$ for every $g \in G$. If the reference code is unknown, the grounded experiment therefore identifies $\hat{A}$ and nothing about $\bar{\theta}$ separately. The stated full identification uses shared referents with a positive-definite second moment. If only one referent vector were supplied, the conditional means would determine $\bar{T}\bar{\theta}^{-1}$ only on one vector. Noise rescales the Fisher information but does not change its kernel.

Remark 18.10 (Compact dictionary groups). Part (iii) gives the largest invisible part. Let $G = O(K)$ or $SO(K)$ and let $\bar{T}$ have rank r. For orthogonal g, $\bar{T}g = \bar{T}$ means g^T, and hence $g = g^{-T}$, fixes the row space of $\bar{T}$ pointwise; conversely such g satisfy $\bar{T}g = \bar{T}$. Thus $\mathrm{Stab}_G(\bar{T})$ is the pointwise stabilizer of $\mathrm{row}\,\bar{T}$, isomorphic to $O(K-r)$ or $SO(K-r)$ through its action on $\ker\bar{T}$, of dimension $(K-r)(K-r-1)/2$, and the identifiable dimension is $r(2K-r-1)/2$. A code of rank $K-1$ identifies an $SO(K)$ dictionary completely, whereas in $\mathrm{GL}(K,\mathbb{R})$ it leaves K invisible directions. The invisible, unreachable, and detectable spaces of section 18.7 shrink accordingly for compact dictionary groups; their counts are computed below only for $\mathfrak{gl}(K,\mathbb{R})$.

Remark 18.11 (Noise laws beyond the Gaussian). Gaussian noise is a computational simplification here. Let the grounded model be the location family $h = \bar{T}\bar{\theta}^{-1}\bar{y} + \varepsilon$ with ε independent of $\bar{y}$ and of a fixed law ν on $\mathbb{R}^r$. Parts (i) and (iii) of theorem 18.9 hold for every ν, because a probability law on $\mathbb{R}^r$ differs from each of its nonzero translates: its characteristic function φ is nonzero near the origin, and $e^{\mathrm{i}\langle t,a\rangle}\varphi(t) = \varphi(t)$ there forces $a = 0$. If ν has a density f with $\sqrt{f} \in H^1(\mathbb{R}^r)$, equivalently finite Fisher information $J_f = 4\int \nabla\sqrt{f}(\nabla\sqrt{f})^T$ (for example a continuously differentiable density with finite J_f, or the Laplace density), then the location family is differentiable in quadratic mean [190, Chapter 7], and so is the referent-averaged model because M is finite. The matrix J_f is then positive definite, since $a^T J_f a = 0$ would make $\sqrt{f}$ constant along a and hence not square

integrable. Part (ii) holds with R^{-1} replaced by J_f,

$$F(\delta,\delta) = \operatorname{tr}\big(J_f \bar{T}\delta C \delta^T \bar{T}^T\big), \tag{18.15}$$

with the same kernel, since the score of the perturbation is $\nabla \log f(\varepsilon)^T \bar{T}\delta\bar{\theta}^{-1}\bar{y}$ and ε is independent of $\bar{y}$. Corollaries 18.12 and 18.13 and the structural results of section 18.7 therefore hold for every such noise law. When ε has finite covariance R and integration by parts gives $\mathbb{E}[\nabla \log f(\varepsilon)(\varepsilon - \mathbb{E}\varepsilon)^T] = -I$, positive semidefiniteness of the joint covariance of the score and ε gives the location Cramér–Rao inequality $J_f \succeq R^{-1}$, with equality exactly when the score is affine, that is, for Gaussian noise. The Gaussian value $J_f = R^{-1}$ is therefore the least Fisher information at a given covariance. For noise laws without differentiability in quadratic mean, such as uniform noise, only parts (i) and (iii) remain. Codes whose messages are not a location family, such as codes into finite alphabets, are open problem 18.36.

18.4 Several modalities

Corollary 18.12 (Pooling modalities through one dictionary). Let modalities $c = 1, \dots, C$ have codes $\bar{T}^c$ and noises R_c, all read through the same dictionary $\bar{\theta}$. Assume the messages are conditionally independent given their referents and dictionary. Then the dictionary is identified up to $\bigcap_c \operatorname{Stab}_G(\bar{T}^c)$, and the Fisher information is $\sum_c F_c$ with kernel $\{\delta \in \mathfrak{g} : \bar{T}^c\delta = 0 \text{ for all } c\}$. For $G = \mathrm{GL}(K,\mathbb{R})$ the identifiable dimension is

$$K \dim\Big(\sum_c \operatorname{row} \bar{T}^c\Big) \le K \min\Big(K, \sum_c r_c\Big), \tag{18.16}$$

If $\sum_c r_c \le K$, equality in the bound holds exactly when the row spaces form a direct sum. If $\sum_c r_c \ge K$, equality holds exactly when their sum is V^*; independence of the row spaces is then unnecessary. The dictionary is fully identified exactly when the row spaces span V^*.

Proof. Identity of the conditional laws of all messages requires identity for each modality, which gives the intersection by theorem 18.9. Fisher information adds over conditionally independent messages, and a sum of positive-semidefinite forms vanishes exactly where each does. For $\mathrm{GL}(K,\mathbb{R})$, $\bar{T}^c\delta = 0$ for all c means that the range of δ lies in $\bigcap_c \ker \bar{T}^c$, whose dimension is K minus the dimension of the sum of the row spaces. □

Modalities that overlap, in the sense that their row spaces share directions, add less than their widths suggest; complementary modalities, such as speech and drawing when they convey different aspects, add fully. The corollary depends on the dictionary being shared across modalities. Without it each modality would have its own effective code to learn, and the modalities could not pool their evidence about the other agent.

Corollary 18.12 assumes that receiver i hears sender j in every modality. A receiver with several senders instead chooses, separately in each modality, whom to listen to. Let $\beta_{ij}^{(c)}$ be the probability that modality c of receiver i carries a message from j, and let $F_c^{(j)}$ denote the form eq. (18.14) for the code $\bar{T}_j^c$, the noise R_c and the dictionary $\bar{\theta}_{ij}$.

Corollary 18.13 (Routed modalities). In each exchange let a referent $\bar{y} \sim \rho$ be drawn, and for each modality c independently let a source label J_c be drawn with $\mathbb{P}(J_c = k) = \beta_{ik}^{(c)}$ and a message with law $\mathcal{N}(\bar{T}_k^c \bar{\theta}_{ik}^{-1} \bar{y}, R_c)$ given $J_c = k$ and $\bar{y}$. Consider the dictionary $\bar{\theta}_{ij} e^{\delta}$ of one edge $j \to i$.

(i) If the labels are observed, the Fisher information about δ per exchange is $\sum_c \beta_{ij}^{(c)} F_c^{(j)}$.

(ii) If the labels are latent, the Fisher information is at most $\sum_c \beta_{ij}^{(c)} F_c^{(j)}$ in the Loewner order.

(iii) With observed labels the information vanishes exactly on $\{\delta \in \mathfrak{g} : \bar{T}_j^c \delta = 0$ for every c with $\beta_{ij}^{(c)} > 0\}$.

Proof. (i) Given $\bar{y}$ and the labels, the messages of different modalities are independent, so the complete-data log likelihood is a sum over c. A term with $J_c = k \neq j$ does not depend on $\bar{\theta}_{ij}$ and contributes nothing. A term with $J_c = j$ contributes the information of theorem 18.9. Averaging the indicator of $J_c = j$ gives the weight $\beta_{ij}^{(c)}$. (ii) The observed-data score is the conditional expectation of the complete-data score given the messages, so by the law of total variance the observed information equals the complete-data information minus the expected conditional covariance of the complete-data score given the messages, which is positive semidefinite; this is the identity of Louis [141] for incomplete-data information. (iii) A sum of positive-semidefinite forms with positive weights vanishes exactly where every form vanishes, and theorem 18.9 identifies the kernel of each. □

A modality that never listens to j teaches i nothing about its dictionary for j, and the directions that the attending modalities' codes discard stay invisible. When the receiver must infer who was heard, it learns less than the routing-weighted sum.

Proposition 18.14 (A receiver's reading of a code is a dictionary modulo the stabilizer). Let $\bar{T}$ and S be $r \times K$ matrices of full row rank r. Then $\{\theta \in \mathrm{GL}(K, \mathbb{R}) : \bar{T}\theta^{-1} = S\}$ is a single left coset $\theta_0 \, \mathrm{Stab}_{\mathrm{GL}(K,\mathbb{R})}(\bar{T})$, and for $r < K$ it meets $\mathrm{GL}^+(K, \mathbb{R})$.

Proof. Complete the rows of $\bar{T}$ to a basis of the dual space and let C_T be the invertible matrix whose first r rows are those of $\bar{T}$; define C_S likewise from S. With $E = (I_r \; 0)$, $\bar{T} = E\mathsf{C}_T$ and $S = E\mathsf{C}_S$, so $\theta_0^{-1} = \mathsf{C}_T^{-1}\mathsf{C}_S$ satisfies $\bar{T}\theta_0^{-1} = S$. For $r < K$, changing the sign of a completing row of C_S changes the sign of $\det \theta_0$ without changing $E\mathsf{C}_S$. Finally $\bar{T}\theta^{-1} = \bar{T}\theta_0^{-1}$ exactly when $\bar{T}\theta_0^{-1}\theta = \bar{T}$, that is, when $\theta_0^{-1}\theta \in \mathrm{Stab}(\bar{T})$. □

A receiver that reads a sender's code through an arbitrary full-rank map S therefore holds exactly the identifiable part of a dictionary, a point of the rK-dimensional quotient of theorem 18.9. The proposition also applies to the stacked code $(\bar{T}^1; \ldots; \bar{T}^C)$ and the stacked readings $(S_1; \ldots; S_C)$. When both stacks have full row rank $\sum_c r_c \leq K$, there is one dictionary $\bar{\theta}$ with $S_c = \bar{T}^c \bar{\theta}^{-1}$ for every c, unique modulo $\bigcap_c \mathrm{Stab}(\bar{T}^c)$: readings learned independently for each modality are then automatically readings of one shared dictionary. For arbitrary stacks, equal ranks alone do not establish compatibility.

Proposition 18.15 (Compatibility of stacked readings). Let T_*, S_* be $m \times K$ matrices, with no full-rank assumption. There is a dictionary $\theta \in \mathrm{GL}(K, \mathbb{R})$ satisfying $T_*\theta^{-1} = S_*$ exactly when

$$\ker(T_*^T) = \ker(S_*^T). \tag{18.17}$$

When this holds, the solution set is $\theta_0 \operatorname{Stab}_{\mathrm{GL}(K,\mathbb{R})}(T_*)$ for any one solution θ_0. If the common rank is less than K, the solution set meets $\mathrm{GL}^+(K,\mathbb{R})$.

Proof. Right multiplication by an invertible matrix preserves every linear relation among rows, proving necessity. Conversely, equality of the kernels makes the map $a^T T_* \mapsto a^T S_*$ a well-defined linear isomorphism between the row spaces. Extend bases of these row spaces to bases of $(\mathbb{R}^K)^*$. Extending that isomorphism to the full dual space gives right multiplication by an invertible matrix A, with $T_* A = S_*$. Put $\theta = A^{-1}$. The coset statement follows as in proposition 18.14. When the common rank is less than K, the extension can reverse one unused basis vector, changing the determinant sign without changing the rows of S_*. □

For example, $T_1 = T_2 = (1, 0)$, $S_1 = (1, 0)$, and $S_2 = (2, 0)$ give two rank-one stacks but no common dictionary: the relation $T_1 - T_2 = 0$ fails for the readings. Sharing also constrains the readings when one dictionary must serve routing and content roles, as discussed in remark 18.16.

Remark 18.16 (Attention heads as modalities). The separation of a sending map, a receiver reading, a decoder, and a routing row provides a structural comparison with multi-head attention [193]. A head reads a finite-width code and routes sources, but the usual query, key, value, and output maps are independently parameterized. Such an architecture is not automatically the Bayesian message model above. Under the full-row-rank hypothesis of proposition 18.14, a reading map can be expressed as a dictionary modulo its stabilizer; several readings share a dictionary precisely under eq. (18.17). Requiring that same dictionary for source selection and content interpretation, and deriving the decoder from the receiver prior as in eq. (18.8), imposes additional restrictions. A learned output matrix alone does not establish those restrictions.

18.5 Curvature as dissonance

Let the communication graph Γ have a directed edge $j \to i$ when i receives from j, with dictionaries $\bar{\theta}_{ij}$. For a directed closed walk $\gamma = (i_0 \to i_1 \to \cdots \to i_k = i_0)$ define its holonomy $H_\gamma = \bar{\theta}_{i_k i_{k-1}} \cdots \bar{\theta}_{i_1 i_0}$, the composite of the receivers' dictionaries along the walk. Reference belief means $(\mu_i)_i$ are in consensus when

$$\mu_i = \bar{\theta}_{ij} \mu_j \qquad \text{for every edge } j \to i, \tag{18.18}$$

so that every receiver, translating each sender's belief through its dictionary, finds agreement.

Theorem 18.17 (Consensus and holonomy invariance). Let Γ be strongly connected and fix i_0. Suppose the dictionaries act as bimeasurable bijections on a standard Borel content space. The map $(q_i)_i \mapsto q_{i_0}$ is a bijection from transported consensus laws, satisfying $q_i = (\bar{\theta}_{ij})_\# q_j$ on every edge, onto laws invariant under every directed closed-walk holonomy based at i_0. In the linear realization, the map $(\mu_i)_i \mapsto \mu_{i_0}$ is a linear bijection from the solutions of eq. (18.18) onto $\bigcap_\gamma \operatorname{Fix}(H_\gamma)$.

Proof. Composing the edge relations around a closed walk proves that a consensus base law is invariant under its holonomy. Conversely, let q_0 have that invariance. For each vertex i, choose a directed path p from i_0 to i and set $q_i = \Pi(p)_\# q_0$, where $\Pi(p)$ is the ordered product of the

dictionaries. If p' is another such path, strong connectivity supplies a directed return path s from i to i_0. Both $\Pi(s)\Pi(p)$ and $\Pi(s)\Pi(p')$ are loop holonomies and preserve q_0. Pushforward by the bimeasurable inverse $\Pi(s)^{-1}$ gives $\Pi(p)_{\#}q_0 = \Pi(p')_{\#}q_0$. The definition is independent of the path. Appending an edge to a path gives the required edge relation, and the base law determines the entire family. For vectors, replace pushforward by matrix multiplication in the same argument. This gives path-independent values $\mu_i = \Pi(p)v$ for every common fixed vector v, and both directions are linear. □

The theorem uses the invertibility of the dictionaries and the directed return paths. A two-cycle is included among its loops and constrains consensus alongside longer loops. If the matrix holonomies have no common nonzero fixed vector, the only consensus mean is zero. This does not exclude a nontrivial invariant law: a centered isotropic Gaussian, for example, is invariant under rotation holonomies. For Gaussian laws, consensus constrains both mean and covariance; for general laws, it constrains every feature moved by the holonomies. In the language of group synchronization, consensus means are the global sections of the flat bundle that the dictionaries define on Γ, and holonomy invariance is the cycle-consistency condition of that problem [86, 179]; the directed version here keeps independent opposite dictionaries.

Corollary 18.18 (Frustration). For positive weights w_{ij}, the consensus energy $E(\mu) = \sum_{j\to i} w_{ij}\|\mu_i - \bar{\theta}_{ij}\mu_j\|^2$ is a positive-semidefinite quadratic form whose kernel is the solution set of eq. (18.18). If Γ is strongly connected, its least eigenvalue on configurations of unit norm is positive exactly when the directed closed-walk holonomies based at one vertex i_0 have no common nonzero fixed vector.

Proof. $E(\mu) = \|B_\Gamma \mu\|^2$ for the weighted incidence operator B_Γ with blocks $\sqrt{w_{ij}}(I, -\bar{\theta}_{ij})$, so $E = \mu^T B_\Gamma^T B_\Gamma \mu$ and its kernel is $\ker B_\Gamma$. The least eigenvalue of $B_\Gamma^T B_\Gamma$ is positive exactly when $\ker B_\Gamma = 0$, and under strong connectivity theorem 18.17 identifies $\ker B_\Gamma$ with the common fixed space of the holonomies based at i_0. □

The kernel statement holds on every graph; the holonomy criterion needs the return paths. On the acyclic graph $1 \to 2$, $2 \to 3$, $1 \to 3$ there is no closed walk, so the empty family of holonomies fixes every vector, while generic invertible dictionaries give $\ker B_\Gamma = 0$. Under strong connectivity the holonomy groups at different base points are conjugate, so the choice of i_0 does not matter. For orthogonal links on undirected graphs with the inverse convention this operator is the connection Laplacian of Singer and Wu [180], whose least eigenvalue measures frustration [12]. The directed version above admits independent opposite dictionaries.

Definition 18.19 (Dissonance of a loop). The dissonance of a directed closed walk γ is the conjugacy class of H_γ. A declared elementary cycle, or plaquette in the lattice terminology of Wilson [199], is consonant when its holonomy is the identity and dissonant otherwise. For compact G, a continuous conjugation-invariant defect such as $\|H_\gamma - I\|_F$ is a quantitative label that vanishes only at consonance. For $G = \mathrm{GL}(K, \mathbb{R})$ no continuous conjugation-invariant function of H_γ alone vanishes only at I: conjugating $I + E_{12}$ by $\mathrm{diag}(t, 1, \ldots, 1)$ and letting $t \to 0$ gives I. The quantitative label is then the covariance-dressed defect $Q(H_\gamma; \Sigma_{i_0})$ below, with the covariance of the base agent, which is invariant under joint re-expression of the loop and the base covariance and is not a function

of the conjugacy class alone. It is the loop defect of proposition 2.12, stated there for two-cycles, applied to a walk based at i_0; the proof is unchanged.

Remark 18.20 (A law comparison and a dressed link comparison). The covariance-dressed defect

$$Q(H;\Sigma) = \tfrac{1}{2}\|\Sigma^{-1/2}(I-H)\Sigma^{1/2}\|_F^2, \qquad \Sigma > 0, \tag{18.19}$$

vanishes only at $H = I$ and has a positive-definite tangent form on every matrix Lie algebra. Its nonlinear use does not introduce law-stabilizer directions. In contrast, the law-comparison penalty $D_{\mathrm{KL}}(q\|H_{\#}q)$ vanishes whenever $H_{\#}q = q$ and can have a larger null space. Its local information geometry requires a declared regular family of laws. For a quarter-turn H and $q = \mathcal{N}(0, I_2)$, the law-comparison penalty is zero while $Q(H; I_2) = 2$.

Observability depends on the record experiment. Comparing coded means tests the weaker condition $\bar{T}_j^c(\mu_j - \bar{\theta}_{ij}^{-1}\mu_i) = 0$ on each edge and modality. Grounded messages with a positive-definite referent second moment instead identify the whole effective code, and hence the dictionary modulo its code stabilizer, as in theorem 18.9. The tangent statement proposition 18.25 concerns this latter experiment. A perturbation can fix one operating content while changing the code on other referents, so invisibility at one content does not imply invisibility to grounded messages. Separately, the law-comparison cost $D_{\mathrm{KL}}(q\|H_{\#}q)$ is blind to any holonomy stabilizing q. The covariance-dressed link defect eq. (18.19) is an observable of the link and covariance together and vanishes only at identity. These are different comparisons.

18.6 Bandwidth and source rows

Proposition 18.21 (Coded messages give Gibbs rows at temperature one over the message count). Let receiver i consider finitely many candidate sources j with prior $\pi_j > 0$, each modeled as emitting independent messages with density ν_j on the message space. Let $h_1, h_2, \ldots$ be independent with density ν, and suppose $\mathbb{E}_\nu|\log \nu_j| < \infty$ for every j. Then, almost surely,

$$\frac{1}{n}\log\frac{\mathbb{P}(J = j \mid h_{1:n})}{\mathbb{P}(J = k \mid h_{1:n})} \longrightarrow \mathbb{E}_\nu[\log \nu_j - \log \nu_k]. \tag{18.20}$$

If both $D_{\mathrm{KL}}(\nu\|\nu_j)$ and $D_{\mathrm{KL}}(\nu\|\nu_k)$ are finite, this limit is $D_{\mathrm{KL}}(\nu\|\nu_k) - D_{\mathrm{KL}}(\nu\|\nu_j)$. Integrability of $\log \nu$ under ν, in addition to the candidate-log assumptions, is a sufficient condition for that representation for every pair.

Proof. By Bayes' rule the left side is $\frac{1}{n}\log(\pi_j/\pi_k) + \frac{1}{n}\sum_{t\le n}[\log \nu_j(h_t) - \log \nu_k(h_t)]$. The strong law of large numbers sends the average to the integrable expected log-likelihood ratio in eq. (18.20). If both relative entropies are finite, subtract their defining logarithmic integrals; the common $\log \nu$ term cancels. Without that finiteness, subtracting two infinite relative entropies is undefined. □

The finite-sample posterior is exactly the Gibbs row $\pi_j \exp(-nC_{j,n})/Z_n$, with empirical cost $C_{j,n} = -n^{-1}\sum_{t=1}^n \log \nu_j(h_t)$. When all candidate relative entropies are finite, its limiting relative exponential rates are those of $\pi_j e^{-nD_{\mathrm{KL}}(\nu\|\nu_j)}/Z$, a Gibbs row at temperature $1/n$, the posterior concentration of a possibly misspecified model [21]. The costs are divergences between message laws. When the true source and candidate j emit through one common kernel $\mathcal{K}$, data processing gives

$D_{\mathrm{KL}}(q\mathcal{K}\|q_j\mathcal{K}) \le D_{\mathrm{KL}}(q\|q_j)$, with equality exactly when $\mathcal{K}$ is sufficient for the pair; when a common noiseless injective code is used, the two divergences are equal by invariance under a measurable injection. For different codes no such comparison holds: with $q = q_j = \mathcal{N}(0,1)$, true code 1 and candidate code 2, the message-law divergence is positive while the content laws agree. This identifies a cost, not a population dynamics: the normalized global model, the per-agent model, and the reciprocal potential remain distinct objectives. Finite message count and selective coding control different aspects of the source posterior.

18.7 Learning the dictionaries

Dictionary learning requires a likelihood and a prior in addition to the communication maps. Theorem 18.9 says what messages can teach: a coset of the stabilizer of the code. This section asks what a prior that rewards agreement around loops adds to that, and answers the question exactly in the tangent model, the linearization at the identity dictionaries. Centering the prior at identity is a declared alignment hypothesis: before observing messages, the receiver expects the sender's dictionary to take its bundle-identification value $\bar{\theta} = I$. In interpretive terms the anchor formalizes a principle of charity in radical interpretation [52]; the correspondence is interpretive.

The tangent model is a computational simplification in two respects: it linearizes at the identity dictionaries, and it takes Gaussian noise and a Gaussian anchor. Its conclusions split accordingly. The invisible directions, the unreachable directions and their sheaf count, the split of the loop space into filled and detectable directions as a dimension count, and the exact identification results propositions 18.29 and 18.30 depend only on the identifiability kernel and on the loops; by remark 18.11 they hold for every additive noise law and every prior. So does the disintegration statement of theorem 18.23(i): at every sample size the recognition law keeps the prior conditional law of the invisible coordinates given the identified ones. The limiting law as a conditioned Gaussian, the shrinkage and bias formulas, and the variances of remark 18.32 are features of the Gaussian family.

18.7.1 The tangent model

Definition 18.22 (Tangent learning model). Let Γ be a finite communication graph with vertex set $\mathcal{V}$ and edge set $\mathcal{E}$, and let the edge $e : j \to i$ carry receiver i's generator $\omega_e = \omega_{ij} \in \mathfrak{g} \subseteq \mathfrak{gl}(K,\mathbb{R})$, so that $\omega = (\omega_e)_e \in X = \mathfrak{g}^{\mathcal{E}}$. Write $\bar{T}_e$ for the stacked code of the modalities with $\beta_{ij}^{(c)} > 0$ in corollary 18.13. Let $\mathcal{L}$ be a finite set of directed closed walks of Γ that contains every two-cycle, each walk listed once up to the cyclic choice of its starting vertex, so that $i \to j \to i$ and $j \to i \to j$ are one element of $\mathcal{L}$, with one form W_γ and, when that form is dressed by a covariance, one declared base agent. Define the linearized holonomy $B : X \to \mathfrak{g}^{\mathcal{L}}$ by $(B\omega)_\gamma = \sum_{e\in\gamma} \omega_e$, an edge being counted as often as γ traverses it. The tangent learning model consists of the prior law with density proportional to

$$\exp\Big(-\tfrac{1}{2}\omega^T\Pi_0\omega - \tfrac{\lambda}{2}\sum_{\gamma\in\mathcal{L}}\|(B\omega)_\gamma\|^2_{W_\gamma}\Big), \tag{18.21}$$

where Π_0 is positive definite on X, $\lambda \ge 0$, and each W_γ is a positive-definite quadratic form on $\mathfrak{g}$, with $W = \bigoplus_\gamma W_\gamma$; and of the messages that i attributes to j, with observed labels as in corollary 18.13(i),

each produced from a referent $\bar{y} \sim \rho$ with finite second moment M positive definite as

$$h = \bar{T}_e^c(I - \omega_e)\bar{y} + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, R_c). \tag{18.22}$$

The prior is Gaussian with precision $\Lambda_0 = \Pi_0 + \lambda B^T W B$, which is positive definite, so the prior is normalized for every $\lambda \geq 0$. The message model is the first-order form of eq. (18.13) at $\bar{\theta}_e = e^{\omega_e}$, and $(B\omega)_\gamma$ is the first-order form of $\log H_\gamma$. By the Baker–Campbell–Hausdorff formula both errors are of second order in ω. Inside the tangent model the likelihood is Gaussian and linear in ω. The recognition law, here the exact posterior, is therefore Gaussian after any number of messages and is updated one message at a time by the information filter

$$\Lambda \leftarrow \Lambda + J_t^T R_c^{-1} J_t, \qquad \eta \leftarrow \eta + J_t^T R_c^{-1}\big(h_t - \bar{T}_e^c \bar{y}_t\big), \qquad J_t\omega = -\bar{T}_e^c \omega_e \bar{y}_t, \tag{18.23}$$

started at $(\Lambda_0, 0)$, with mean $\Lambda^{-1}\eta$ and covariance Λ^{-1}. This is one normalized joint over the dictionaries of all receivers, so the tangent learning model is the global picture of chapter 4. The receiver-owned restriction of its recognition law is remark 18.32; per-agent models in which peers' laws are data are a different objective, treated in chapter 19.

The loop prior is a Gaussian, noncompact lattice gauge action in the linearized holonomies [199] and a cycle-consistency penalty in the sense of group synchronization [86]. Its anchor removes the zero modes $\ker B$ that the loop term alone leaves free. The kernels, ranks and dimension counts below do not depend on how often a walk is listed in $\mathcal{L}$, whereas the shrinkage factors, the bias and the detectable dissonance do; proposition 18.28 lists the two-cycle once.

A change of the common reference coordinates by $k \in G$ acts by $\omega_e \mapsto k^{-1}\omega_e k$ under reference consistency, and then $(B\omega)_\gamma \mapsto k^{-1}(B\omega)_\gamma k$. The loop term is invariant when each W_γ is. The Frobenius norm is invariant under orthogonal k. For $\mathrm{GL}(K, \mathbb{R})$ the dressed norm $\|X\|_\Sigma^2 = \mathrm{tr}(X^T\Sigma^{-1}X\Sigma)$, with Σ the reference covariance of the belief of the agent at which γ is based, satisfies $\|k^{-1}Xk\|^2_{k^{-1}\Sigma k^{-T}} = \|X\|_\Sigma^2$. A law is needed here because $\mathfrak{gl}(K, \mathbb{R})$ carries no conjugation-invariant positive-definite form for $K \geq 2$: conjugation by $\mathrm{diag}(t, 1, \ldots, 1)$ multiplies E_{12} by t. In this model Σ is fixed data at the operating point. If it were a variable of an objective, the log normalizer $\frac{1}{2}\log\det(\Pi_0 + \lambda B^T W(\Sigma) B)$ of the loop prior would be a belief-dependent term that must be retained.

The anchor Π_0 and the forms W_γ are declared in the common reference presentation of chapter 1; reference coordinates are common to all agents, not owned by the receiver. A fixed non-invariant anchor, such as $\sigma^{-2}I$, is therefore a choice of that presentation modulo the subgroup preserving it, which is $O(K)$ for a Frobenius form. The quantitative conclusions below, namely theorem 18.23(iii)–(v), proposition 18.25(ii) and proposition 18.28, hold in the presentation in which the anchor and loop forms take their displayed values and are invariant only under reference changes preserving those forms. With the covariance-dressed form used for both the anchor and the loop term they are covariant under every reference change. The kernels, ranks and zero sets are invariant in either case.

For the statements below, let $N_e = \{\delta \in \mathfrak{g} : \bar{T}_e\delta = 0\}$, the Lie algebra of the stabilizer of the stacked code, and let $N = \bigoplus_e N_e \subseteq X$, the invisible directions. Let Z be a matrix whose columns form an

orthonormal basis of N for the Frobenius inner product on X, and Y one for $N^{\perp}$. Put

$$S = Z^{T}\Lambda_{0}Z, \qquad \Pi_{N} = ZS^{-1}Z^{T}\Lambda_{0}, \qquad L_{N} = Z^{T}B^{T}WBZ, \tag{18.24}$$

so that Π_{N} is the Λ_{0}-orthogonal projection of X onto N and L_{N} is the loop operator on the invisible directions. Let $\bar{F}_{e} = \sum_{c} \beta_{ij}^{(c)} F_{c}^{(j)}$ be the information per exchange of corollary 18.13(i), evaluated at $\bar{\theta}_{e} = I$, where $C = M$.

18.7.2 Concentration and selection

Theorem 18.23 (Learning with a loop prior). In the tangent learning model let the truth be $\omega^{\star} \in X$, let each edge e receive $n_{e} = \lceil \upsilon_{e} n \rceil$ independent exchanges with sampling rate $\upsilon_{e} > 0$, and write $z^{\star} = Z^{T}\omega^{\star}$.

(i) For every set of messages the likelihood is constant on each coset $\omega + N$, so the density of the recognition law relative to the prior depends on ω only through $Y^{T}\omega$. Consequently, for every prior and every additive noise law for which the messages have densities, the conditional law of $Z^{T}\omega$ given $Y^{T}\omega$ is the same under the recognition law as under the prior.

(ii) Almost surely, as $n \to \infty$, the mean of the recognition law converges to

$$\omega_{\infty} = (I - \Pi_{N})\omega^{\star}, \tag{18.25}$$

its covariance is $ZS^{-1}Z^{T} + O(1/n)$, and n times its covariance in the coordinates $Y^{T}\omega$ converges to $(Y^{T}F_{\upsilon}Y)^{-1}$ with $F_{\upsilon} = \bigoplus_{e} \upsilon_{e}\bar{F}_{e}$. Equivalently, the recognition law converges to the prior conditioned on the coset $\omega^{\star} + N$. Its precision grows at rate n in the identified directions, and within the coset it is the loop prior restricted to the coset.

(iii) Let $\Pi_{0} = \sigma^{-2}I$. In the coordinates $Z^{T}\omega$ the limiting covariance is $\sigma^{2}(I + \lambda\sigma^{2}L_{N})^{-1}$. It equals σ^{2} on $\ker L_{N} = Z^{T}(N \cap \ker B)$, which is the prior covariance there. For $\lambda > 0$ it is strictly smaller than σ^{2} on the orthogonal complement, with eigenvalues $\sigma^{2}/(1 + \lambda\sigma^{2}\mu)$ for the eigenvalues $\mu > 0$ of L_{N}, and it tends to zero there as $\lambda \to \infty$. It never exceeds the prior marginal covariance $Z^{T}\Lambda_{0}^{-1}Z$ of the invisible coordinates and can equal it. The rank of L_{N} is $\dim B(N) = \dim N - \dim(N \cap \ker B)$.

(iv) If in addition the truth is flat, $B\omega^{\star} = 0$, then

$$Z^{T}(\omega_{\infty} - \omega^{\star}) = -(I + \lambda\sigma^{2}L_{N})^{-1}z^{\star}. \tag{18.26}$$

The loop term recovers the invisible part of the truth along an eigenvector of L_{N} with eigenvalue $\mu > 0$ up to the factor $\lambda\sigma^{2}\mu/(1 + \lambda\sigma^{2}\mu)$, and as $\lambda \to \infty$ it recovers all of it except the component in $\ker L_{N}$, which stays at the prior mean zero.

(v) For a general truth, as $\lambda \to \infty$,

$$Z^{T}(\omega_{\infty} - \omega^{\star}) \longrightarrow -P_{\ker}z^{\star} - L_{N}^{+}Z^{T}B^{T}WB\omega^{\star}, \tag{18.27}$$

with $P_{\ker}$ the orthogonal projection onto $\ker L_{N}$ and L_{N}^{+} the pseudoinverse.

(vi) For every positive-definite anchor Π_0, as $\lambda \to \infty$ the coordinate $Z^T(\omega_\infty - \omega^\star)$ converges to the unique minimizer of $\|\omega^\star + Zz\|^2_{\Pi_0}$ over the minimizers of $\|B(\omega^\star + Zz)\|^2_W$. If the truth is flat and $N \cap \ker B = 0$, then $\omega_\infty \to \omega^\star$.

Proof. (i) By eq. (18.22) the law of a message on edge e depends on ω only through $\bar{T}^c_e \omega_e$ for attending modalities c, which does not change when ω_e changes by an element of N_e. Bayes' rule gives the second statement.

(ii) After the messages the precision is $\Lambda_n = \Lambda_0 + nF_n$ with $nF_n = \sum_t J_t^T R_c^{-1} J_t$, and since $h_t - \bar{T}^c_e \bar{y}_t = J_t \omega^\star + \varepsilon_t$, the information vector is $\eta_n = nF_n\omega^\star + \xi_n$ with $\xi_n = \sum_t J_t^T R_c^{-1} \varepsilon_t$. Every J_t vanishes on N, so $F_n = Y F_{Y,n} Y^T$ and $Z^T \xi_n = 0$. By the strong law of large numbers $F_{Y,n} \to Y^T F_v Y$ almost surely, and this limit is positive definite because the kernel of F_v is N by corollary 18.13(iii). In the basis (Y, Z) the matrix Λ_n has blocks $A_n = Y^T \Lambda_0 Y + nF_{Y,n}$, $D = Y^T \Lambda_0 Z$ and S, and block inversion gives

$$\Lambda_n^{-1} = Z S^{-1} Z^T + (Y - Z S^{-1} D^T)(A_n - D S^{-1} D^T)^{-1} (Y - Z S^{-1} D^T)^T, \tag{18.28}$$

with $n(A_n - DS^{-1}D^T)^{-1} \to (Y^T F_v Y)^{-1}$. This proves the statements about the covariance, since $Y^T \Lambda_n^{-1} Y = (A_n - DS^{-1}D^T)^{-1}$. For the mean, $nF_n\omega^\star = \Lambda_n \omega^\star - \Lambda_0 \omega^\star$ gives

$$\Lambda_n^{-1} \eta_n = \omega^\star - \Lambda_n^{-1} \Lambda_0 \omega^\star + \Lambda_n^{-1} \xi_n. \tag{18.29}$$

The second term converges to $Z S^{-1} Z^T \Lambda_0 \omega^\star = \Pi_N \omega^\star$. Because $Z^T \xi_n = 0$, the third term equals $(Y - ZS^{-1}D^T)(A_n - DS^{-1}D^T)^{-1} Y^T \xi_n$, which tends to zero since $\xi_n / n \to 0$ almost surely by the strong law. Finally, the Gaussian law with precision Λ_0 conditioned on the affine set $\omega^\star + N$, in the coordinate z of $\omega^\star + Zz$, has precision S and the mean that minimizes $\tfrac{1}{2}(\omega^\star + Zz)^T \Lambda_0 (\omega^\star + Zz)$, which is $z = -S^{-1} Z^T \Lambda_0 \omega^\star$. Its mean is ω_∞ and its covariance is $Z S^{-1} Z^T$.

(iii) Here $S = \sigma^{-2} I + \lambda L_N$, and the eigenvalues of S^{-1} are $\sigma^2/(1 + \lambda \sigma^2 \mu)$ for the eigenvalues $\mu \geq 0$ of L_N. Since W is positive definite, $L_N z = 0$ exactly when $BZz = 0$, so $\ker L_N = Z^T(N \cap \ker B)$ and the rank of L_N is that of BZ.

(iv) From (ii) and $Z^T \Lambda_0 \omega^\star = \sigma^{-2} z^\star + \lambda Z^T B^T W B \omega^\star$,

$$Z^T(\omega_\infty - \omega^\star) = -S^{-1}\big(\sigma^{-2} z^\star + \lambda Z^T B^T W B \omega^\star\big), \tag{18.30}$$

which is eq. (18.26) when $B\omega^\star = 0$.

(v) The vector $Z^T B^T W B \omega^\star$ lies in the range of $(W^{1/2} BZ)^T$, which is the range of $L_N = (W^{1/2}BZ)^T (W^{1/2} BZ)$. As $\lambda \to \infty$, $\lambda S^{-1} \to L_N^+$ and $S^{-1} \to 0$ on the range of L_N, while $\sigma^{-2} S^{-1} = I$ on $\ker L_N$. Applying this to eq. (18.30) gives eq. (18.27).

(vi) By (ii), $z_\lambda = Z^T(\omega_\infty - \omega^\star)$ minimizes $f_\lambda = g + \lambda h$ with $g(z) = \tfrac{1}{2}\|\omega^\star + Zz\|^2_{\Pi_0}$ and $h(z) = \tfrac{1}{2}\|B(\omega^\star + Zz)\|^2_W$. The set $\mathcal{A}$ of minimizers of h is a nonempty affine set, and g is strictly convex and coercive, so it has a unique minimizer $z_{\mathcal{A}}$ on $\mathcal{A}$. From $f_\lambda(z_\lambda) \leq f_\lambda(z_{\mathcal{A}})$ one obtains $g(z_\lambda) \leq g(z_{\mathcal{A}})$ and $\lambda(h(z_\lambda) - \min h) \leq g(z_{\mathcal{A}})$. Hence z_λ is bounded, $h(z_\lambda) \to \min h$, and every limit point lies in $\mathcal{A}$ with g at most $g(z_{\mathcal{A}})$, so it equals $z_{\mathcal{A}}$. For a flat truth $\min h = 0$ is attained at $z = 0$, and $N \cap \ker B = 0$ makes BZ injective, so $\mathcal{A} = \{0\}$. □

Parts (i) and (ii) are the linear Gaussian instance of the limiting posterior of a partially identified model: the identified coordinates concentrate, and the nonidentified ones keep their conditional prior [99, 163]; the underlying conditional-independence statement is that of Dawid [56]. The loop prior acts only through that conditional prior, which (iii) to (vi) compute. For a general prior and location noise the same disintegration holds at every sample size by (i); its limit requires posterior consistency of the identified coordinates and continuity of the prior conditional, which are not proved here.

With an isotropic anchor and no loop term, the recognition law on the invisible coordinates stays at its anchor marginal. For a general anchor, messages can condition these coordinates through prior correlations, while leaving their conditional law given the identified coordinates unchanged. The loop term changes the geometry in which the identified part of the truth is completed, from the anchor to the anchor plus the loop energy, and eq. (18.25) is the resulting projection.

18.7.3 Filling and detection

Definition 18.24 (Detectable dissonance). The detectable dissonance of a configuration ω is

$$\mathcal{D}(\omega) = \min_{\nu \in N} \|B(\omega + \nu)\|_W^2, \tag{18.31}$$

the least linearized loop energy among the configurations that no message distinguishes from ω.

Proposition 18.25 (Filling and detection are complementary). In the tangent learning model let $Q_{\mathcal{D}}$ be the W-orthogonal projection of $\mathfrak{g}^{\mathcal{L}}$ onto the W-orthogonal complement of $B(N)$.

(i) $\mathcal{D}(\omega) = \|Q_{\mathcal{D}} B\omega\|_W^2$. In particular $\mathcal{D}$ is constant on cosets of N, so messages determine it.

(ii) For every positive-definite anchor, as $\lambda \to \infty$ the limit mean ω_∞ satisfies $B\omega_\infty \to Q_{\mathcal{D}} B\omega^\star$, so its loop energy tends to $\mathcal{D}(\omega^\star)$.

(iii) The loop space $B(X)$ is the W-orthogonal sum of $B(N)$, of dimension $\dim N - \dim(N \cap \ker B)$, and of $B(X) \cap B(N)^{\perp_W}$, of dimension $\operatorname{rank} B - \dim B(N)$. The loop energy splits as $\|B\omega\|_W^2 = \|(I - Q_{\mathcal{D}})B\omega\|_W^2 + \|Q_{\mathcal{D}} B\omega\|_W^2$. The first term is the only one that depends on the invisible coordinates, and a dissonance in $B(N)$ is not detectable; the second is a function of the identified coordinates alone.

Proof. (i) Since $B(\omega + Zz) = B\omega + BZz$, the minimum over z leaves the W-orthogonal residual of $B\omega$ from the range of BZ, which is $B(N)$; this residual is $Q_{\mathcal{D}} B\omega$. For $\nu \in N$, $Q_{\mathcal{D}} B\nu = 0$. (ii) By theorem 18.23(vi), $Z^T(\omega_\infty - \omega^\star)$ converges to a minimizer z of $\|B\omega^\star + BZz\|_W^2$. Every such minimizer satisfies $BZz = -(I - Q_{\mathcal{D}})B\omega^\star$, because $I - Q_{\mathcal{D}}$ is the W-orthogonal projection onto the range of BZ. Hence $B\omega_\infty \to Q_{\mathcal{D}} B\omega^\star$. For the isotropic anchor the same limit follows from eq. (18.27). (iii) The dimension of $B(N)$ is the rank of BZ, computed in theorem 18.23(iii). The energy splits because $Q_{\mathcal{D}}$ is a W-orthogonal projection, and $Q_{\mathcal{D}} B\omega$ does not depend on the component of ω in N by (i). □

The proposition states the cost of the loop prior. Every loop direction in which agreement around loops completes a dictionary is a direction in which the community assumes an agreement it cannot check. If the truth is dissonant there, a strong loop prior can select a flat representative of the same

message laws, as quantified by the bias term of eq. (18.27). Dissonance is detectable only in the loop directions that the invisible directions cannot reach.

18.7.4 Which directions loops cannot reach

Proposition 18.26 (The unreachable directions). Let Γ be connected, and suppose that the chains of the walks in $\mathcal{L}$ span the cycle space of Γ, regarded as an undirected multigraph in which the edge $e : j \to i$ is oriented from j to i. Then

$$\ker B = \big\{(a_i - a_j)_{e:j\to i} : a \in \mathfrak{g}^{\mathcal{V}}\big\}, \tag{18.32}$$

and $N \cap \ker B$ consists of the configurations in eq. (18.32) with $\bar{T}_e(a_i - a_j) = 0$ on every edge $e : j \to i$. For $\mathfrak{g} = \mathfrak{gl}(K, \mathbb{R})$,

$$\begin{aligned} \dim(N \cap \ker B) &= K\big(\dim H^0 - K\big), \\ H^0 &= \big\{u \in (\mathbb{R}^K)^{\mathcal{V}} : \bar{T}_e(u_i - u_j) = 0 \text{ for every edge } e : j \to i\big\}. \end{aligned} \tag{18.33}$$

Proof. The value $(B\omega)_\gamma$ is the pairing of ω, read as a $\mathfrak{g}$-valued cochain on edges, with the chain of γ, and that chain is a cycle because a closed walk leaves each vertex as often as it enters it. If these chains span the cycle space, $\omega \in \ker B$ exactly when ω vanishes on every cycle. Choose a spanning tree and a root, let a_v be the sum of ω along the tree path from the root to v, each edge taken with the sign of its orientation, and put a equal to zero at the root. Then $\omega_e - (a_i - a_j)$ vanishes on tree edges and, on any other edge, equals the value of ω on its fundamental cycle, which is zero. Conversely every configuration of the form eq. (18.32) vanishes on cycles. The condition for membership in N is the definition of N_e. For $\mathfrak{gl}(K, \mathbb{R})$ the condition $\bar{T}_e(a_i - a_j) = 0$ holds column by column, so $a \mapsto (a_i - a_j)_e$ maps the families whose K columns lie in H^0, a space of dimension $K \dim H^0$, onto $N \cap \ker B$. Its kernel consists of the constant families, of dimension K^2, because Γ is connected. □

The configurations in eq. (18.32) are the coboundary, or vertex-wise, configurations: what the dictionaries would become if each agent's reading were re-expressed by e^{a_v} with codes and contents re-read accordingly. Relative to the observable algebra of chapter 1 they are not gauge. Passive changes leave reference dictionaries unchanged, and grounded messages distinguish a coboundary configuration from the identity unless the codes absorb it. The space $N \cap \ker B$ consists of the coboundary configurations that no attending code registers. They create no holonomy, so no loop term reaches them, and no message reveals them. Equation (18.32) is the coboundary part of the combinatorial Hodge decomposition of edge cochains with Lie-algebra coefficients [138]. The space H^0 is the space of global sections of a cellular sheaf on Γ whose stalk at an agent is $\mathbb{R}^K$, whose stalk at an edge is the message space, and whose two restriction maps at an edge are both the attending code [101]. Such sheaves are used for discourse by Hansen and Ghrist [102]; here the code sheaf arises from identifiability.

Corollary 18.27 (Complete exchange). Let every pair of agents exchange in both directions, and let $\mathcal{L}$ contain every two-cycle and every directed triangle. Then the hypothesis of proposition 18.26 holds and

$$H^0 = \big\{u : u_i - u_j \in \ker \bar{T}_{j\to i} \cap \ker \bar{T}_{i\to j} \text{ for every pair } \{i, j\}\big\}, \tag{18.34}$$

where $\bar{T}_{j\to i}$ is the code on the edge $j \to i$. If the pairs with $\ker \bar{T}_{j\to i} \cap \ker \bar{T}_{i\to j} = 0$ connect all agents, then H^0 consists of constant sections, $N \cap \ker B = 0$, and for a flat truth in the Gaussian tangent model with any positive-definite anchor the recognition law recovers every dictionary in the limits $n \to \infty$ and then $\lambda \to \infty$.

Proof. Fix an agent 0. The edges $0 \to i$ form a spanning tree. The fundamental cycle of $i \to 0$ is the two-cycle through 0 and i, and that of $j \to i$ with $i, j \neq 0$ is the directed triangle $0 \to j \to i \to 0$ minus the two-cycle through 0 and i. The two edges between i and j give $\bar{T}_{j\to i}(u_i - u_j) = 0$ and $\bar{T}_{i\to j}(u_j - u_i) = 0$. If the pairs with trivial intersection connect the agents, u is constant along each of them, hence constant. The last statement is theorem 18.23(vi). □

The sufficient condition is not necessary. For three agents in complete exchange, with every two-cycle and both directed triangles in $\mathcal{L}$, write $I_{ij} = \ker \bar{T}_{j\to i} \cap \ker \bar{T}_{i\to j}$. A section is fixed by u_0 and by differences $d_{01} \in I_{01}$, $d_{12} \in I_{12}$, $d_{20} \in I_{20}$ with zero sum, so the kernel of the sum map gives

$$\dim(N \cap \ker B) = K\big[\dim I_{01} + \dim I_{12} + \dim I_{20} - \dim(I_{01} + I_{12} + I_{20})\big]. \tag{18.35}$$

In $K = 3$, three linearly independent lines I_{ij} give zero, although no pairwise intersection is zero: triangles remove invisible flat directions that remain for each pair separately. Three distinct coplanar lines leave K such directions, and a common unread line leaves $2K$.

Proposition 18.28 (Reciprocal pair). Let two agents exchange in both directions with $\mathcal{L}$ the two-cycle, W the Frobenius norm and $\mathfrak{g} = \mathfrak{gl}(K, \mathbb{R})$. Write $\bar{T}_1$ for the code on $1 \to 0$ and $\bar{T}_0$ for the code on $0 \to 1$, $\mathcal{K}_a = \ker \bar{T}_a$, $\mathcal{R}_a = \mathcal{K}_a^{\perp}$ for the row space of $\bar{T}_a$, and $P_\cap$ for the orthogonal projection onto $\mathcal{R}_0 \cap \mathcal{R}_1$. Then

$$\begin{aligned} \dim B(N) &= K \dim(\mathcal{K}_0 + \mathcal{K}_1), \qquad \dim(N \cap \ker B) = K \dim(\mathcal{K}_0 \cap \mathcal{K}_1), \\ \operatorname{rank} B - \dim B(N) &= K \dim(\mathcal{R}_0 \cap \mathcal{R}_1), \end{aligned} \tag{18.36}$$

and $\mathcal{D}(\omega) = \|P_\cap(\omega_{01} + \omega_{10})\|_F^2$.

Proof. N consists of the pairs (ν_{01}, ν_{10}) with the range of ν_{01} in $\mathcal{K}_1$ and the range of ν_{10} in $\mathcal{K}_0$, and $B\nu = \nu_{01} + \nu_{10}$. Hence $B(N)$ is the set of matrices with range in $\mathcal{K}_0 + \mathcal{K}_1$, of dimension $K \dim(\mathcal{K}_0 + \mathcal{K}_1)$. Its Frobenius-orthogonal complement in $B(X) = \mathfrak{gl}(K, \mathbb{R})$ is the set of matrices with range in $(\mathcal{K}_0 + \mathcal{K}_1)^{\perp} = \mathcal{R}_0 \cap \mathcal{R}_1$, and the orthogonal projection onto it is $X \mapsto P_\cap X$. Thus $Q_{\mathcal{D}} = P_\cap$ and proposition 18.25(i) gives $\mathcal{D}$. The space $N \cap \ker B$ consists of the pairs with $\nu_{01} = -\nu_{10}$ and range in $\mathcal{K}_0 \cap \mathcal{K}_1$. □

A pair detects its mutual misunderstanding only in $\mathcal{R}_0 \cap \mathcal{R}_1$, what both codes can say. The loop term acts on holonomy components with range in $\mathcal{K}_0 + \mathcal{K}_1$, completing the invisible part of each dictionary from what the other code says. Of what neither code can say, only the antisymmetric part $(\nu, -\nu)$ with range in $\mathcal{K}_0 \cap \mathcal{K}_1$ stays at the prior. When the two row spaces are independent and together have dimension K, the invisible flat-truth coordinates are recovered as the loop weight grows, while every tangent loop discrepancy can also be absorbed by an invisible change. Recovery under the flat-truth hypothesis therefore differs from detecting whether that hypothesis holds.

18.7.5 Without linearization

The next results use unrestricted group-valued dictionaries and are exact without linearization. They identify the configurations consistent with grounded messages and exactly flat loops. Restricting dictionaries to a single-exponential image requires intersecting the stated solution sets with that image. The following two statements hold for the grounded model eq. (18.13) itself, in the constraint form of the loop term.

Proposition 18.29 (Exact reciprocal identification). Let $G = \mathrm{GL}(K, \mathbb{R})$ and let the true dictionaries of two agents be reciprocal, $\bar\theta^\star_{10} = (\bar\theta^\star_{01})^{-1}$. The pairs $(\bar\theta_{01}, \bar\theta_{10})$ that reproduce the message laws of both edges for every referent and satisfy $\bar\theta_{01}\bar\theta_{10} = I$ are exactly

$$\bar\theta_{01} = \bar\theta^\star_{01} s, \qquad \bar\theta_{10} = s^{-1}\bar\theta^\star_{10}, \qquad s \in \mathrm{Stab}(\bar T_1) \cap \mathrm{Stab}(\bar T_0 \bar\theta^\star_{01}). \tag{18.37}$$

The pair is determined uniquely exactly when the stacked map $(\bar T_1; \bar T_0\bar\theta^\star_{01})$ is injective; otherwise the solutions form a set of dimension $K \dim\ker(\bar T_1; \bar T_0\bar\theta^\star_{01})$.

Proof. By theorem 18.9(i), the edge $1 \to 0$ reproduces its law exactly when $\bar\theta_{01} = \bar\theta^\star_{01} s$ with $s \in \mathrm{Stab}(\bar T_1)$. Reciprocity forces $\bar\theta_{10} = s^{-1}\bar\theta^\star_{10}$, and the edge $0 \to 1$ reproduces its law exactly when $(\bar\theta^\star_{10})^{-1} s^{-1} \bar\theta^\star_{10} = \bar\theta^\star_{01} s^{-1} (\bar\theta^\star_{01})^{-1} \in \mathrm{Stab}(\bar T_0)$, that is, when $\bar T_0 \bar\theta^\star_{01} s^{-1} = \bar T_0 \bar\theta^\star_{01}$. The intersection of the two stabilizers is the stabilizer of the stacked map, the set of s with $s - I$ mapping $\mathbb{R}^K$ into its kernel, which is open in an affine space of dimension K times the dimension of that kernel. □

The map $\bar T_0 \bar\theta^\star_{01}$ is agent 0's code expressed in agent 1's reference coordinates, so the pair is identified exactly when, between them, the two agents can say everything.

Proposition 18.30 (Exact flat identification). Let $G = \mathrm{GL}(K, \mathbb{R})$ and Γ be connected, and suppose that every configuration whose holonomies on $\mathcal{L}$ are the identity is a coboundary configuration, $\bar\theta_{ij} = k_i k_j^{-1}$ for some $k \in G^{\mathcal{V}}$. Let the truth be a coboundary configuration, $\bar\theta^\star_{ij} = g_i g_j^{-1}$, and put $\bar T'_e = \bar T_e g_j$ for $e : j \to i$. The configurations that reproduce every message law and have identity holonomy on $\mathcal{L}$ are exactly

$$\bar\theta_{ij} = g_i u_i u_j^{-1} g_j^{-1}, \qquad u \in G^{\mathcal{V}}, \qquad \bar T'_e (u_i - u_j) = 0 \text{ for every edge } e : j \to i, \tag{18.38}$$

and two families u, u' give the same configuration exactly when $u'_v = u_v c$ for all v and one $c \in G$. The truth is the only such configuration exactly when the space H^0 of eq. (18.33), formed with the codes $\bar T'_e$, consists of constant sections. The hypothesis holds for the complete exchange with every two-cycle and every directed triangle.

Proof. By hypothesis $\bar\theta_{ij} = k_i k_j^{-1}$; write $k_v = g_v u_v$. By theorem 18.9(i), the edge e reproduces its law exactly when $(\bar\theta^\star_{ij})^{-1}\bar\theta_{ij} = g_j u_i u_j^{-1} g_j^{-1} \in \mathrm{Stab}(\bar T_e)$, that is, when $\bar T'_e u_i u_j^{-1} = \bar T'_e$, that is, when $\bar T'_e u_i = \bar T'_e u_j$. Two families give the same configuration exactly when $(u'_i)^{-1} u_i = (u'_j)^{-1} u_j$ on every edge, which on a connected graph means $u' = uc$. If H^0 consists of constant sections, every column of u is constant, so $u_v = u_0$ for all v and the configuration is the truth. If $s \in H^0$ is not constant, $u_v = I + t s_v w^T$ with $w \neq 0$ and small $t \neq 0$ is an invertible family satisfying the condition, and it is not of the form $(c)_v$, so its configuration differs from the truth. For the complete exchange

put $k_0 = I$ and $k_i = \bar{\theta}_{i0}$. The two-cycle through 0 and i gives $\bar{\theta}_{0i} = k_i^{-1}$, and the directed triangle $0 \to j \to i \to 0$, whose holonomy is $\bar{\theta}_{0i}\bar{\theta}_{ij}\bar{\theta}_{j0}$, gives $\bar{\theta}_{ij} = \bar{\theta}_{0i}^{-1}\bar{\theta}_{j0}^{-1} = k_i k_j^{-1}$. □

The residual set of eq. (18.38) has dimension $K(\dim H^0 - K)$ at the truth, the count of proposition 18.26. In the noiseless, fully observed case the hypothesis of proposition 18.30, that flat holonomy forces a coboundary configuration, is the synchronizability criterion of group synchronization [86]. What the proposition adds is that each relative element is observed only through a code, so that uniqueness is governed by the sections of the code sheaf.

Remark 18.31 (A loop family for exact flat identification). The real cycle-spanning hypothesis of proposition 18.26 suffices for the tangent count but not for the nonlinear hypothesis of proposition 18.30. On the directed triangle $0 \to 1 \to 2 \to 0$ let $\mathcal{L} = \{\gamma^2\}$, whose chain 2γ spans the cycle space, and let all codes have rank one and annihilate e_3. With $a^T e_3 = -2$, the reflection $s = I + e_3 a^T$ satisfies $s^2 = I$ and $\bar{T}s = \bar{T}$, so the configuration with dictionary s on one edge and I on the others reproduces every message law of the identity truth, has $H_{\gamma^2} = I$, and has $H_\gamma = s \neq I$; it is not a coboundary configuration. A checkable family exists on every strongly connected graph. Fix a root r, an out-arborescence T_{out} and an in-arborescence T_{in} at r, and for every edge $e : j \to i$ let $\mathcal{L}$ contain the closed walk from r to j along T_{out}, across e, and back to r along T_{in}. Let k_v be the ordered product along the T_{out} path from r to v, with $k_r = I$, and $P_{\mathrm{in}}(v)$ the product along the T_{in} path from v to r. Identity holonomy on the walk of e gives $\bar{\theta}_{ij} = P_{\mathrm{in}}(i)^{-1} k_j^{-1}$. For the edge of T_{out} entering $i \neq r$, $k_i = \bar{\theta}_{ij} k_j$, so $P_{\mathrm{in}}(i)^{-1} = k_i$; this also holds at r. Hence $\bar{\theta}_{ij} = k_i k_j^{-1}$ on every edge, which is the hypothesis of proposition 18.30. For the complete exchange with $r = 0$ this family consists of the two-cycles and triangles through agent 0 used in its proof.

18.7.6 Receiver-owned recognition laws

Remark 18.32 (Ownership and the loop term). Under a variational restriction of the global posterior with factors owned by receivers, each receiver infers its own dictionaries, so its dictionary recognition law is one factor $\mathfrak{q}_i(\omega_{i\cdot})$ of a law factorized over receivers. A two-cycle couples two receivers and a directed triangle three, so the loop term couples the factors. In the tangent model the factorized law closest to the posterior in $D_{\mathrm{KL}}(\mathfrak{q} \| \text{posterior})$ has the exact posterior means and, as covariances, the inverses of the diagonal blocks of the posterior precision [23, Section 10.1.2]. These are smaller in the Loewner order than the exact marginal covariances. For this one Gaussian target, the factorization therefore leaves the estimates of the dictionaries unchanged and makes each receiver overconfident about them. This is the global free energy restricted to a factorized family. The law-level per-agent model of chapter 19, in which each receiver integrates its loop factors against the peers' dictionary laws, is a different objective: its fixed-point means need not equal the posterior means, and its variances differ as well (proposition 19.13). Exact means are a feature of Gaussian targets: for a non-Gaussian posterior a factorized law generally has means different from the exact ones, so for general laws the factorization can change the estimates as well. To update its factor, receiver i needs the current means of the dictionaries that other agents hold along its loops: a loop prior can be used only if agents communicate about how they read each other, in addition to communicating about the world.

18.8 Extensions beyond the declared experiments

Open problem 18.33 (Learning the dictionaries beyond the tangent model). Theorem 18.23 and proposition 18.25 give the exact recognition law of the tangent model at every number of exchanges and its limit as $n \to \infty$. Their $\lambda \to \infty$ statements hold for every positive-definite anchor and take $n \to \infty$ first. The two limits do not commute when the truth has detectable dissonance: imposing the flat constraint before the data defines a misspecified model, whose posterior concentrates on the F_υ-closest points of $\ker B$ to the truth, with ties broken by the prior [21], and distorts the identified coordinates. Propositions 18.29 and 18.30 treat the constraint form of the loop term for a flat truth without linearization, under the explicit loop hypothesis of proposition 18.30. The detectable dissonance vanishes exactly when some flat configuration reproduces every message law, so it decides whether flatness can be rejected; the rate at which a test of flatness rejects is governed by the Fisher metric F_υ rather than by the loop metric W, and is not computed here. The remaining questions have separate scopes. For the full grounded model eq. (18.13) with a prior on G or on a declared exponential chart, establish the limiting conditional law on the identified fiber $\prod_e \bar{\theta}_e^\star \operatorname{Stab}(\bar{T}_e)$, or on its preimage in the chart, and determine the selection within the fiber for a dissonant truth and a finite loop weight. The nonlinear covariance-dressed defect and the law-comparison penalty defined above lead to different learning problems; the latter requires its own regularity and stabilizer analysis. Another extension treats latent routing labels, which lower the information below the routed sum of corollary 18.13, and loop weights built from the routing weights. For location-family noise and a non-Gaussian anchor, determine the limit of the recognition law on the fiber, which is no longer a conditioned Gaussian, and whether the filled directions are still recovered exactly as the loop weight grows.

Open problem 18.34 (Codes as part of the species model). Let codes be learned as well, so that a shared language is a joint fixed point of code and dictionary learning. The expression-map dynamics of Hansen and Ghrist [102] is the vector-valued analogue on a sheaf. Determine when the learned codes of a species become related by one frame change, as the shared-language hypothesis of definition 18.1 assumes, and when learning drives code-visible dissonance to zero while leaving dissonance in unread directions. When senders choose codes under objectives different from the receiver's, code choice is an information-design problem, of Bayesian-persuasion type when the sender commits to its code [116]; the receiver-side results of this chapter then describe one side of a signaling game.

Open problem 18.35 (Dictionaries in the free-energy pictures). The Gaussian tangent learning calculation fixes a normalized likelihood and an anchored loop prior. Extending it to fully nonlinear receiver-owned dictionaries and learned codes requires separate global, per-agent, and reciprocal objectives, with every law dependence differentiated in the chosen objective. Whether those dynamics reduce code-visible dissonance is a further dynamical question; posterior conditioning within the tangent model alone does not establish it.

Open problem 18.36 (Beyond Gaussian codes). Corollaries 18.4 and 18.5 and proposition 18.21 already hold for general laws and measurable codes, including codes into finite alphabets such as words, and the consensus theorem holds for laws as well as means. Remark 18.11 extends identifiability to every location family of noise. Extend the identifiability, learning and dissonance results to codes whose message laws are not a location family, such as codes into finite alphabets,

where the Fisher form is replaced by the information in the induced discrete channel and the stabilizer by the subgroup that leaves the channel unchanged, and to agents of different kinds, whose dictionaries are not invertible.

The relation to discourse sheaves is the use of private content spaces and restriction maps into shared message spaces [102]. The reconstruction step additionally specifies a receiver conditional law, while invertible dictionaries supply loop products. Unknown frame conventions have also been studied as communication constraints, with frame changes treated as gauge data [17, 191]. Linear interpretation maps between agents' value spaces, with beliefs lost in their null spaces, are proposed by Amornbunchornvej [6], and meaning as maps between individual conceptual spaces, with agreement as a fixed point, by Warglien and Gärdenfors [196]. Communication as active inference between agents whose shared generative model makes their exchange converge is developed by Friston and Frith [81] and Friston et al. [82], and language as efficient compression by Zaslavsky et al. [202]. Relative to these sources the present chapter adds frame-relative laws on standard Borel spaces, receiver reconstruction with the exact loss of corollary 18.4, identification of receiver-owned dictionaries modulo code stabilizers, directed loop products with independent opposite dictionaries, and the exact split of a loop prior into filled and detectable directions. These comparisons do not identify graph loops at one context with transport between contexts.

Chapter 19

Learning Directed Links under Three Objectives

The receiver-owned dictionaries of definition 18.2 can be uncertain and learned together with the contents they interpret. Their dynamics depends on which of the three objectives in chapter 4 is chosen. A global model differentiates every comparison record at both endpoints. A per-agent model updates only the receiver's variables while treating the source law as data. A reciprocal potential differentiates both arguments of a divergence between communicated laws. These distinctions concern the objective; the recognition family supplies the Fisher metric that turns its derivatives into velocities.

The general dependency statements and the pointwise record invariance below do not require Gaussian laws. Explicit mean flows, grounding rates, and confidence formulas use a linear-Gaussian tangent family with fixed operating contents. The fixed-covariance reciprocal mean theorem is followed by the coupled flow of all content and dictionary means and covariances, with either relaxing or Gibbs-optimized source rows. Its global-existence theorem and exact stability example concern the restricted family of definition 4.10; they do not assert that a full law-space flow preserves Gaussianity.

19.1 The tangent communication model

Two agents who exchange messages about something they both represent can register whether a message matches what the receiver expected. A comparison record is a noisy linear function $A_i y_i - B_{ij} y_j + \epsilon$ of the two agents' contents. With codes and receiver-owned dictionaries the natural choice is $B_{ij} = \bar{T}_e$, the sender's code, and $A_i = \bar{T}_e \bar{\theta}_e^{-1}$, the receiver's prediction of the message from its own content translated into the sender's coordinates. The following construction linearizes that record and places it in the two free energies.

Definition 19.1 (Tangent communication model). Let $\mathcal{V}$ be a finite set of agents and $\mathcal{E}$ a set of directed edges $e : j \to i$ with receiver i and sender j, and write $\mathrm{in}(i)$ for the edges received by i. Agent i has content $y_i \in \mathbb{R}^K$ in its reference coordinates. Edge e carries the receiver's dictionary $\bar{\theta}_e = e^{\omega_e}$ with $\omega_e \in \mathfrak{g} = \mathfrak{gl}(K, \mathbb{R})$, the sender's attending code $\bar{T}_e : \mathbb{R}^K \to H_e = \mathbb{R}^{n_e}$ of full row rank $n_e \le K$ and a positive-definite noise covariance R_e. Fix operating contents $c_i \in \mathbb{R}^K$ and write $\kappa_i \omega = \omega c_i$. The dictionary is a receiver-owned hypothesis about the sender's coordinates; the independently owned opposite dictionary need not be its inverse. Define $N_e = \{\delta \in \mathfrak{g} : \bar{T}_e \delta = 0\}$, the directions invisible to grounded referents that span the content space. These can be fewer than the directions invisible at a single operating content, which satisfy only $\bar{T}_e \delta c_i = 0$. The comparison

record of e is

$$o_e = \bar{T}_e\big(y_i - y_j - \kappa_i\omega_e\big) + \varepsilon_e, \qquad \varepsilon_e \sim \mathcal{N}(0, R_e), \tag{19.1}$$

the first-order form of $\bar{T}_e\bar{\theta}_e^{-1}y_i - \bar{T}_e y_j + \varepsilon_e$ at $\bar{\theta}_e = I$ and $y_i = c_i$. Each agent has a self term $\mathcal{N}(m_i, P_i^{-1})$ with P_i positive definite, the Gaussian form at the operating point of its own generative model and its own records. Each edge has a grounded term $\exp(-\frac{1}{2}\|\omega_e - \hat{\omega}_e\|^2_{G_e})$ with G_e positive semidefinite, the information of grounded messages, with $\ker G_e = N_e$ when their referents identify every coded direction. The dictionaries have an anchored loop prior with anchor $\sigma^{-2}I$ on each edge, a finite set $\mathcal{L}$ of simple directed loops containing every two-cycle, each listed once as in definition 18.22, linearized holonomy $(B\omega)_\gamma = \sum_{e\in\gamma}\omega_e$, positive-definite forms W_γ and weight $\lambda \geq 0$. Put $W = \bigoplus_\gamma W_\gamma$ and use $\|v\|^2_A = v^T A v$. All codes, operating contents, grounding data, and prior forms are fixed. The Frobenius anchor and loop forms are declared reference-coordinate data, not an assertion of a positive-definite conjugation-invariant form on $\mathfrak{gl}(K, \mathbb{R})$. The quantitative values below, including the shares of proposition 19.6, the ring coupling, the echo variances and the coupled threshold, therefore hold in the reference presentation in which these forms take their displayed values, and are invariant only under reference changes preserving them, as in definition 18.22.

Definition 19.2 (Global joint). Write $x = (y, \omega)$ and $L_e x = \bar{T}_e(y_i - y_j - \kappa_i\omega_e)$. The global joint has posterior density proportional to $e^{-E(x)}$ with

$$\begin{aligned} E(x) = {} & \sum_i \tfrac{1}{2}\|y_i - m_i\|^2_{P_i} + \sum_e \Big(\tfrac{1}{2\sigma^2}\|\omega_e\|^2 + \tfrac{1}{2}\|\omega_e - \hat{\omega}_e\|^2_{G_e} + \tfrac{1}{2}\|o_e - L_e x\|^2_{R_e^{-1}} \Big) \\ & + \tfrac{\lambda}{2}\sum_{\gamma\in\mathcal{L}} \|(B\omega)_\gamma\|^2_{W_\gamma}. \end{aligned} \tag{19.2}$$

Its precision J is positive definite because the self terms and the anchor are, so the joint is normalized for every value of the registered records. Write η for its information vector and $x^\star = J^{-1}\eta$ for the posterior mean.

The linearization drops the product $\omega_e(y_i - c_i)$, which is of second order in the deviations from the operating point. Registering $o_e = 0$ registers agreement. The record is invariant under passive changes of frame when the displayed contents, codes, and dictionaries are transformed together, because the dictionary sits in the receiver's matrix.

Definition 19.3 (Per-agent tangent model). Fix agent i and the current peer laws, $q_k = \mathcal{N}(\mu_k, \Sigma_k)$ for contents and $r_f = \mathcal{N}(a_f, V_f)$ for the dictionaries that i does not own. Agent i's model has the variables $x_i = (y_i, \omega_{\mathrm{in}(i)})$ and negative log density, up to a constant,

$$\begin{aligned} E_i = {} & \tfrac{1}{2}\|y_i - m_i\|^2_{P_i} + \sum_{e\in\mathrm{in}(i)} \Big(\tfrac{1}{2\sigma^2}\|\omega_e\|^2 + \tfrac{1}{2}\|\omega_e - \hat{\omega}_e\|^2_{G_e} + \tfrac{1}{2}\big\|o_e - \bar{T}_e(y_i - \mu_j - \kappa_i\omega_e)\big\|^2_{\tilde{R}_e^{-1}} \Big) \\ & + \sum_{\gamma\in\mathcal{L}_i} \tfrac{1}{2}\Big\| \sum_{e\in\gamma\cap\mathrm{in}(i)} \omega_e + \sum_{f\in\gamma\setminus\mathrm{in}(i)} a_f \Big\|^2_{\Omega_{i\gamma}}, \end{aligned} \tag{19.3}$$

where $\mathcal{L}_i$ is the set of loops through an edge of i and

$$\tilde{R}_e = R_e + \bar{T}_e \Sigma_j \bar{T}_e^T, \qquad \Omega_{i\gamma} = \Big((\lambda W_\gamma)^{-1} + \sum_{f \in \gamma \setminus \mathrm{in}(i)} V_f\Big)^{-1}. \tag{19.4}$$

For $\lambda > 0$, these are the record eq. (19.1) and the loop factor integrated against the peers' laws; we call this the law-level model. At $\lambda = 0$ the loop terms are omitted, equivalently $\Omega_{i\gamma} = 0$ by continuity. The convolution uses the declared product of peer dictionary marginals for each loop factor. Replacing $\tilde{R}_e$ by R_e and $\Omega_{i\gamma}$ by λW_γ gives the plug-in model, in which peers enter only through their means.

The integration is Gaussian convolution: $\int \mathcal{N}(o; \bar{T}(y_i - y_j - \kappa_i\omega), R) q_j(dy_j) = \mathcal{N}(o; \bar{T}(y_i - \mu_j - \kappa_i\omega), \tilde{R})$, and $\int e^{-\frac{\lambda}{2}\|u+s\|_W^2} \mathcal{N}(s; \bar{a}, \bar{V}) ds$ is proportional to $e^{-\frac{1}{2}\|u+\bar{a}\|_\Omega^2}$ with $\Omega = ((\lambda W)^{-1} + \bar{V})^{-1}$. The model is normalized for every value of the peer laws, which are parameters of i's model and not variables over which it infers, as in a per-agent generative model. The records with peers enter through the record term of the per-agent free energy, and the loop factors enter i's prior over its own dictionaries. Agent i's model contains no record in which i is the sender; such records belong to the receivers' models.

The recognition laws are Gaussian. In the global picture they are factorized by agent, $Q = \bigotimes_i Q_i(x_i)$, each agent's factor holding its content and the dictionaries it owns; each local update varies one factor of a single global free energy. In the per-agent picture agent i holds a Gaussian Q_i on x_i and lowers its own free energy $\mathcal{F}_i$. This per-agent model is a record-likelihood model: peers enter through comparison-record likelihoods and loop factors, not as mixture sources with latent labels and Gibbs rows. It is therefore not the mixture-source model of definition 4.19, and the results of section 4.6 are neither used nor transferred. Each factor is averaged separately against the declared peer marginals; integrating the product of a record and a loop factor that read the same peer against that peer's joint recognition would define a different model.

19.2 What conversation determines

Remark 19.4 (The interpretation gauge without linearization or Gaussian noise). Let the records be $o_e = \bar{T}_e(\bar{\theta}_e^{-1} y_i - y_j) + \varepsilon_e$ with noise of any law, independent of everything else, or more generally let each record kernel read $(y, \bar{\theta})$ only through the coded discrepancy $\bar{T}_e(\bar{\theta}_e^{-1} y_i - y_j)$. Let $K \geq 2$ and let G act transitively on $\mathbb{R}^K \setminus \{0\}$, as $\mathrm{GL}^+(K, \mathbb{R})$, $\mathbb{R}_{>0} \times SO(K)$ and $SL(K, \mathbb{R})$ do. For every configuration $(y, \bar{\theta})$ and every belief displacement $y \mapsto y'$ with $y'_i \neq 0$ and $v_e = \bar{\theta}_e^{-1} y_i + y'_j - y_j \neq 0$ on every edge $e : j \to i$, transitivity gives dictionaries $\bar{\theta}'_e \in G$ with $\bar{\theta}'_e v_e = y'_i$. Then $\bar{\theta}'^{-1}_e y'_i - y'_j = v_e - y'_j = \bar{\theta}_e^{-1} y_i - y_j$, so $(y', \bar{\theta}')$ gives every record the same law as $(y, \bar{\theta})$. The interdependence of belief and meaning is therefore not a feature of the Gaussian tangent model: it holds pointwise, without linearization, for every record kernel of this form, and only terms other than the records can separate the two readings. It is a property of the action of the dictionary group. For $G = SO(K)$ the vector $\bar{\theta}'^{-1}_e y'_i$ ranges over the sphere of radius $|y'_i|$, so a displacement is absorbable on e exactly when $\bar{T}_e v_e \in \bar{T}_e(|y'_i| S^{K-1})$. For a full-row-rank code of rank below K this is the inequality $|P_{\mathrm{row}\,\bar{T}_e} v_e| \leq |y'_i|$, and for an injective code it is the equality $|v_e| = |y'_i|$; with compact dictionary groups the records therefore separate belief from meaning in part. In the

tangent model with $\mathfrak{g} = \mathfrak{so}(K)$, where $\mathfrak{so}(K)c = c^{\perp}$ for $c \neq 0$, part (i) of theorem 19.5 holds exactly when $\bar{T}_e(c_i^{\perp}) = H_e$ on every edge.

Theorem 19.5 (Belief and meaning are interdependent). In the tangent communication model the law of the comparison records depends on (y, ω) only through the coded discrepancies $L_e x = \bar{T}_e(y_i - y_j - \kappa_i \omega_e)$.

(i) The displacements $(\delta y, \delta\omega)$ that leave the law of every record unchanged form the subspace

$$\mathcal{G} = \big\{(\delta y, \delta\omega) : \bar{T}_e(\delta y_i - \delta y_j - \kappa_i \delta\omega_e) = 0 \text{ for every edge } e : j \to i\big\}. \tag{19.5}$$

If $c_i \neq 0$ for every receiver, the projection of $\mathcal{G}$ onto belief displacements is all of $(\mathbb{R}^K)^{\mathcal{V}}$: for every δy, the choice $\delta\omega_e = (\delta y_i - \delta y_j)c_i^T/|c_i|^2$ completes it to an element of $\mathcal{G}$.

(ii) The Fisher information of the records about (y, ω) vanishes exactly on $\mathcal{G}$, so along every direction of $\mathcal{G}$ the posterior precision is the sum of the contributions of the self terms, the grounded terms and the dictionary prior.

(iii) Among the displacements with $\delta\omega_e \in N_e$ for every edge, the belief components of $\mathcal{G}$ are exactly the sections $\{\delta y : \bar{T}_e(\delta y_i - \delta y_j) = 0 \text{ for every edge}\}$ of the code sheaf of eq. (18.33), whose edge condition is the displayed equality. They contain one independent common shift $\delta y_i = v$ on each weakly connected component of the graph.

Proof. (i) Each record is $L_e x$ plus independent noise, so its law is unchanged exactly when L_e annihilates the displacement. For the stated choice, $\kappa_i \delta\omega_e = (\delta y_i - \delta y_j)c_i^T c_i/|c_i|^2 = \delta y_i - \delta y_j$. (ii) The Fisher information of Gaussian records with fixed covariances is $\sum_e L_e^T R_e^{-1} L_e$, whose kernel is $\bigcap_e \ker L_e = \mathcal{G}$; the posterior precision is J, the sum of this form and of the other terms of eq. (19.2). (iii) For $\delta\omega_e \in N_e$, $\bar{T}_e \kappa_i \delta\omega_e = \bar{T}_e \delta\omega_e c_i = 0$, so the condition reduces to $\bar{T}_e(\delta y_i - \delta y_j) = 0$. □

An exchange that disagrees with the receiver's belief can be explained by changing the belief or by changing the dictionary. The comparison record alone does not choose between these responses. The self terms, grounded records, and anchored prior supply distinct information. Even when grounding removes dictionary freedom modulo N_e, common content shifts remain invisible to every pairwise discrepancy. Theorem 19.5 is a formal counterpart, in the tangent model, of the interdependence of belief and meaning in radical interpretation [52, 53], and the stabilizer directions are an instance of the indeterminacy of translation [168]; these correspondences are interpretive.

Proposition 19.6 (Persuasion and reinterpretation). Take one edge $e : j \to i$, independent Gaussian priors $y_i \sim \mathcal{N}(m_i, \Sigma_i^0)$, $y_j \sim \mathcal{N}(m_j, \Sigma_j^0)$ and $\omega_e \sim \mathcal{N}(\bar{\omega}_e, V_e)$, and one record o_e. Put

$$r = o_e - \bar{T}_e(m_i - m_j - \kappa_i \bar{\omega}_e), \qquad C_\rho = \kappa_i V_e \kappa_i^T. \tag{19.6}$$

(i) In the global joint, with $S = \bar{T}_e(\Sigma_i^0 + \Sigma_j^0 + C_\rho)\bar{T}_e^T + R_e$, the posterior mean moves by

$$\delta y_i = \Sigma_i^0 \bar{T}_e^T S^{-1} r, \qquad \delta y_j = -\Sigma_j^0 \bar{T}_e^T S^{-1} r, \qquad \kappa_i \delta\omega_e = -C_\rho \bar{T}_e^T S^{-1} r. \tag{19.7}$$

(ii) In the law-level per-agent model of the receiver, with the sender's law $\mathcal{N}(m_j, \Sigma_j^0)$ as data, the record covariance is $\bar{T}_e(\Sigma_i^0 + C_\rho)\bar{T}_e^T + (R_e + \bar{T}_e \Sigma_j^0 \bar{T}_e^T) = S$. The receiver's mean therefore

moves by $\delta y_i = \Sigma_i^0 \bar{T}_e^T S^{-1} r$ and $\kappa_i \delta\omega_e = -C_\rho \bar{T}_e^T S^{-1} r$, as in (i), and the sender does not move. In the plug-in model the covariance S is replaced by $\bar{T}_e(\Sigma_i^0 + C_\rho)\bar{T}_e^T + R_e$.

(iii) For the isotropic anchor $V_e = \sigma^2 I$, $C_\rho = \sigma^2 |c_i|^2 I$.

Proof. The quantity $u = y_i - y_j - \kappa_i \omega_e$ has prior mean $m_i - m_j - \kappa_i \bar{\omega}_e$ and covariance $\Sigma_i^0 + \Sigma_j^0 + C_\rho$, and $o_e = \bar{T}_e u + \varepsilon_e$. The covariances of y_i, y_j and $\kappa_i \omega_e$ with o_e are $\Sigma_i^0 \bar{T}_e^T$, $-\Sigma_j^0 \bar{T}_e^T$ and $-C_\rho \bar{T}_e^T$, and that of o_e is S, so Gaussian conditioning gives (i). In (ii) the term $\bar{T}_e y_j$ is distributed as $\mathcal{N}(\bar{T}_e m_j, \bar{T}_e \Sigma_j^0 \bar{T}_e^T)$ and is absorbed into the noise, which gives the displayed covariance, equal to S. For (iii), $\kappa_i^T v = v c_i^T$ and $\kappa_i (v c_i^T) = |c_i|^2 v$. □

The Gaussian conditional update assigns a discrepancy to receiver adjustment, sender adjustment, and reinterpretation through the three cross-covariances in eq. (19.7). Observation noise leaves a residual gap, so these adjustments need not reproduce the record exactly. For one record, the receiver's shifts in the law-level per-agent model coincide with the global ones, because the sender's uncertainty, which widens the receiver's likelihood, enters S in the same way; only the sender's shift differs. A plug-in receiver uses the smaller covariance and hence a larger gain. Here persuasion means adjustment of a content in response to a record; it is not Bayesian persuasion in the sense of information design [116]. The interpretation variance grows as $|c_i|^2$ under the isotropic anchor; at $c_i = 0$ the first-order dictionary term cannot absorb a discrepancy. These linear shares are features of the Gaussian family.

Corollary 19.7 (Grounding turns reinterpretation into persuasion). Let Π_e be the orthogonal projection of $\mathfrak{g}$ onto $N_e^\perp$. Then $\bar{T}_e C_\rho \bar{T}_e^T = \bar{T}_e \kappa_i \Pi_e V_e \Pi_e \kappa_i^T \bar{T}_e^T$, so only the uncertainty of the identified part of the dictionary can absorb a disagreement. If the grounded information on edge e is $g\bar{G}_e$ with $\ker \bar{G}_e = N_e$, then $\Pi_e V_e \Pi_e = O(1/g)$ and the reinterpretation share of a record is $O(1/g)$.

Proof. For $\nu \in N_e$, $\bar{T}_e \kappa_i \nu = \bar{T}_e \nu c_i = 0$, so $\bar{T}_e \kappa_i = \bar{T}_e \kappa_i \Pi_e$. With a positive-definite prior precision Λ on ω_e and grounded precision $g\bar{G}_e$, block inversion in the orthonormal bases of $N_e^\perp$ and N_e shows, by taking the Schur complement of the fixed invisible block, that the covariance restricted to $N_e^\perp$ is $O(1/g)$. □

Grounding therefore suppresses the part of the response attributable to uncertainty about the identified dictionary coordinates.

19.3 The two flows

Proposition 19.8 (The global flow). Let the global recognition be $Q = \bigotimes_i \mathcal{N}(\mu_i, S_i)$ on the agent blocks $x_i = (y_i, \omega_{\mathrm{in}(i)})$, let J_{ii} be the diagonal blocks of J and $D = \bigoplus_i J_{ii}$.

(i) Up to a constant, $\mathcal{F}_G(Q) = \frac{1}{2}(\mu - x^\star)^T J (\mu - x^\star) + \frac{1}{2}\sum_i [\mathrm{tr}(J_{ii} S_i) - \log\det S_i]$, which is minimized by $S_i = J_{ii}^{-1}$ and $\mu = x^\star$.

(ii) At these covariances the Fisher-metric gradient flow of the means is $\dot{\mu} = -D^{-1} J (\mu - x^\star)$. The matrix $D^{-1}J$ is similar to the positive-definite matrix $D^{-1/2} J D^{-1/2}$, so its spectrum is real and positive, $\mathcal{F}_G$ decreases strictly away from $x^\star$, and $\mu \to x^\star$ exponentially.

(iii) Each agent's block of the force is the gradient of its local free energy, the global free energy as a function of its own factor. A record therefore moves the receiver's content, the sender's content and the receiver's dictionary through one functional.

Proof. (i) Since the factors are independent, $\mathbb{E}_Q[E] = \frac{1}{2}(\mu - x^\star)^T J(\mu - x^\star) + \frac{1}{2}\sum_i \operatorname{tr}(J_{ii}S_i) + \text{const}$, and the entropy is $\frac{1}{2}\sum_i \log\det S_i + \text{const}$. (ii) The Fisher metric of $\mathcal{N}(\mu_i, S_i)$ in μ_i is $S_i^{-1} = J_{ii}$, and $\nabla_\mu \mathcal{F}_G = J(\mu - x^\star)$. The similarity is $D^{-1}J = D^{-1/2}(D^{-1/2}JD^{-1/2})D^{1/2}$, and $\frac{d}{dt}\mathcal{F}_G = -\nabla\mathcal{F}_G^T D^{-1}\nabla\mathcal{F}_G$. (iii) follows by varying one factor in the same global functional; every incident likelihood contributes to that variation. □

The optimized factor means equal the Gaussian posterior mean, whereas J_{ii}^{-1} is no larger in the Loewner order than the exact marginal covariance. Indeed, the exact marginal precision is the Schur complement $J_{ii} - J_{i,-i}J_{-i,-i}^{-1}J_{-i,i} \preceq J_{ii}$. Exact factor means are a Gaussian property; the coordinate-descent identity holds more generally, without implying exact moments [23].

Proposition 19.9 (Law-level covariance updates). In the law-level per-agent model let Φ send a family $X = (X_i)_i$ of positive-semidefinite peer covariances, consisting of the content blocks Σ_j and the dictionary blocks V_f, to the corresponding marginal blocks of $(A_i(X)^{-1})_i$, where $A_i(X) = \nabla^2_{x_i} E_i$. The map Φ has exactly one fixed point X^*. The iterations $\Phi^n(0)$ increase to X^*, the iterations from the blocks U of $(A_i^0)^{-1}$ decrease to X^*, and $\Phi^n(X) \to X^*$ from every positive-semidefinite start. Here $A_i^0 = P_i \oplus \bigoplus_{e\in\mathrm{in}(i)}(\sigma^{-2}I + G_e)$ is the peer-independent part of A_i.

Proof. Write $A_i(X) = A_i^0 + G_i(X)$, where $G_i(X)$ collects the terms $L_e^T(R_e + \bar{T}_e\Sigma_j\bar{T}_e^T)^{-1}L_e$ of the received records and, for $\lambda > 0$, the loop terms $B_\gamma^T((\lambda W_\gamma)^{-1} + \sum_f V_f)^{-1}B_\gamma$; here L_e and B_γ are the matrices that read the record discrepancy and the owned part of the loop sum from x_i. Since $Y \mapsto (R+Y)^{-1}$ reverses the Loewner order, $X \preceq X'$ gives $G_i(X) \succeq G_i(X')$, hence $A_i(X)^{-1} \preceq A_i(X')^{-1}$, and principal blocks preserve the order: Φ is order preserving. For $0 < t < 1$, $R+tY \succeq t(R+Y)$ gives $(R+tY)^{-1} \preceq t^{-1}(R+Y)^{-1}$, so $A_i(tX) \preceq t^{-1}A_i(X) - (t^{-1}-1)A_i^0 \prec t^{-1}A_i(X)$ and $\Phi(tX) \succ t\Phi(X)$, with strict order on positive-definite principal blocks. Every image satisfies $\Phi(0) \preceq \Phi(X) \preceq U$ because $G_i \succeq 0$. The sequence $\Phi^n(0)$ is increasing and bounded by U, so it converges, and its limit is a fixed point by continuity; likewise $\Phi^n(U)$ decreases to a fixed point. If X and Y are fixed points, both are positive definite; let t^* be the largest t with $tX \preceq Y$. If $t^* < 1$, then $Y = \Phi(Y) \succeq \Phi(t^*X) \succ t^*X$, so a slightly larger t is admissible, a contradiction. Hence $X \preceq Y$, and by symmetry $X = Y$. Any start X gives $\Phi^n(0) \preceq \Phi^{n+1}(X) \preceq \Phi^n(U)$, and both bounds converge to X^*. □

This is the standard argument for order-preserving, strictly subhomogeneous cone maps and uses only the Gaussian tangent family. In the plug-in model the covariances do not depend on the peers, so the statement is trivial there.

Proposition 19.10 (The per-agent flow). Hold the peers' covariances at the unique fixed point of proposition 19.9, and let agent i hold $\mathcal{N}(\mu_i, S_i)$ on x_i.

(i) Up to a constant, $\mathcal{F}_i = \frac{1}{2}(\mu_i - x_i^\dagger)^T A_i(\mu_i - x_i^\dagger) + \frac{1}{2}[\operatorname{tr}(A_iS_i) - \log\det S_i]$, where $A_i = \nabla^2_{x_i}E_i$ depends on the peers' covariances but not on their means, and $A_i x_i^\dagger = b_i + \sum_{k\neq i} C_{ik}\mu_k$ is affine in the peers' means.

(ii) At $S_i = A_i^{-1}$ the Fisher flow of the means is $\dot{\mu} = -A^{-1}(H\mu - b)$ with $A = \bigoplus_i A_i$ and $H = A - C$. Its fixed points solve $H\mu = b$.

(iii) In the plug-in model, the rows of H and b that belong to dictionaries equal the corresponding rows of J and η. This equality concerns objective derivatives before Fisher preconditioning. The rows that belong to the content of agent i equal those of J and η minus the contributions $L_e^T R_e^{-1} L_e$ and $L_e^T R_e^{-1} o_e$ of the records e in which i is the sender. In the law-level model R_e is in addition replaced by $\tilde{R}_e$ and λW_γ by $\Omega_{i\gamma}$.

(iv) For an edge $e : j \to i$ the block $H_{y_i y_j}$ contains $-\bar{T}_e^T \tilde{R}_e^{-1} \bar{T}_e$, while e contributes nothing to $H_{y_j y_i}$. Hence H need not be symmetric, and its fixed points, which are unique when H is invertible, need not equal $x^\star$.

Proof. (i) and (ii) follow as in proposition 19.8, with the peers' means as data in E_i. For (iii), compare the terms of eqs. (19.2) and (19.3). A dictionary ω_e appears in the anchor, in its grounded term, in the record of e and in the loop factors, and all of these belong to the model of its owner, with the same precisions in the plug-in model; the loop factor of γ in the owner's model has the same gradient in ω_e as the single loop term of the global energy. The content y_i appears in its self term, in the records of $\mathrm{in}(i)$, which are in i's model, and in the records in which i is the sender, which are in the receivers' models only. (iv) is read off from (iii). □

Ownership removes sender-role content derivatives from the per-agent update. The equality of dictionary derivative rows holds only in the plug-in model: the law-level model uses both $\tilde{R}_e$ and $\Omega_{i\gamma}$. These derivatives are covectors. Their Fisher velocities are obtained using the block mobilities J_{ii}^{-1} and A_i^{-1}, which can differ and mix content with owned dictionary coordinates. Equal dictionary derivative rows consequently do not establish equal dictionary velocities. The dependency statement survives for general laws whenever the declared model contains only received records and the dictionary appears only in its owner's factors; equality of derivatives after marginalizing peers does not follow from ownership.

Proposition 19.11 (A directed ring). Let $K = 1$ and let $\ell \geq 3$ agents form a directed ring in which agent i receives from agent $i + 1$, indices modulo ℓ, with $\bar{T}_e = 1$, record precision $\tilde{q}$, self precision p, anchor precision $s = \sigma^{-2}$, common operating content c, and no grounded or loop term. With the own-block Gaussian covariances held at their optima, the per-agent mean flow has the eigenvalue -1 with multiplicity ℓ and the eigenvalues $-(1 - \zeta w)$ for the ℓ-th roots of unity ζ, where

$$w = \frac{\tilde{q}s}{ps + p\tilde{q}c^2 + \tilde{q}s} = \frac{\tilde{q}_{\mathrm{eff}}}{p + \tilde{q}_{\mathrm{eff}}}, \qquad \tilde{q}_{\mathrm{eff}} = \left(\frac{1}{\tilde{q}} + \frac{c^2}{s}\right)^{-1}. \tag{19.8}$$

The eigenvalues with $\zeta \notin \mathbb{R}$ are not real, so the per-agent flow is not a gradient flow for any Riemannian metric, whereas the global flow of the same ring has real spectrum. As $s \to \infty$, $w \to \tilde{q}/(p + \tilde{q})$; as $s \to 0$ with $c \neq 0$, $w \to 0$. If $c = 0$, then $w = \tilde{q}/(p + \tilde{q})$ for every $s > 0$.

Proof. Agent i's energy is $\frac{1}{2}p(y_i - m_i)^2 + \frac{1}{2}s\omega_i^2 + \frac{1}{2}\tilde{q}(o_i - y_i + y_{i+1} + c\omega_i)^2$. In the variables (y_i, ω_i) its Hessian is $A_i = \left(\begin{smallmatrix} p+\tilde{q} & -\tilde{q}c \\ -\tilde{q}c & s+\tilde{q}c^2 \end{smallmatrix}\right)$ with $\det A_i = ps + p\tilde{q}c^2 + \tilde{q}s$, and its coupling to (y_{i+1}, ω_{i+1}) is $C' = \left(\begin{smallmatrix} -\tilde{q} & 0 \\ \tilde{q}c & 0 \end{smallmatrix}\right)$. The flow matrix is block circulant, so on the vectors $x_k = \zeta^k v$ it acts as $-(I + \zeta A_i^{-1} C')$. The matrix $A_i^{-1} C'$ has zero second column, and its first diagonal entry is $\det A_i^{-1}[(s + \tilde{q}c^2)(-\tilde{q}) +$

$(\tilde{q}c)^2] = -w$. Hence the eigenvalues are -1 and $-(1-\zeta w)$. The linearization of a gradient flow at an equilibrium is self-adjoint for the metric there and has real spectrum, since its Jacobian is the negative metric inverse times the symmetric objective Hessian. The global statement is proposition 19.8(ii). The second form of w and the limits are algebra. □

For $c \neq 0$, uncertainty about the dictionary acts in the mean equations as additional record noise of variance c^2/s. The directed oscillatory modes retain angular frequency $w \operatorname{Im}\zeta$, while $0 < w < 1$ keeps every mode stable. In the limit $s \downarrow 0$ the receiver can absorb the discrepancy through reinterpretation and the coupling vanishes. At $c = 0$ the dictionary does not enter the tangent record, so weakening its anchor does not weaken persuasion. Here $\tilde{q} = R^{-1}$ in the plug-in model and $\tilde{q} = (R+\Sigma)^{-1}$ in the law-level model, with Σ the content variance at the unique law-level covariance fixed point of proposition 19.9. The complex spectrum is one explicit instance of non-reciprocal dynamics [85]; it rules out a general gradient-flow interpretation of the per-agent picture.

19.4 The grounded limit

Theorem 19.12 (Separation in the grounded limit). Let $G_e = g\bar{G}_e$ with $\bar{G}_e$ positive semidefinite and $\ker \bar{G}_e = N_e$ for every edge, keep the grounded centers $\hat{\omega}_e$ fixed, and let $g \to \infty$.

(i) In the global joint the posterior mean converges to (y^G, ω^∞), where y^G minimizes

$$E_{\text{known}}(y) = \sum_i \tfrac{1}{2}\|y_i - m_i\|^2_{P_i} + \sum_e \tfrac{1}{2}\big\|o_e + \bar{T}_e\kappa_i\hat{\omega}_e - \bar{T}_e(y_i - y_j)\big\|^2_{R_e^{-1}}, \tag{19.9}$$

the identified part of ω_e^∞ is that of $\hat{\omega}_e$, and the invisible part minimizes the dictionary prior on the coset $\hat{\omega} + \bigoplus_e N_e$. In the limit the beliefs do not depend on the dictionary prior.

(ii) In the plug-in per-agent model, if the content equations

$$P_i(y_i - m_i) - \sum_{e\in\text{in}(i)} \bar{T}_e^T R_e^{-1}\big(o_e + \bar{T}_e\kappa_i\hat{\omega}_e - \bar{T}_e(y_i - y_j)\big) = 0 \qquad (i \in \mathcal{V}) \tag{19.10}$$

have a unique solution y^P, the fixed point converges to it in the contents, and the identified dictionary parts converge to those of $\hat{\omega}_e$.

(iii) In both cases the error is $O(1/g)$.

Proof. Write $\omega_e = \hat{\omega}_e + Y_e\alpha_e + Z_e\nu_e$ in orthonormal bases Y_e of $N_e^\perp$ and Z_e of N_e. Since $\bar{T}_e\kappa_i Z_e = 0$, the records depend on (y, α) only. The energy is $\frac{g}{2}\sum_e \alpha_e^T\Gamma_e\alpha_e + \tilde{E}(y,\alpha,\nu)$ with $\Gamma_e = Y_e^T\bar{G}_eY_e$ positive definite and $\tilde{E}$ independent of g. In both pictures the stationarity equations are linear and have the block form $(g\Gamma + H_{\alpha\alpha})\alpha + H_{\alpha z}z = b_\alpha$ and $H_{z\alpha}\alpha + H_{zz}z = b_z$ with $z = (y,\nu)$ and H, b independent of g. Eliminating $\alpha = (g\Gamma + H_{\alpha\alpha})^{-1}(b_\alpha - H_{\alpha z}z) = O(1/g)$ gives $(H_{zz} + O(1/g))z = b_z + O(1/g)$, so $z \to H_{zz}^{-1}b_z$ at rate $1/g$ whenever H_{zz} is invertible. At $\alpha = 0$ the records involve y alone and the dictionary prior involves ν alone, so H_{zz} is block diagonal. Its content block gives eq. (19.9) in the global joint, where J is positive definite, and eq. (19.10) in the plug-in model, invertible by hypothesis. Its invisible-dictionary block is the restriction of the dictionary prior precision $\sigma^{-2}I + \lambda B^TWB$ to the invisible directions, positive definite, in both pictures by proposition 19.10(iii). □

The theorem identifies two different known-dictionary limits. The global mean minimizes a symmetric quadratic comparison energy, while the plug-in per-agent fixed point solves receiver-only equations. The identified dictionaries agree in the limit, but the contents need not. The theorem makes no claim about law-level covariance updates that vary with g; controlling those updates is an additional obligation.

19.5 Confidence under ownership

Mean agreement does not imply agreement about uncertainty. The following example isolates one coordinate that records and grounding do not read, so all information coupling it to other dictionaries comes from one declared loop factor.

Proposition 19.13 (Echo around a loop). Let $\lambda > 0$ and let $\ell \geq 2$ edges form one loop, with distinct receivers and one coordinate z_e per edge that no record or grounded term reads, with anchor $\mathcal{N}(0, \sigma^2)$ and loop factor $\exp(-\frac{\lambda}{2}(\sum_e z_e)^2)$. Every recognition below has mean zero, and the variances of z_e are as follows.

(i) Exact joint posterior: $\sigma^2(1 + (\ell - 1)\lambda\sigma^2)/(1 + \ell\lambda\sigma^2)$, which tends to $\sigma^2(\ell - 1)/\ell$ as $\lambda \to \infty$.

(ii) Receiver-factorized global recognition: $(\sigma^{-2} + \lambda)^{-1}$, which tends to zero.

(iii) Per-agent law-level recognition at its covariance fixed point, which is unique by proposition 19.9 and therefore symmetric: the unique positive solution v of

$$v^{-1} = \sigma^{-2} + \big(\lambda^{-1} + (\ell - 1)v\big)^{-1}, \tag{19.11}$$

which tends to $\sigma^2(\ell - 2)/(\ell - 1)$ for $\ell \geq 3$ and satisfies $v = \sigma\lambda^{-1/2}(1 + o(1))$ for $\ell = 2$.

(iv) Cavity recognition, in which each receiver's loop source uses its partners' laws with its own contribution removed, so that those laws are the anchors: $(\sigma^{-2} + (\lambda^{-1} + (\ell - 1)\sigma^2)^{-1})^{-1}$, which equals (i) for every λ.

Proof. All terms are even in z, so the means vanish. (i) The precision is $\sigma^{-2}I + \lambda\mathbf{1}\mathbf{1}^T$, and the Sherman–Morrison formula gives the diagonal entry $\sigma^2 - \sigma^4\lambda/(1 + \ell\lambda\sigma^2)$. (ii) The factorized covariance is the inverse of the diagonal of the precision. (iii) By symmetry each receiver's loop source is the convolution of the loop factor with $\ell - 1$ partner laws of variance v, which gives eq. (19.11). With $\epsilon = \lambda^{-1}$, clearing denominators gives $(\ell-1)\sigma^{-2}v^2 + (\epsilon\sigma^{-2} + 2 - \ell)v - \epsilon = 0$, whose roots have product $-\epsilon\sigma^2/(\ell - 1) < 0$, so exactly one is positive. As $\epsilon \to 0$ it tends to $\sigma^2(\ell - 2)/(\ell - 1)$ for $\ell \geq 3$, and for $\ell = 2$ it satisfies $v^2 + \epsilon v - \epsilon\sigma^2 = 0$, so $v = \sigma\epsilon^{1/2}(1 + o(1))$. (iv) With $\epsilon = \lambda^{-1}$, $(\sigma^{-2} + (\epsilon + (\ell - 1)\sigma^2)^{-1})^{-1} = \sigma^2(\epsilon + (\ell - 1)\sigma^2)/(\epsilon + \ell\sigma^2)$, which is (i) after multiplying numerator and denominator by λ. □

All four constructions have the same zero mean and different variances. Factorized reverse-KL recognition suppresses the correlations by which the exact law satisfies the sum constraint. The per-agent update instead treats a partner's current marginal as new independent information, even though that marginal already contains information received from the same loop. For a two-cycle its fixed-point variance tends to zero while the exact marginal tends to $\sigma^2/2$. Removing the loop contribution from the peer messages gives the anchor laws and restores the exact marginal here.

This is belief propagation on the tree consisting of one factor connected to ℓ variables [160]; it is not a general exactness theorem for overlapping loop factors. Extended to overlapping Gaussian loop factors, the cavity construction of (iv) is Gaussian belief propagation on the factor graph of anchors, grounded terms and loop factors, whose means are exact whenever it converges while its variances are in general not [198]. The per-agent echo is the double counting of information that distributed estimation calls data incest; in opinion dynamics, treating a partner's current opinion as new evidence is persuasion bias [59].

The recycled-information mechanism can occur for non-Gaussian laws as well. The displayed fixed points, rates, and equality of factor means are conclusions only for this Gaussian single-loop model.

19.6 The reciprocal potential

The reciprocal potential is a function of agent laws. It moves each source through the costs of receivers that attend to it. The following extension includes dictionary laws as configuration variables. Registered comparison records are absent from this potential; adding persistent likelihood terms would define another objective.

Definition 19.14 (Dictionary laws and source rows). Consider one content sector, with singleton model-presentation spaces. Let q_i and r_e be laws on standard Borel content and dictionary spaces, let p_i be fixed proper content references, and let π_ω be a fixed proper joint dictionary reference. Recognition is restricted to the product $r = \bigotimes_e r_e$, even when π_ω correlates edges. Fixed normalized kernels give laws $A_e(\omega_e) = q_i \mathcal{K}_e^{\mathrm{rec}}(\omega_e)$ and $B_e = q_j \mathcal{K}_e^{\mathrm{src}}$ on a common message space for each $e : j \to i$. The receiver kernel is a normalized kernel from the product of the dictionary and content spaces to the message space, jointly measurable in (ω_e, k); then $\omega_e \mapsto A_e(\omega_e)$ is measurable by a monotone-class argument and $\omega_e \mapsto D_{\mathrm{KL}}(A_e(\omega_e)\|B_e)$ is Borel, as for the law-dependent channels of chapter 4. Put

$$\begin{aligned} C_e &= \int \Delta_e(\omega_e) r_e(d\omega_e), & \Delta_e(\omega_e) &= D_{\mathrm{KL}}(A_e(\omega_e)\|B_e), \\ F_0 &= \sum_i D_{\mathrm{KL}}(q_i\|p_i) + D_{\mathrm{KL}}(r\|\pi_\omega). \end{aligned} \tag{19.12}$$

All displayed relative entropies and expected costs are required to be finite. Each nonempty candidate set $\mathcal{I}_i = \mathrm{in}(i)$ has a positive normalized prior $\pi_i = (\pi_e)_{e\in\mathcal{I}_i}$, a temperature $T_i > 0$, and a positive normalized row β_i. Agent-dependent temperatures extend the reciprocal potential of definition 4.2. Rows with no candidates are omitted. The variable-row potential and its minimum over rows are

$$\begin{aligned} \mathcal{E}_T &= F_0 + \sum_i \left[\sum_{e\in\mathcal{I}_i} \beta_e C_e + T_i D_{\mathrm{KL}}(\beta_i\|\pi_i) \right], \\ Z_i &= \sum_{e\in\mathcal{I}_i} \pi_e e^{-C_e/T_i}, \qquad \beta_e^* = \frac{\pi_e e^{-C_e/T_i}}{Z_i}, \qquad E_T = F_0 - \sum_i T_i \log Z_i. \end{aligned} \tag{19.13}$$

All reference laws, kernels, priors, and temperatures are fixed in the variation. The dictionary expectation is outside the message divergence. Averaging receiver laws before taking the divergence defines a different cost.

On regular dominated charts with positive message densities and integrable envelopes justifying differentiation, eq. (4.10) gives

$$dC_e = \int \Delta_e dr_e + \int r_e(d\omega_e) \left[\int \log(A_e/B_e) dA_e - \int (A_e/B_e) dB_e \right]. \tag{19.14}$$

Here $dA_e = (dq_i)\mathcal{K}_e^{\mathrm{rec}}(\omega_e)$ and $dB_e = (dq_j)\mathcal{K}_e^{\mathrm{src}}$. The term in dr_e accounts for changing dictionary recognition. In density notation its own-block first variation is $\delta\mathcal{E}_T/\delta r_e = \log r_e - \mathbb{E}_{r_{-e}} \log \pi_\omega + \beta_e \Delta_e$, up to a sample-independent constant, whenever the displayed conditional expectation and split variation are integrable. This follows by differentiating the product entropy and the expected log joint prior, and retains the prior's correlations. For a parametric Fisher flow the family must additionally have finite positive-definite Fisher tensor. None of these regularity conditions follows from standard Borel measurability alone.

Proposition 4.3 applies to these expected costs and gives

$$\mathcal{E}_T = E_T + \sum_i T_i D_{\mathrm{KL}}(\beta_i \| \beta_i^*), \qquad dE_T = dF_0 + \sum_e \beta_e^* dC_e. \tag{19.15}$$

The second identity differentiates the complete row objective, including entropy. Substitution of $\beta^*(x)$ into the entropy-free weighted sum does not give this derivative.

Definition 19.15 (Tangent reciprocal potential). Let a configuration consist of content laws $q_i = \mathcal{N}(\mu_i, \Sigma_i)$ and a factorized dictionary law $r = \bigotimes_e \mathcal{N}(a_e, V_e)$. For an edge $e : j \to i$ put $S_{i,e} = \bar{T}_e \Sigma_i \bar{T}_e^T + R_e$ and $S_{j,e} = \bar{T}_e \Sigma_j \bar{T}_e^T + R_e$, and define the message laws

$$m_e^{\mathrm{rec}}(\omega_e) = \mathcal{N}\big(\bar{T}_e(\mu_i - \kappa_i \omega_e), S_{i,e}\big), \qquad m_e^{\mathrm{src}} = \mathcal{N}\big(\bar{T}_e \mu_j, S_{j,e}\big), \tag{19.16}$$

the law of the message the receiver would send about its own content read in the sender's coordinates, and the law of the sender's message. With fixed positive rows β_e, the tangent reciprocal potential is

$$\mathcal{E} = \sum_i D_{\mathrm{KL}}\big(q_i \| \mathcal{N}(m_i, P_i^{-1})\big) + \sum_e \beta_e \mathbb{E}_r \big[D_{\mathrm{KL}}\big(m_e^{\mathrm{rec}}(\omega_e) \| m_e^{\mathrm{src}}\big)\big] + D_{\mathrm{KL}}(r \| \pi_\omega), \tag{19.17}$$

where π_ω is the normalized dictionary prior with density proportional to $\exp(-\frac{1}{2}\omega^T(\sigma^{-2}I + \lambda B^T W B)\omega - \frac{1}{2}\sum_e \|\omega_e - \hat{\omega}_e\|_{G_e}^2)$.

The receiver's law comes first and the source's second, as in a directed peer divergence, and the costs are divergences between message laws, rather than comparisons between independent content samples. The self term replaces the local term $\mathcal{E}_0$ by its tangent form. The last term makes the dictionary laws configuration variables whose part of the potential is a recognition free energy over dictionaries. For normalized variable rows, use $\mathcal{E}_T = \mathcal{E} + \sum_i T_i D_{\mathrm{KL}}(\beta_i \| \pi_i)$ from eq. (19.13). Its Gibbs reduction has the pointwise derivative eq. (19.15), so the derivative statements below

apply with the current Gibbs weights inserted. State-dependent weights do not preserve the constant-coefficient mean-flow theorem.

Proposition 19.16 (Forces in the reciprocal potential). Hold the covariances Σ_i and V_e fixed. Let E^S be the global energy eq. (19.2) with every R_e replaced by $S_{j,e}/\beta_e$ and every registered value o_e by zero, with precision J^S and information vector η^S.

(i) $\mathcal{E} = E^S(\mu, a) + c(\Sigma, V)$, where c does not depend on the means.

(ii) The force on a dictionary mean a_e comes only from the cost of its own edge and from π_ω, because no other cost contains ω_e. There is no additional sender-role edge derivative for the dictionary. The force on a content mean contains the terms of the edges on which the agent is the sender, as in the global joint.

(iii) The Fisher gradient flow of the means is $(\dot\mu, \dot a) = -MJ^S\big((\mu, a) - x^S\big)$ with $M = (\bigoplus_i \Sigma_i) \oplus (\bigoplus_e V_e)$ and $x^S = (J^S)^{-1}\eta^S$. Its spectrum is real and negative, and its stationary point is the posterior mean of the global joint in which every record has noise $S_{j,e}/\beta_e$ and registered value zero.

(iv) For $\beta_e = 1$, the content rows of J^S minus the sender terms are the content rows of the per-agent law-level model with $o_e = 0$ at the same covariances. The dictionary rows of J^S differ from the per-agent law-level ones only in that the loop sources of the latter are widened as in eq. (19.4).

Proof. (i) For Gaussian laws of dimension n, $D_{\mathrm{KL}}(\mathcal{N}(\alpha, A)\|\mathcal{N}(\beta', B)) = \frac{1}{2}[\mathrm{tr}(B^{-1}A) - n + (\alpha - \beta')^T B^{-1}(\alpha - \beta') + \log\det B - \log\det A]$. Under r the mean of m_e^{rec} is $\bar T_e(\mu_i - \kappa_i a_e) - \bar T_e\kappa_i(\omega_e - a_e)$, so the averaged cost is

$$\tfrac{1}{2}\big[\mathrm{tr}(S_{j,e}^{-1}S_{i,e}) - n_e + d_e^T S_{j,e}^{-1} d_e + \mathrm{tr}(S_{j,e}^{-1}\bar T_e\kappa_i V_e\kappa_i^T\bar T_e^T) + \log\det S_{j,e} - \log\det S_{i,e}\big] \tag{19.18}$$

with $d_e = \bar T_e(\mu_i - \mu_j - \kappa_i a_e) = L_e(\mu, a)$. The self term is $\frac{1}{2}(\mu_i - m_i)^T P_i(\mu_i - m_i)$ plus covariance terms. Since r is a product and $\mathbb{E}_r[\omega^T A\omega] = a^T Aa + \mathrm{tr}(AV)$, $D_{\mathrm{KL}}(r\|\pi_\omega)$ is $\frac{1}{2}a^T(\sigma^{-2}I + \lambda B^T W B)a + \frac{1}{2}\sum_e(a_e - \hat\omega_e)^T G_e(a_e - \hat\omega_e)$ plus covariance terms and a constant. The mean-dependent terms sum to E^S. (ii) In eq. (19.17) the variable ω_e appears only in the cost of e and in π_ω. (iii) The Fisher metric of q_i in μ_i is Σ_i^{-1} and that of $\mathcal{N}(a_e, V_e)$ in a_e is V_e^{-1}, so the Fisher gradient is $M\nabla E^S = MJ^S((\mu, a) - x^S)$, and MJ^S is similar to $M^{1/2}J^S M^{1/2}$, which is positive definite because the self terms and the anchor are. (iv) In the per-agent law-level model $\tilde R_e = R_e + \bar T_e\Sigma_j\bar T_e^T = S_{j,e}$; the rest is proposition 19.10(iii). □

At fixed covariances and fixed rows, the reciprocal potential has the same mean-dependent quadratic as a particular widened Gaussian global model. This identity does not identify their covariance objectives, their recognition families, or their full dynamics. A dictionary occurs only in its own edge cost and in the declared prior, so other dictionary means enter its derivative only through the loop term. This locality is a dependency property of the declared fixed codes and prior; it persists for other laws with the same factor structure. The explicit quadratic and spectral conclusions use Gaussian laws and fixed covariances.

Proposition 19.17 (Disagreement and certainty).

(i) In the Gaussian tangent forms of the global joint and of both per-agent models, the optimizing recognition covariances do not depend on the registered records or on any mean. This is a feature of linear-Gaussian models, whose posterior covariance never depends on the data, and not a property of the free energies.

(ii) In the tangent reciprocal potential, the derivative of the unweighted cost of an edge $e : j \to k$ with respect to the sender's covariance Σ_j is

$$\tfrac{1}{2}\bar{T}_e^T S_{j,e}^{-1}\big[S_{j,e} - S_{k,e} - d_e d_e^T - \bar{T}_e \kappa_k V_e \kappa_k^T \bar{T}_e^T\big] S_{j,e}^{-1}\bar{T}_e, \tag{19.19}$$

so the source-role force on the sender's covariance vanishes exactly when the sender's message covariance equals the receiver's predicted message covariance plus the outer product of the coded disagreement and the reinterpretation covariance.

(iii) Let $K = 1$, $\bar{T}_e = 1$, $\beta_e = 1$, and hold the dictionary variance $V_e \geq 0$ fixed. Hold the means, $R > 0$, and receiver variance v_k fixed, and let the sender's variance-dependent objective consist only of its self term with precision p_j and one outgoing edge cost. With $a = v_k + R + d^2 + c_k^2 V_e$, the stationary sender variance v solves $f(v) = p_j - v^{-1} + (v + R)^{-1} - a(v + R)^{-2} = 0$. This equation has exactly one positive root, which is the unique global minimizer of the variance-dependent objective, and it increases strictly with $d^2 + c_k^2 V_e$.

Proof. (i) The global covariances are J_{ii}^{-1} and the per-agent ones are A_i^{-1}; these matrices depend on the precisions and on the peers' covariances only. (ii) The terms of the averaged cost that contain $S_{j,e}$ are $\frac{1}{2}[\mathrm{tr}(S_{j,e}^{-1}C) + \log\det S_{j,e}]$ with $C = S_{k,e} + d_e d_e^T + \bar{T}_e \kappa_k V_e \kappa_k^T \bar{T}_e^T$; their derivative with respect to $S_{j,e}$ is $\frac{1}{2}(S_{j,e}^{-1} - S_{j,e}^{-1} C S_{j,e}^{-1})$, and $S_{j,e} = \bar{T}_e \Sigma_j \bar{T}_e^T + R_e$ gives eq. (19.19) by the chain rule. (iii) The function f is twice the derivative of $\mathcal{E}$ in v; the dictionary variance enters through the term $\bar{T}_e \kappa_k V_e \kappa_k^T \bar{T}_e^T = c_k^2 V_e$ of the edge cost. Over the common denominator $v^2(v + R)^3$,

$$\partial_v f = \frac{R^3 + 3R^2 v + 4Rv^2 + 2(a - R)v^2}{v^2(v + R)^3} > 0 \tag{19.20}$$

for every $v > 0$, since $a - R \geq 0$. As $f \to -\infty$ when $v \downarrow 0$ and $f \to p_j > 0$ as $v \to \infty$, f has exactly one root, where the objective changes from decreasing to increasing. Its derivative with respect to a is $-(v + R)^{-2} < 0$, so the implicit function theorem gives $dv/da > 0$. □

In the potential the source law is compared only through its message law. For a source q_η in a regular exponential family with statistic t, compared through a fixed kernel $\mathcal{K}$ with positive message densities, $\partial_\eta \log(q_\eta \mathcal{K})(h) = \mathbb{E}_{q_\eta}[t \mid h] - \mathbb{E}_{q_\eta} t$, so that

$$\partial_\eta D_{\mathrm{KL}}(p \| q_\eta \mathcal{K}) = \mathbb{E}_{q_\eta} t - \mathbb{E}_{\tilde{p}_\eta} t, \qquad \tilde{p}_\eta = \int p(dh) q_\eta(\cdot \mid h), \tag{19.21}$$

whenever the moments exist and differentiation under the integral is justified. Source-role stationarity is therefore the fixed-point condition $\mathbb{E}_{q_\eta} t = \mathbb{E}_{\tilde{p}_\eta} t$ of the M-step of the EM algorithm [60]: the source matches, in its sufficient statistics, its own posterior reconstruction $\tilde{p}_\eta$ of the receiver's message law, the noisy reconstruction of corollary 18.5 with the source law as decoder prior. For an identity kernel it is the moment match $\mathbb{E}_{q_\eta} t = \mathbb{E}_p t$. The Gaussian covariance formula eq. (19.19) is the second-moment instance at fixed mean, including the squared discrepancy about the sender's

mean; it reduces to an equality of message covariances because the linear-Gaussian code maps the Gaussian family onto an open set of Gaussian message laws. Over the outgoing edges of a source these conditions are pooled with the weights β_e of the full potential. The scalar result proves increasing stationary variance with disagreement only while the other specified quantities are frozen. Competing self terms, receiver roles, other source roles, moving means, and changing rows determine the full covariance velocity, so no monotone loss of certainty is asserted for that flow. General non-Gaussian Bayesian posteriors may also change their uncertainty when records change.

For product dictionary laws in definition 19.15, covariance stationarity gives $V_e^{-1} = \sigma^{-2}I + \lambda(B^TWB)_{ee} + G_e + \beta_e\kappa_i^T\bar{T}_e^TS_{j,e}^{-1}\bar{T}_e\kappa_i$. The grounded and record information vanish on N_e. When $W_\gamma = I$, the loop diagonal block is the number of simple loops containing e times the identity, so N_e is invariant under this precision. In the single-loop setting of proposition 19.13, the variance on an unread coordinate is $(\sigma^{-2}+\lambda)^{-1}$, the same value as factorized global recognition. No general marginal-variance formula for overlapping loops is implied.

19.7 Coupled means, covariances, and rows

All content and dictionary parameters now vary within the Gaussian product family. The codes and operating contents remain fixed. Identify $\mathfrak{g}$ with its $d_\omega = K^2$ reference-coordinate vector space and write

$$\begin{aligned} &A_\omega = \sigma^{-2}I + \lambda B^TWB + \operatorname{diag}(G_e), \quad b_\omega = (G_e\hat{\omega}_e)_e, \quad a_0 = A_\omega^{-1}b_\omega, \\ &K_e = \bar{T}_e\kappa_i, \qquad\qquad J_e = S_{j,e}^{-1}, \qquad Q_e = S_{i,e} + d_ed_e^T + K_eV_eK_e^T. \end{aligned} \tag{19.22}$$

Thus $\pi_\omega = \mathcal{N}(a_0, A_\omega^{-1})$ is a normalized joint prior. Its recognition covariance is $V = \operatorname{diag}(V_e)$ and its mean is $a = (a_e)_e$. The expected edge cost and dictionary KL are

$$\begin{aligned} C_e &= \tfrac{1}{2}[\operatorname{tr}(J_eQ_e) - n_e + \log\det S_{j,e} - \log\det S_{i,e}], \\ D_{\mathrm{KL}}(r\|\pi_\omega) &= \tfrac{1}{2}[(a-a_0)^TA_\omega(a-a_0) + \operatorname{tr}(A_\omega V) - \log\det(A_\omega V) - d_{\mathrm{tot}}], \end{aligned} \tag{19.23}$$

where $d_{\mathrm{tot}} = |\mathcal{E}|d_\omega$. Since V is block diagonal, $\operatorname{tr}(A_\omega V) = \sum_e \operatorname{tr}((A_\omega)_{ee}V_e)$ and $\log\det(A_\omega V) = \log\det A_\omega + \sum_e \log\det V_e$. Off-diagonal prior blocks still couple the dictionary means.

Proposition 19.18 (Complete Gaussian Fisher equations). Use the fixed data of definition 19.1 and positive rows and temperatures of definition 19.14. Let g_{μ_i}, g_{a_e} be mean derivatives and G_{Σ_i}, G_{V_e} symmetric covariance derivatives under the trace pairing. In the full flow β is independent;

in the reduced flow it is $\beta^*(x)$. Then

$$\begin{aligned} g_{\mu_i} &= P_i(\mu_i - m_i) + \sum_{e\in\mathrm{in}(i)} \beta_e \bar{T}_e^T J_e d_e - \sum_{e\in\mathrm{out}(i)} \beta_e \bar{T}_e^T J_e d_e, \\ g_{a_e} &= (A_\omega a - b_\omega)_e - \beta_e K_e^T J_e d_e, \\ 2G_{\Sigma_i} &= P_i - \Sigma_i^{-1} + \sum_{e\in\mathrm{in}(i)} \beta_e \bar{T}_e^T (J_e - S_{i,e}^{-1}) \bar{T}_e \\ &\quad + \sum_{e\in\mathrm{out}(i)} \beta_e \bar{T}_e^T (J_e - J_e Q_e J_e) \bar{T}_e, \\ 2G_{V_e} &= (A_\omega)_{ee} - V_e^{-1} + \beta_e K_e^T J_e K_e. \end{aligned} \tag{19.24}$$

Here $\mathrm{out}(i)$ consists of edges on which i is the source. The unit-mobility product Fisher equations are

$$\begin{aligned} \dot{\mu}_i &= -\Sigma_i g_{\mu_i}, \quad \dot{\Sigma}_i = -2\Sigma_i G_{\Sigma_i} \Sigma_i, \\ \dot{a}_e &= -V_e g_{a_e}, \quad \dot{V}_e = -2V_e G_{V_e} V_e. \end{aligned} \tag{19.25}$$

For full row relaxation, append

$$h_e = C_e + T_i(\log(\beta_e/\pi_e) + 1), \qquad \dot{\beta}_e = -\beta_e \left(h_e - \sum_{f\in\mathcal{I}_i} \beta_f h_f \right). \tag{19.26}$$

For Gibbs reduction this row ODE is absent. Its kinetic metric is the matter-family Fisher metric, not the pullback of the full metric to $x \mapsto (x, \beta^*(x))$.

Proof. Differentiate eq. (19.23). Its mean derivatives are $\bar{T}_e^T J_e d_e$, $-\bar{T}_e^T J_e d_e$, and $-K_e^T J_e d_e$ for receiver, source, and dictionary. Varying $S_{i,e}$ gives $(J_e - S_{i,e}^{-1})/2$. Using $dJ_e = -J_e(dS_{j,e})J_e$ gives $(J_e - J_e Q_e J_e)/2$ in the source covariance; varying V_e gives $K_e^T J_e K_e/2$. Add the self KL derivatives and the prior derivatives $A_\omega a - b_\omega$ and $((A_\omega)_{ee} - V_e^{-1})/2$ to obtain eq. (19.24). For a Gaussian perturbation (u, U) the Fisher quadratic form is $u^T\Sigma^{-1}u + \mathrm{tr}(\Sigma^{-1}U\Sigma^{-1}U)/2$, obtained by averaging the product of the Gaussian scores. Inverting these blocks gives eq. (19.25), as in eq. (4.17). The categorical score metric is $\sum_e (d\beta_e)^2/\beta_e$ on zero-sum tangents, whose gradient centers h_e as in eq. (19.26). Finally eq. (19.15) gives the reduced derivatives without an extra $d\beta^*$ term. □

The full energy dissipation is

$$\begin{aligned} -\dot{\mathcal{E}}_T = &\sum_i [g_{\mu_i}^T \Sigma_i g_{\mu_i} + 2\,\mathrm{tr}(\Sigma_i G_{\Sigma_i} \Sigma_i G_{\Sigma_i})] \\ &+ \sum_e [g_{a_e}^T V_e g_{a_e} + 2\,\mathrm{tr}(V_e G_{V_e} V_e G_{V_e})] + \sum_i \mathrm{Var}_{\beta_i}(h). \end{aligned} \tag{19.27}$$

Each covariance trace is nonnegative: $\mathrm{tr}(\Sigma G \Sigma G) = \mathrm{tr}((\Sigma^{1/2} G \Sigma^{1/2})^2)$. For E_T the same identity omits the last sum and uses the current Gibbs weights. This does not give a sign to an individual edge cost or covariance velocity.

Since every Fisher block is positive definite, stationarity sets all four derivatives in eq. (19.24) to zero, and full row stationarity also imposes $\beta = \beta^*$. In particular,

$$\begin{aligned} V_e^{-1} &= (A_\omega)_{ee} + \beta_e K_e^T J_e K_e, \\ [A_\omega + \operatorname{diag}(\beta_e K_e^T J_e K_e)]a &= b_\omega + (\beta_e K_e^T J_e \bar{T}_e(\mu_i - \mu_j))_e. \end{aligned} \tag{19.28}$$

The latter system determines a unique dictionary mean for fixed content means, content covariances, and rows. It does not prove uniqueness of the complete stationary configuration, because the rows and source precisions depend on that configuration.

Theorem 19.19 (Global flow in the Gaussian tangent family). Fix a finite graph, fixed codes and operating maps κ_i, proper Gaussian content references with $P_i \succ 0$, a fixed normalized dictionary prior with $A_\omega \succ 0$, message noises $R_e \succ 0$, and positive fixed normalized source priors and temperatures. From every finite initial mean, positive-definite content and dictionary covariance, and interior initial row, eqs. (19.25) and (19.26) has a unique solution for every $t \geq 0$. The means remain bounded, all covariance eigenvalues remain bounded above and away from zero, and all row entries have positive uniform lower bounds depending on the initial state and fixed data. The Gibbs-reduced convention has the same conclusions. Both potentials have an interior stationary point; along either flow the squared Fisher-gradient norm is integrable and tends to zero, and every trajectory converges to a single stationary point.

Proof. The vector fields are smooth on the product of mean spaces, positive-definite covariance cones, and simplex interiors, so they have unique maximal interior solutions. All terms of $\mathcal{E}_T$ are nonnegative. Since $C_e \geq 0$ and the priors are normalized, $0 < Z_i \leq 1$, so $E_T \geq F_0 \geq 0$ as well. Energy dissipation therefore bounds each content self KL and the dictionary KL by the initial energy.

If λ_k are the eigenvalues of $P_i^{1/2}\Sigma_i P_i^{1/2}$, the content KL is

$$\tfrac{1}{2}\|\mu_i - m_i\|_{P_i}^2 + \tfrac{1}{2}\sum_k (\lambda_k - \log\lambda_k - 1). \tag{19.29}$$

The scalar summand is nonnegative and diverges at zero and infinity. Thus the means are bounded and the covariances are spectrally bounded in the interior. The same argument applies to the eigenvalues of $A_\omega^{1/2} V A_\omega^{1/2}$ and the mean $a - a_0$ in eq. (19.23). Fixed positive-definite A_ω converts these to bounds on V, and hence on every V_e. The product-covariance restriction is closed within these bounds. The matter variables therefore stay in a compact interior set $\mathcal{K}$, on which every continuous cost lies in $[0, M]$ for some finite M.

For Gibbs rows, $\beta_e^* \geq \pi_e e^{-M/T_i}$. For a full row put $\eta_{ef} = \log(\beta_e/\pi_e) - \log(\beta_f/\pi_f)$. Subtracting eq. (19.26) in logarithmic form gives

$$\dot{\eta}_{ef} = -T_i \eta_{ef} - (C_e - C_f), \qquad |\eta_{ef}(t)| \leq \max\{|\eta_{ef}(0)|, M/T_i\}. \tag{19.30}$$

The bound follows by variation of constants. Let L_i be the maximum of these bounds over the row. With $u_e = \beta_e/\pi_e$, one has $u_f \leq e^{L_i} u_e$ and $\sum_f \pi_f u_f = 1$, so $\beta_e \geq \pi_e e^{-L_i} > 0$. Singleton rows are identically one. The full trajectory stays in a compact subset of its smooth domain; ordinary finite-dimensional continuation excludes a finite maximal time. The reduced field is smooth on $\mathcal{K}$ and has the same conclusion.

The reduced potential attains a global minimum on a nonempty compact sublevel by eq. (19.29); the minimum lies in the interior. Its Gibbs rows give a full stationary point by eq. (19.15). Finally let $\mathcal{D}$ be the right side of eq. (19.27), with the row term omitted in the reduced convention. Integrating dissipation gives $\int_0^\infty \mathcal{D}(t)dt \le E_{\mathrm{init}}$. On the compact trajectory closure $\mathcal{D}$ and its derivative along the smooth vector field are bounded, so $\mathcal{D}$ is uniformly continuous. A nonnegative uniformly continuous integrable function tends to zero: otherwise disjoint intervals of fixed positive width would each contribute a fixed positive amount to its integral. Continuity and the positive-definite Fisher metric then make every accumulation point stationary.

Both potentials are real analytic on the open product of mean spaces, positive-definite cones and simplex interiors: they are built from Gaussian relative entropies, log determinants, traces rational in the covariances, $\beta \log \beta$, and log-sum-exp. The Fisher metric is analytic there and, on the compact trajectory closure, uniformly equivalent to the Euclidean metric of the affine coordinates, with constants $0 < m \le M$. Along the flow $\dot{y} = -g^{-1}\nabla V$ one therefore has $-\frac{d}{dt}V \ge (m/M)|\nabla V||\dot{y}|$. At an accumulation point the Łojasiewicz gradient inequality holds, so the trajectory has finite length and converges to that point [2, 140], as in proposition 4.7. □

These bounds need not be uniform as temperature tends to zero or reference precisions become singular. The theorem concerns a fixed finite Gaussian family; it does not prove uniqueness of the stationary point, and the limit of a trajectory depends on its initial state. Its hypotheses exclude moving operating contents and changes of the reference laws.

Proposition 19.20 (Source adaptation and the complete Hessian). On any regular finite-dimensional matter family where F_0 and all C_e are twice differentiable, let $c_e = \nabla C_e$ and $\bar{c}_i = \sum_{e\in\mathcal{I}_i} \beta_e^* c_e$. The reduced Hessian is

$$H_{\mathrm{red}} = \nabla^2 F_0 + \sum_e \beta_e^* \nabla^2 C_e - \sum_i T_i^{-1} \left[\sum_{e\in\mathcal{I}_i} \beta_e^* c_e c_e^T - \bar{c}_i \bar{c}_i^T \right]. \tag{19.31}$$

At a critical point the full and reduced Hessians have the same negative and zero indices. With either positive-definite Fisher metric, a positive-definite Hessian gives local exponential stability, while a negative direction gives instability. The linearized eigenvalues are real; a zero Hessian mode requires a separate nonlinear analysis.

Proof. Differentiating the normalized Gibbs row gives $d\beta_e^*[\xi] = -\beta_e^*(dC_e[\xi] - \sum_f \beta_f^* dC_f[\xi])/T_i$. Differentiate eq. (19.15) once more to obtain eq. (19.31). For a row tangent v_i with $\sum_e v_e = 0$, write $z_e = dC_e[\xi]$ and $\bar{z}_i = \sum_e \beta_e^* z_e$. At a critical point, direct completion of squares gives

$$\delta^2 \mathcal{E}_T[(\xi, v), (\xi, v)] = \xi^T H_{\mathrm{red}} \xi + \sum_i T_i \sum_{e\in\mathcal{I}_i} \frac{[v_e + \beta_e^*(z_e - \bar{z}_i)/T_i]^2}{\beta_e^*}. \tag{19.32}$$

The shift of v_i has zero sum, so this is an invertible tangent change of variables plus a positive row form. It is the Schur-complement identity behind proposition 4.9 and Haynsworth's inertia relation [104]. At a stationary point the flow Jacobian is $-g^{-1}H$, because the metric derivative multiplies a vanished gradient. It is similar to the symmetric matrix $-g^{-1/2}Hg^{-1/2}$. Positive definiteness gives

a negative spectrum and local exponential stability; a negative Hessian direction gives a positive eigenvalue and instability. A zero eigenvalue is not classified by this argument. □

The last term of eq. (19.31) lowers the quadratic form relative to rows frozen at the same state and weights, by $-\sum_i T_i^{-1}\operatorname{Var}_{\beta_i^*}(dC_e[\xi])$ in direction ξ. This is not a general ordering in temperature: changing temperature usually changes the weights and the stationary point. Nor does finite row relaxation create a Hopf bifurcation at an interior stationary point of this exact gradient system.

19.8 An exact stability threshold with changing uncertainty

The following example keeps every content and dictionary mean and variance dynamic. It shows why the fixed-covariance mean theorem does not decide stability after source selection and uncertainty are allowed to respond.

Theorem 19.21 (A three-agent coupled threshold). Take three scalar content laws $q_i = \mathcal{N}(\mu_i, v_i)$, $i = 0, 1, 2$, and two scalar dictionary laws $r_j = \mathcal{N}(a_j, V_j)$ on edges $j \to 0$, $j = 1, 2$. Set $\bar{T}_e = 1$, $R_e = 1$, $\kappa_0 = 1$, and $\lambda = 0$. Use content reference precisions $(P_0, P_1, P_2) = (1, 5/4, 5/4)$ and means $(m_0, m_1, m_2) = (0, -23/15, 23/15)$, and independent dictionary references $\mathcal{N}(0, 4/3)$. Receiver 0 has exactly the two peer candidates, with prior $(1/2, 1/2)$ and temperature $T > 0$; no identity self candidate is added to this row. The source agents may have singleton identity self rows, with zero contribution. In the coordinate order

$$x = (\mu_0, \mu_1, \mu_2, a_1, a_2, v_0, v_1, v_2, V_1, V_2), \tag{19.33}$$

the point

$$x_* = (0, -4/3, 4/3, 1/3, -1/3, 1, 1, 1, 1, 1), \qquad \beta_* = (1/2, 1/2) \tag{19.34}$$

is stationary for every $T > 0$. For both the Gibbs-reduced flow and full unit-mobility row relaxation it is locally exponentially stable when $T > T_c = 3269/10118$, and a hyperbolic saddle with one unstable direction when $0 < T < T_c$. At $T = T_c$ the Hessian has exactly one zero mode. If all five variances are instead held at one, the mean-only stability threshold is $T_{\text{mean}} = 31/122 < T_c$.

Proof. Put $S_i = v_i + 1$ and $d_j = \mu_0 - \mu_j - a_j$. The costs and full objective are

$$\begin{aligned} C_j &= \tfrac{1}{2}[(S_0 + d_j^2 + V_j)/S_j - 1 + \log S_j - \log S_0], \\ \mathcal{E}_T(x, b) &= F_0(x) + bC_1 + (1-b)C_2 + T[b\log(2b) + (1-b)\log(2(1-b))], \end{aligned} \tag{19.35}$$

where F_0 is the sum of the five Gaussian self KL terms with the references just specified. At x_*, $S_i = 2$, $d_1 = 1$, $d_2 = -1$, and $C_1 = C_2 = 1/2$, so $b_* = 1/2$. The two source self mean derivatives are $(1/4, -1/4)$ and their weighted edge derivatives are $(-1/4, 1/4)$; the dictionary mean derivatives have the same cancellation, and the two receiver mean forces cancel each other. Each source variance has self and weighted edge derivatives $1/8$ and $-1/8$, each dictionary variance has $-1/8$ and $1/8$, and the receiver variance derivatives vanish. Thus all ten matter derivatives vanish.

Let $H = \nabla^2(F_0 + (C_1 + C_2)/2)(x_*)$ and $g = \nabla(C_1 - C_2)(x_*)$. Separate the first and last five coordinates by a semicolon; then

$$g = (1, -1/2, -1/2, -1/2, -1/2; 0, -1/4, 1/4, 1/4, -1/4)^T. \tag{19.36}$$

For an arbitrary perturbation $\xi = (u_0, u_1, u_2, b_1, b_2, h_0, h_1, h_2, k_1, k_2)$, set $z_j = u_0 - u_j - b_j$ and $\epsilon_1 = 1$, $\epsilon_2 = -1$. Direct differentiation of eq. (19.35) gives

$$\begin{aligned} \xi^T H \xi = {} & u_0^2 + \tfrac{5}{4}(u_1^2 + u_2^2) + \tfrac{3}{4}(b_1^2 + b_2^2) + \tfrac{1}{2}h_0^2 \\ & + \sum_{j=1}^{2} \left[\tfrac{1}{4}(z_j - \epsilon_j h_j/2)^2 + \tfrac{1}{16}(h_0 - h_j)^2 + \tfrac{71}{128}h_j^2 + \tfrac{1}{2}(k_j - h_j/8)^2 \right]. \end{aligned} \tag{19.37}$$

For completeness, one weighted edge contributes $z_j^2/4 - \epsilon_j z_j h_j/4 + h_0^2/16 - h_0 h_j/8 + 3h_j^2/16 - h_j k_j/8$. Adding the self mean terms and the five self variance terms $\sum_i h_i^2/2 + \sum_j k_j^2/2$ and completing squares gives eq. (19.37). Its vanishing forces all ten components of ξ to vanish, proving $H \succ 0$ without restricting the tangent space.

Solving the linear equations specified by this quadratic form gives, in the same two blocks,

$$\begin{aligned} (H^{-1}g)_{\mu,a} &= (2400, -960, -960, -1600, -1600)^T/5059, \\ (H^{-1}g)_{v,V} &= (0, -716, 716, 2440, -2440)^T/5059, \\ g^T H^{-1} g &= 6538/5059. \end{aligned} \tag{19.38}$$

These rational values are checked by multiplying with the symmetric matrix of eq. (19.37). At the equal-cost point, eq. (19.31) becomes $H_T = H - gg^T/(4T)$. Congruence by $H^{-1/2}$ gives the identity minus one rank-one matrix, with nine positive eigenvalues and the remaining sign determined by $1 - g^T H^{-1} g/(4T)$. Hence $T_c = g^T H^{-1} g/4 = 3269/10118$, with the stated positive, negative, and zero indices.

The inverse matter Fisher metric is

$$M(x) = \operatorname{diag}(v_0, v_1, v_2, V_1, V_2, 2v_0^2, 2v_1^2, 2v_2^2, 2V_1^2, 2V_2^2). \tag{19.39}$$

For full row relaxation the inverse metric is $\operatorname{diag}(M(x), b(1-b))$ and the stationary Hessian is

$$\mathbb{H}_T = \begin{pmatrix} H & g \\ g^T & 4T \end{pmatrix}, \qquad (\xi, \delta b)^T \mathbb{H}_T (\xi, \delta b) = \xi^T H_T \xi + 4T(\delta b + g^T \xi/(4T))^2. \tag{19.40}$$

It has the same negative and zero indices, with an additional positive direction. Proposition 19.20 therefore gives the asserted stability classifications in all ten or eleven coordinates.

To freeze the variances, set $h_i = k_j = 0$ in eq. (19.37). Put $t = u_0 - (u_1 + u_2 + b_1 + b_2)/2 = g_{\mathrm{mean}}^T \xi_{\mathrm{mean}}$. Convexity and exchange of the two sources put the minimum of the mean quadratic at fixed t in $u_1 = u_2 = y$, $b_1 = b_2 = z$. Minimizing there gives $(u_0, y, z) = (15, -6, -10)t/31$ and value $61t^2/62$. The minimum of $\xi^T H_{\mathrm{mean}} \xi$ under $g_{\mathrm{mean}}^T \xi = t$ is also $t^2/(g_{\mathrm{mean}}^T H_{\mathrm{mean}}^{-1} g_{\mathrm{mean}})$ by completing a positive quadratic square. Thus the inverse quadratic form is $62/61$ and $T_{\mathrm{mean}} = 31/122$. □

The exact inequality $31/122 < 3/10 < 3269/10118$ means that at $T = 3/10$ the frozen-variance mean system is stable while both complete coupled systems have one unstable direction. At T_c the reduced null vector is $H^{-1}g$, and the full null vector is $(H^{-1}g, -1)$. Source and dictionary uncertainty participate in the zero linear mode. The following nonlinear calculation classifies that mode within this Gaussian tangent model.

19.9 The local subcritical pitchfork

Retain all hypotheses, references, coordinates, and Fisher mobilities of theorem 19.21. Write $s = b - 1/2$, $\delta = T - T_c$, and

$$\begin{aligned} A(x) &= F_0(x) + (C_1(x) + C_2(x))/2, \qquad \Delta(x) = C_1(x) - C_2(x), \\ G(x,s) &= A(x) + s\Delta(x), \\ h(s) &= (1/2+s)\log(1+2s) + (1/2-s)\log(1-2s). \end{aligned} \tag{19.41}$$

Thus $\mathcal{E}_T = G + Th$. The scalar s is the actual row imbalance for full relaxation, and the Gibbs row imbalance $s = b^*(x,T) - 1/2$ for the reduced flow. Both reductions below use the original time variable. In this section A, G and h denote the functions just displayed, c the cubic coefficient of the center field (the operating content is $\kappa_0 = 1$), k the response vector of eq. (19.47), D the squared matter length of eq. (19.52), and B the vector defined there; these local meanings replace the earlier uses of the same letters.

The stationary elimination below is a Lyapunov–Schmidt reduction of a $\mathbb{Z}_2$-equivariant gradient problem, and the conditions $\Phi_s = \Phi_{ss} = 0$, $\Phi_{ssss} \neq 0$ and $\partial_T \Phi_{ss} \neq 0$ at $(0, T_c)$ are the recognition conditions of the pitchfork [92]. In the language of phase transitions, $\Phi(s,T) - \Phi(0,T) = 2(T - T_c)s^2 + \frac{c}{4}s^4 + O(s^6)$ is a Landau expansion in the order parameter s. Temperature-controlled breaking of an exchange symmetry between information sources through a pitchfork is also the mechanism of nonlinear opinion dynamics [25]; here the bifurcation parameter is a source-row temperature, and the center direction mixes matter means and uncertainties.

Theorem 19.22 (Critical dynamics and nearby branches). There are local parameter-dependent invariant center manifolds on which

$$\dot{s} = -\frac{4\delta s + cs^3}{d} + O(\delta^2|s| + |\delta||s|^3 + |s|^5), \qquad d = \begin{cases} D, & \text{Gibbs reduction,} \\ D+4, & \text{full row relaxation,} \end{cases} \tag{19.42}$$

where the remainder is uniform near $(s,\delta) = (0,0)$ and

$$c = -\frac{657534393590529472}{6223404588394626861} < 0, \qquad D = \frac{19189456}{25593481} > 0. \tag{19.43}$$

In sufficiently small state and temperature neighborhoods, the symmetric equilibrium is the only equilibrium for $\delta \leq 0$. For each sufficiently small $\delta > 0$ there are exactly two additional equilibria, common to the full and reduced systems, with

$$s_\pm(\delta) = \pm\sqrt{-4\delta/c} + O(\delta^{3/2}). \tag{19.44}$$

Each asymmetric equilibrium has one unstable direction, with eigenvalue $8\delta/d + O(\delta^2)$, and nine reduced or ten full stable directions. At $T = T_c$ the symmetric equilibrium is nonlinearly unstable. With destabilizing parameter $T_c - T$, this is a subcritical pitchfork for these reference data; it creates no locally stable asymmetric branch.

Proof. The reflection and source exchange

$$Rx = (-\mu_0, -\mu_2, -\mu_1, -a_2, -a_1, v_0, v_2, v_1, V_2, V_1) \tag{19.45}$$

fixes x_*, preserves A, and sends Δ to $-\Delta$. The mobility satisfies $M(Rx) = RM(x)R^T$. Since $H = A_{xx}(x_*) \succ 0$, the implicit-function theorem gives a unique analytic stationary matter curve $X(s)$ near x_* with

$$G_x(X(s), s) = 0, \qquad X(0) = x_*, \qquad p = X'(0) = -H^{-1}g. \tag{19.46}$$

Uniqueness gives $X(-s) = RX(s)$. The stationary scalar potential $\Phi(s, T) = G(X(s), s) + Th(s)$ is therefore even. This curve solves the matter stationary equations at each fixed row; invariance under either flow has not been asserted.

All tensor derivatives in the next calculation are at x_*. Put $k = A_{xxx}[p, p] + 2\Delta_{xx}p$. Differentiating eq. (19.46) twice gives $X''(0) = -H^{-1}k$. Four derivatives of Φ then give

$$\Phi_{ssss}(0, T_c) = A_{xxxx}[p, p, p, p] + 4\Delta_{xxx}[p, p, p] - 3k^T H^{-1}k + 32T_c = 6c. \tag{19.47}$$

For the explicit costs in eq. (19.35) and the tangent p in eq. (19.38), the three matter contractions are

$$\begin{aligned} A_{xxxx}[p, p, p, p] &= \frac{236914859731152}{655026269697361}, \\ \Delta_{xxx}[p, p, p] &= -\frac{31633266640}{129477420379}, \\ 3k^T H^{-1}k &= \frac{21485600170644766720}{2074468196131542287}. \end{aligned} \tag{19.48}$$

These values follow by differentiating the rational costs and Gaussian self terms; together with $32T_c$ they give eq. (19.43). Restricting the potential to the straight line $x_* + ps$ omits the negative stable-response term: along that line the fourth derivative is $A_{xxxx}[p, p, p, p] + 4\Delta_{xxx}[p, p, p] + 32T_c = 6368971071507328/655026269697361 \approx 9.72 > 0$, which would predict a supercritical pitchfork. The response term $3k^T H^{-1}k \approx 10.357$ exceeds $32T_c \approx 10.339$ by less than 0.2 percent and reverses that classification. Since $\Phi_{ss}(0, T) = 4(T - T_c)$ and h is even,

$$\Phi_s(s, T_c + \delta) = 4\delta s + cs^3 + O(|\delta||s|^3 + |s|^5). \tag{19.49}$$

Analyticity and evenness give $\Phi_s = sQ(s^2, \delta)$ with $Q(0, \delta) = 4\delta$ and $Q_z(0, 0) = c < 0$. A second implicit-function argument gives the unique small root $z(\delta) = -4\delta/c + O(\delta^2)$. It is positive precisely for small positive δ, proving the complete local equilibrium count and eq. (19.44). At these points G_{xx} remains positive definite, whereas $\Phi_{ss} = -8\delta + O(\delta^2) < 0$. The latter is its Schur complement in the full stationary Hessian. Thus that Hessian has one negative and ten positive directions. Eliminating the positive row block gives one negative and nine positive directions in the reduced Hessian. Positive Fisher mobilities preserve these stability indices.

It remains to compute the actual center velocity. The fields are analytic on the interior and equivariant under the orthogonal reflection above. At T_c each has a simple zero eigenvalue and all remaining eigenvalues are negative. The parameter-dependent equivariant center-manifold theorem [173, Theorem 1.2] therefore supplies a local C^7 invariant reduction containing the nearby equilibria; no analytic center manifold is required. Equivariance of the manifold is not needed for the oddness used below: the reflection maps a local C^7 center manifold to another one, and the Taylor jets of center manifolds at an equilibrium are unique. The full coordinate $s = b - 1/2$ is regular on its center tangent $(p, 1)$. For the reduced flow,

$$D_x s^*(x_*, T_c)p = -g^T p/(4T_c) = 1, \tag{19.50}$$

so its Gibbs imbalance also supplies a regular coordinate. Reflection makes the scalar field $f(s, \delta) = \dot{s}$ odd in s.

Write $Y(s, \delta)$ for the matter coordinates on either invariant manifold. Its matter equation is

$$Y_s f = -M(Y)G_x(Y, s). \tag{19.51}$$

At $\delta = 0$, $Y_s(0, 0) = p$ and $f = O(s^3)$, so comparison through quadratic order gives $Y_{ss}(0, 0) = X''(0)$. Set $B = H^{-1}M_*^{-1}p$, with $M_* = M(x_*)$, and let $w = Y - X$. Then

$$D = p^T M_*^{-1} p = \frac{19189456}{25593481}, \qquad g^T B = -D. \tag{19.52}$$

If $f = \alpha\delta s + \eta s^3 + O(\delta^2|s| + |\delta||s|^3 + |s|^5)$, the coefficients of eq. (19.51) give $w_{\delta s} = -\alpha B$ and $w_{s^3} = -\eta B$. Variation of M contributes only at higher orders. Consequently the row force $\Delta(Y) + (T_c + \delta)h'(s)$ has coefficients $4 + D\alpha$ and $c + D\eta$ at δs and s^3.

For the reduced flow this force is identically zero, so $\alpha = -4/D$ and $\eta = -c/D$. Full relaxation instead obeys

$$f = -(1/4 - s^2)[\Delta(Y) + (T_c + \delta)h'(s)], \tag{19.53}$$

giving $\alpha = -(4 + D\alpha)/4$ and $\eta = -(c + D\eta)/4$. These yield the denominator $D + 4$ in eq. (19.42). Odd C^7 regularity gives its stated uniform remainder. In particular,

$$Y(s, 0) - X(s) = \frac{c}{d}Bs^3 + O(s^4), \tag{19.54}$$

which exhibits the nonzero difference between stationary elimination and dynamical invariance.

At either asymmetric branch, differentiation of the center field gives $8\delta/d + O(\delta^2) > 0$. At criticality it gives $\dot{s} = (-c/d)s^3 + O(s^5)$, pointing away from zero on both sides. Arbitrarily small nonzero center data therefore leave a fixed small interval, proving nonlinear instability at the threshold. Instability at T_c also follows without a center manifold, for both flows at once. By eq. (19.47), $\Phi(s, T_c) - \Phi(0, T_c) = \frac{c}{4}s^4 + O(s^6) < 0$, so the points $(X(s), s)$ have lower full energy arbitrarily close to the equilibrium, and the reduced potential, a minimum over rows, is at most $\Phi(s, T_c)$ at $X(s)$. For a gradient flow with positive-definite metric, an equilibrium that is isolated among equilibria and has points of lower energy arbitrarily close to it is Lyapunov unstable: a trajectory started at such a point cannot approach it, and has no other equilibrium nearby to approach. The stable transverse spectrum persists from the Hessian calculation. □

The additional 4 is the Fisher squared length of the row tangent at $b = 1/2$. Thus full relaxation has $D/(D+4) = 4797364/30390845$ times the leading center speed of matter-metric Gibbs reduction, or a leading time scale about 6.34 times longer. The same computation applies to Gibbs reduction with any positive kinetic metric g on the matter coordinates, with $B = H^{-1}g_*p$ and D replaced by $p^T g_* p$. For the pullback of the full metric to the graph $x \mapsto (x, b^*(x))$, the derivative $D_x b^*(x_*, T_c)p = 1$ of eq. (19.50) adds $1/(b_*(1-b_*)) = 4$, so pullback-metric Gibbs reduction has the leading center coefficients of full relaxation. The factor $(D+4)/D$ therefore measures the kinetic cost of the row that matter-metric Gibbs reduction omits; it is not an additional slowness of row relaxation relative to a correctly reduced flow. This comparison identifies the leading scalar polynomials after a constant time rescaling; it does not identify higher-order trajectories or stable transients.

The subcritical type is a property of the declared reference data, not of the construction. Equation (19.47) is the general formula for any symmetric two-candidate configuration with a positive-definite fixed-row Hessian, and for the present data its sign rests on the near cancellation noted after eq. (19.48). A float64 continuation that is not interval validated finds $c \approx +0.023$, with $T_c \approx 0.319$, when the dictionary reference precision is 1 instead of $3/4$ and all other data are kept; for such data the pitchfork is supercritical, with locally stable asymmetric branches below T_c. In Landau language a surface $c = 0$ in the reference data, a tricritical point in the sense of Griffiths [96], separates the two types near the published data. Distant stable equilibria, folds, and the destinations of trajectories leaving this neighborhood require separate arguments. The local result establishes neither monotone uncertainty growth, hysteresis, grouping, nor formation of an inherited parent.

19.10 Scope and open problems

Open problem 19.23 (Coupled dynamics beyond the finite example). Proposition 19.18 and theorem 19.19 give the complete Gaussian tangent flow, global forward existence and convergence of each trajectory to a stationary point, and theorem 19.22 resolves the local critical mode of the three-agent example, for its reference data, as a subcritical pitchfork with nearby asymmetric saddles. Determine its distant stationary branches and long-time destinations, the corresponding behavior on graphs with cycles, and well-posedness beyond regular finite-dimensional law families. Neither the scalar certainty result nor this local instability establishes grouping or formation.

Open problem 19.24 (Convergence of per-agent flows). Find conditions on the self, record, and loop precisions under which the per-agent fixed point is unique and its mean flow converges. The law-level covariance updates have a unique attracting fixed point by proposition 19.9; the open part concerns the means. Stability can fail. Take two agents exchanging in both directions with $K = 2$, codes $(1, 0)$ on $1 \to 0$ and $(0, 1)$ on $0 \to 1$, record precision q, $c_0 = c_1 = 0$, no loop or grounded terms, and self precisions $P_0 = \left(\begin{smallmatrix} 10 & 3 \\ 3 & 1 \end{smallmatrix}\right)$, $P_1 = \left(\begin{smallmatrix} 1 & 3 \\ 3 & 10 \end{smallmatrix}\right)$. The dictionaries then decouple, and the content block of the plug-in mean flow has the eigenvalues $-1 \pm 3q/(1+q)$, since $(P_0 + qe_1e_1^T)^{-1}_{21} = (P_1 + qe_2e_2^T)^{-1}_{12} = -3/(1+q)$. The flow is unstable for $q > 1/2$, and H is singular at $q = 1/2$, so the solvability hypothesis of theorem 19.12(ii) is not vacuous, whereas the global flow of the same model is stable by proposition 19.8. The receiver's own prior regresses the unread coordinate on the read one, and the partner reads exactly that coordinate. The directed ring supplies one stable non-gradient example, not a general convergence theorem.

Open problem 19.25 (Beyond the tangent model). Keep the product $\omega_e(y_i - c_i)$ and let the operating contents follow the beliefs. Factorized recognition then acquires second-moment terms, and the interpretation gauge of theorem 19.5 becomes a nonlinear orbit. Determine whether grounding still separates belief from meaning at the nonlinear level.

Open problem 19.26 (General laws). The Gaussian tangent model is a computational simplification. Extend its conclusions to general laws, with record kernels that read the coded discrepancy and location-family noise. Four parts are open: the attribution of a coded disagreement among receiver persuasion, sender persuasion and reinterpretation when contents and dictionary priors are not Gaussian; the grounded limit, in which the reinterpretation share should vanish as grounding identifies the dictionaries; whether disagreement changes certainty in the two free energies, and how that Bayesian effect compares with the source-role stationarity eq. (19.21) of the reciprocal potential; and the echo and cavity statements for non-Gaussian loop factors, where cavity messages are belief propagation without Gaussian closure and the factorized means are no longer exact.

Open problem 19.27 (What agents must exchange). A receiver using a loop prior needs information about dictionaries owned elsewhere. An echo-free protocol must specify which prior contributions are removed from each transmitted law. Formulate that exchange through declared codes and record kernels, and determine the loop information it makes observable.

Chapter 20

Curvature Costs and Supplied Hierarchies

Integrating out contents can couple dictionaries even when the model contains no loop prior. The resulting action is the negative log evidence of the observed records at fixed dictionaries. In a Gaussian tangent model its quadratic part charges coded reinterpretations; in the weak-anchor or small-noise limit it charges exactly those that changes of contents cannot absorb. Its linear part depends on the records, so this calculation does not imply that flat dictionaries are always preferred. A second construction restricts a supplied hierarchy to rigid clusters. It gives a parent density under explicit normalization hypotheses and a Gaussian minimizer limit, without selecting the clusters or establishing inherited responses and dynamics.

20.1 Contents, records, and supplied parents

Work in one linear sector with reference content space $\mathbb{R}^K$. The graph has finite vertex set V and directed edge set E. An edge $e : j \to i$ has receiver-owned dictionary generator $\omega_e \in \mathfrak{gl}(K, \mathbb{R})$, code $\bar{T}_e : \mathbb{R}^K \to H_e$, and positive-definite noise covariance R_e on the finite-dimensional message space H_e. Fix operating contents c_i and put $\kappa_i \omega = \omega c_i$. The contents $y_i \in \mathbb{R}^K$ have independent self laws $\mathcal{N}(m_i, P_i^{-1})$ with P_i positive definite. These self laws summarize the declared local models and records in this Gaussian example. They are not general definitions of an agent.

The comparison records and global posterior energy are

$$o_e = \bar{T}_e(y_i - y_j - \kappa_i \omega_e) + \varepsilon_e, \qquad \varepsilon_e \sim \mathcal{N}(0, R_e), \tag{20.1}$$

$$\begin{aligned} E(y, \omega) = {} & \frac{1}{2} \sum_i \|y_i - m_i\|^2_{P_i} + \frac{1}{2\sigma^2} \sum_e \|\omega_e\|^2_F \\ & + \frac{1}{2} \sum_e \|o_e - \bar{T}_e(y_i - y_j - \kappa_i \omega_e)\|^2_{R_e^{-1}}, \end{aligned} \tag{20.2}$$

where $\sigma > 0$ and the noises are independent. The matrix anchor is isotropic for the declared reference inner product. Coordinate changes carry this metric with the other constitutive data, so the quantitative statements below hold in the reference presentation in which the anchor is isotropic, and are invariant under the reference changes that preserve it, as in definition 18.22. There are no grounded dictionary factors and no explicit loop factor: the loop weight is $\lambda = 0$. The tangent approximation drops the terms of second and higher order in $(\omega_e, y_i - c_i)$ in $\bar{T}_e(e^{-\omega_e} y_i - y_j)$, among them the product $\omega_e(y_i - c_i)$. Its operating contents remain fixed. The approximation is consistent

only while the posterior spread of $y_i - c_i$ is small compared with $|c_i|$; remark 20.4 records what happens when it is not.

Let $Y = (\mathbb{R}^K)^{\mathsf{V}}$, $X = \mathfrak{gl}(K,\mathbb{R})^{\mathsf{E}}$, and $H = \bigoplus_e H_e$. Define

$$(\delta y)_e = \bar{T}_e(y_i - y_j), \qquad (\mathcal{V}\omega)_e = \bar{T}_e\omega_e c_i, \qquad P = \bigoplus_i P_i, \qquad R = \bigoplus_e R_e. \tag{20.3}$$

The record term is $\frac{1}{2}\|o - \delta y + \mathcal{V}\omega\|^2_{R^{-1}}$. The map $\delta : Y \to H$ is the coboundary of the code sheaf: the two restrictions to the edge stalk H_e are the same code $\bar{T}_e$. Its first cohomology on the graph, which has no two-cells, is $H^1 = H/\operatorname{im}\delta$ [101, Section 2.2]. Cellular sheaves of this kind are used for discourse by Hansen and Ghrist [102]; that paper works with H^0.

For a directed closed walk γ, let $(B\omega)_\gamma = \sum_{e\in\gamma}\omega_e$, counting traversals with multiplicity. The corresponding product of dictionaries $\bar{\theta}_e = e^{\omega_e}$ is ordered along the walk, and $B\omega$ is its first-order term at the identity. Opposite attending edges remain independent; traversing a particular transport in geometric reverse would instead use its inverse. Write $\Delta_e = y_i - y_j$. Then $\sum_{e\in\gamma}\Delta_e = 0$ on a directed closed walk.

Definition 20.1 (A supplied hierarchy). Partition the child agents into clusters I, introduce a parent A for each cluster, and give it a self law $\mathcal{N}(m_A, P_A^{-1})$. Add the cross-scale comparison factor

$$\exp\left(-\frac{\tau}{2}\|y_i - y_A - \omega_{iA}c_A\|^2\right), \qquad i \in I, \qquad \tau > 0, \tag{20.4}$$

with an isotropic dictionary anchor of variance σ^2. This is the tangent form of the conditional law $y_i \mid y_A, \omega_{iA} \sim \mathcal{N}(e^{\omega_{iA}}y_A, \tau^{-1}I)$ at $y_A = c_A$. It is a comparison factor with identity code and fixed operating vector c_A. Edges between different child clusters are called bridges. The partition, parents, and cross-scale factors are supplied model data.

20.2 The action induced by contents

Remark 20.2 (Evidence gives an induced action for general laws). Let $\pi(d\omega)$ be a dictionary prior and $p(dy)$ a content law on standard Borel spaces, with contents and dictionaries independent a priori, so that the joint prior is $p \otimes \pi$. Let λ_o be a sigma-finite measure on the record space that dominates every record law, and let $\ell(o \mid y, \omega)$ be a jointly measurable version of the record density. Suppose, for the registered record o,

$$Z(\omega) = \int \ell(o \mid y, \omega)p(dy), \qquad 0 < \int Z(\omega)\pi(d\omega) < \infty. \tag{20.5}$$

Bayes' theorem for this dominated model gives, for λ_o-almost every record, the posterior

$$\pi(d\omega \mid o) = \frac{Z(\omega)\pi(d\omega)}{\int Z(\omega')\pi(d\omega')}. \tag{20.6}$$

Where $0 < Z(\omega) < \infty$, the contents induce the action $-\log Z(\omega)$. At one registered record the formula defines a version of the posterior: two jointly measurable versions of the density that differ on a null set of records can give different posteriors at that record. The induced action is therefore

defined for a fixed version, canonically the version continuous in o when it exists, as in theorem 20.3. This identity requires no Gaussian family. If the record kernels use the dictionaries only through coded discrepancies, that dependence is inherited by Z. Neither monotonicity in a loop defect nor invariance under arbitrary re-readings of contents follows: the content law and the registered records also enter the integral.

Theorem 20.3 (The Gaussian action and its cohomological quadratic part). In the model of eq. (20.2), integrating out y leaves the complete dictionary action

$$S_{\mathrm{marg}}(\omega) = \frac{1}{2\sigma^2}\|\omega\|^2 + \frac{1}{2}\|o + \mathcal{V}\omega - \delta m\|^2_{C_P^{-1}} + \mathrm{const}, \qquad C_P = R + \delta P^{-1}\delta^T. \tag{20.7}$$

Thus the dictionary marginal is Gaussian with precision

$$\Lambda_\omega = \sigma^{-2} I + \mathcal{V}^T C_P^{-1} \mathcal{V}. \tag{20.8}$$

Within this tangent model, if $P = \epsilon \bar{P}$ with fixed positive-definite $\bar{P}$ and $\epsilon \downarrow 0$, its record precision tends to $\mathcal{V}^T \Pi_H \mathcal{V}$, where

$$\Pi_H = R^{-1} - R^{-1}\delta(\delta^T R^{-1}\delta)^+ \delta^T R^{-1}. \tag{20.9}$$

The superscript $+$ denotes the Moore–Penrose inverse. If instead $R = \rho \bar{R}$ with fixed positive-definite $\bar{R}$, fixed P and $\rho \downarrow 0$, then ρ times the record precision tends to $\mathcal{V}^T \Pi_H(\bar{R}) \mathcal{V}$, with $\Pi_H(\bar{R})$ formed from $\bar{R}$ in place of R. For every ω,

$$(\mathcal{V}\omega)^T \Pi_H \mathcal{V}\omega = \min_{a \in Y} \|\mathcal{V}\omega - \delta a\|^2_{R^{-1}}. \tag{20.10}$$

This is the squared quotient norm of the class of $\mathcal{V}\omega$ in H^1, the squared norm of its harmonic representative in the R^{-1}-weighted discrete Hodge decomposition in degree one [101, Theorem 3.1], whose graph form goes back to Eckmann [69]; see also Lim [138]. It vanishes exactly when $\bar{T}_e \omega_e c_i = \bar{T}_e(a_i - a_j)$ for one $a \in Y$ on every edge. If every receiver has $c_i \neq 0$, the rank of the limiting quadratic form on X is

$$\sum_e \operatorname{rank} \bar{T}_e - K|\mathsf{V}| + \dim H^0, \tag{20.11}$$

where $H^0 = \ker \delta$; for codes of full row rank this is $\dim H^1$.

If every code is injective and the underlying undirected multigraph is connected, the rank is $K(|\mathsf{E}| - |\mathsf{V}| + 1)$; within the full-row-rank class of definition 19.1 injective codes are square and invertible. Suppose in addition that all operating contents equal one $c \neq 0$ and the chains of the declared directed closed walks span the real cycle space of that multigraph. Opposite attending edges count as separate edges of it. Then the limiting quadratic part vanishes exactly when $(B\omega)_\gamma c = 0$ for every declared walk. For positive P, an absorbable reinterpretation obeys

$$(\delta a)^T C_P^{-1} \delta a \leq a^T P a. \tag{20.12}$$

Proof. Under the content prior, $\delta y + \varepsilon$ has mean δm and covariance C_P. The registered value in that Gaussian experiment is $o + \mathcal{V}\omega$. Its density gives eq. (20.7), including the affine term, and hence eq. (20.8). The determinant of C_P is independent of ω in this tangent model.

Set $A = R^{-1/2}\delta\bar{P}^{-1/2}$. Then $C_P^{-1} = R^{-1/2}(I + \epsilon^{-1}AA^T)^{-1}R^{-1/2}$. On the range of A the middle factor tends to zero; on its orthogonal complement it is identity. The limit is eq. (20.9), the residual quadratic form for weighted least squares. For the small-noise limit write $\bar{A} = \bar{R}^{-1/2}\delta P^{-1/2}$; then $\rho C_P^{-1} = \bar{R}^{-1/2}\rho(\rho I + \bar{A}\bar{A}^T)^{-1}\bar{R}^{-1/2}$, and $\rho(\rho I + \bar{A}\bar{A}^T)^{-1}$ tends to the orthogonal projection onto the complement of the range of $\bar{A}$, which is the same as the range of $\bar{R}^{-1/2}\delta$. Minimizing over a proves eq. (20.10) and its zero criterion. The kernel of Π_H is $\operatorname{im}\delta$. When every $c_i \neq 0$, $\omega_e \mapsto \omega_e c_i$ is onto $\mathbb{R}^K$, so $\operatorname{im}\mathcal{V} = \bigoplus_e \operatorname{im}\bar{T}_e \supseteq \operatorname{im}\delta$, and the rank of $\mathcal{V}^T\Pi_H\mathcal{V}$ is $\dim\operatorname{im}\mathcal{V} - \dim\operatorname{im}\delta$, which is eq. (20.11).

For injective codes and a common c that criterion becomes $\omega_e c = a_i - a_j$. A vector-valued edge cochain has this form exactly when it pairs to zero with every cycle. To see the converse directly, choose a spanning tree and define a_i by signed sums along its paths from a root. Each remaining edge closes a fundamental cycle, whose zero sum gives its required difference. The cycle-spanning hypothesis makes the declared directed-walk tests sufficient for these signed cycle tests; it does not identify an opposite attending edge with geometric reversal. For injective codes on a connected graph $\dim H^0 = K$ and $\operatorname{rank}\bar{T}_e = K$, so eq. (20.11) is $K|\mathsf{E}| - K(|\mathsf{V}| - 1)$.

Finally, Gaussian elimination, or completion of the square, gives $u^T C_P^{-1} u = \min_y [y^T P y + \|u - \delta y\|^2_{R^{-1}}]$. Set $u = \delta a$ and evaluate the right side at $y = a$ to obtain eq. (20.12). □

The cycle-spanning hypothesis is stronger than connectedness. Strong connectivity, with all directed closed walks admitted, is sufficient: a strictly positive circulation is obtained by summing directed cycles containing each edge. Adding a sufficiently large multiple of that circulation to any signed cycle makes it nonnegative. Both nonnegative circulations decompose into directed cycles, so their difference lies in the directed-cycle span. More generally, with all directed closed walks admitted, the directed cycles span the cycle space exactly when the condensation multigraph, obtained by contracting each strong component and keeping the edges between components with multiplicity, is a tree. Directed cycles lie inside strong components and span the cycle space of each [91], and the cycle rank of the whole graph exceeds the sum of those of its strong components by the cycle rank of the condensation.

For a connected graph where the hypothesis fails, take scalar edges $1 \to 2$, $2 \to 3$, $1 \to 3$, unit codes and noise, $c = 1$, and $\mathcal{V}\omega = (0, 0, 1)$. There is no nonconstant directed closed walk, but

$$\delta y = (y_2 - y_1, y_3 - y_2, y_3 - y_1), \qquad \min_a \|(0, 0, 1) - \delta a\|^2 = \frac{1}{3}. \tag{20.13}$$

Indeed, $(1, 1, -1)$ spans the orthogonal complement of $\operatorname{im}\delta$, and the squared projection onto it is $1/3$. The coded obstruction exists even though this directed graph has no loop product that detects it.

The theorem describes the Hessian of an evidence cost, not a record-independent preference for flatness. With weak anchors the full record action tends to $\frac{1}{2}(o + \mathcal{V}\omega)^T\Pi_H(o + \mathcal{V}\omega)$. At zero records it charges the cohomology class of the coded reinterpretation. At nonzero records it can prefer a nonzero class that compensates the observed residual. For finite anchors, the shift by δm

in eq. (20.7) must also be retained. At finite anchors the quadratic part also charges absorbable reinterpretations, bounded by eq. (20.12): one scalar edge with unit code, noise and precisions has $H^1 = 0$ and record precision $c^2/3$. Only the limits are cohomological. The precision Λ_ω and the coupling of dictionaries through shared contents belong to the exact dictionary marginal. Under the agent-factorized global recognition of chapter 19 the optimal recognition covariance has no cross-owner blocks, and only the stationary means carry the induced action.

Remark 20.4 (The omitted determinant at weak anchors). The weak-anchor limit of theorem 20.3 is a statement inside the declared tangent model, and in that limit the tangent model ceases to describe the record it linearizes. Take injective codes on a connected graph. The common mode $y_i = u$ for all i, which spans $\ker \delta$, is constrained by no record, so its posterior precision is $\epsilon \iota^T \bar{P} \iota$, where $\iota u = (u, \dots, u)/\sqrt{|\mathrm{V}|}$. The spread of $y_i - c_i$ is then of order $\epsilon^{-1/2}$, and the dropped product $\omega_e(y_i - c_i)$ is not small. For fixed dictionaries the record that keeps it, $\bar{T}_e(e^{-\omega_e} y_i - y_j) + \varepsilon_e$, is still linear in y, with matrix $L(\omega)$ and $L(0) = \delta$; it is the coboundary of the code sheaf twisted by the dictionaries. Gaussian integration of the contents is therefore exact at fixed ω, and the evidence is the expression of open problem 20.13 with the dictionary-dependent content precision $J(\omega) = \epsilon \bar{P} + L(\omega)^T R^{-1} L(\omega)$. Since $L(\omega)\iota u = -\mathcal{V}_u \omega/\sqrt{|\mathrm{V}|} + O(|\omega|^2)$, with $(\mathcal{V}_u \omega)_e = \bar{T}_e \omega_e u$, a Schur complement on the common mode gives

$$\tfrac{1}{2} \log\det J(\omega) = \mathrm{const} + \tfrac{1}{2} \log\det\big(\epsilon \iota^T \bar{P} \iota + \mathcal{G}(\omega)\big) + r(\omega), \qquad \mathcal{G}(\omega)_{uu'} = \frac{(\mathcal{V}_u \omega)^T \Pi_H (\mathcal{V}_{u'} \omega)}{|\mathrm{V}|}, \tag{20.14}$$

where $\mathcal{G}$ is read as a quadratic form in $u, u' \in \mathbb{R}^K$, higher-order terms in ω are omitted, and $r(\omega)$ has curvature bounded as $\epsilon \downarrow 0$. For $|\omega|^2 \ll \epsilon$ the dictionary dependence is $\frac{1}{2\epsilon} \operatorname{tr}\big((\iota^T \bar{P} \iota)^{-1} \mathcal{G}(\omega)\big)$, of order ϵ^{-1}, whereas the tangent quadratic part stays bounded. This term charges ω whenever $\mathcal{V}_u \omega \notin \operatorname{im} \delta$ for some content direction u, including directions with $\omega c = 0$: with $K = 2$, $\bar{T}_e = I_2$, $c = e_1$ and $\omega_e = E_{22}$ on every edge of a graph with a directed cycle, the tangent penalty vanishes while $\mathcal{G}(\omega) \neq 0$. At weak anchors the omitted determinant therefore dominates the tangent quadratic part. In the small-noise limit $R = \rho \bar{R}$ at fixed anchors both terms scale as ρ^{-1}, and the same Schur computation, with P in place of $\epsilon \bar{P}$, makes the ratio of the determinant term to the tangent term of the order of the common-mode prior variance divided by $|c|^2$. The cohomological part is therefore the leading dictionary dependence of the evidence in the window noise variance $\ll$ content prior variance $\ll |c|^2$, which is also the regime in which the tangent approximation is consistent.

Remark 20.5 (Which experiment sees a dictionary). The matter term at one fixed operating content is blind to a perturbation A when $\bar{T}_e A c_i = 0$. Grounded messages with a referent law of positive-definite second moment instead test $\bar{T}_e A$, as their means range over the referents. These are different kernels. With $K = 2$, $\bar{T} = I_2$, $c = (1, 0)^T$, and $A = E_{22}$, the first experiment sees no change, whereas grounded referents identify A. A loop prior can constrain directions left free by a particular matter experiment, but those directions need not be untestable in another admitted experiment.

There is a further distinction between law comparisons and link observables. For a loop map H and law q, $D_{\mathrm{KL}}(q \| H_\# q)$ vanishes whenever H preserves q. For positive-definite Σ, the covariance-dressed link defect

$$Q(H; \Sigma) = \frac{1}{2} \| \Sigma^{-1/2} (I - H) \Sigma^{1/2} \|_F^2 \tag{20.15}$$

vanishes exactly at $H = I$, since the similarity transform is invertible. A quarter-turn preserves $q = \mathcal{N}(0, I_2)$, giving zero law divergence, while $Q(H; I_2) = 2$. The choice of defect therefore does not inherit the law stabilizer automatically.

A shared-sample relay around a loop is one way to probe its action. Another is to identify the constituent dictionaries from sufficiently informative grounded edge records and compose them. The evidence calculation requires neither direct loop observations nor such an identification procedure. Its conclusion is that records couple dictionaries through shared contents in the declared joint.

20.3 Stationary reinterpretation

Proposition 20.6 (One operating content gives a rank-one stationary mean). At a stationary point of eq. (20.2), each dictionary satisfies

$$\omega_e = \sigma^2 \bar{T}_e^T \big(R_e + \sigma^2 |c_i|^2 \bar{T}_e \bar{T}_e^T\big)^{-1} (\bar{T}_e \Delta_e - o_e) c_i^T. \tag{20.16}$$

It has rank at most one. For registered agreement, $o_e = 0$,

$$\omega_e c_i = F_e \Delta_e, \qquad F_e = \sigma^2 |c_i|^2 \bar{T}_e^T \big(R_e + \sigma^2 |c_i|^2 \bar{T}_e \bar{T}_e^T\big)^{-1} \bar{T}_e. \tag{20.17}$$

These are also equations for the stationary Gaussian recognition means. In the plug-in per-agent model the same equation holds with $\Delta_e = \mu_i - \mu_j$. In its law-level version, with peer covariance Σ_j fixed at the considered state, replace R_e by $\widetilde{R}_e = R_e + \bar{T}_e \Sigma_j \bar{T}_e^T$. For the tangent reciprocal potential at fixed positive covariances and positive row weights β_e, use $o_e = 0$ and replace R_e by $(R_e + \bar{T}_e \Sigma_j \bar{T}_e^T)/\beta_e$. In all cases the operating contents are fixed, the anchor is isotropic, and there are no loop or grounded terms. The replacements do not assert equality of dictionary forces across these objectives.

Proof. Only the anchor and the edge record depend on ω_e. Its normal equation is

$$(\sigma^{-2} I + \kappa_i^T \bar{T}_e^T R_e^{-1} \bar{T}_e \kappa_i) \omega_e = \kappa_i^T \bar{T}_e^T R_e^{-1} (\bar{T}_e \Delta_e - o_e). \tag{20.18}$$

The adjoint is $\kappa_i^T v = v c_i^T$ for the Frobenius inner product, and $\kappa_i \kappa_i^T = |c_i|^2 I$. Applying the push-through identity gives eq. (20.16); multiplication by c_i gives eq. (20.17). A quadratic Gaussian free energy has the same mean normal equation as its energy. In the per-agent objectives only the owner's record and anchor contain this dictionary, with the stated convolution covariance in the law-level case. In the reciprocal objective its mean-dependent edge cost is $\frac{\beta_e}{2} \|\bar{T}_e(\mu_i - \mu_j - \omega_e c_i)\|^2_{(R_e + \bar{T}_e \Sigma_j \bar{T}_e^T)^{-1}}$, which gives the last replacement. □

Corollary 20.7 (Equal gains cancel the linearized loop sum). Let the operating content be a common $c \neq 0$, register $o = 0$, and suppose the stationary equations of proposition 20.6 hold, with the effective noise of the chosen objective. Then $(B\omega)_\gamma = g_\gamma c^T$ for a vector g_γ, and

$$(B\omega)_\gamma c = \sum_{e \in \gamma} F_e \Delta_e = \sum_{e \in \gamma} (F_e - \bar{F}) \Delta_e \tag{20.19}$$

for every fixed matrix $\bar{F}$. Thus the linearized loop map annihilates $c^{\perp}$. The exact ordered product has the same property. Every stationary dictionary has the form $\omega_e = v_e c^T$; since $(vc^T)^n = (c\cdot v)^{n-1} vc^T$, one has $e^{vc^T} = I + \varphi(c\cdot v)vc^T$ with $\varphi(t) = (e^t - 1)/t$, and $(I + uc^T)(I + hc^T) = I + (h + u(1 + c\cdot h))c^T$. Hence $H_\gamma = I + h_\gamma c^T$ exactly, where h is updated along the walk by $h \leftarrow h + u_e(1 + c\cdot h)$ with $u_e = \varphi(c\cdot v_e)v_e$, and the residual dissonance of the loop is the single vector $h_\gamma = (H_\gamma c - c)/|c|^2$. It vanishes when all gains on the loop agree, or when every disagreement on the loop is zero. On a two-cycle $e : j \to i$, $e' : i \to j$, its action on c is $(F_e - F_{e'})\Delta_e$.

Proof. Every term of eq. (20.16) has right factor c^T. Summing the equations and applying them to c gives the first equality in eq. (20.19). The second uses $\sum_\gamma \Delta_e = 0$. A matrix gc^T with $c \neq 0$ is zero exactly when its action on c is zero, which proves the cancellation statements. □

Nonzero stationary loop action therefore requires both disagreement and variation of the gains in this fixed-topic tangent experiment; those conditions are not sufficient. The gains depend on code, noise, and anchor strength. If all self means coincide and all records register agreement, the strictly convex global energy has the common-content, zero-dictionary minimizer. Conflicting self means or other registered records change this conclusion. For general laws the evidence depends on the complete content distributions, and one operating vector need not determine a stationary dictionary. Even within the Gaussian example, cancellation of $B\omega$ is a first-order statement: equal gains make the generator sum vanish but need not make the second-order part of h_γ vanish.

20.4 Rigid clusters and their boundary dictionaries

Remark 20.8 (Reinterpretation limits the effective top-down precision). Integrating out the free Gaussian cross-scale dictionary in eq. (20.4) leaves a factor in $y_i - y_A$ with precision

$$(\tau^{-1} + \sigma^2 |c_A|^2)^{-1} I. \tag{20.20}$$

This follows because $\omega_{iA} c_A$ has covariance $\sigma^2 |c_A|^2 I$ and adds to the cross-scale record noise. For $c_A \neq 0$ the precision is bounded above by $(\sigma^2 |c_A|^2)^{-1} I$ as τ increases. For $c_A = 0$ it equals τI and there is no such finite cap. Pinning the cross-scale dictionaries removes this source of reinterpretation. This elimination identity alone does not prove that disagreement concentrates on a cluster boundary.

Proposition 20.9 (A normalized rigid restriction of a supplied hierarchy). Let child and parent contents lie in one standard Borel space E with a common sigma-finite reference measure μ, and let G be a measurable dictionary group with identity I. Specify measurable self densities p_i, p_A with respect to μ. Let the record likelihoods have the form $\ell_e(o_e \mid d_e(z_i, z_j, \theta_e))$, where $d_e : E \times E \times G \to H_e$ is a declared measurable discrepancy with $d_e(z, z, I) = 0$ and $\ell_e(o_e \mid \cdot)$ is Borel on H_e. Fix internal and cross-scale dictionaries at the identity. Suppose every internal or cross-scale likelihood at its registered record and zero discrepancy is a finite positive constant. For every supplied cluster require

$$0 < Z_A = \int_E p_A(z) \prod_{i \in I} p_i(z) \mu(dz) < \infty. \tag{20.21}$$

Evaluate the joint density on $y_i = y_A$ and use μ once for each parent coordinate, together with a declared bridge-dictionary prior π_b. If the dictionary prior couples bridge dictionaries with pinned internal or cross-scale ones, as the anchored loop prior of definition 19.1 does around loops containing both, π_b is the normalized restriction of the joint prior density at the pinned values, assumed to have a finite positive normalizer; for the Gaussian loop prior this restriction retains the wall terms $\frac{\lambda}{2}\| \sum_{e\in\gamma\cap \mathsf{E}_b} \omega_e\|^2_{W_\gamma}$, where E_b is the set of bridges, which couple bridge dictionaries. Assume that the bridge factors, integrated against the parent self laws eq. (20.22) and π_b, have a finite positive integral, $0 < \int \prod_{\text{bridges}} \ell_e \prod_A p'_A(z_A)\mu(dz_A)\pi_b(d\theta) < \infty$. Then the resulting normalized rigid restriction is a communication model with parent self density

$$p'_A(z) = Z_A^{-1} p_A(z) \prod_{i\in I} p_i(z) \tag{20.22}$$

and one record factor for each original bridge, retaining its code, dictionary, and registered value. The parent density is a product of the child and parent self densities, a logarithmic pool with unit weights [88].

For the Gaussian tangent family, with μ Lebesgue measure on $\mathbb{R}^K$, its parent self parameters are

$$P'_A = P_A + \sum_{i\in I} P_i, \qquad P'_A m'_A = P_A m_A + \sum_{i\in I} P_i m_i. \tag{20.23}$$

A bridge retains its fixed operating vector c_i; these vectors can be bridge dependent at the parent level unless they are common within each receiving cluster. With positive-definite self precisions, identity internal and cross-scale dictionaries, and the Gaussian cross-scale factors of precision τ, the normalized soft Gaussian law converges to the rigid restriction with Lebesgue measure, with mean and covariance errors $O(\tau^{-1})$; in particular the global stationary minimizer tends to the minimizer of this rigid parent model at that rate. A reference measure $g(z)dz$ instead multiplies the parent density by $g^{-|I|}$, so eq. (20.23) requires constant g, and the finiteness of Z_A can depend on that choice.

Proof. On equal contents an internal discrepancy is zero, and the hypotheses replace each internal and cross-scale record by a finite positive constant. The self factors at a cluster multiply at their common argument. Dividing by Z_A gives eq. (20.22); the assumed remaining normalizer makes the entire parent law a probability law. Each bridge retains its original likelihood evaluated at the two parent contents. In the tangent model this argument is $\bar{T}_e(y_{A(i)} - y_{A(j)} - \omega_e c_i)$. Completing the square in the Gaussian self factors gives eq. (20.23).

For the minimizer limit collect all free Gaussian variables, including bridge dictionaries, in x. The energy is $E_\tau(x) = \frac{\tau}{2}\|Cx\|^2 + \frac{1}{2}x^T H x - h^T x + \text{const}$, where Cx lists the child-parent differences and H is positive definite. Choose orthonormal bases U for $\ker C$ and V for its orthogonal complement, and write $x = Ua + Vb$. With $H_{aa} = U^T H U$ and the analogous blocks, the first normal equation gives $a = a_0 - H_{aa}^{-1} H_{ab} b$, where $a_0 = H_{aa}^{-1} U^T h$ is the constrained minimizer. The second gives

$$\left[\tau V^T C^T C V + H_{bb} - H_{ba} H_{aa}^{-1} H_{ab}\right] b = V^T h - H_{ba} a_0. \tag{20.24}$$

Both the first coefficient and the Schur complement are positive definite on the b space. Hence $b = O(\tau^{-1})$ and $a - a_0 = O(\tau^{-1})$. The same block decomposition of the precision $\tau C^T C + H$

gives its inverse as $UH_{aa}^{-1}U^T + O(\tau^{-1})$, and Ua_0 and $UH_{aa}^{-1}U^T$ are the mean and covariance of the Gaussian law restricted to $\ker C$ with Lebesgue measure there, which is the rigid restriction because $\ker C$ is parameterized linearly by the parent contents and the bridge dictionaries. This proves the claims. □

The reference measure and normalizers are part of this restriction. It is neither conditioning a continuous law on a probability-zero diagonal without a specified convention nor integrating out children. Individually normalized self densities do not guarantee eq. (20.21): two copies of $p(z) = \frac{1}{3}z^{-2/3}$ on $(0, 1)$ have a product proportional to $z^{-4/3}$, whose integral is infinite. Disjoint supports can instead make the product integral zero. The Gaussian convergence statement does not extend by itself to arbitrary laws. In the Gaussian case the normalized cross-scale kernel tends to a Dirac mass with respect to Lebesgue measure, which is why that measure appears; for general laws the cross-scale kernels would have to select the reference measure.

Proposition 20.10 (Flat wall walks and a common boundary dictionary). Consider two clusters I, J whose internal dictionaries are identity. Assume that bridges exist in both directions. For every bridge $e : j \to i$ from J to I and every bridge $g : i' \to j'$ from I to J, require directed internal paths $i \to i'$ in I and $j' \to j$ in J. Internal strong connectivity is sufficient. Every edge of these connecting paths, including any cross-scale edge they use, carries the identity dictionary. Let the bridge dictionaries be arbitrary elements of a group G.

Every directed walk that uses e, such internal paths, and g has holonomy $\bar{\theta}_g\bar{\theta}_e$. All these walks are flat exactly when there is one $\bar{\theta}_D \in G$ such that all bridges $J \to I$ carry $\bar{\theta}_D$ and all bridges $I \to J$ carry $\bar{\theta}_D^{-1}$. At first order the corresponding condition is $\omega_e = D$, $\omega_g = -D$.

For rigid clusters with common operating content c and zero bridge records, if the free bridge dictionaries satisfy proposition 20.6, the linearized wall loop acts on c by $(F_e - F_g)(y_I - y_J)$. More generally, with identity internal dictionaries but nonzero internal disagreements, that action is

$$(\omega_e + \omega_g)c = (F_e - F_g)\Delta_e + F_g(\Delta_e + \Delta_g). \tag{20.25}$$

Proof. The internal factors are identity, so the ordered product is $\bar{\theta}_g\bar{\theta}_e$. The path hypothesis realizes every opposite bridge pair as a walk of this kind. Fix one bridge in each direction, which is possible by nonemptiness. The equations $\bar{\theta}_g\bar{\theta}_e = I$ for all pairs imply that every forward bridge is the inverse of the fixed backward bridge, and every backward bridge is the inverse of that common forward value. The converse follows by multiplication. Replacing products by their first-order sums gives the generator condition. The stationary bridge equations give $\omega_e c + \omega_g c = F_e\Delta_e + F_g\Delta_g$, which is eq. (20.25); rigidity makes $\Delta_g = -\Delta_e$. □

The identity requirement on the connecting paths is needed: with $I = \{a_1, a_2\}$, $J = \{b\}$, bridges $e : b \to a_1$ with $\bar{\theta}_e = 3$ and $g : a_2 \to b$, and a path $a_1 \to A \to a_2$ through a parent with scalar dictionaries 2 and $1/4$, the choice $\bar{\theta}_g = 2/3$ makes the only wall walk flat while $\bar{\theta}_g\bar{\theta}_e = 2$, so no common $\bar{\theta}_D$ exists. The all-pairs path hypothesis can be weakened. Form the bipartite pairing graph whose vertices are the bridges and whose edges join e and g when an admissible walk of the stated kind closes them. If this graph is connected and both directions are nonempty, the conclusion holds, by propagating $\bar{\theta}_g = \bar{\theta}_e^{-1}$ along its edges; if it is disconnected and G is nontrivial, different components can carry different values of $\bar{\theta}_D$.

Without the path hypothesis, separate reciprocal bridge pairs can each be flat while carrying different dictionaries. For $I = \{a_1, a_2\}$ and $J = \{b_1, b_2\}$ with no internal directed paths, let $b_1 \leftrightarrow a_1$ carry scalar dictionaries $1, 1$ and $b_2 \leftrightarrow a_2$ carry $2, 1/2$. Every available directed loop is flat, but the bridges from J to I disagree. Parent-to-child edges alone supply no return paths between these pairs.

When the proposition applies, all records in one bridge direction read a common dictionary through their own codes. Grounded referents with positive-definite second moment identify it up to the intersection of those code stabilizers; an injective stacked code identifies it completely for the location-family experiments. The group-level flatness statement is independent of Gaussianity. Its stationary gain formula is a Gaussian tangent statement. Neither licenses replacing distinct dictionaries by one merely because a single-topic experiment fails to distinguish them.

Remark 20.11 (Repeating a supplied rigid restriction). The density factorization can be repeated along a supplied hierarchy whenever the required measures and normalizers exist. In the Gaussian family the parent self term pools child information through eq. (20.23), while the bridge factors retain the boundary records. At finite coupling, integrating out children instead gives a Schur complement; representing it within the prescribed comparison-record family is an additional closure requirement.

These statements do not select a partition, prove its persistence, or construct an inherited response and update kernel for a predeclared probe class. They also do not establish compression on a post-handoff invariant set, interaction necessity against a null flow, or a second selection depending on the first. Those are separate requirements of the formation standard. A supplied rigid parent density therefore does not by itself establish recursive formation or complete dynamical inheritance.

20.5 Remaining extensions

Open problem 20.12 (Formation with top-down feedback). Determine whether a specified dynamics can select persistent blocks and create parents whose feedback supports further selection. The construction must retain the laws, records, evaluators, and complete response and update data required by its admitted protocols, and must meet the compression, null-flow, and counterfactual-recursion conditions. The supplied partitions and minimizer limits above do not settle these questions. This is the specialization of open problem 6.38 to constructions with declared top-down factors. One candidate in which the link sector does selection work is a wall across a frustrated cut: loops inside two candidate clusters admit flat assignments while cross-cluster loops do not, and the source rows would have to concentrate inside the flat clusters above a loop-weight threshold while the null flow and the zero-loop-weight control form no wall. The record dependence of theorem 20.3 decides the sign of such an effect, and inheritance of a learned link sector is open.

Open problem 20.13 (Fluctuations beyond the fixed tangent experiment). In the present tangent model C_P is dictionary independent, so its determinant contributes no dictionary force. For a nonlinear link model with Gaussian conditional content precision $J(\omega)$, information vector $b(\omega)$, and scalar energy term $k(\omega)$, Gaussian integration gives $k(\omega) - \frac{1}{2} b(\omega)^T J(\omega)^{-1} b(\omega) + \frac{1}{2} \log\det J(\omega)$ up to a constant, provided $J(\omega)$ is positive definite. At weak anchors the log-determinant is the leading dictionary dependence by remark 20.4, and it charges the directions with $\omega c = 0$. For unit codes and $U(1)$-valued links the content precision is P plus the connection Laplacian of the

multigraph of directed links, and Forman's theorem expands its determinant over cycle-rooted spanning forests with nonnegative cycle weights $2 - 2\cos$ of the holonomies [76]; Kenyon [120] extends the expansion to vector bundles. For codes of rank below K the operator is a sheaf Laplacian, for which no such expansion is derived here. Determine the complete action in a declared nonlinear communication model, including its off-topic directions and the signs of the resulting loop terms. The quadratic penalty of theorem 20.3 alone does not supply that calculation.

Open problem 20.14 (General content laws). Identify hypotheses under which the evidence action for general laws admits a useful decomposition into absorbable reinterpretations and residual incompatibility. Such a result must preserve record dependence and distinguish an invariance of the likelihood from an invariance of the content measure. Establishing a normalized soft-to-rigid limit also requires more than the finite product normalizers used for the static restriction; in the Gaussian family it holds with Lebesgue measure by proposition 20.9, and for general laws the reference measure would be selected by the cross-scale kernels. The stationary rank-one and gain formulas are not claimed outside the Gaussian tangent family.

Open problem 20.15 (Learning gains from loop records). For a declared grounded or relay experiment, determine which combinations of code, noise, and dictionary-anchor strength can be inferred from observed loop effects. The gain-weighted formula is an identity at a stationary tangent state; an identifiability theorem requires the full admitted record experiment and cannot be inferred from that identity alone.

Chapter 21

Beyond the Existence Constructions

The communication and link constructions of chapters 18 to 20 add a declared observation model to the dictionary sector. General-law reconstruction and identifiability statements distinguish a code, its stabilizer, and the receiver's conditional reconstruction; identification of dictionaries modulo stabilizers assumes a known code (theorem 18.9). The normalized tangent prior and Gaussian recognition formulas give a bounded model of learning those dictionaries. The global, plug-in per-agent, law-level per-agent, and reciprocal calculations retain different forces and metrics; equality of one derivative does not identify their flows. The per-agent link model of definition 19.3 is a record-likelihood model, distinct from the mixture-source per-agent model of definition 4.19, so the manuscript contains two per-agent constructions that are not yet related. The induced curvature cost is a quadratic cost of an affine coded residual and includes its record-dependent linear term; its cohomological part is stated in the small-noise limit at fixed anchors and, within the tangent model, in the weak-anchor limit, where the omitted log-determinant dominates outside that model (remark 20.4). Rigid-cluster calculations assume a supplied partition and do not establish its dynamical formation. The Gaussian tangent extension in proposition 19.18 and theorem 19.19 includes joint covariance and source-row evolution with global forward existence under fixed proper references and convergence of each trajectory to one stationary point. Its three-agent example has, for its reference data, the local subcritical pitchfork of theorem 19.22, with nearby asymmetric saddles; the sign of the quartic coefficient depends on those data. Nonlinear dictionary learning, distant branches and destinations on general graphs, unrestricted law-space dynamics, and inheritance of a learned link sector remain open.

The theory distinguishes three objectives for interacting agents. In the global picture one normalized joint covers the contents and records of every agent, and local updates are coordinate descent of one free energy (proposition 4.16). In the per-agent picture each agent has its own generative model in which its own evaluator and its peers' communicated laws compete as sources of prior information (proposition 4.20); with fixed channels its forces agree with the mean-field block forces of a global model on an open set only when every attended peer channel is constant in its input (proposition 4.30). Under Gaussian references, affine evaluators, linear-Gaussian channels, and log-quadratic records both free-energy flows are tangent to the Gaussian product family (proposition 4.12), whereas the reciprocal flow is not (proposition 4.11). The reciprocal law potential is a third object. With fixed positive rows on finite alphabets, proposition 4.24 shows that it is not an affine function of the free energy of any finite normalized model under any recognition rule when an attended edge is an identity channel, and that with fixed input-dependent channels it is not, on any open set, an affine function of the free energy of a normalized model with product recognition by the agent laws. Remark 4.25 records exact identifications outside these hypotheses: constant-output channels, countable auxiliary latents, and coupled recognition on atomless spaces. Whether the obstruction extends to the Gibbs-reduced potential, to law-dependent channels, or to declared

charts is open. Every construction of chapter 7 to chapter 13 concerns the reciprocal potential or, in chapter 7, a nonsmooth modification of it. Chapters 14 and 15 concern the normalized per-agent free energies under their Gibbs-reduced dynamics. No construction states a selection result for the global free energy, and no objective is declared primary; which objective carries the formation program, and whether any conclusion transfers between objectives, is part of open problem 4.31.

Appendix C supplies a different composition: exact correlated inference under a common preparation, private records, and a known posterior-sampling communication protocol. Its joint private-record posterior is a product tilt of the fixed preparation coupling (proposition C.6): every message multiplies the factor of its speaker, the cross-agent dependence of the coupling never changes, and with the preparation fixed the two private-record marginals determine the posterior. Together with the fixed conditional reconstruction it preserves the declared future messages; complete local beliefs alone need not do so. The affine singleton-presentation Gaussian case closes, whereas multiple presentations need not remain a finite Gaussian mixture. Smaller sufficient representations, new private measurements over time, adaptation of the common model, and interactions between agents using different models remain open. None of those extensions follows merely from composing normalized local evaluators. The generative models of chapter 1 contain no action, policy, or learning rule, so every free energy in the manuscript is perceptual; agents that choose their queries or adapt their models are a further construction.

Measured against the formation standard definition 6.36, with the nulls and the frozen-environment control of definition 6.33 and the counterfactual class of definition 6.35, no construction in this manuscript is proved to satisfy every clause. The calibration example proposition 6.37 passes interaction necessity and fails the counterfactual clause. The nonsmooth fusion construction of chapter 7 is a two-stage selection with exact inheritance: its fused children are identical, so compression fails; it meets interaction necessity only through its exact-equality selector, since on the common-offset family its partition sequence is the normalized-centroid linkage of the initial data for every coupling (proposition 7.6) and a threshold version of the selector gives the same sequence along the nulls; and its second stage forms from every counterfactual handoff. The gated erasure construction of chapter 9 is a single-stage selection with exact inheritance whose interaction-necessity verdict is not determined, since the proved results fix only its first and last partitions (section 9.5). The Wishart construction of chapter 12 is a two-stage selection with exact inheritance: its membership schedule is the centroid dendrogram of the initial belief shapes at every coupling, the renormalized null and the frozen-environment control reproduce it, the label-preserving null has agent-dependent clocks already at zero coupling (proposition 12.8) and integrations indicate that it changes the schedule on an open set, so the verdict of interaction necessity is split and the clause fails, and the union forms from every counterfactual handoff. The six-agent construction of chapter 13 admits a full open neighborhood of positive laws and inherits a moving-representative flow, but its theorem establishes neither interaction necessity nor counterfactual dependence; integrations of its limit field indicate that the null flow forms the same blocks (remark 13.6), its block relations are transient, the selector eventually merges the exterior pair (corollary 13.10), and its handoff loss is created by the response interface rather than by the selection. It is not shown to be a recursive formation. The angular per-agent construction of chapter 14 and the collective construction of chapter 15 are single-stage selections with exact inheritance whose retentions fail compression. In chapter 14 the persistent grouping passes interaction necessity against both

nulls only for ratio thresholds in $(1/8, 1/3)$, the frozen-environment control reproduces it at every such threshold, and the four-state and persistent-record groupings carry no compressive parent. The retention constructions of chapters 8, 10 and 11 make no selection claim, and the smooth constructions supply their compression architectures. Their quotients exist before membership occurs, so the dynamics selects when a description is used without creating a new operational equivalence. Every exact compression realized here erases data that no agent, channel, or energy term reads, and no mechanism is given by which an interaction creates an exact symmetry or decoupling; a graded notion of inheritance measured by the exact defect functionals of the inheritance chapter is not developed.

The supplied-architecture results retain their scopes. The profile hierarchy of chapter 11 is an equivariant symmetry reduction with a supplied tree; the data it discards are invisible to the fine energy, metric, and admitted records. Its two nulls coincide and preserve the same family, descriptors, and descended interface, so the exact hierarchy is a property of the prepared family and its symmetry rather than of the interaction. The collective phase of chapter 10 is an asymptotic-phase projection that loses ordinary child sampling. The erasure construction of chapter 9 keeps public child states and its heterogeneity is transient. The per-agent parent of proposition 14.7 preserves complete admitted histories and persistent records, while proposition 14.8 and theorem 14.9 preserve the actual collective geometry on sections with context-constant discarded phases. Its phase quotient is supplied before grouping, and the phase is gauge relative to every algebra the chapter declares. Its grouping holds against both nulls for ratio thresholds in $(1/8, 1/3)$ and fails above $1/3$ (proposition 14.4). Adding a future hidden-angle probe destroys that quotient, and joining already canonical parents erases no new coordinate. Response sufficiency depends on the interface: an unrestricted identity-channel Gibbs interface identifies distinct exposed laws, and child-specific probes, exact constituent-law records, or feedback can require discarded distinctions.

The collective alternative of chapter 15 exposes every proper block through fixed relative-angle probes and permits one common orientation, gauge relative to every declared algebra, to be erased only by the whole-block retention. Its reference-memory realization closes a local two-jet in the original time and preserves the actual normalized-law Fisher cost. A finite preparation generator yields a complete conditional whole-patch realization, with the full chronological record retained. The causal accumulator of proposition 15.4 instead integrates the local angular velocity from the actual preparation time. It preserves the original native dynamics and every admitted joint history, and corollary 15.5 proves actual geometric inheritance under initial collective-phase coherence, a sufficient condition that is not shown to be necessary. That condition is not generated by the memory. Equation (15.30) identifies the missing initial contextual phase relation for arbitrary preparations. Theorem 15.11 and corollary 15.12 exclude order-zero rotation-invariant value-path retentions and smooth bounded-jet causal memories on the class of definition 15.8, in addition to the finite-order and finite global-encoding obstructions. They refer to the declared marked structure group; with the common rotation added to the group they lose their force (remark 15.7). They leave the unrestricted causal-local-parent question inconclusive (open problem 15.13). More general admissible memory mechanisms, interaction-created invisibility and a new second compressive handoff remain unresolved.

The remaining problems are more specific than an existence question. Open problem 6.38 asks for a construction that meets every clause of the standard with a derived selector, and whether type

closure and an action of the parent on its children should be required. The per-agent example meets interaction necessity at one stage for a range of selector constants, but still needs a compressive retention of a distinction that some declared probe resolves, a selection that the frozen-environment control does not reproduce, and a second compressive handoff with counterfactual dependence. Under the reciprocal potential with shared targets and additive moment channels, every pairwise law comparison relaxes on a common clock (proposition 6.31) and selectors invariant under common tilts fail interaction necessity (corollary 6.32); interaction-sensitive selection read by such selectors requires that a hypothesis of proposition 6.31 fail, while selectors that threshold a divergence between agent laws can pass. Open problem 4.31 asks how the three objectives are related and which conclusions survive a change of objective. Within the formation constructions the gauge structure supplies the orbits that a parent forgets and does no dynamical work, and the connection remains diagnostic. Chapter 16 separately gives it a dynamical role through auxiliary alignment: the finite graph has a regular-branch Fisher ODE whose reduced cost is the minimized cost only while the branch stays globally minimizing (proposition 16.5), and a flat periodic base has a locally well-posed shape PDE coupled to frame and source-row ODEs. After elimination no locally gauge-invariant contextual energy diffuses frames; independent smoothing of rows, frames, and communication links, which would need a declared row or link energy, global continuation, and graph refinement remain open. Chapter 17 supplies displacement maps and their connection from regular contextual agent data with declared comparison rules, including a curved reciprocal-link example; the rectangular selector has holonomy orthogonal for its induced metric (corollary 17.9). The message or target metric remains a declared comparison rule. Selecting that rule, continuing across rank or orbit degeneracy, and proving dynamical or parent inheritance of the resulting link sector remain open. Full permutation symmetry still constrains deterministic law-only selection as in proposition 6.29. Persistent membership under the fixed per-agent record episode does not extend its geometric theorem beyond the declared finite window. None of the results establishes an indefinite spacetime metric, a thermodynamic limit, typicality under random initialization, or unbounded recursive formation.

The open problems stated in the chapters and appendices are collected below by label.

Open problem	Question
open problem 3.12	Extensions of the contextual Fisher flow.
open problem 4.31	Relating the global, per-agent, and reciprocal objectives.
open problem 6.20	Type-closed parents.
open problem 6.38	A recursive formation with a derived selector.
open problem 11.12	Minimality of the unordered profile tree.
open problem 12.14	Approximate inheritance off the common-precision family.
open problem 13.19	Interaction-necessary selection in the bit-circle architecture.
open problem 15.13	Causal local parents on arbitrary preparations.
open problem 18.33	Learning the dictionaries beyond the tangent model.
open problem 18.34	Codes as part of the species model.
open problem 18.35	Dictionaries in the free-energy pictures.
open problem 18.36	Beyond Gaussian codes.
open problem 19.23	Coupled link dynamics beyond the finite example.
open problem 19.24	Convergence of per-agent link flows.
open problem 19.25	Link dynamics beyond the tangent model.
open problem 19.26	Link dynamics for general laws.

Open problem	Question
open problem 19.27	What agents must exchange.
open problem 20.12	Formation with top-down feedback.
open problem 20.13	Fluctuations beyond the fixed tangent experiment.
open problem 20.14	Curvature costs for general content laws.
open problem 20.15	Learning gains from loop records.
open problem D.4	Selecting and continuing interaction-derived transport.
open problem C.13	Smaller sufficient states, asymptotic agreement, and changing preparations.

Appendix A

A Normalized Internal Record Process

This appendix constructs a finite-history law from an internal interaction rule. The rule is supplied; normalization does not select it. Let S and O be standard Borel spaces. A state x includes an accessible record register. Write $\rho : \mathsf{S} \to \mathsf{O}$ for its readout.

Definition A.1 (Event and registration). Choose a normalized kernel $R : \mathsf{S} \rightsquigarrow \mathsf{O}$ and a measurable update $A : \mathsf{S} \times \mathsf{O} \to \mathsf{S}$ satisfying

$$\rho(A(x, o)) = o, \tag{A.1}$$

for $R(x, \cdot)$-almost every o, for every x in a Borel admitted set $\mathsf{S}_{\mathrm{adm}} \subseteq \mathsf{S}$. Define

$$J(x, dx', do) = R(x, do)\delta_{A(x,o)}(dx'). \tag{A.2}$$

If necessary, the state can be explicitly augmented by a register so that eq. (A.1) holds. The admitted set is assumed to contain μ_0-almost every initial state and to be invariant, $J(x, \mathsf{S}_{\mathrm{adm}} \times \mathsf{O}) = 1$ for $x \in \mathsf{S}_{\mathrm{adm}}$, so that the registration condition holds at every visited state. The map $x \mapsto J(x, B)$ is measurable for every measurable $B \subseteq \mathsf{S} \times \mathsf{O}$, because $x \mapsto \int f(x, o)R(x, do)$ is measurable for every bounded jointly measurable f, by a monotone-class argument from products. When the event kernel is generated by an interaction transition Γ and the readout, R is the record kernel of proposition 1.9, and the record marginal of J reproduces it: $R(x, B) = \int \mathbf{1}_B(\rho(A(x, o)))R(x, do)$. The state marginal of J is $\int R(x, do)\delta_{A(x,o)}$, which differs from $\Gamma(x, \cdot)$ in general. It reproduces Γ when the deterministic update is replaced by the stochastic conditional update $A(x, o; dx') = \Gamma(x, dx' \mid \rho(\mathrm{pr}\, x') = o)$, a regular conditional law that exists on standard Borel spaces; then $J(x, dx', do) = \Gamma(x, dx')\delta_{\rho(\mathrm{pr}\, x')}(do)$.

Proposition A.2 (Consistent finite histories). For an initial law μ_0 on S, the iterated kernel product

$$\mathbb{P}_n(dx_0, do_1, dx_1, \ldots, do_n, dx_n) = \mu_0(dx_0) \prod_{t=1}^{n} J(x_{t-1}, dx_t, do_t), \tag{A.3}$$

is a probability law. Marginalizing its last pair gives $\mathbb{P}_{n-1}$, and every record agrees with the readout of the resulting state almost surely.

Proof. The product denotes successive kernel integration, not independent factors. Integrating the last pair gives one because $J(x, \mathsf{S} \times \mathsf{O}) = 1$. Repeating this argument gives total mass one and the stated consistency. Each transition is supported on $x' = A(x, o)$ and hence on $\rho(x') = o$. For a finite history, the intersection of these probability-one events still has probability one. □

By the Ionescu-Tulcea theorem [114], the consistent family $(\mathbb{P}_n)_n$ extends to a unique law on infinite histories, on which persistence statements about later readouts can be posed.

Definition A.3 (Adaptive histories and record-interrupted flows). The construction extends to kernels that depend on the whole past. Write $\mathsf{H}_n = \mathsf{S} \times (\mathsf{O} \times \mathsf{S})^n$ for the space of histories of length n. A policy is a sequence of Borel kernels $\pi_n : \mathsf{H}_n \rightsquigarrow \mathsf{A}$ into a standard Borel space of admitted actions, which may include waiting times and the choice of the next interaction. The event kernel and the update at a stage may depend on the action, as $R(x, a; do)$ and $A(x, a, o; dx')$, jointly measurably in all arguments. The iterated kernel product of μ_0, π_0, R, A, π_1, and so on is again a consistent family of probability laws, and the Ionescu-Tulcea theorem extends it to a unique law on infinite histories. When a flow (Φ_t) acts between events, with waiting time τ chosen by the policy, the stage kernel is

$$J_n(x, a; dx', do) = R(\Phi_\tau(x), a; do)A(\Phi_\tau(x), a, o; dx'). \tag{A.4}$$

This requires $(t, x) \mapsto \Phi_t(x)$ to be jointly Borel on its domain, every record density to be jointly Borel in the outcome and the state it reads, and the policies to be Borel kernels as above. If the flow has a maximal existence time, the history is stopped at the first stage at which the requested waiting time exceeds it, and if the event times accumulate, the history is stopped at the accumulation time; no further records are defined on these events. Statements about finite adaptive history laws in later chapters concern kernels of this kind.

State-level and objective-level persistence are different statements. A record persists in the state when a later readout recovers it, in the sense of definition 1.8. It persists in an objective when its likelihood term remains in the free energy that drives later updates, as in the global and per-agent free energies of chapter 4; this requires the record value itself to remain accessible in the state, because the update rule must read it. In the reciprocal potential a record enters only as a declared assimilation event and leaves no term in the potential. When each record is binary, the records of a finite history carry at most $n \log 2$ nats about any variable, since $I(X; O_{1:n}) \le H(O_{1:n}) \le n \log 2$.

Nothing in this construction requires a variable external to the agents. It does require a declared accessible register and an update. A stochastic A is handled by replacing its Dirac kernel with a normalized kernel supported on the registration condition. A deterministic transition is recovered when R is a Dirac kernel as well.

A.1 Prediction followed by assimilation

One possible internal state contains a model law s and a record register. Fix a normalized density $k(o \mid m)$ with respect to a reference measure on O. Let

$$R(s, do) = \left[\int k(o \mid m)s(dm)\right] do, \qquad s^o(dm) = \frac{k(o \mid m)s(dm)}{\int k(o \mid m')s(dm')}. \tag{A.5}$$

The update replaces s by s^o and stores o. The denominator is positive and finite for predictive-almost every record. Defining an arbitrary probability law on the remaining null set completes the measurable rule under the usual conditional-law hypotheses.

Proposition A.4 (The Bayesian history identity). For a fixed normalized observation density k and initial law s_0, sequential prediction and Bayesian assimilation give the record density

$$p(o_1, \dots, o_n) = \int \prod_{t=1}^{n} k(o_t \mid m) s_0(dm). \tag{A.6}$$

Proof. After $t-1$ records, the posterior is proportional to $s_0(dm) \prod_{r<t} k(o_r \mid m)$. The next predictive density is the integral with the product extended to t, divided by the integral up to $t-1$. Multiplying these conditional densities cancels every intermediate normalizer and leaves eq. (A.6). □

Let the presentations m be the laws on a finite record alphabet, let $k(o \mid m) = m(o)$, and let s_0 be a Dirichlet law with parameter α and total concentration α_+. The predictive law after n records is $\bar{m}_n = (\alpha + c_n)/(\alpha_+ + n)$, with c_n the record counts. Then $\bar{m}_n - \bar{m}_{n-1} = (e_{o_n} - \bar{m}_{n-1})/(\alpha_+ + n)$, so the total-variation step is at most $1/(\alpha_+ + n)$: a slow model sector produced by accumulated records rather than by an inserted rate.

This identity applies to the stated fixed observation model. If the likelihood itself is changed using the current state, the product of the actual predictive kernels still defines a normalized process, but it need not equal the fixed-latent expression. Likewise, replacing Bayes' rule by a finite natural-gradient step defines a different assimilation mechanism unless the updates coincide. The normalization of a process and its interpretation as Bayesian conditioning are separate statements. Their difference has an exact measure.

Proposition A.5 (Departure from Bayesian assimilation). Let $\mathbb{P}_n$ be the record law of n steps of eq. (A.5) and let $\tilde{\mathbb{P}}_n$ be the record law generated from the same s_0 and k by any measurable assimilation rule, with model laws s_{t-1} and $\tilde{s}_{t-1}$ after the same history $o_{1:t-1}$. Then

$$D_{\mathrm{KL}}(\mathbb{P}_n \| \tilde{\mathbb{P}}_n) = \sum_{t=1}^{n} \mathbb{E}_{\mathbb{P}} D_{\mathrm{KL}}(s_{t-1}k \| \tilde{s}_{t-1}k) \le \sum_{t=1}^{n} \mathbb{E}_{\mathbb{P}} D_{\mathrm{KL}}(s_{t-1} \| \tilde{s}_{t-1}), \tag{A.7}$$

where $s_{t-1}k$ denotes the predictive record law.

Proof. Both processes predict the next record by $s_{t-1}k$ and $\tilde{s}_{t-1}k$ respectively, so the chain rule for relative entropy over the successive records gives the equality, and the data-processing inequality for the kernel k gives each bound. □

The identity applies to record laws. The joint laws of states and records of the two processes are generally mutually singular, because their state updates are different deterministic functions of the same history.

Appendix B

Conditional Prediction and Generative Memory

A generative model can relate a cue to a different result. Evidence about the cue then changes the predicted result through that relation. This appendix realizes the general agent model of definition 1.11 with correlated recognition, derives its exact local inference, and separates that inference from a subsequent compression of the posterior. The construction begins with general probability laws. Gaussian mixture regression supplies an explicit finite-dimensional example [35]; its use here makes the model-presentation sector and the retained information explicit.

B.1 A normalized cue and result model

Definition B.1 (Conditional prediction model). Fix a context. Let $\mathsf{M}, \mathsf{X}, \mathsf{Y}, \mathsf{O}$ be standard Borel spaces. Choose a proper presentation reference $r(dm)$, a normalized cue kernel $\mu_m(du)$, a normalized result kernel $K_m(u, dv)$, and a normalized record kernel $\mathcal{L}(u, do)$. All kernels are jointly measurable in their conditioning variables. Define

$$P(dm, du, dv, do) = r(dm)\mu_m(du)K_m(u, dv)\mathcal{L}(u, do). \tag{B.1}$$

Successive integration gives total mass one. In the agent notation, take $\mathsf{K} = \mathsf{X} \times \mathsf{Y}$, evaluator $G(m, d(u, v)) = \mu_m(du)K_m(u, dv)$, and record kernel $\ell((u, v), m; do) = \mathcal{L}(u, do)$. Thus eq. (B.1) is precisely $rG\ell$. It declares conditional independence of the result and record given the cue and presentation.

Assume $\mathcal{L}(u, do) = l(o \mid u)\lambda(do)$ for a fixed sigma-finite reference measure λ and a jointly measurable density l. At a specified record o, put $Z_m(o) = \int l(o \mid u)\mu_m(du)$ and suppose

$$0 < Z(o) := \int Z_m(o)r(dm) < \infty. \tag{B.2}$$

Then $Z_m(o) < \infty$ for r-almost every m, and the set where $Z_m(o) = 0$ has posterior weight zero, so the component conditionals below are needed only off a w^o-null set. Bayes' rule gives the presentation posterior and component conditional

$$w^o(dm) = \frac{Z_m(o)}{Z(o)}r(dm), \qquad \Pi^o_m(du, dv) = \frac{l(o \mid u)\mu_m(du)K_m(u, dv)}{Z_m(o)}. \tag{B.3}$$

The joint recognition law is $\zeta^o(dm, du, dv) = w^o(dm)\Pi^o_m(du, dv)$. It has the general disintegration of eq. (1.3). Dependence between presentation and contents is retained. Since $K_m(u, \mathsf{Y}) = 1$, the

presentation evidence depends on the cue and record mechanism, while the conditional result kernel determines what each selected presentation predicts.

Since $Z(o)$ involves only the cue kernel and the record mechanism, no cue record carries information about the result kernels K_m: the evidence of every cue-record history is the same for all result kernels, and learning the result mechanism requires records of results.

The represented content can instead be the result alone. Disintegrate the joint law $r(dm)\mu_m(du)K_m(u,dv)$ along (M,V); this gives the presentation-indexed result law $G_Y(m,dv)$ and a kernel $R : \mathsf{M} \times \mathsf{Y} \rightsquigarrow \mathsf{X}$, jointly measurable in (m,v), with $\mu_m(du)K_m(u,dv) = G_Y(m,dv)R_m(v,du)$ for r-almost every m, and set

$$\ell_Y(v,m;do) = \int \mathcal{L}(u,do)R_m(v,du). \tag{B.4}$$

Disintegration on standard Borel spaces supplies these measurable versions, with fixed normalized extensions on null sets [114]. The kernel ℓ_Y is normalized, and $rG_Y\ell_Y$ is the (M,Y,O) marginal of eq. (B.1). The two representations give the same result posterior at every record satisfying eq. (B.2), not only almost everywhere: by Tonelli's theorem, $r(dm)G_Y(m,dv)l_Y(o \mid v,m) = \int_u r(dm)\mu_m(du)K_m(u,dv)l(o \mid u)$ for the density l_Y of ℓ_Y, for every version of R. A presentation-dependent likelihood is allowed by definition 1.11.

Proposition B.2 (Correlated local free energy). At a record satisfying eq. (B.2), let a candidate joint recognition law be $\mathcal{Q}(dm,dk) = s(dm)q_m(dk)$, where $k = (u,v)$, and define $\mathcal{F}$ by eq. (1.16). Then, for every candidate, in $(-\infty,\infty]$,

$$\mathcal{F}(\mathcal{Q};o) = D_{\mathrm{KL}}(s\|w^o) + \int D_{\mathrm{KL}}(q_m\|\Pi_m^o)s(dm) - \log Z(o). \tag{B.5}$$

Consequently the unrestricted minimum is $-\log Z(o)$, attained uniquely by the posterior up to null sets.

Proof. For this model $P(o) = Z(o)$, because each K_m is normalized, and $P(\cdot \mid o) = \zeta^o$, so $\mathcal{F} = D_{\mathrm{KL}}(\mathcal{Q}\|\zeta^o) - \log Z(o)$. The chain rule for relative entropy in $[0,\infty]$ splits the joint relative entropy into the two terms of eq. (B.5); their nonnegativity and equality conditions prove the minimum statement. □

The energy and entropy form $D_{\mathrm{KL}}(s\|r) + \int[D_{\mathrm{KL}}(q_m\|G_m) - \mathbb{E}_{q_m}\log l(o \mid X)]s(dm)$ of the same quantity requires the separated terms to be finite. It can fail at the posterior itself when the record likelihood is unbounded, in which case it reads $\infty - \infty$ while eq. (B.5) is finite. For a finite presentation space with $r(a) = \hat{\pi}_a > 0$, fixing the content conditionals gives

$$s_a \propto \hat{\pi}_a e^{-f_a}, \qquad f_a = D_{\mathrm{KL}}(q_a\|\Pi_a^o) - \log Z_a(o) \in (-\infty,\infty], \tag{B.6}$$

and $f_a = D_{\mathrm{KL}}(q_a\|G_a) - \mathbb{E}_{q_a}\log l(o \mid X)$ whenever $D_{\mathrm{KL}}(q_a\|G_a)$ and $\mathbb{E}_{q_a}|\log l(o \mid X)|$ are finite. At the exact conditionals, $f_a = -\log Z_a(o)$. These are Gibbs weights for optimized component free energies. They are distinct from the peer-source rows of chapter 4: the label here chooses a model presentation.

The product restriction changes the inference problem. For the structured family $s_a q_X(du)K_a(u,dv)$, the result conditional cancels from the density ratio, and the label update is

$$s_a \propto \hat{\pi}_a \exp[-D_{\mathrm{KL}}(q_X \| \mu_a)]. \tag{B.7}$$

The record term is common to the labels at fixed q_X. With cue densities $\mu_a(u)$, varying q_X gives $q_X(u) \propto l(o \mid u)\exp(\sum_a s_a \log \mu_a(u))$. By the weighted arithmetic and geometric mean inequality the normalizer is at most $\sum_a s_a Z_a(o) < \infty$; the update is defined when the normalizer is positive, which fails when the cue components charged by s have no common part of positive $l(o \mid \cdot)$-weighted measure, and then every candidate of the structured family at that s has infinite free energy. Alternating the two block updates lowers the structured free energy by exactly the relative entropy from the old block to the new one, since each block objective is a relative entropy to its minimizer up to a constant; on finite positive tables every limit point of the alternation is a stationary point of the structured family, which need not be its minimizer. This is a product over (A, X), while retaining the result conditional; it is not a product over $(A,(X,Y))$. It illustrates the expected-log restriction described in remark 4.1 and classical mean-field inference [28]. Exact conditioning, this structured approximation, and the product-family Fisher flow are separately specified operations. The Fisher flow of $\mathcal{F}(\cdot; o)$ on the unrestricted joint law is a fourth. On finite positive tables its first variation is $\phi = \log(\zeta/\zeta^o)$ up to constants, so $\dot{\phi} = -(\phi - \mathbb{E}_\zeta \phi)$ and $\zeta_t \propto \zeta_0^{e^{-t}}(\zeta^o)^{1-e^{-t}}$; from the prior $\zeta_0 = rG$ this is the tempered posterior $\zeta_t \propto rG\, l(o \mid \cdot)^{1-e^{-t}}$, which reaches exact conditioning only as $t \to \infty$.

B.2 A Gaussian realization and its response

Let $\mathsf{M} = \{1, \dots, M\}$ and $\mathsf{X} = \mathsf{Y} = \mathsf{O} = \mathbb{R}^d$. Fix $\hat{\pi}_a > 0$, $\sum_a \hat{\pi}_a = 1$, vectors ν_a, ω_a, positive-definite matrices L_a, Q_a, C, and arbitrary matrices F_a. Set

$$\begin{aligned} X \mid A = a &\sim \mathcal{N}(\nu_a, L_a), \\ Y \mid X = u, A = a &\sim \mathcal{N}(\omega_a + F_a(u - \nu_a), Q_a), \\ O \mid X = u &\sim \mathcal{N}(u, C). \end{aligned} \tag{B.8}$$

The record covariance is fixed side information for this model. The constitutive data $(\hat{\pi}_a, \nu_a, L_a, \omega_a, F_a, Q_a)_a$ are called the bank of the model. Each evaluator is the joint Gaussian with mean (ν_a, ω_a) and covariance

$$\begin{pmatrix} L_a & L_a F_a^T \\ F_a L_a & F_a L_a F_a^T + Q_a \end{pmatrix}. \tag{B.9}$$

Its Schur complement is $Q_a \succ 0$, so arbitrary, including singular, regressions are allowed.

Proposition B.3 (Exact component and result posteriors). At $O = \xi$, put $W_a = L_a + C$. Then

$$\begin{aligned} Z_a(\xi) &= \mathcal{N}(\xi; \nu_a, W_a), & w_a &= \frac{\hat{\pi}_a Z_a(\xi)}{\sum_b \hat{\pi}_b Z_b(\xi)}, \\ m_{X,a} &= \nu_a + L_a W_a^{-1}(\xi - \nu_a), & P_{X,a} &= (L_a^{-1} + C^{-1})^{-1}, \\ m_{Y,a} &= \omega_a + F_a(m_{X,a} - \nu_a), & P_{Y,a} &= Q_a + F_a P_{X,a} F_a^T. \end{aligned} \tag{B.10}$$

The result marginal is $p_Y^\xi = \sum_a w_a \mathcal{N}(m_{Y,a}, P_{Y,a})$, with moments

$$\bar{m} = \sum_a w_a m_{Y,a}, \qquad S = \sum_a w_a \left[P_{Y,a} + (m_{Y,a} - \bar{m})(m_{Y,a} - \bar{m})^T\right]. \tag{B.11}$$

Proof. The cue and independent record noise sum to a Gaussian with covariance W_a, giving Z_a. Completing the cue's quadratic log density gives precision $L_a^{-1} + C^{-1}$ and information vector $L_a^{-1}\nu_a + C^{-1}\xi$. Solving for the mean gives the displayed gain. Applying the affine result kernel gives its conditional moments and cross-covariance $\mathrm{Cov}(X, Y \mid \xi, a) = P_{X,a}F_a^T$. Total expectation and total covariance give eq. (B.11). These are conditional-mixture identities, as in Gaussian mixture regression [35]. □

For the result-only representation eq. (B.4), let $V_a = F_a L_a F_a^T + Q_a$, $B_a = L_a F_a^T V_a^{-1}$, and $S_a = L_a - L_a F_a^T V_a^{-1} F_a L_a$. Then

$$\begin{aligned} G_Y(a, dv) &= \mathcal{N}(dv; \omega_a, V_a), \\ \ell_Y(v, a; do) &= \mathcal{N}(do; \nu_a + B_a(v - \omega_a), C + S_a). \end{aligned} \tag{B.12}$$

Conditioning the joint evaluator on $Y = v$ gives cue covariance $S_a \succ 0$, and adding independent record noise proves the likelihood formula.

Proposition B.4 (Cross-prediction response and posterior Fisher tensor). Hold the prior joint law of (A, X, Y) and $C \succ 0$ fixed, with record $O = X + \epsilon$ for independent $\epsilon \sim \mathcal{N}(0, C)$. Suppose the posterior moments used below exist and differentiation can be passed through their integrals on an open set of records. Then

$$D_\xi \mathbb{E}[Y \mid \xi] = \mathrm{Cov}(Y, X \mid \xi) C^{-1}, \qquad I_\xi = C^{-1}\,\mathrm{Cov}(X \mid \xi) C^{-1}, \tag{B.13}$$

where I_ξ is the Fisher tensor of the complete posterior family in record coordinates. These assumptions hold for the finite Gaussian model above.

Proof. The gradient of the log record likelihood is $C^{-1}(X - \xi)$. Subtracting its posterior expectation, the gradient of the log normalizer, gives posterior score $C^{-1}(X - \mathbb{E}[X \mid \xi])$. Multiplying the score by Y and integrating proves the response identity. Its score second moment gives I_ξ. Finite Gaussian mixtures have smooth finite component moments and locally dominated Gaussian integrands, which justify these operations. □

For same-state prediction $Y = X$, the whitened Jacobian is $C^{-1/2}\,\mathrm{Cov}(X \mid \xi) C^{-1/2}$, positive definite in the Gaussian example. For a distinct result the cross-covariance can have either sign and need not be symmetric. Already with one component, $D_\xi \mathbb{E}[Y \mid \xi] = FL(L + C)^{-1}$. For any prescribed dense response H, choosing $F = H(L + C)L^{-1}$ gives H exactly. The result center sets the affine intercept. This conditional response is not a gradient step on the cue, and its sign places no conflict with the positive Fisher tensor. The tensor does bound its magnitude. The proof gives $D_\xi \mathbb{E}[Y \mid \xi] = \mathrm{Cov}(Y, S \mid \xi)$ and $I_\xi = \mathrm{Cov}(S \mid \xi)$ for the posterior score S, so positivity of the conditional covariance of (Y, S) gives, when $\mathrm{Cov}(X \mid \xi)$ is nonsingular,

$$(D_\xi \mathbb{E}[Y \mid \xi]) I_\xi^{-1} (D_\xi \mathbb{E}[Y \mid \xi])^T \preceq \mathrm{Cov}(Y \mid \xi), \tag{B.14}$$

with equality exactly when Y is posterior-almost surely affine in X. The identity is a partial derivative at fixed bank and record covariance; it does not differentiate a larger computation in which those data also vary.

B.3 Retained moments and statistical geometry

Proposition B.5 (Gaussian projection and the loss of dependence). For the positive finite Gaussian mixture p^{ξ}_{Y} above, $q_G = \mathcal{N}(\bar{m}, S)$ uniquely minimizes $D_{\mathrm{KL}}(p^{\xi}_{Y}\|q)$ over nondegenerate Gaussians. If $\Pi^{\xi}_{A,Y}$ is the exact label/result joint, then

$$D_{\mathrm{KL}}(\Pi^{\xi}_{A,Y}\|w \otimes q_G) = I_{\Pi^{\xi}}(A;Y) + D_{\mathrm{KL}}(p^{\xi}_{Y}\|q_G). \tag{B.15}$$

For the diagonal carrier $q_D = \mathcal{N}(\bar{m}, \operatorname{diag} S)$,

$$D_{\mathrm{KL}}(p^{\xi}_{Y}\|q_D) - D_{\mathrm{KL}}(p^{\xi}_{Y}\|q_G) = \frac{1}{2}\log\frac{\prod_k S_{kk}}{\det S}. \tag{B.16}$$

Proof. The expectation of a Gaussian log density depends only on the mean and covariance of the sampling law. Consequently, for every nondegenerate Gaussian q, $D_{\mathrm{KL}}(p^{\xi}_{Y}\|q) - D_{\mathrm{KL}}(p^{\xi}_{Y}\|q_G) = D_{\mathrm{KL}}(q_G\|q)$. This proves the minimum and its uniqueness. Inserting $w \otimes p^{\xi}_{Y}$ into the joint logarithmic ratio proves eq. (B.15). Subtracting the two Gaussian log densities and using $\mathbb{E}[(Y-\bar{m})(Y-\bar{m})^T] = S$ gives eq. (B.16). All entropies and relative entropies here are finite for the stated positive finite mixtures. □

This inclusive-divergence projection is the local moment-matching operation used in assumed-density filtering [146]. It generally differs from minimizing the reverse divergence of variational free energy. For example, let $p_a = \frac{1}{2}\mathcal{N}(-a,1) + \frac{1}{2}\mathcal{N}(a,1)$ and $q_a = \mathcal{N}(0, 1+a^2)$. The component $\mathcal{N}(a,1)$ has reverse divergence at most $\log 2$ from p_a, whereas

$$D_{\mathrm{KL}}(q_a\|p_a) \geq a^2 - a\sqrt{\frac{2(1+a^2)}{\pi}} - \frac{1}{2}\log(1+a^2) \longrightarrow \infty. \tag{B.17}$$

Indeed, $p_a(y) \leq (2\pi)^{-1/2}e^{-(|y|-a)^2/2}$; integrate its negative logarithm under q_a and subtract the Gaussian entropy. Thus moment matching does not even minimize the reverse divergence for sufficiently separated components. Already for $a \geq \sqrt{2/(\pi-2)} \approx 1.3236$ the moment carrier is not stationary for the reverse divergence over centered Gaussians $\mathcal{N}(0,\sigma^2)$. Indeed, the Gaussian heat identity $\partial_{\sigma^2}\mathbb{E}f(Y) = \frac{1}{2}\mathbb{E}f''(Y)$ and Stein's identity give $\partial_{\sigma^2}D_{\mathrm{KL}}(\mathcal{N}(0,\sigma^2)\|p_a) = \frac{a}{2\sigma^2}\big(a - \mathbb{E}[Y\tanh(aY)]\big)$ at $\sigma^2 = 1+a^2$, with $Y \sim \mathcal{N}(0,1+a^2)$, and $\mathbb{E}[Y\tanh(aY)] < \mathbb{E}|Y| = \sqrt{2(1+a^2)/\pi} \leq a$.

Proposition B.6 (Moments need not preserve later Gaussian records). For $0 < t < 1$, the admissible result mixture

$$p_t = \tfrac{1}{2}\mathcal{N}(t, 1-t^2) + \tfrac{1}{2}\mathcal{N}(-t, 1-t^2) \tag{B.18}$$

has constant Gaussian carrier $q_G = \mathcal{N}(0,1)$, but a later record kernel $K_c(y, do) = \mathcal{N}(do; y, c)$, $c > 0$, distinguishes p_t from q_G.

Proof. Identical cue components, equal weights, $F_1 = F_2 = 0$, result centers $\pm t$, and result noise $1 - t^2$ realize p_t. Its mean is zero and variance one. Convolution with the record noise gives densities at zero

$$(p_t K_c)(0) = \frac{e^{-t^2/[2(1-t^2+c)]}}{\sqrt{2\pi(1-t^2+c)}}, \qquad (q_G K_c)(0) = \frac{1}{\sqrt{2\pi(1+c)}}. \tag{B.19}$$

They agree at $t = 0$, while the derivative of the logarithm of the first is $-t^3/(1-t^2+c)^2 < 0$. Their densities therefore differ for every $t > 0$; continuity gives a differing probability on a small interval about zero. □

The specific computation is not needed. The characteristic function of pK_c is that of p times $e^{-cs^2/2}$, which has no zeros, so for every $c > 0$ the later Gaussian record law of a result law p equals that of q_G exactly when $p = q_G$, that is, exactly when p is itself Gaussian. Moments preserve expectations of affine and quadratic readouts. Proposition B.6 shows that this does not imply preservation of every later observation law. It concerns the chosen projection, not an obstruction to richer retained states or to general-law parents. Exact inheritance still requires the complete configuration retention and interface of definition 6.1 and proposition 6.3.

For a regular family $\zeta_\theta(a, dv) = w_a \mathcal{N}(dv; m_a, P_a)$, the score split underlying proposition 3.2 gives

$$\begin{aligned} g^F_\zeta(u, v) = &\sum_a \frac{(d_u w_a)(d_v w_a)}{w_a} \\ &+ \sum_a w_a \left[(d_u m_a)^T P_a^{-1} (d_v m_a) + \frac{1}{2} \operatorname{tr}(P_a^{-1} d_u P_a P_a^{-1} d_v P_a) \right]. \end{aligned} \tag{B.20}$$

The mixed term vanishes because each conditional Gaussian score has conditional mean zero. Null parameter directions must be removed before calling this tensor a metric. For the label/result family producing eq. (B.18), its coefficient along t is $(1+t^2)/(1-t^2)^2 > 0$, while the Gaussian carrier has zero coefficient. The result marginal itself also varies, with fourth moment $3 - 2t^4$. Joint statistical geometry therefore cannot be recovered from the moment carrier in general. These are admissible model families, not assertions about a particular evolution.

B.4 Frames, sharing, and sequential records

One group $\mathrm{GL}(d)$ acts on the doubled content by $h \cdot (u, v) = (hu, hv)$ and trivially on the finite labels. Re-expressing the evaluator and record data requires

$$\begin{gathered} \nu'_a = h\nu_a, \quad \omega'_a = h\omega_a, \quad L'_a = hL_a h^T, \quad Q'_a = hQ_a h^T, \\ F'_a = hF_a h^{-1}, \quad C' = hCh^T, \quad \xi' = h\xi. \end{gathered} \tag{B.21}$$

The evidence densities acquire the same factor $|\det h|^{-1}$, so the weights are invariant; the conditional means and covariances transform by h and congruence. This follows directly by substitution into eq. (B.10). If agent i uses a frame $U_i \in \mathrm{GL}(d)$ and a Gaussian reference cue with mean m_i and covariance Σ_i in shared coordinates, the frame-reduced pair $(U_i^{-1} m_i, U_i^{-1} \Sigma_i U_i^{-T})$ is invariant

under the simultaneous passive change $U_i \mapsto hU_i$, $m_i \mapsto hm_i$, $\Sigma_i \mapsto h\Sigma_i h^T$. A singular F_a still defines a normalized response kernel; it is not a group-valued dictionary.

Sharing a bank supplies common reference and evaluator data. Sharing a generative model in the sense of definition 1.13 also requires matched record kernels under the declared frame maps. Different covariances C_i instead give a shared family of conditional mechanisms indexed by supplied side information. That family is a literal shared model once the covariance is part of the record: take the record (O, C) on $\mathsf{O} \times S_d^{++}$ with kernel $\kappa(dC)\mathcal{N}(do; u, C)$ for a fixed law κ independent of the represented variables. Agents with different C_i then share one generative model, their covariances are different record values, and the posterior given (ξ_i, C_i) is the C_i-indexed posterior because κ factors out; the ancillarity of the covariance becomes an explicit modeling assumption. The distinction must be stated if uncertainty varies between agents. If the bank covariances L_a, Q_a are restricted to diagonal matrices, this sub-parametrization is not closed under arbitrary rebasing: if hDh^T is diagonal for every positive diagonal D, its off-diagonal entries force $h_{ik}h_{jk} = 0$ for $i \neq j$, so invertibility makes h a permutation times an invertible diagonal matrix. This restriction on the reference parametrization does not remove local passive covariance with that reference fixed.

Definition B.7 (An exact carried-presentation filter). Let the finite label evolve by a fixed row-stochastic matrix T. At step l, draw a fresh cue from μ_{A_l}, a result from K_{A_l}, and a record from $\mathcal{L}_{C_l}$, with draws at distinct steps conditionally independent given the label sequence. For fixed positive record covariances, the finite law is

$$\begin{aligned} &P(a_{0:n}, du_{0:n}, dv_{0:n}, do_{0:n}) \\ &\quad = \hat{\pi}_{a_0} \prod_{l=1}^{n} T_{a_{l-1}a_l} \prod_{l=0}^{n} \left[\mu_{a_l}(du_l) K_{a_l}(u_l, dv_l) \mathcal{L}_{C_l}(u_l, do_l)\right]. \end{aligned} \tag{B.22}$$

Integration and summation of the last step give one conditional on the previous label, proving normalization and consistency. If $w_l(a) = P(A_l = a \mid O_{0:l})$, marginalization and Bayes' rule give

$$v_l(b) = \sum_a w_{l-1}(a) T_{ab}, \qquad w_l(b) = \frac{v_l(b) Z_b(o_l; C_l)}{\sum_c v_l(c) Z_c(o_l; C_l)}. \tag{B.23}$$

This is the forward filter of a hidden Markov model [20]. Because the label sequence is Markov and the draws at each step are fresh given the label, the future after step l is conditionally independent of the past given A_l, so w_l is sufficient for every future label, cue, result, and record of the declared process. Current readouts need the last record as well: $P(V_l \in \cdot \mid O_{0:l}) = \sum_b w_l(b) \Pi_b^{o_l}(\mathsf{X} \times \cdot)$, so (w_l, o_l, C_l) with the fixed bank determines the current result law. A later interaction that reads the result of one step as the cue of the next is a different model and requires its own filter.

A modified prior proportional to $\hat{\pi}_b v_l(b)^\vartheta$ has a separate one-step interpretation. With baseline prior $\hat{\pi}$ and a record $o \mid B = b \sim \mathrm{Dir}(\alpha \mathbf{1} + \vartheta e_b)$, where $\alpha > 0$ and $\vartheta \geq 0$, its normalizer is independent of b, and the density at an interior simplex value $o = v_l$ is proportional to $v_l(b)^\vartheta$. Conditioning gives the modified prior. It equals the prediction in eq. (B.23) for uniform $\hat{\pi}$ and $\vartheta = 1$. Combining this record with a cue requires declaring their joint conditional law; the algebra alone does not supply conditional independence.

Appendix A distinguishes normalization of a history from exact Bayesian assimilation under its record mechanism. That distinction applies here. If $O_2 = f(O_1)$ and O_1 is already registered, conditioning on (O_1, O_2) is the same as conditioning on O_1; an independent-replicate likelihood would count the evidence again. Communication may bring information that is new to one receiver even in a deterministic population computation. A repeated conditional-prediction agent must therefore specify its accessible history, communication-generated record, retained joint law, and any projection. The local identities above establish neither a common free energy for an arbitrary composition nor a new formation result.

Appendix C

Interacting Conditional-Prediction Agents

The conditional model of appendix B becomes an interacting system once a transmitted result has an actual record law. This appendix supplies one such law. Two agents share a normalized preparation and a communication protocol, receive separate private records, and exchange fresh samples from their current result predictions. The receiver conditions on the sender's sampling mechanism, including its inaccessible private evidence. For every finite history outside the usual conditioning null sets, the resulting posterior can be reconstructed from the joint posterior of the private records and one fixed conditional kernel, and that joint posterior is itself a product tilt of the prepared private-record law by one evidence function per agent. This is an exact representation before any moment projection.

The common preparation, its cross-agent coupling, the private sensors, and reliable registration of the transcript are constitutive assumptions. Sharing a local bank does not supply them. The distinction between common information and privately evaluated prescriptions is familiar from decentralized control [152]; the common posterior Γ_t below is its common-information belief for the fixed randomized prescriptions of the protocol. Here the prescription is fixed to posterior-predictive transmission; the results below concern its normalized law and Bayesian conditioning, without an optimal-control assertion. Repeated exchange of posterior-derived messages under a common prior is the setting of the agreement theorems of Aumann [8] and Geanakoplos and Polemarchakis [87]. Communication in which each agent sends a fresh sample from its current exact posterior and receivers condition exactly on the known sampling rule is studied on networks by Deshpande et al. [62], who prove long-run agreement and efficiency results and show that receivers who treat incoming samples as independent signal draws become overconfident. The protocol here is a two-agent variant with distinct represented states joined by a supplied coupling, a fixed known schedule, and general encoders, and the results are exact finite-history identities rather than limits. All variables are defined at one context. No connection, continuous-time flow, or identification with the reciprocal law potential is assumed.

C.1 Preparation, information, and transmission

For $i \in \{0,1\}$, let $\mathsf{A}_i, \mathsf{X}_i, \mathsf{Y}_i$, and D_i be standard Borel spaces of model presentations, cues, results, and private records. Write $Z_i = (A_i, X_i, Y_i)$ and $Z = (Z_0, Z_1)$. A proper presentation law r_i, a normalized cue kernel $\mu_{i,a}$, and a normalized result kernel $K_{i,a}$ define the local bank

$$\pi_i(da, du, dv) = r_i(da)\mu_{i,a}(du)K_{i,a}(u, dv). \tag{C.1}$$

Choose a probability coupling P_0 of π_0 and π_1. For example, any coupling Ξ of the two presentation/cue laws gives

$$P_0(dz_0, dz_1) = \Xi(da_0, du_0, da_1, du_1) K_{0,a_0}(u_0, dv_0) K_{1,a_1}(u_1, dv_1). \tag{C.2}$$

Each result factor integrates to one, so this is normalized and has the required local marginals. More general couplings with those marginals are allowed in the general construction.

Let $L_i : \mathsf{X}_i \rightsquigarrow \mathsf{D}_i$ be normalized private-record kernels, and put $D = (D_0, D_1)$ and $E = (Z, D)$. The preparation is

$$\Lambda_0(dz, dd) = P_0(dz_0, dz_1) L_0(u_0, dd_0) L_1(u_1, dd_1). \tag{C.3}$$

Thus the private records are conditionally independent given Z. All integrations in eqs. (C.1) to (C.3) are kernel integrations; no Lebesgue density or Gaussian support is required. Agent i observes its value d_i, knows the preparation and protocol, and initially has the correlated recognition law

$$\zeta_{i,0}(da_i, du_i, dv_i) = P(Z_i \in \cdot \mid D_i = d_i) = s_{i,0}(da_i) q_{i,0}(du_i, dv_i \mid a_i). \tag{C.4}$$

Marginalizing the other agent before conditioning gives exactly the local bank posterior. Presentation/content independence is not imposed.

The preparation may be realized by concealed internal values Z followed by their records. Alternatively, its record marginal can be sampled directly after integrating out Z. These realizations have the same communication law below; statements about represented variables refer to the augmented model eq. (C.3). Normalization does not require an external ontology for those variables.

Let H be a standard Borel message space, let j_t be a fixed known sequence of speakers, alternating in the running examples, and let

$$\mathcal{H}_t = \sigma(H_1, \dots, H_t), \qquad \mathcal{I}_{i,t} = \sigma(D_i, H_1, \dots, H_t). \tag{C.5}$$

The common and private posteriors, and the local recognition, are

$$\Gamma_t = P(E \in \cdot \mid \mathcal{H}_t), \qquad \Pi_{i,t} = P(E \in \cdot \mid \mathcal{I}_{i,t}), \qquad \zeta_{i,t} = (Z_i)_{\#}\Pi_{i,t}. \tag{C.6}$$

The common posterior is determined from the common transcript; it does not disclose either private value. Regular conditional probabilities exist on these spaces [114]. Choose their measurable versions in the history and private-record variables, with fixed normalized extensions on conditioning null sets. Each finite history is constructed before its conditional kernels are used to generate the next message. The resulting history laws do not depend on the chosen versions, since versions that differ on a null set of the current history law give the same next stage; pathwise formulas below refer to the versions defined recursively by eqs. (C.12) and (C.20). An autonomous map on law-valued states needs a jointly measurable choice of disintegration on the family of such states. One always exists on all of $\mathrm{Prob}(\mathsf{D}_0 \times \mathsf{D}_1)$, by the disintegration theorem for kernels applied to the identity kernel from that space of laws [114]; finite tables and the dominated formulas below also provide it directly.

Definition C.1 (Posterior-sampled communication). Fix normalized encoders $C_j : \mathsf{Y}_j \rightsquigarrow \mathsf{H}$. At the next step $j = j_{t+1}$, let

$$S_{j,t}(d_j, dy) = P(Y_j \in dy \mid D_j = d_j, \mathcal{H}_t), \qquad \kappa_{j,t}(d_j, dh) = \int C_j(y, dh) S_{j,t}(d_j, dy). \tag{C.7}$$

The speaker draws a fresh $\widetilde{Y}_{t+1}$ with law $S_{j,t}(D_j, \cdot)$ and encodes it using C_j. Conditional on $(D_j, \mathcal{H}_t)$, both randomizations are independent of E. Both agents observe the delivered H_{t+1} and append it to their persistent registers. Future policies read their private record and common transcript through the stated conditional laws; no additional access to old transmission randomization is included. This exclusion loses nothing: because later policies read H_s and not the sample $\widetilde{Y}_s$, the joint law of $(E, \widetilde{Y}_s)$ given $(D_j, \mathcal{H}_t)$ factors into a function of E and a function of $(\widetilde{Y}_s, d_j)$, so a speaker that retained its own past samples would have the same posterior and the same policy.

The sampled $\widetilde{Y}_{t+1}$ and represented Y_j have the same conditional distribution but are different variables. In particular,

$$H_{t+1} \perp (Z, D_{1-j}) \mid (D_j, \mathcal{H}_t). \tag{C.8}$$

A direct sensor $C_j(Y_j, dh)$ changes this conditional independence and therefore changes the experiment.

Proposition C.2 (Public calibration). For every stage of the declared protocol, the public predictive law is

$$P(H_{t+1} \in dh \mid \mathcal{H}_t) = \int C_j(y, dh) \Gamma_t^{Y_j}(dy). \tag{C.9}$$

Proof. Average eq. (C.7) over the common marginal of D_j. The integral of the conditional law $P(Y_j \in dy \mid D_j, \mathcal{H}_t)$ is $\Gamma_t^{Y_j}(dy)$. Substitution gives eq. (C.9). □

This equality concerns the message marginal. Its joint law with E is $\Gamma_t(de)\kappa_{j,t}(d_j(e), dh)$, which generally differs from $\Gamma_t(de)C_j(Y_j(e), dh)$. Equal public output laws need not give equal receiver posteriors.

C.2 The history law and exact assimilation

Theorem C.3 (Normalized histories and Bayesian recursion). For every nonnegative integer n, the preparation and definition C.1 define the consistent probability law

$$\mathbb{P}_n(de, dh_{1:n}) = \Lambda_0(de) \prod_{t=1}^{n} \kappa_{j_t,t-1}(h_{1:t-1}, d_{j_t}(e); dh_t). \tag{C.10}$$

Here the history argument, suppressed in eq. (C.7), specifies the posterior used by that policy. Suppose, for density formulas, that $C_j(y, dh) = c_j(h \mid y)\tau(dh)$ for a fixed sigma-finite τ. Set

$$k_{j,t}(h \mid d_j) = \int c_j(h \mid y) S_{j,t}(d_j, dy), \qquad \bar{k}_{j,t}(h) = \int k_{j,t}(h \mid d_j(e)) \Gamma_t(de). \tag{C.11}$$

At every realized prefix where $0 < \bar{k}_{j,t}(h) < \infty$, the exact common posterior update is

$$\Gamma_{t+1}(de) = \frac{k_{j,t}(h_{t+1} \mid d_j(e))}{\bar{k}_{j,t}(h_{t+1})}\Gamma_t(de). \tag{C.12}$$

The denominator condition holds for predictive-almost every next message. Equivalently, the update multiplies Γ_t by the density of the message kernel $\kappa_{j,t}(d_j(e), \cdot)$ with respect to the predictive law of the next message, independently of τ; a density form of the update exists exactly when that density exists for Γ_t-almost every prepared value, and an encoder density with respect to τ is one sufficient condition. Without domination, the corresponding update is the disintegration of $\Gamma_t(de)\kappa_{j,t}(d_j(e), dh)$.

Proof. Start from $\mathbb{P}_0 = \Lambda_0$. At stage t, the previously chosen conditional kernels make the next policy a measurable normalized kernel of $(e, h_{1:t})$. Integrating its last message gives one. Induction proves normalization of eq. (C.10) and consistency with each preceding marginal. Conditional on the existing transcript, the joint law of the prepared variables and next message is $\Gamma_t(de)\kappa_{j,t}(d_j(e), dh)$. Its message marginal is eq. (C.9). Dividing the joint density by eq. (C.11) proves eq. (C.12). A normalized nonnegative density is finite almost everywhere under its base measure and positive almost surely under its own law. This gives the denominator claim. Disintegrating the same joint measure proves the statement without densities. □

For clarity, private continuous records can also be handled by explicit densities. If $L_i(u, dd_i) = l_i(d_i \mid u)\sigma_i(dd_i)$, with sigma-finite σ_i, take $\eta = P_0 \otimes \sigma_0 \otimes \sigma_1$. Then $g_0 = d\Gamma_0/d\eta = l_0 l_1$, and

$$g_{t+1}(z, d) = \frac{g_t(z, d)k_{j,t}(h_{t+1} \mid d_j)}{\int g_t(z', d')k_{j,t}(h_{t+1} \mid d'_j)\eta(dz', dd')}. \tag{C.13}$$

For a private value d_i, divide the slice $g_t(z, d_i, d_{1-i})P_0(dz)\sigma_{1-i}(dd_{1-i})$ by its integral. This gives $\Pi_{i,t}$ wherever that integral is positive and finite, without placing a density assumption on P_0 itself.

Proposition C.4 (Sender invariance and the receiver likelihood). If agent j sends at step $t+1$, then $\Pi_{j,t+1} = \Pi_{j,t}$ almost surely. For the receiving agent $i = 1 - j$, define

$$R_{ij,t}(dd_j \mid z_i) = P(D_j \in dd_j \mid Z_i = z_i, D_i = d_i, \mathcal{H}_t). \tag{C.14}$$

Under the dominated hypotheses of theorem C.3, and wherever the receiver's own predictive density $\int \ell_{i\leftarrow j,t}(h_{t+1} \mid z'_i)\zeta_{i,t}(dz'_i)$ is positive and finite, which holds for almost every $(d_i, h_{1:t+1})$, its actual local likelihood and update are

$$\begin{aligned} \ell_{i\leftarrow j,t}(h \mid z_i) &= \int k_{j,t}(h \mid d_j)R_{ij,t}(dd_j \mid z_i), \\ \zeta_{i,t+1}(dz_i) &= \frac{\ell_{i\leftarrow j,t}(h_{t+1} \mid z_i)\zeta_{i,t}(dz_i)}{\int \ell_{i\leftarrow j,t}(h_{t+1} \mid z'_i)\zeta_{i,t}(dz'_i)}. \end{aligned} \tag{C.15}$$

Proof. Condition eq. (C.12) further on $D_j = d_j$. Its message factor is constant in every remaining variable, so it cancels from the conditional numerator and denominator. This proves invariance of the entire private posterior; eq. (C.8) gives the same conclusion without densities. For the receiver, first condition the actual message kernel on $(Z_i, D_i, \mathcal{H}_t)$. The remaining private coordinate has law

eq. (C.14); integrating it gives the first equation of eq. (C.15). Marginalizing the private joint update then gives the second. □

The conditional law $R_{ij,t}$ carries the relationship between the receiver's represented state and the sender's evidence. A local marginal $\zeta_{i,t}$ does not in general specify it. If a message policy reads only common information, its factor is independent of both private values and cancels from the common posterior as well. A deterministic echo of an already registered message has this property in the kernel formulation. A fresh posterior sample can instead convey the speaker's private evidence to its peer, while leaving the speaker's own posterior unchanged.

There is an exact information restriction for the fixed preparation. Every factor in eq. (C.10) depends on E only through D at a fixed transcript. Thus Z and $H_{1:n}$ are conditionally independent given D, and, whenever $I(Z; D_{1-i} \mid D_i) < \infty$,

$$I(Z; H_{1:n} \mid D_i) = I(Z; D_{1-i} \mid D_i) - I(Z; D_{1-i} \mid D_i, H_{1:n}). \tag{C.16}$$

Indeed, expand $I(Z; (D_{1-i}, H_{1:n}) \mid D_i)$ in the two possible orders by the chain rule for mutual information [46]; the term conditioned on both private records vanishes, so the expansion equals $I(Z; D_{1-i} \mid D_i)$ and both remaining terms are finite. In particular $I(Z; H_{1:n} \mid D_i) \le I(Z; D_{1-i} \mid D_i)$. The protocol redistributes evidence from the preparation. New measurements require an additional observation mechanism.

C.3 A fixed reconstruction and the retained state

Disintegrate the preparation once as

$$\Lambda_0(dz, dd) = W(d, dz)\lambda_0(dd), \qquad \lambda_0 = P(D \in \cdot), \quad W(d, \cdot) = P(Z \in \cdot \mid D = d). \tag{C.17}$$

The null-set version of W is fixed with the model. Write W_i and W^{Y_j} for its local-state and result marginals, and let

$$V_j(d, dh) = \int C_j(y, dh) W^{Y_j}(d, dy). \tag{C.18}$$

Theorem C.5 (Exact private-record reduction). For every finite stage of the protocol,

$$\Gamma_t(dz, dd) = W(d, dz)\lambda_t(dd), \qquad \lambda_t = P(D \in \cdot \mid \mathcal{H}_t). \tag{C.19}$$

For its chosen conditional versions the local recognition and transmission law are

$$\begin{aligned} \zeta_i^{\lambda_t, d_i}(dz_i) &= \int W_i(d, dz_i)\lambda_t(dd \mid D_i = d_i), \\ \kappa_j^{\lambda_t}(d_j, dh) &= \int V_j(d, dh)\lambda_t(dd \mid D_j = d_j). \end{aligned} \tag{C.20}$$

If the latter kernel has density $k_j^{\lambda_t}(h \mid d_j)$ relative to τ, the common state obeys the closed recursion

$$\lambda_{t+1}(dd) = \frac{k_j^{\lambda_t}(h_{t+1} \mid d_j)\lambda_t(dd)}{\int k_j^{\lambda_t}(h_{t+1} \mid d'_j)\lambda_t(dd')}. \tag{C.21}$$

Reconstruction by W makes this recursion exactly equivalent to eq. (C.12) for the complete joint posterior, and hence for every finite future sequence of protocol messages.

Proof. At a fixed history, its likelihood $\mathcal{L}_t(d)$ is a function of D alone. In the dominated case, for bounded measurable f and a the posterior integral is

$$\begin{aligned}\int f(z)a(d)\Gamma_t(dz, dd) &= \frac{\int a(d)\mathcal{L}_t(d)[\int f(z)W(d, dz)]\lambda_0(dd)}{\int \mathcal{L}_t(d)\lambda_0(dd)} \\ &= \int a(d)\left[\int f(z)W(d, dz)\right]\lambda_t(dd).\end{aligned} \tag{C.22}$$

Products of bounded measurable functions determine the joint law by a monotone-class argument. Without domination, the same identity follows by integrating $f(z)$ first in eq. (C.10); its conditional history kernel depends on d alone. This proves eq. (C.19). Conditioning it on the observed private coordinate and taking the indicated marginals gives eq. (C.20). Multiplication by the message density changes only λ_t, proving eq. (C.21). Conversely, marginalizing eq. (C.12) gives that update. At each future step the reconstruction supplies the same message kernel and conditional update, so induction proves equality of every finite future joint law. □

The sufficient state for an agent includes the fixed preparation, reconstruction and encoders, its private value, the joint λ_t, and the time index with the fixed schedule. When queries of actual past records are admitted, take the retained state to be

$$\mathcal{R}_{i,t} = \big(P_0, L_0, L_1, W, C_0, C_1, (j_s)_s, d_i, \lambda_t, t, H_{1:t}\big). \tag{C.23}$$

Equality of these states gives the same current private posterior and, by theorem C.5, the same joint conditional laws for all future protocol messages and represented-variable readouts, for the policy versions chosen as functions of the reduced state in eq. (C.20). The last coordinate preserves the literal transcript; its values need not be recoverable from λ_t. This separates inferential sufficiency from persistence of the full admitted record interface.

The changing part of this state is smaller than a general law on $\mathsf{D}_0 \times \mathsf{D}_1$, because each message factor reads only its speaker's private record. In the dominated case write, at a fixed history,

$$\mathcal{L}_t(d; h) = \prod_{s=1}^{t} k_{j_s,s-1}(h_s \mid d_{j_s}; h_{1:s-1}), \qquad F_i^t(d_i) = \prod_{s\le t:\ j_s=i} k_{i,s-1}(h_s \mid d_i; h_{1:s-1}). \tag{C.24}$$

Proposition C.6 (Product-tilt representation). Under the dominated hypotheses of theorem C.3, at every history with $0 < Z_t = \int \mathcal{L}_t\, d\lambda_0 < \infty$,

$$\mathcal{L}_t(d; h) = F_0^t(d_0)F_1^t(d_1), \qquad \lambda_t(dd) = \frac{F_0^t(d_0)F_1^t(d_1)}{Z_t}\lambda_0(dd). \tag{C.25}$$

Consequently: (i) the private posterior of agent i depends on the transcript only through the peer factor F^t_{1-i}, the own factor cancelling; (ii) a message from agent j multiplies F^t_j and leaves F^t_{1-j} unchanged; (iii) the cross-agent dependence of λ_t is that of λ_0: on finite tables every cross-ratio $\lambda_t(a,b)\lambda_t(a',b')/(\lambda_t(a,b')\lambda_t(a',b))$ equals that of λ_0, and in the nondegenerate Gaussian case of theorem C.10 the off-diagonal block of the precision of λ_t equals that of λ_0. Moreover, two product tilts $P = f_0 f_1 \lambda_0/Z$ and $P' = g_0 g_1 \lambda_0/Z'$ with the same two marginals coincide whenever the private-record alphabets are finite, or more generally whenever $\log(g_0/f_0)$ and $\log(g_1/f_1)$ are integrable under the marginals.

Proof. Each factor of $\mathcal{L}_t$ depends on d only through d_{j_s}, since the speaker's policy reads its own record and the common transcript; grouping the factors by speaker gives the first identity, and Bayes' formula the second. Conditioning λ_t on $D_i = d_i$ divides out $F^t_i(d_i)$, which proves (i); (ii) is the grouping itself, together with (i) for the speaker's kernel. A product tilt multiplies each cross-ratio by $f_0(a)f_1(b)f_0(a')f_1(b')/(f_0(a)f_1(b')f_0(a')f_1(b)) = 1$, and in the Gaussian case each message adds $A_t^T R_t^{-1} A_t$ to the precision with A_t supported on the speaker coordinate, which proves (iii). For uniqueness, let $S_i = \{f_i > 0\}$. Since $P'(S_0^c \times \mathsf{D}_1) = P(S_0^c \times \mathsf{D}_1) = 0$ and likewise for S_1, and P has a positive density with respect to λ_0 on $S_0 \times S_1$, $P' \ll P$, and symmetrically $P \ll P'$. Then $dP'/dP = a(d_0)b(d_1)$ with a, b positive and finite almost everywhere, and under the integrability hypothesis, automatic on finite alphabets, $D_{\mathrm{KL}}(P'\|P) = \mathbb{E}_{\mu_0} \log a + \mathbb{E}_{\mu_1} \log b = -D_{\mathrm{KL}}(P\|P')$ because the marginals agree. Both divergences are nonnegative, so both vanish and $P' = P$. □

The factorization of the history likelihood is the rectangle property of two-party private-coin protocols [14, Lemma 6.7]. On the reachable family the exact changing state can therefore be taken as the pair of evidence classes $[F^t_0], [F^t_1]$, up to reciprocal positive scaling, with λ_0 and the other fixed data of eq. (C.23). Equivalently, the two private-record marginals of λ_t together with the fixed λ_0 determine λ_t and every conditional pair law in eq. (C.20) in all cases treated in this appendix, including the Gaussian case, where the log tilts are quadratic and integrable. The product of the two marginals alone does not, since it discards the fixed cross-agent dependence. Whether a smaller state serves a particular admitted interface is a separate minimality question, which this appendix does not settle.

On the family $\{W\lambda\}$, the map $W\lambda \mapsto \lambda$ is invertible by the fixed reconstruction. It is the sample compression of the common posterior by the coordinate channel $E \mapsto D$, and as a configuration retention in the sense of definition 6.1 it is injective on this family, so it removes redundant changing coordinates and is not a lossy retention. The fixed W can be large, and a law on continuous private-record spaces need not have a finite description.

The supplied coupling has a separate role. At a fixed transcript the history likelihood is always a product of a function of D_0 and a function of D_1, by proposition C.6. If the preparation also factorizes across (Z_0, D_0) and (Z_1, D_1), the common posterior consequently remains a product across the two agents. After conditioning on its own private record, an agent's own represented-state posterior cannot change when its peer speaks. It may learn about the peer, but information transfer to its own target requires a cross-agent relation supplied by the model.

C.4 Free energies for the declared information

Fix a dominated observed history $h_{1:n}$, and define

$$\mathcal{L}_n(d;h)=\prod_{t=1}^{n}k_{j_t,t-1}(h_t\mid d_{j_t};h_{1:t-1}),\qquad \mathcal{Z}_n(h)=\int\mathcal{L}_n(d;h)\Lambda_0(dz,dd). \tag{C.26}$$

The policy values at each observed prefix are those determined from the actual preparation and protocol. They remain fixed when a candidate recognition law is varied. Assume $0<\mathcal{Z}_n(h)<\infty$. For a candidate probability $Q\ll\Lambda_0$ supported where $\mathcal{L}_n>0$, define

$$\begin{aligned}\mathcal{F}_n(Q;h)&=\mathbb{E}_Q\left[\log\frac{dQ}{d\Lambda_0}(E)-\log\mathcal{L}_n(D;h)\right]\\&=D_{\mathrm{KL}}(Q\|\Lambda_0)-\mathbb{E}_Q\log\mathcal{L}_n(D;h).\end{aligned} \tag{C.27}$$

In the statements below, the relative entropy and the absolute expectation of the displayed log likelihood are finite. These conditions prevent an undefined subtraction of infinities and imply the finite posterior relative entropy used in each identity. The statements extend to any other admissible domain on which the defining logarithmic integral and chain rules are well defined, but no such extension is needed here.

Proposition C.7 (Full, private, and local inference objectives). Under these conditions,

$$\mathcal{F}_n(Q;h)=D_{\mathrm{KL}}(Q\|\Gamma_n)-\log\mathcal{Z}_n(h). \tag{C.28}$$

For agent i, let $\Lambda_{0,i}^{d_i}=P(E\in\cdot\mid D_i=d_i)$ and $\mathcal{Z}_{i,n}=\int\mathcal{L}_n(D;h)\Lambda_{0,i}^{d_i}(dE)$, assumed positive and finite. Its private objective is

$$\begin{aligned}\mathcal{F}_{i,n}(Q;d_i,h)&=\mathbb{E}_Q\left[\log\frac{dQ}{d\Lambda_{0,i}^{d_i}}(E)-\log\mathcal{L}_n(D;h)\right]\\&=D_{\mathrm{KL}}(Q\|\Pi_{i,n})-\log\mathcal{Z}_{i,n}.\end{aligned} \tag{C.29}$$

Let $\zeta_{i,0}$ denote its initial local marginal, write $\zeta_{i,n}(da,dk)=s_{i,n}(da)q_{i,n}(dk\mid a)$ for the disintegration of the local posterior over the presentation, and put

$$b_{i,n}(z_i)=\mathbb{E}_{\Lambda_{0,i}^{d_i}}[\mathcal{L}_n(D;h)\mid Z_i=z_i]. \tag{C.30}$$

For a candidate local correlated law $\mathcal{Q}_i(da,dk)=s_i(da)q_i(dk\mid a)$, where $k=(u,v)$, define

$$\mathcal{F}_{i,n}^{\mathrm{loc}}(\mathcal{Q}_i)=\mathbb{E}_{\mathcal{Q}_i}\left[\log\frac{d\mathcal{Q}_i}{d\zeta_{i,0}}(Z_i)-\log b_{i,n}(Z_i)\right]. \tag{C.31}$$

With the same absolute-continuity and integrability hypotheses on this local space,

$$\begin{aligned}\mathcal{F}_{i,n}^{\mathrm{loc}}(\mathcal{Q}_i)&=D_{\mathrm{KL}}(s_i\|s_{i,n})\\&\quad+\int D_{\mathrm{KL}}(q_i(\cdot\mid a)\|q_{i,n}(\cdot\mid a))s_i(da)-\log\mathcal{Z}_{i,n}.\end{aligned} \tag{C.32}$$

Whenever the corresponding posterior satisfies the stated finiteness conditions, each unrestricted minimum is its negative log evidence, attained only by that posterior up to null sets.

Proof. Bayes' formula gives $d\Gamma_n/d\Lambda_0 = \mathcal{L}_n/\mathcal{Z}_n$. Subtract its logarithm from $\log(dQ/d\Lambda_0)$ inside eq. (C.27); integrating yields eq. (C.28). The same argument on the conditional base $\Lambda_{0,i}^{d_i}$ proves eq. (C.29). Marginalization gives $\zeta_{i,n}(dz_i) = b_{i,n}(z_i)\zeta_{i,0}(dz_i)/\mathcal{Z}_{i,n}$, so the local objective equals $D_{\mathrm{KL}}(\mathcal{Q}_i\|\zeta_{i,n}) - \log \mathcal{Z}_{i,n}$. To split that relative entropy, disintegrate both laws over A_i. Their logarithmic density ratio is the marginal label ratio plus the conditional content ratio. Integration gives eq. (C.32). Relative entropy is nonnegative and vanishes precisely for equal probability laws, proving the constrained minima over normalized laws. When the posterior belongs to the stated finite-objective domain, these minima are attained there. □

The last integrability qualification matters for general laws. Finiteness of the evidence alone does not imply finite expected log likelihood at the posterior. The normalized posterior and its relative-entropy characterization still exist under the evidence condition; the separated finite expectations in eq. (C.27) require the additional hypothesis stated above.

The conditional information omitted by a local objective is explicit. For an admissible full Q, disintegration over Z_i gives

$$\begin{aligned} D_{\mathrm{KL}}(Q\|\Pi_{i,n}) = D_{\mathrm{KL}}(Q^{Z_i}\|\zeta_{i,n}) \\ + \int D_{\mathrm{KL}}\big(Q(\cdot \mid Z_i = z_i)\|\Pi_{i,n}(\cdot \mid Z_i = z_i)\big)Q^{Z_i}(dz_i). \end{aligned} \tag{C.33}$$

The proof is the same logarithmic split. Optimizing only the local term does not identify the conditional law of the peer's private record required in eq. (C.15). Likewise, a product restriction on presentation and content changes the admitted variational family; it is not part of the exact posterior construction.

Proposition C.8 (Exact elimination in free energy). Let $Q(dz, dd) = Q^Z(d, dz)\nu(dd)$, with the absolute-continuity and finiteness conditions of proposition C.7. Then

$$\mathcal{F}_n(Q; h) = D_{\mathrm{KL}}(\nu\|\lambda_n) + \int D_{\mathrm{KL}}(Q^Z(d,\cdot)\|W(d,\cdot))\nu(dd) - \log \mathcal{Z}_n(h). \tag{C.34}$$

For fixed ν, the minimizing conditional is W almost everywhere, whenever that candidate is admissible. The remaining objective is exactly

$$\begin{aligned} \mathcal{F}_n^D(\nu; h) &= \mathbb{E}_\nu\left[\log \frac{d\nu}{d\lambda_0}(D) - \log \mathcal{L}_n(D; h)\right] \\ &= D_{\mathrm{KL}}(\nu\|\lambda_n) - \log \mathcal{Z}_n(h). \end{aligned} \tag{C.35}$$

Proof. Substitute $\Gamma_n = W\lambda_n$ into eq. (C.28). Splitting its logarithmic ratio over D gives the marginal relative entropy and the averaged conditional relative entropy in eq. (C.34). The latter is minimized uniquely by $Q^Z = W$, ν-almost everywhere. Because $\mathcal{L}_n$ depends only on D, the defining expectation for $Q = W\nu$ reduces to the first line of eq. (C.35). Bayes' formula for λ_n gives its second line. □

These are unit-temperature inference objectives for a fixed observed history, instances of the Gibbs variational principle with the chain rule for relative entropy [66]. The local objective eq. (C.31) is a marginal objective, minimized by the exact marginal posterior $\zeta_{i,n}$; it is not the local free energy of proposition 4.16, which is a global free energy as a function of one mean-field block. Their evidence terms are constant in the candidate law within that optimization block. The agents condition on different private information, so their objectives have different minimizers. Recomputing historical policies from the candidate Q inside eq. (C.26) would also vary the observation model and define a different optimization problem. No natural-gradient update, monotone sum during communication, or bank-learning rule follows from the variational identities alone.

C.5 An exact two-way binary exchange

Let every variable be a bit. Draw X_0 fairly, set $P(X_1 = X_0 \mid X_0) = 3/4$, and, independently conditional on the cues, prepare

$$P(A_i = X_i \mid X_i) = \frac{3}{4}, \qquad P(Y_i = 1 - X_i \mid X_i) = \frac{3}{4}, \qquad P(D_i = X_i \mid X_i) = \frac{3}{4}. \tag{C.36}$$

All base measures are counting measures, and all preparation atoms are positive. Since each cue is fair, A_i is fair and $P(X_i = A_i \mid A_i) = 3/4$. This is a local bank of the form eq. (C.1), with a negative cue/result response and correlated presentation/content variables. Use identity encoders on the sampled result bits, with agent 1 speaking first.

At any information state of this protocol, if $p_i = P(X_i = 1 \mid \mathcal{I}_i)$, then

$$\begin{aligned} P(A_i = 1 \mid \mathcal{I}_i) &= \frac{1}{4} + \frac{1}{2}p_i, & P(Y_i = 1 \mid \mathcal{I}_i) &= \frac{3}{4} - \frac{1}{2}p_i, \\ \mathrm{Cov}(A_i, X_i \mid \mathcal{I}_i) &= \frac{1}{2}p_i(1-p_i), & \mathrm{Cov}(Y_i, X_i \mid \mathcal{I}_i) &= -\frac{1}{2}p_i(1-p_i). \end{aligned} \tag{C.37}$$

Indeed, conditioning further on the cues leaves the independent presentation and result kernels in eq. (C.36) unchanged: the message likelihood depends on the preparation only through the private records. Total expectation and covariance then give eq. (C.37). In particular, the presentation/content covariance is nonzero at every finite history.

Initially the speaker's result-sampling probabilities are $k_{1,0}(1 \mid 0) = 5/8$ and $k_{1,0}(1 \mid 1) = 3/8$. Integrating its private sensor and then the cue coupling gives

$$\begin{aligned} P(H_1 = 1 \mid X_1 = 0) &= \frac{9}{16}, & P(H_1 = 1 \mid X_1 = 1) &= \frac{7}{16}, \\ P(H_1 = 1 \mid X_0 = 0) &= \frac{17}{32}, & P(H_1 = 1 \mid X_0 = 1) &= \frac{15}{32}. \end{aligned} \tag{C.38}$$

For example, the first probability is $(3/4)(5/8)+(1/4)(3/8)$, and the third is $(3/4)(9/16)+(1/4)(7/16)$. At $D_0 = 1, H_1 = 1$, Bayes' rule gives

$$P(X_0 = 1 \mid D_0 = 1, H_1 = 1) = \frac{(3/4)(15/32)}{(3/4)(15/32) + (1/4)(17/32)} = \frac{45}{62}. \tag{C.39}$$

At $D_0 = 0$ the same calculation gives $5/22$. The reply therefore uses

$$k_{0,1}(1 \mid D_0 = 0, H_1 = 1) = \frac{7}{11}, \qquad k_{0,1}(1 \mid D_0 = 1, H_1 = 1) = \frac{12}{31}. \tag{C.40}$$

Both values differ from the initial sampling probabilities. The reception changes the outgoing law.

The four private-record weights make the second reception explicit. In the order $(00, 01, 10, 11)$, direct marginalization of eq. (C.36), followed by the two reweightings, yields

$$\begin{aligned} \lambda_0 &= \frac{1}{32}(9, 7, 7, 9), \\ \lambda_1 \,|_{H_1=1} &= \frac{1}{128}(45, 21, 35, 27), \\ \lambda_2 \,|_{H_1=H_2=1} &= \left(\frac{105}{242}, \frac{49}{242}, \frac{70}{341}, \frac{54}{341}\right). \end{aligned} \tag{C.41}$$

The first normalization is $P(H_1 = 1) = 1/2$. The second is $(66/128)(7/11) + (62/128)(12/31) = 33/64$; multiplying the corresponding rows of λ_1 by eq. (C.40) and dividing by this value proves the last line. The fixed reconstruction supplies

$$\begin{aligned} P(X_0 = 1 \mid D) &= \left(\frac{1}{6}, \frac{5}{14}, \frac{9}{14}, \frac{5}{6}\right), \\ V_0(D; \{1\}) &= \left(\frac{2}{3}, \frac{4}{7}, \frac{3}{7}, \frac{1}{3}\right), \\ V_1(D; \{1\}) &= \left(\frac{2}{3}, \frac{3}{7}, \frac{4}{7}, \frac{1}{3}\right). \end{aligned} \tag{C.42}$$

For the first table, each entry is the normalized sum over X_1 of the cue prior and two private-sensor factors. The corresponding table for X_1 exchanges D_0, D_1; the last two tables then follow from $P(Y_i = 1 \mid X_i) = 3/4 - X_i/2$.

Take actual private values $D_0 = 1, D_1 = 0$ and delivered bits $H_1 = H_2 = 1$. Each agent still conditions only on its own private value. The common weights give

$$P(D_0 = 1 \mid D_1 = 0, H_1 = H_2 = 1) = \frac{44}{137}, \tag{C.43}$$

and hence agent 1's final cue probability is

$$P(X_1 = 1 \mid D_1 = 0, H_1 = H_2 = 1) = \frac{93}{137}\frac{1}{6} + \frac{44}{137}\frac{5}{14} = \frac{437}{1918}. \tag{C.44}$$

The accessible cue probabilities are therefore

Registered messages	Agent 0, $D_0 = 1$	Agent 1, $D_1 = 0$
None	$3/4$	$1/4$
$H_1 = 1$	$45/62$	$1/4$
$H_1 = H_2 = 1$	$45/62$	$437/1918$.

(C.45)

The sender invariance, visible in each column at that agent's own transmission, follows from proposition C.4; the receiver changes at both receptions. Presentation and result probabilities follow from eq. (C.37). The fixed reconstruction and the four masses of λ_t reproduce the full joint law, including its local correlations. By proposition C.6 these masses are a redundant form of two changing numbers: the odds ratio of every λ_t equals that of λ_0, namely $81/49$, so with λ_0 fixed the two marginals determine λ_t.

The same preparation distinguishes two errors in message interpretation. A direct observation of represented $Y_1 = 1$ has cue likelihoods $5/8$ and $3/8$ conditional on $X_0 = 0, 1$, respectively. It gives $P(X_0 = 1 \mid D_0 = 1, Y_1 = 1) = 9/14$, in place of $45/62$. If agent 1 instead sends two fresh samples before a reply, its private posterior is unchanged, so the samples are independent conditional on D_1. Their probabilities conditional only on X_1 are

$$P(H_1 = H_2 = 1 \mid X_1 = 0) = \frac{21}{64}, \qquad P(H_1 = H_2 = 1 \mid X_1 = 1) = \frac{13}{64}. \tag{C.46}$$

These equal $(3/4)(5/8)^2 + (1/4)(3/8)^2$ and its exchanged version. Integrating the cue coupling and conditioning on $D_0 = 1$ gives $45/64$. Squaring the once-marginalized likelihoods $9/16, 7/16$ instead gives $171/244$, because that operation deletes their shared private cause. A deterministic echo $H_2 = H_1$ has conditional likelihood one at the registered bit and causes no further update.

C.6 Gaussian communication and an exact affine closure

Take finite presentation sets, vector cues $X_i \in \mathbb{R}^{d_i}$ and results $Y_i \in \mathbb{R}^{e_i}$, and a strictly positive label table p_{ab} of total mass one, with marginals π_{0a}, π_{1b}. For each pair choose the conditional cue law

$$\begin{pmatrix} X_0 \\ X_1 \end{pmatrix} \,\Big|\, A_0 = a, A_1 = b \sim \mathcal{N}\left(\begin{pmatrix} \nu_{0a} \\ \nu_{1b} \end{pmatrix}, \begin{pmatrix} L_{0a} & S_{ab} \\ S_{ab}^T & L_{1b} \end{pmatrix} \right), \tag{C.47}$$

where the block covariance is positive definite. For instance, $S_{ab} = L_{0a}^{1/2} J_{ab} L_{1b}^{1/2}$ with $\|J_{ab}\| < 1$ satisfies this condition. Given the cues and labels, generate the results independently by

$$Y_i \mid X_i = u, A_i = a \sim \mathcal{N}\big(\omega_{ia} + F_{ia}(u - \nu_{ia}), Q_{ia}\big), \qquad Q_{ia} \succ 0. \tag{C.48}$$

The matrices F_{ia} are arbitrary. Summing the label table and integrating the normalized cue and result kernels proves normalization. The marginal cue law for agent i does not depend on the other label, so its local bank is precisely $\pi_{ia}\mathcal{N}(du; \nu_{ia}, L_{ia})K_{ia}(u, dv)$. The label coupling and cross-covariances are additional model data.

Let the private sensors be $D_i = X_i + \epsilon_i$, where $\epsilon_i \sim \mathcal{N}(0, N_i)$, $N_i \succ 0$, are mutually independent and independent of the prepared variables. This notation distinguishes sensor covariance from the encoder kernel C_i. Before any message, the sender, agent 1, has

$$\begin{aligned} T_b &= L_{1b} + N_1, & w_b(d) &= \frac{\pi_{1b}\mathcal{N}(d; \nu_{1b}, T_b)}{\sum_c \pi_{1c}\mathcal{N}(d; \nu_{1c}, T_c)}, \\ G_b &= L_{1b}T_b^{-1}, & P_b &= L_{1b} - L_{1b}T_b^{-1}L_{1b}, \\ m_b(d) &= \omega_{1b} + F_{1b}G_b(d - \nu_{1b}), & \Sigma_b^Y &= Q_{1b} + F_{1b}P_bF_{1b}^T. \end{aligned} \tag{C.49}$$

Completing the cue quadratic gives $P_b = (L_{1b}^{-1} + N_1^{-1})^{-1} \succ 0$. Applying the affine result kernel gives the mixture with weights w_b, means m_b, and covariances $\Sigma_b^Y \succ 0$, as in proposition B.3.

Use the encoder $H = B\widetilde{Y}_1 + \eta$ with independent $\eta \sim \mathcal{N}(0, R)$, $R \succ 0$. Its actual first-message density conditional on the sender's private record is

$$k(h \mid d) = \sum_b w_b(d)\mathcal{N}(h; Bm_b(d), R + B\Sigma_b^Y B^T). \tag{C.50}$$

The sample and encoder noises are independent of the prepared variables conditional on D_1. Conditional on a complete represented value z, the likelihood is therefore

$$P(H \in dh \mid Z = z)/dh = \int \mathcal{N}(d; u_1, N_1)k(h \mid d)dd. \tag{C.51}$$

At private value d_0 and received h, the augmented posterior is

$$\Pi(dz, dd_1 \mid d_0, h) = \frac{P_0(dz)\mathcal{N}(d_0; u_0, N_0)\mathcal{N}(d_1; u_1, N_1)k(h \mid d_1)}{\mathcal{Z}(d_0, h)}dd_1. \tag{C.52}$$

The normalizer is positive and finite for every finite (d_0, h). Positivity follows from the strictly positive Gaussian factors. Finiteness follows because the private sensor density is bounded and the finite mixture in eq. (C.50) is bounded by the largest of its component density maxima, independently of d. All integrands are nonnegative, so their order can be exchanged. Integrating h, then the private records and prepared variables, gives total mass one.

Proposition C.9 (The first receiver likelihood for Gaussian banks). For the preparation above, put

$$\begin{aligned} u_{ab}(x) &= \nu_{1b} + S_{ab}^T L_{0a}^{-1}(x - \nu_{0a}), \\ U_{ab} &= N_1 + L_{1b} - S_{ab}^T L_{0a}^{-1} S_{ab}, \\ \ell_0(h \mid a, x) &= \sum_b \frac{p_{ab}}{\pi_{0a}} \int \mathcal{N}(d; u_{ab}(x), U_{ab})k(h \mid d)dd. \end{aligned} \tag{C.53}$$

Then the exact correlated local posterior is

$$\zeta_0(a, dx, dy \mid d_0, h) \propto \pi_{0a}\mathcal{N}(dx; \nu_{0a}, L_{0a})K_{0a}(x, dy)\mathcal{N}(d_0; x, N_0)\ell_0(h \mid a, x). \tag{C.54}$$

Proof. Completing the joint cue quadratic conditional on $(A_0, A_1, X_0) = (a, b, x)$ gives the mean $u_{ab}(x)$ and covariance $L_{1b} - S_{ab}^T L_{0a}^{-1} S_{ab}$ for X_1. Adding the private sensor noise gives $U_{ab} \succ 0$; the Schur complement is positive by eq. (C.47). Since the marginal law of X_0 at label a does not depend on b, conditioning on $X_0 = x$ leaves the label weights p_{ab}/π_{0a} unchanged. Integrating the sender's private record against eq. (C.50) therefore gives eq. (C.53). The own-private likelihood and this message likelihood are conditionally independent of Y_0 at fixed (A_0, X_0). Multiplying by the local bank and normalizing proves eq. (C.54), equivalently by marginalizing eq. (C.52). □

The integral includes the sender's posterior-dependent weights $w_b(d)$. It cannot generally be replaced by a Gaussian observation of Y_1 or by evaluating those weights at a message mean.

The scalar, one-presentation case makes the different observation mechanism explicit. Write $\mathrm{Cov}(X_0, X_1) = S$, with $S^2 < L_0 L_1$, result slope f_1, and encoder slope b. Let

$$g = \frac{L_1}{L_1 + N_1}, \qquad P = \frac{L_1 N_1}{L_1 + N_1}, \qquad a = b f_1 g. \tag{C.55}$$

A fresh posterior sample has the realization $\widetilde{Y}_1 = \omega_1 + f_1 g(D_1 - \nu_1) + \xi$, where ξ is independent Gaussian with variance $Q_1 + f_1^2 P$. Substituting $D_1 = X_1 + \epsilon_1$ gives

$$\begin{aligned} H &= b\omega_1 + a(X_1 - \nu_1) + a\epsilon_1 + b\xi + \eta, \\ N_{\mathrm{eff}} &= a^2 N_1 + b^2(Q_1 + f_1^2 P) + R, \\ H \mid X_1 = x &\sim \mathcal{N}\big(b\omega_1 + a(x - \nu_1), N_{\mathrm{eff}}\big). \end{aligned} \tag{C.56}$$

The direct sensor $H_{\mathrm{dir}} = bY_1 + \eta$ instead has gain bf_1 and conditional variance $b^2 Q_1 + R$. Nevertheless,

$$\begin{aligned} &\mathrm{Var}(H) = a^2 L_1 + N_{\mathrm{eff}} = b^2(f_1^2 L_1 + Q_1) + R = \mathrm{Var}(H_{\mathrm{dir}}), \\ &\mathrm{Cov}(X_0, H) = aS, \qquad\qquad \mathrm{Cov}(X_0, H_{\mathrm{dir}}) = bf_1 S. \end{aligned} \tag{C.57}$$

The variance equality uses $g^2(L_1 + N_1) + P = L_1$; both means equal $b\omega_1$. Thus even equal Gaussian message marginals leave different cross-agent evidence.

For completeness, first conditioning on $D_0 = d_0$ gives

$$\begin{aligned} m_0 &= \nu_0 + \frac{L_0}{L_0 + N_0}(d_0 - \nu_0), \quad v_0 = \frac{L_0 N_0}{L_0 + N_0}, \\ m_1^0 &= \nu_1 + \frac{S}{L_0 + N_0}(d_0 - \nu_0), \quad v_1^0 = L_1 - \frac{S^2}{L_0 + N_0}, \\ c_{01}^0 &= \frac{S N_0}{L_0 + N_0}. \end{aligned} \tag{C.58}$$

Let $m_H = b\omega_1 + a(m_1^0 - \nu_1)$ and $V_H = a^2 v_1^0 + N_{\mathrm{eff}}$. Conditioning the remaining Gaussian on $H = h$ gives

$$\begin{aligned} m_0^+ &= m_0 + \frac{a c_{01}^0}{V_H}(h - m_H), \qquad v_0^+ = v_0 - \frac{a^2 (c_{01}^0)^2}{V_H}, \\ Y_0 \mid d_0, h &\sim \mathcal{N}\big(\omega_0 + f_0(m_0^+ - \nu_0), Q_0 + f_0^2 v_0^+\big). \end{aligned} \tag{C.59}$$

Both conditioning operations are completions of positive Gaussian quadratics. Positive sensor and encoder noise make their Schur complements positive. These formulas therefore give exact conditioning without a moment projection.

Theorem C.10 (Repeated affine Gaussian closure). Suppose each agent has one presentation, the joint cue covariance is positive definite, and all result kernels, private sensors, and encoders are affine Gaussian with positive-definite noise covariances. For every finite protocol history, the common private-record posterior λ_t is a nondegenerate joint Gaussian. Its mean and full covariance, together with the fixed Gaussian reconstruction W, determine every posterior and future message law in theorem C.5. The covariance Σ_t^D is a deterministic function of t, independent of the realized messages, and only the mean carries the history.

Proof. After the fixed singleton labels are suppressed, the preparation makes (Z, D) jointly Gaussian and λ_0 nondegenerate. Completing its quadratic gives $W(d, dz)$ an affine mean in d and a covariance independent of d. Suppose $\lambda_t = \mathcal{N}(m_t, \Sigma_t^D)$ with $\Sigma_t^D \succ 0$. Then $W\lambda_t$ is jointly Gaussian. Conditioning its Y_j coordinate on $D_j = d_j$ gives an affine mean and constant covariance. After fresh sampling and encoding, the message therefore has a representation

$$H_{t+1} = b_t + A_t D + \eta_t, \qquad \eta_t \sim \mathcal{N}(0, R_t), \quad R_t \succ 0, \tag{C.60}$$

where A_t reads only the speaker coordinate and η_t is independent of D at the fixed prefix. The slope A_t and the noise covariance R_t depend on the current common state only through Σ_t^D, while b_t depends on m_t. Multiplying the Gaussian prior density by this likelihood completes a quadratic with

$$\begin{aligned} (\Sigma_{t+1}^D)^{-1} &= (\Sigma_t^D)^{-1} + A_t^T R_t^{-1} A_t, \\ m_{t+1} &= \Sigma_{t+1}^D \left[(\Sigma_t^D)^{-1} m_t + A_t^T R_t^{-1} (h_{t+1} - b_t) \right]. \end{aligned} \tag{C.61}$$

The precision remains positive definite, and its recursion involves no message, so Σ_t^D is history independent. Induction gives the claimed closure for every finite stage; theorem C.5 gives the reconstruction and future-law assertion. □

Equation (C.61) is the information-form measurement update of the Kalman filter [115] for the static state D observed through eq. (C.60), and the history independence of Σ_t^D is the data independence of its Riccati recursion. The retained covariance in this theorem is the joint covariance of (D_0, D_1); replacing it by separate-agent covariance blocks changes the conditional averages in eq. (C.20). In information coordinates the picture is different: since $A_t^T R_t^{-1} A_t$ lives on the speaker's diagonal block, the off-diagonal precision block is fixed model data, as proposition C.6 requires, and only the diagonal precision blocks and the information vector change. The Gaussian family is an exact special case of the general filter, rather than an assumed closure for multiple presentations.

C.7 Failure of finite-mixture closure

Proposition C.11 (A two-presentation nonclosure example). There is a preparation of the form eqs. (C.47) and (C.48), with one receiver presentation, two sender presentations, and strictly positive Gaussian noises, whose receiver cue posterior after one message, at every receiver private value d_0 and every received $h \neq 0$, is not any finite positive mixture of nondegenerate Gaussian laws. At $h = 0$ the posterior is the Gaussian prior.

Proof. Let the sender label be $b \in \{-1, 1\}$ with probabilities $1/2$, receiver cue mean zero, sender cue means $b\alpha$, $\alpha > 0$, common variances $L_0, L_1 > 0$, and cross-covariance $S > 0$ with $S^2 < L_0 L_1$. Choose sender result means bm, $m > 0$, common result variance $Q > 0$, and slope $F_{1b} = 0$. Use the identity encoder with noise variance $R > 0$, and positive private-sensor variances N_0, N_1. The sender's positive-label posterior weight is

$$w_+(d) = \frac{1}{1 + \exp[-2\alpha d/(L_1 + N_1)]}. \tag{C.62}$$

This follows by taking the ratio of the equal-variance Gaussian label evidences. For any received $h > 0$, let $c_\pm = \mathcal{N}(h; \pm m, Q + R)$; for $h < 0$ the roles of c_+ and c_- exchange and the argument is the same, while at $h = 0$ they coincide, the likelihood below is constant, and the posterior equals the prior. Both constants are positive and $c_+ > c_-$, so

$$k(h \mid d) = c_- + (c_+ - c_-)w_+(d). \tag{C.63}$$

Conditional on $X_0 = x$ and b, the sender-private record has mean $b\alpha + (S/L_0)x$ and variance $L_1 - S^2/L_0 + N_1 > 0$. The label weights remain $1/2$ because the receiver cue marginal is identical under both labels. Represent this conditional record as its moving mean plus a fixed centered Gaussian. As $x \to +\infty$ or $x \to -\infty$, the bounded function w_+ tends pointwise to one or zero. Bounded convergence inside eq. (C.53) gives

$$\lim_{x \to +\infty} \ell_0(h \mid x) = c_+, \qquad \lim_{x \to -\infty} \ell_0(h \mid x) = c_-. \tag{C.64}$$

Before the message, $X_0 \mid D_0 = d_0$ has a nondegenerate Gaussian density g_0. Afterwards its density is

$$p^+(x) = \frac{g_0(x)\ell_0(h \mid x)}{Z_h}, \qquad Z_h = \int g_0(x)\ell_0(h \mid x)dx. \tag{C.65}$$

The likelihood lies between c_- and c_+, so Z_h is positive and finite. Consequently p^+/g_0 has two distinct positive finite tail limits.

Suppose instead $p^+ = \sum_{r=1}^{M} a_r \mathcal{N}(\mu_r, v_r)$, with $a_r > 0$ and $v_r > 0$, and write $g_0 = \mathcal{N}(\mu_0, v_0)$. A component with $v_r > v_0$ has density ratio to g_0 diverging at both tails, because its logarithm has positive quadratic coefficient. A component with $v_r = v_0$ and $\mu_r \neq \mu_0$ diverges at one tail, because its logarithmic ratio is nonconstant affine. Positive mixture weights prevent cancellation, so neither type is compatible with finite limits. Every component with $v_r < v_0$ has ratio tending to zero at both tails. The only remaining components with nonzero finite limits are identical to g_0, and their total weight contributes the same limit at both tails. This contradicts eq. (C.64). Zero-weight components can be removed, so the argument covers every finite Gaussian mixture. □

The same example concerns the common state of theorem C.10 as well. The D_0-marginal of λ_1 is $\lambda_0^{D_0}(d_0)P(H = h \mid D_0 = d_0)/P(H = h)$, where $\lambda_0^{D_0} = \mathcal{N}(0, L_0 + N_0)$ is a single Gaussian because the receiver cue mean is zero under both labels, and $P(H = h \mid D_0 = d_0) = \mathbb{E}[\ell_0(h \mid X_0) \mid D_0 = d_0]$ tends to $c_\pm$ as $d_0 \to \pm\infty$ by bounded convergence. The tail argument above therefore shows that this marginal, and hence λ_1, is not a finite positive Gaussian mixture, since marginals of finite Gaussian mixtures are finite Gaussian mixtures. The D_1-marginal of λ_1 is instead an exact two-component mixture, so the nonclosure enters through the cross-agent conditional of D_0 given D_1.

This counterexample concerns closure in a specified finite-mixture family. The exact normalized posterior remains eq. (C.65); its integral representation and the general private-record filter remain valid. Finite Gaussian sums are closed under likelihoods that are themselves finite Gaussian sums in the state [184], and $k(h \mid d) = c_- + (c_+ - c_-)w_+(d)$ is not one. Gaussian prior components, sensors, and encoders alone do not imply a finite Gaussian-mixture posterior when the transmission policy contains posterior-dependent mixing weights.

C.8 Local beliefs do not generally determine the next update

Proposition C.12 (A reachable collision of local-only retention). Enlarge the information fields of eq. (C.5) to $\sigma(\chi, D_i, H_{1:t})$ by a registered public preparation record χ, with policies conditioned on it, in the sense of definition A.1; equivalently, regard the two conditional preparations given χ as two initializations of the declared protocol with the same local banks. There is one fixed finite preparation, such a public record, and a posterior-sampling policy with two positive-probability registered histories having the same pair of complete local recognition laws and private values, but different receiver updates under the same next message. Thus retaining the local joint laws while deleting the distinguishing public record and cross-agent dependence does not define an exact closed assimilation rule.

Proof. Draw X_0 fairly and an independent bit B with $P(B = 0) = \rho$, $0 < \rho < 1$, and set $X_1 = X_0 \oplus B$. Conditional on the cues, generate presentations and results as in eq. (C.36). Each local bank is unchanged. Register the deterministic public preparation readout

$$\chi = X_0 \oplus X_1 = B. \tag{C.66}$$

Let $D_0 = *$ be uninformative and let $P(D_1 = X_1 \mid X_1) = 3/4$, with independent sensor noise. Both parity records occur with positive probability. Given either parity, each individual cue remains fair. The local laws of (A_i, X_i, Y_i) are consequently identical under $\chi = 0$ and $\chi = 1$. The entire local law of (A_1, X_1, Y_1, D_1) is independent of χ, so the sender's local posterior at either fixed value $D_1 = d$ is also identical. Agent 0's local posterior and the actual private values are the same in these two histories as well.

Agent 1 sends a fresh posterior sample H of Y_1. Its probabilities are

$$P(H = 1 \mid D_1 = 0, \chi) = \frac{5}{8}, \qquad P(H = 1 \mid D_1 = 1, \chi) = \frac{3}{8}. \tag{C.67}$$

Integrating the private sensor gives $P(H = 1 \mid X_1 = 0, \chi) = 9/16$ and $P(H = 1 \mid X_1 = 1, \chi) = 7/16$. Thus $P(H = 1 \mid \chi) = 1/2$ for either parity. At the same received $h = 1$, however,

$$P(X_0 = 1 \mid \chi = 0, H = 1) = \frac{7}{16}, \qquad P(X_0 = 1 \mid \chi = 1, H = 1) = \frac{9}{16}. \tag{C.68}$$

For $\chi = 0$ the cues agree; for $\chi = 1$ they are opposite. Applying eq. (C.37) yields distinct presentation and result marginals,

	$\chi = 0$	$\chi = 1$
$P(A_0 = 1 \mid \chi, H = 1)$	$15/32$	$17/32$
$P(Y_0 = 1 \mid \chi, H = 1)$	$17/32$	$15/32.$

(C.69)

Let $\mathcal{T}$ retain the pair of local joint recognition laws, fixed local banks, private values, and schedule, while deleting χ and the corresponding cross-agent dependence. The two histories have equal $\mathcal{T}$. Their exact updates after the same admissible message have different $\mathcal{T}$ by eqs. (C.68) and (C.69). They therefore violate the retained-fiber condition of proposition 6.3, which is necessary for an update defined solely on $\mathcal{T}$. □

Keeping χ resolves this particular collision. In the private-record representation, one may first condition the preparation on χ and retain $W_\chi = P(Z \in \cdot \mid D, \chi)$ together with $P(D \in \cdot \mid \chi, H)$. Alternatively, one may include the preparation record among the retained coordinates. A complete common posterior also preserves its effect. The counterexample proves neither minimality of that posterior nor failure for every fixed initialization; it gives two reachable histories of the specified preparation. It uses exact local label/content laws throughout, so its lost information is cross-agent and historical even before any moment projection.

For comparison, if a local label/result posterior $\Pi_{A,Y}$ has label marginal w, finite second moments with nondegenerate covariance, and Gaussian result moment match q_G, then, whenever the relative entropies are finite,

$$D_{\mathrm{KL}}(\Pi_{A,Y} \| w \otimes q_G) = I_\Pi(A;Y) + D_{\mathrm{KL}}(\Pi_Y \| q_G). \tag{C.70}$$

Insert $w \otimes \Pi_Y$ into the logarithmic density ratio and integrate to obtain the identity. This separates local label/result dependence from non-Gaussian result information, as in proposition B.5. It does not supply the additional private-evidence conditional in eq. (C.14). An approximation to the common state must specify that reconstruction as well. A Gaussian output stored alongside a complete reconstructible transcript is a different retention from one that deletes the distinguishing information.

Open problem C.13 (Smaller sufficient states and changing preparations). For multiple presentations, determine a compression of the two one-agent evidence functions F_0^t, F_1^t of proposition C.6 that preserves every admitted future message law and receiver update, or specify an approximation together with the conditional reconstruction it changes. The exact product-tilt representation does not establish minimality for a given admitted interface or finite descriptive complexity on continuous spaces. Determine also whether posterior-sampling dialogues reveal the private evidence asymptotically, so that both agents' predictions of a common target agree in the limit, and under which encoders and schedules. New private observations, adaptation of the shared model, differing agent models, or unknown transmission policies require their own normalized mechanisms. These extensions, and any inherited recursive formation, are separate from the fixed-preparation communication theorem proved here.

Appendix D

Connections in a Moving Presentation

Links between agents at a single context do not determine transport between neighboring contexts. A connection requires a smooth dependence on the direction and size of a base displacement. This appendix records, in matrix form, the transformation law of connection data supplied in a moving presentation: given a presentation U_i and a one-form a_i, it computes the displayed connection and curvature and their behavior under changes of coordinates. The one-form a_i is supplied for the transformation calculation in this appendix. Chapter 17 separately constructs it from regular contextual laws, interfaces, and links. The remaining selection and continuation questions are open problem D.4. Collective least-squares selection in proposition 3.9 is a separate construction. The context-local and formation flows use it diagnostically. The added energy in chapter 16 instead makes collective alignment enter the law velocity through a contextual force.

Fix one sector and one agent, suppressing the sector superscript. On a coordinate neighborhood of $\mathcal{C}$, let $U_i(c)$ be a twice continuously differentiable invertible presentation and let $a_i = a_{i,\mu}(c)dc^\mu$ be a continuously differentiable matrix-valued one-form. For a principal G-connection its values must lie in the represented Lie algebra. General endomorphism-valued forms instead describe connections for the extended general linear frame bundle.

Definition D.1 (Infinitesimal directional transport). The proposed transport family has first-order behavior

$$\Theta_i(c+\delta c, c) = U_i(c+\delta c)\exp[-a_{i,\mu}(c)\delta c^\mu]U_i(c)^{-1} + O(|\delta c|^2). \tag{D.1}$$

The first-order generator is linear in the tangent displacement. A single finite matrix at c supplies neither this linear dependence nor the identity value at zero displacement.

Proposition D.2 (Connection and curvature). With the convention $D_i = d + B_i$, eq. (D.1) has connection and curvature

$$B_i = U_i a_i U_i^{-1} - dU_i U_i^{-1}, \qquad F_i = U_i(da_i + a_i \wedge a_i)U_i^{-1}. \tag{D.2}$$

Proof. Expanding the two factors at c gives $\Theta_i = I + (dU_iU_i^{-1} - U_ia_iU_i^{-1})(\delta c) + O(|\delta c|^2)$, which identifies B_i. Acting on a vector-valued function gives the operator identity $d + B_i = U_i(d + a_i)U_i^{-1}$. Squaring it yields $(d + B_i)^2 = U_i(d + a_i)^2U_i^{-1}$, and the squares act by the curvature two-forms in eq. (D.2). □

For an explicit curved example, take $U_i = I$ over an open subset of $\mathbb{R}^2$ and set

$$a_{i,x} = E_{12}, \qquad a_{i,y} = E_{21}. \tag{D.3}$$

Both are supplied rank-one receiving–sending products, while their derivatives vanish. Nevertheless,

$$F_{i,xy} = [E_{12}, E_{21}] = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \neq 0. \tag{D.4}$$

Thus even constant directional interfaces can give curvature through noncommutativity. The example is supplied. Its curvature has the real eigenvalues ± 1, and the spectrum of a curvature value is unchanged by every change of frame. A connection that preserves a fiber metric M, $dM = Ma + a^T M$, has curvature skew for M and therefore purely imaginary spectrum, so this example is not produced by any selector whose output preserves a fiber metric, such as the rectangular selector of chapter 17 with its induced metric eq. (17.25). By contrast, replacing a finite pointwise map by a new frame $V(c)$ and setting $B = -dVV^{-1}$ gives zero curvature: it is the case $a = 0$ of eq. (D.2).

Finite paths use the solution of

$$\frac{d}{dt}T_i(t) = -B_i(\dot{\gamma}(t))T_i(t), \qquad T_i(0) = I. \tag{D.5}$$

Uniqueness of this linear equation gives composition under concatenation and inversion under path reversal. The solution is the path-ordered exponential; the usual link and Wilson-line constructions use this ordering [187]. Independently declared opposite attending edges are not two orientations of this same path and need not be inverses.

Proposition D.3 (Compatibility under re-expression). Under an agent-coordinate change $U_i' = g_i U_i$ with the same reference form,

$$B_i' = g_i B_i g_i^{-1} - dg_i g_i^{-1}. \tag{D.6}$$

Under a change of reference presentation $U_i' = U_i h$, which sends reference laws to their h^{-1} pushforwards and is the inverse of a target change written $p' = h \cdot p$, keeping the same intrinsic transport instead requires

$$a_i' = h^{-1} a_i h + h^{-1} dh. \tag{D.7}$$

Proof. Insert $g_i U_i$ into eq. (D.2) and apply the product rule. Inserting $U_i h$ and eq. (D.7) cancels the two terms containing dh, leaving the original B_i. □

The same affine law must hold between overlapping coordinate patches. A directional interface product transforming only by conjugation is a tensorial one-form. It can describe the difference between two connections, but it is not by itself the full affine connection data. Constructing displacement links with the endpoint transformation law and then differentiating them is one way to obtain the required inhomogeneous term.

If patch coordinates satisfy $v_\alpha = g_{\alpha\beta} v_\beta$, their transition functions must obey $g_{\alpha\beta} g_{\beta\gamma} = g_{\alpha\gamma}$ on triple overlaps. The displayed affine law then makes the local connections compatible. A nontrivial bundle need not admit a single global presentation U_i. Even on a trivial bundle, zero curvature does not imply path independence: the scalar connection $B = b d\theta$ on the circle is flat but has holonomy

$e^{-2\pi b}$, which differs from one for real $b \neq 0$. Global path independence requires trivial holonomy as well as local flatness.

Different agents may be assigned different connection data in their own presentations. Since all agents are sections of bundles associated to one principal bundle P, and in the linear realization of one associated vector bundle, these data define one connection on P only when, after re-expression in a common frame, their local forms agree and satisfy its overlap relations; otherwise they are several connections on the same bundle, each used by one agent. Neither case requires an averaging construction.

Open problem D.4 (Selecting and continuing interaction-derived transport). The full-rank construction in theorems 17.3 and 17.7 supplies displacement identity, a linear first jet, compatible endpoint laws, and path transport from contextual law and link messages. The regular alignment construction in proposition 17.11 gives another selector, and its reciprocal-link example has nonzero curvature. Both require a declared comparison rule. The directed backtrack sector can enter the data and is not identified with the reversal of a contextual path.

The remaining problem is to select the message labels, column weights, or augmented comparison metric from a stated agent model, and to establish continuation when rank or infinitesimal identifiability fails. Link-only data cannot identify a continuous gauge subgroup that fixes every input, as proposition 17.12 proves. A selector also needs new dynamical analysis before it can replace the law-Fisher auxiliary field in chapter 16: its full constitutive derivative must enter the energy variation. Neither construction derives the contextual base, its topology, or its dimension.

Appendix E

A Conditional Determinant Geometry

The collective construction of proposition 3.9 does not require a Lorentzian target. This appendix gives one additional choice for a complex rank-two representation. Its indefinite form is supplied by a determinant; the Fisher metric used to choose the connection remains positive.

Let $W \to \mathcal{U}$ be a complex rank-two vector bundle with structure group $\mathrm{GL}(2,\mathbb{C})$. In a frame, identify the real rank-four bundle $\mathrm{Herm}(W)$ with Hermitian matrices transforming by $H' = gHg^\dagger$. Suppose the collective target laws p_i on W have finite second moments and that their moment maps are smooth on the admitted regular families. With the same declared positive weights as in the collective metric, set

$$M = \mathcal{M}(\boldsymbol{p}) = \sum_i w_i \int zz^\dagger p_i(dz), \qquad e(X) = d\mathcal{M}_{\boldsymbol{p}} r_X. \tag{E.1}$$

Equivariance of the moment map gives

$$e = dM + A^F M + M(A^F)^\dagger, \qquad e' = geg^\dagger. \tag{E.2}$$

The first expression is independent of the stabilizer-valued representative of A^F by proposition 3.13.

Proposition E.1 (Determinant metric). Assume $\mathcal{U}$ is four dimensional and the actual one-form $e : T\mathcal{U} \to \mathrm{Herm}(W)$ is a bundle isomorphism. Supply a smooth positive determinant-scale section $\kappa_{\det}$ with local transformation $\kappa'_{\det} = |\det g|^2 \kappa_{\det}$. Then

$$\eta_{\det}(H,K) = \tfrac{1}{2}[\mathrm{tr}(HK) - \mathrm{tr}(H)\,\mathrm{tr}(K)], \qquad \gamma(X,Y) = \kappa_{\det}^{-1}\eta_{\det}(eX,eY). \tag{E.3}$$

defines a gauge-invariant metric with one negative and three positive directions. The positive-definite Hermitian cone determines a time orientation through e.

Proof. For a Hermitian matrix $H = \begin{pmatrix} t+z & x-iy \\ x+iy & t-z \end{pmatrix}$, direct evaluation gives

$$\eta_{\det}(H,H) = -\det H = -t^2 + x^2 + y^2 + z^2. \tag{E.4}$$

Multiplicativity of the determinant under congruence gives $-\det(gHg^\dagger) = |\det g|^2(-\det H)$. Polarizing proves the same scaling law for $\eta_{\det}(H,K)$. The compensator cancels that factor in eq. (E.3), and the isomorphism e preserves its inertia. Positive Hermitian matrices have $t > 0$ and $t^2 > x^2 + y^2 + z^2$, so they form one timelike cone, preserved by congruence. □

The positive determinant scale has a canonical choice built from the same data. Where M is positive definite, $\det(gMg^\dagger) = |\det g|^2 \det M$, so $\kappa_{\det} = \det M$ obeys the required transformation

law. For 2×2 matrices $\operatorname{tr}(B^2) = (\operatorname{tr} B)^2 - 2 \det B$; applying this to $B = M^{-1}H$ and polarizing gives

$$\frac{\eta_{\det}(H,K)}{\det M} = \tfrac{1}{2}\big[\operatorname{tr}(M^{-1}HM^{-1}K) - \operatorname{tr}(M^{-1}H)\operatorname{tr}(M^{-1}K)\big]. \tag{E.5}$$

With this scale,

$$\gamma = \tfrac{1}{2}\big(h^{\mathcal{CN}} - \vartheta \otimes \vartheta\big), \qquad h^{\mathcal{CN}}(X,Y) = \operatorname{tr}\big(M^{-1}e(X)M^{-1}e(Y)\big), \qquad \vartheta(X) = \operatorname{tr}\big(M^{-1}e(X)\big). \tag{E.6}$$

Here $h^{\mathcal{CN}}$ is the Fisher metric of the proper complex Gaussian law with covariance M, pulled back by e, and $\vartheta = d\log\det M + 2\operatorname{Re}\operatorname{tr} A^F$ is the covariant derivative of its entropy $\log\det(\pi e M)$. The indefinite signature is a rank-one subtraction from a Fisher-type metric: a direction is timelike exactly when the squared entropy rate exceeds the squared Fisher speed, and the dilation $e = cM$ has $\gamma = -c^2$. The form is invariant under constant rescaling of M, so any physical scale remains a calibration. For non-Gaussian targets $h^{\mathcal{CN}}$ is the Fisher metric of the moment projection, not of the actual laws. The scale also changes the geometry. For an exact soldering $e = dM$ with constant $\kappa_{\det}$, γ is the pullback of a flat Minkowski form. With $\kappa_{\det} = \det M$ it is the pullback of the $\mathrm{GL}(2,\mathbb{C})$-invariant form $-\det(dM)/\det M$ on the positive Hermitian cone, which is a Lorentzian product of a line with hyperbolic three-space and has scalar curvature -6.

The rank hypothesis is imposed on the actual contextual first jet. It constrains the number and configuration of the target laws.

Proposition E.2 (Rank of the aggregate jet for proper complex Gaussian targets). Let p_i be centered proper complex Gaussian laws with positive-definite covariances M_i, $i = 1, \dots, n$, with the collective metric $\sum_i w_i \operatorname{tr}(M_i^{-1}h_i M_i^{-1}k_i)$ and the action $M \mapsto gMg^\dagger$. Then

$$H = \Big\{(h_i)_i : \sum_i w_i M_i^{-1} h_i = 0\Big\}, \qquad d\mathcal{M}(h) = \sum_i w_i h_i. \tag{E.7}$$

For $n = 1$, $H = 0$. For $n = 3$ at triples whose orbit has the maximal dimension seven, the rank of $d\mathcal{M}|_H$ is at most three, so e cannot be an isomorphism on a four-dimensional base there. For $n = 4$, at $M_1 = I$, $M_2 = 2I$, $M_3 = \operatorname{diag}(1,2)$, $M_4 = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ with unit weights, the orbit has dimension seven and $d\mathcal{M}|_H$ is surjective.

Proof. An orbit direction is $(\xi M_i + M_i \xi^\dagger)_i$ for $\xi \in \mathfrak{gl}(2,\mathbb{C})$. By cyclicity of the trace its pairing with $(h_i)_i$ is $2\operatorname{Re}\operatorname{tr}(B\xi)$ with $B = \sum_i w_i h_i M_i^{-1}$. This vanishes for all ξ exactly when $B = 0$, equivalently when $B^\dagger = \sum_i w_i M_i^{-1} h_i = 0$; this proves eq. (E.7). For $n = 1$ it forces $h_1 = 0$. Put $k_i = w_i h_i$. For $n = 3$, $\dim H = 12 - 7 = 5$, and the rank of $d\mathcal{M}|_H$ is 5 minus the dimension of $\{k : \sum_i k_i = 0,\ \sum_i M_i^{-1} k_i = 0\}$. Eliminating k_3 leaves $\mathsf{A}k_1 + \mathsf{B}k_2 = 0$ with $\mathsf{A} = M_1^{-1} - M_3^{-1}$ and $\mathsf{B} = M_2^{-1} - M_3^{-1}$. When A, B are invertible, $(k_1, k_2) = (\mathsf{A}^{-1}, -\mathsf{B}^{-1})$ and $(\mathsf{A}^{-1}\mathsf{B}\mathsf{A}^{-1}, -\mathsf{A}^{-1})$ are Hermitian solutions, independent unless B is a multiple of A. So the rank is at most three on the complement of a proper real-algebraic subset, which is dense in the open set of triples with seven-dimensional orbits. The rank of a smoothly varying linear map is lower semicontinuous, so the set where it is at least four is open; being disjoint from a dense set, it is empty. For the four-agent tuple, a unitary g fixing $\operatorname{diag}(1,2)$ is diagonal, and a diagonal unitary fixing M_4 is scalar,

so the stabilizer is the central circle and the orbit has dimension seven. Given a Hermitian S, the choice $k_1 = -S$, $k_2 = 2S$, $k_3 = k_4 = 0$ satisfies $\sum_i M_i^{-1} k_i = 0$ and $\sum_i k_i = S$. □

For $n \leq 3$ centered proper complex Gaussian targets, no section over an open four-dimensional base has an aggregate jet e that is a continuous bundle isomorphism. The stabilizer of a tuple is conjugate to the unitaries commuting with every $M_1^{-1/2} M_j M_1^{-1/2}$, so the orbit dimension is seven, six, or four according as these Hermitian matrices fail to commute, commute without all being scalar, or are all scalar. The orbit dimension along a section is lower semicontinuous, hence locally constant on an open dense set, and on each component of that set the section stays in one orbit-type stratum. On the maximal stratum the proposition gives rank $e \leq 3$. On the jointly diagonalizable stratum, in a frame with $M_1 = I$ and every M_j diagonal, the non-orbit tangent directions are diagonal tuples, and their horizontal projections are diagonal because diagonal and off-diagonal Hermitian matrices are orthogonal for the collective metric at diagonal base points; so e takes diagonal values and has rank at most two. On the proportional stratum it takes values in the multiples of M and has rank at most one, and for $n = 1$, $H = 0$. Thus $\det e = 0$ on a dense open set and, by continuity, everywhere. At an isolated tuple of smaller orbit dimension $d\mathcal{M}|_H$ can be onto, for example for two agents with covariances M and $2M$, whose orbit and horizontal space are four dimensional; but a section realizing a full-rank first jet there leaves the stratum, and its e is discontinuous at that point. Four agents are the minimum in this family: at the four-agent tuple of the proposition a section whose residuals span horizontal preimages of a basis gives an invertible e at that point, and hence on a neighborhood. For improper complex laws the central phase acts nontrivially on the pseudo-covariance $\mathbb{E}[zz^T]$, which transforms as gPg^T, so the normal equation can then determine the phase component of the connection; we do not develop that case. Surjectivity of the horizontal differential of $\mathcal{M}$ establishes the availability of such a jet at a law tuple, but does not select a section or prove persistence under inference. If the base has dimension greater than four and e is surjective, the same formula gives a nondegenerate form on the quotient vector bundle by $\ker e$. A quotient-manifold interpretation still requires a regular leaf quotient and basicness, as in proposition 3.6.

Proposition E.3 (Circle completion). Let $\mathscr{Q}_{\det} \to \mathcal{U}$ be the unit circle bundle of $\det W$ for the line metric whose local coefficient is $\kappa_{\det}^{-1}$. Write $\pi_{\det}$ for its projection. Supply a real principal circle connection $\omega_{\det}$ normalized to one on the vertical generator of period 2π, and a smooth positive radius $r_{\det} : \mathcal{U} \to (0, \infty)$. Then

$$G_{\mathrm{KK}} = \pi_{\det}^* \gamma + (r_{\det} \circ \pi_{\det})^2 \omega_{\det} \otimes \omega_{\det} \tag{E.8}$$

is circle invariant and has one negative and four positive directions.

Proof. The connection splits each total tangent space into its horizontal subspace and the one-dimensional vertical subspace. On the first, eq. (E.8) is the pullback of γ; on the second, it is positive because $r_{\det} > 0$. The two subspaces are orthogonal. The base pullback, radius, and principal connection are invariant under the circle action, proving invariance of the total metric. □

A supplied global $\mathrm{GL}(2, \mathbb{C})$ connection A gives the local expression $\omega_{\det} = d\theta_{\det} + \operatorname{Im} \operatorname{tr} A$ on this circle bundle. The coordinate transformation $\theta'_{\det} = \theta_{\det} + \arg \det g$ cancels $\operatorname{Im} \operatorname{tr} A' = \operatorname{Im} \operatorname{tr} A - d \arg \det g$. When the collective minimization determines only a class, its lift to a full connection is

further data. For instance, central phase rotations act trivially on centered proper complex Gaussian laws and their covariance matrices, so that family cannot fix the corresponding phase connection from its Fisher normal equations. Neither the positive circle radius nor equations of motion for γ follow from the construction. The determinant and circle choices give this conditional geometric example without identifying it with physical spacetime dynamics.

Appendix F

Notation

Symbol	Meaning
$\mathcal{C}, c_*$	Context base and a selected context.
P, G	Principal bundle and its structure group, one group acting on both sectors.
ϱ_q, ϱ_s	Representations of G on the belief space V_q (dimension K) and the model space V_s (dimension $\tilde{K}$).
$\mathcal{E}_q, \mathcal{E}_s$	Associated content bundles $P \times_{\varrho_q} V_q$ and $P \times_{\varrho_s} V_s$.
$\Phi, \tilde{\Phi}$	Belief and model content-bundle morphisms induced by one principal bundle map.
$\theta_{ij}, \tilde{\theta}_{ij}$	Group-valued belief and model links, with $\Theta^q_{ij} = \varrho_q(\theta_{ij})$ and $\Theta^s_{ij} = \varrho_s(\tilde{\theta}_{ij})$; tied when equal.
$\omega_{ij}, \tilde{\omega}_{ij}$	Their generators in the Lie algebra $\mathfrak{g}$.
Ξ_{ij}	Sector mismatch $\tilde{\theta}_{ij}\theta_{ij}^{-1}$, covariant under conjugation at the receiver.
$\mathsf{K}_i, \mathsf{M}_i, \mathsf{Y}_i$	Represented state contents, model presentations, and $\mathsf{Y}_i = \mathsf{K}_i \times \mathsf{M}_i$ of agent i.
$\mathcal{B}_i, E_i$	Statistical bundle $P \times_G \mathcal{P}(\mathsf{Y}_i)$ and sample bundle $P \times_G \mathsf{Y}_i$.
$\mathcal{P}(X)$	Probability measures on a measurable space X.
ζ_i, q_i, s_i	Joint agent law (recognition law) and its belief and model marginals.
r_i, G_i, ℓ_i	Presentation prior, evaluator, and record kernel of agent i's generative model. The term reference law is reserved for the laws $\bar{q}_i, \bar{s}_i$ in reference coordinates.
$l_i, \lambda_{\mathsf{O}_i}$	Declared record density of ℓ_i and the declared sigma-finite reference measure on O_i.
$K_i^{(n)}$	Composite kernel of n records given a presentation, with $K_i^{(1)} = K_i^{\mathrm{rec}}$.
$\mathbf{g}$	A passive change, the tuple $(g_i)_i$ of elements of G acting on the agents' frames.
$\mathfrak{A}_{\mathrm{pop}}, \mathfrak{A}_{\mathrm{prot}}$	Population and protocol observable algebras. A distinction is gauge relative to an algebra when no element separates it at any time along the declared dynamics.
$P_i, \mathcal{F}_i$	Normalized generative model of agent i and its single-agent or per-agent free energy.
$R_i, \mathcal{E}_0$	Reference joint $r_i G_i$ and the sum of local terms $D_{\mathrm{KL}}(q_i \otimes s_i \| R_i)$.
ϕ, ψ	Presentation and content maps relating two agents that share a generative model.
Z, P_0, Q	Coefficient space, prior, and recognition law of a content law on fields.
f_i, F_D, Q_D	Sample-section maps, a finite-design evaluation map, and its law.
L_o, Π_o, Z_o	Likelihood, posterior, and evidence of a fixed normalized model at record value o.
g^F, ℓ_v	Fisher metric of an admitted regular family and the score in tangent direction v.
$D^\omega\zeta, h^\omega_\zeta$	Covariant statistical derivative and its Fisher pullback on the context base.
$J_{\boldsymbol{p}}, H_{\boldsymbol{p}}$	Infinitesimal collective group action and its Fisher-orthogonal complement.
$\mathfrak{k}, A^F$	Collective orbit metric and a representative of the minimizing connection class.
$\theta^{\mathrm{mech}}, \Omega^{\mathrm{mech}}$	Mechanical connection of the declared comparison data and its curvature.
u_i, σ, U_i	Frame of agent i in P_{c_*}, a reference point, and U_i with $u_i = \sigma U_i^{-1}$.
V_a, E_i^a	Reference vector space of sector $a \in \{q, s\}$ and agent i's coordinate copy of it.
U_i^q, U_i^s	Frame representatives in the two sectors.
$\bar{q}_i, \bar{s}_i$	Laws in common reference coordinates on a trivializing patch.
L_{ij}^q, L_{ij}^s	Law-dependent interaction generators in the vector realization.

Symbol	Meaning
$\Theta^q_{ij}, \Theta^s_{ij}$	Invertible source-to-receiver interaction maps.
$N_{ij,j}$	Backtrack loop $\Theta_{ji}\Theta_{ij}$ at j of a directed pair.
$\mathcal{K}^q_{ij}, \mathcal{K}^s_{ij}$	Normalized source-to-receiver comparison channels.
ν^q_{ij}, D^q_{ij}	Compared law $q_j\mathcal{K}^q_{ij}$ and peer divergence $D_{\mathrm{KL}}(q_i\|\nu^q_{ij})$; model sector analogous.
T^a_j, O^a_i	Sending and receiving maps in sector $a \in \{q, s\}$.
μ_j	Message law $(T_j)_{\#}q_j$.
β_i, γ_i	Belief-sector and model-sector source rows.
$\pi^q_i, \pi^s_i, T_q, T_s$	Source priors and row temperatures.
H_i	State-to-record kernel.
K^{rec}_i	Normalized model-to-record predictive evaluator.
S, x	Complete agent-configuration space and a configuration.
O_i, ρ_i	Record alphabet and internal record readout.
$O_{ij,t}, o_{ij,t}$	Record random variable and one realized value.
$\Gamma_{ij}, \mathcal{R}_{ij}$	Interaction transition on configurations and the record kernel it induces.
$\mathcal{E}, \mathcal{E}_{\mathrm{red}}$	Reciprocal law potential and its row-reduced form.
$\mathcal{E}_o$	Evidence-augmented reciprocal potential.
$\mathcal{F}$	Variational free energy for a specified normalized probabilistic model.
Φ	The flow of a construction, on its declared chart or invariant family.
$\Phi^0_{\mathrm{lp}}, \Phi^0_{\mathrm{rn}}$	Label-preserving and renormalized null flows: every peer cost is deleted, with the peer labels and priors kept, or with the deleted labels removed so that a self source has weight one.
β^0_{ii}	Self weight of a Gibbs row along the label-preserving null, below one when the self prior is below one.
Φ^{fr}	Frozen-environment control: nothing is deleted, and the laws of the other agents are held at their initial values in every force.
$H(x), P(x)$	Membership history with event times and the partition sequence without them, modulo exchange of consecutive merges of disjoint agent sets; $H^0_{\mathrm{lp}}, P^0_{\mathrm{lp}}, H^0_{\mathrm{rn}}, P^0_{\mathrm{rn}}$, and $H^{\mathrm{fr}}, P^{\mathrm{fr}}$ along the controls.
S_c, C	A declared selector family and its constant set, fixed before the flows are analyzed.
A_I, B_I, R_I	In the formation standard, the parent protocol class, the child-level probe class, and the retention of a block I.
$\Omega^{\mathrm{reach}}_I$	Reachable post-handoff set of a block I.
$\mathcal{C}$,$\mathcal{U}$	In the counterfactual clause, the declared counterfactual handoff class and the class of initial configurations that generates it.
(F1)–(F6)	Clauses of the formation standard; (F5) is interaction necessity, $P \neq P^0$ pointwise on a nonempty open admitted set, against both nulls and for every constant in C.
$W_2(\nu, \mathsf{P})$	Matrix Wishart law with shape $\nu > 1/2$ and precision P in the conditional formation example.
n_I, A_I, B_I	Primitive count and belief/model shape sums of a retained block in that example.
β_i	In that example, the self weight of agent i along the label-preserving null.
Λ_q, C_q	Moment-dependent comparison precision and its complete force coefficient; the model sector uses Λ_s, C_s.
B^q_{IJ}, B^s_{IJ}	Exact receiver-averaged, source-summed native rows of current blocks in the open-law construction.
H_I, ϕ_I	Discarded constant relative rotations and the retained local mean anchor phase.
F, R, F^c	Fine deterministic update, retention map, and inherited update, with $R \circ F = F^c \circ R$.
a_i, B_i, F_i	Reference connection form, its local representative, and curvature.
X, U, Z, Y	Source, encoded representation, decoder information, and prediction target in the coding construction.
$\mathcal{T}, \Delta$	Encoder kernel and predictive decoder in the coding construction.

Symbol	Meaning
α, α_+	Dirichlet parameter of the record-process example and its total concentration.
$\bar{\sigma}$	Magnetization of the Ising example, solving $\bar{\sigma} = \tanh(J\bar{\sigma})$.
λ_t, F_0^t, F_1^t	Joint private-record posterior after t messages and its product-tilt factors, $\lambda_t \propto F_0^t F_1^t \lambda_0$.
Σ_t^D	Covariance of the private records in the Gaussian closure of the interacting protocol.
$\ell_\psi, \mathcal{X}_{\mathrm{rich}}$	Score of the common ψ rotation and the rich preparation class of the collective construction.
$\mathcal{R}, \mathcal{R}_{\mathrm{glob}}$	Regular critical set of the finite contextual flow and its subset on which every transport is the unique global minimizer.
X, H_X, Q_X, P_X	Collective message matrix, $H_X = XX^T$, the right inverse $Q_X = X^T H_X^{-1}$, and the projector $P_X = Q_X X$; H_X^{-1} is the induced fiber metric.
$C(c', c), T(c', c)$	Cross-context map $X(c')Q_X(c)$ and the direct rotation between message row spaces.
$\bar{\Theta}_{ij}, R_{ij}$	A link and its reciprocal part in reference coordinates, $U_i^{-1}\Theta_{ij}U_j$ and $U_i^{-1}\Omega_{ij}U_j$.
$J_Z, \mathcal{H}$	Infinitesimal action on contextual link data and the locked form of the alignment selector.
$\bar{T}, \bar{\theta}, \hat{A}$	Code and dictionary in reference coordinates and the effective code $\hat{A} = \bar{T}\bar{\theta}^{-1}$ that messages identify.
O_0, Q_D	Noiseless gain of the Gaussian decoder and the projection onto the directions that a code reads.

The symbol F for a discrete update and F_i for curvature are used in separate constructions, and the calligraphic $\mathcal{F}_i$ denotes a free energy. The letter B denotes a connection representative B_i, block shape sums B_I in the Wishart construction, the child-level probe class B_I of the formation standard, and native rows B_{IJ}^q, each in its own construction; A_I is likewise a shape sum in the Wishart construction and a protocol class in the standard, and $\mathcal{C}$ denotes a counterfactual class only inside the counterfactual clause. The symbol ν denotes a compared law ν_{ij} in the dynamics, a sector marginal ν_i^a in the interaction chapter, and a Wishart shape. A frame U_i^a is an invertible map; the coding variable U is a random representation. The letter Γ denotes an interaction transition Γ_{ij} and, in the causal-memory obstruction of the collective construction, the evolution map of a memory. A Markov kernel returns a probability measure, whereas an ordinary map returns one value. The symbol $\circ$ composes maps and juxtaposition composes kernels, in the order in which a law traverses them.

Bibliography

[1] Ali E. Abbas. A Kullback-Leibler view of linear and log-linear pools. *Decision Analysis*, 6(1):25–37, 2009. doi: 10.1287/deca.1080.0133.

[2] P.-A. Absil, R. Mahony, and B. Andrews. Convergence of the iterates of descent methods for analytic cost functions. *SIAM Journal on Optimization*, 16(2):531–547, 2005.

[3] Shun-ichi Amari. Natural gradient works efficiently in learning. *Neural Computation*, 10(2):251–276, 1998. doi: 10.1162/089976698300017746.

[4] Shun-ichi Amari and Hiroshi Nagaoka. *Methods of Information Geometry*, volume 191 of *Translations of Mathematical Monographs*. American Mathematical Society and Oxford University Press, Providence, RI, 2000.

[5] Luigi Ambrosio, Nicola Gigli, and Giuseppe Savaré. *Gradient Flows in Metric Spaces and in the Space of Probability Measures*. Lectures in Mathematics ETH Zürich. Birkhäuser, Basel, 2 edition, 2008.

[6] Chainarong Amornbunchornvej. Interpretation as linear transformation: A cognitive-geometric model of concepts and meaning. *Minds and Machines*, 36(3):36, 2026. doi: 10.1007/s11023-026-09789-6.

[7] Karl Johan Åström. Optimal control of Markov processes with incomplete state information. *Journal of Mathematical Analysis and Applications*, 10(1):174–205, 1965. doi: 10.1016/0022-247X(65)90154-X.

[8] Robert J. Aumann. Agreeing to disagree. *The Annals of Statistics*, 4(6):1236–1239, 1976. doi: 10.1214/aos/1176343654.

[9] Nihat Ay, Jürgen Jost, Hông Vân Lê, and Lorenz Schwachhöfer. Information geometry and sufficient statistics. *Probability Theory and Related Fields*, 162:327–364, 2015. doi: 10.1007/s00440-014-0574-8.

[10] Nihat Ay, Jürgen Jost, Hông Vân Lê, and Lorenz Schwachhöfer. Parametrized measure models. *Bernoulli*, 24(3):1692–1725, 2018. doi: 10.3150/16-BEJ910. URL https://arxiv.org/abs/1510.07305.

[11] Paul Baird and John C. Wood. *Harmonic Morphisms Between Riemannian Manifolds*, volume 29 of *London Mathematical Society Monographs, New Series*. Oxford University Press, Oxford, 2003.

[12] Afonso S. Bandeira, Amit Singer, and Daniel A. Spielman. A Cheeger inequality for the graph connection Laplacian. *SIAM Journal on Matrix Analysis and Applications*, 34(4):1611–1630, 2013.

[13] Masako Bando, Taichiro Kugo, and Koichi Yamawaki. Nonlinear realization and hidden local symmetries. *Physics Reports*, 164(4–5):217–314, 1988. doi: 10.1016/0370-1573(88)90019-1.

[14] Ziv Bar-Yossef, T. S. Jayram, Ravi Kumar, and D. Sivakumar. An information statistics approach to data stream and communication complexity. *Journal of Computer and System Sciences*, 68(4):702–732, 2004. doi: 10.1016/j.jcss.2003.11.006.

[15] Lionel Barnett and Anil K. Seth. Dynamical independence: Discovering emergent macroscopic processes in complex dynamical systems. *Physical Review E*, 108:014304, 2023. doi: 10.1103/PhysRevE.108.014304.

[16] Nix Barnett and James P. Crutchfield. Computational mechanics of input–output processes: Structured transformations and the ϵ-transducer. *Journal of Statistical Physics*, 161(2):404–451, 2015. doi: 10.1007/s10955-015-1327-5.

[17] Stephen D. Bartlett, Terry Rudolph, and Robert W. Spekkens. Reference frames, superselection rules, and quantum information. *Reviews of Modern Physics*, 79(2):555–609, 2007. doi: 10.1103/RevModPhys.79.555.

[18] Peter W. Bates, Kening Lu, and Chongchun Zeng. Invariant foliations near normally hyperbolic invariant manifolds for semiflows. *Transactions of the American Mathematical Society*, 352(10):4641–4676, 2000. doi: 10.1090/S0002-9947-00-02503-4.

[19] Giorgio Battistelli and Luigi Chisci. Kullback–Leibler average, consensus on probability densities, and distributed state estimation with guaranteed stability. *Automatica*, 50(3):707–718, 2014.

[20] Leonard E. Baum and Ted Petrie. Statistical inference for probabilistic functions of finite state Markov chains. *The Annals of Mathematical Statistics*, 37(6):1554–1563, 1966. doi: 10.1214/aoms/1177699147.

[21] Robert H. Berk. Limiting behavior of posterior distributions when the model is incorrect. *Annals of Mathematical Statistics*, 37(1):51–58, 1966.

[22] Wolf-Jürgen Beyn and Vera Thümmler. Freezing solutions of equivariant evolution equations. *SIAM Journal on Applied Dynamical Systems*, 3(2):85–116, 2004. doi: 10.1137/030600515.

[23] Christopher M. Bishop. *Pattern Recognition and Machine Learning*. Springer, New York, 2006.

[24] Pier Giovanni Bissiri, Chris C. Holmes, and Stephen G. Walker. A general framework for updating belief distributions. *Journal of the Royal Statistical Society: Series B*, 78(5):1103–1130, 2016.

[25] Anastasia Bizyaeva, Alessio Franci, and Naomi Ehrich Leonard. Nonlinear opinion dynamics with tunable sensitivity. *IEEE Transactions on Automatic Control*, 68(3):1415–1430, 2023. doi: 10.1109/TAC.2022.3159527.

[26] Åke Björck and Gene H. Golub. Numerical methods for computing angles between linear subspaces. *Mathematics of Computation*, 27(123):579–594, 1973. doi: 10.1090/S0025-5718-1973-0348991-3.

[27] David Blackwell. Equivalent comparisons of experiments. *The Annals of Mathematical Statistics*, 24 (2):265–272, 1953. doi: 10.1214/aoms/1177729032.

[28] David M. Blei, Alp Kucukelbir, and Jon D. McAuliffe. Variational inference: a review for statisticians. *Journal of the American Statistical Association*, 112(518):859–877, 2017.

[29] Sergey G. Bobkov and Prasad Tetali. Modified logarithmic Sobolev inequalities in discrete settings. *Journal of Theoretical Probability*, 19(2):289–336, 2006. doi: 10.1007/s10959-006-0016-3.

[30] Giovanni A. Bonaschi, José A. Carrillo, Marco Di Francesco, and Mark A. Peletier. Equivalence of gradient flows and entropy solutions for singular nonlocal interaction equations in 1D. *ESAIM: Control, Optimisation and Calculus of Variations*, 21(2):414–441, 2015. doi: 10.1051/cocv/2014032.

[31] Haïm Brézis. *Opérateurs Maximaux Monotones et Semi-Groupes de Contractions dans les Espaces de Hilbert*, volume 5 of *North-Holland Mathematics Studies*. North-Holland, Amsterdam, 1973.

[32] Peter Buchholz. Exact and ordinary lumpability in finite Markov chains. *Journal of Applied Probability*, 31(1):59–75, 1994. doi: 10.2307/3215235.

[33] Nazmi Burak Budanur, Predrag Cvitanović, Ruslan L. Davidchack, and Evangelos Siminos. Reduction of SO(2) symmetry for spatially extended dynamical systems. *Physical Review Letters*, 114:084102, 2015. doi: 10.1103/PhysRevLett.114.084102.

[34] C. J. Burke and M. Rosenblatt. A Markovian function of a Markov chain. *The Annals of Mathematical Statistics*, 29(4):1112–1122, 1958. doi: 10.1214/aoms/1177706444.

[35] Sylvain Calinon. A tutorial on task-parameterized movement learning and retrieval. *Intelligent Service Robotics*, 9(1):1–29, 2016. doi: 10.1007/s11370-015-0187-9. URL https://calinon.ch/papers/Calinon-JIST2015.pdf.

[36] Miguel Á. Carreira-Perpiñán. Fast nonparametric clustering with Gaussian blurring mean-shift. In *Proceedings of the 23rd International Conference on Machine Learning*, pages 153–160, 2006. doi: 10.1145/1143844.1143864.

[37] José A. Carrillo, Marco Di Francesco, Alessio Figalli, Thomas Laurent, and Dejan Slepčev. Global-in-time weak measure solutions and finite-time aggregation for nonlocal interaction equations. *Duke Mathematical Journal*, 156:229–271, 2011. doi: 10.1215/00127094-2010-211.

[38] Jonas Cassel, Bastian Boll, Stefania Petra, Peter Albers, and Christoph Schnörr. Sigma flows for image and data labeling and learning structured prediction. *Journal of Mathematical Imaging and Vision*, 67(6):54, 2025. doi: 10.1007/s10851-025-01270-w.

[39] N. N. Čencov. *Statistical Decision Rules and Optimal Inference*, volume 53 of *Translations of Mathematical Monographs*. American Mathematical Society, Providence, RI, 1982.

[40] Chung Chan, Ali Al-Bashabsheh, Qiaoqiao Zhou, Tarik Kaced, and Tie Liu. Info-clustering: A mathematical theory for data clustering. *IEEE Transactions on Molecular, Biological and Multi-Scale Communications*, 2(1):64–91, 2016. doi: 10.1109/TMBMC.2016.2630054.

[41] Yifan Chen, Daniel Zhengyu Huang, Jiaoyang Huang, Sebastian Reich, and Andrew M. Stuart. Sampling via gradient flows in the space of probability measures, 2023.

[42] Yizong Cheng. Mean shift, mode seeking, and clustering. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 17(8):790–799, 1995. doi: 10.1109/34.400568.

[43] Frank H. Clarke. *Optimization and Nonsmooth Analysis*. Wiley, New York, 1983.

[44] Jorge Cortés. Finite-time convergent gradient flows with applications to network consensus. *Automatica*, 42(11):1993–2000, 2006. doi: 10.1016/j.automatica.2006.06.015.

[45] Thomas A. Courtade and Tsachy Weissman. Multiterminal source coding under logarithmic loss. *IEEE Transactions on Information Theory*, 60(1):740–761, 2014. doi: 10.1109/TIT.2013.2288257.

[46] Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory*. Wiley, Hoboken, NJ, second edition, 2006.

[47] Michael G. Crandall and Thomas M. Liggett. Generation of semi-groups of nonlinear transformations on general Banach spaces. *American Journal of Mathematics*, 93(2):265–298, 1971. doi: 10.2307/2373376.

[48] Imre Csiszár. I-divergence geometry of probability distributions and minimization problems. *The Annals of Probability*, 3(1):146–158, 1975.

[49] Imre Csiszár. Sanov property, generalized I-projection and a conditional limit theorem. *The Annals of Probability*, 12(3):768–793, 1984.

[50] Walter J. Culver. On the existence and uniqueness of the real logarithm of a matrix. *Proceedings of the American Mathematical Society*, 17(5):1146–1151, 1966. doi: 10.1090/S0002-9939-1966-0202740-6.

[51] A. D'Adda, M. Lüscher, and P. Di Vecchia. A $1/n$ expandable series of non-linear σ models with instantons. *Nuclear Physics B*, 146(1):63–76, 1978. doi: 10.1016/0550-3213(78)90432-7.

[52] Donald Davidson. Radical interpretation. *Dialectica*, 27(3–4):313–328, 1973.

[53] Donald Davidson. Belief and the basis of meaning. *Synthese*, 27(3–4):309–323, 1974. doi: 10.1007/BF00484597.

[54] Chandler Davis and W. M. Kahan. The rotation of eigenvectors by a perturbation. III. *SIAM Journal on Numerical Analysis*, 7(1):1–46, 1970. doi: 10.1137/0707001.

[55] M. H. A. Davis. Piecewise-deterministic Markov processes: a general class of non-diffusion stochastic models. *Journal of the Royal Statistical Society: Series B*, 46(3):353–388, 1984.

[56] A. P. Dawid. Conditional independence in statistical theory. *Journal of the Royal Statistical Society: Series B (Methodological)*, 41(1):1–31, 1979. doi: 10.1111/j.2517-6161.1979.tb01052.x.

[57] Gaspard Riche de Prony. Essai expérimental et analytique: sur les lois de la dilatabilité des fluides élastiques et sur celles de la force expansive de la vapeur de l'alkool, à différentes températures. *Journal de l'École Polytechnique*, 1(22):24–76, 1795.

[58] Morris H. DeGroot. Reaching a consensus. *Journal of the American Statistical Association*, 69(345):118–121, 1974.

[59] Peter M. DeMarzo, Dimitri Vayanos, and Jeffrey Zwiebel. Persuasion bias, social influence, and unidimensional opinions. *The Quarterly Journal of Economics*, 118(3):909–968, 2003. doi: 10.1162/00335530360698469.

[60] A. P. Dempster, N. M. Laird, and D. B. Rubin. Maximum likelihood from incomplete data via the EM algorithm. *Journal of the Royal Statistical Society: Series B*, 39(1):1–22, 1977. doi: 10.1111/j.2517-6161.1977.tb01600.x.

[61] Salem Derisavi, Holger Hermanns, and William H. Sanders. Optimal state-space lumping in Markov chains. *Information Processing Letters*, 87(6):309–315, 2003. doi: 10.1016/S0020-0190(03)00343-0.

[62] Yash Deshpande, Elchanan Mossel, and Youngtak Sohn. Agreement and statistical efficiency in Bayesian perception models, 2022. Version 3, 9 August 2023.

[63] Persi Diaconis and Sandy L. Zabell. Updating subjective probability. *Journal of the American Statistical Association*, 77(380):822–830, 1982.

[64] Jean Dieudonné. *Foundations of Modern Analysis*. Academic Press, New York, 1960.

[65] John D. Dixon and Brian Mortimer. *Permutation Groups*, volume 163 of *Graduate Texts in Mathematics*. Springer, New York, 1996. doi: 10.1007/978-1-4612-0731-3.

[66] Monroe D. Donsker and S. R. Srinivasa Varadhan. Asymptotic evaluation of certain Markov process expectations for large time, I. *Communications on Pure and Applied Mathematics*, 28(1):1–47, 1975.

[67] Paul Dupuis and Richard S. Ellis. *A Weak Convergence Approach to the Theory of Large Deviations*. Wiley, New York, 1997.

[68] E. B. Dynkin. *Markov Processes, Volume I*, volume 121 of *Grundlehren der mathematischen Wissenschaften*. Springer, Berlin, 1965.

[69] Beno Eckmann. Harmonische Funktionen und Randwertaufgaben in einem Komplex. *Commentarii Mathematici Helvetici*, 17(1):240–255, 1944. doi: 10.1007/BF02566245.

[70] James Eells, Jr. and J. H. Sampson. Harmonic mappings of Riemannian manifolds. *American Journal of Mathematics*, 86(1):109–160, 1964. doi: 10.2307/2373037.

[71] Jaap Eldering, Matthew Kvalheim, and Shai Revzen. Global linearization and fiber bundle structure of invariant manifolds. *Nonlinearity*, 31(9):4202–4245, 2018. doi: 10.1088/1361-6544/aaca8d. URL https://arxiv.org/abs/1711.03646.

[72] Mark Fels and Peter J. Olver. Moving coframes: I. A practical algorithm. *Acta Applicandae Mathematicae*, 51(2):161–213, 1998. doi: 10.1023/A:1005878210297.

[73] Neil Fenichel. Asymptotic stability with rate conditions. *Indiana University Mathematics Journal*, 23 (12):1109–1137, 1974. doi: 10.1512/iumj.1974.23.23090.

[74] Michael J. Field. *Dynamics and Symmetry*, volume 3 of *ICP Advanced Texts in Mathematics*. Imperial College Press, London, 2007.

[75] Aleksei F. Filippov. *Differential Equations with Discontinuous Righthand Sides*, volume 18 of *Mathematics and Its Applications (Soviet Series)*. Kluwer Academic Publishers, Dordrecht, 1988.

[76] Robin Forman. Determinants of Laplacians on graphs. *Topology*, 32(1):35–46, 1993. doi: 10.1016/0040-9383(93)90035-T.

[77] Cédric Fournel, Jordan François, Serge Lazzarini, and Thierry Masson. Gauge invariant composite fields out of connections, with examples. *International Journal of Geometric Methods in Modern Physics*, 11(3):1450016, 2014. doi: 10.1142/S0219887814500169.

[78] Jordan François. Artificial versus substantial gauge symmetries: A criterion and an application to the electroweak model. *Philosophy of Science*, 86(3):472–496, 2019. doi: 10.1086/703571.

[79] Noah E. Friedkin and Eugene C. Johnsen. Social influence and opinions. *The Journal of Mathematical Sociology*, 15(3–4):193–206, 1990. doi: 10.1080/0022250X.1990.9990069.

[80] Karl Friston. The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience*, 11(2): 127–138, 2010.

[81] Karl Friston and Christopher Frith. A duet for one. *Consciousness and Cognition*, 36:390–405, 2015.

[82] Karl J. Friston, Thomas Parr, Yan Yufik, Noor Sajid, Catherine J. Price, and Emma Holmes. Generative models, linguistic communication and active inference. *Neuroscience and Biobehavioral Reviews*, 118: 42–64, 2020.

[83] Karl J. Friston, Erik D. Fagerholm, Tahereh S. Zarghami, Thomas Parr, Inês Hipólito, Loïc Magrou, and Adeel Razi. Parcels and particles: Markov blankets in the brain. *Network Neuroscience*, 5(1):211–251, 2021. doi: 10.1162/netn_a_00175.

[84] Tobias Fritz. A synthetic approach to Markov kernels, conditional independence and theorems on sufficient statistics. *Advances in Mathematics*, 370:107239, 2020.

[85] Michel Fruchart, Ryo Hanai, Peter B. Littlewood, and Vincenzo Vitelli. Non-reciprocal phase transitions. *Nature*, 592:363–369, 2021.

[86] Tingran Gao, Jacek Brodzki, and Sayan Mukherjee. The geometry of synchronization problems and learning group actions. *Discrete & Computational Geometry*, 65(1):150–211, 2021. doi: 10.1007/s00454-019-00100-2.

[87] John D. Geanakoplos and Heraklis M. Polemarchakis. We can't disagree forever. *Journal of Economic Theory*, 28(1):192–200, 1982. doi: 10.1016/0022-0531(82)90099-0.

[88] Christian Genest and James V. Zidek. Combining probability distributions: A critique and an annotated bibliography. *Statistical Science*, 1(1):114–135, 1986. doi: 10.1214/ss/1177013825.

[89] Borjan Geshkovski, Cyril Letrouit, Yury Polyanskiy, and Philippe Rigollet. A mathematical perspective on Transformers. *Bulletin of the American Mathematical Society*, 62(3):427–479, 2025. arXiv:2312.10794.

[90] S. G. Gindikin. Invariant generalized functions in homogeneous domains. *Functional Analysis and Its Applications*, 9(1):50–52, 1975. doi: 10.1007/BF01078179.

[91] Petra M. Gleiss, Josef Leydold, and Peter F. Stadler. Circuit bases of strongly connected digraphs. *Discussiones Mathematicae Graph Theory*, 23(2):241–260, 2003. doi: 10.7151/dmgt.1200.

[92] Martin Golubitsky and David G. Schaeffer. *Singularities and Groups in Bifurcation Theory, Volume I*, volume 51 of *Applied Mathematical Sciences*. Springer, New York, 1985. doi: 10.1007/978-1-4612-5034-0.

[93] Martin Golubitsky and Ian Stewart. *The Symmetry Perspective: From Equilibrium to Chaos in Phase Space and Physical Space*, volume 200 of *Progress in Mathematics*. Birkhäuser, Basel, 2002. doi: 10.1007/978-3-0348-8167-8.

[94] Martin Golubitsky, Ian Stewart, and David G. Schaeffer. *Singularities and Groups in Bifurcation Theory, Volume II*, volume 69 of *Applied Mathematical Sciences*. Springer-Verlag, New York, 1988.

[95] Robert M. Gray. *Entropy and Information Theory*. Springer-Verlag, New York, 1990. URL https://www-ee.stanford.edu/~gray/it.pdf. Corrected first edition, revised June 26, 2023.

[96] Robert B. Griffiths. Thermodynamics near the two-fluid critical mixing point in He^3-He^4. *Physical Review Letters*, 24(13):715–717, 1970. doi: 10.1103/PhysRevLett.24.715.

[97] John Guckenheimer. Isochrons and phaseless sets. *Journal of Mathematical Biology*, 1(3):259–273, 1975. doi: 10.1007/BF01273747.

[98] A. Guichardet. On rotation and vibration motions of molecules. *Annales de l'I.H.P. Physique théorique*, 40(3):329–342, 1984. URL https://www.numdam.org/item/AIHPA_1984__40_3_329_0/.

[99] Paul Gustafson. Bayesian inference for partially identified models. *The International Journal of Biostatistics*, 6(2):Article 17, 2010. doi: 10.2202/1557-4679.1206.

[100] Paul R. Halmos and L. J. Savage. Application of the Radon–Nikodym theorem to the theory of sufficient statistics. *The Annals of Mathematical Statistics*, 20(2):225–241, 1949.

[101] Jakob Hansen and Robert Ghrist. Toward a spectral theory of cellular sheaves. *Journal of Applied and Computational Topology*, 3(4):315–358, 2019. doi: 10.1007/s41468-019-00038-7.

[102] Jakob Hansen and Robert Ghrist. Opinion dynamics on discourse sheaves. *SIAM Journal on Applied Mathematics*, 81(5):2033–2060, 2021.

[103] Philip Hartman. *Ordinary Differential Equations*, volume 38 of *Classics in Applied Mathematics*. Society for Industrial and Applied Mathematics, Philadelphia, PA, second edition, 2002. doi: 10.1137/1.9780898719222.

[104] Emilie V. Haynsworth. Determination of the inertia of a partitioned Hermitian matrix. *Linear Algebra and its Applications*, 1(1):73–81, 1968.

[105] Rainer Hegselmann and Ulrich Krause. Opinion dynamics and bounded confidence: models, analysis and simulation. *Journal of Artificial Societies and Social Simulation*, 5(3), 2002.

[106] Robert Hermann and Arthur J. Krener. Nonlinear controllability and observability. *IEEE Transactions on Automatic Control*, 22(5):728–740, 1977. doi: 10.1109/TAC.1977.1101601.

[107] Nicholas J. Higham. *Functions of Matrices: Theory and Computation*. Society for Industrial and Applied Mathematics, Philadelphia, 2008. doi: 10.1137/1.9780898717778.

[108] Toby Dylan Hocking, Armand Joulin, Francis Bach, and Jean-Philippe Vert. Clusterpath: An algorithm for clustering using convex fusion penalties. In *Proceedings of the 28th International Conference on Machine Learning*, pages 745–752, 2011. URL https://icml.cc/2011/papers/419_icmlpaper.pdf.

[109] Josef Hofbauer and Karl Sigmund. *Evolutionary Games and Population Dynamics*. Cambridge University Press, Cambridge, 1998.

[110] Leonhard Horstmeyer and Fatihcan M. Atay. Characterization of exact lumpability for vector fields on smooth manifolds. *Differential Geometry and its Applications*, 48:46–60, 2016. doi: 10.1016/j.difgeo.2016.06.001.

[111] Ali Jadbabaie, Pooya Molavi, Alvaro Sandroni, and Alireza Tahbaz-Salehi. Non-Bayesian social learning. *Games and Economic Behavior*, 76(1):210–225, 2012. doi: 10.1016/j.geb.2012.06.001.

[112] Michael I. Jordan, Zoubin Ghahramani, Tommi S. Jaakkola, and Lawrence K. Saul. An introduction to variational methods for graphical models. *Machine Learning*, 37(2):183–233, 1999.

[113] Abram Kagan and Lawrence A. Shepp. A sufficiency paradox: An insufficient statistic preserving the Fisher information. *The American Statistician*, 59(1):54–56, 2005. doi: 10.1198/000313005X21041.

[114] Olav Kallenberg. *Foundations of Modern Probability*, volume 99 of *Probability Theory and Stochastic Modelling*. Springer, Cham, 3 edition, 2021. doi: 10.1007/978-3-030-61871-1.

[115] R. E. Kalman. A new approach to linear filtering and prediction problems. *Journal of Basic Engineering*, 82(1):35–45, 1960.

[116] Emir Kamenica and Matthew Gentzkow. Bayesian persuasion. *American Economic Review*, 101(6): 2590–2615, 2011. doi: 10.1257/aer.101.6.2590.

[117] Alexander S. Kechris. *Classical Descriptive Set Theory*, volume 156 of *Graduate Texts in Mathematics*. Springer, New York, 1995.

[118] John G. Kemeny and J. Laurie Snell. *Finite Markov Chains*. Springer, New York, 1976. ISBN 9780387901923. Reprint of the 1960 edition, with a new appendix.

[119] David G. Kendall. Shape manifolds, Procrustean metrics, and complex projective spaces. *Bulletin of the London Mathematical Society*, 16(2):81–121, 1984. doi: 10.1112/blms/16.2.81.

[120] Richard Kenyon. Spanning forests and the vector bundle Laplacian. *The Annals of Probability*, 39(5): 1983–2017, 2011. doi: 10.1214/10-AOP596.

[121] Michael Kirchhoff, Thomas Parr, Ensor Palacios, Karl Friston, and Julian Kiverstein. The Markov blankets of life: Autonomy, active inference and the free energy principle. *Journal of the Royal Society Interface*, 15(138):20170792, 2018. doi: 10.1098/rsif.2017.0792.

[122] Shoshichi Kobayashi and Katsumi Nomizu. *Foundations of Differential Geometry, Volume I*. Interscience Publishers, New York, 1963.

[123] Solomon Kullback. *Information Theory and Statistics*. Wiley, New York, 1959.

[124] Solomon Kullback and Richard A. Leibler. On information and sufficiency. *The Annals of Mathematical Statistics*, 22(1):79–86, 1951. doi: 10.1214/aoms/1177729694.

[125] Martin Kummer. On the construction of the reduced phase space of a Hamiltonian system with symmetry. *Indiana University Mathematics Journal*, 30(2):281–291, 1981. doi: 10.1512/iumj.1981.30.30022.

[126] G. N. Lance and W. T. Williams. A general theory of classificatory sorting strategies. I. Hierarchical systems. *The Computer Journal*, 9(4):373–380, 1967. doi: 10.1093/comjnl/9.4.373.

[127] Kim G. Larsen and Arne Skou. Bisimulation through probabilistic testing. *Information and Computation*, 94(1):1–28, 1991. doi: 10.1016/0890-5401(91)90030-6.

[128] Joseph P. LaSalle. Some extensions of Liapunov's second method. *IRE Transactions on Circuit Theory*, 7(4):520–527, 1960. doi: 10.1109/TCT.1960.1086720.

[129] Steffen L. Lauritzen. *Graphical Models*, volume 17 of *Oxford Statistical Science Series*. Clarendon Press, Oxford, 1996.

[130] Lieven Le Bruyn and Claudio Procesi. Semisimple representations of quivers. *Transactions of the American Mathematical Society*, 317(2):585–598, 1990. doi: 10.1090/S0002-9947-1990-0958897-0.

[131] Lucien Le Cam and Grace Lo Yang. *Asymptotics in Statistics: Some Basic Concepts*. Springer, New York, second edition, 2000. doi: 10.1007/978-1-4612-1166-2.

[132] John M. Lee. *Introduction to Smooth Manifolds*, volume 218 of *Graduate Texts in Mathematics*. Springer, New York, 2 edition, 2013. doi: 10.1007/978-1-4419-9982-5.

[133] John M. Lee. *Introduction to Riemannian Manifolds*, volume 176 of *Graduate Texts in Mathematics*. Springer, Cham, 2 edition, 2018. doi: 10.1007/978-3-319-91755-9.

[134] Melvin Leok, Jerrold E. Marsden, and Alan D. Weinstein. A discrete theory of connections on principal bundles, 2005. URL https://arxiv.org/abs/math/0508338.

[135] Gérard Letac and Hélène Massam. The Laplace transform $(\det s)^{-p} \exp \operatorname{tr}(s^{-1}w)$ and the existence of non-central Wishart distributions. *Journal of Multivariate Analysis*, 163:96–110, 2018. doi: 10.1016/j.jmva.2017.10.005.

[136] Debra Lewis, Nilima Nigam, and Peter J. Olver. Connections for general group actions. *Communications in Contemporary Mathematics*, 7(3):341–374, 2005. doi: 10.1142/S0219199705001787.

[137] Genyuan Li and Herschel Rabitz. A general analysis of exact lumping in chemical kinetics. *Chemical Engineering Science*, 44(6):1413–1430, 1989. doi: 10.1016/0009-2509(89)85014-6.

[138] Lek-Heng Lim. Hodge Laplacians on graphs. *SIAM Review*, 62(3):685–715, 2020. doi: 10.1137/18M1223101.

[139] Robert G. Littlejohn and Matthias Reinsch. Gauge fields in the separation of rotations and internal motions in the n-body problem. *Reviews of Modern Physics*, 69(1):213–275, 1997. doi: 10.1103/RevModPhys.69.213.

[140] Stanisław Łojasiewicz. Une propriété topologique des sous-ensembles analytiques réels. In *Les Équations aux Dérivées Partielles*, pages 87–89. Éditions du Centre National de la Recherche Scientifique, Paris, 1963.

[141] Thomas A. Louis. Finding the observed information matrix when using the EM algorithm. *Journal of the Royal Statistical Society: Series B*, 44(2):226–233, 1982.

[142] Kanti V. Mardia and Peter E. Jupp. *Directional Statistics*. Wiley, Chichester, 2000.

[143] Jerrold E. Marsden, Richard Montgomery, and Tudor Ratiu. *Reduction, Symmetry, and Phases in Mechanics*, volume 88 of *Memoirs of the American Mathematical Society*. American Mathematical Society, Providence, RI, 1990. doi: 10.1090/memo/0436.

[144] Fortuné Massamba and George Thompson. The universal connection and metrics on moduli spaces. *Communications in Mathematical Physics*, 253(2):323–335, 2005. doi: 10.1007/s00220-004-1191-7.

[145] Alexandre Mauroy, Igor Mezić, and Jeff Moehlis. Isostables, isochrons, and Koopman spectrum for the action-angle representation of stable fixed point dynamics. *Physica D: Nonlinear Phenomena*, 261:19–30, 2013. doi: 10.1016/j.physd.2013.06.004.

[146] Thomas P. Minka. Expectation propagation for approximate Bayesian inference. In *Proceedings of the Seventeenth Conference on Uncertainty in Artificial Intelligence*, pages 362–369. Morgan Kaufmann, 2001. URL https://tminka.github.io/papers/ep/minka-ep-uai.pdf.

[147] Pierre Molino. *Riemannian Foliations*, volume 73 of *Progress in Mathematics*. Birkhäuser, Boston, 1988. doi: 10.1007/978-1-4684-8670-4.

[148] Dov Monderer and Lloyd S. Shapley. Potential games. *Games and Economic Behavior*, 14(1):124–143, 1996.

[149] Robb J. Muirhead. *Aspects of Multivariate Statistical Theory*. John Wiley & Sons, New York, 1982.

[150] M. S. Narasimhan and S. Ramanan. Existence of universal connections. *American Journal of Mathematics*, 83(3):563–572, 1961. doi: 10.2307/2372896.

[151] M. S. Narasimhan and S. Ramanan. Existence of universal connections II. *American Journal of Mathematics*, 85(2):223–231, 1963. doi: 10.2307/2373211.

[152] Ashutosh Nayyar, Aditya Mahajan, and Demosthenis Teneketzis. Decentralized stochastic control with partial history sharing: A common information approach. *IEEE Transactions on Automatic Control*, 58(7):1644–1658, 2013. doi: 10.1109/TAC.2013.2239000. URL https://cim.mcgill.ca/~adityam/projects/info-structures/journal/partial-sharing.pdf.

[153] Radford M. Neal and Geoffrey E. Hinton. A view of the EM algorithm that justifies incremental, sparse, and other variants. In Michael I. Jordan, editor, *Learning in Graphical Models*, pages 355–368. Kluwer, Dordrecht, 1998.

[154] Barrett O'Neill. The fundamental equations of a submersion. *Michigan Mathematical Journal*, 13(4):459–469, 1966. doi: 10.1307/mmj/1028999604.

[155] Brad E. Paden and Shankar S. Sastry. A calculus for computing Filippov's differential inclusion with application to the variable structure control of robot manipulators. *IEEE Transactions on Circuits and Systems*, 34(1):73–82, 1987. doi: 10.1109/TCS.1987.1086038.

[156] Ensor Rafael Palacios, Adeel Razi, Thomas Parr, Michael Kirchhoff, and Karl Friston. On Markov blankets and hierarchical self-organisation. *Journal of Theoretical Biology*, 486:110089, 2020. doi: 10.1016/j.jtbi.2019.110089.

[157] Richard S. Palais. On the existence of slices for actions of non-compact Lie groups. *Annals of Mathematics*, 73(2):295–323, 1961. doi: 10.2307/1970335.

[158] Richard S. Palais. The principle of symmetric criticality. *Communications in Mathematical Physics*, 69 (1):19–30, 1979. doi: 10.1007/BF01941322.

[159] George J. Pappas, Gerardo Lafferriere, and Shankar Sastry. Hierarchically consistent control systems. *IEEE Transactions on Automatic Control*, 45(6):1144–1160, 2000. doi: 10.1109/9.863598.

[160] Judea Pearl. *Probabilistic Reasoning in Intelligent Systems: Networks of Plausible Inference*. Morgan Kaufmann, San Mateo, CA, 1988.

[161] Dénes Petz. Sufficiency of channels over von Neumann algebras. *The Quarterly Journal of Mathematics*, 39(1):97–108, 1988.

[162] Julianna Pinele, João E. Strapasson, and Sueli I. R. Costa. The Fisher–Rao distance between multivariate normal distributions: Special cases, bounds and applications. *Entropy*, 22(4):404, 2020. doi: 10.3390/e22040404.

[163] Dale J. Poirier. Revising beliefs in nonidentified models. *Econometric Theory*, 14(4):483–509, 1998. doi: 10.1017/S0266466698144043.

[164] David Pollard. A note on insufficiency and the preservation of Fisher information. In *From Probability to Statistics and Back: High-Dimensional Models and Processes*, volume 9 of *IMS Collections*, pages 266–275. Institute of Mathematical Statistics, 2013. doi: 10.1214/12-IMSCOLL919. URL https://arxiv.org/abs/1107.3797.

[165] George R. Price. Selection and covariance. *Nature*, 227:520–521, 1970. doi: 10.1038/227520a0.

[166] Claudio Procesi. The invariant theory of $n \times n$ matrices. *Advances in Mathematics*, 19(3):306–381, 1976. doi: 10.1016/0001-8708(76)90027-X.

[167] J. P. Provost and G. Vallee. Riemannian structure on manifolds of quantum states. *Communications in Mathematical Physics*, 76(3):289–301, 1980. doi: 10.1007/BF02193559.

[168] Willard Van Orman Quine. *Word and Object*. MIT Press, Cambridge, MA, 1960.

[169] Bruce L. Reinhart. Foliated manifolds with bundle-like metrics. *Annals of Mathematics*, 69(1):119–132, 1959. doi: 10.2307/1970097.

[170] L. C. G. Rogers and J. W. Pitman. Markov functions. *The Annals of Probability*, 9(4):573–582, 1981. doi: 10.1214/aop/1176994363.

[171] Clarence W. Rowley and Jerrold E. Marsden. Reconstruction equations and the Karhunen–Loève expansion for systems with symmetry. *Physica D: Nonlinear Phenomena*, 142(1–2):1–19, 2000. doi: 10.1016/S0167-2789(00)00042-7.

[172] Clarence W. Rowley, Ioannis G. Kevrekidis, Jerrold E. Marsden, and Kurt Lust. Reduction and reconstruction for self-similar dynamical systems. *Nonlinearity*, 16(4):1257–1275, 2003. doi: 10.1088/0951-7715/16/4/304.

[173] David Ruelle. Bifurcations in the presence of a symmetry group. *Archive for Rational Mechanics and Analysis*, 51:136–152, 1973. URL https://www.ihes.fr/~ruelle/PUBLICATIONS/[39].pdf.

[174] Daniil Ryabko. Independence clustering (without a matrix), 2017. URL https://arxiv.org/abs/1703.06700.

[175] Peter H. Schönemann. A generalized solution of the orthogonal Procrustes problem. *Psychometrika*, 31(1):1–10, 1966. doi: 10.1007/BF02289451.

[176] Siavash Shahshahani. *A New Mathematical Framework for the Study of Linkage and Selection*, volume 17 of *Memoirs of the American Mathematical Society*. American Mathematical Society, Providence, RI, 1979.

[177] Cosma Rohilla Shalizi and James P. Crutchfield. Computational mechanics: Pattern and prediction, structure and simplicity. *Journal of Statistical Physics*, 104(3–4):817–879, 2001. doi: 10.1023/A:1010388907793.

[178] Claude E. Shannon. Coding theorems for a discrete source with a fidelity criterion. *IRE International Convention Record*, 7(4):142–163, 1959.

[179] A. Singer. Angular synchronization by eigenvectors and semidefinite programming. *Applied and Computational Harmonic Analysis*, 30(1):20–36, 2011. doi: 10.1016/j.acha.2010.02.001.

[180] Amit Singer and Hau-Tieng Wu. Vector diffusion maps and the connection Laplacian. *Communications on Pure and Applied Mathematics*, 65(8):1067–1144, 2012.

[181] Morris Skibinsky. Adequate subfields and sufficiency. *The Annals of Mathematical Statistics*, 38(1): 155–161, 1967.

[182] Stephen Smale. Topology and mechanics. I. *Inventiones Mathematicae*, 10:305–331, 1970. doi: 10.1007/BF01418778.

[183] Robert R. Sokal and Charles D. Michener. A statistical method for evaluating systematic relationships. *The University of Kansas Science Bulletin*, 38(22):1409–1438, 1958.

[184] H. W. Sorenson and D. L. Alspach. Recursive Bayesian estimation using Gaussian sums. *Automatica*, 7(4):465–479, 1971. doi: 10.1016/0005-1098(71)90097-5.

[185] Michael E. Taylor. *Partial Differential Equations III: Nonlinear Equations*, volume 117 of *Applied Mathematical Sciences*. Springer, New York, 2 edition, 2011. doi: 10.1007/978-1-4419-7049-7.

[186] Naftali Tishby, Fernando C. Pereira, and William Bialek. The information bottleneck method, 1999. URL https://arxiv.org/abs/physics/0004057. Manuscript dated September 30, 1999; arXiv:physics/0004057, posted 2000.

[187] David Tong. Gauge theory. University of Cambridge lecture notes, 2018. URL https://www.damtp.cam.ac.uk/user/tong/gaugetheory/gt.pdf.

[188] Flemming Topsøe. Information-theoretical optimization techniques. *Kybernetika*, 15(1):8–27, 1979.

[189] Erik Torgersen. *Comparison of Statistical Experiments*, volume 36 of *Encyclopedia of Mathematics and its Applications*. Cambridge University Press, Cambridge, 1991.

[190] A. W. van der Vaart. *Asymptotic Statistics*, volume 3 of *Cambridge Series in Statistical and Probabilistic Mathematics*. Cambridge University Press, Cambridge, 1998. doi: 10.1017/CBO9780511802256.

[191] S. J. van Enk. Quantum communication, reference frames, and gauge theory. *Physical Review A*, 73: 042306, 2006.

[192] Tim van Erven and Peter Harremoës. Rényi divergence and Kullback–Leibler divergence. *IEEE Transactions on Information Theory*, 60(7):3797–3820, 2014. doi: 10.1109/TIT.2014.2320500.

[193] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, pages 5998–6008, 2017.

[194] N. N. Vorob'ev. Consistent families of measures and their extensions. *Theory of Probability and Its Applications*, 7(2):147–163, 1962. doi: 10.1137/1107014.

[195] Peter Walters. *An Introduction to Ergodic Theory*, volume 79 of *Graduate Texts in Mathematics*. Springer-Verlag, New York, 1982.

[196] Massimo Warglien and Peter Gärdenfors. Semantics, conceptual spaces, and the meeting of minds. *Synthese*, 190(12):2165–2193, 2013.

[197] James Wei and James C. W. Kuo. A lumping analysis in monomolecular reaction systems: Analysis of the exactly lumpable system. *Industrial & Engineering Chemistry Fundamentals*, 8(1):114–123, 1969. doi: 10.1021/i160029a019.

[198] Yair Weiss and William T. Freeman. Correctness of belief propagation in Gaussian graphical models of arbitrary topology. In *Advances in Neural Information Processing Systems*, volume 12, 1999.

[199] Kenneth G. Wilson. Confinement of quarks. *Physical Review D*, 10(8):2445–2459, 1974. doi: 10.1103/PhysRevD.10.2445.

[200] Aaron D. Wyner and Jacob Ziv. The rate-distortion function for source coding with side information at the decoder. *IEEE Transactions on Information Theory*, 22(1):1–10, 1976. doi: 10.1109/TIT.1976.1055508.

[201] Ram Zamir. A proof of the Fisher information inequality via a data processing argument. *IEEE Transactions on Information Theory*, 44(3):1246–1250, 1998. doi: 10.1109/18.669301.

[202] Noga Zaslavsky, Charles Kemp, Terry Regier, and Naftali Tishby. Efficient compression in color naming and its evolution. *Proceedings of the National Academy of Sciences*, 115(31):7937–7942, 2018.